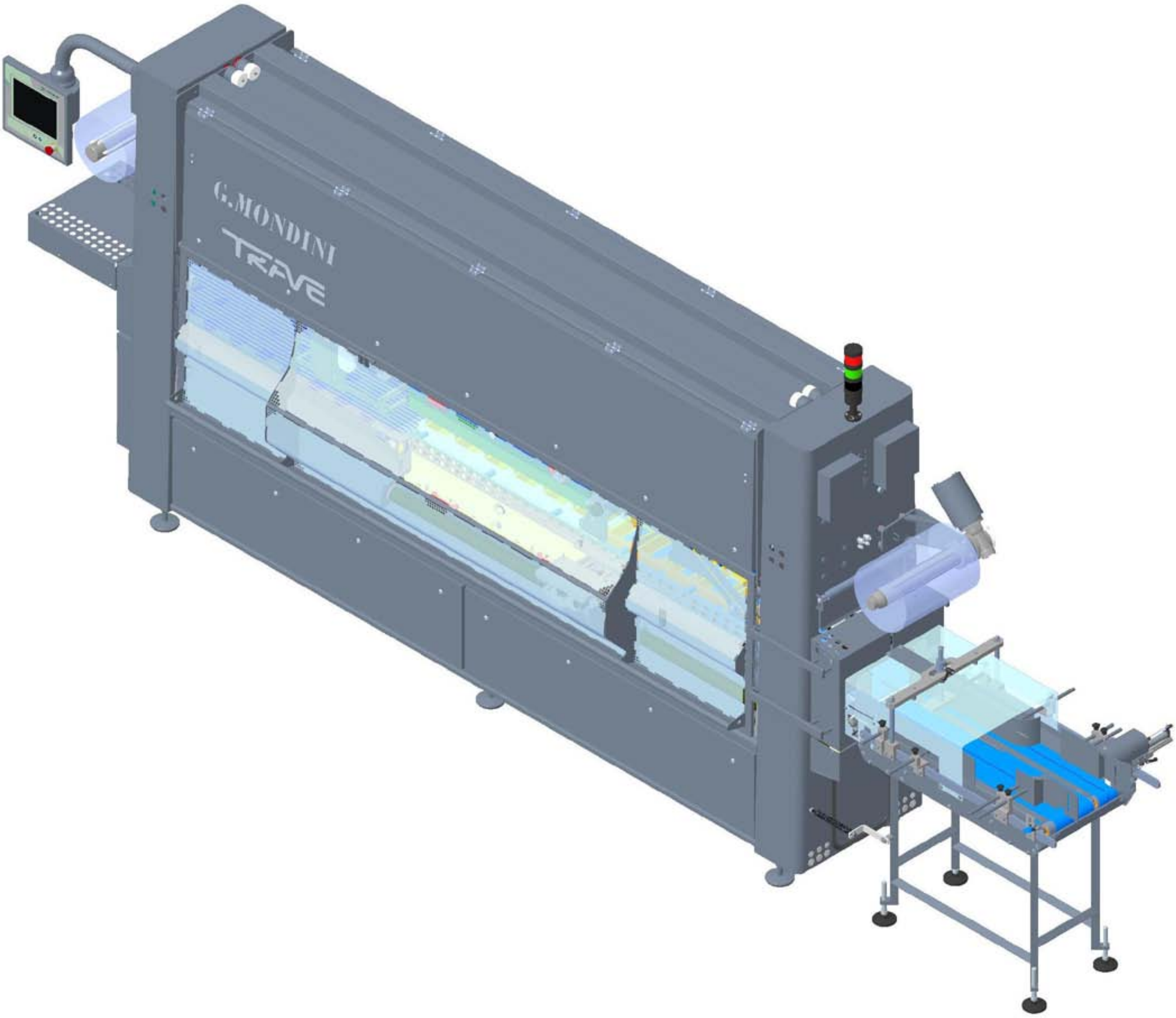
G. MONDINI S.p.A.
Via Brescia, 5-7
25033 COLOGNE (BS) – ITALIA

CLOSING MACHINE
TRAVE 1000 VG



User and maintenance manual

ORIGINAL DOCUMENTS

Construction year: 2020

Serial number: A/20/A-12309

GENERAL INDEX

1. FOREWORD

2. TECHNICAL SPECIFICATIONS

3. SAFETY

4. TRANSPORT AND INSTALLATION

5. CONTROL SYSTEM

6. OPERATION

7. DIAGNOSTICS

8. MAINTENANCE

9. DISPOSAL

ANNEX A. SEALING TOOL TROLLEY CPS/R

ANNEX B. TRAYS ROTATION BELT

1 FOREWORD

CONTENTS

1.1 OVERVIEW

This handbook contains all the information needed for correct installation and use, and appropriate maintenance of the line involved.

G.MONDINI S.p.A. insists that this document be read by the persons assigned to the running and the maintenance of the machine, as well as by the persons executing the transport and assembly operations.

This document is the instructions handbook for the **CLOSING MACHINE TRAVE 1000 VG** and has been drawn up in conformity with directive 2006/42/EC annex I paragraph 1.7.4. The Use and Maintenance Handbook is to be considered an integral part of the line and is to be kept until the final dismantling. The handbook is to be kept by the person in charge of the system after its installation.

The handbook reflects the technical state at the time of the line marketing and shall not be considered inadequate if later versions are updated on the basis of new experience, furthermore G.MONDINI S.p.A. reserves the right to update production and handbooks with no obligation to update previous production or handbooks.

The drawings and other documents that accompany the system are the property of G.MONDINI S.p.A. that reserves the rights and underlines that these documents are not to be handed over to third parties. It is therefore forbidden to duplicate them in any way, either electronically or mechanically, for any type of use, without first receiving the written permission from G.MONDINI S.p.A.

G.MONDINI S.p.A. shall not be held in any way liable for any direct or indirect damage to persons, goods or animals caused by the use of this documentation and/or the line under any conditions that differ from those foreseen.

1.1.1 Spare parts

It is recommended to use only ORIGINAL SPARE PARTS.

To order spare components send a form like that shown in figure A to G.MONDINI.

The system is marked with a code and a serial number, indicated on the ID plate.

The component description and code, as well as the number of the assembly and the table the component belongs to can be traced in the lists attached to the handbook.

MODULO ORDINAZIONE RICAMBI

Spett.le: **G.MONDINI S.p.A.**

Dosatrici - confezionatrici automatiche
- 25033 COLOGNE (Brescia) ITALIA, Via Brescia 5/7 -
Tel. +39 030 705600 - Fax +39 030 7056250
E-mail: - **info@gmondini.com** - **www.gmondini.com** - **spare_parts@gmondini.com.**

Richiediamo la spedizione dei seguenti ricambi:

Nº MATRICOLA: ____________ **TIPO DI MACCHINA:** ____________

Q.tà	Nº codice pezzo	Nº gruppo	Nº tavola	Descrizione pezzo

DITTA: .... p.I.V.A. Nº:..

Via: .... Nº: ....

Loc.: .... Città: .... C.a.p.:.............. Prov.: ....

INVIO A ½: ☐ Corriere ☐ Corriere espresso ☐ Posta
☐ ....
PAGAMENTO: ☐ Contrassegno ☐ Bonifico bancario (v. ricevuta allegata)
☐ ....

La richiesta è stata compilata da: COGNOME .... NOME ....

In qualità di: ....

Le richieste non saranno evase se mancanti di tutte le indicazioni della ditta richiedente nonché delle generalità complete e di firma del compilatore. La merce viaggia a rischio del committente. Non si accettano restituzioni di materiale in caso di errori di compilazione del modulo.

TIMBRO e FIRMA

Figure A: Spare parts order form.

1.1.2 How to request services

For any type of information or explanations regarding the use, maintenance , installation etc., the G. MONDINI S.p.A. technical department is always at the disposal of the Customer.

With regard to requests, the questions should be set out in clear terms , with reference to this handbook and always indicating the data found on the ID plate of the line.

Any request for after-sales service in the Customer's plant, or questions regarding technical aspects of this document are to be addressed to:

G.MONDINI S.p.A.
Dosatrici – confezionatrici automatiche
25033 COLOGNE (Brescia), ITALIA, Via Brescia 5/7
Tel. +39 030 705600 Fax. +39 030 7056250
E-mail: info@gmondini.com – www.gmondini.com –
spare_parts@gmondini.com

1.1.3 " CE" Declaration of Conformity

" CE" Declaration of Conformity

(in conformity with Machines Directive, annex II letter A)

TRANSLATION OF ORIGINAL DOCUMENTS

The Manufacturer	**G.MONDINI S.p.A.**	
Address	Via Brescia, 5-7 – 25033 COLOGNE (BS) – Italy	
Tel.	+39-030-705600	
Telefax	+39-030-7056250	

HEREBY DECLARES THAT THE FOLLOWING MACHINE

Machine type	N/R	CPS/R
	TRAVE 1000 VG	CPS/R
Serial number	A/20/A-12309.30	A/20/A-12309.50
	A/20/A-12309.40	A/20/A-12309.51
Product handled	FOOD PRODUCTS	
Type of machine	DOSING AND PACKAGING LINE	
G. MONDINI S.p.A. job number	A/20/A-12309	
Year of built	2020	

COMPLIES WITH THE PROVISIONS OF LAWS IMPLEMENTING MACHINERY DIRECTIVE 2006/42/EC AND SUBSEQUENT AMENDMENTS

The manufacture declares that the machine conforms to these other European directives: 2014/35/UE, 2014/30/UE.

The manufacture also declares that the machine conforms to the following other harmonized standards: UNI EN ISO 12100:2010, UNI EN ISO 13849-1:2016, EN 60204-1:2018.

The manufacturer strictly prohibits the use of the above machine/fixture in any manner other than that indicated in the Instructions for Use Handbook.

Representative: Patrizio Chiari

Office: Responsible of production **Signature** ....

Cologne, dated 22/05/2020

The technical documentation has been prepared by **G. MONDINI S.p.A.,** a legal entity with head office at 5/7 Via Brescia, 25033 Cologne, Brescia, Italy.

1.1.4 System marking

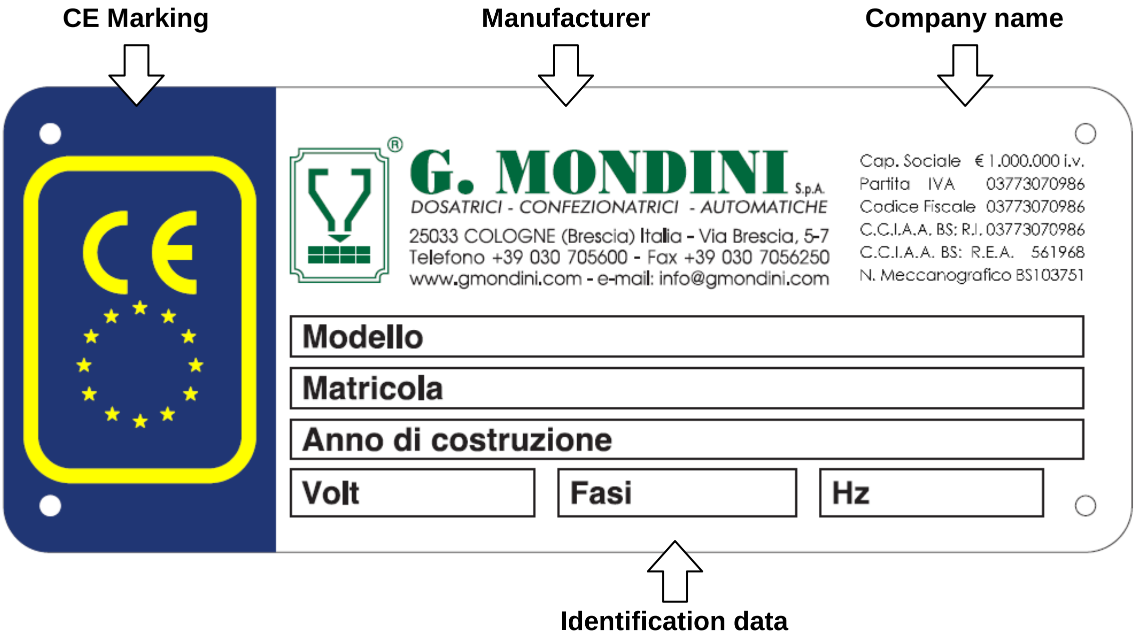


Figure B: CE Plate.

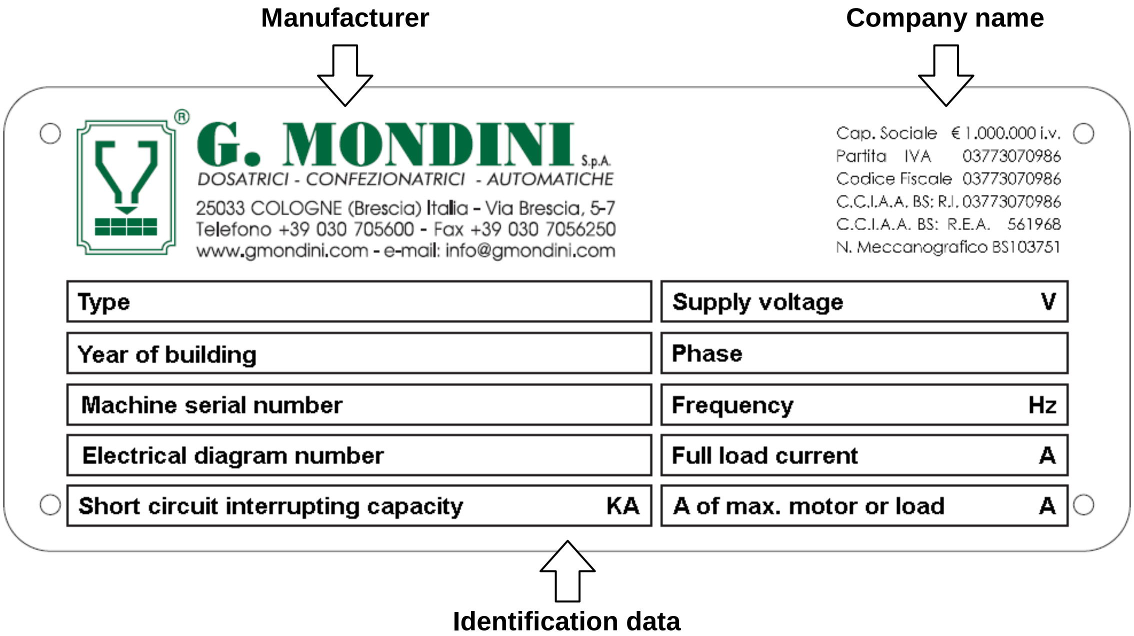


Figure B: USA Plate.

1.1.5 Normative references

DIRECTIVE / STANDARD	EDITION	TITLE
2006/42/CE	DEC. 2009	Machinery Directive
2014/68/EU “PED”	MAY 2014	Approximation of the laws of the Member States concerning pressure equipment.
2014/34/EU “ATEX”	FEB. 2014	Approximation of the laws of the Member States concerning equipment and protective systems intended for use in potentially explosive atmospheres.
1999/92/CE	JAN. 2000	Minimum requirements for improving the safety and health protection of workers potentially at risk from explosive atmospheres.
2014/30/EU	APRIL 2016	Concepts regarding electromagnetic compatibility issues.
2014/35/EU	APRIL 2016	Low-voltage directive
UNI EN 349	NOV. 2008	Safety of machinery. Minimum gaps to avoid crushing of parts of the human body.
UNI EN 415-3	FEB. 2010	Safety of packaging machines. Part 3: Molders, fillers and sealers. Security requirements.
UNI EN 415-10	JAN. 2014	Safety of packaging machines. Part 10: General requirements.
UNI EN 547-1	MAR. 2009	Safety of machinery. Human body measurements. Part 1: Principles for determining the dimensions required for openings for whole body access into machinery.
UNI EN 547-2	MAR. 2009	Safety of machinery. Human body measurements. Part 2: Principles for determining the dimensions required for access openings.
UNI EN 574	NOV. 2008	Safety of machinery. A two-hand control device. Functional aspects. Principles for design.
UNI EN 842	MAR. 2009	Safety of machinery. Visual danger signals. General requirements, design and testing.
UNI EN 981	MAR. 2009	Safety of machinery. System of auditory and visual danger and information signals.
UNI EN 1037	SEPT. 2008	Safety of machinery. Prevention of unexpected start-up.

DIRECTIVE / STANDARD	EDITION	TITLE
UNI EN 1672-2	MAR. 2009	Machines for the foods industry. Basic concepts. Part 2: hygiene requirements.
UNI EN ISO 4413	FEB. 2012	Hydraulic fluid power. General rules and safety requirements for systems and their components.
UNI EN ISO 4414	FEB. 2012	Pneumatic fluid power. General rules and safety requirements for systems and their components.
UNI EN ISO 11161	JULY 2010	Safety of machinery. Integrated manufacturing systems. Basic requirements.
UNI EN ISO 11202	JUNE 2010	Acoustics. Noise issued by machinery and equipment. Determination of emission sound pressure levels at a work station and at other specified positions applying approximate environmental corrections.
UNI EN ISO 12100	NOV. 2010	Safety of machinery. General principles for design. Risk assessment and risk reduction.
UNI EN ISO 13849-1	JUNE 2016	Safety of machinery. Safety-related parts of control systems Part 1: General principles for design.
UNI EN ISO 13849-2	NOV. 2013	Safety of machinery. Safety-related parts of control systems Part 2: Validation.
UNI EN ISO 13850	MAY 2016	Safety of machinery. Emergency stop devices. Functional aspects. Principles for design.
UNI EN ISO 13855	NOV. 2010	Safety of machinery. Positioning of protective equipment in respect of approach speeds of parts of the human body.
UNI EN ISO 13857	MAY 2008	Safety of machinery. Safety distances to prevent hazard zones being reached by upper and lower limbs.
UNI EN ISO 14119	NOV. 2013	Safety of machinery. Interlocking devices associated with guards. Principles for design and selection
UNI EN ISO 14120	JUNE 2016	Machinery safety systems. Guards. General requirements for the design and engineering of fixed and mobile guard systems.
CEI EN 60204-1	MAR. 2018	Safety of machinery. Electrical equipment of machines Part 1: General rules.

1.2 HANDBOOK STRUCTURE

This handbook is divided into 9 chapters.

CHAPTER 1 – GENERAL INFORMATION

This chapter contains the general descriptions regarding the structure of the handbook, the topics dealt with, the symbols, guarantee and responsibilities.

CHAPTER 2 – DESCRIPTION

This chapter contains the description of the system operating principles, the general technical data and the description of the mechanical, electrical and fluidic assemblies that it comprises.

CHAPTER 3 – SAFETY

This chapter contains a description regarding the standards, working environment conditions, ergonomics, accident prevention devices used, residual risks, warning plates on the system and the Manufacturer's Declaration of Conformity .

CHAPTER 4 – INSTALLATION

This chapter contains the instructions for correct packing, handling, transport, unloading, installation in the user's plant, connections to the plant energy sources, checks, controls and any adjustments to be made before the start-up.

CHAPTER 5 – CONTROL SYSTEM

This chapter contains the description of all the controls available for the operator.

CHAPTER 6 –OPERATION

This chapter is addressed to those running the system and contains the instructions for the start-up, use and stopping of the system in the different operating cycles. It also describes all the video pages displayed on the control panel.

CHAPTER 7 –DIAGNOSTICS

This chapter describes all the alarms of the system, the causes that generated them and the procedures to solve them .

CHAPTER 8 – MAINTENANCE

This chapter is for the maintenance technicians and contains the machine maintenance schedule. It includes the warnings, precautions and instructions to carry out the maintenance operations correctly on the system.

CHAPTER 9 – DISPOSAL

This chapter describes how to dispose of the system and how to dismantle it.

1.3 SAFETY STANDARDS CONTAINED IN THIS HANDBOOK

The instructions, indications, standards and safety notes described in the various chapters of the handbook have the purpose of defining how to behave and the obligations to be respected when carrying out the various activities. They form the methods of use for the line, so as to operate under safe conditions for the personnel, the fixtures and the surrounding environment. The safety standards contained are directed to all the skilled persons who are authorised and assigned to carry out the various operations that include:

- transport
- installation
- operation
- use
- management
- maintenance
- cleaning
- putting out of service and dismantling

that constitute the methods of use foreseen for the system.

IMPORTANT NOTE

The system must not be modified in any way and the safety guards must not be removed or disactivated without first informing the Manufacturer. If these instructions are not complied with, the Manufacturer DECLINES ALL RESPONSIBILITY for any situations that might occur. The Manufacturer cannot be held liable for:

A) THE USER'S SAFETY.
B) PROPER OPERATION of the machine supplied.

- IMPORTANT: If the system supplied is to be linked to an existing installation, the costumer must - unless agreed otherwise - provide adequate protection for the area where the two systems are linked. Mondini S.p.a. declines all liability if damage or injury is caused because this protection has not been provided.

1.4 SYMBOLS USED IN THIS HANDBOOK

This handbook uses certain symbols to call the attention of the reader and point out some particularly important features.

The table below contains a list of the different symbols used in the handbook and describes their meaning.

SYMBOL	MEANING	NOTES
	Danger	Indicates a hazard with accident risk, even death, for the user. Pay very careful attention to the texts indicated by this symbol.
	Warning	A warning of possible downgrading or damage to the line, and/or the fixtures. Pay attention to the texts indicated by this symbol.
	Warning Note	A warning or note regarding key functions or useful information. Pay attention to the texts indicated by this symbol.
	Additional information	Texts that contain additional information are indicated by this symbol. This information does not relate directly to the description of an operation or to the development of a procedure.. It may give reference to other supplementary documentation such as attached instructions for use handbooks, technical documentation or other sections of this handbook.
	Avoid damage to material	Indicates there is a high risk of damaging a part, for example using a wrong tool or assembling following an incorrect procedure
	Special tool	Indicates that for this operation a special tool or fixture is necessary.
	Visual check	Informs the reader that a visual check is to be made. This symbol will also be found in the instructions for use. The user is told to read a measure value, to check an indication, etc.
	Sound check	Informs the reader that a sound check is to be made. This symbol will also be found in the instructions for use. The user is told to listen to an operating noise.
	See the maintenance charts	Indicates that a special maintenance chart is to be consulted.

1.5 RESPONSIBILITIES OF THE MANUFACTURER

G. MONDINI S.p.A. shall not be held liable for situations deriving from the incorrect or improper use of the system described herein or any damage caused by the use of spare parts other than those recommended, by maintenance operations not carried out correctly or by tampering with circuits, components and/or system software.

The responsibility to ensure that the safety instructions indicated in this handbook are applied is assigned to the technical supervisor responsible for the foreseen activities on the line. He shall ascertain that the operatives authorised to carry out the required activities are qualified , that they respect and are aware of the requirements contained in this handbook and the general safety standards applied to the system.

Not observing the safety standards could cause injuries to the personnel and damage to the fixtures.

1.6 GUARANTEE

For the construction of the system G. MONDINI S.p.A. has used materials deemed most suitable in their final judgement.

G. MONDINI S.p.A. guarantees that the system is free of machining flaws and ensures the quality of the material for a period of time and in accordance with the conditions agreed upon when drawing up the contract with the customer.

1.7 LINE MANAGEMENT

The system management is only allowed to authorised operators who have received sufficient instruction, or who have appropriate technical experience.

The operators assigned to the running and the maintenance of the system are to be aware that the knowledge and application of the safety standards is an integrating part of their job.

Operators who are not assigned to activities on the system are not to have access to the operating area and/or the control panels.

Before starting up the system carry out these operations:

- read this handbook carefully
- know which protections and emergency stop devices are installed on the line, where they are located and how they function.

It is forbidden to remove, even only partially, protections and safety devices installed to safeguard the personnel in the system hazardous zones.

The same regulation applies to the warning plates.

It is strictly forbidden to open the doors of the electric cabinet when the system is running and immediately after it has stopped.

The protections and safety devices are to be kept in perfect order to ensure the correct functioning. In the case of malfunctioning or failure on these devices, they are to be immediately repaired or replaced.

The use of commercial components that are not those specified for safety devices and protections could cause malfunctioning or the generation of hazardous situations for the operators working on the line.

1.7.1 Plates on the system

PLATE 1	
	Use gloves to protect from cuts and burns.
PLATE 2	
	Wear safety shoes.
PLATE 3	
	See the use and maintenance handbook.
PLATE 4	
	**Danger!** Mechanical parts in motion.
PLATE 5	
	**Danger!** Hazardous voltages present; Danger of fulguration.
PLATE 6	
	**Warning!** In this zone there are parts that work at high temperatures.
PLATE 7	
	**Warning!** There are protruding obstacles in this area that could cause head injuries.

PLATE 8	
	**Warning!** Danger of limbs being crushed.
PLATE 9	
	**Warning!** In this zone of the system parts are operating that could cause deep cuts and serious amputations.
PLATE 10	
	**Warning!** This indication is usually placed near the closing zone of the protective casing, where its closure affects the resumption of production.
PLATE 11	
	**Warning!** Constant hazard pay attention when handling.
PLATE 12	
	Do not remove the safety devices and protections.
PLATE 13	
	Do not operate on moving parts.
PLATE 14	
	This symbol indicates the direction of rotation of the part .

1.7.2 Glossary and symbols

Some terminology and symbols are used in this handbook to call the attention of the reader and to emphasise certain aspects of particular importance .

The tables that follow list them and describe their significance.

Table 1 - Glossary

Term	Description
Emergency stop function	Function to: - prevent, upon occurrence, hazards for persons, damage to machines or machining in progress, or otherwise reduce them. - be activated by a single human action when the normal stop function is inadequate for the purpose. In accordance with this standard, the hazards are those that could arise from: - operation irregularities (machine fault, inadequate processing material, human error, …) - normal operation Note – *Functions such as inversion, limiting movements, deviation, shielding, braking, sectioning, etc, may be part of emergency stop functions. This standard (EN 13850) does not cover these functions.* (Item 3.1 of EN 13850)
Machine actuator	A power mechanism used to start the movement of a machine. (Item 3.3 of EN 13850)
Control circuits	Circuit used to control the machine operation and to protect the power circuits.
Manual control (actuator)	The component of the control device that, when activated, activates the actual control device, and is designed for actuation by a person (see item 4.4.1 of EN 13850). (Item 3.2 of EN 13850)
Customer/ Purchaser	The person who orders and/or purchases.
Machines directive	The machines Directive is the ruling indicated with the Executive Order of the President of the Republic 459/96.
Emergency stop device	Together with the components designed to execute the emergency stop (see fig. 2 of EN 13850, that shows the parts of a machine to which these components may belong). (Item 3.2 of EN 13850)
Enable device (control	Additional control device activated manually and used together with a start command, to permit the machine to operate permanently when activated. (Item 3.26.2 of EN 12100)

Term	Description
Control device	Component of the emergency stop device that generates an emergency stop signal when the associated manual control (actuator) is activated. (Item 3.2 of EN 13850)
Interlock device (interlock)	Mechanical, electrical or other type of device that has the purpose of preventing machine parts operating under certain conditions (usually before the guard is closed). (Item 3.26.1 of EN 12100)
Safety device	Device (not a guard) that eliminates or reduces the risk, either alone or associated to a guard. (Item 3.26 of EN 12100)
Limiting device	Device that prevents the machine or its parts from exceeding the set limit (distance, pressure, etc.). (Item 3.26.8 of EN 12100)
Bill of Materials	List of the components that are part of the mechanical assemblies, fluidic or electrical systems, indicated with the quantity, code and name of supplier.
Manufacturer	Manufacturer of the machine
Supplier	Body (for example, manufacturer, installer agent, system integrator) that supplies the equipment or associated services of the line (the user may also act as his/her own manufacturer).
Automatic operation	Mode in which the entire system executes its operations autonomously, with gates and barriers closed and inserted in the safety circuits.
Manual operation	Operating mode that cuts out automatic running and that allows manual handling activities under the control of the operator.
Installation	Installation is the mechanical and electrical integration of the machine in a production system, in conformity with the Directive safety requirements.
Instructions for use	Safety measures that consist of a set of information, such as texts, words, signs, symbols or diagrams that are used either separately or in combination, to convey the instructions to the user. They are intended for professional and/or non professional users. Note – *Item 6 of EN 12100 deals with instructions for use.* (Item 3.21 of EN 12100)

Term	Description
Machinery/ Machine	A group of parts or components, of which at least one is movable, connected together with appropriate actuators, control and power circuits etc. of the machine, integrally connected for a specific application, in particular for conversion, treatment, handling or packaging of a material. The term ”machinery” also covers a group of machines that, to obtain the same result, are arranged and controlled so as to have an integrated operation. Appendix A of EN 12100 contains a general schematic diagram of a machine. (Item 3.1 of EN 12100)
Maintenance and Repair	Maintenance and repair operations are periodical checking activities and/or replacement of mechanical, electrical parts, software or machine components that serve to identify the cause of a fault, that terminate with the machine being returned to the project functioning condition.
Marking	Signs or writing that identify the type of component or equipment, applied by the manufacturer of the component or equipment.
Commissioning or Putting into service	Commissioning is the functional checking activity on the installed system.
Safety measures	A means that eliminates or reduces a hazard.
Operator	Person or persons assigned to the installation, operation, regulation, maintenance , cleaning, repairing or transport of the machine (EN 12100).
Exposed person	Any person that is completely or partially in a hazardous zone (RES 1.1.1 – C of 2006/42/EC).
Skilled person	A person with sufficient technical knowledge or experience to be able to avoid hazards that could be present.
Protections	Safety measures that consist in the use of specific technical measures called protections (guards, safety devices) to protect persons against hazards that cannot be reasonably eliminated or sufficiently limited by design. Note – *Item 5 of EN 12100 deals with protections.* (Item 3.20 of EN 12100)
Contact person	Person responsible for the running of certain operations or assessments that could arise during work or maintenance .
Redundancy	Application of more than one device or system, or part of a device or system, so as to ensure that in the case of a failure in the functioning of one of them, another is available to execute that function.

Term	Description
Guard	A part of a machine used in a specific manner to give protection by means of a physical barrier. According to its construction, a guard may be called hood, cover, shield, screen, door, fencing, enclosure, etc. Note 1 – *A guard may act:* - *alone; it is therefore effective only when it is closed,* - *associated to an interlock device with or without locking of the guard; in this case the protection is insured whatever the position of the guard.* Note 2 – *“Closed” means, for the fixed guard, “kept in position”.* (Item 3.25 of EN 12100)
Fixed guard	Guard kept in position (closed): - either permanently (by welding, etc.), - or by fastening elements (screws, bolts, etc.) that make it impossible to remove/open it without the use of tools. (Item 3.25.1 of EN 12100)
Movable guard	Guard usually mechanically connected to the frame of the machine or to a fixed element nearby (for example by hinges or guides), and that can be opened without the use of tools. (Item 3.25.2 of EN 12100)
Interlocked guard	Guard associated to an interlock device (see item 3.26.1 of EN 12100), so that: - the hazardous functions of the machine “involved” with the guard cannot be carried out until the guard has been closed, - if the guard is opened while machine hazardous operations are running, the order to stop is given, - the closing of the guard allows the machine to run the hazardous operations with which the guard is “involved”, but does not give the start commands. (Item 3.25.4 of EN 12100)
Risk	Combination of probabilities and seriousness of possible injuries or harm to health in a hazardous situation. (Item 3.11 of EN 12100)
Machine safety	Capability of a machine to carry out its functions, to be transported, installed, adjusted, serviced, dismantled and removed under the foreseen use conditions (see Item 3.22 of EN 12100) specified in the instructions handbook (and, in some cases, in a certain time indicated in the handbook) without causing injuries or harm to health. (Item 3.19 of EN 12100)

Term	Description
Dismantling	Dismantling is the demolition and disposal of the parts constituting the machine.
Protected space	Protected space is the area limited off by the protection barriers and assigned for the installation of the machine.
Environment temperature	Temperature of the air or other agent where the equipment is used.
Transport	The series of operations to transfer the work centre from the assembly site of the manufacturer to the final work site of the customer .
Foreseen use of a machine	Use for which a machine has been designed, in conformity with the indications supplied by the manufacturer, or that is considered standard in relation to its design, construction and function. Foreseen use implies also the observance of the technical instructions contained in the instructions handbook (see Item 6.5 of EN 12100), and taking into consideration the incorrect use that it is feasible to foresee. Note – R*egarding incorrect use in the assessment of risks, these types of behaviour should be given particular consideration*: - *incorrect behaviour that results from normal negligence and not a deliberate intention to use the machine improperly,* - *the instinctive reaction of a person during use, in the case of malfunctioning, accidents, failures, etc.* - *behaviour that derives from "the line of least resistance" when performing a task,* - *probable behaviour of some persons, such as children or disabled persons for certain machines (especially those not for professional use).* *See also* Item *3.21 of EN 12100.* (Item 3.22 of EN 12100)
Incorrect or improper use	Use of the machine out of the limits specified in the technical specifications .
User	Body that uses the machine and the associated equipment.
Hazardous zone	Any zone inside and/or near to a machine in which a person is exposed to the risk of injuries or harm to health. Note – *The hazard that causes the risk considered in this definition*: - *permanent presence during the foreseen use of the machine (motion of hazardous moving parts, electric arc during welding, etc.), or* - *that can unexpectedly occur (sudden /unexpected start, etc.)* (Item 3.10 of EN 12100)

2 TECHNICAL FEATURES

CONTENTS

2.1 GENERAL FEATURES

The **TRAVE Tray Sealer** is designed and engineered to seal food packaging containers and trays with an appropriate film.

The sealing process is performed via the following sequence:

1. The trays are fed onto the motor-driven feed belt.
2. The trays are positioned on the servo-assisted feed belt.
3. The trays are picked up by the pusher unit and placed in the lower tool.
4. The tool closes and seals the tray.
5. The tool opens.
6. The sealed trays are picked up by the pusher unit and placed on the delivery belt.

The sealing film is then prepared for the next sealing cycle, and the waste is recovered.

ADDITIONAL INFORMATION

For further details of correct machine operating sequence, please refer to Chapter 6 “OPERATION”.

2.2 MAKE-UP OF THE MACHINE

The **TRAVE Tray Sealer** comprises the following assemblies:

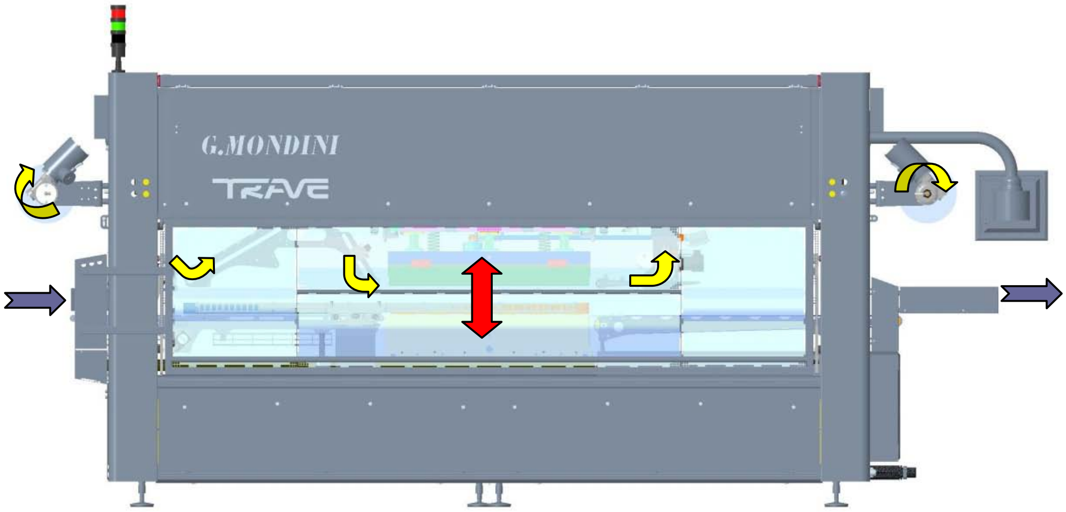
Assembly	Sub-section
FRAME AND COVER CASING	2.5.1
SCREW-OPERATED SEALING ASSEMBLY	2.5.2
PUSHER ASSEMBLY	2.5.3
FEED BELT	2.5.4
DELIVERY BELT	2.5.5
FRONT MOBILE ROLLER	2.5.6
FILM UNWINDER ROLLER	2.5.7
FILM DRIVE AND PULL ROLLERS AND TRANSMISSIONS	2.5.8
WINDING SHAFT	2.5.9
MOVING PLATE	2.5.10
TOP TOOL PLATE	2.5.11
GUARDS	2.5.12
VACUUM CIRCUIT	2.5.13
LUBRICATING SYSTEM	2.5.14
PNEUMATIC SYSTEM	2.5.15
TOOL TROLLEY	2.5.16
SEALING TOOL	2.5.17

2.3 TECHNICAL FEATURES

The machine’s main technical features are listed below.

General features	
Approximate outside dimensions Length x Width x Height	6077 x 1065 x 2535
Total machine weight	6630 kg
Vibration	Non detectable / not an operator health hazard
Noise level	< 75 dB(A)
Operating temperature range	0-45°C
Ambient humidity	Must not exceed 90%
Production	
Cycle time	6 seconds
Output	3600 trays/hour
Output at 80%	2880 trays/hour
Pneumatic system	
Hook-up pressure	Max 8 bar
Operating pressure	6 bar
Air consumption	451 Nl/min
Connection fitting Ø	10/12 mm
Electrical system	
Power supply	415 V – 3-phase N + earth – 50 Hz
Auxiliary voltages	24V DC
Socket power supply	220V AC
Total rated power	37 kVA

2.4 MACHINE FLOWS



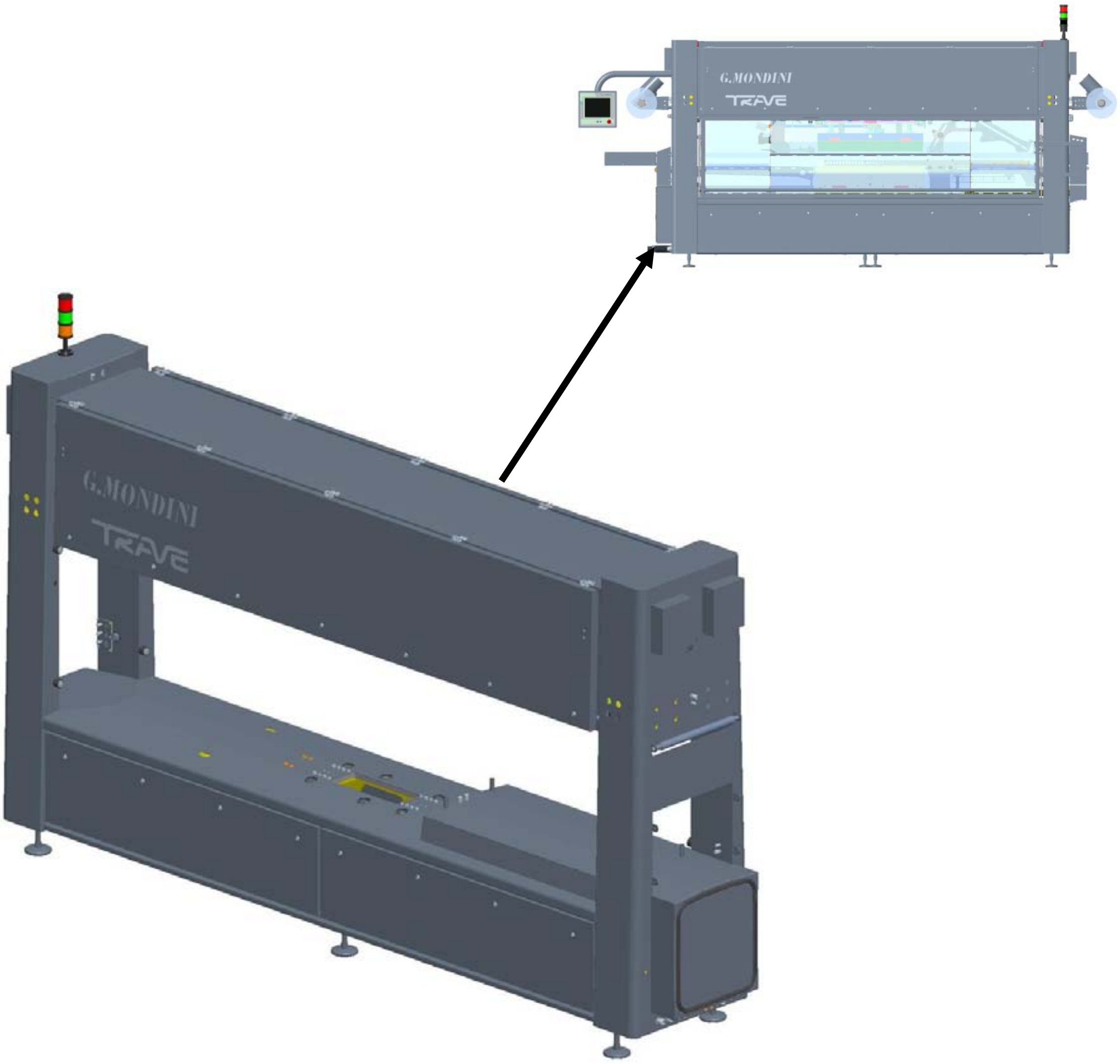
	Tray flow
	Tool motion
	Film flow

Figure 2.1 –TRAVE tray sealer flows

2.5 DESCRIPTION OF THE ASSEMBLIES

Below is a description of the various machine assemblies.

2.5.1 Frame and cover casing



Function

Supports the assemblies constituting the tray sealer, with adjustable legs.

Guarantees safe running of the machine.

Make-up

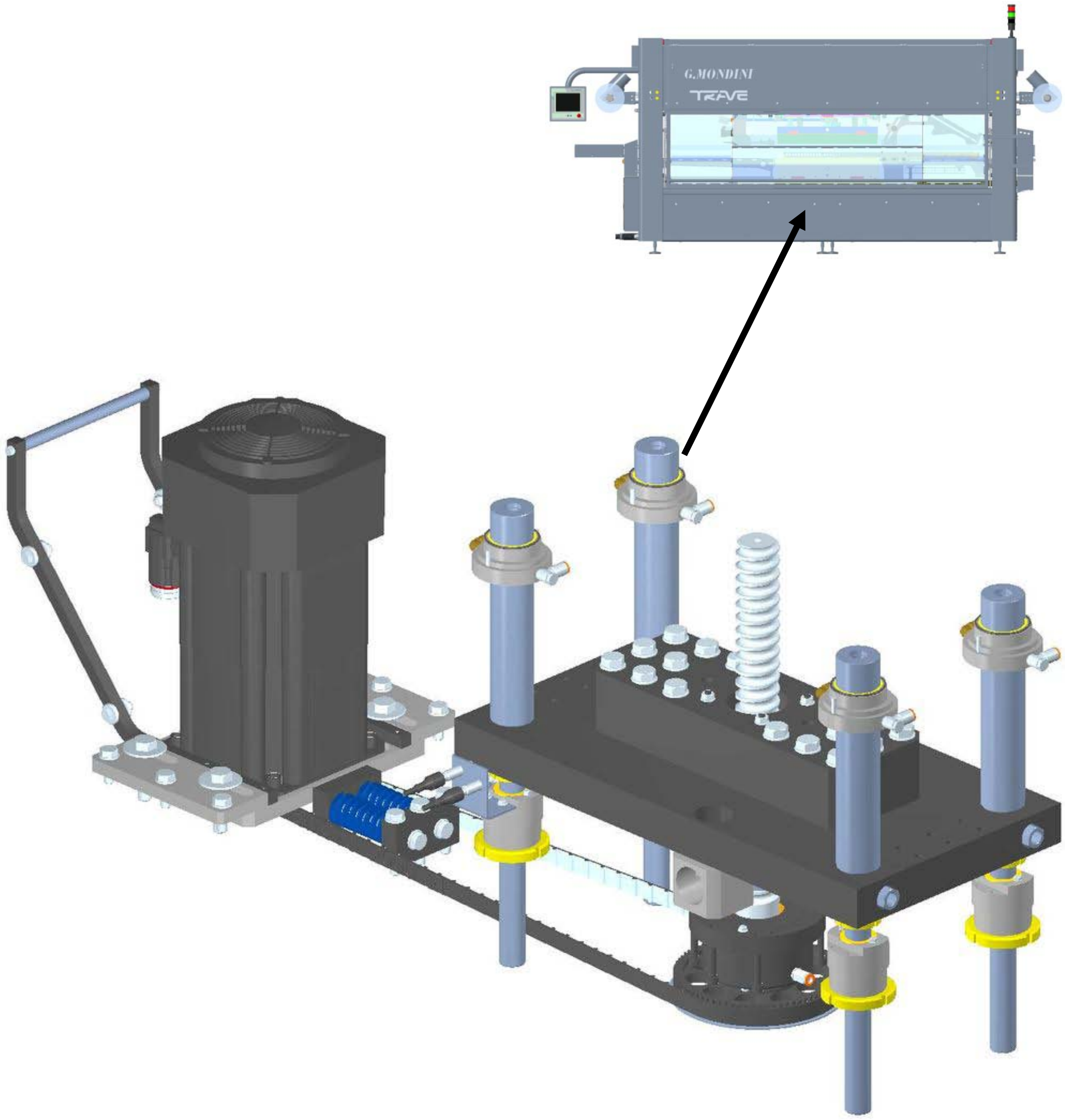
The frame is made up of a welded steel structure.

The lower part houses the pneumatic system, a lubricating system, the feed belt assembly, the pusher assembly, the sealing assembly, the delivery belt and the element driving the pusher arms and the lower tool.

The upper part houses the upper tool, the automatic reel-change assembly, the mobile roller, the winding shaft and the film rollers.

The entire area id protected by guards that are not fitted with safety micro-switches as there is no need to access the internal elements for any reason, except for maintenance and adjustable operations by qualified personnel.

2.5.2 Screw-operated sealing assembly



.

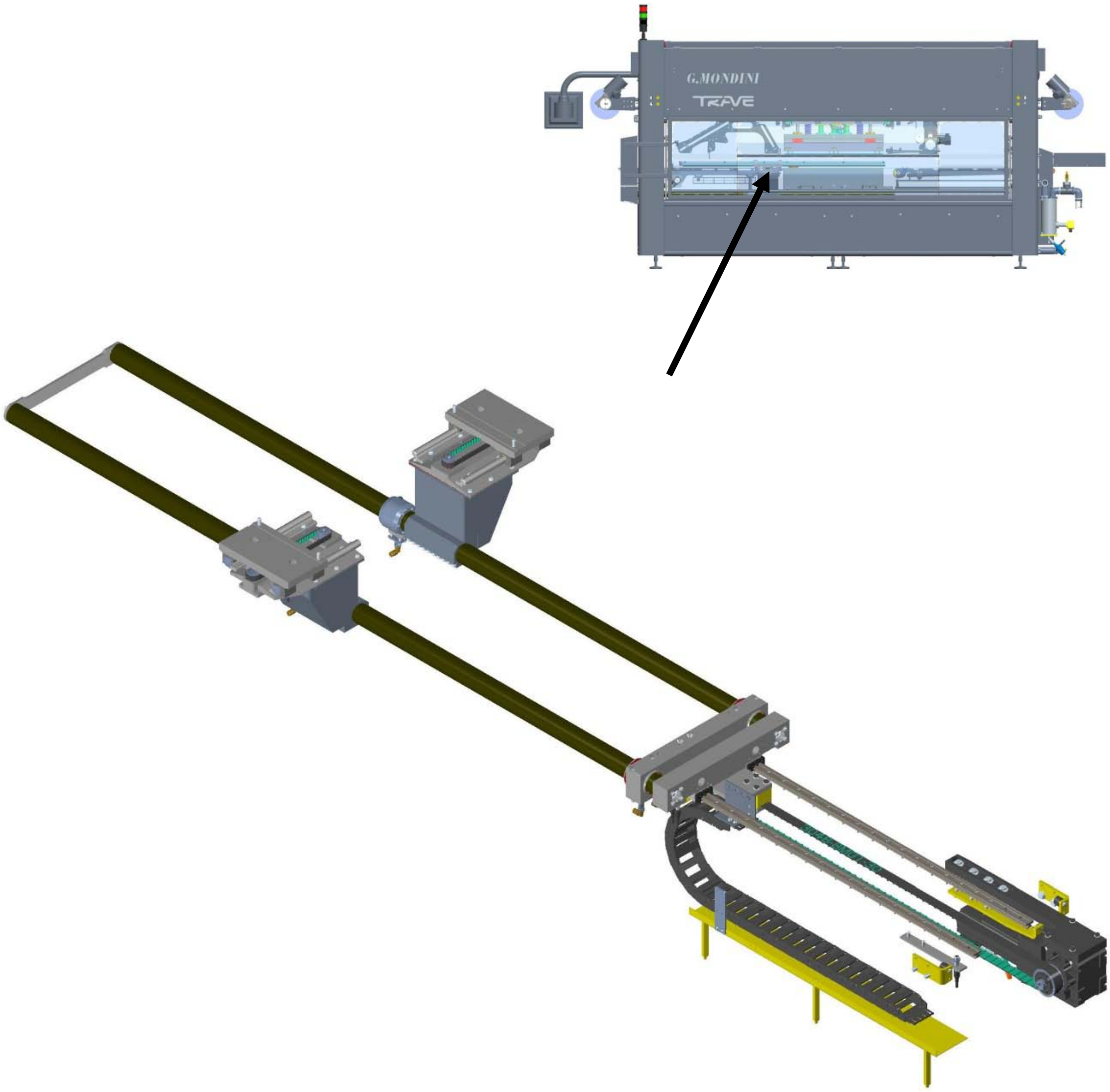
Function

Moves the lower tool for sealing the trays.

Make-up:

This assembly comprises 4 uprights, which are screw operated and move the lower tool up and down.

2.5.3 Pusher assembly



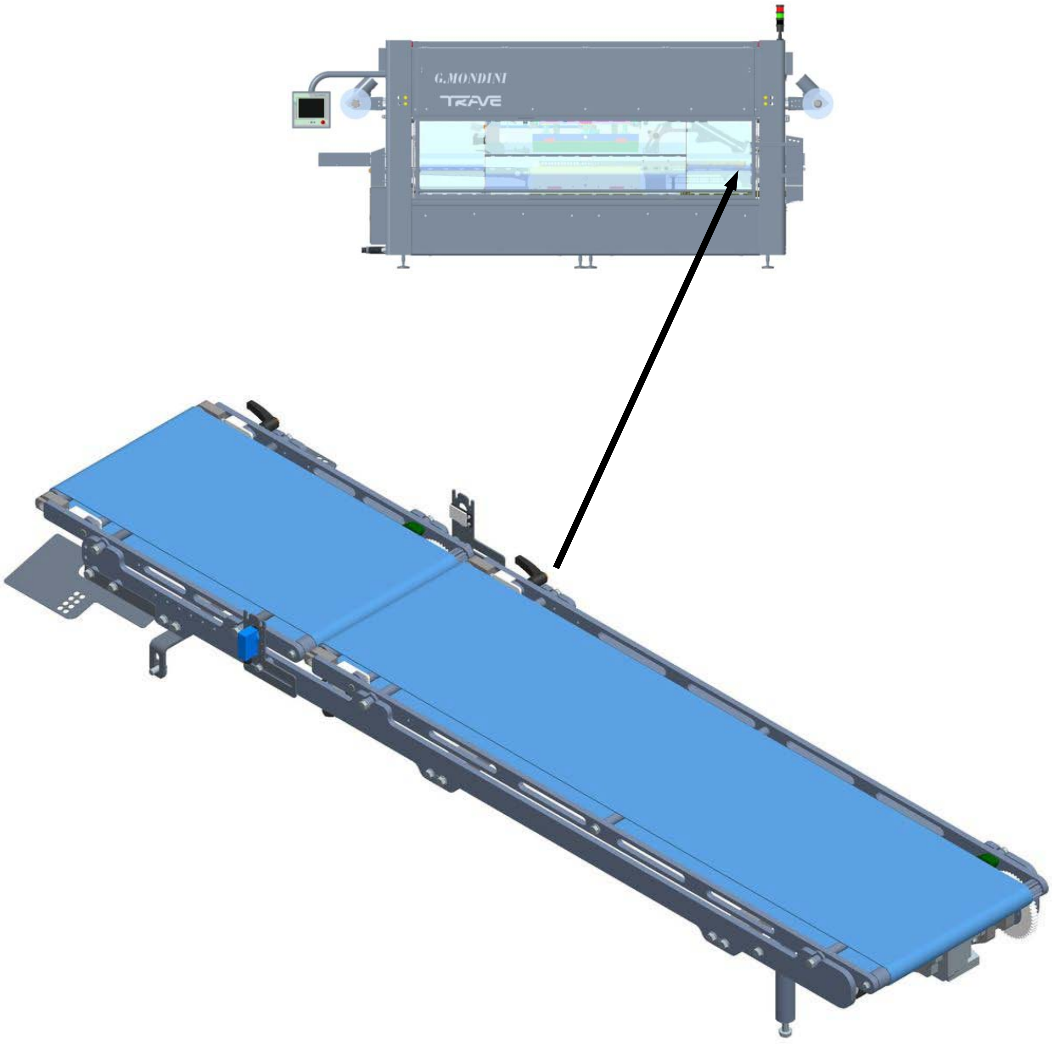
Function

Simultaneously moves empty trays from the feed belt to the tool, and sealed trays from the tool to the delivery belt.

Make-up:

This assembly comprises two horizontally moving columns onto which are fixed the pushers that move the trays.

2.5.4 Feed belt



Function:

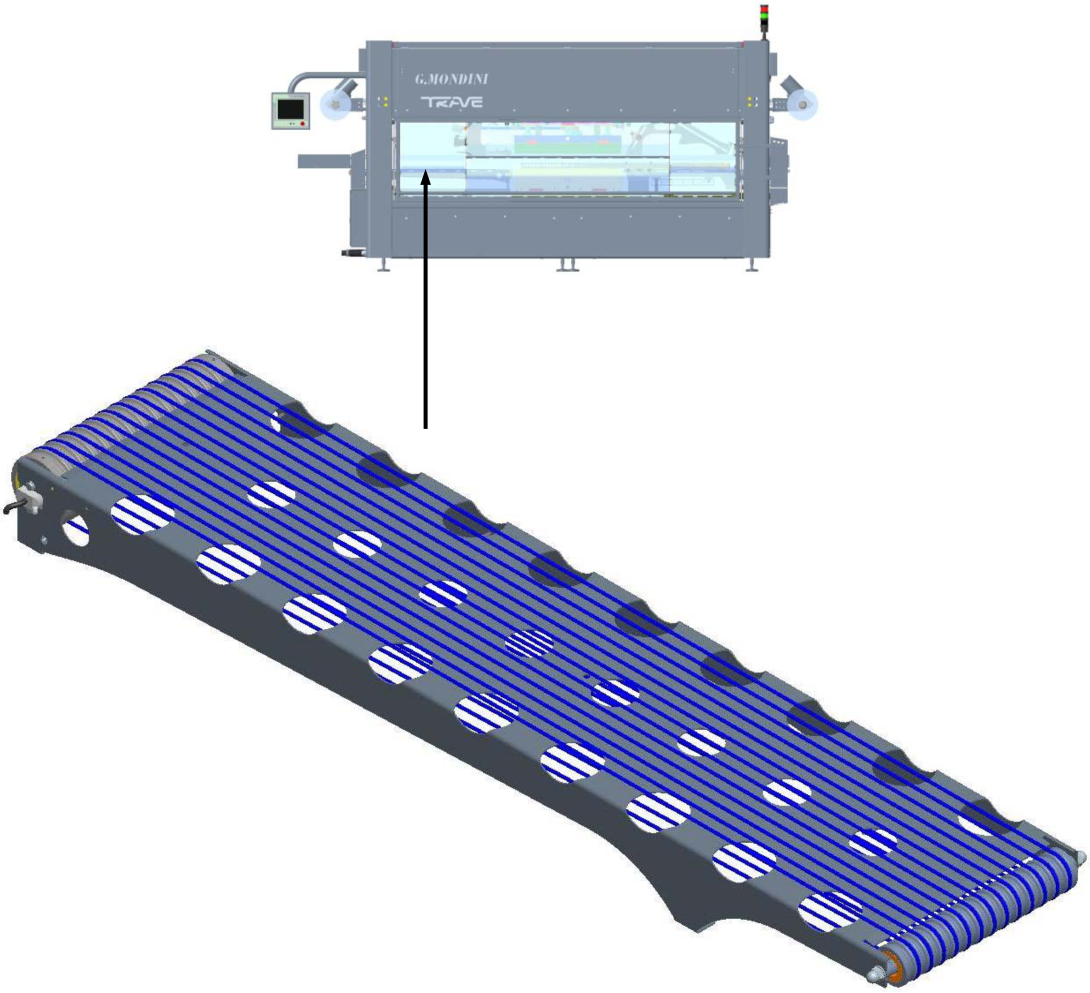
Loads trays and moves them towards the pusher assembly.

Make-up:

This assembly comprises two sections of conveyor belt, one motorized with continuous operation, and one controlled by a limit switch and driven by an inverter.

Based on the type of production, the sensor counts the number of trays to feed into the tool and sends them to the third conveyor, which via a stepping motion places the trays in the right position for pick-up by the pusher assembly.

2.5.5 Delivery belt



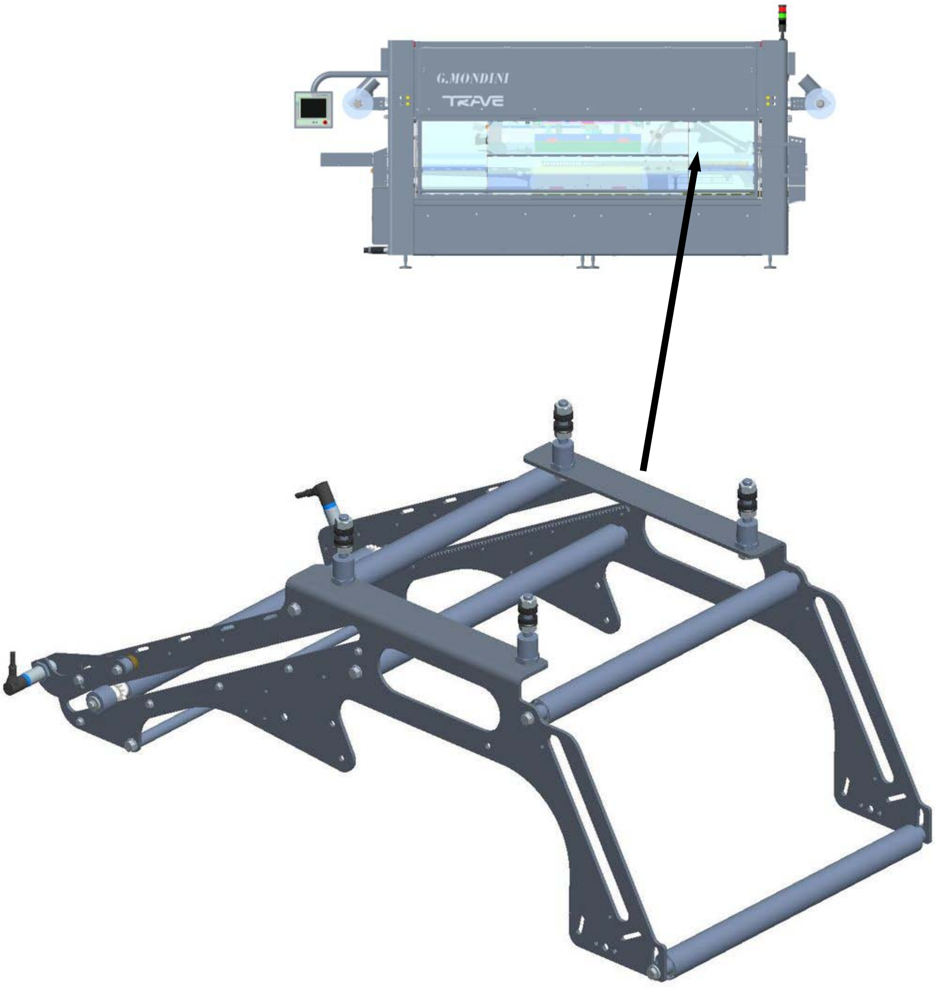
Function

Conveys sealed trays out of the machine.

Make-up

This assembly comprises a motorized drum and polyurethane belts for discharging the trays.

2.5.6 Front mobile roller



Function

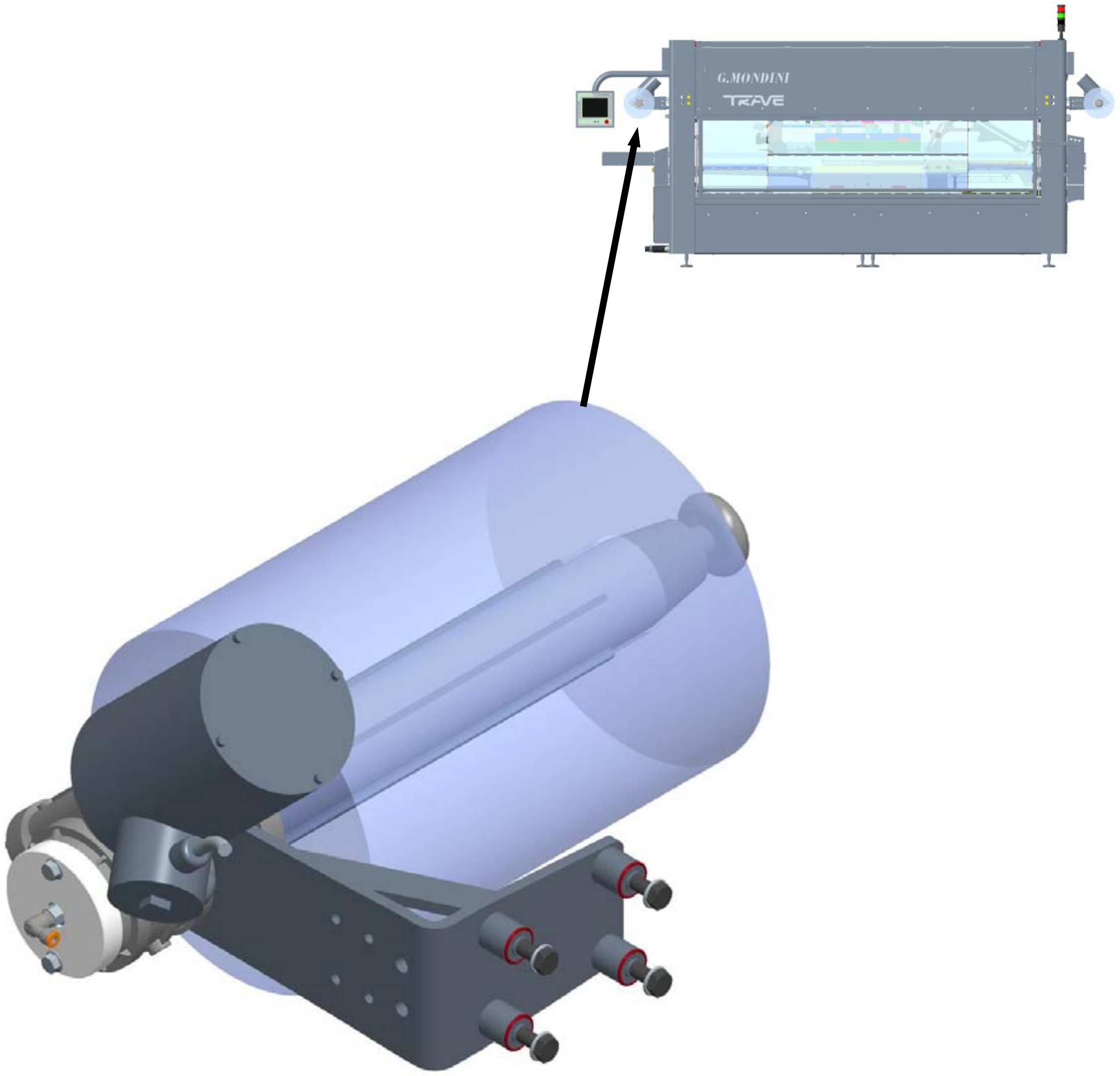
Feeds the sealing film towards the tool area.

Make-up

This assembly comprises a frame with rollers and a retention system to ensure correct unwinding of the film from the reel.

Sensors detect reel centring and when the reel is empty.

2.5.7 Film unwinder roller



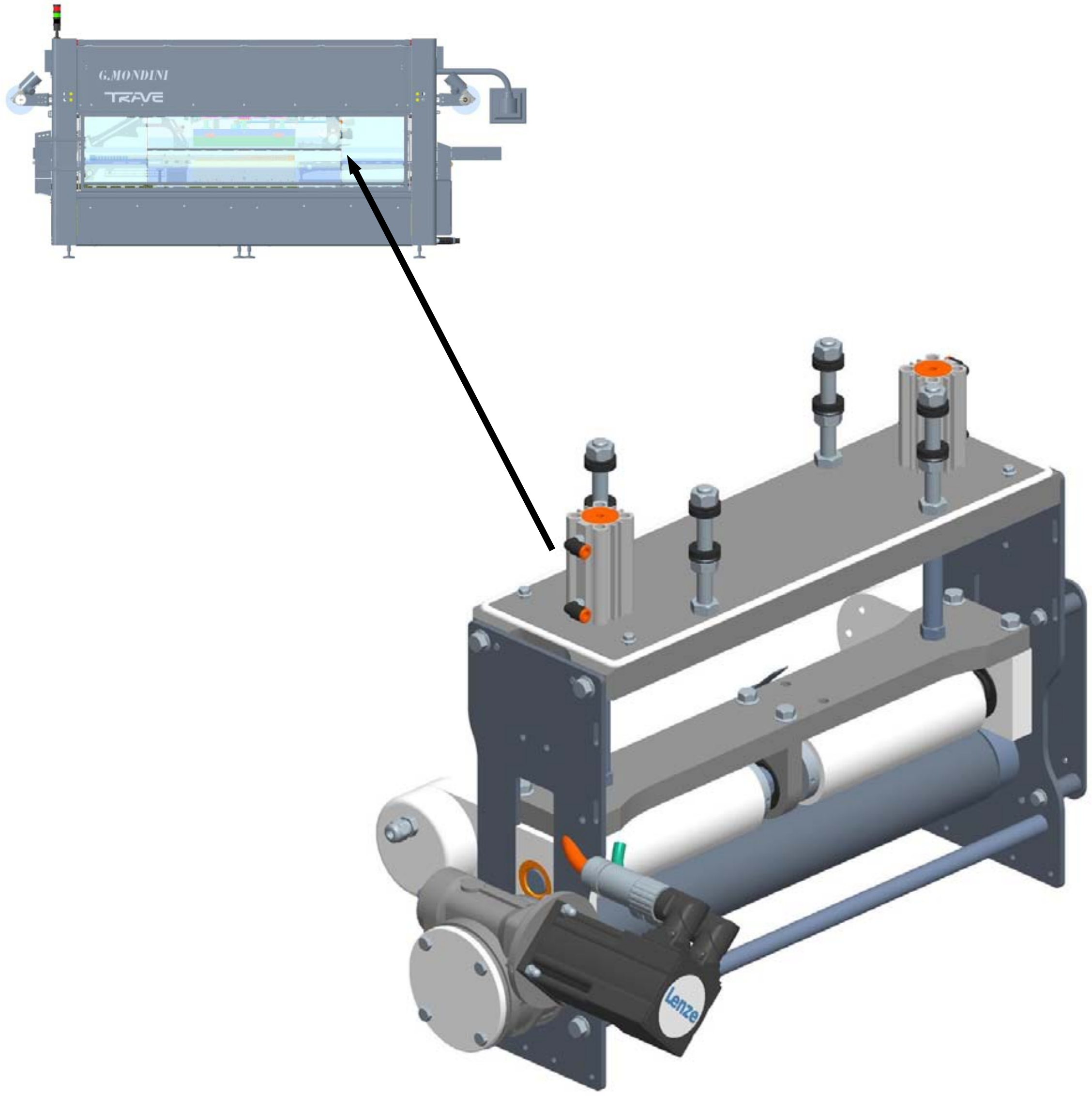
Function

Unwinds the film in the direction of the sealing tool.

Make-up

This unit assembly is composed of a motor-driven shaft that the sealing film bobbin is fitted onto.

2.5.8 Film drive and pull roller and transmissions



Function

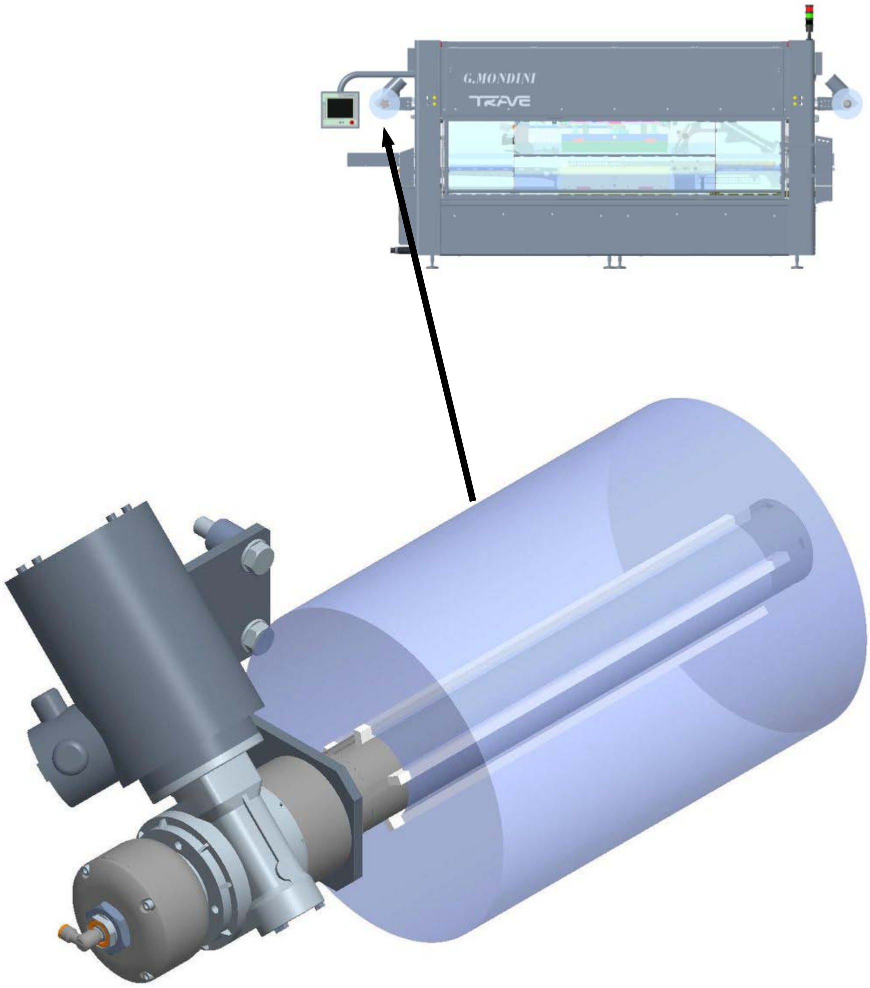
Moves the sealing film waste from the tool area to the waste collecting shaft.

Make-up

This assembly is made up of a frame with rollers that wind up the waste, keeping it taut.

Waste breakage monitoring is provided by a blade system mounted on one of the rollers.

2.5.9 Winding shaft



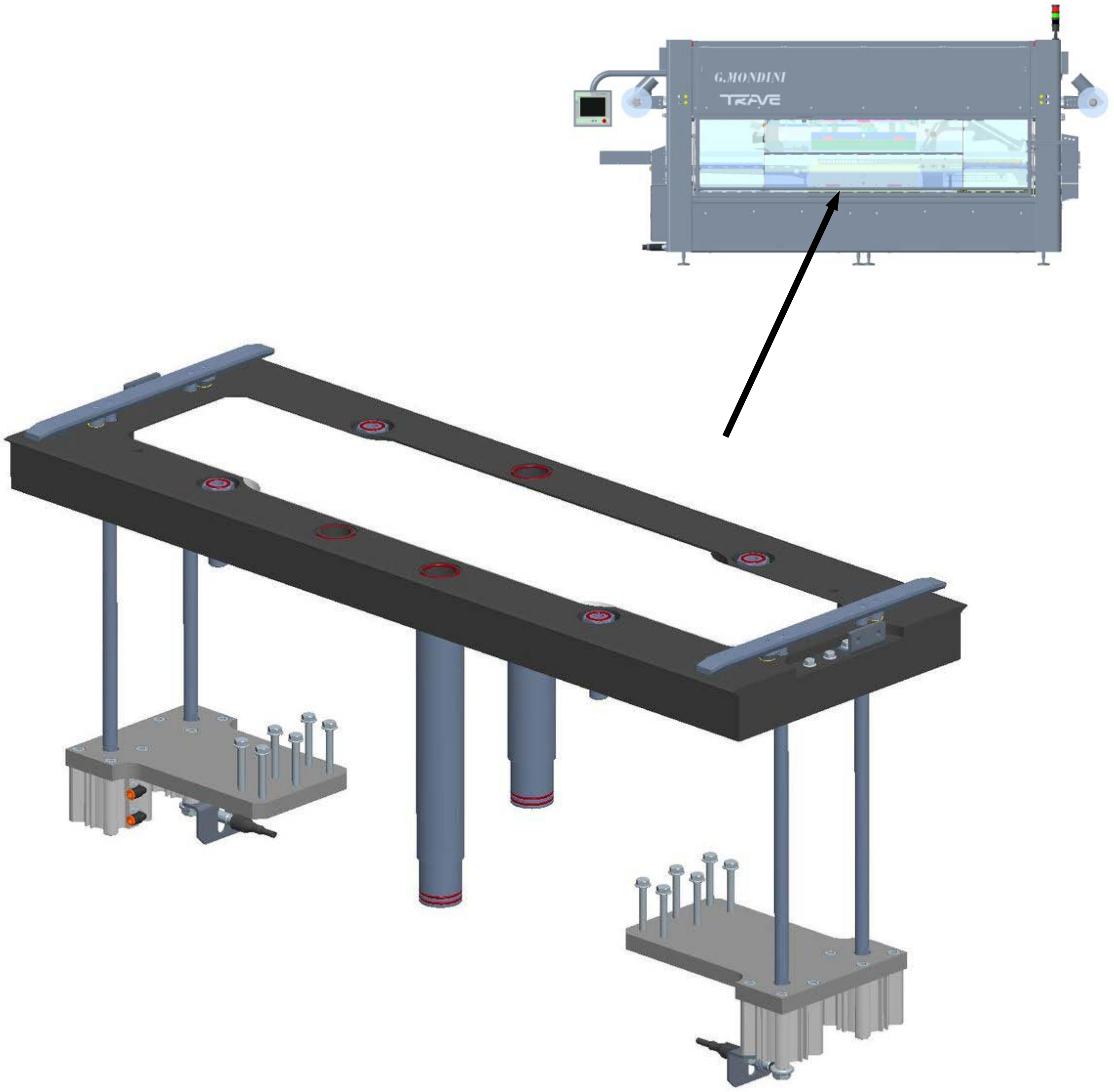
Function

Winds up the waste film from the sealing tool.

Make-up

This assembly is made up of a motor-driven shaft that winds up the waste (either directly onto the shaft or by means of a collector cylinder).

2.5.10 Mobile plate



Function

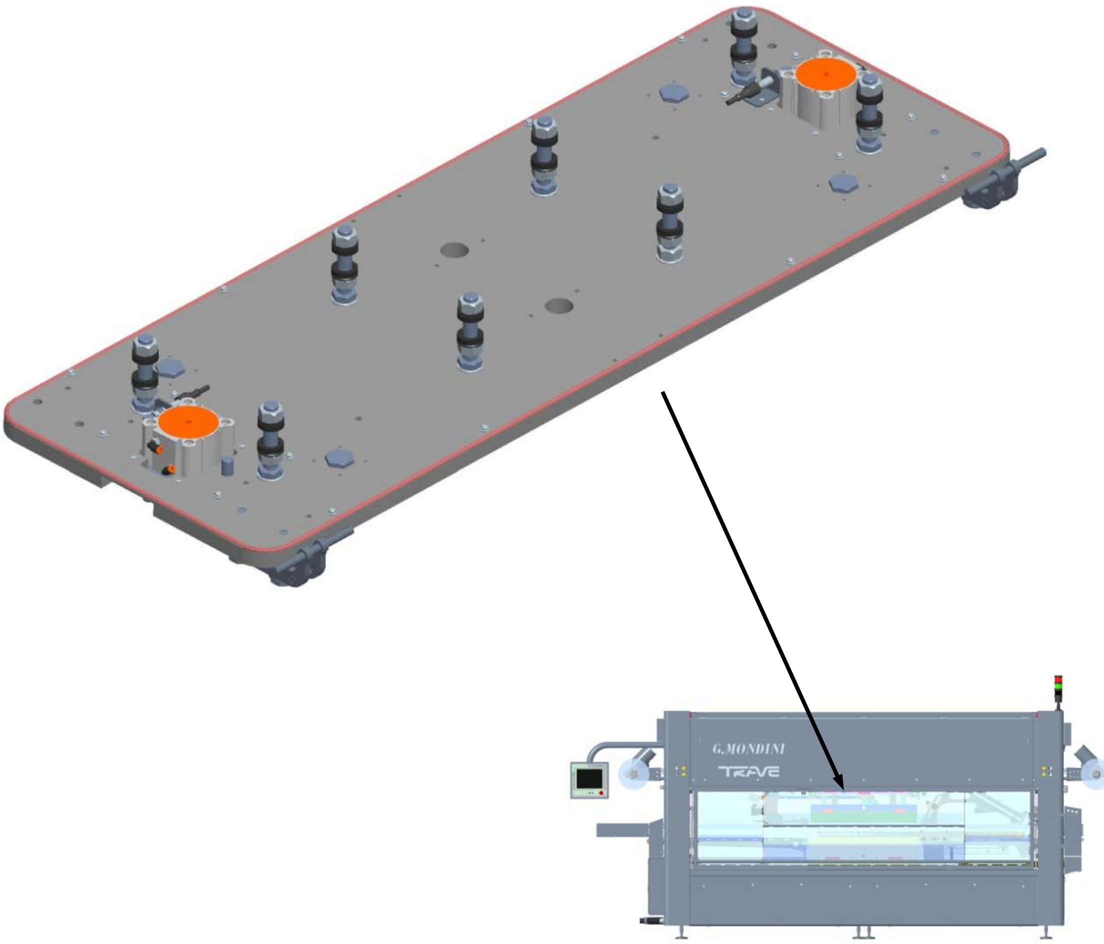
Locks onto the lower tool and moves it towards the upper tool.

Make-up

This assembly is made up of a plate with two guides and centring and top devices to make sure the lower tool fix on to the upper one correctly.

The position of the plate is detected by sensors.

2.5.11 Top tool plate



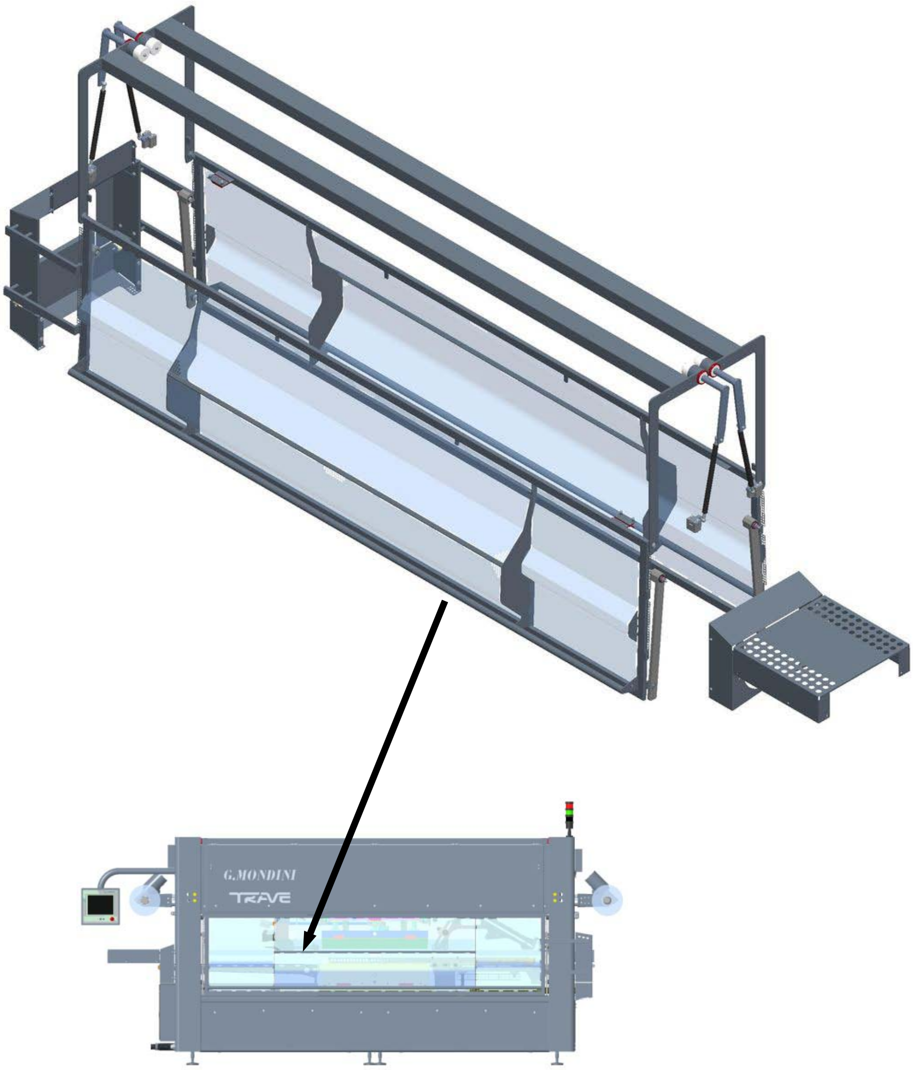
Function:

Enables lock-on of the top half tool.

Make-up:

This unit assembly is composed of a plate fitted with two guides, as well as appropriate centring and stop devices that allow for correct assembly of the top half tool.

2.5.12 Guards



Function

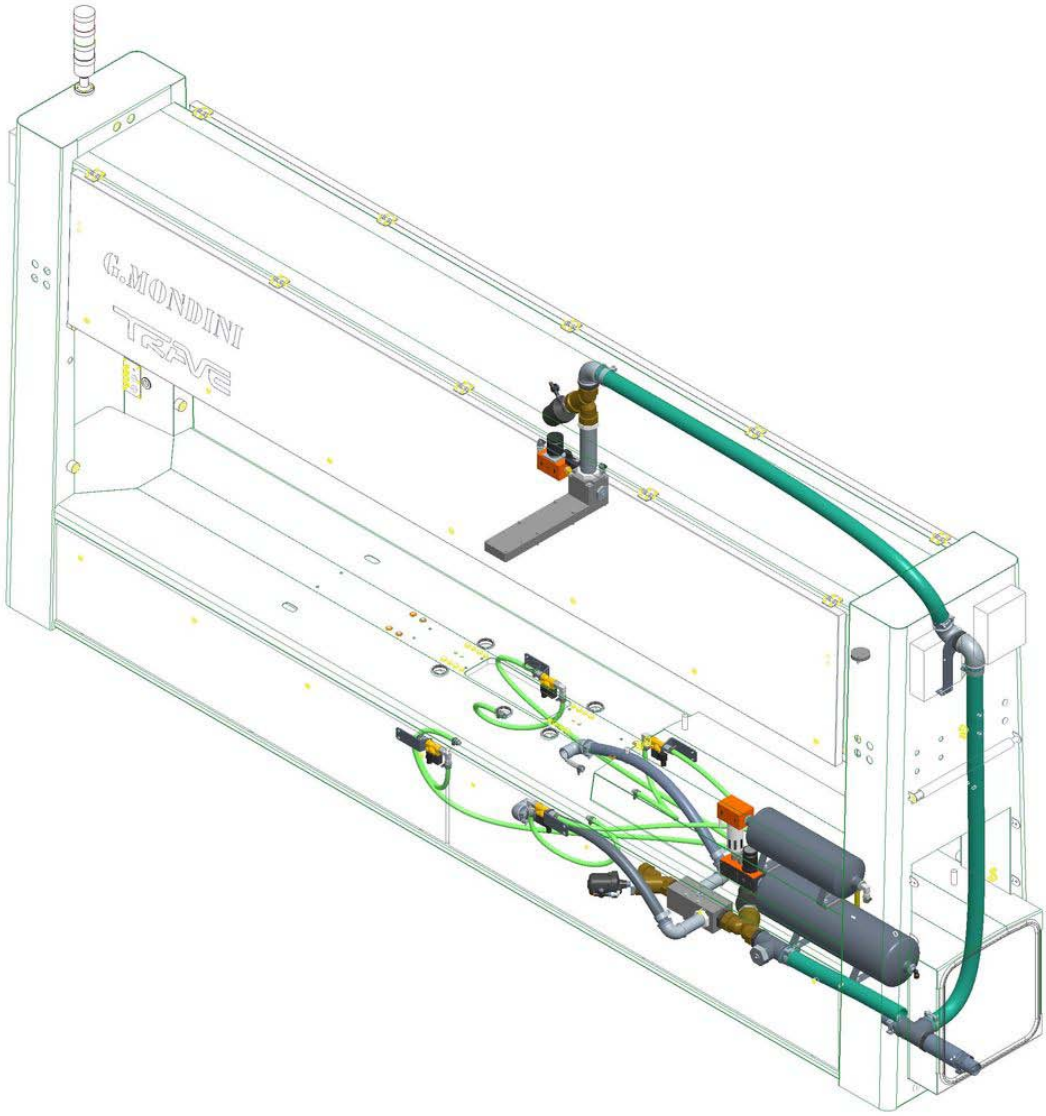
Protects the machine operator.

Make-up

These guards are made up of stainless steel panels that cover the entire tray entry area, the sealing area (where all the most dangerous mechanical operations take place), the exit area and all other areas where there are moving parts.

If any of the guards is opened whilst the machine is running, the work cycle stops instantly and the guard-open alarm goes off.

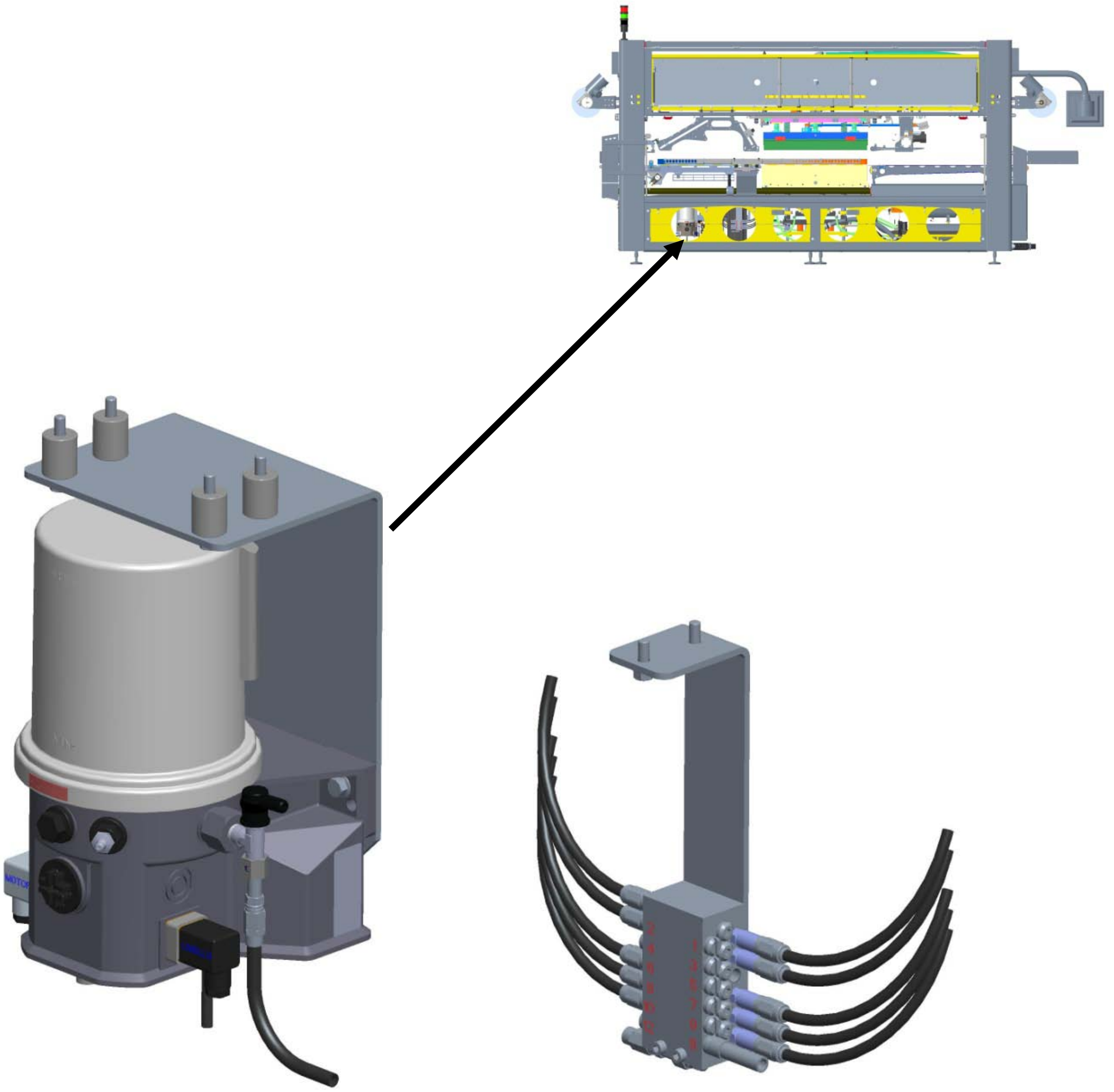
2.5.13 Vacuum circuit



Function

Creates a vacuum inside the tray during sealing.

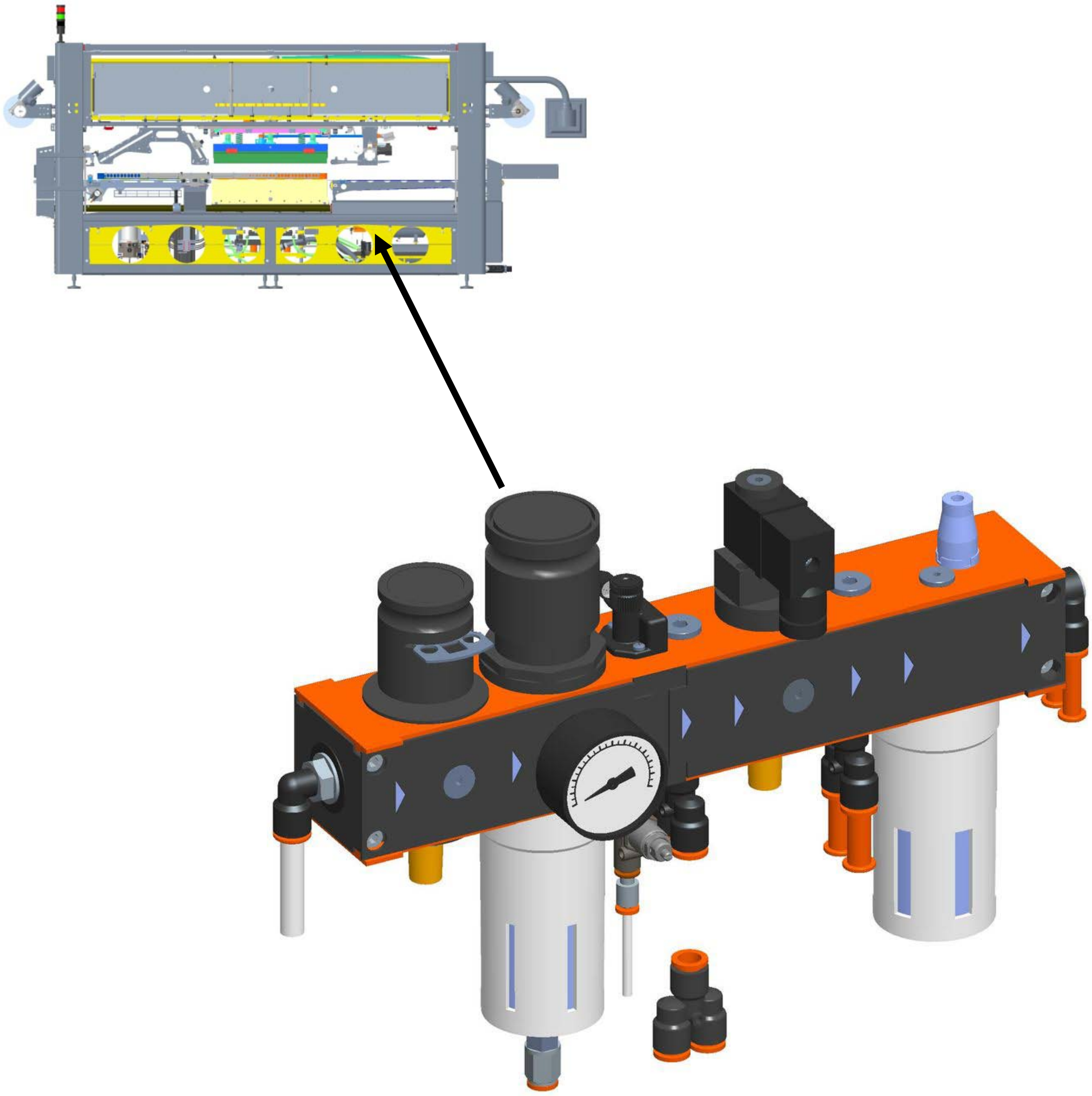
2.5.14 Lubricating system



Function

Automatically lubricates the main components of the machine.

2.5.15 Pneumatic system



Function

Establishes and regulates operation of the pneumatic actuators of the machine.

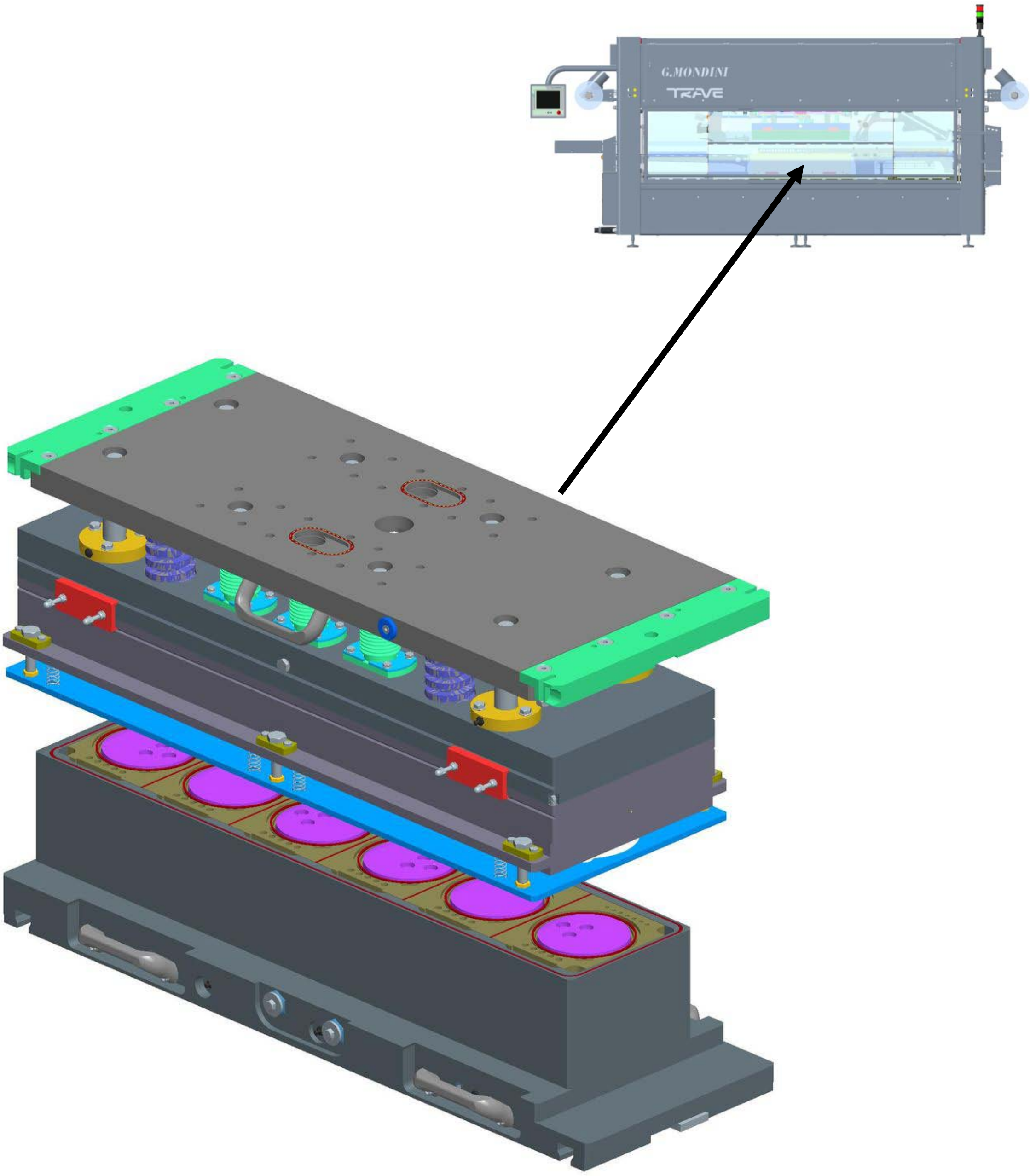
2.5.16 Tool trolley

Function

This castor-mounted trolley allows all the tool-change elements to be kept ready at hand.

It hooks onto the machine so that tools can be removed or placed on it quickly and in complete safety.

2.5.17 Sealing tool



Function

This component (which varies with the type of tray) comprises two parts, an upper tool and a lower tool. It the part of the machine that actually seals the film onto the tray.

2.6 OPERATING CONDITIONS

In order to provide optimal operating conditions, the **Customer** must install the machine and associated systems in such a way as adequate ventilation is guaranteed and they can operate in full compliance with CEI EN60204-1 (Italian Electrotechnical Committee standards). The tray sealer must not operate in atmospheric conditions where there is the risk of explosion or fire.

2.6.1 Lighting

The existing lighting system is deemed appropriate for consignment and commissioning of the machine, and the area involved does not require specific lighting.

The user must provide adequate lighting for access to the work area.

2.6.2 Vibration

The machine does not generate or transmit vibration at frequencies that can affect operator safety.

2.6.3 Stability

The machine does not comprise any elements of precarious stability.

2.6.4 Emissions

This machine is designed for industrial use and the level of emissions lies within the permitted range.

2.6.5 Noise

As a noise prevention measure, G.MONDINI S.p.A. has designed this machine using the best available technology. The noise level at the operator control position is guaranteed to be <75 dB(A).

2.6.6 Fume extraction

The machine does not produce any fumes requiring the installation of an extraction system.

3 SAFETY

CONTENTS

3.1 GENERAL SAFETY STANDARDS

To ensure the maximum working reliability, G. MONDINI S.p.A. has made a careful choice of the materials and components to be used to manufacture the **Closing Machine TRAVE**, subjecting them to inspection before delivery.

Good performance over time also depends on correct use and adequate preventive maintenance, following the instructions contained in this handbook.

The materials are of the best quality and the arrival in the company, storage and use in the workshop are constantly checked to ensure there is no damage, deterioration or malfunctioning.

However, it should be remembered:

- The line is never to be used, nor any intervention be made on it, before carefully reading and fully understanding this handbook, from cover to cover.
- It is forbidden to use the system under conditions, or for a use, that is different to that indicated in the handbook. G. MONDINI S.p.A. shall not be held liable for breakdowns, faults or accidents caused by ignoring this prohibition.

NOTE:
It is forbidden to tamper with, alter or change, even partially, the fixtures dealt with in this handbook, especially the guards provided for safety of persons. It is also forbidden to operate in any way that does not comply to the indications or to neglect operations that are necessary for safety.

It is the task of the operator to ensure that the line is kept clean and free of foreign matter such as scraps, oil or other.

If air or water is used under pressure for cleaning operations, the operator is to wear protective goggles and mask and make sure that there is nobody near the line that could be hit by material or dust. Inflammable liquids are not to be used for cleaning operations.

Periodically check the condition of the plates and if necessary restore them to the proper condition.

After cleaning the line, check there are no worn or damaged parts (if there are, call immediately for the maintenance engineer) or parts that are not securely fixed (intervening within the competencies of the operator).

Special attention is to be paid to the intactness of flexible hoses and other parts subject to wear.

If these situations are found at the end of the work shift, before leaving, the operator is to place a warning signboard on the control console indicating that the machine is under maintenance and is not to be started up (these signboards can usually be found on the market in the form approved by EU standards).

The operator is to keep the line and the surrounding area free of all loose items that are not integrating parts (for example tools left after a maintenance operation, ropes used to handle parts, personal belongings, etc.).

The operator, the assistant and the maintenance engineer are to wear suitable protective clothing according to the need and the characteristics of the work executed on the line component. It is also important not to wear loose clothes or chains, bracelets or other items that could become entangled in moving mechanical parts. Long hair is to be gathered in a net to avoid becoming caught up in the machinery.

Line protections and safety devices are not to be removed, except where necessary for repair and/or maintenance.

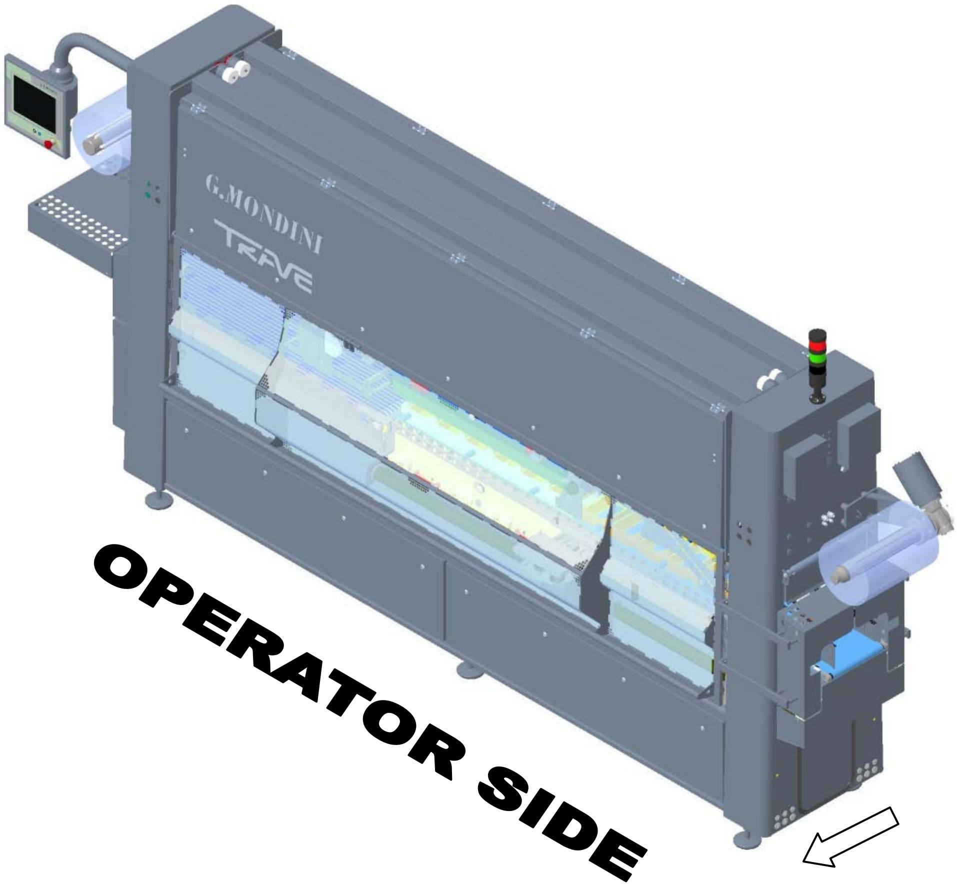
They are to be restored to normal functioning as soon as the reason for their temporary removal has been solved, and in any case before starting up the line.

3.2 DESCRIPTION OF THE WORK STATION

G.MONDINI S.p.A. has always designed and manufactured their machines with explicit focus on the safety of the operator, who shall perform his tasks safely and with full control at every stage of the operating cycle.

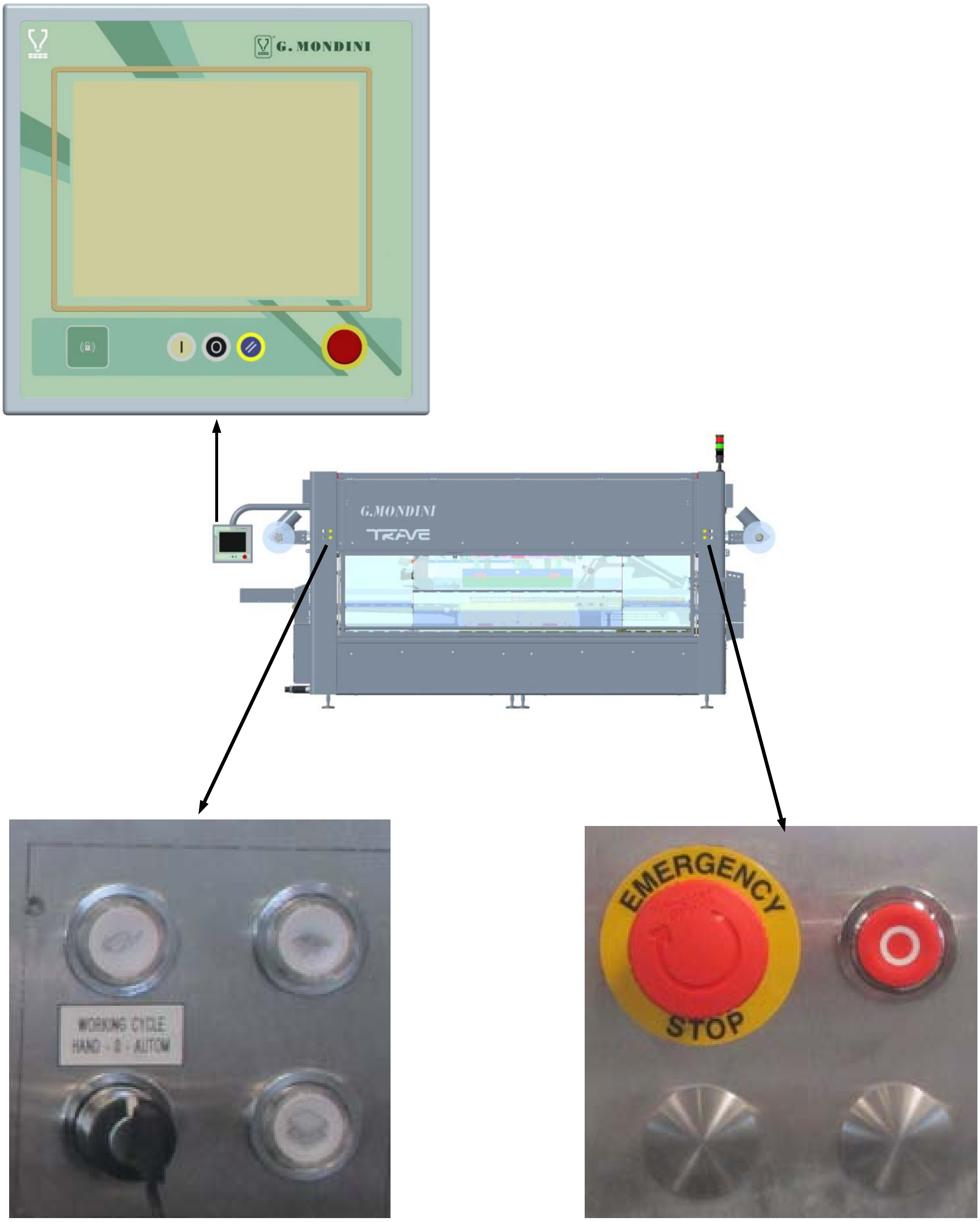
The machine was designed and built by identifying an "*operator side*" that is agreed jointly with the customer, where the following main operations can be carried out:

- installation and removal of the closing tool;
- replacement of sealing film coils;
- removal of sealing trimmings.



Nevertheless, the operator can move along the machine perimeter and have access to all controls during operation.

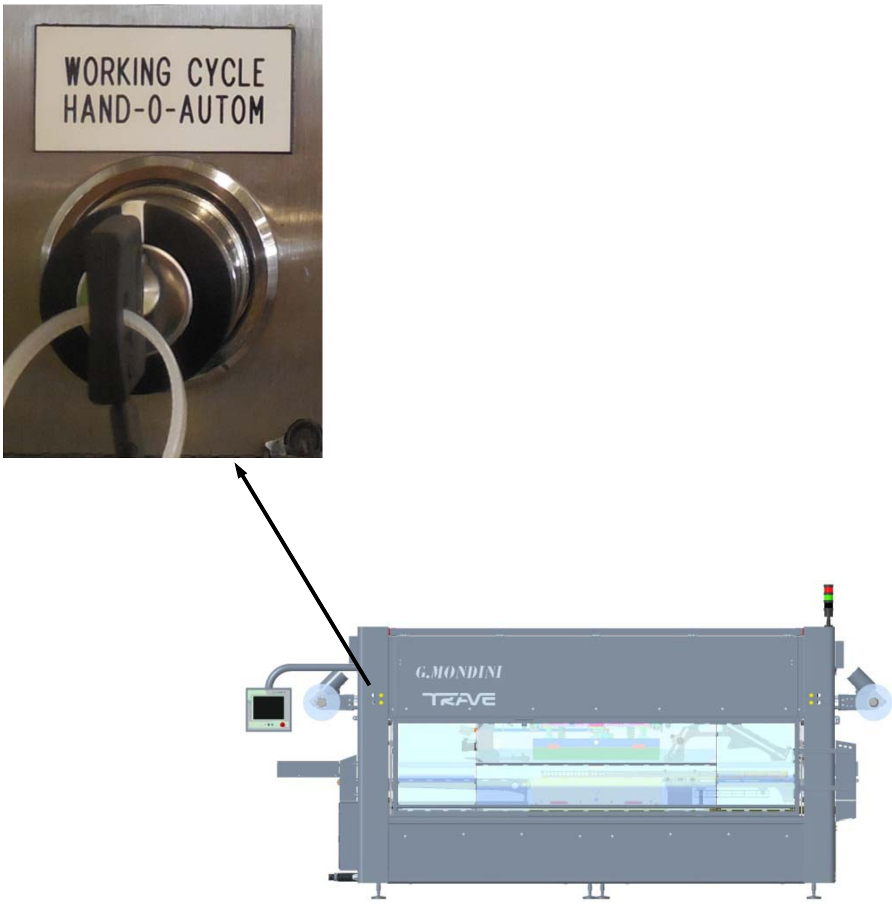
All the control devices required for machine operation are accessible from the *"operator side"*. All the commands are clearly visible and identifiable with plates with a description explaining their use. The controls are located at a safety distance from hazardous areas, including the emergency stop buttons that cause the immediate halt of the machine in case of hazard or as required.



The machine cannot be operated by more than one person at a time, despite the operator can, from the operating position, ensure at any time that there is no one else operating the rest of the installation. A sound and/or visual warning activates on turning the machine to allow anyone present to leave the hazardous area or to prevent the machine from starting up.

The machine can be started and restarted after a stop only by the operator in charge of operation using the appropriate commands.

The machine can be run either in “automatic” or “manual” mode, which can be selected by the selector switch knob situated on the “operator side”. The selection of one operating mode excludes the other.



3.3 FORESEEN, UNFORESEEN AND IMPROPER USE

The line described herein has been designed to execute packing operations on certain foodstuff products.

DANGER

The use of the line for purposes and processes that are not described in this handbook is **IMPROPER USE**. G. MONDINI S.p.A. shall not be held in any way liable for damages caused to goods and / or persons and shall consider any type of guarantee of the line as null and void.

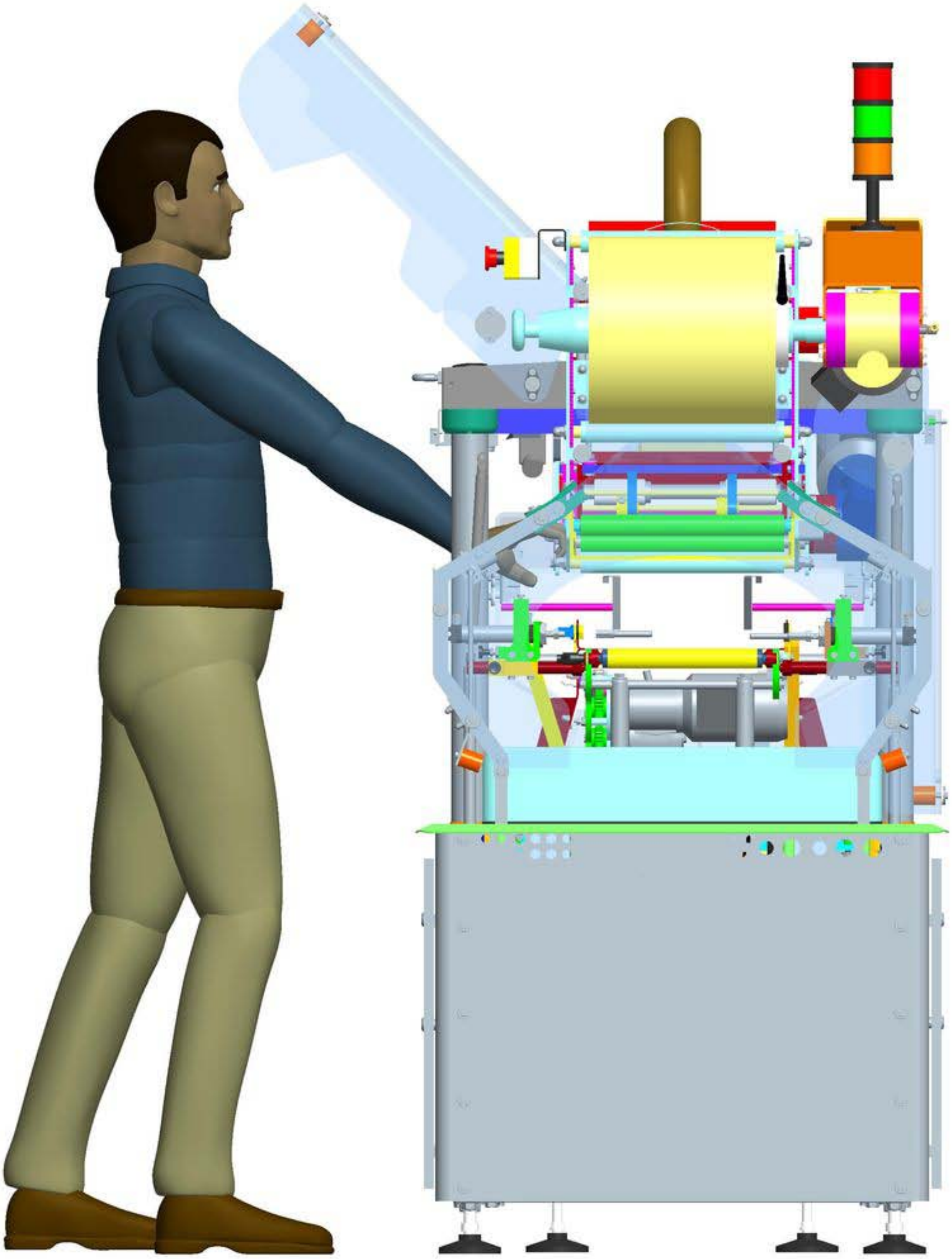
It is forbidden to use the line with a number of workers over the number foreseen in this instructions handbook.

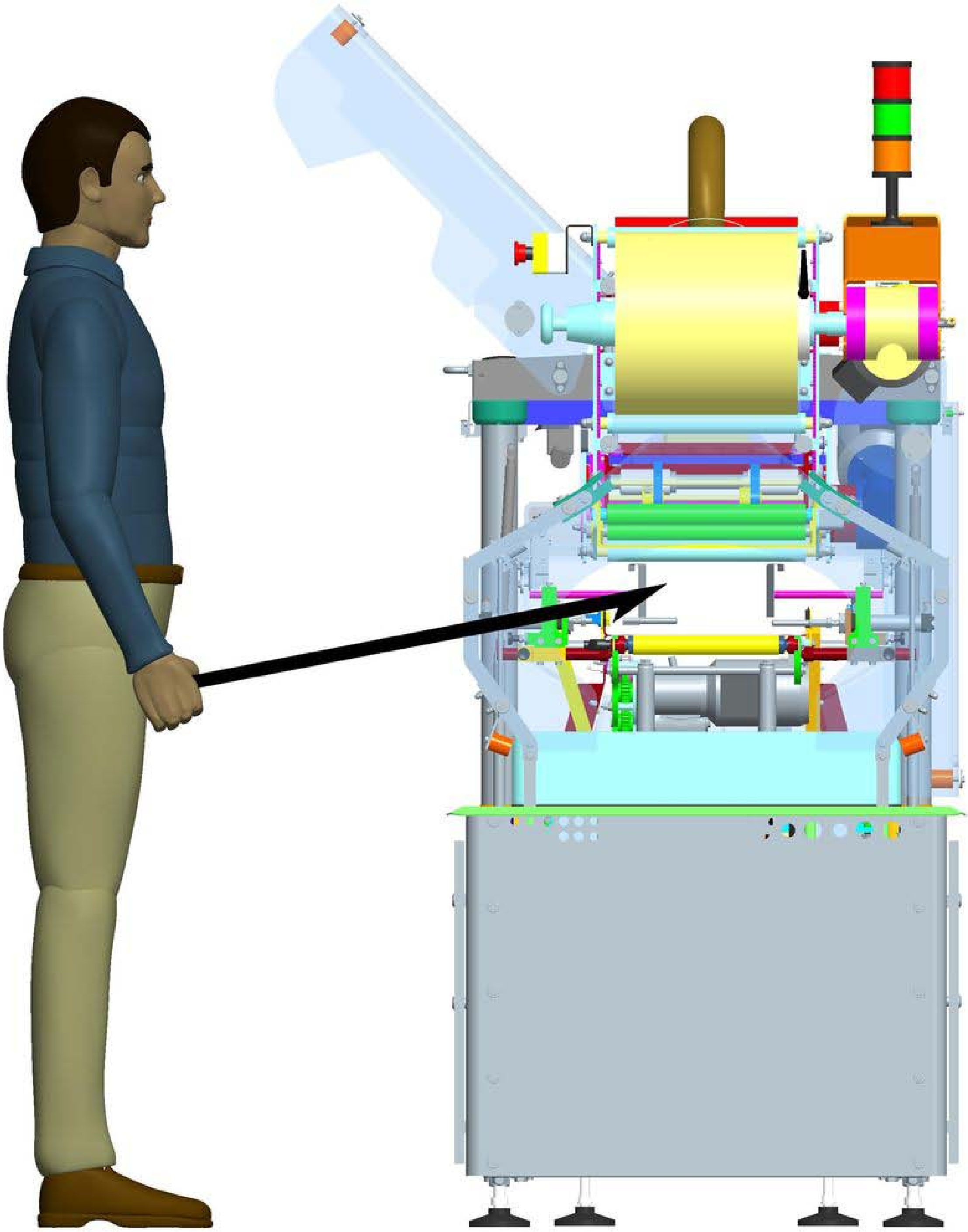
As a function to the use of the fixture, the presence of others in its vicinity when working is not foreseen.

It is the responsibility of the user to install the line so it is not adjacent to passageways; otherwise the user is to provide appropriate barriers to prevent others accidentally entering the work zone.

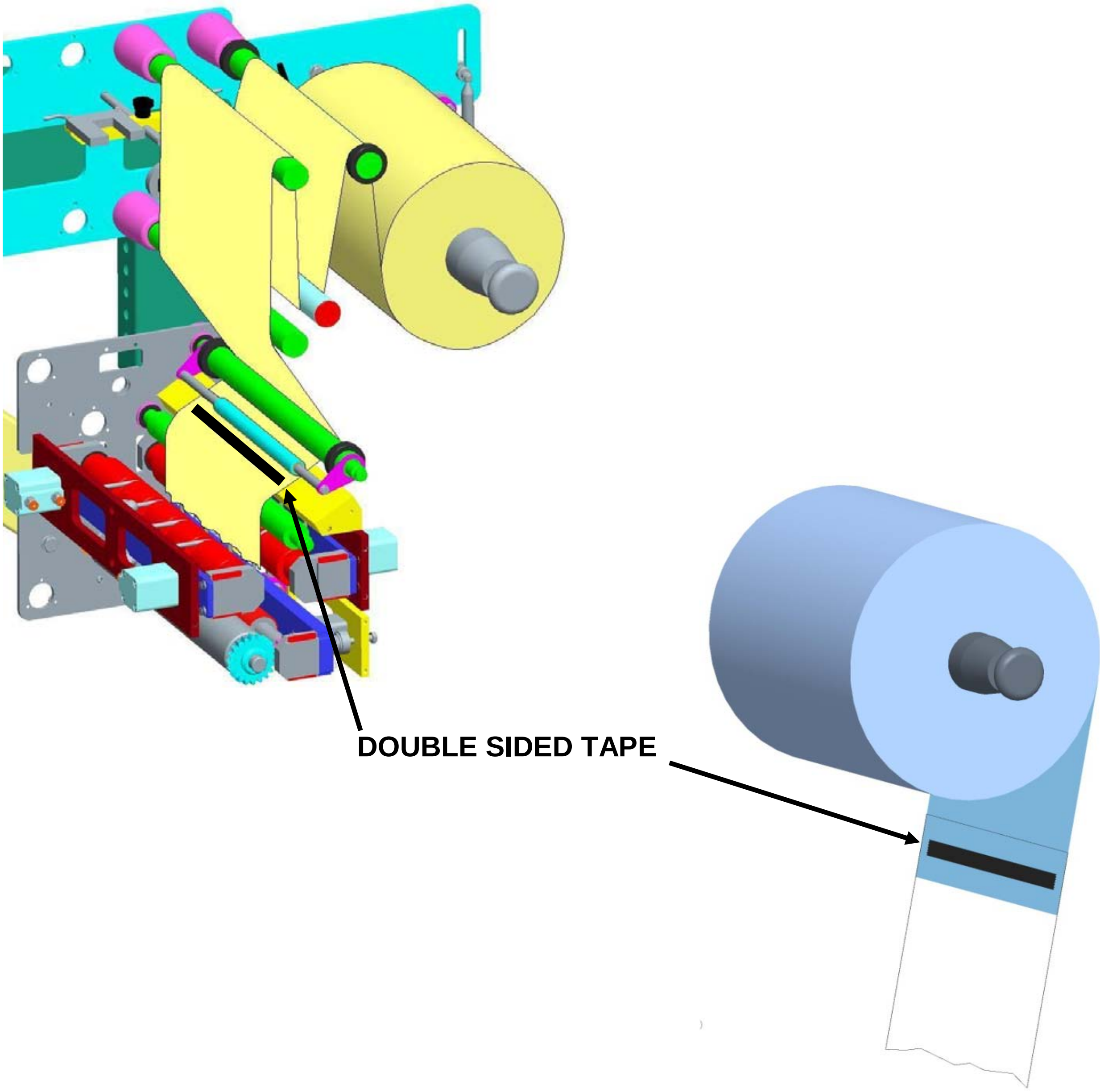
The following is not allowed:

1. Use of materials that are not specifically foreseen by the manufacturer.
2. Handling, installation, inspection, power-on, preparation, use, shut-down, maintenance methods that are not those indicated.
3. Use as a means of transport or to move in the space with the line moving parts running .
4. Allow unskilled persons to use the line.
5. Run the line without protections or with only partial protections.
6. Install them in a way that is not as indicated in the Installation and the Commissioning chapters (in the case of re-installation).
7. Connect the power supplies to the line inadequately or in manner that does not comply to the instructions.
8. Serious neglect in maintenance .
9. Make unauthorised changes or interventions (especially on safety devices).
10. Use of spare parts that are not originals or not for that model.
11. Total or partial non observance of the instructions.
12. Use of the fixture in a manner that does not comply with the methods described in the handbook.
13. Modification of system programs without serious assessment and authorisation from the manufacturer.
14. Put your hands into any section of the system, even when it is disconnected from the electricity. Both standard and reactive machine cleaning, maintenance and the sealing film bobbin change operations must be performed using appropriate tools, to ensure complete and overall operator safety.
15. Before resuming production after any type of maintenance, always check that the nuts and bolts have been tightened correctly. Every six months it is advisable to check all the nuts and bolts on the machine, especially those subjected to vibration like the whole sealing tool group.
16. Automatic solenoid valve operating mode.
17. More than one operator running the machine at the same time.





We suggest replacing the old film reel sticking the end of the film with the new one, using a self double sided tape.



3.4 RESIDUAL RISKS

3.4.1 General information

During the design, all the zones and parts that incur risks have been assessed and as a result, all the necessary precautions have been taken to avoid risks to persons and damage to **Closing Machine** components.

> **WARNING!**
>
> Periodically check that all safety devices function correctly. Never remove the fixed or movable guards from the system. Do not leave objects or tools in the working area.

3.4.2 Residual risks and precautions

After carefully studying all the possible risks on the line, all the necessary solutions have been applied to remove the risks and limit the dangers for exposed persons.

On the line, although fitted with these safety systems, some risks remain that can be eliminated or reduced by the relevant precautions.

Table 3.1 summarises the list of residual risks that remain on the line.

Table 3.1 - Summary of residual risks

Type of risk	Description of the risk	Precautions
	The safety circuits could be subject to a lowering of the safety level.	Make periodical checks (once a month) on the serviceability of the safety circuits.
	Connections between safety circuits and control relays could be subject to failures that are not immediately detected .	Make periodical checks (once a month) on the preservation and efficiency of the control relays.
	Crushing risk following access to moving parts caused by the removal of fixed parts, or bypassing safety devices when an automatic or manual cycle is running.	The exposure of persons to moving parts of the machine can cause seriously hazardous situations. The machine is not to be started without all the fixed and movable protections correctly installed.
	Accident risk caused by deliberate tampering with safety devices or machine parts.	It is forbidden to make any type of mechanical, electrical or fluidic modification, to avoid creating additional hazards and consequent risks that have not been foreseen.
	Electrical hazards caused by unauthorised access to electric cabinets.	The keys giving access to electrical equipment are not to be given to unauthorised operators.

Type of risk	Description of the risk	Precautions
	Breaking or damage risks, with possible lowering of the safety level, of electrical equipment components following a short circuit .	Before making the connection check that the d.c. current at the installation point is not higher than that indicated on the protection switches on the electric panel; otherwise the user is to provide the appropriate limiting devices.
	Accident risk caused by poor lighting in the area where the machine is installed.	The user customer is to provide lighting in the area surrounding the machine, especially at the operator work stations, to comply to the relevant standards in force in the user country and according to the EU Directives.
	Accident risk caused by unexpected movements or contact with live parts.	Before starting maintenance activities on the machine that require the replacement of assemblies or components, cut-out and padlock the energy sources using the specific cut-out devices.

3.4.3 Grease families NLGI 00

Composition, information on the components

Components in concentrations equal to or over 0.1% of the weight (classified as toxic or very toxic) or 1% (classified as harmful, irritating or corrosive).

DANGEROUS COMPONENTS	DANGEROUS CONCENTRATION
Not classified	

Type of risk

Product consisting of basic oils at high degree of refining and additives. The product is characterised by low oral and skin toxicity values and under normal use conditions should not present health hazards. Scrupulously observe the recommendations for use.

First Aid

INHALING

In the case of over exposure to mist, vapours and fumes, take the injured person immediately into the open and, if necessary give artificial respiration.

Call a doctor immediately.

SKIN CONTACT

Carefully wash with abundant water, and if available use neutral soap.

Remove shoes and contaminated clothing. If the irritation persists, go to the doctor. If the material, due to improper use of the high pressure greaser, has been injected under the skin, call a doctor immediately.

EYE CONTACT

Wash immediately with abundant water until the irritation ceases. If the irritation continues go to the doctor.

SWALLOWING

If swallowed, do NOT cause vomiting. Keep the injured person resting.

Call a doctor immediately.

Fire fighting measures

Extinguishing agents: foam, chemical powder, carbon dioxide.

FIRE AND EXPLOSION HAZARDS

Low risk combustible material. The product may form inflammable mixtures and burn only is heated to temperatures over its flash point. However, the presence of small quantities of more volatile hydrocarbons could increase the risk.

SPECIAL FIRE FIGHTING MEASURES

Use nebulized water to cool the surfaces exposed to the fire and to protect the personnel putting out the fire. Fire fighters exposed to fumes and vapours are to wear adequate protection (respirators and masks).

DANGEROUS COMBUSTIBLE PRODUCTS

In the case of incomplete combustion, smoke, sulphur oxides and carbon monoxide could develop.

Measures to control accidental spillage

PRECAUTIONS FOR THE PERSONNEL

See EXPOSURE CONTROL AND PERSONAL PROTECTION

SPILLAGE ON THE GROUND

Collect up the residual material; spread absorbing substances on the area then collect up in containers. If necessary dispose of the absorbed residue. Inform the competent Authorities if the product has reached water ways or sewers or if it has contaminated the ground or vegetation. Take measures to minimise the effects on waterbeds.

SPILLAGE ON WATER

Not available.

Handling and Storage

Keep the product in cool ventilated areas, away from heat sources. Use suitable equipment for lifting and transporting the bins and heavy packs under safe conditions. The electrical equipment used ids to conform to local ruling concerning fire prevention for these types of materials:

Loading/unloading temperature	°C : ambient
Storage temperature	°C : ambient

SPECIAL PRECAUTIONS

Keep the containers closed.

Exposure control and personal protection

EXPOSURE LIMITS

Oil mists: 5mg/m^3 - limit ACGHIH TLV-TWA, 8 h

Personal protection

In cases of potential contact, use safety goggles, clothing and gloves resistant to chemical agents. If only accidental contact is probable, wear safety goggles with side protection. Other special protection is not necessary if contact with the skin and eyes has been safeguarded. Should the concentration of the product in the air exceed the exposure limits and if the systems, operating methods and other means to reduce the exposure of the workers are not adequate, protection for the respiratory organs is necessary

Stability and reactivity

Stable.

CONDITIONS TO BE AVOIDED

Keep away from heat sources, naked flames and all other sources of ignition.

MATERIALS TO BE AVOIDED

Avoid contact with oxidisers such as liquid chlorine and concentrated oxygen.

DANGEROUS DECOMPOSITION PRODUCTS

The product is stable at ambient temperature.

Toxicity information

OVEREXPOSURE EFFECTS

Inhaling:

Negligible risk at ambient temperature or with normal handling. At high temperatures it may form high concentrations of vapours and mist that could irritate the eyes and respiratory organs. Do not breathe in the vapours and mist.

Skin contact:

Low level of acute toxicity. Frequent or prolonged contact could cause irritation or dermatitis: in the case of wounds caused by jets of grease at high pressure and should this become injected into the skin or other part of the body, since in time it could cause serious damage to the soft tissues, surgical treatment is necessary immediately.

Eye contact:

Slightly irritating: the optical tissues are not damaged.

Swallowing:

Low level of acute/systematic toxicity.

Chronic effects

The basic oils contained in this product have not shown cancerous activity in tests on guinea pigs (long term skin tests).

TOXICITY DATA

Acute toxicity:

Experimental data on the finished product is not available. The heath risks indicated derive from the current knowledge on the toxicity of the basic mineral oils and additives used, taking into consideration the concentration in the finished product. The general effects of this type of mineral oils are well known and indicated in many scientific papers and in the CONCAWE report "Health Aspect of Lubricants" - May 1987.

Chronic toxicity

Not having specific data on all the basic oils used in the product and by comparison with basic oils of similar composition and refining potential cancerous effects are not expected. The basic oils assessed have undergone biological tests according to the Exxon standard protocol. These basic oils have not shown cancerous effects.

Ecological information

Not having specific environmental data concerning this product, it has been assessed on information relevant to the hydrocarbons contained in the lubricating oils of mineral origin. When spilt they remain mainly on the soil or water surface. On the basis of informative literature, this product should not present particular hazards for the earth and/or water habitat. The product is poorly biodegradable and persists in the atmosphere. It may contain additives for which environmental data is not available. Therefore this assessment only concerns the basic mineral oils contained.

Remarks concerning disposal

Collect and dispose of the refuse products in authorised dumps according to the current national and EEC ruling.

Information concerning transport

CONTAINERS FOR TRANSPORT

Bins, buckets, cartridges.

Transport/storage temperature °C: ambient

Other information

TYPE/USE OF PRODUCT

Multipurpose lubricating grease of high quality for industrial and marine use.

Sources of information

The information contained herein refers only to the specific product and may not be valid should the product be used in combination with other products or in other types of processing. The information has been updated to the best of current knowledge at the date of the last revision. However we give no guarantee regarding the accuracy, reliability or completeness of this information. In fact, it is the responsibility of the user to ascertain the suitability and completeness of the information given, according to the particular use to be made.

3.5 SAFETY DEVICES

The closing machine **TRAVE** described herein is fitted with the following safety devices/solutions:

Type of device/solution	Function	Description in paragraph
Main switch	Cuts out the machine electrical power	3.5.1
Air cut-out valve	Cuts out the pneumatic supply	3.5.2
Fixed and movable protections	Confine the machine hazardous zones	3.5.3
Emergency stop	Immediate stopping of the line.	3.5.4
Safety limit switches with interlock on protection doors	Controls the closing condition of the access doors inside the machine and locks them when the machine is running.	3.5.5
Automatic circuit breakers	Cuts out electric supply in the case of overload or short circuit	3.5.6
Individual protection devices	Protection for operators when carrying out the job	3.5.7

3.5.1 Power supply sectioning and padlocking

Function:

Cuts out power supply sources to the machine.

Method:

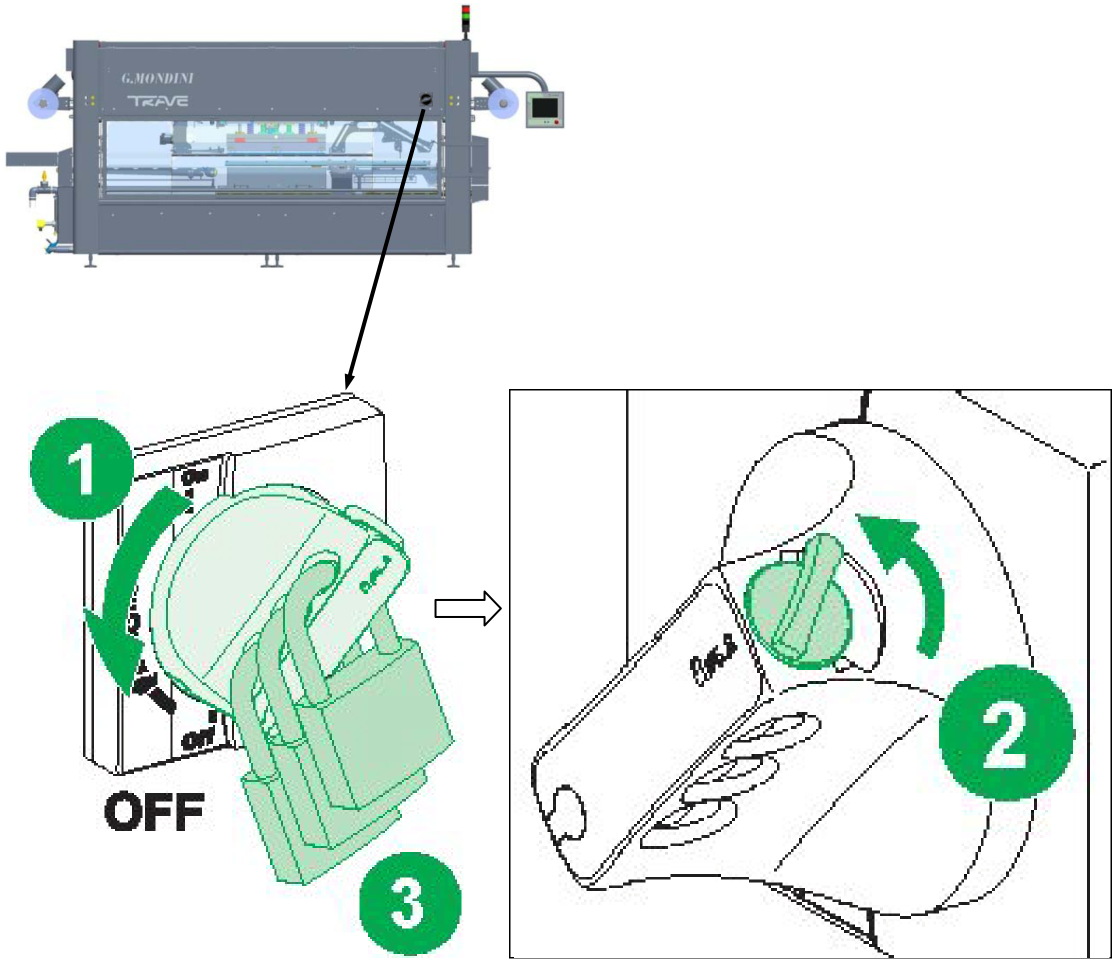
Before carrying out any type of maintenance operation on the machine the energy sources must be cut out and any accumulated energy discharged.

The electric power is cut out through the main switch on the bench.

The master switch on the machine can be padlocked to prevent the risk of injury due to unexpected start-up or contact with live parts.

For a description of risks and precautions, please refer to table 3.1 in this chapter.

G. Mondini machines are normally not supplied with a padlock. This accessory must be requested specifically on finalisation of the contract.



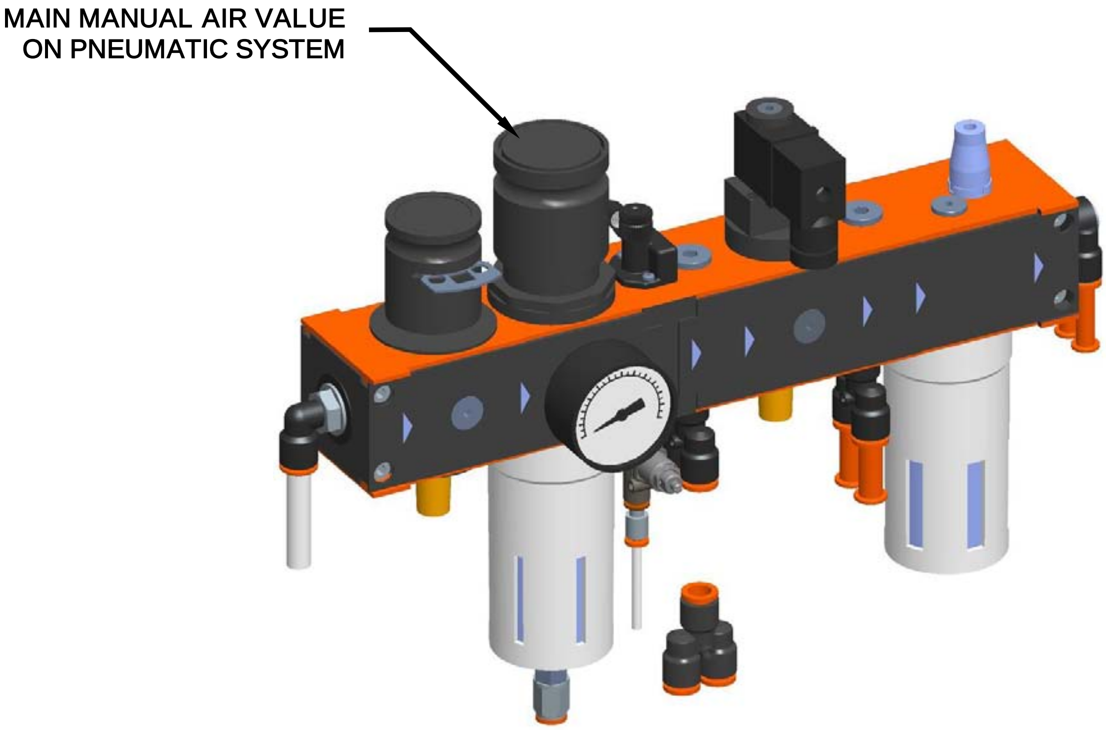
3.5.2 Air cut-out valve

Function:

Cuts out the air supply

Method:

The pneumatic supply is to be cut out through the manual valve on the air treatment assembly.



Manual pneumatic valves are circuit selector valves and their activation inhibits the system air supply.

They shall be used only in maintenance phases, while the machine is not in operation.

Their aim is to increase safety during maintenance operations and they shall be padlocked to prevent involuntary activation.

Pressure restoration through these valves shall occur in full safety, with the guards closed, since it restores only air supply ignoring the state of the solenoid valves that control pistons; the machine shall not be in operation.

These devices are not intended to be emergency devices (they are not connected with any safety unit) and, therefore, cannot be used during the normal production cycle.

Their correct use includes the following steps:

- Guard closing
- Circuit breaker activation
- Disabling of the restoration control with a padlock (the key must be kept by the maintenance person)
- Interruption of power supplies
- Guard opening
- Maintenance operations
- Restoration of safety conditions (guard closing and fitting)
- Restoration of power supplies
- Removal of the padlock from the breaker
- Restoration of the network pressure

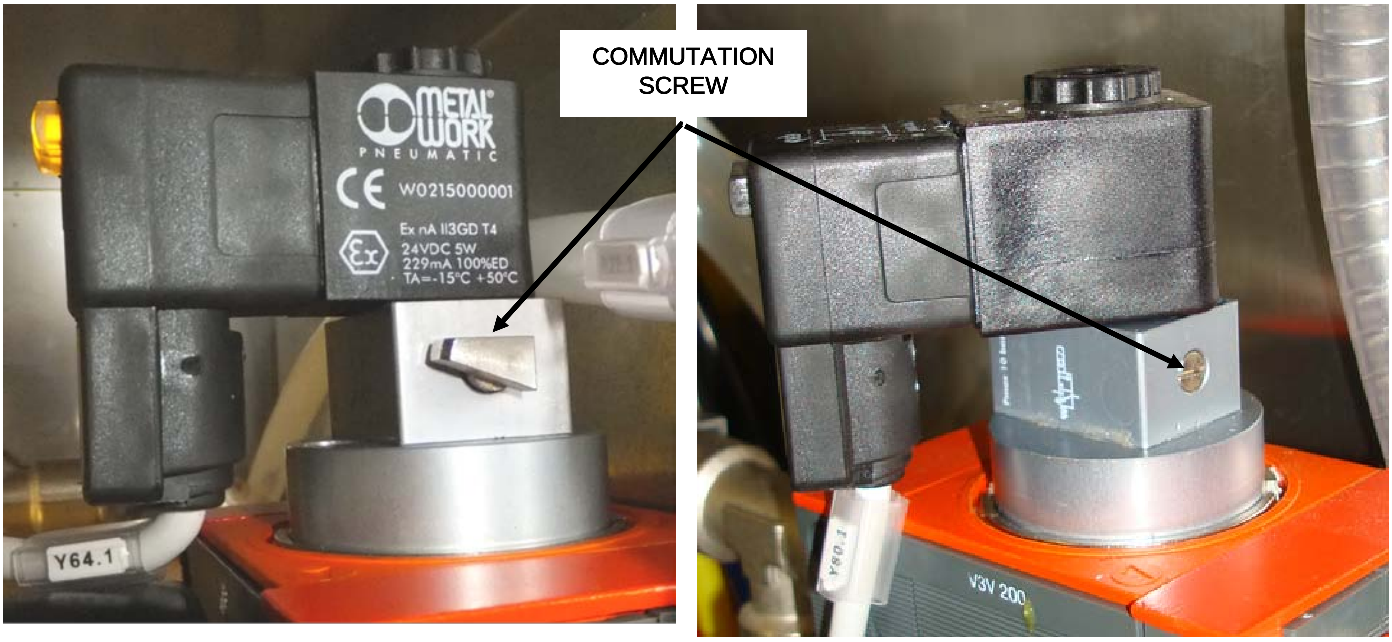
The solenoid valves can work on automatic or manual, depending on the position of the commutation screw on the coil solenoid.

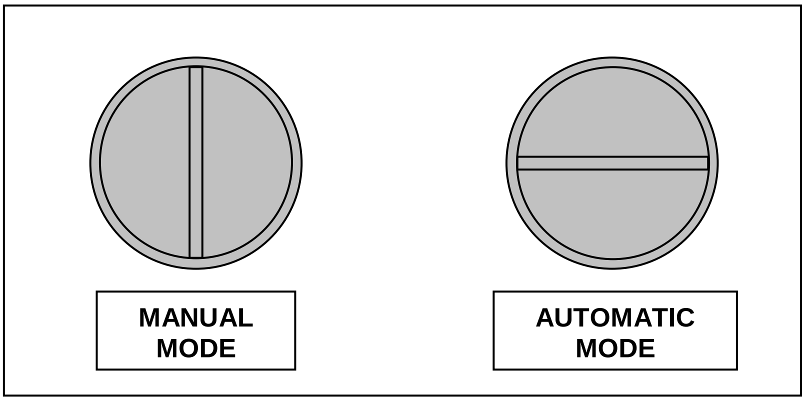
IT IS MANDATORY FOR THE SOLENOID VALVES TO WORK IN AUTOMATIC MODE.

To make sure the solenoid valves are on automatic, check that the commutation screw is in the horizontal position, as shown below. Before switching on the machine, make sure all the solenoid valves are in the automatic position.

Switching to manual mode (screw slot vertical), bypasses the safety circuit and is hence considered as tampering with the system. If this is done, G.Mondini S.p.A. disclaims all liability for damage or injury and any warranties will be invalidated.

NEVER SWITCH SOLENOID VALVES FROM AUTOMATIC TO MANUAL MODE.





3.5.3 Fixed and movable protections

Function:

Confinement of the line hazardous zones of the closing machine **TRAVE**.

Method:

Fixed protections

The fixed protections surround the line working area, to prevent contact with moving components.

They consist of plexiglass and stainless steel structures that are fixed onto the machine and stainless steel structures at the base of the machine.

The fixed protections are not controlled and their removal is subordinated exclusively to maintenance operations, with the machine energy sources cut-out.

After maintenance work, always remember to lock the hinged guards and the drop-down ones at the base of the machine.

Movable protections

The area where the belt of the closing machine runs is protected by a door fitted with an interlocking limit switch. This device stops the machine cycle when the door is opened.

3.5.4 Emergency stop

Function:

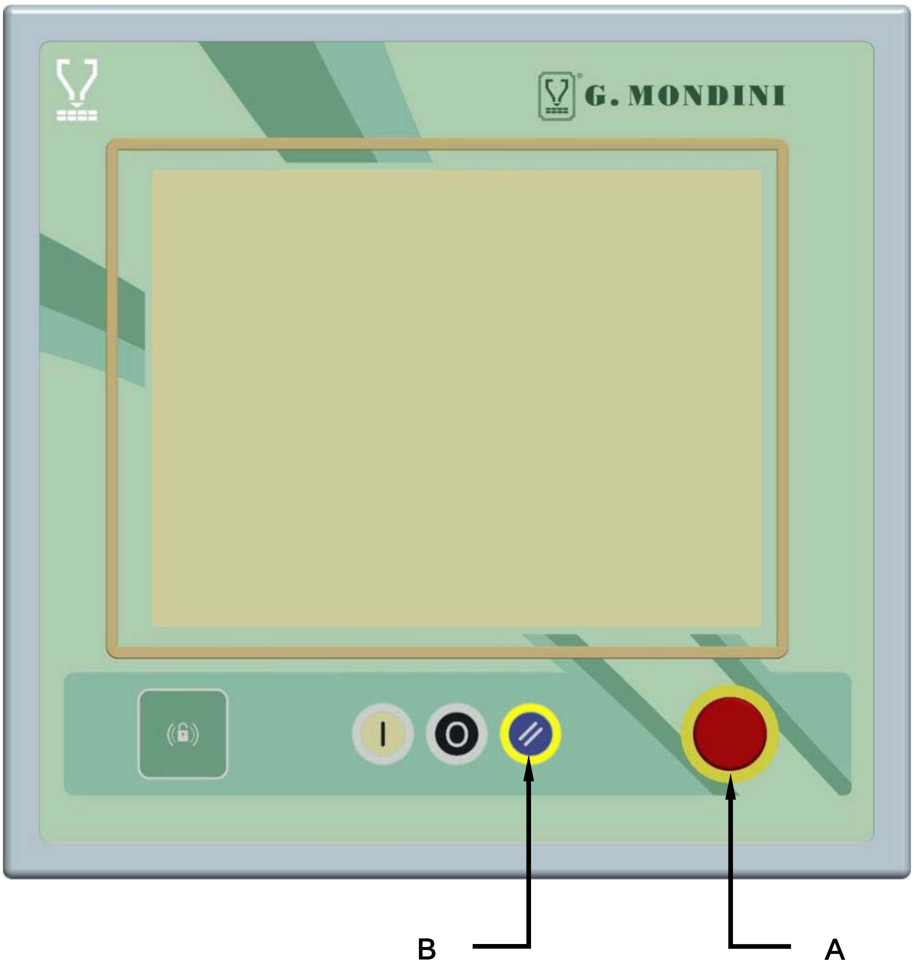
Immediate stop of the closing machine **TRAVE** processing cycle.

Method:

If one of the red "mushroom" buttons (**A**) on the operator panels is pressed, the line processing cycle is stopped, because the power to the control PLC outputs of all the line actuators is deactivated.

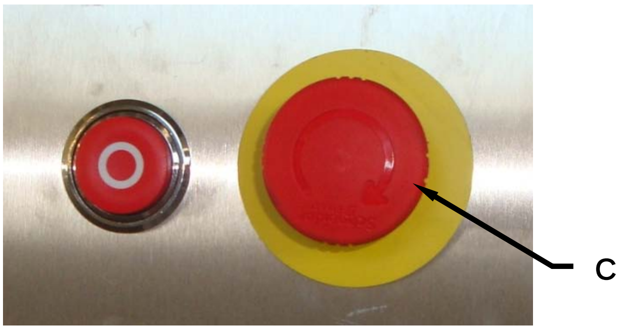
If pressed, the button is lowered and remains down in this position. To rearm, turn the button clockwise.

To restore the power is necessary press the EMERGENCY RESET pushbutton (**B**) on the panel.



If one of the red “mushroom” buttons (**C**) on the operator panels is pressed, the line processing cycle is stopped, because the power to the control PLC outputs of all the line actuators is deactivated.

To restore the power is necessary press the EMERGENCY RESET pushbutton (**B**) on the panel.



DANGER
Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

The emergency circuits are to be checked periodically to ensure the correct functioning.

3.5.5 Safety limit switch with interlock on access door

Function:

Belt sliding protection door closure control.

Method:

The closure of the access door is controlled by safety limit switch.

The limit switch has a mechanical lock to prevent the door opening during the operating cycle.

DANGER
Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

The safety device described above is to be checked periodically to ensure the correct functioning.

3.5.6 Automatic circuit breakers

Function:

Control the electric circuits and cut out the supply before dangerous overloads or short circuits are generated

Method:

If there is an overload and/or a short circuit, the switches cut out the supply to the circuits, causing the machine to stop immediately.

These switches are in the main electric cabinet.

3.5.7 Individual protection devices

Function:

To protect the operator when working.

Method:

The operators who are qualified and authorised to operate on the line are to use the individual protection devices to reduce risks and keep them unharmed.

WARNING!

The clothing of those who run the line or carry out maintenance on it is to conform the safety requirements specified in the EU directive 89/656/EEC and the laws in force in the country where the line is installed.

DANGER!

To avoid mechanical risks, such as dragging, trapping and others, do not wear bracelets, wristwatches, rings or chains during operations in processing cycles and maintenance.

4 TRANSPORT AND INSTALLATION

TABLE OF CONTENTS

4.1 TRANSPORT AND MARKING

The operations described in the following pages are only to be carried out by authorised persons.

The persons assigned to all the installation operations described in this chapter regarding packing, handling, transport, unloading, positioning, connections, checks and controls of the Closing Machine, are to be well trained and have thorough knowledge of accident prevention standards.

Unauthorised persons must remain outside the area of these operations.

The accident prevention precautions and the operations to be carried out that are contained in this chapter are to be always strictly observed during the various operations, to avoid accidents to persons and damage to the fixtures.

NOTE:
For information regarding the handling and lifting of commercial components that are installed on the line, see the specific use and maintenance handbooks.

4.2 PACKING

The size and structure of the line does not permit the shipping of the same fully assembled; therefore the structure and the enclosures have to be dismantled.

All machined surfaces are to be covered with a film of lubricant.

Before starting operations, it is best to define with certainty all the lifting points on the line components, so that this stage is carried out in safety.

If the line is installed for preliminary tests in the manufacturer’s plant, the packing and re-assembly in the Customer’s plant will be carried out by G. MONDINI S.p.A.

The lifting of the line assemblies can be by means of lift truck or with a crane/hoist.

In the first case, position the assembly on an appropriate wooden pallet. The entry points for the trolley forks should be properly identified so that the assembly is lifted in a stable and safe manner (see figure 4-1).

In the second case, sling the assembly with the ropes, taking care to protect parts that could become damaged by the rubbing of the cords with rags and wooden blocks and balance the part before lifting it (see figure 4-2).

WARNING!

When the load has been lifted to a height of more than 30 cm, the operators assigned to the lifting must remain at a safety distance of more than 2 metres from the perimeter of the load. A breakdown on the lifting equipment or an uncontrolled movement of the load is a serious hazard for the safety of the persons.

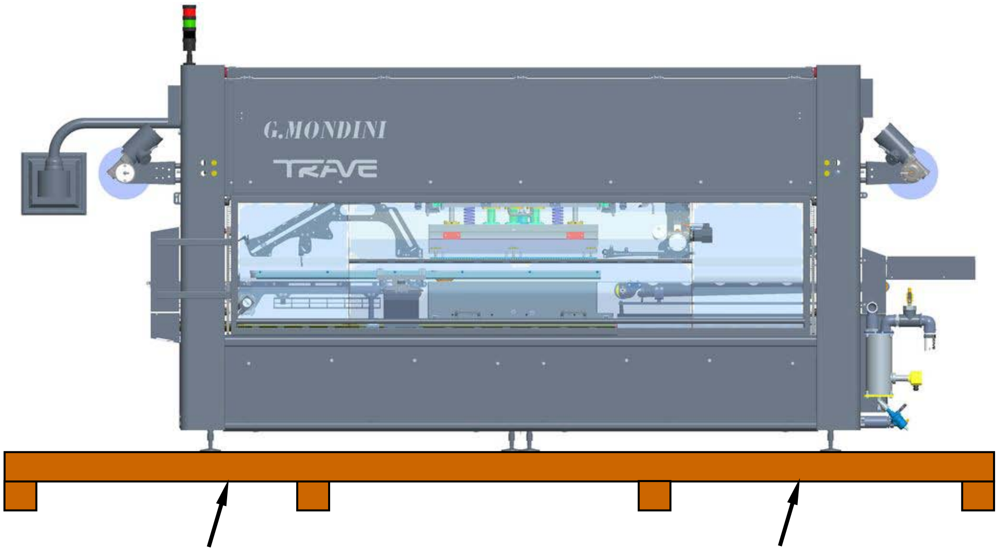


Figure 4-1: Lifting using a forklift truck.

NOTE:
When transporting the machine on a pallet, make sure that the forks are properly inserted at the points shown by the arrows.

Take care to lift the machine keeping the centre of the machine as a reference and make sure it is properly balanced.

If the machine is lifted without a pallet, make sure that the forks are placed at machine's bearing points and they do not damage any parts of the machine that protrude or collapse.

Do not place any wooden inserts, shims or cloths between the forks and the machine to prevent it from slipping.

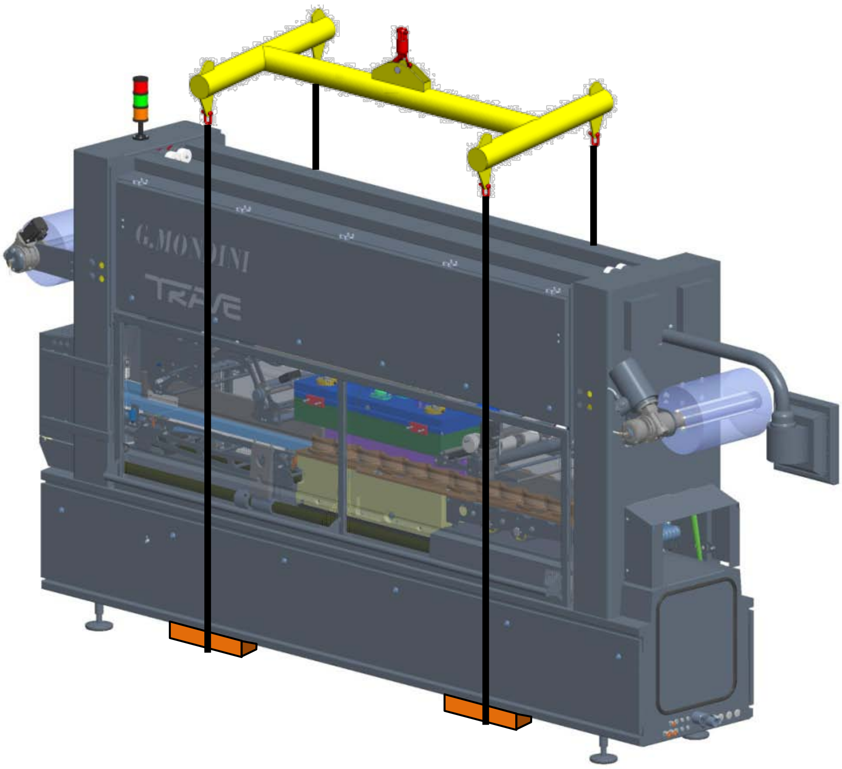
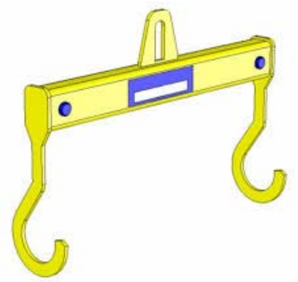
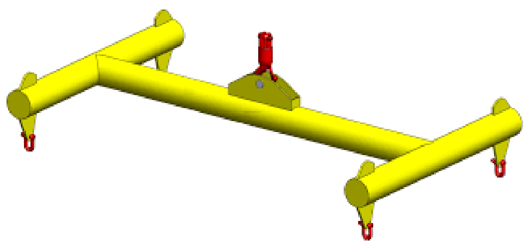


Figure 4-2: Lifting using ropes, cables or belts (crane/hoist)

NOTE:
We recommend using a lifting sling based on one of the systems shown in the pictures below so as to keep the ropes detached from the external cover of the machine.

NOTE:
It is recommended to place wooden inserts between the machine and the ropes in order to prevent sheet metal from being crushed as a result of the stress generated by the ropes.





4.3 TRANSPORT

NOTE:
G. MONDINI S.p.A. shall not be held liable for any damage to persons or goods caused by wrong handling of the loads, executed by unskilled persons and/or using inadequate lifting equipment.

The line parts are shipped, according to the destination, in these ways:

BY SEA ⇒ the assembly is packed in a flat bottom crate and anchored by tie-bars. The crate is lined with tarred paper and has a flap for Customs inspections; it contains bags with drying salts to protect against humidity and salt.

BY AIR ⇒ the assembly is packed in a flat bottom crate and anchored by tie-bars. The crate is lined with tarred paper and has a flap for Customs inspections; it contains bags with drying salts to protect against humidity and other atmospheric agents.

BY LAND ⇒ Land transport can be divided into two types:

LONG DISTANCE TRANSPORT ⇒ the assembly is covered with protective sheets, enclosed in a flat bottom wooden cage and clamped with tie-rods to the loading bed of the semi-trailer.

To lift the crate, scrupulously follow the instructions stamped on the outside of the packing. The packaging can be kept for further use; it is therefore a good practice to keep it in a protected area so that it does not become damaged and hence not reliable. If it is to be disposed of, it will be the responsibility of the **Customer** to dispose of it in compliance with the laws in force in the country.

MEDIUM AND SHORT DISTANCE TRANSPORT⇒ each separate component of the line is fastened to a bed and covered with protective sheets.

4.4 CHARACTERISTICS OF THE INSTALLATION AREA

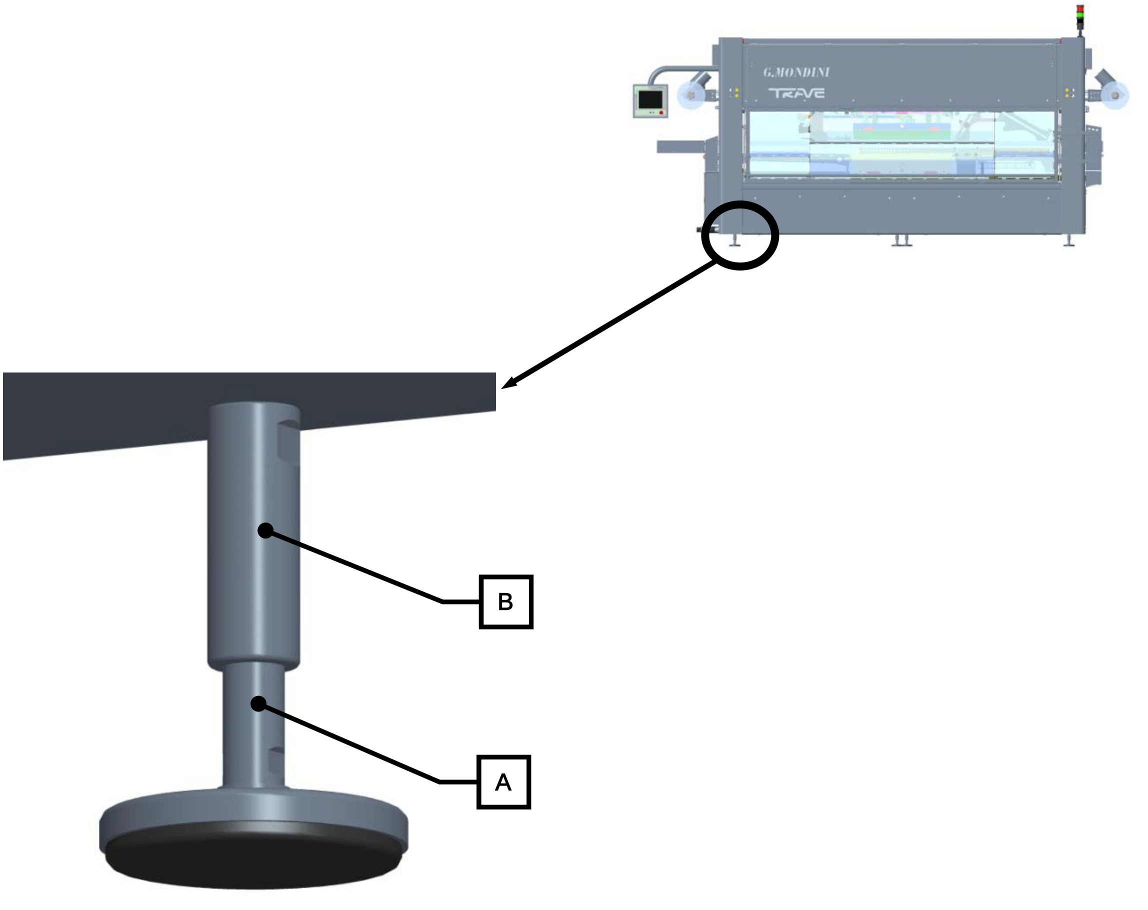
Check that the area selected for the installation is compatible with the overall dimensions on the lay-out and that there are the required environmental conditions.

There must be sufficient space around the line to permit operations and especially maintenance operations.

The floor must be level and flat, dimensioned to sustain the weight of the line to be installed and without ruts or steps.

4.5 POSITIONING

After the site has been decided, the line assemblies are to be positioned, placing them on their feet, to which the rubber vibration dampers are added.



Once the line is in position, the planarity adjustment operations have to be performed.

Make a first alignment with the level adjustment screws (**A**) on the feet (screw to lower and unscrew to raise).

Place a level (of adequate precision) on a line bearing surface and with the alignment screws adjust the planarity, both lengthwise and crosswise. After adjustment has been made, lock all the fastening feet nuts (**B**).

Longitudinal direction

4.6 CONNECTION TO EXTERNAL ENERGY SOURCES

4.6.1 Electrical circuit connection

WARNING!

The machine power supply connection operations are only to be carried out by skilled personnel who are to use all the necessary personal protection devices.

The electrical power supply connection is to be carried out in compliance with the safety standards of the country where the line is to be used.

WARNING!

Before starting the electrical connection, check that the power supply cut-out switch is locked in open position.

Check that the power line is dimensioned to the system total load.

Connect the system to earth before making any other connection to the mains.

Check the correct voltage value indicated on the line ID plate and the type of connection on the power supply transformer on the electric panel.

It is advised to install an automatic cut-out device at the beginning of the supply line for easier maintenance operations.

Make sure that a residual-current circuit breaker is fit up in the electrical distribution line that interrupts the power supply when a ground fault is detected.

Make sure that a residual-current circuit breaker is fit up in the electrical distribution line that interrupts the power supply when a ground fault is detected.

Upstream of the main electric panel, use cut-out devices with an adequate cut-out capacity for the switches on the panel.

Before powering the system, make the checks indicated in the “ELECTRICAL CONNECTION CHECK” paragraph.

The point of connection is located inside the main console.

This position is safe as there is no interference with the moving parts and does not affect the insulation of the console and all the parts connected to it.

The cables are to be connected directly on to the terminals of the main switch. The connection of the earth conductor is to be fastened on the earth terminal as specified on the electrical drawings.

After all connections have been made, refit the terminal covers.

4.6.2 Pneumatic connection

WARNING!

Connection to the machine supplies are only to be carried out by skilled personnel and they are to use all the necessary individual protection devices.

WARNING!

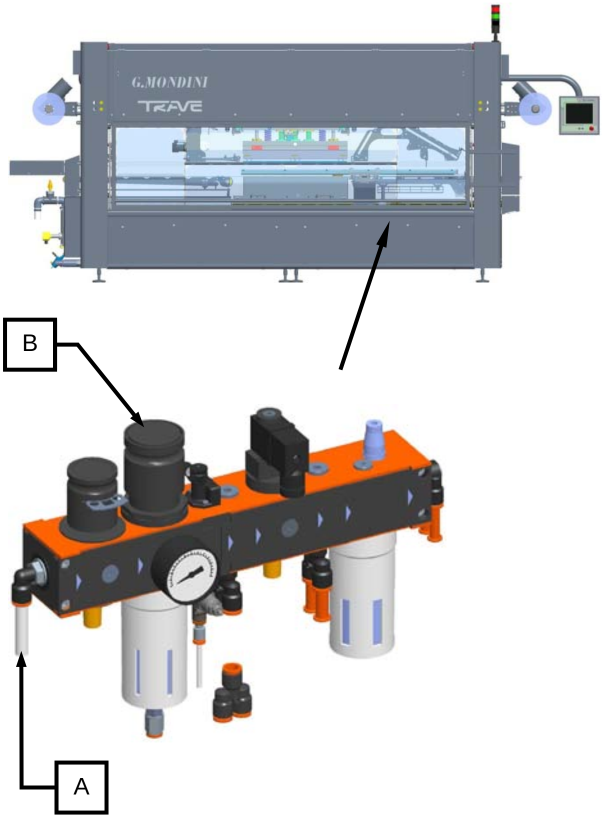
Make the machine pneumatic connections correctly, checking the correct pressure of the pneumatic system.

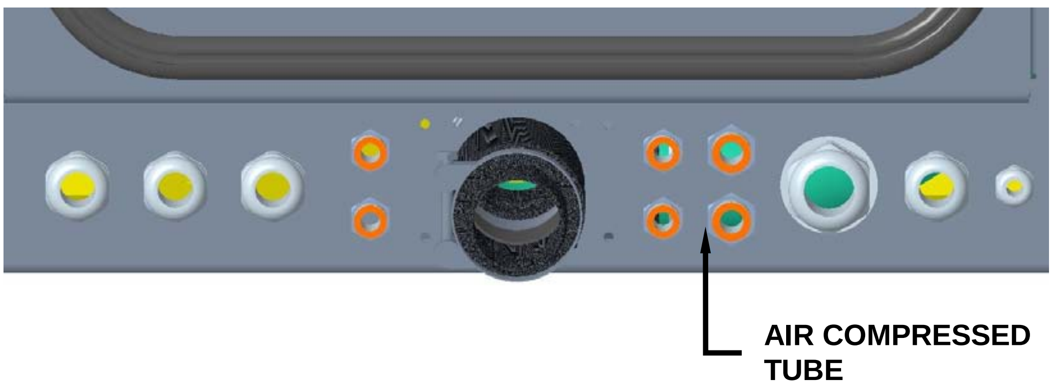
Make sure that the manual cut-off valve is closed.

Before switching on the system pneumatic supply, make the checks indicated in the “PNEUMATIC CONNECTION CHECK” paragraph.

When the pneumatic supply is activated, there could be uncontrolled movements of actuators. Therefore when carrying out the operations be sure there is nobody in the hazardous working zone.

The connection (**A**) of the compressed air tube is to be made on the main air treatment unit of the actuators supply branch, upstream of the manual cut-out valve (**B**).





Use fittings and tubes that have a higher resistance than the maximum compressed air network pressure.

4.6.3 Vacuum circuit connection

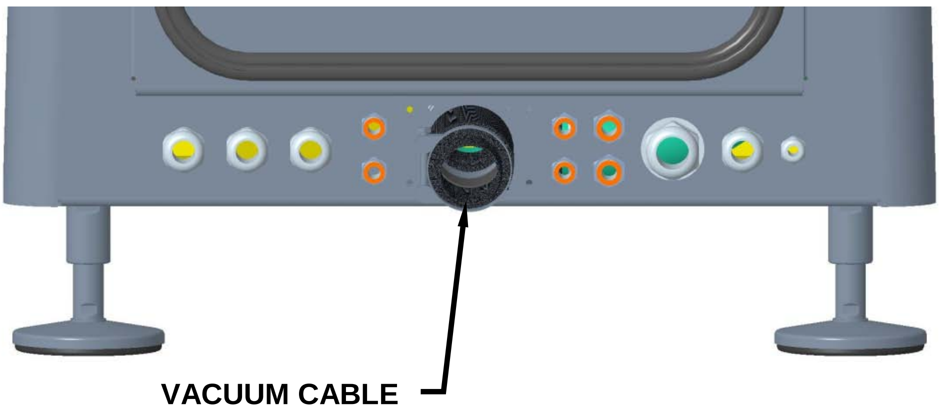
WARNING!

Connection to the machine supplies are only to be carried out by skilled personnel and they are to use all the necessary individual protection devices.

WARNING!

Make the machine vacuum connections correctly, checking the correct pressure of the vacuum system.

The connection point of the closing unit is inside the vacuum panel (behind the electric panel). To reach it , use the hole in the pedestal that sustains the vacuum panel, as shown in the next figure.



4.6.4 Gas circuit connection

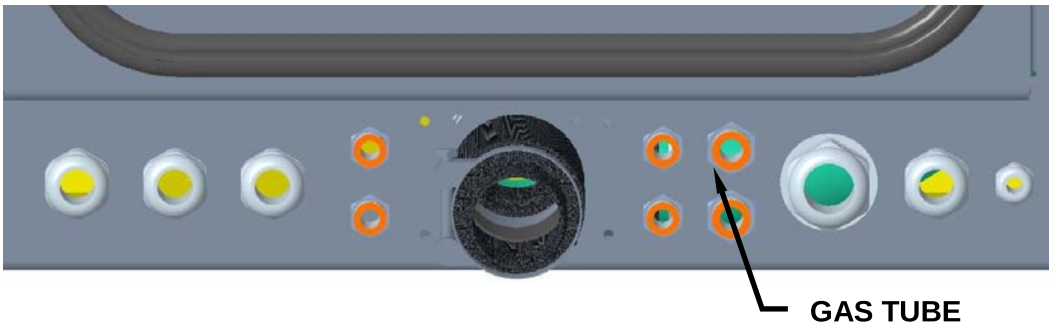
WARNING!

Connection to the machine supplies are only to be carried out by skilled personnel and they are to use all the necessary individual protection devices.

WARNING!

Make the machine gas connections correctly, checking the correct pressure of the gas system **(MAX 8 bar)**.

To connect to the gas circuit, the tube coming from the plant mains line is to be inserted in the machine mask, as shown in the next figure.



4.6.5 Hydraulic system connection (if provided)

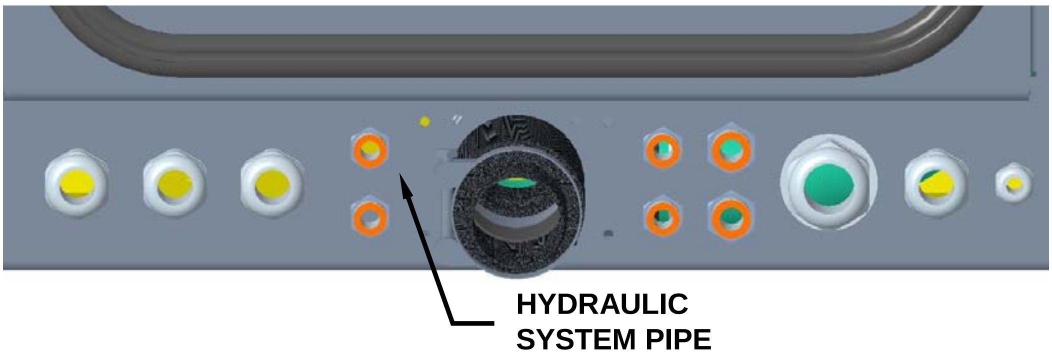
WARNING!

The connection operations of the various machine supplies must be performed solely by appropriately specialised and qualified personnel and are subject to compliance with personal protection equipment requirements.

WARNING!

Proceed with correct hook-up of the machine's hydraulic system connections by checking to ensure that the system pressure levels are appropriate **(MAX 4 bar)**.

For hook-up to the hydraulic system, insert the pipe leading from the factory supply circuit into the machine mask, as illustrated in the figure hereunder.



4.7 FIRST START-UP

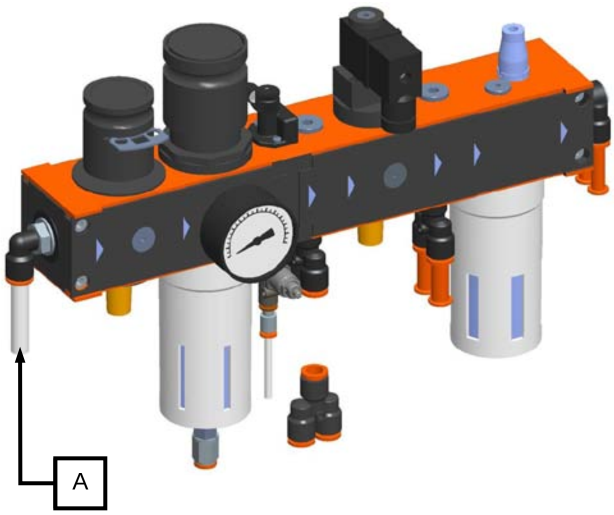
4.7.1 Preparation for the first start-up

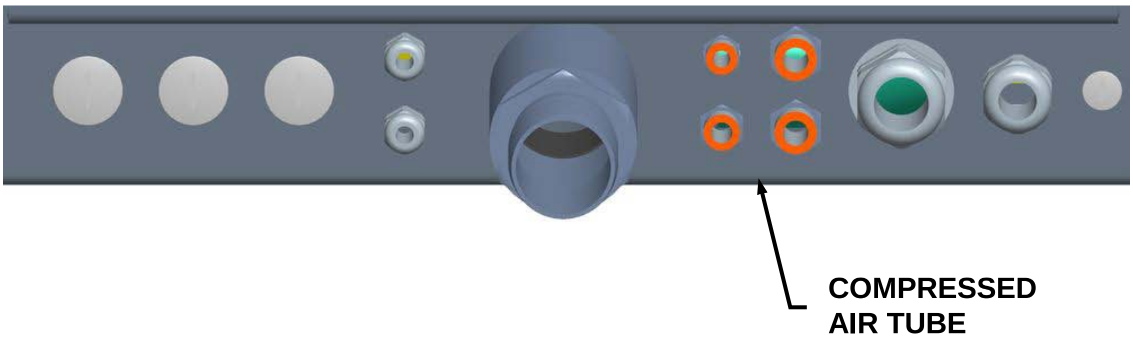
Before starting up the line, check the serviceability of all the connections made and described in the previous paragraphs, and make the checks indicated below.

4.7.2 Pneumatic connection check

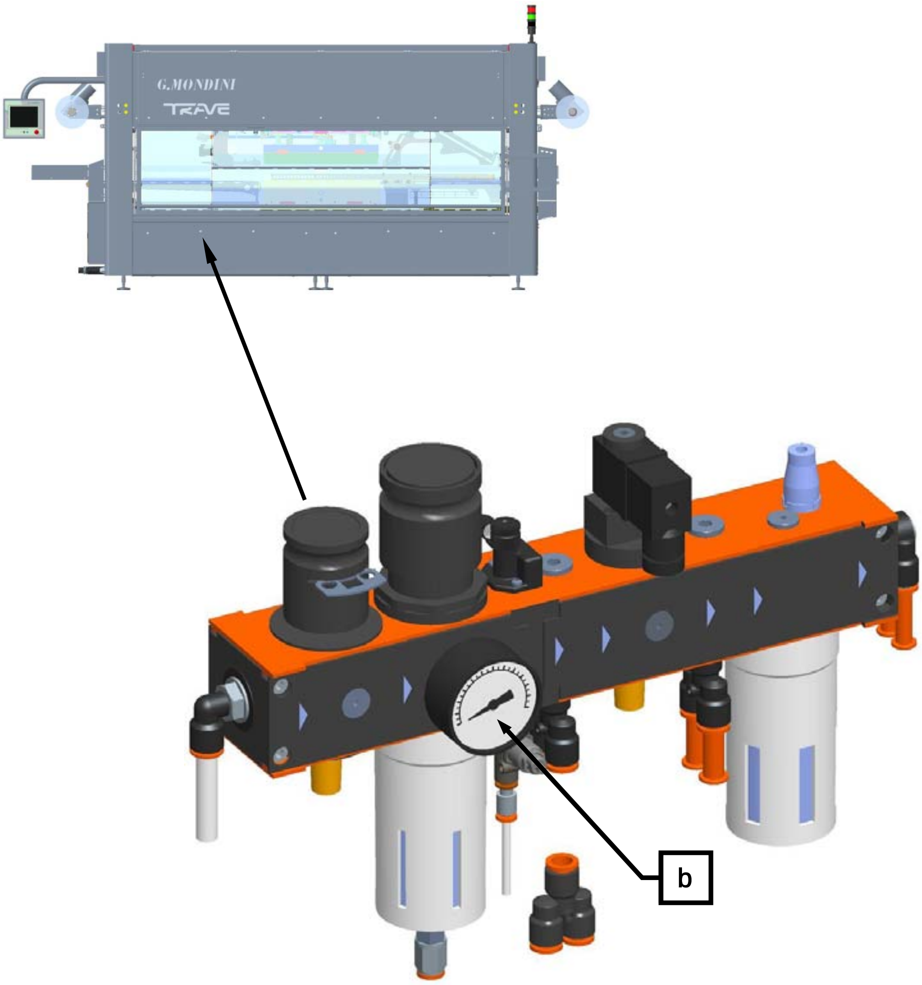
Proceed as follows:

1. Check that the mains air tube (a) has been connected correctly.





2. Check that the pressure value indicated on the air treatment unit gauge (b) is not lower than 6 bar.

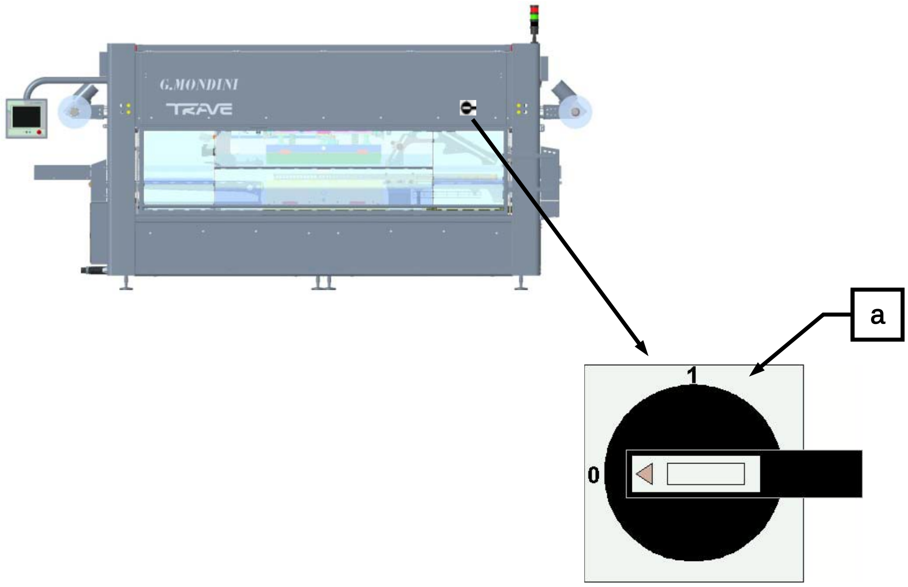


3. Check that the air supply tube fittings to the components of the pneumatic system have been tightened correctly.

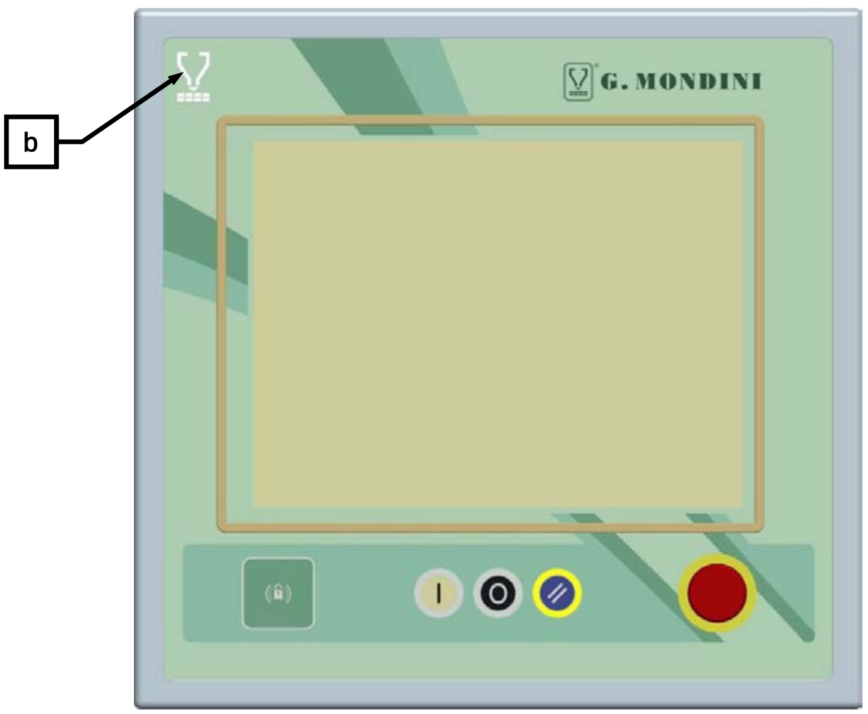
4.7.3 Electrical connection check

Proceed as follows:

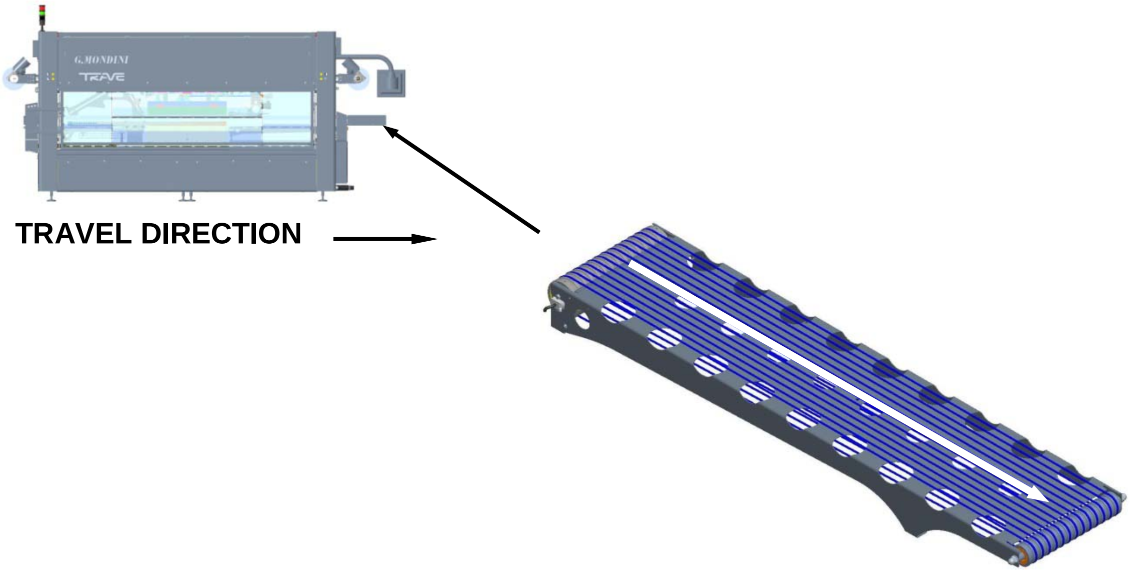
1. Supply the line by positioning the selector switch (**a**) on the electric power panel to 1.



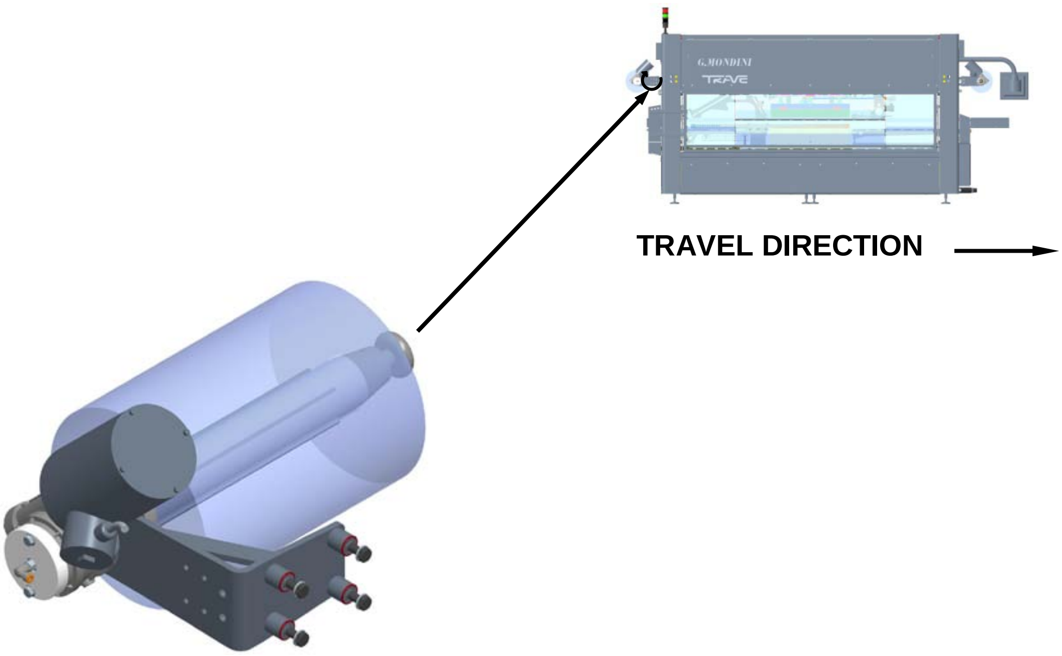
2. Check that on the control panel the “CURRENT INSERTED” (**b**) warning light switches on.



3. Check the closing machine exit belts move in the direction indicated by the arrow in the next figure.



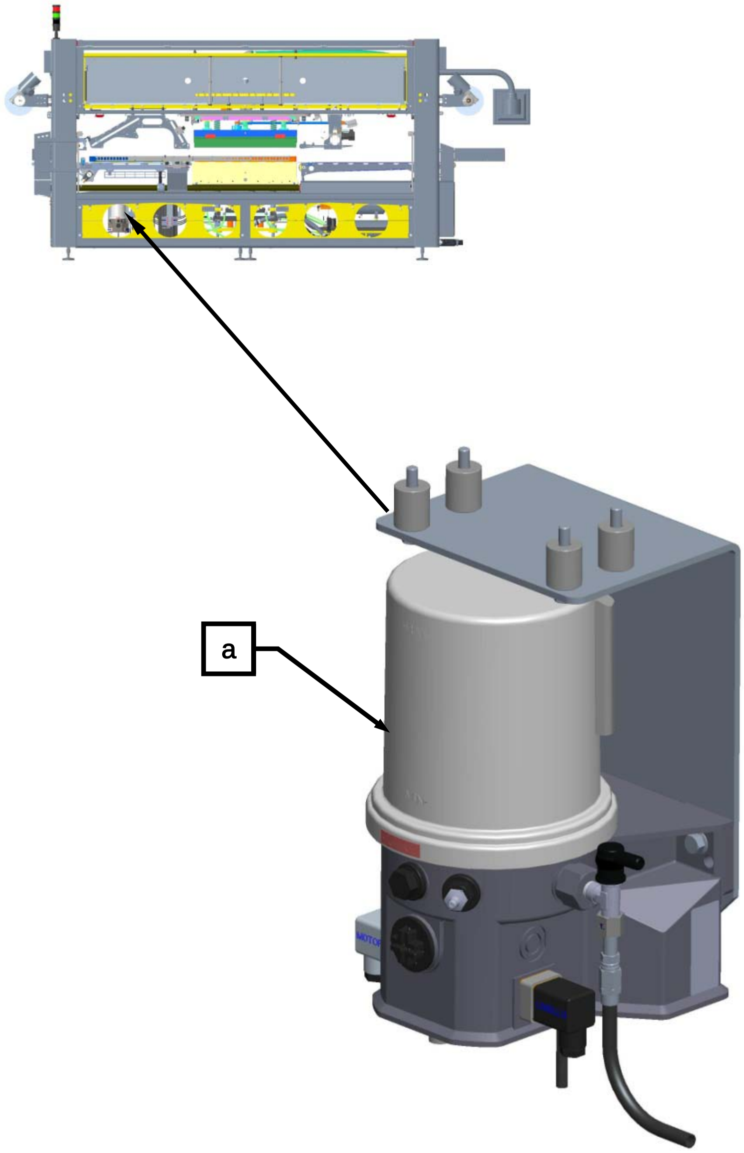
A further check can be made on the direction of the bobbins rotation, which is to be as indicated on the shaft itself.



If one of these components should rotate in the wrong direction, shut off the machine immediately and reverse the cables on the terminals of the main switch.

4.7.4 Lubrication grease level check

Check that the grease level (**a**) is sufficient, if it is not, top up with **NILS ITALIA FOOD NLGI 1** multi-purpose grease.

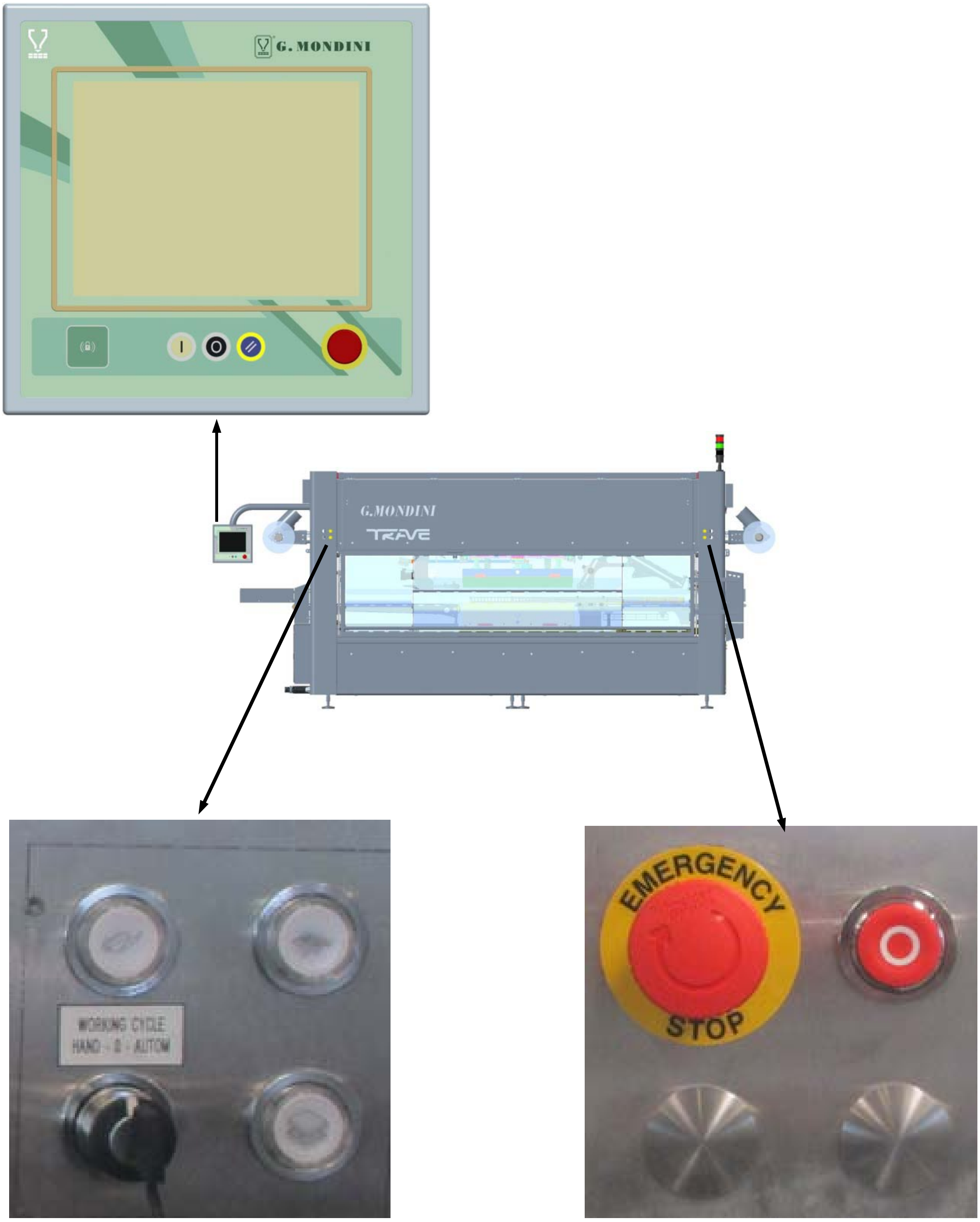


5 CONTROL SYSTEM

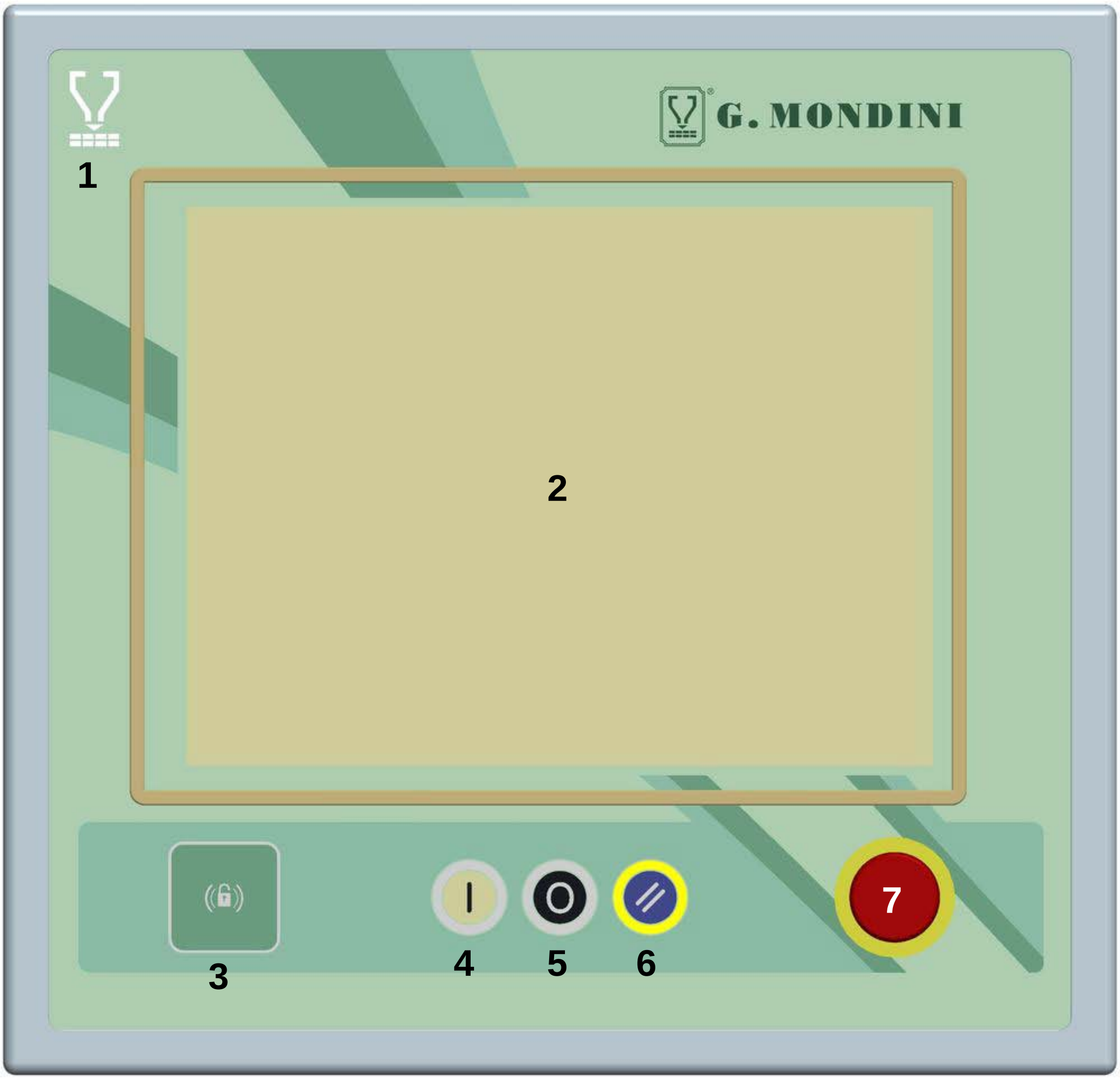
INDEX

5.1 GENERAL DATA

The Sealing Machine's Control System is constituted by a control panel comprising all the commands needed to guarantee control and operation of the machine. Assembled onto a swivel arm, it is fixed onto the machine's top panel.



WARNING! If you touch the panel when switching the machine on you will access the touch screen calibration mode.



1	When it lights up, the white lamp signals that current is inserted to the machine, i.e. that electric power supply is provided.
2	This operator panel is connected up to the PLC module through which all necessary production parameters are set and entered.
3	This device allows to effect the "Login" to a determined level of password. For this machine this device is optional.

4	This button has the function to activate the machine when ready for the production. When this button is pressed with the machine in condition of alarm the machine it doesn't work.
5	The stop cycle pushbutton is used for immediate interruption of machine operations. The peculiarity of this pushbutton is that it provides for “focused” intervention, i.e. the intervention takes place only at the end of the cycle underway. This allows the machine, prior to stopping, to position all the elements subject to cyclic operation, back into their cycle start positions. As mentioned, the stop command does not involve all the machine elements, but only those elements that operate according to subsequent closed cycle chains. An example thereof is the pusher unit assembly motion. Elements subject to continuous operations such as the sealing or snap-on lid outfeed belt or the film waste winding device motor, remain active. For the machine to resume operations after this pushbutton has been pressed, simply press on the “Start” pushbutton again.
6	Button of zero resetting of the alarms in progress. This button is used before using the button of march to take back the production.
7	Pressing this pushbutton will instantly stop all operations and movement of the sealing machine’s in-line elements, regardless of the position they are in.

8	The “manual cycle lid unrolling/decoiler pushbutton” will activate the machine’s relative unrolling cycle. If this pushbutton is set accordingly, it will work both in the “Automatic” cycle and in the “Manual” cycle modes.
9	The “manual cycle pusher arms” pushbutton will only work in the “Manual” cycle mode. Pressing this pushbutton continuously will execute a complete cycle by the pusher arms, that will come to a stop in cycle-start position. If this pushbutton is instead pushed in intermittently, the same cycle is executed but in a fragmented manner. This pushbutton, furthermore, also has a function when the machine is set in the “Automatic” cycle mode. This option is implemented when, at the end of production, the very last containers sealed by the machine needs to be removed from the sealing area. As there are no other containers being fed in along the line, the machine stops in standby mode. Pressing this pushbutton will provide for one complete cycle enabling the sealing area to be totally emptied.
10	The position of this switch determines the type of machine operation mode required: When this selector is in “Manual” position, the machine will run only if the relative commands are active, i.e. only if the manual operation pushbuttons have been activated. When this selector is in “Automatic” position, once the start pushbutton has been pressed, the machine will operate completely independently and automatically. When this selector is in “0” position, the machine goes into standby mode. To reset the machine operating conditions, simply switch this selector either to the “Manual” or to the “Automatic” position and then proceed as if to start up operations.

11	The “manual cycle tool closure” pushbutton will only work in the “Manual” cycle mode. Holding this pushbutton pressed in continuously will execute a complete cycle by the sealing tool, that will come to a stop in cycle-start position. If this pushbutton is instead pushed in intermittently, the same cycle is executed but in a fragmented manner. This pushbutton does not have an active function during machine operations and it is only used for check-ups and verification of the tool and of the various tool movements.
12	Pressing this pushbutton will instantly stop all operations and movement of the sealing machine’s in-line elements, regardless of the position they are in.
13	The stop cycle pushbutton is used for interruption of machine operations at the end of cycle.

5.2 NAVIGATION KEYS

Each page on the operator panel displays a bar of navigation keys.

The arrow keys at the ends of the bar can be used to scroll through the various icons.

Touching any icons displays related sub-pages containing parameters or a description of the associated functions.

MACHINE SYNOPTIC PAGE (HOME PAGE)

SAFETY PAGE (SAFETY DEVICES)

PLU PRODUCT PAGE

RECIPE PAGE

LOCKING PAGE

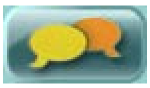
COMMANDS PAGE

MAINTENANCE PAGE

TOOL HEATING PAGE

MOST USED PAGE

FAST ICONS



CHAT FOR TECHNICAL ASSISTANCE WITH G.MONDINI FIRM

LINE CONTROL PAGE (WHERE AVAILABLE)

POWER PACK PAGE (WHERE AVAILABLE)

FILLER PAGE (WHERE AVAILABLE)

DEVIATOR PAGE (WHERE AVAILABLE)

TRAY TURNER PAGE (WHERE AVAILABLE)

SNAP ON LID MACHINE PAGE (WHERE AVAILABLE)

MOBILE HEAD MACHINE PAGE (WHERE AVAILABLE)


NOT USED KEY

NOT USED KEY

NOT USED KEY


PRODUCTION DATA PAGE


TRENDS PAGE


QUALITY PAGE


NOT USED KEY

FILE CONTROL PAGE (USB)

MULTI-LANGUAGE TEXT PAGE

USER AND PASSWORD PAGE

WORKSHIFT PAGE

NOT USED KEY

PANEL CLEANING PAGE

HELP PAGE

WINDOWS EXIT (ONLY WITH "MONDINI" USER AUTHORISATION)

ACTIVE ALARMS PAGE

5.3 PAGE HEADING

Access to machine mode setting (maintenance or pause) and shift display.

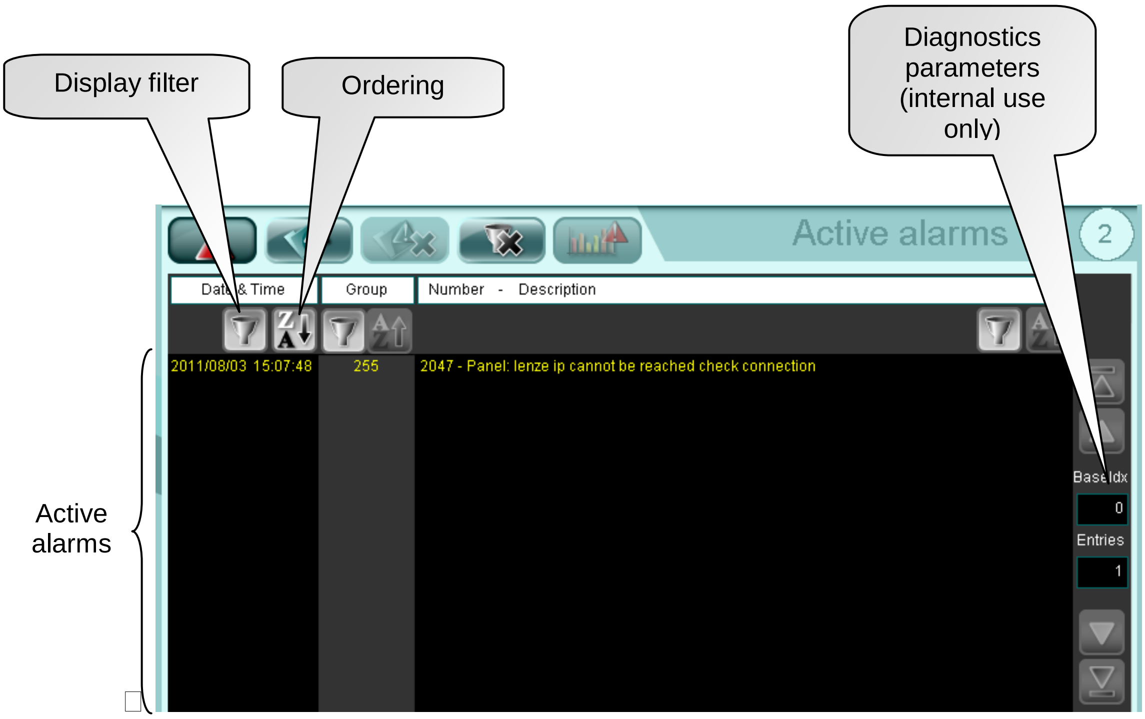
Date and time setting and display

User login

Language selection

5.4 ALARMS

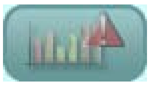
This page displays a list of active alarms occurring during production due to system malfunctions.



 ACTIVE ALARMS PAGE

 ALARM HISTORY PAGE

 SEARCH FILTER REMOVAL

 EXTERNAL DEVICE ALARM TIME GRAPHIC

 ALARM LIST ORDER

 ALARM LIST FILTER

5.5 ALARM HISTORY

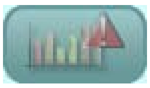
This page displays a list of the alarms that have occurred during production since the last reset.

ACTIVE ALARMS PAGE

ALARM HISTORY PAGE

ERASE ALARM HISTORY

DISABLE SEARCH FILTER

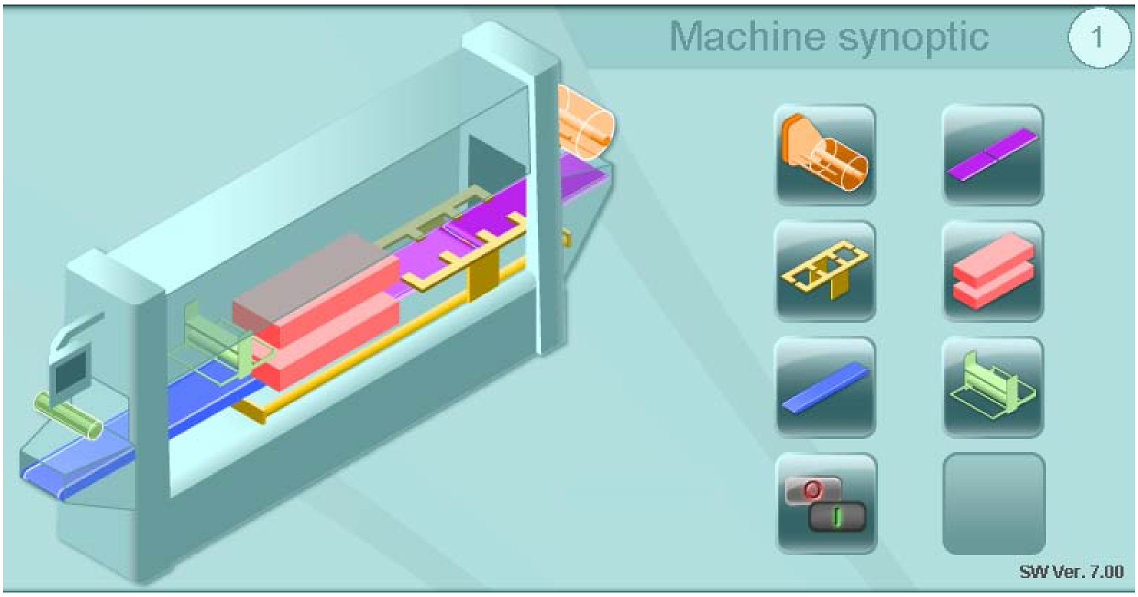


EXTERNAL DEVICE ALARM TIME GRAPHIC

ORDER ALARM LIST

ALARM LIST SEARCH FILTER

5.6 HOME PAGE – MACHINE SYNOPTIC



FILM UNWINDER PAGE

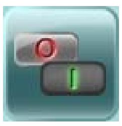
SPACER BELT, FEED BELT AND SYNCRHONIZER (WHERE AVAILABLE) PAGE

PUSHER PAGE

TOOL PAGE

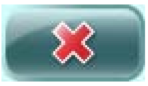
DELIVERY BELT PAGE

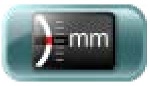
FILM UNWINDER



ENABLES PAGE

5.6.1 Tool

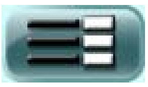

RETURN TO MACHINE SYNOPTIC (HOME PAGE)


TOOL AXIS PAGE


PNEUMATIC SYSTEM PAGE

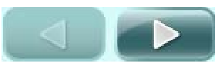

GAS MIXER PAGE (WHERE AVAILABLE)


RETURN TO TOOL PAGE

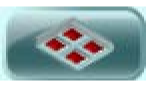

PARAMETERS LIST PAGE


ENABLES LIST PAGE

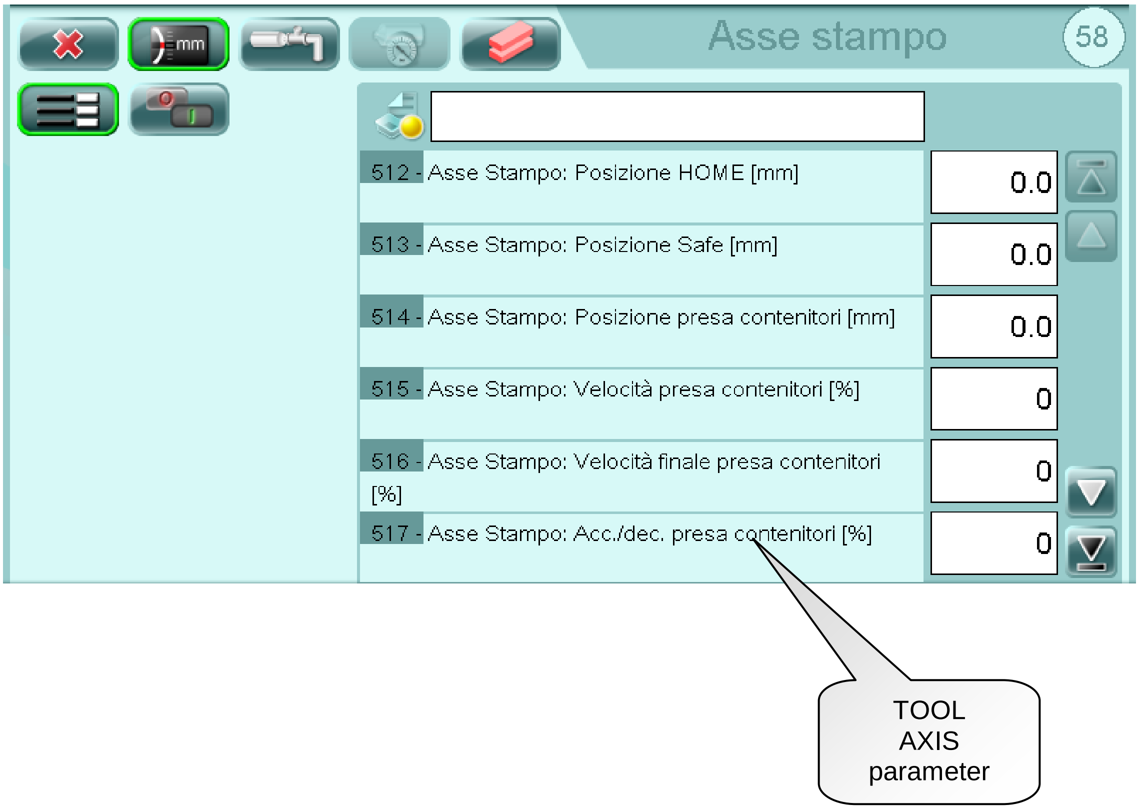

TOOL MAINTENANCE PAGE

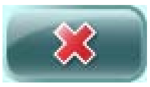

TOOL PARAMETERS TYPE CHANGE


TOOL CAVITIES HEATING/COOLING PAGE


DOT PLATES HEATING PAGE

5.6.1.1 Tool axis



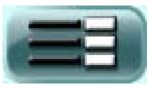

RETURN TO MACHINE SYNOPTIC (HOME PAGE)

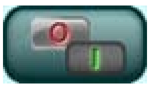

TOOL AXIS PAGE


PNEUMATIC SYSTEM PAGE

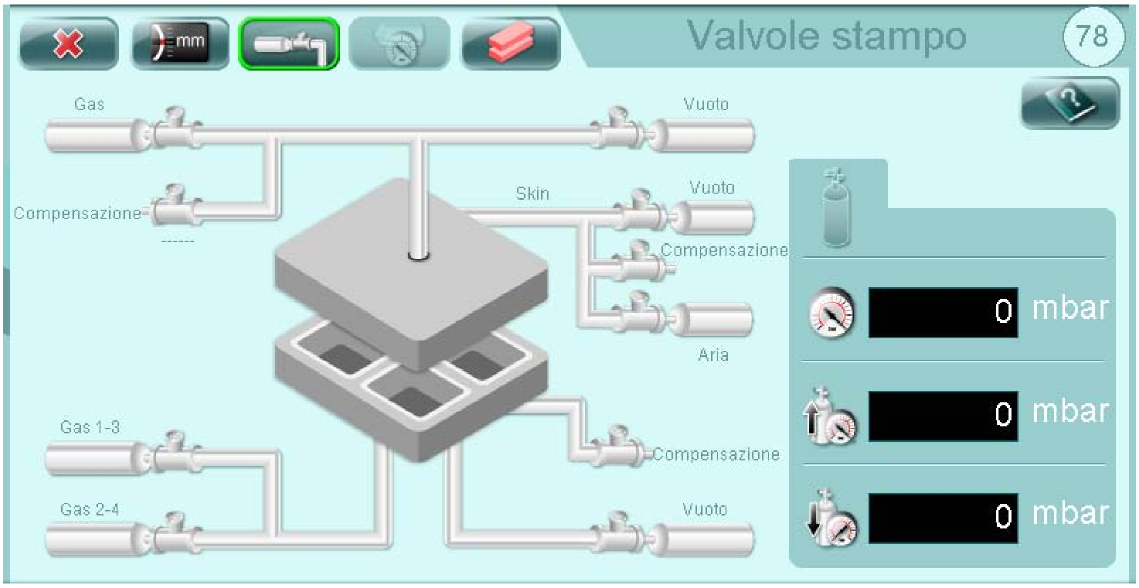

GAS MIXER PAGE (WHERE AVAILABLE)

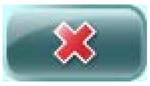

RETURN TO TOOL PAGE


PARAMETERS LIST PAGE

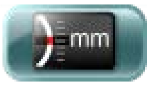

ENABLES LIST PAGE

5.6.1.2 Valve layout





RETURN TO MACHINE SYNOPTIC (HOME PAGE)



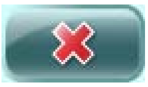
TOOL AXIS PAGE

PNEUMATIC SYSTEM PAGE

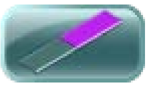
GAS MIXER PAGE (WHERE AVAILABLE)

RETURN TO TOOL PAGE

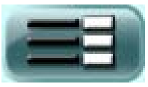
5.6.2 Spacer belt



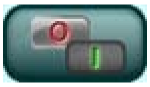
RETURN TO MACHINE SYNOPTIC (HOME PAGE)



FEED BELT PAGE



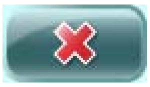
PARAMETERS LIST PAGE



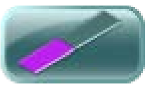
ENABLES LIST PAGE

BELT MAINTENANCE PAGE

5.6.3 Inlet belt



RETURN TO MACHINE SYNOPTIC (HOME PAGE)

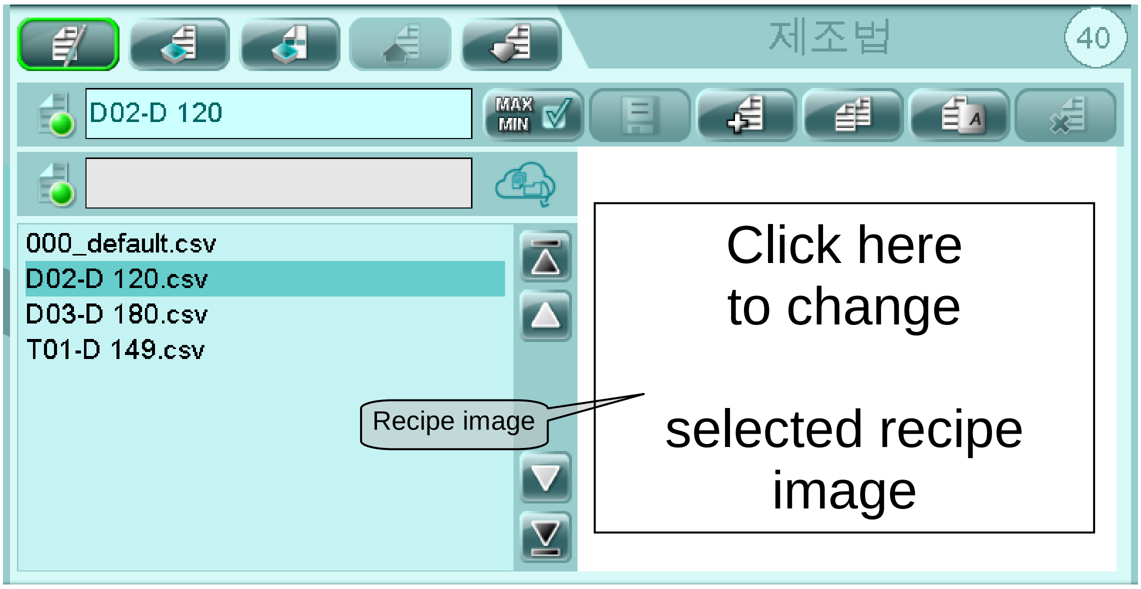


SPACER BELT PAGE

ENABLES LIST PAGE

BELT MAINTENANCE PAGE

5.7 RECIPE MANAGEMENT



IMPORTANT! The “000_default.csv” recipe cannot be deleted or modified as it is the default recipe used at the test stage at the Mondini plant.

Scroll through the recipes to display images of them.
Selection is automatic and the parameters are loaded with a slight delay.

After changing a parameter, it is necessary to save the recipe (for an OP level change) or load the data in a recipe and save it (for a PLC level change).

Data overwritten in the PLC cannot be recovered, so it is advisable to create a new recipe when you wish to change more than one parameter.

Before putting a recipe into production, it is very important for the machine to be stopped and all the mechanical parts ready to receive a new setting.

When the recipe is sent to the PLC, the current one is overwritten. So, when the current recipe parameters have not been saved in a recipe, it is advisable to save them before loading a new recipe.


RECIPE MANAGEMENT PAGE

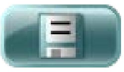

RECIPE PARAMETER PAGE


RECIPE ENABLE PAGE


DOWNLOAD RECIPE FROM PLC (PLC -> OP) Download PLC parameters to the OP parameters of the current recipe


UPLOAD RECIPE TO PLC (OP -> PLC) Copy OP parameters to PLC parameters

Parameter limit SETTING and consents

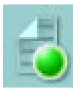

SAVE SELECTED RECIPE DATA (OP parameters)


NEW RECIPE (a copy of the “000_default” recipe is created and must be renamed)

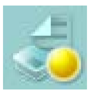

COPY RECIPE (a copy of the selected recipe is created and must be renamed)

RENAME SELECTED RECIPE

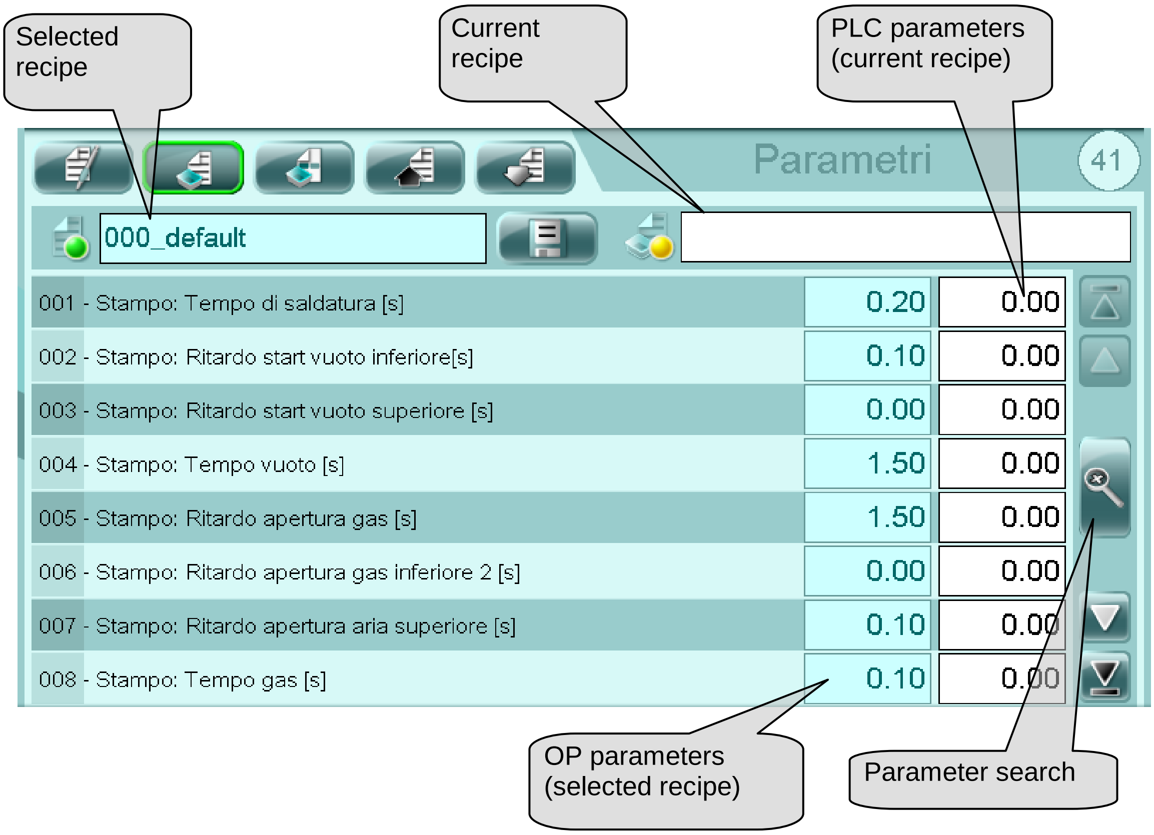

DELETE SELECTED RECIPE


RECIPE STATUS (MODIFIED OR SAVED)


RECIPE REQUESTED BY THE SUPERVISOR

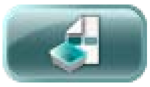

RECIPE DISPLAYED DIFFERENT FROM THOSE IN THE PLC

5.7.1 Recipe parameters

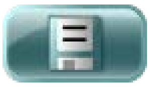


RECIPE MANAGEMENT PAGE

RECIPE PARAMETER PAGE

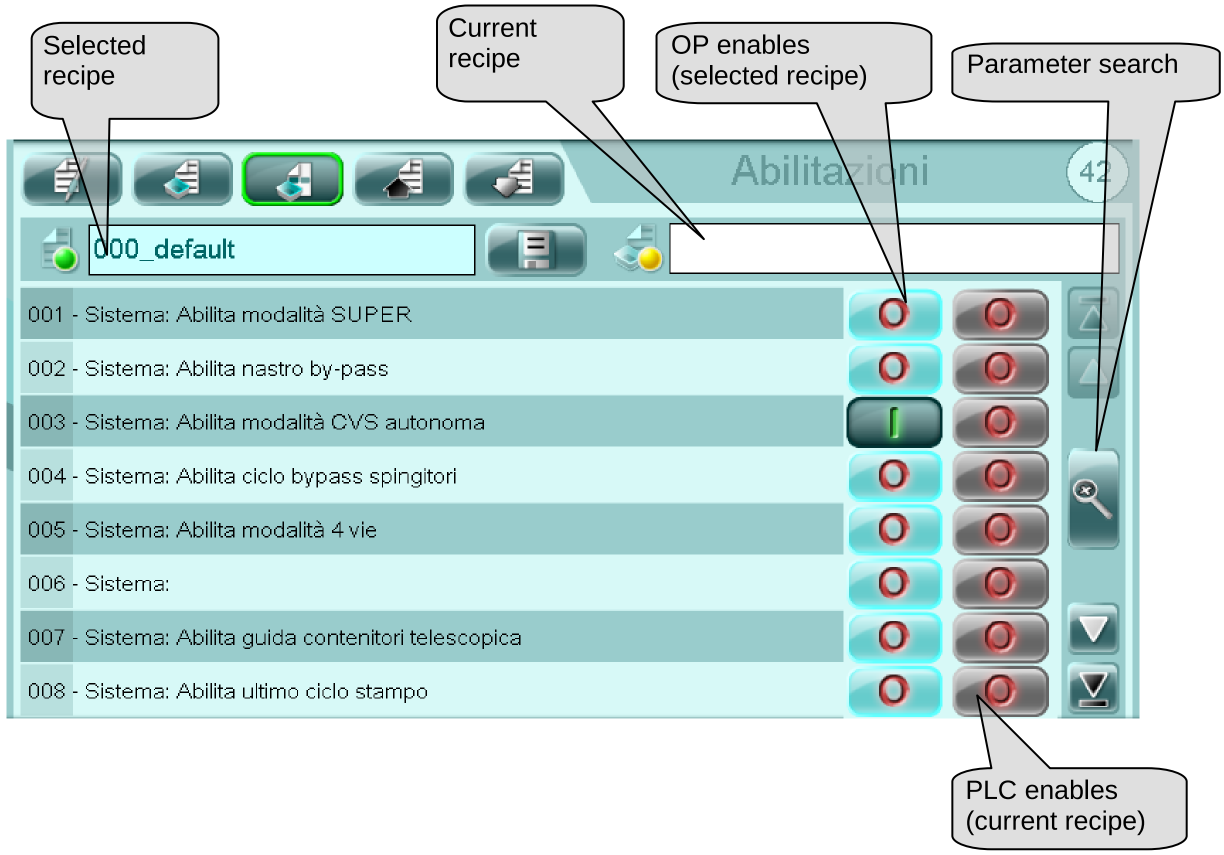


RECIPE ENABLE PAGE



SAVE SELECTED RECIPE DATA (OP parameters)

5.7.2 Enables recipe



RECIPE MANAGEMENT PAGE

RECIPE PARAMETER PAGE

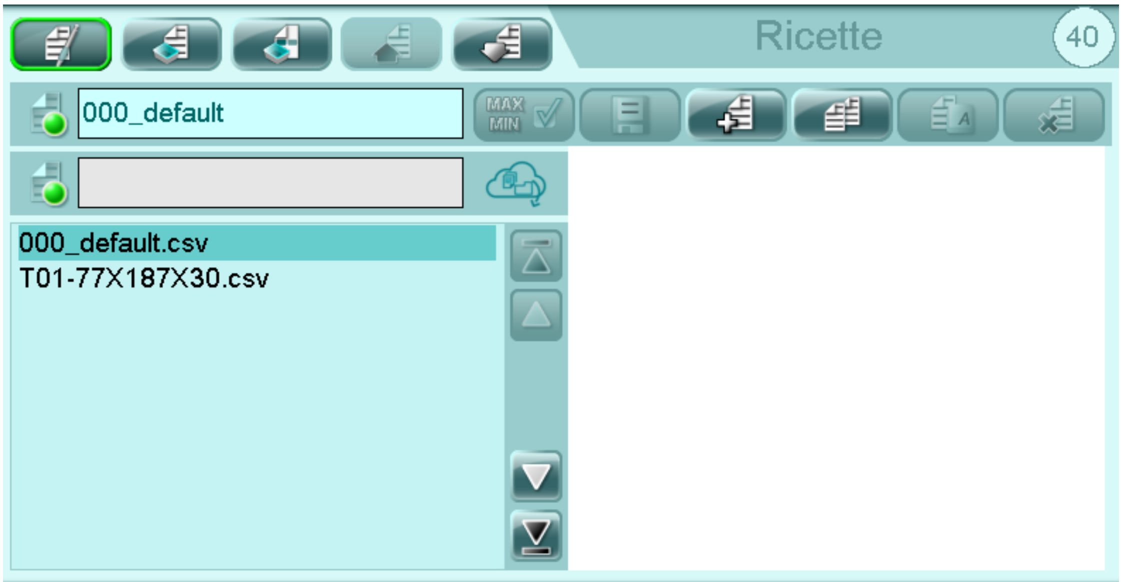
ENABLES RECIPE PAGE

SAVE SELECTED RECIPE DATA

5.7.3 Creating a new recipe

Proceed as follows to create a new recipe.

Touch  from the main menu bar to display the main recipe page.



Select  to open the new Recipe window.

Touch  to confirm the name of the recipe.

G. MONDINI
DOSATRICI - CONFEZIONATRICI AUTOMATICHE

After the new recipe has been created, select it from the list of available recipes on the right-hand side of the panel display and press  to copy the current recipe data to the newly-created recipe.

Touch  to confirm.

Press  to save.

A

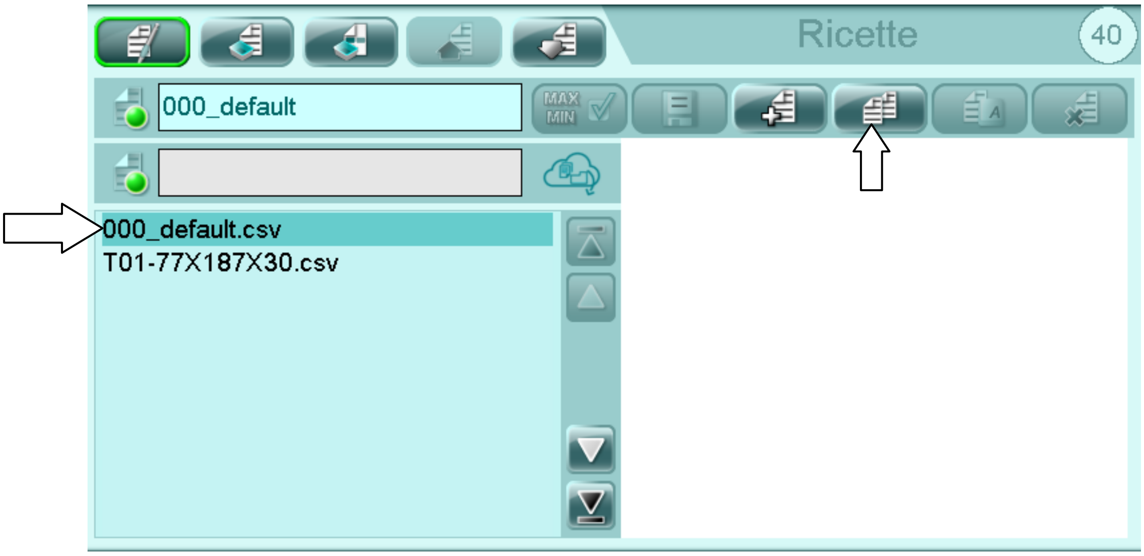
Touch  to confirm.

In order to verify whether the recipe has been correctly saved, you need to check that icon (**A**) light next to the recipe name is green and not orange.

5.7.4 Creating a new recipe from an existing one

Proceed as follows to create a copy of an existing recipe.

Select the recipe you wish to copy and touch .



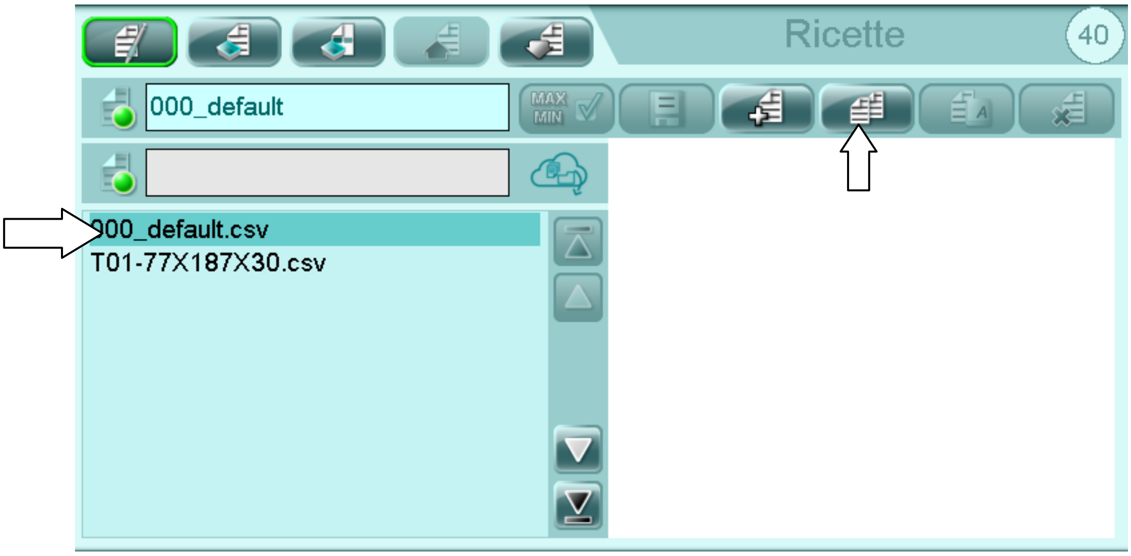
Insert the name of the new recipe using the alphanumeric keypad.

Touch  to confirm.

5.7.5 Renaming an existing recipe

Proceed as follows to rename an existing recipe.

Select the recipe that you wish to rename and touch .



Insert the new name of the selected recipe using the alphanumeric keypad.

Touch  to confirm.

5.7.6 Recipe image

If you want insert a image into recipe:

- Copy the image into a USB stick (supplied with machine) into a "root" not in a folder, ex. E:/image).
- Insert USB stick on the rear of the panel.
- Login with right password level.
- Scroll through the recipes and click for select recipe.
- Click on the image list saved on USB stick.
- Select the correct image and confirm.
- Now it's possible remove the USB stick.

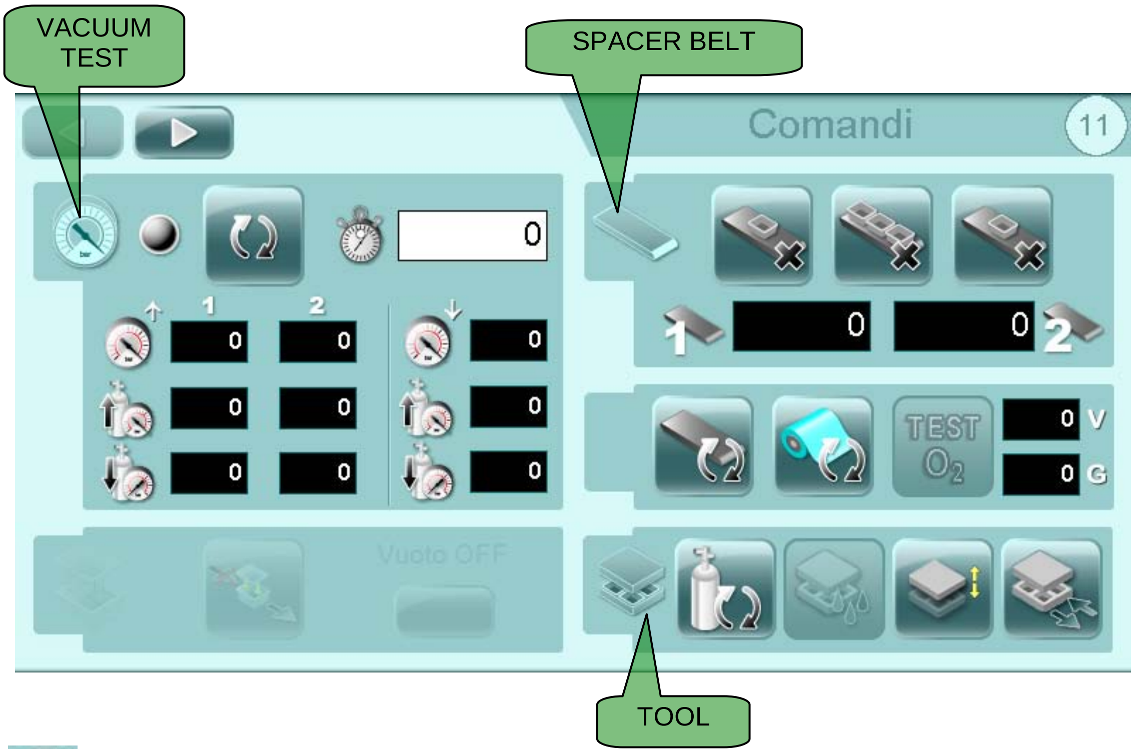
IMPORTANT!

Image properties:

- Format: PNG
- Dimension: max 100-150 Kb
- Size recommended: 370 x 290 pixel.

Default recipe image is not changeable!

5.8 COMMANDS



VACUUM TEST ON

START VACUUM TEST

VACUUM TEST TIME

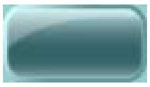
UPPER INSTALLATION CURRENT PRESSURE

LOWER INSTALLATION CURRENT PRESSURE

GAS VALUE REACHED

VACUUM VALUE REACHED


LINE EMPTYING


VACUUM OFF IN DARFRESH PLATES

TRAYS ON SPACER BELT 1

TRAYS ON SPACER BELT 2


REMOVE ONE TRAY FROM SPACER BELT


REMOVE ALL TRAYS FROM SPACER BELT


START BELT TEST


START FILM TEST

OXYGEN TEST

VALUE OF VACUUM TEST

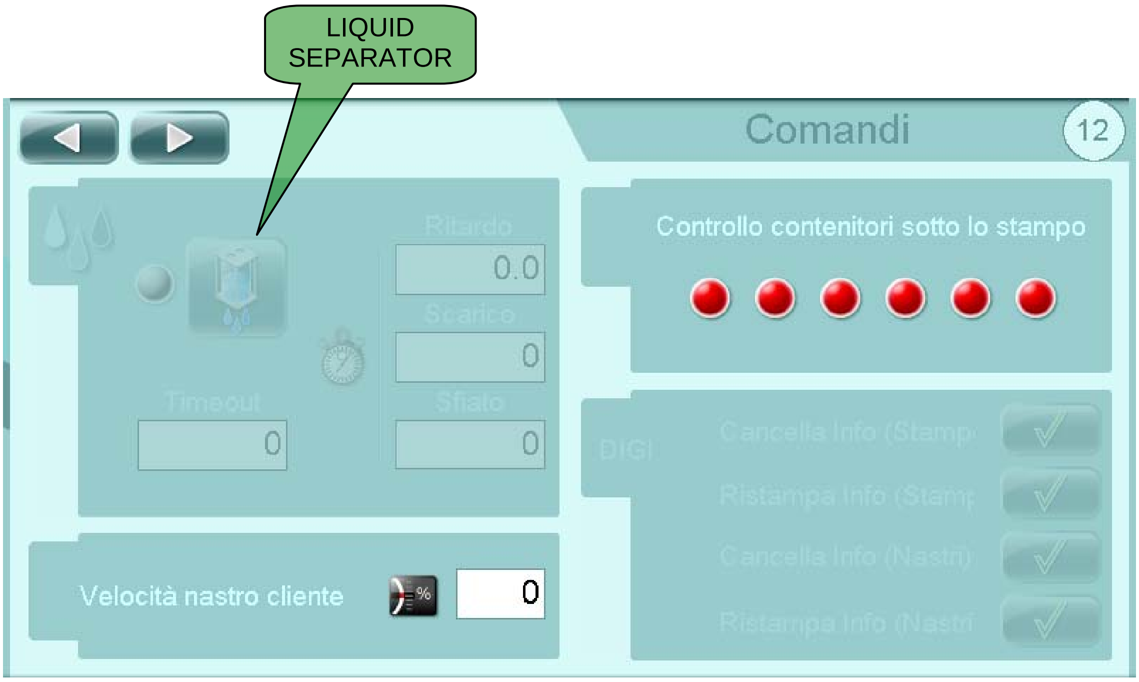
VALUE OF GAS TEST


WASHING WITH GAS

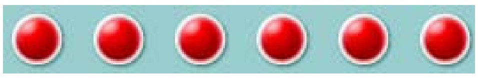

TOOL WASHING CYCLE (TOOL IN VACUUM/GAS POSITION)

UPPER TOOL UP/DOWN MOVEMENT (DOT and SP tools)

TOOL IN TOOL-CHANGE POSITION



LIQUID SEPARATOR MANUAL START (WHERE AVAILABLE)



TRAYS CONTROL UNDER THE TOOL

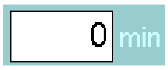
CUSTOMER BELT SPEED

CUT FILM CYCLE FOR DARFRESH APPLICATION

DARFRESH SHUTTLE IN CHANGE POSITION

VACUUM OFF IN DARFRESH PLATES

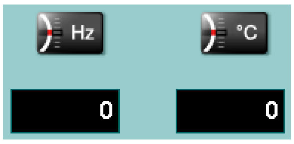
DARFRESH SHUTTLE UNDER THE SEALING TOOL



SHUTTLE POSITIONING TIME UNDER THE SEALING TOOL (IN MINUTES)

TIME REMAINING FOR THE POSITIONING OF THE SHUTTLE UNDER THE SEALING TOOL (IN SECONDS)

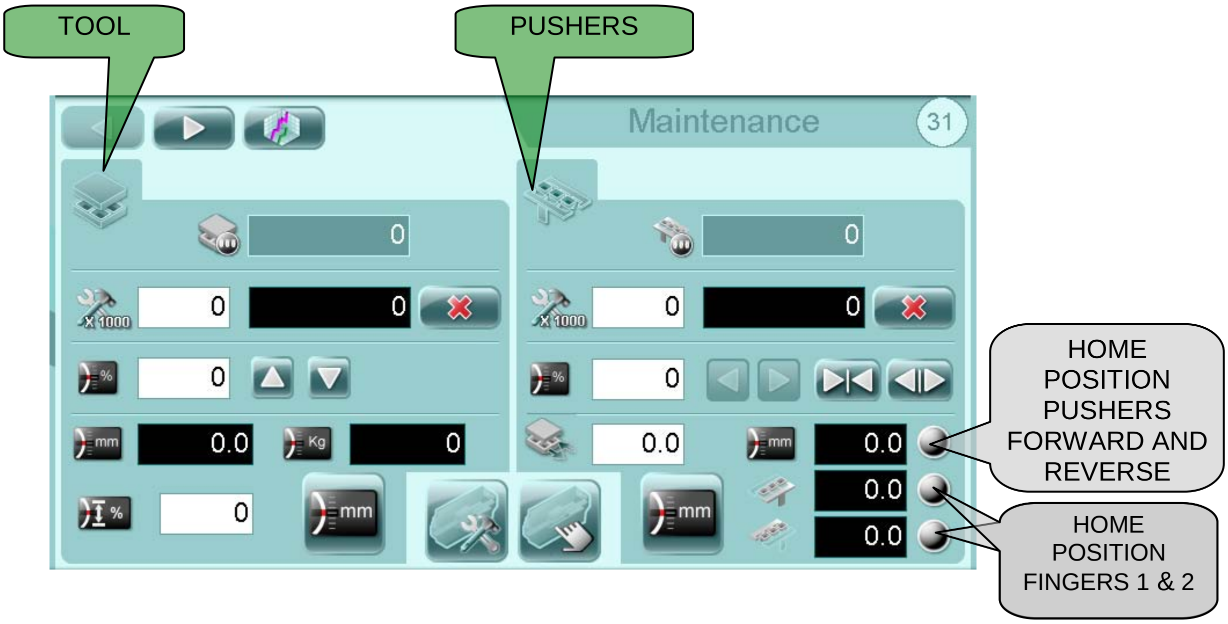
LINE EMPTYING



SPEED AND TEMPERATURE OF PUMP

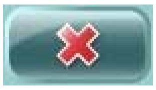
5.9 MAINTENANCE

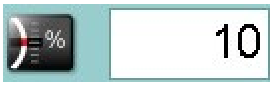
IMPORTANT! This page is available only if the user has the "maintenance" authorisation.




TOOL CYCLES COUNTER

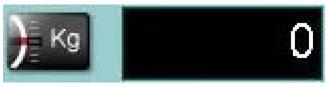

FIXED PROGRAMMED MAINTENANCE

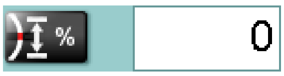

RESET PARTIAL COUNTER


% SPEED MANUAL MOVEMENT

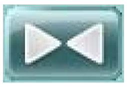

UP AND DOWN MANUAL COMMAND


ACTUAL TOOL POSITION

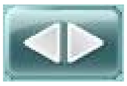

PRESSURE


TORQUE LIMITATION DURING TOOL

FORWARD AND REVERSE MANUAL COMMAND



CLOSE FINGERS



OPEN FINGERS

ACTUAL PUSHERS POSITION

DESCENT

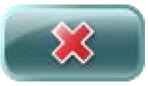
FINGER 1 POSITION

FINGER 2 POSITION

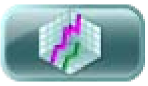
HOME KEYS (TOOL AND PUSHERS)

ENABLE MAINTENANCE (JOG MODE)

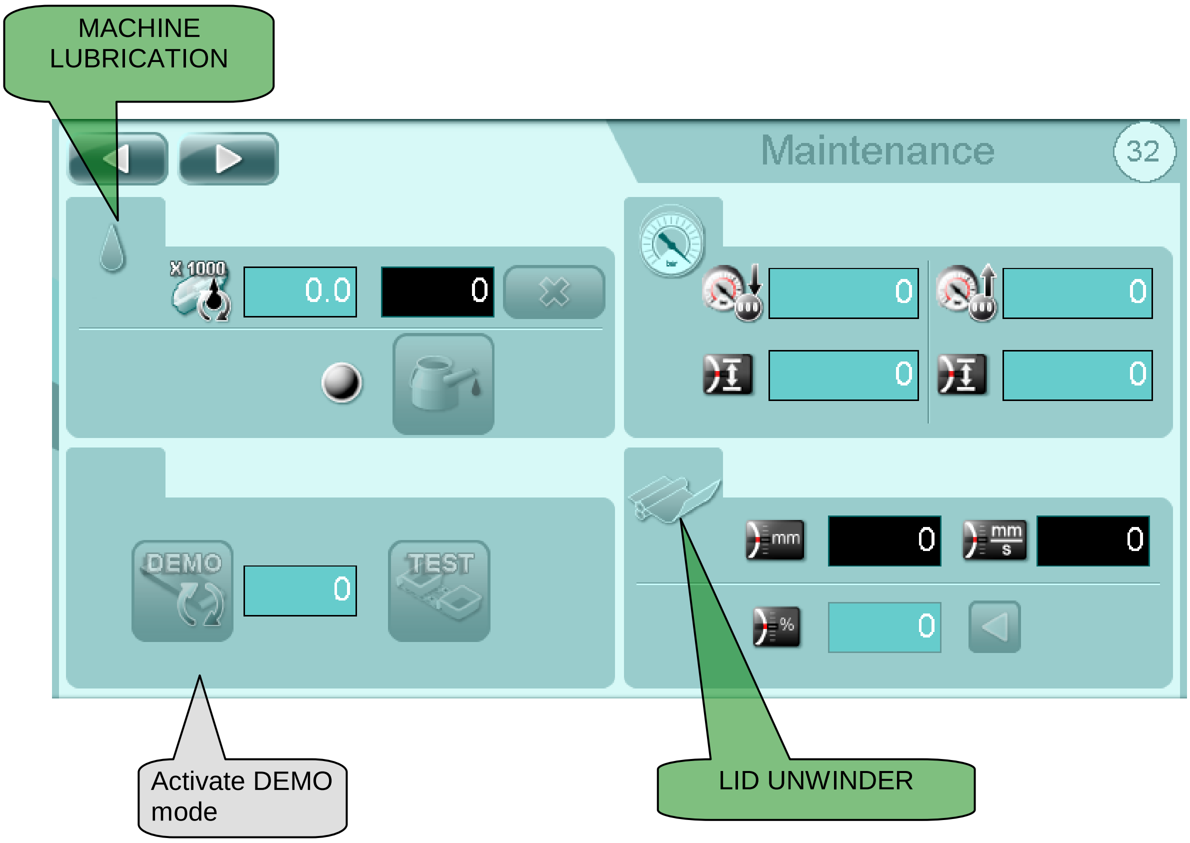
ENABLE MANUAL CYCLES

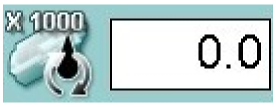


RESET TOOL MAINTENANCE CYCLES



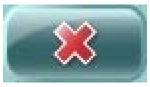
TRENDS PAGE





LUBRICATION THRESHOLD

LUBRICATION CYCLES

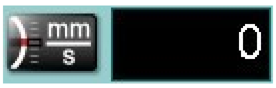

RESET LUBRICATION CYCLES


LUBRICATION IN PROGRESS


MANUAL LUBRICATION

ENTER NUMBER OF TRAYS TO SIMULATE

CURRENT POSITION

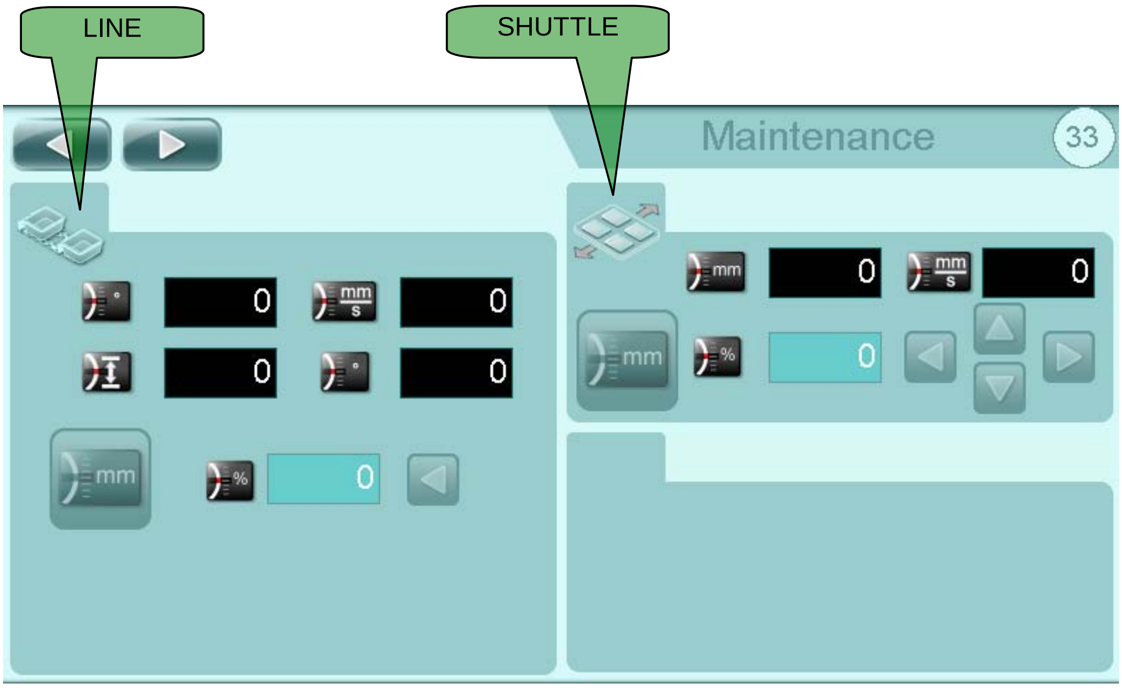


CURRENT SPEED

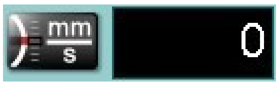
MANUAL SPEED AND FORWARD COMMAND

FEED BELT DEMO MODE (IF AVAILABLE). Used to simulate tray feed at a specified rate.

TEST (IF AVAILABLE): Allows to enable the transport chain test.

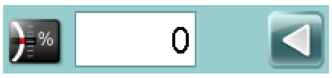


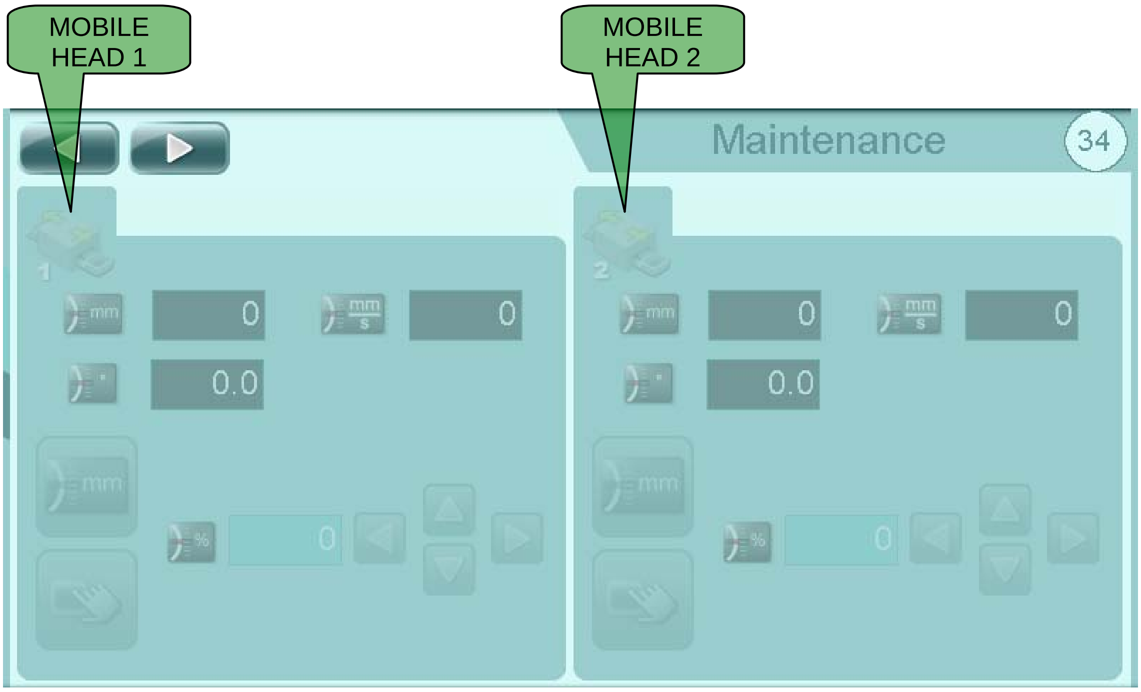

CURRENT POSITION


CURRENT SPEED


CURRENT POSITION


HOME POSITION


MANUAL SPEED AND COMMAND



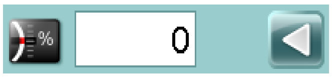
CURRENT MOBILE HEAD POSITION

CURRENT MOBILE HEAD SPEED

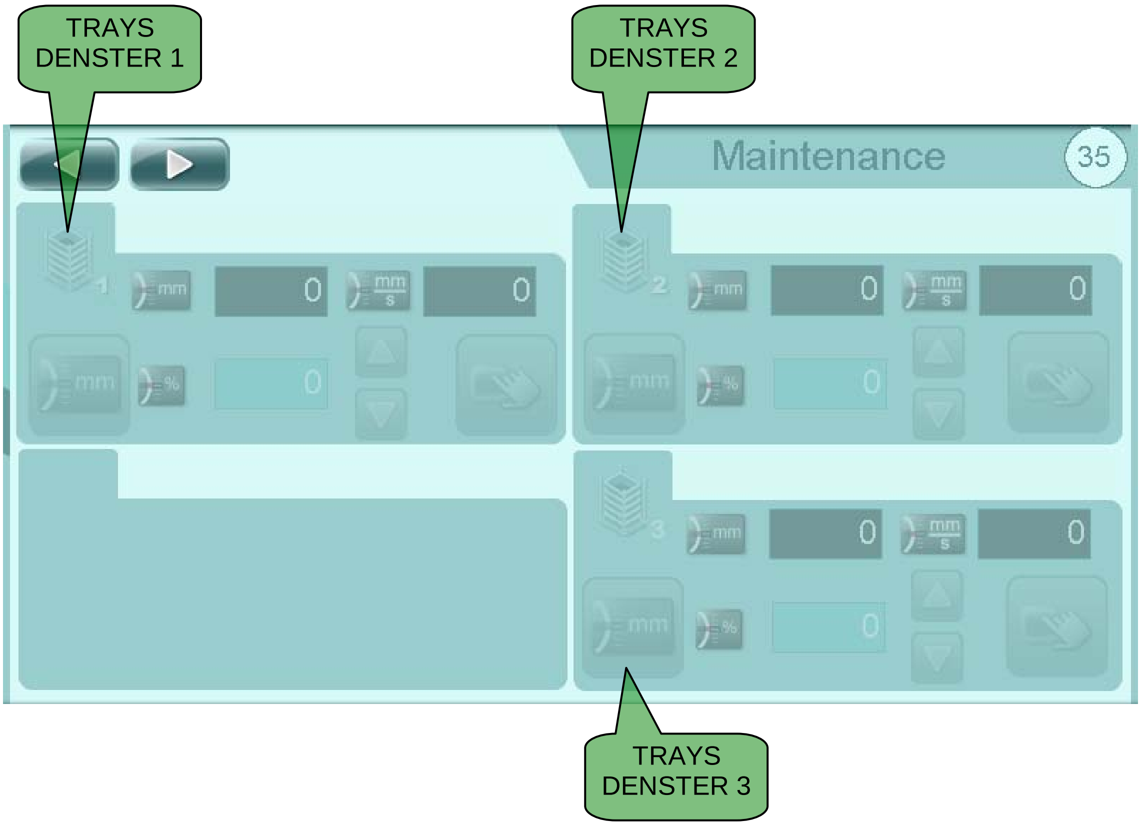
CURRENT MOBILE HEAD POSITION

MOBILE HEAD HOME


MANUAL CYCLE COMMAND (WHERE AVAILABLE)



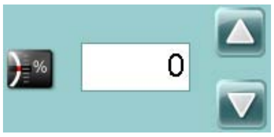
MANUAL SPEED AND COMMAND



CURRENT POSITION

CURRENT SPEED

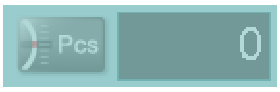
TRAYS DENESTER HOME



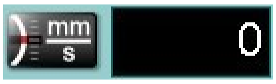
MANUAL SPEED AND COMMAND


MANUAL CYCLE COMMAND (WHERE AVAILABLE)

CURRENT DEVIATOR POSITION



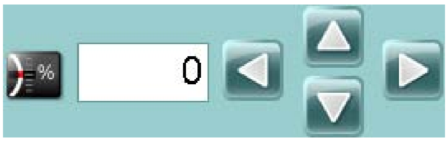
NUMBER OF CONTAINERS TO DEVIATION



CURRENT SPEED

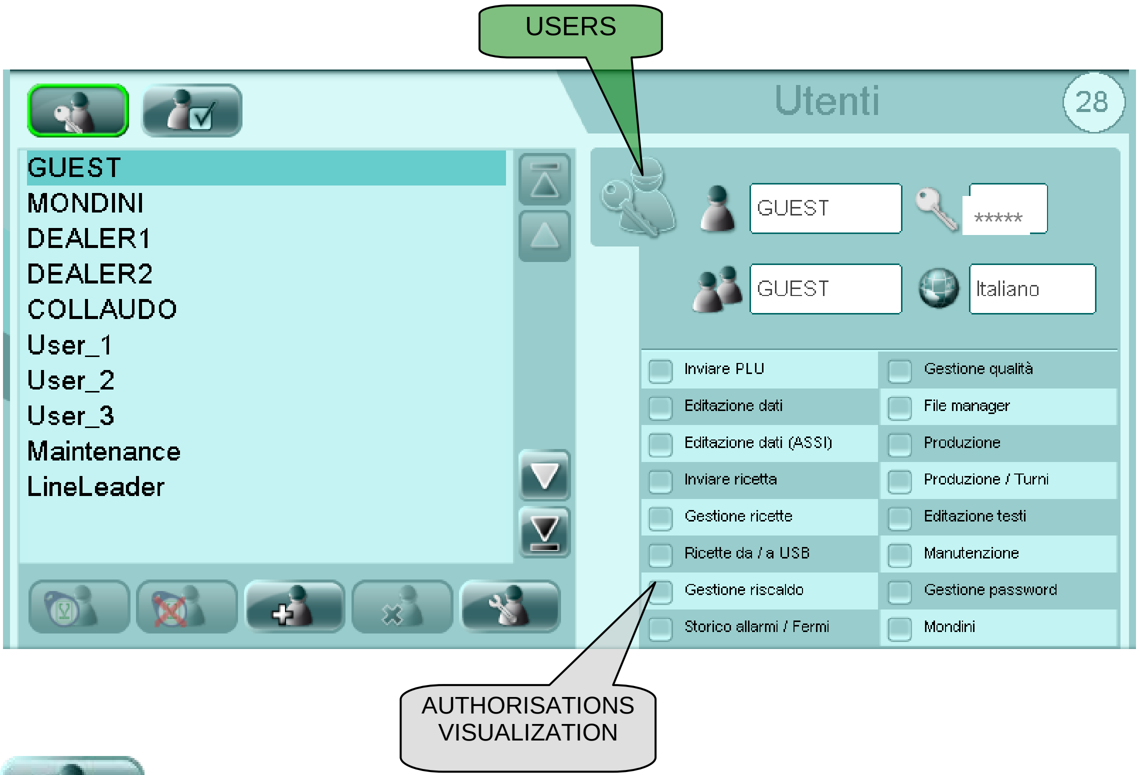
SHUTTLE HOME

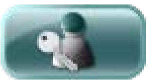
CURRENT POSITION



MANUAL SPEED AND COMMAND

5.10 USER AND PASSWORD MANAGEMENT

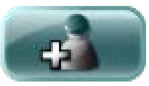


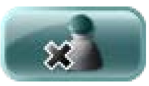

USER AND PASSWORD MANAGEMENT PAGE

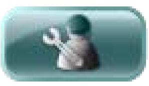

CONSENT PAGE


IT ASSOCIATES A TAG TO THE USER


IT CANCELS ASSOCIATION USER TO A TAG


NEW USER


DELETE USER


MODIFY USER (VIEW PASSWORD)

USER

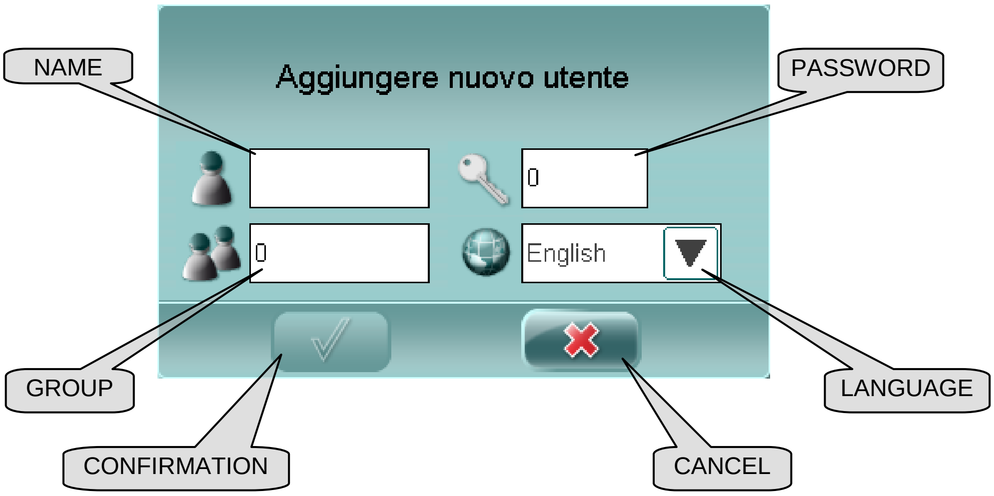
ENCRIPTED PASSWORD (NEARBY THE SYMBOL MEANS IF THE LOGGED USER HAS A TAG ASSOCIATED)

GROUP

LANGUAGE

IT ASSOCIATES THE TAG TO THE USER

IT CANCELS THE ASSOCIATION OF THE TAG WITH THE USER



IMPORTANT! GUEST and MONDINI users cannot be modified or deleted.

To log in a user, touch the icon at the top of the screen .

Users in possession of higher level passwords are advised to log out (GUEST) on completion of the modification.

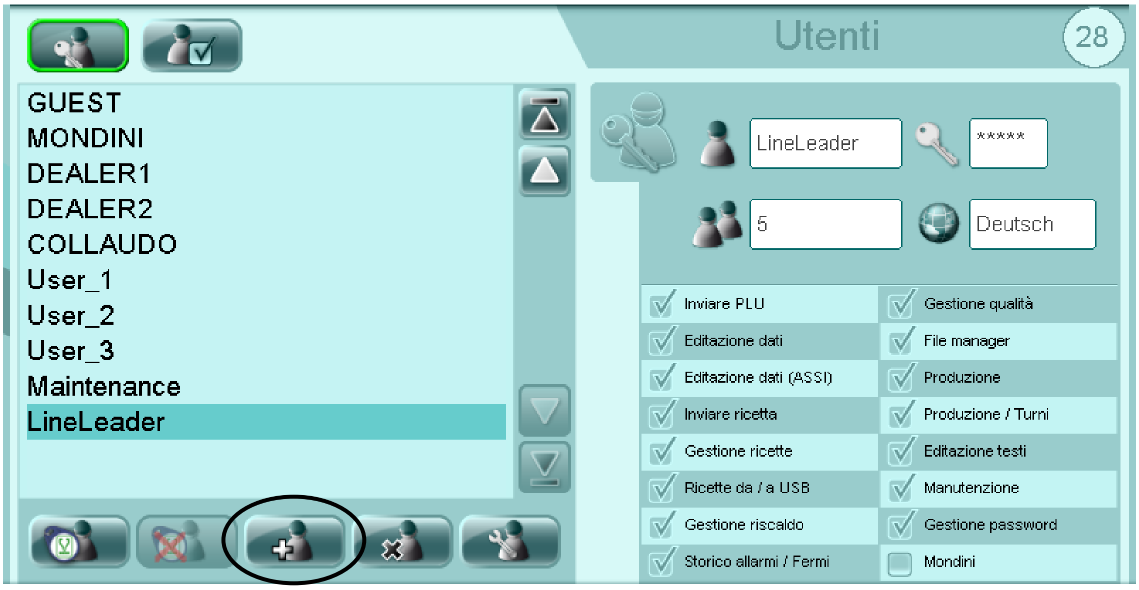
If the panel is not used for a set time, the active password expires (GUEST).

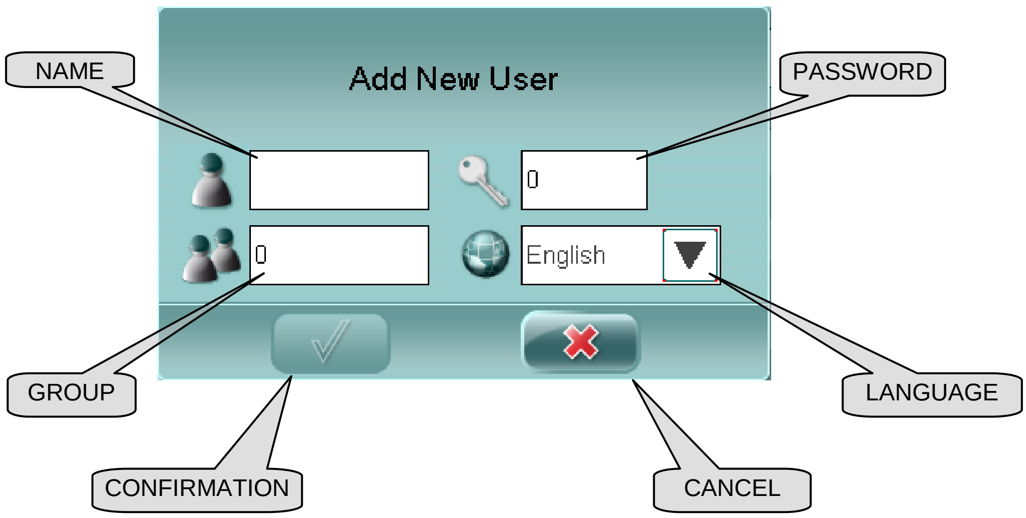
Each password gives access to different modification and processing levels. It is there advisable to keep access passwords secret and only communicate them to authorised personnel.

Authorisations are linked to a group and the user belongs to a group. So first create a group with specific authorisations and then create the users belonging to this group (same logic as Windows).

5.10.1 Creation new user

To create a new user touch the button  and compile the fields in the window.

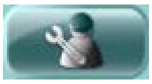


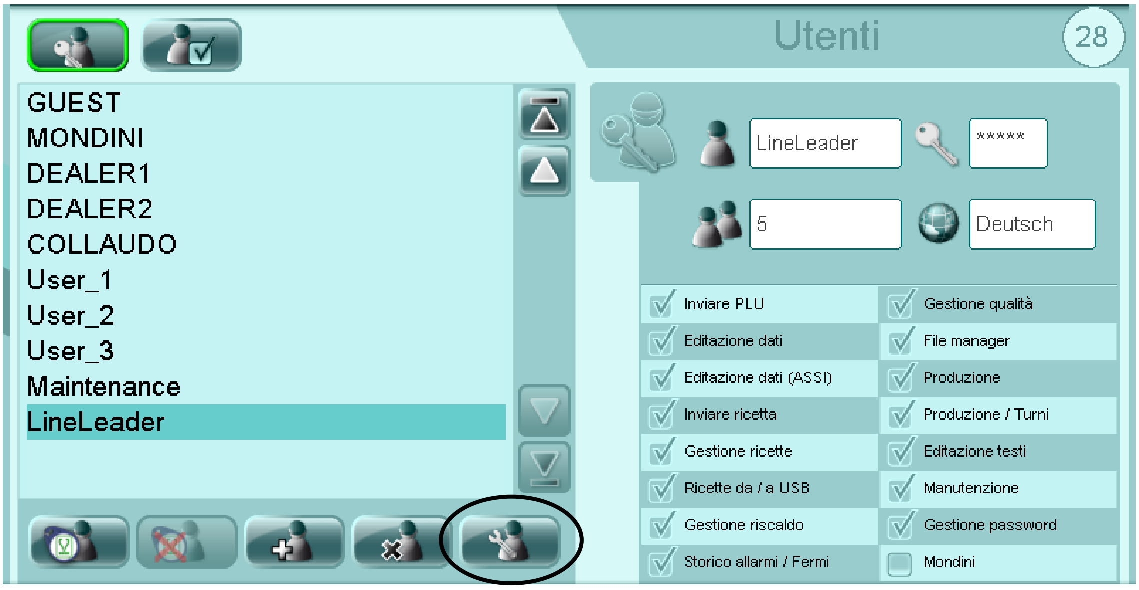


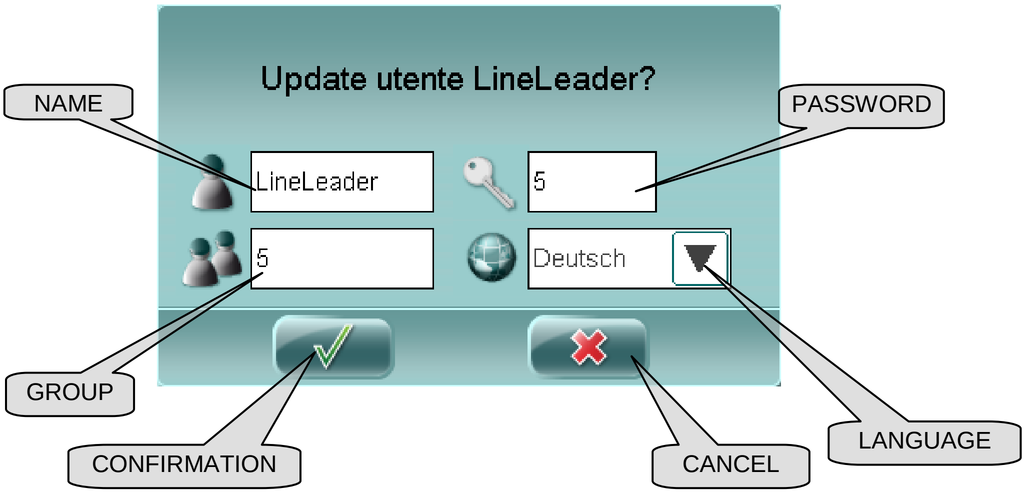
ATTENTION!

The authorizations are connected to the group and the user is join to a group. Then first a group is created (if it is not already existing) with some specific authorizations and then the users are created that have to join to this group (same philosophy of windows).

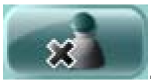
5.10.2 Modification existing user

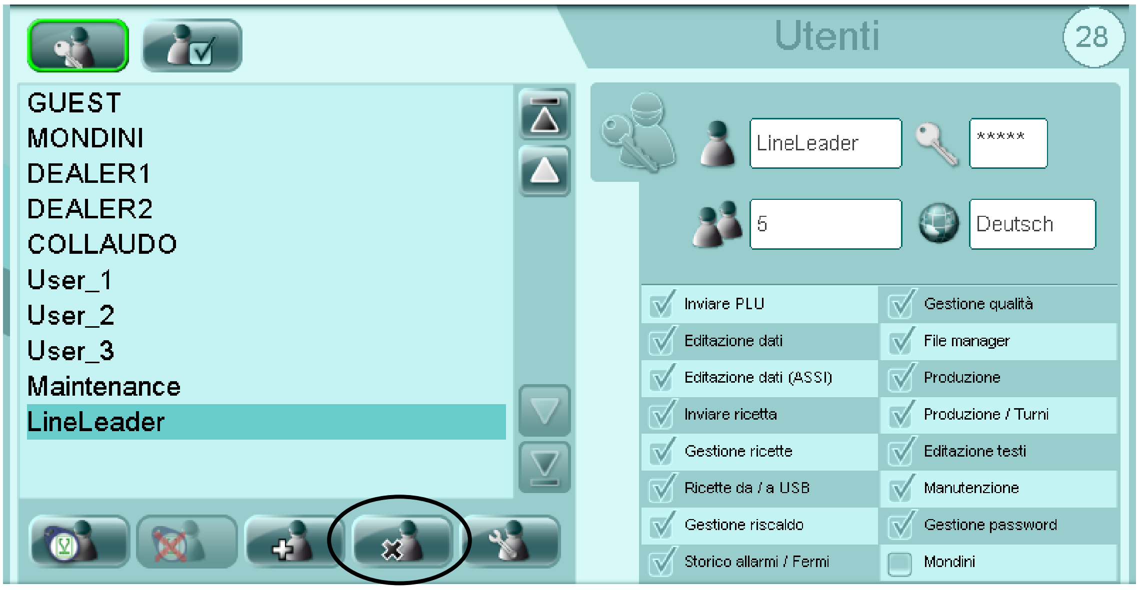
To modify an existing user touch the user to modify from the list of the users, touch the button  and to modify the data desired in the window t.





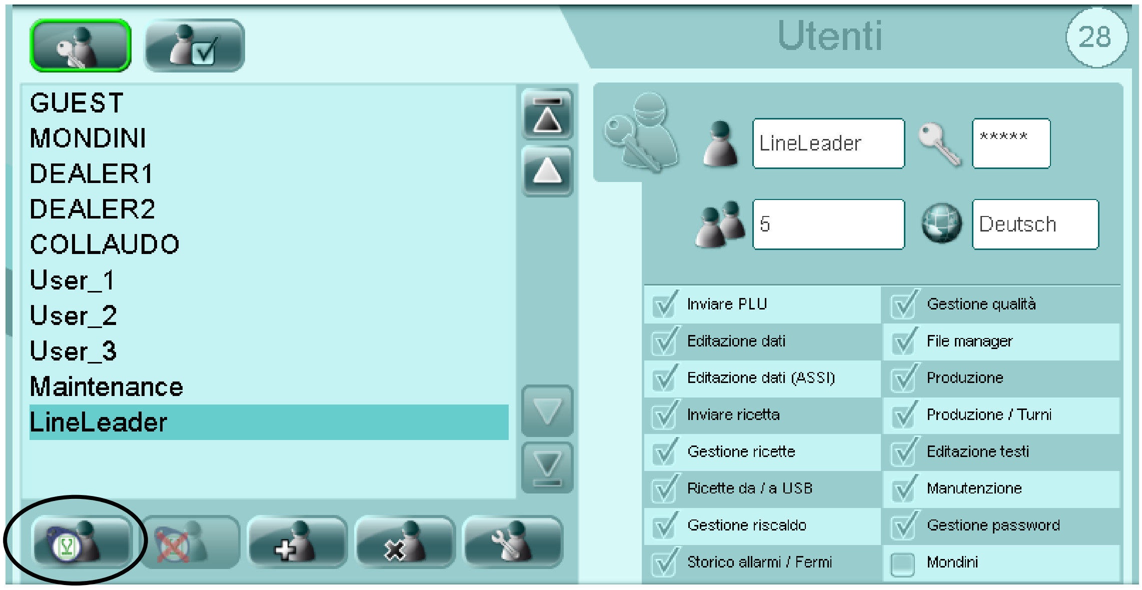
5.10.3 Eliminate existing user

To eliminate an existing user touch the user to eliminate from the list of the users and touch the button .

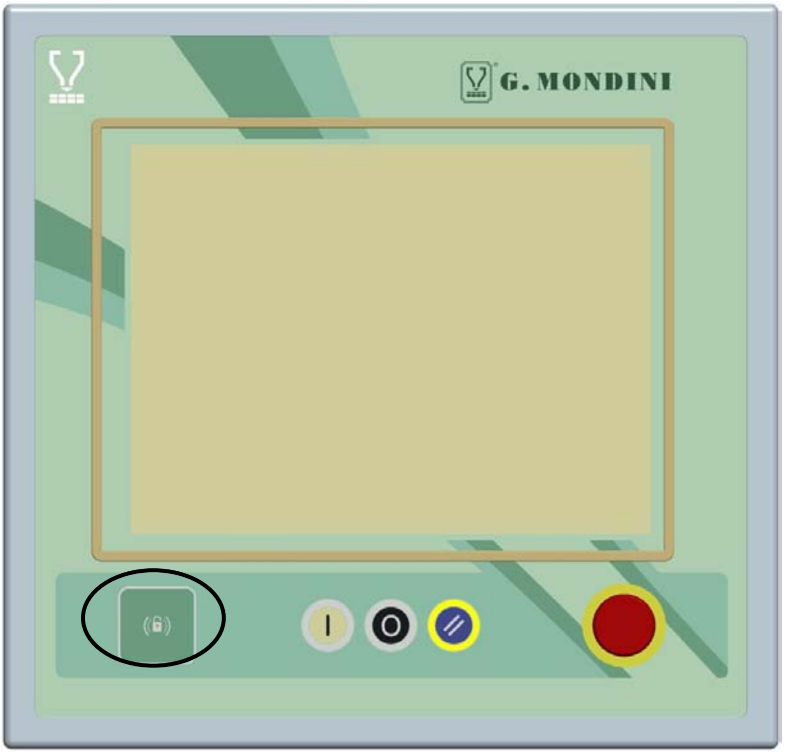


5.10.4 Associate user to TAG

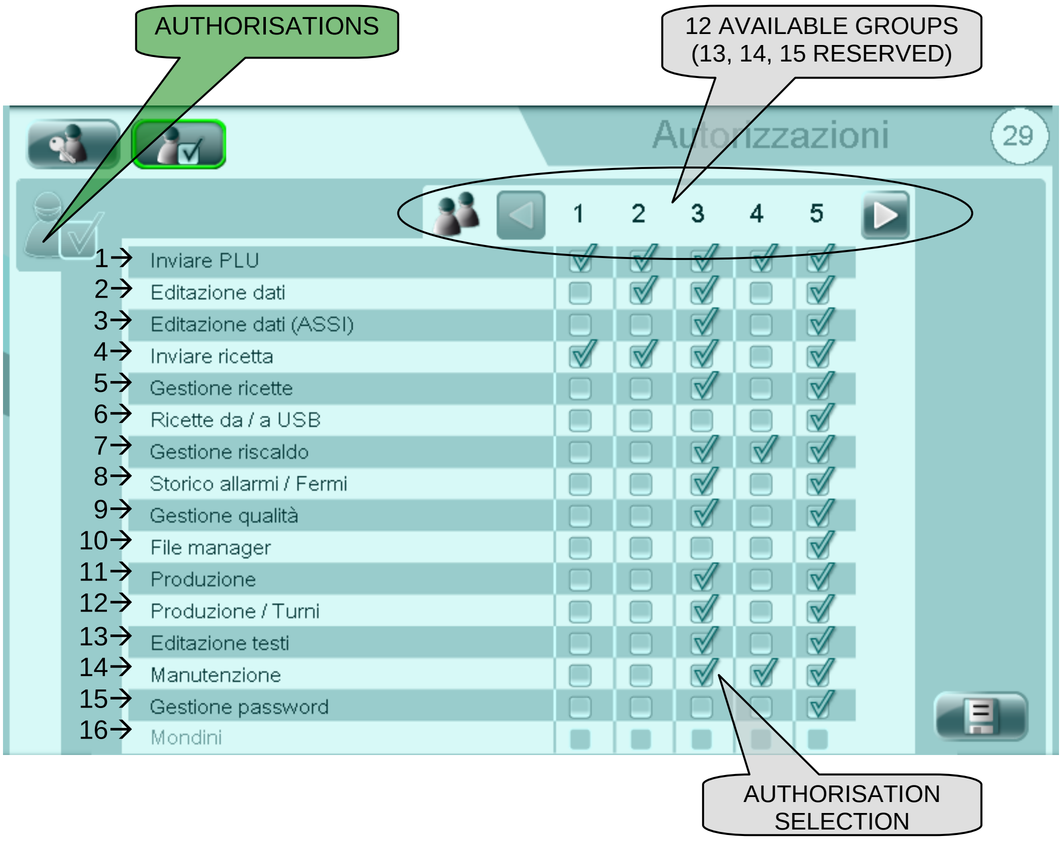
To associate an existing user to a TAG touch the user to associate from the list of the users and touch the button .



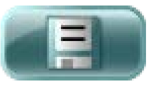
Position the TAG in correspondence of the device on the left bottom side of the operator panel  and confirm on the screen of the operator panel.



5.10.5 Authorisations



To train an authorization in a group is enough to touch the corresponding box so that the symbol ☑ appears.



SAVE CHANGES TO AUTHORISATIONS

1 Send PLU: basic authorization for the use of the page "most used";

2 Edit Data: it allows to modify all the "not critical parameters" of the machine;

3 Edit Data (AXIS): it allows to modify the advanced parameters;

4 Send Recipe: it allows to send the recipe from the panel to the PLC;

5 Recipe Management: it allows to create, to cancel and to effect changes to the recipes;

6 Recipes from/to USB: it allows to effect the transfer of the recipes from/to USB;

7 Heating Management: it allows to effect changes to the heating of the sealing tool;

8 History Alarms: it allows to visualize the "alarm history" page;

9 Quality Management: it allows to visualize the "quality" page;

10 File Manager - Exit: it allows to effect the management of the files for the export and the importation through USB;

11 Production: it allows the visualization of the production data of the machine;

12 Production/Shift: it allows the production shifts management;

13 Edit texts: it allows the texts management in the different languages;

14 Maintenance: it allows to access the maintenance page;

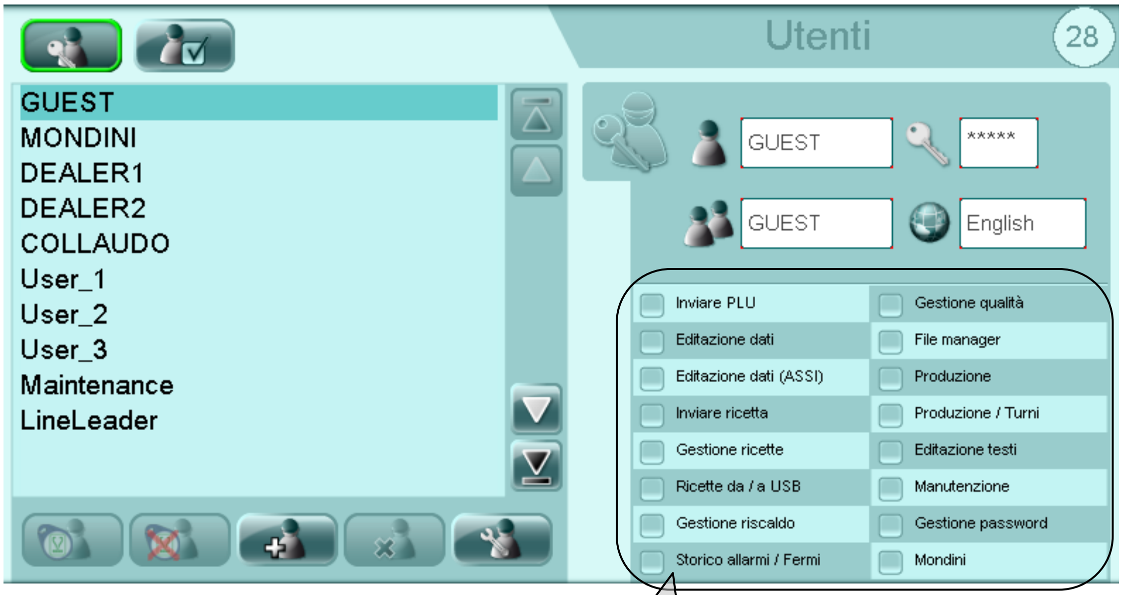
15 Password management: it allows the creation, the elimination and the change of the users and of the user groups;

16 Mondini: this authorization is accessible only to the "G.Mondini" technicians.

5.10.6 Standard "G.Mondini" users

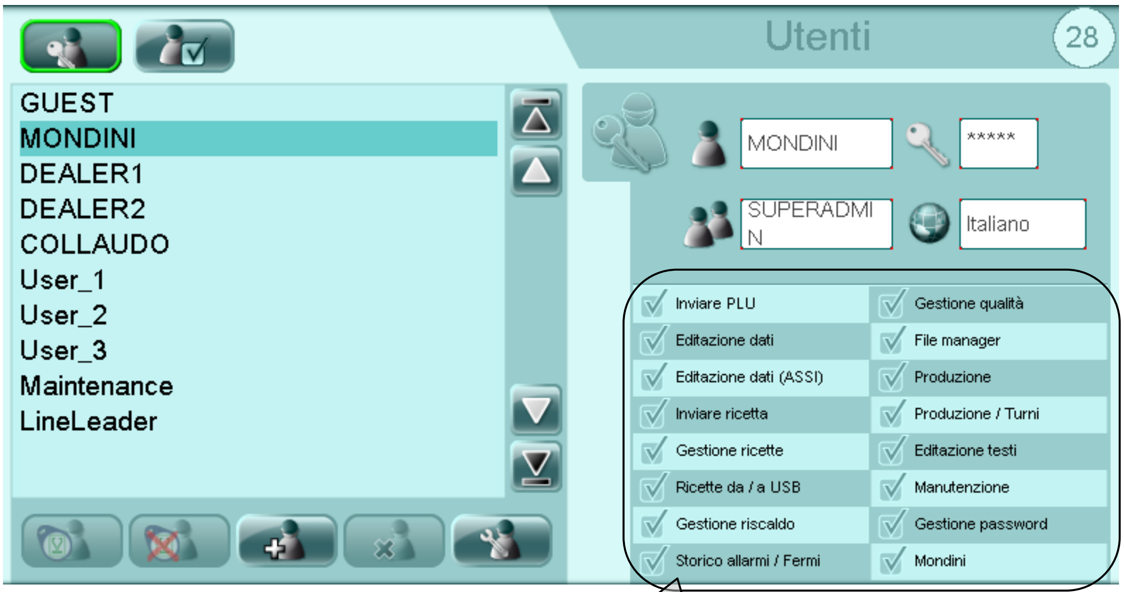
In this paragraph the users and the relative activated authorizations are described as they have been thought and realized by the "G.Mondini" firm.

- **"GUEST" user**



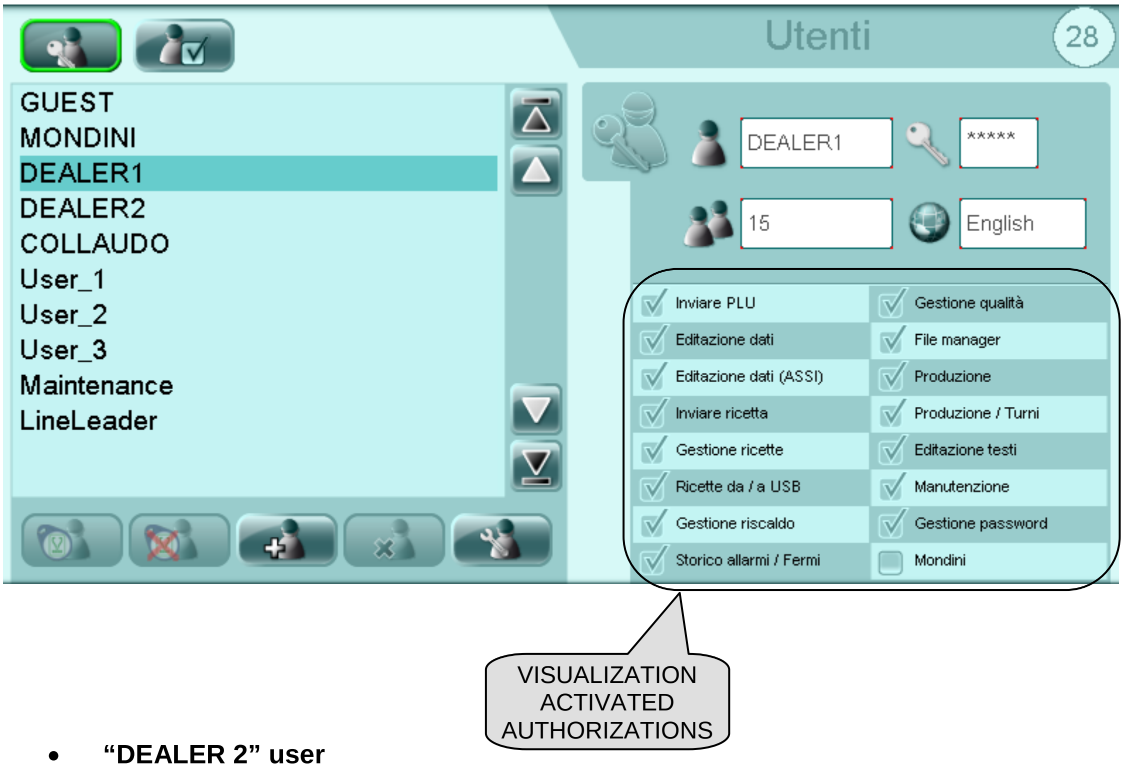
VISUALIZATION ACTIVATED AUTHORIZATIONS

- **"MONDINI" user**

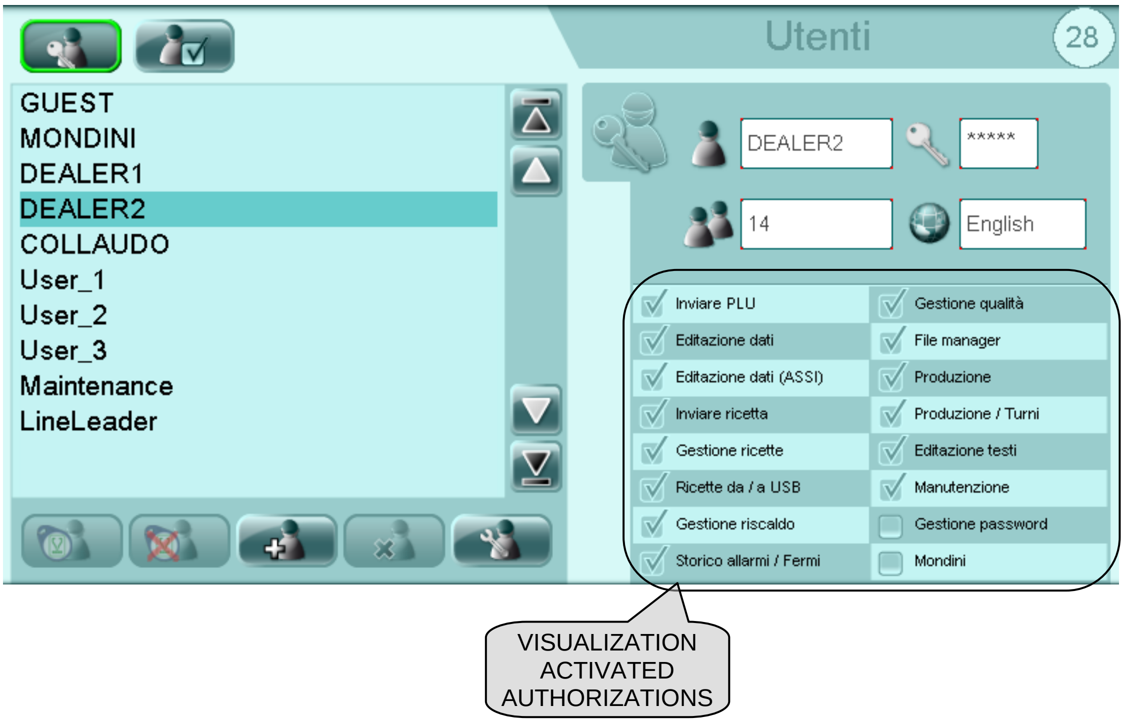


VISUALIZATION ACTIVATED AUTHORIZATIONS

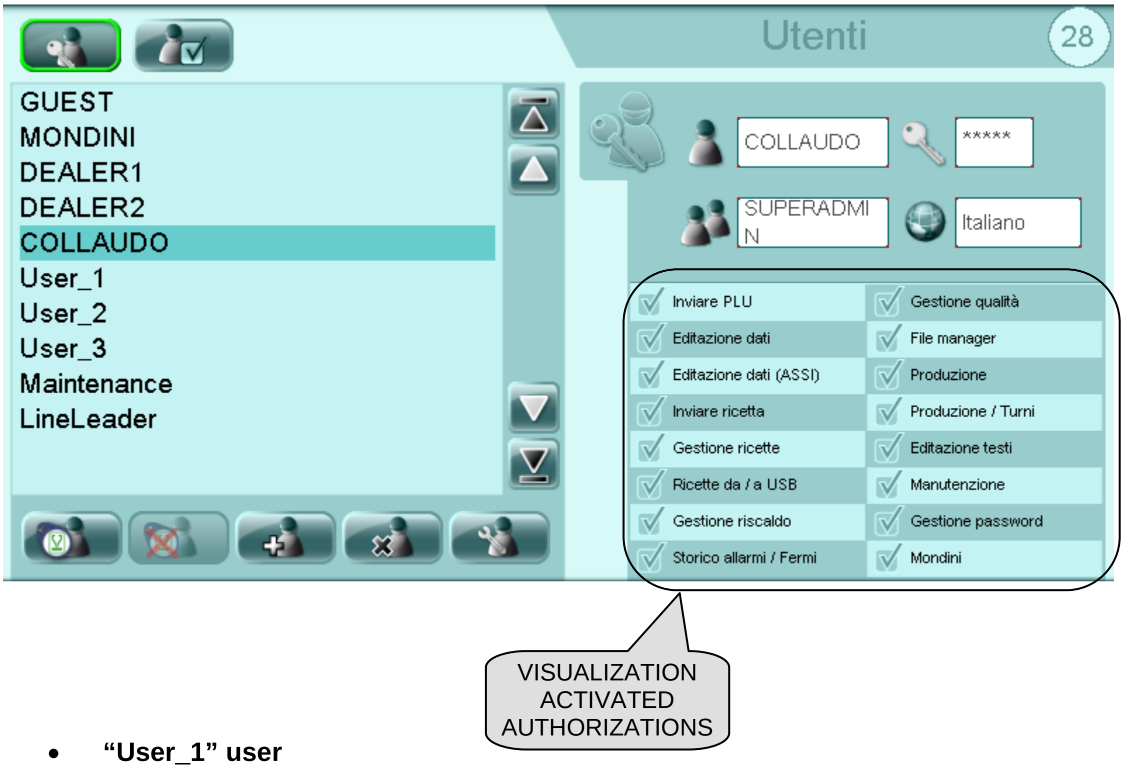
- **“DEALER 1” user**



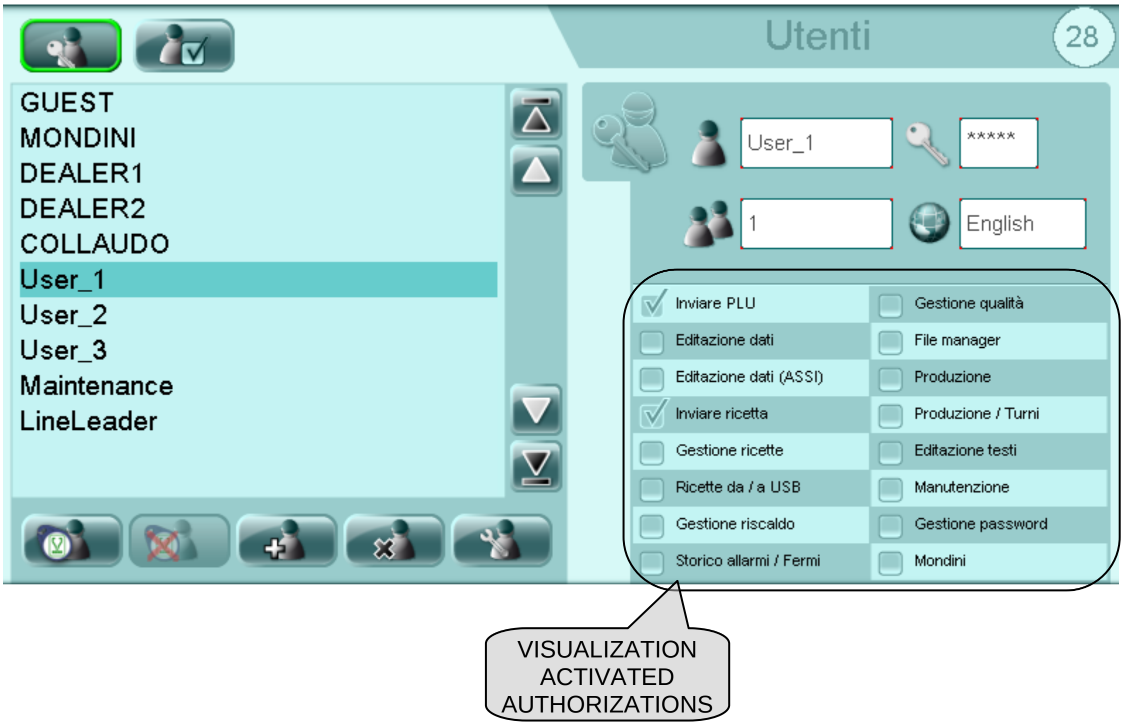
- **“DEALER 2” user**



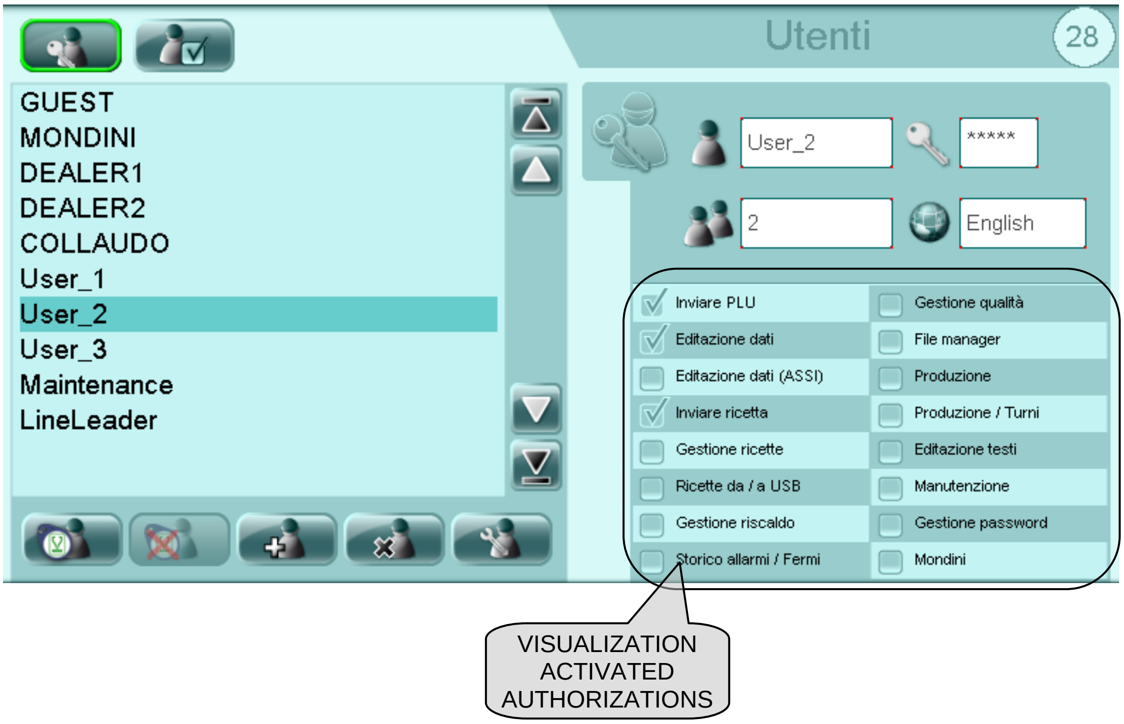
- **“COLLAUDO” user**



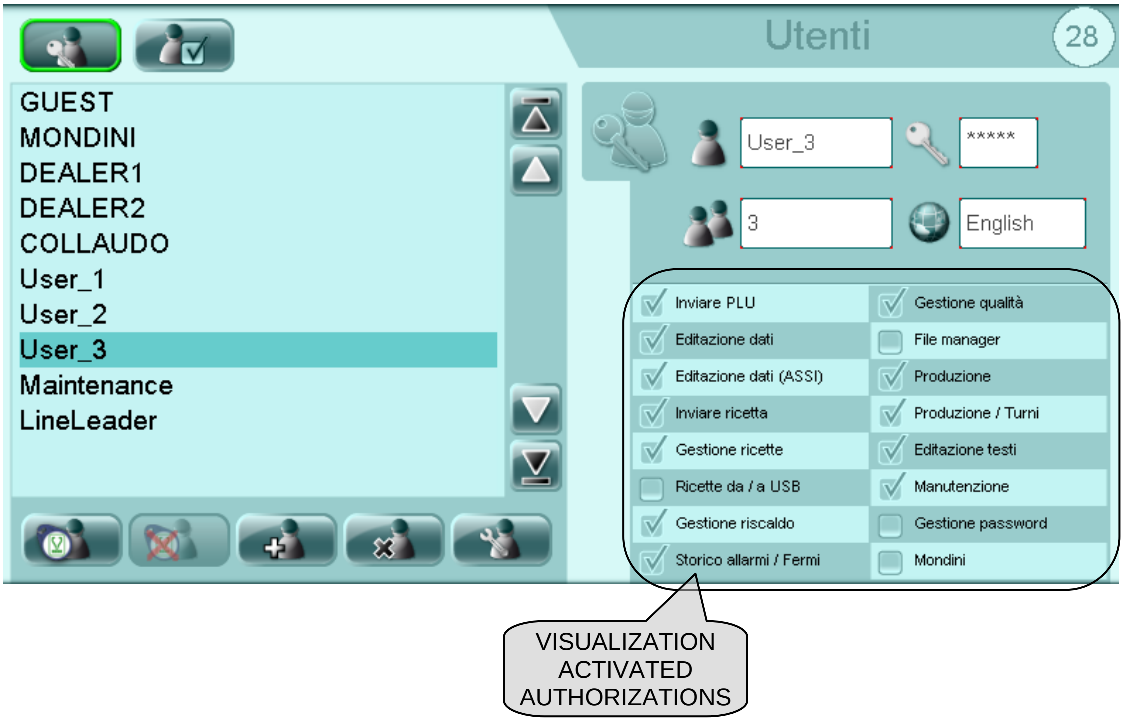
- **“User_1” user**



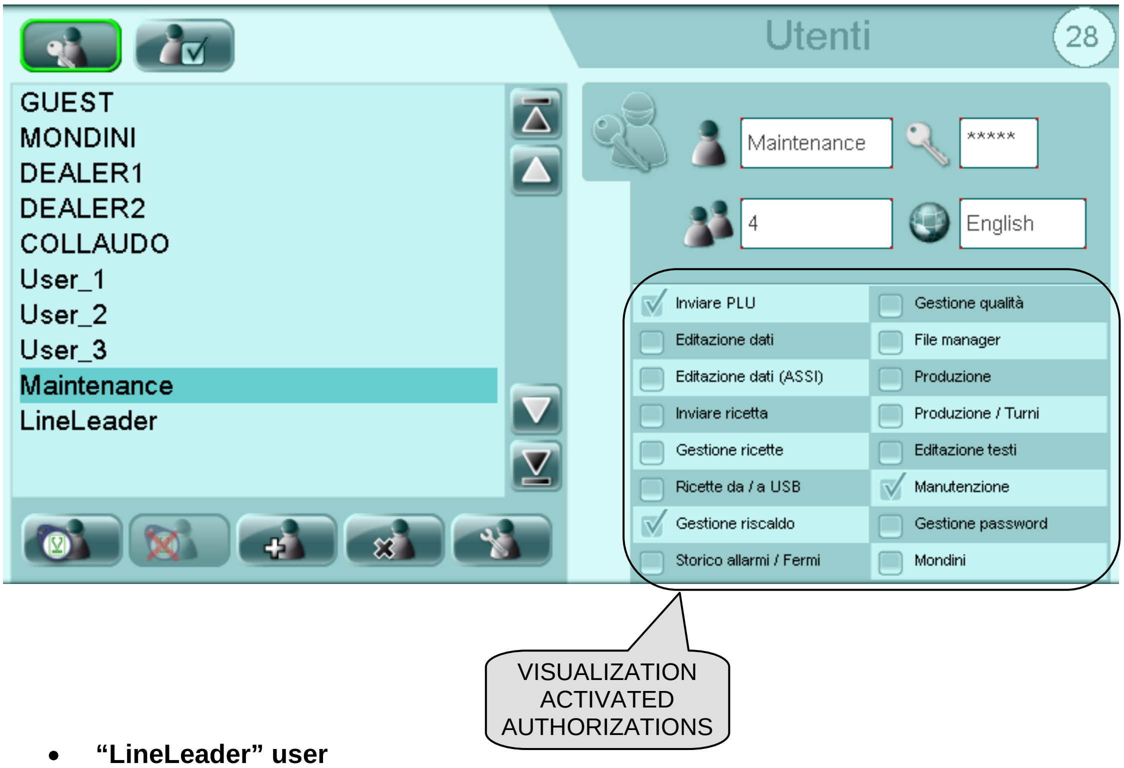
- **“User_2” user**



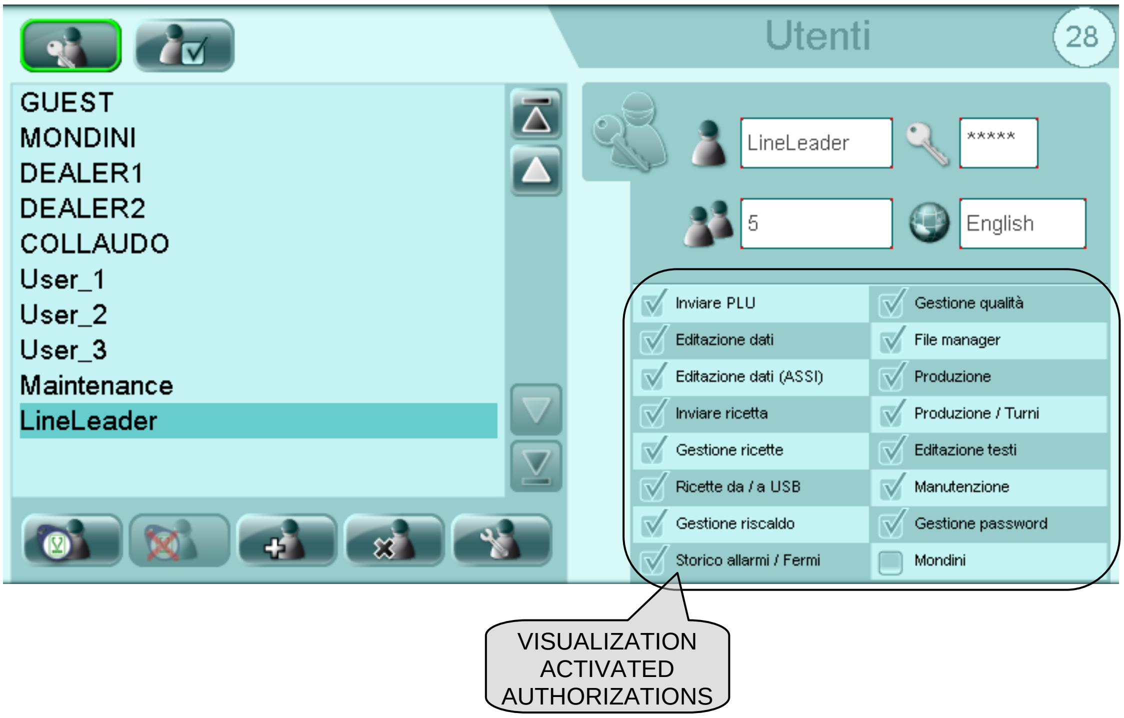
- **“User_3” user**



- **“Maintenance” user**

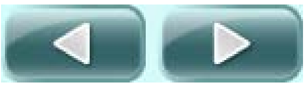


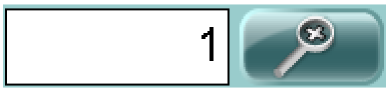
- **“LineLeader” user**



5.11 MULTILANGUAGE TEXT MANAGEMENT

ATTENTION! This page is accessible only if the user is logged with the authorization "Edit texts."

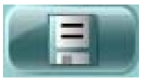
Scroll the kind of text (parameter, enables, alarm and stops) with  pushbuttons.

Search the text number toching the search text 

area and insert the text number or scroll the text with 

pushbuttons.

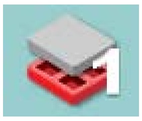
Write the new text.

Save the changes with  pushbutton.

5.12 TOOL HEATING

UPPER TOOL 1 HEATING

UPPER TOOL 2 HEATING

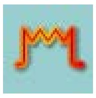


LOWER TOOL 1 HEATING

LOWER TOOL 2 HEATING

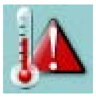
CURRENT TEMPERATURE

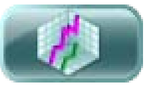
SET POINT TEMPERATURE (RECIPE DATA)

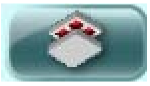

RESISTANCE HEATING POWER (%)

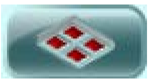

LIGHT SHOWS WHEN HEATING ON

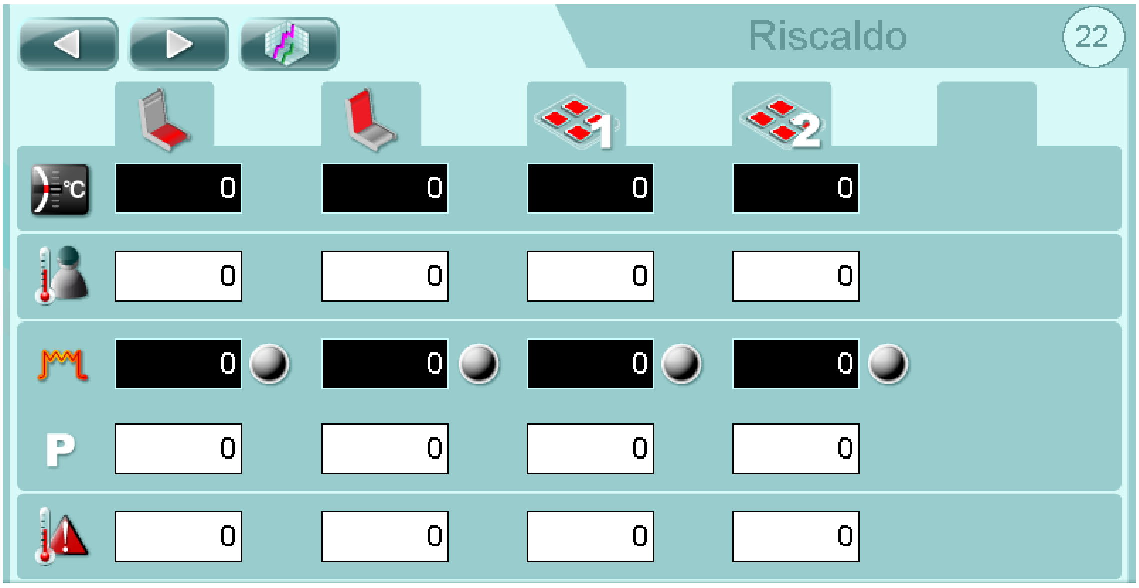
PROPORTIONAL PARAMETER FOR HEATING SYSTEM

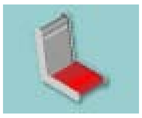

TEMPERATURE ALARM (RECIPE DATA)


TRENDS PAGE

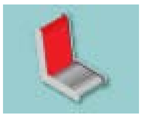

TOOL CAVITIES HEATING/COOLING PAGE


DOT PLATES HEATING PAGE

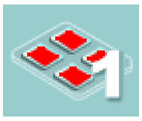




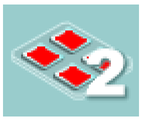
SKIN CYCLE HORIZONTAL PREHEATING



SKIN CYCLE VERTICAL PREHEATING



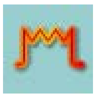
SHUTLLE PLATES 1 HEATING



SHUTLLE PLATES 2 HEATING

CURRENT TEMPERATURE

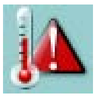
SET POINT TEMPERATURE (RECIPE DATA)



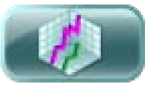
RESISTANCE HEATING POWER (%)

LIGHT SHOWS WHEN HEATING ON

PROPORTIONAL PARAMETER FOR HEATING SYSTEM



TEMPERATURE ALARM (RECIPE DATA)



TRENDS PAGE

ENABLE UPPER TOOL 1 HEATING (RECIPE DATA)

ENABLE UPPER TOOL 2 HEATING (RECIPE DATA)

ENABLE ONE SHOT HEATING (RECIPE DATA)

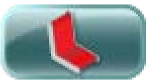
ENABLE PROGRAMMED HEATING CYCLE

NOTE: To change the time for switching on and off of the heating cycle programmed tool simply enter the time of ignition in the "From 00:00" and the turn-off time in the "To 00:00".

ENABLE LOWER TOOL 1 HEATING (RECIPE DATA)

ENABLE LOWER TOOL 2 HEATING (RECIPE DATA)

ENABLE LOWER TOOL ONE SHOT HEATING (RECIPE DATA)



ENABLE VERTICAL AND HORIZONTAL PREHEATING FOR SKIN CYCLE (IF AVAILABLE)

ENABLE THE PLATE PREHEATING MOUNTED ON THE SHUTTLE (IF AVAILABLE)

5.13 LOCKING

REEL 1

REEL 2 (WHEN IT IS AVAILABLE)

DRIVE ROLLER (WHEN AUTOMATIC REEL CHANGE IS AVAILABLE)

TOOL

LID UNWINDER

DISABLE THE COMPRESSED AIR CIRCUIT

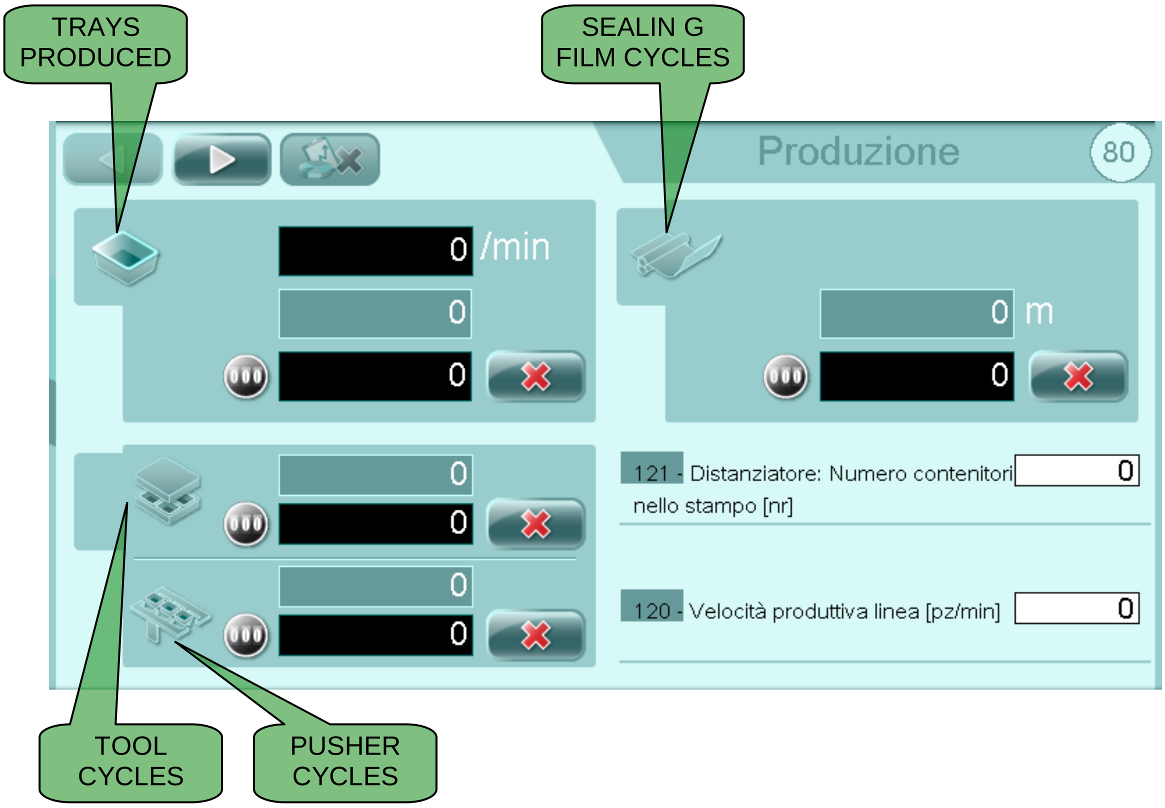
DAR TOOL

WASTE WINDER

WASTE WINDER WASTE WINDER CYCLE

5.14 PRODUCTION DATA

IMPORTANT! This page is available only if the user has the "production data" authorisation.



CURRENT PRODUCTION

TOTAL MACHINE OUTPUT

PARTIAL MACHINE OUTPUT

RESET PIECE COUNTER

PRODUCTION RESET (NOT BY SHIFT)

PRODUCTION DATA (NOT BY SHIFT)

PRODUCTION DATA SEARCH BY SHIFT

Trays: number of items produced by the machine.

Tool cycles: number of cycles effected by the tool.

Pusher cycles: number of cycles effected by the pusher arms.

Actual Performance %: items produced in consideration to the items to produce.

Efficiency/min: Items / time OPERATION.

Productivity/min: Items / RUN time.

Performance/min: Items / time

Efficiency %: (run time x theoretical production x 100) / ietms.

Productivity %: (function time x theoretical production x 100) / items.

Performance %: (time x theoretical production x 100) / items.

USED TO DISPLAY OUTPUT ON A PARTICULAR DATE

5.15 FILE MANAGER

IMPORTANT! This page is available only if the user has the "File manager" authorisation.

Imports from USB the .csv file for a recipe

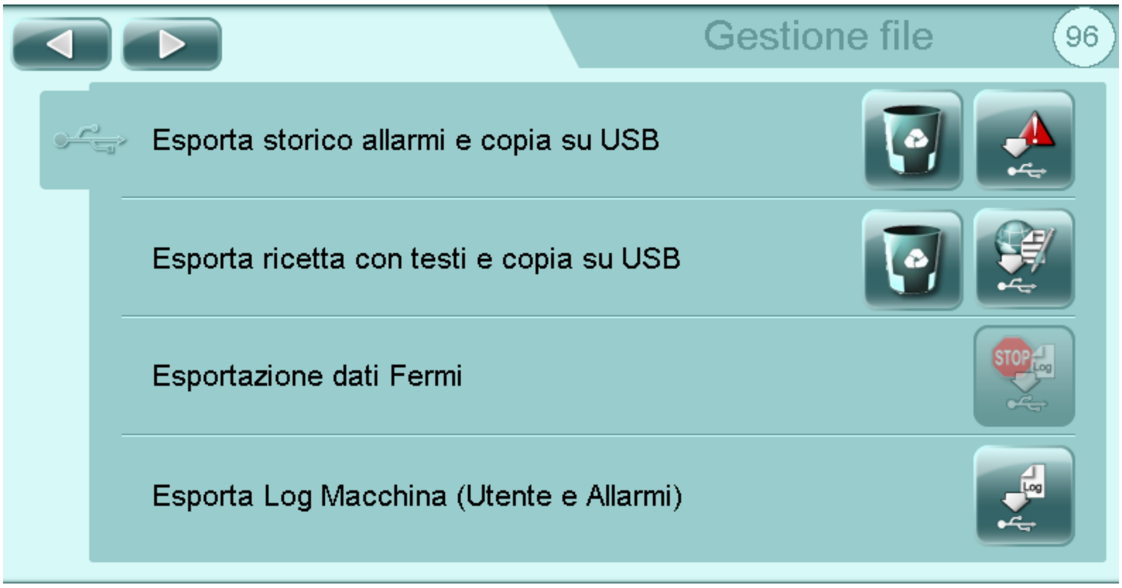
It allows to export on USB or to care from USB on her or from the folder "RECIPE" the file ". csv" for every recipe and a folder with the same name of the recipe containing the image.

It allows to export on USB a file ". csv" in the folder "PRODUCTION" where the data of the production are copied separated in the modality "with shifts" and in the modality "without shifts".

It allows to export on USB a file ". csv" in the folder "QUALITY" for every effected recording.

Imports from USB, parameter, enable, alarm and stoppage texts

It allows to export a file ". csv" in the folder "TEXTS" of the USB with parameters, enable, alarms and locks.



Exports to USB a .csv file in the EXP_AL_STORY folder with alarms in the selected language for a given time period.

It allows to export on USB a file ". csv" with the parameters of the recipe, the texts in language and the image related to the recipe.

It allows to export on USB the LOG of the users in a file ". csv" in the folder "LOGGER."

Imports from the USERS folder in USB the list of users with passwords and groups.

Exports on USB a file ". csv" in the folder "USERS" the users list with the relative passwords and the groups.

Imports from the AL_CONF folder in USB the alarm configuration for the alarm group management

Exports on USB a file ". csv" in the folder "AL_CONF" with the configuration of the alarms for the management of the groups alarm.

Imports from the USB the inverters parameters file.

Exports in the USB the inverters parameters file.

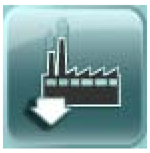
Enable the option to request a recipe by the Supervisor.

Complete system restart, every temporary data will be lost!

Restores the files of the last back up saved.

Exports on USB a back up copy of the whole files used byu the application with the exclusion of manuals and video.

It allows the selective restoration of the factory settings.



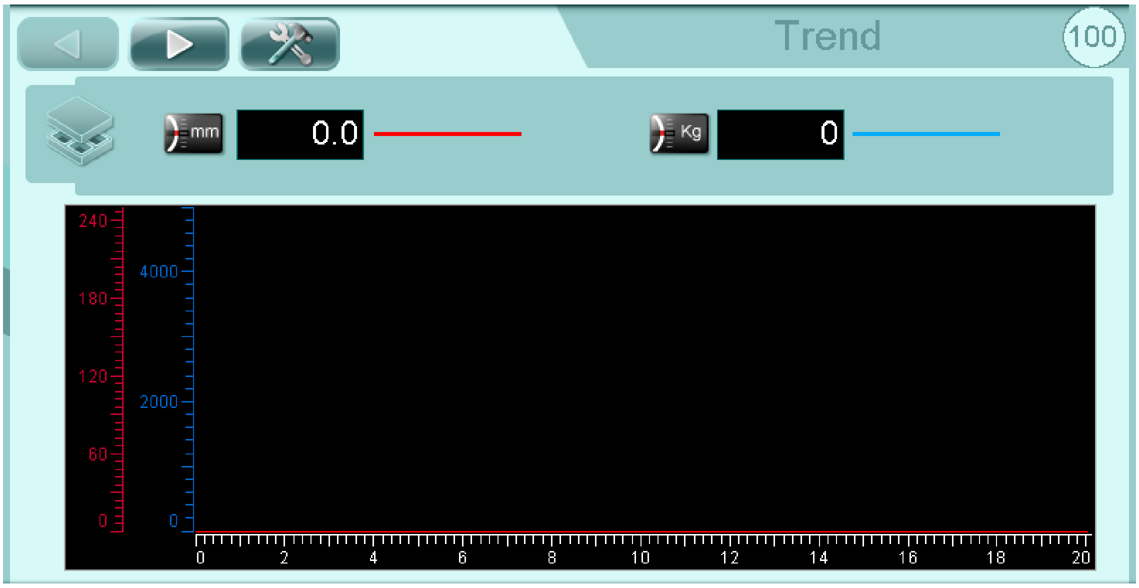
It allows to create the factory settings (this operation must be executes only after having contacted the "G.Mondini" technicians).

In the lower part of the page there are the exit ethernet data of the panel.

5.15.1 Restore factory defaults

5.16 TRENDS

These pages display graphs representing tool data trends during the operating cycle.

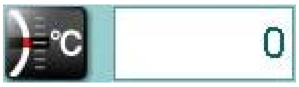


ACCESS TO THE MANAGEMENT PAGE

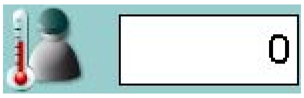
TOOL POSITION IN MILLIMETRES

PRESSURE CLOSING TOOL

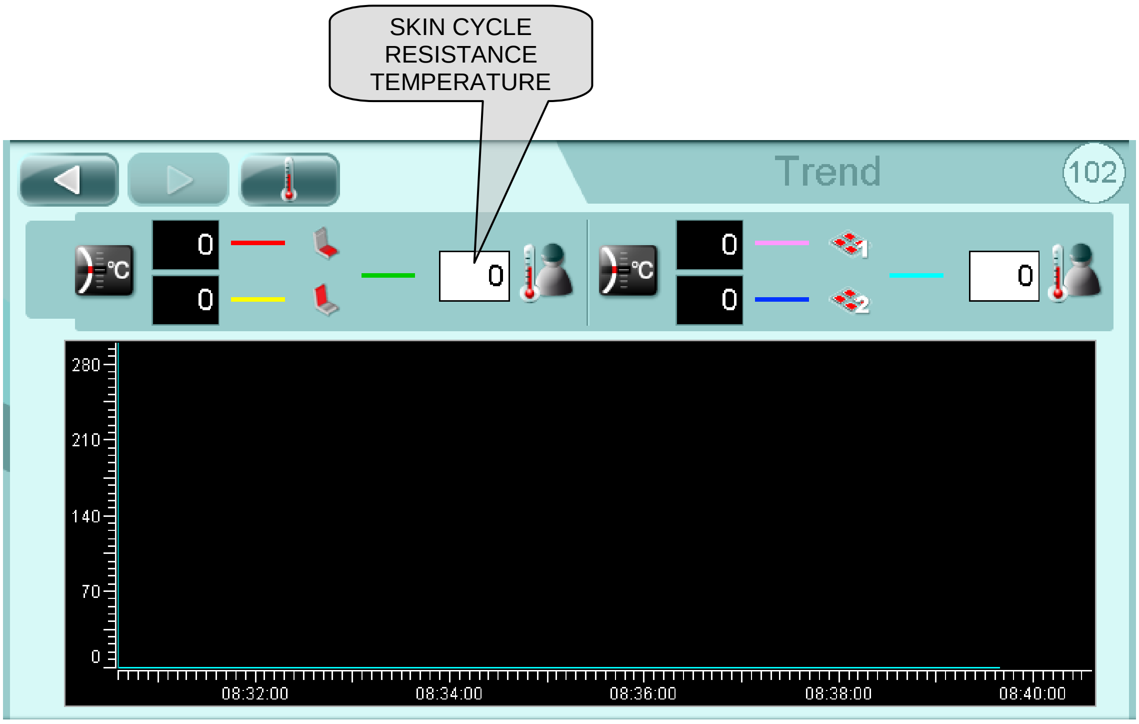
ACCESS TO TEMPERATURE PAGE

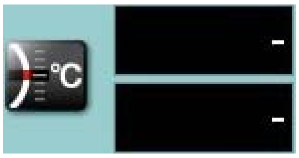


(CURRENT) TOOL TEMPERATURE

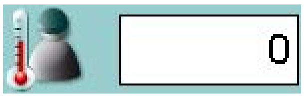


SET POINT TEMPERATURE





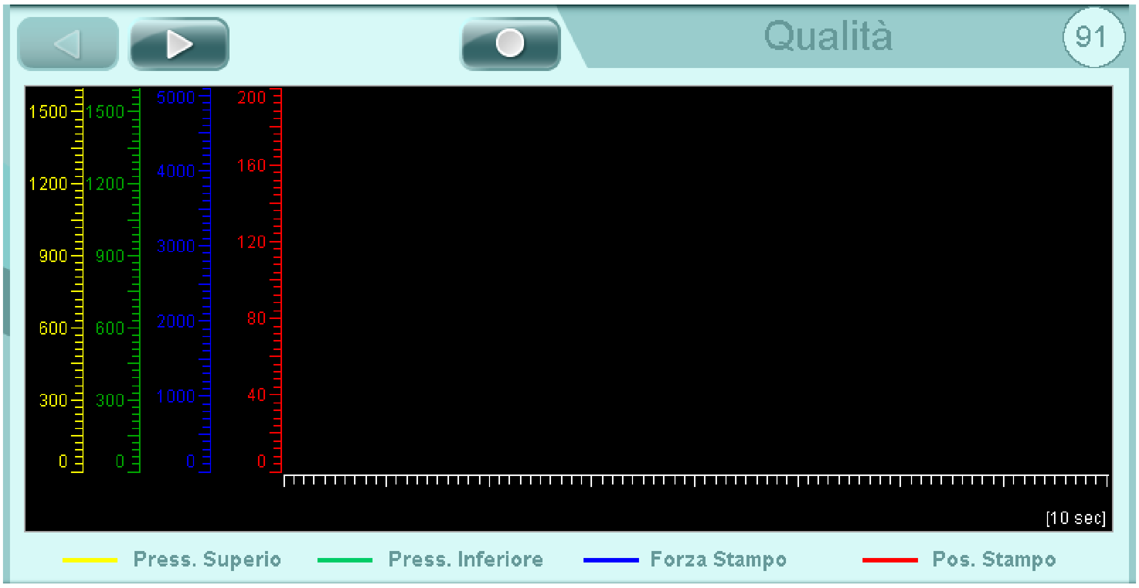
SKIN CYCLE RESISTANCE TEMPERATURE



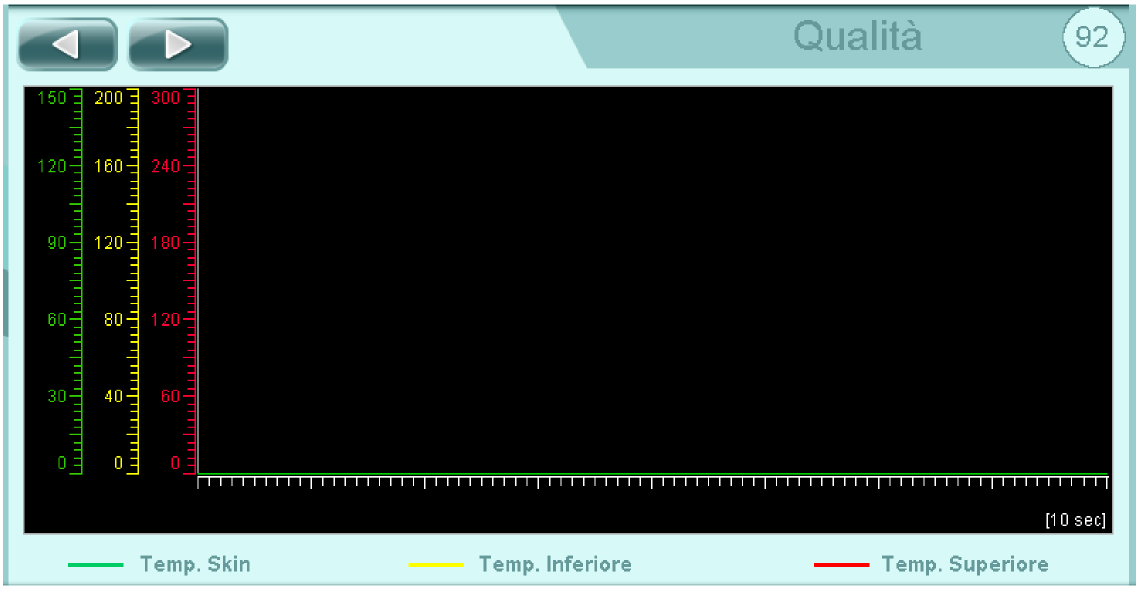
SKIN CYCLE RESISTANCE SET POINT TEMPERATURE

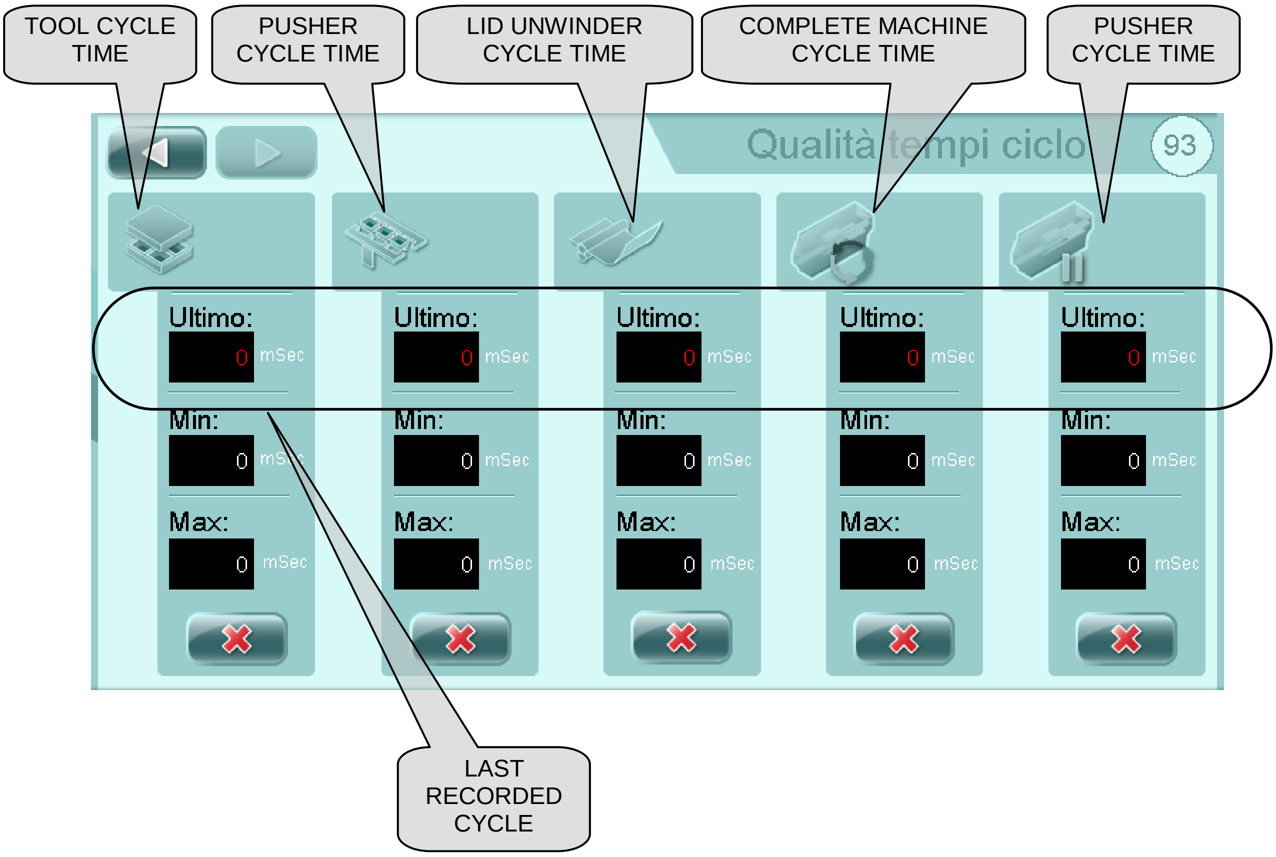
5.17 QUALITY

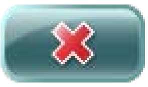
IMPORTANT! This page is available only if the user has the "Quality" authorisation.



START QUALITY DATA RECORDING (1 CYCLE). When this cycle is activated, the operator panel starts to record data as from the first available cycle for a time of 10 seconds. The panel records the operating cycle situation every 10ms and saves the data in an Excel file. The data recorded by the operator panel refer to the sealing tool, and in particular to the position, force, upper and lower pressure, upper and lower temperature, and skin temperature.

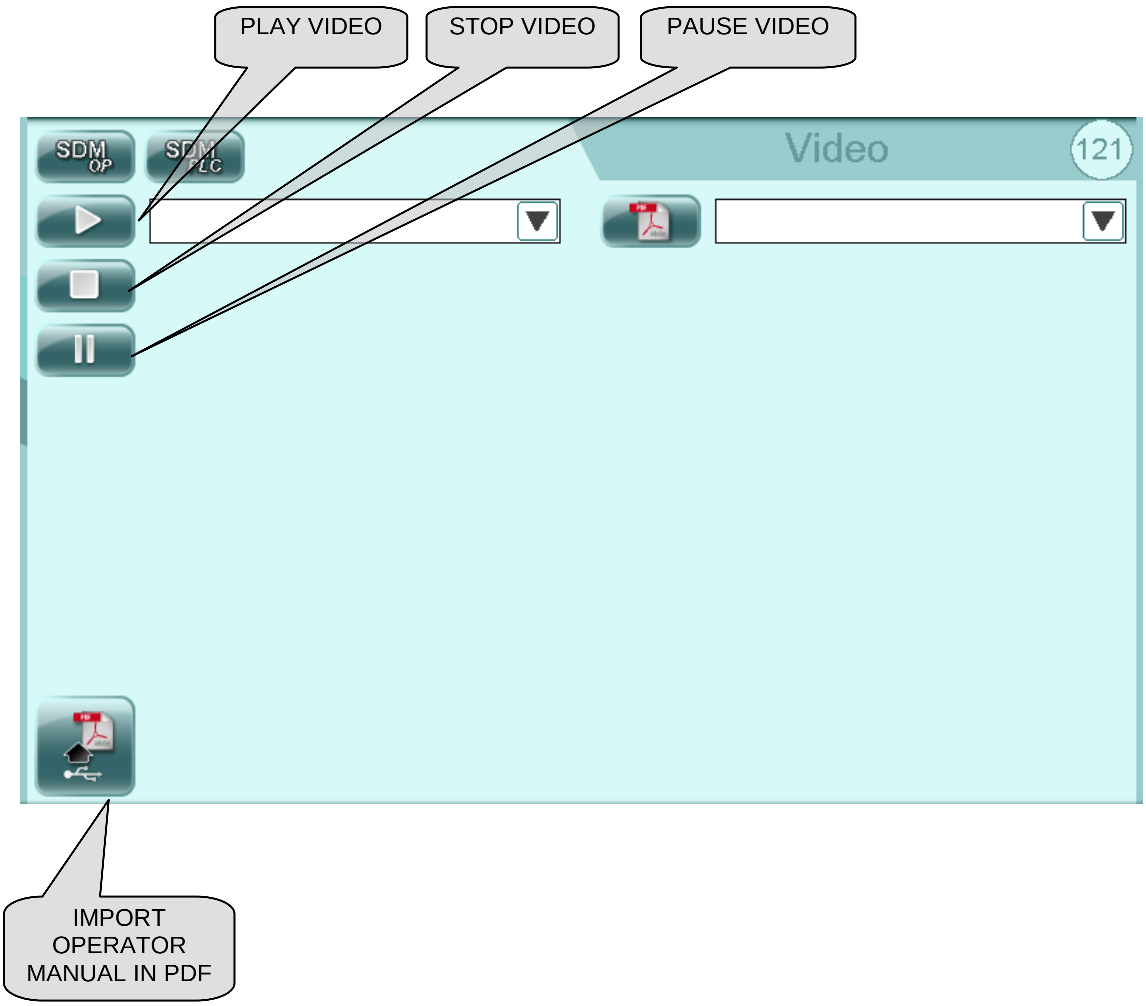

Qualità
92
150
120
90
60
30
0
200
160
120
80
40
0
300
240
180
120
60
0
[10 sec]
Temp. Skin
Temp. Inferiore
Temp. Superiore





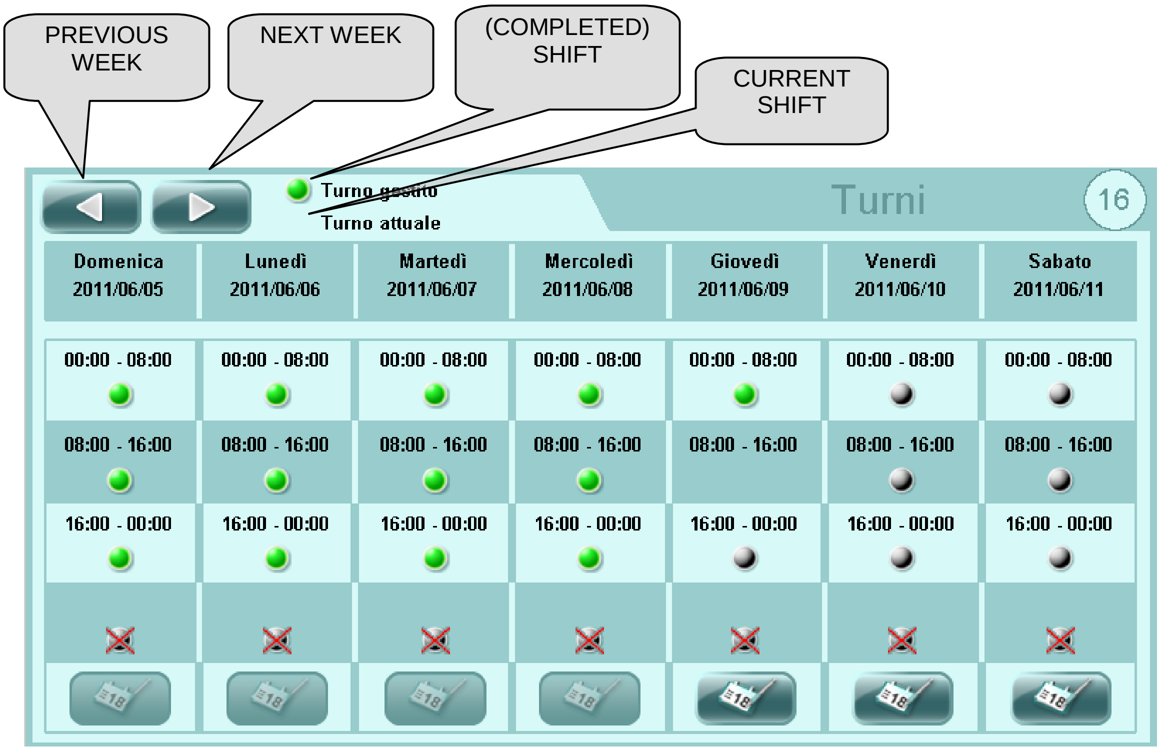
Reset cycle times

5.18 HELP


PLAY VIDEO
STOP VIDEO
PAUSE VIDEO
SDM OP
SDM PLC
Video
121
IMPORT OPERATOR MANUAL IN PDF

5.19 WORK SHIFTS

IMPORTANT! This page is available only if the user has the "Work shifts" authorisation.



5.20 USB ENTRY

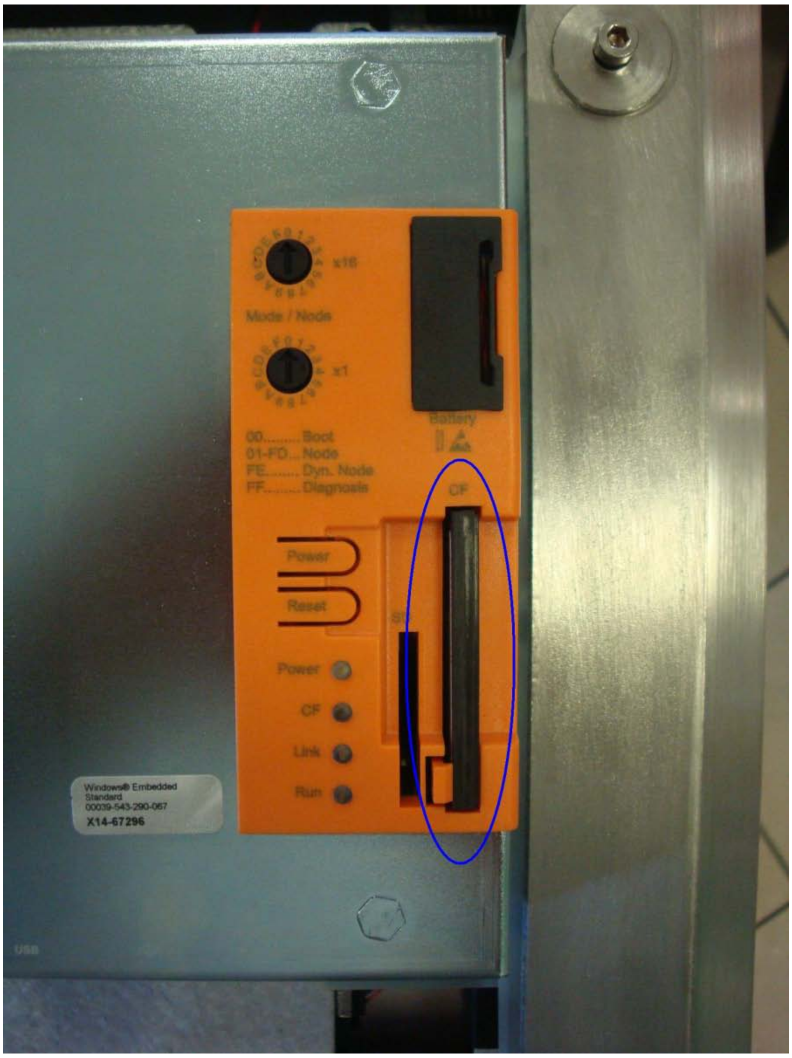
To access to the USB port is necessary remove the cover placed on the back side of the operator panel.

5.21 CF-CARD REPLACEMENT

For CF-CARD replacement, proceed as follows:

1. Save the recipes to have a backup copy.
2. Switch off the machine.
3. Unscrew the TORX - T 20 screws and remove panel screen.

4. Remove the old CF-CARD.



5. Insert the new CF-CARD.

6. Reassemble the screen panel.

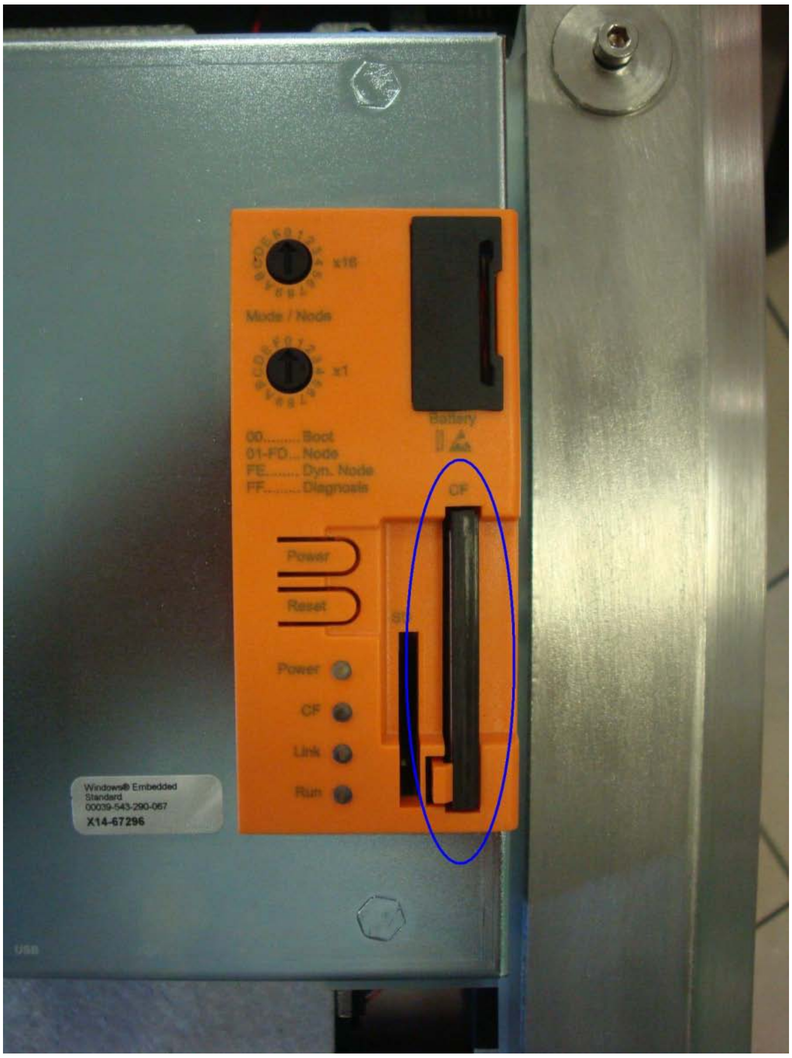
7. Turn on the machine.

5.22 REPLACE B&R PANEL

For replace B&R panel is necessary:

1. Save recipes for back up.
2. Switch OFF the machine.
3. Unscrew 16 TORX - T 20, open the frontal display, unplug the connectors and E-stop; note the right position and numbers.

4. Remove CF-CARD.



5. Remove the rear of panel from arm.

6. Assemble then new rear of panel to the arm.

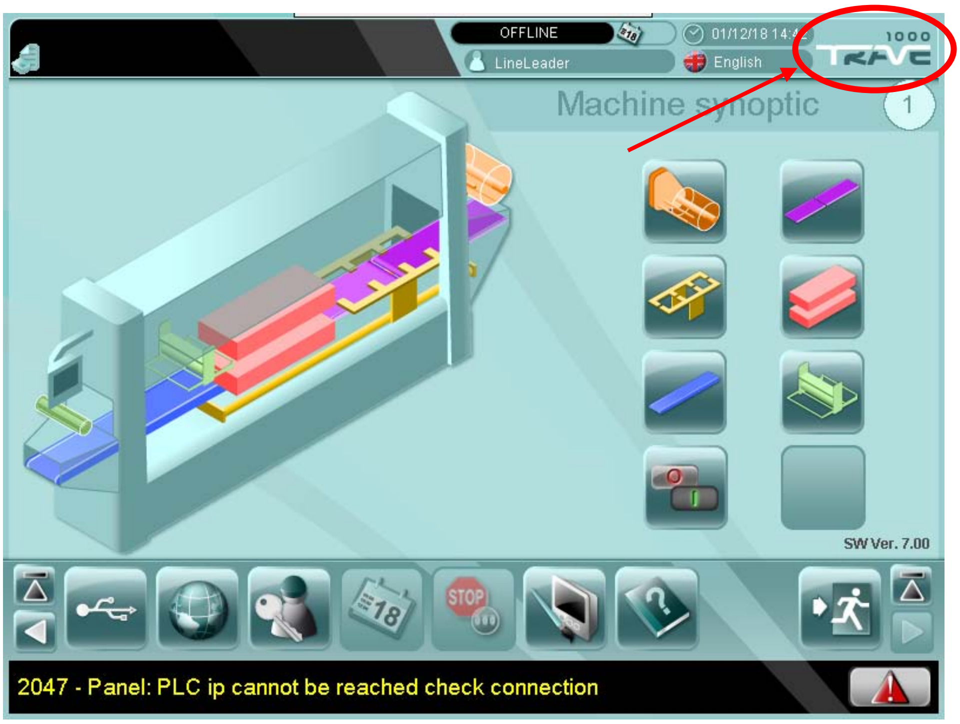
7. Insert the CF-CARD into new panel (or insert the new CF-CARD if available).

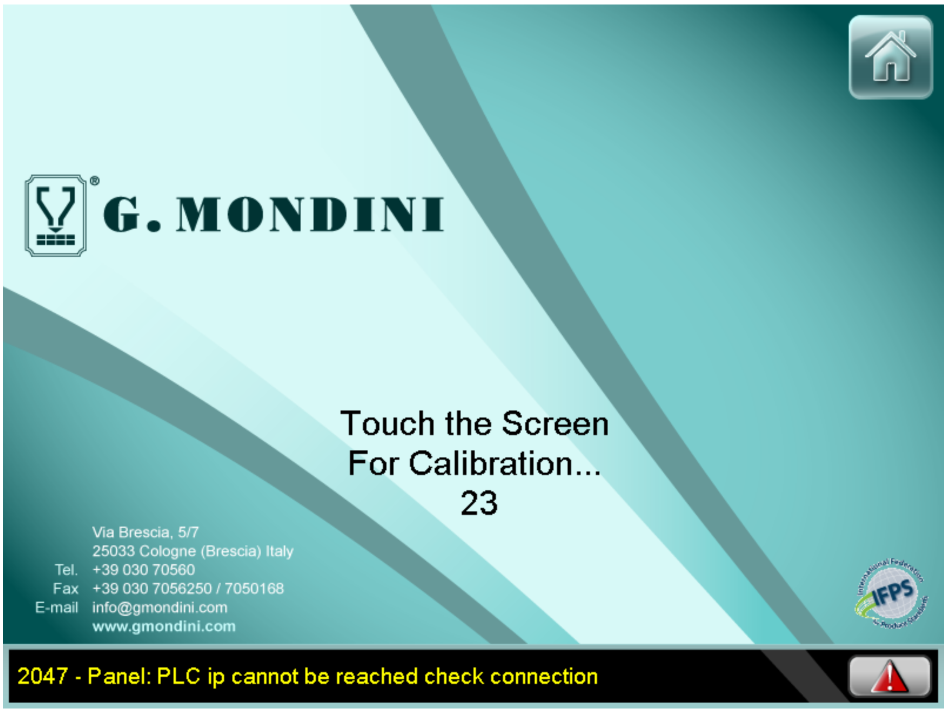
8. Plug the connectors and E-stop.

9. Switch ON the machine.

5.23 MANUAL CALIBRATION OF B&R PANEL

For manual calibration of B&R panel is necessary loggin with “COLLAUDO” and touch the logo machine in the right upper corner.
After this, press the screen for calibration.





5.24 LIST AND DESCRIPTION OF RECIPE PARAMETERS

The various recipe parameters are described in the following pages.

001 - Tool: Sealing time [s]	The tool closing time required to seal the seal on the tray.
002 - Tool: Lower tool start vacuum delay [s]	The delay time regulating the start of the vacuum cycle in the lower tool.
003 - Tool: Upper tool start vacuum delay [s]	The delay time regulating the start of the vacuum cycle in the upper tool.
004 - Tool: Vacuum time [s]	The vacuum time required to remove air from the tray before sealing.
005 - Tool: Lower gas 1 opening delay [s]	The delay time regulating opening of lower gas valve 1.
006 - Tool: Lower gas 2 opening delay [s]	The delay time regulating opening of the lower gas valve 2.
007 - Tool: Upper air opening delay [s]	The delay time regulating opening of the upper gas valve.
008 - Tool: Gas time [s]	The time required to fill the tray with gas.
009 - Tool: Lower gas 2 stop advance time [s]	The advance time regulating closing of lower gas valve 2.
010 - Tool: Upper air stop advance time [s]	The advance time regulating closing of the upper air valve.
011 - Tool: Lower compensation valve opening delay [s]	The delay time regulating opening of the lower tool compensation valve.
012 - Tool: Upper compensation valve opening offset [s]	The time difference for start of upper compensation.
013 - Tool: Tool lift delay [s]	The delay time regulating the lift movement of the tool.
014 - Tool: Vacuum threshold [mBar]	The pressure to reach during the vacuum cycle.
015 - Tool: Gas threshold [mBar]	The pressure to reach during the gas cycle.
016 - Tool: Lower 1 gas stop advance [mBar]	The pressure determining advance closing of lower gas valve 1.
017 - Tool: Lower 2 gas stop advance [mBar]	The pressure determining advance closing of lower gas valve 2.
018 - Tool: Upper air stop advance [mBar]	The pressure determining advance closing of the upper gas valve.
019 - Tool: Low/High vacuum switch threshold [mbar]	The vacuum switch threshold.
020 - Tool: Upper tool temperature set point [°C]	The operating temperature of the sealing tool.
021 - Tool: Upper tool temperature max allowed deviation [°C]	The maximum upper and lower temperature deviation from the upper tool set point temperature beyond which a machine alarm is generated.
022 - Tool: Upper tool One shot heating time [s]	The duration of the upper one-shot heating cycle.

023 - Tool: Upper tool heating 1 offset [°C]	The temperature difference between the set point and the value measured by the sensor of upper tool 1.
024 - Tool: Upper tool heating 2 offset [°C]	The temperature difference between the set point and the value measured by the sensor of upper tool 2.
025 -	
026 - Tool: Lower tool temperature set point [°C]	The operating temperature of the lower tool.
027 - Tool: Lower tool temperature max allowed deviation [°C]	The maximum upper and lower temperature deviation from the lower tool set point temperature beyond which a machine alarm is generated.
028 - Tool: Lower tool One shot heating time [s]	The duration of the lower one-shot heating cycle.
029 - Tool: Lower tool heating 1 offset [°C]	The temperature difference between the set point and the value measured by the sensor of lower tool 1.
030 - Tool: Lower tool heating 2 offset [°C]	temperature deviation between the set point and the value read by the lower tool 2 sensor.
031 - Tool: Cavities temperature max allowed deviation [°C]	maximum set point upper and lower temperature deviation from the set point temperature of the sealing tool resistances beyond which the machine generates an alarm.
032 - Tool: Cavities cooling start threshold [°C]	the temperature of the tool cavities defining the start of the cooling cycle.
033 - Tool: Cavities temperature set point [°C]	temperature deviation between the set point and the value read by the tool cavity sensor.
034 - Tool: Gas mixer - O2 [%]	the percentage of oxygen in the tray after a sealing cycle.
035 - Tool: Gas mixer - CO2 [%]	the percentage of carbon dioxide in the tray after a sealing cycle.
036 - Tool: Gas mixer - N2 [%]	the percentage of nitrogen in the tray after a sealing cycle.
037 - Tool - Skin: Forming compensation 2 start delay [s]	The delay time regulating opening of the tool compensation valve during skin cycle.
038 - Tool: Gas cleaning time when chaging the gas selection [s]	is the cleaning time of the gas circuit during the changing of the gas cycle.
039 - Tool - Skin: Vacuum valve 2 delay [s]	The dealy time for the vacuum valve 2 opening during the skin cycle.

040 - Tool: Speed set point [%]	Tool operating speed in automatic mode.
041 - Tool: Tool lift delay from Vacuum\Gas position [s]	The delay time regulating start of tool ascent.
042 - Tool: Descent pause time [s]	Tool pause during descent.
043 - Tool: Sealing force [kg]	The tool sealing pressure.
044 - Tool: Opening pause time [s]	the dwell time for the tool during the tool opening phase.
045 - Tool - Slice pack: Start delay [s]	The delay time regulating start of the sealing cycle during the slice-pack cycle.
046 - Tool: Blowing time to detach the trays [s]	the blow time it takes to detach the trays in the tool.
047 - Tool - Slice pack: Vacuum release delay after pusher starts [s]	The delay time regulating vacuum cycle stop after pusher activation during a slice-pack cycle.
048 - Tool - Stretch cycle: Pre-forming time [s]	the time required to film pre-forming inside the tool during a “stretch” cycle.
049 - Tool - Stretch cycle: Delay start vacuum-gas cycle [s]	the delay time after which the vacuum-gas cycle starts during a “stretch” cycle.
050 - Tool - Stretch cycle: Air time [s]	the time it takes for the tool air vents to open during a “stretch” cycle.
051 - Tool - Skin: Vacuum+forming delay [s]	The delay required to start the vacuum cycle to form the film in the tool during a skin cycle.
052 - Tool - Skin: Vacuum+forming time [s]	The vacuum time required for film formation in the tool during a skin cycle.
053 - Tool - Skin: Lower tool air time for forming [s]	The opening time of the valve regulating air supply to the lower tool during a skin cycle.
054 - Tool - Skin: Lower vacuum start delay [s]	The delay time regulating opening of the lower vacuum valve during a skin cycle.
055 - Tool - Skin: Upper vacuum start delay [s]	The delay time regulating opening of the upper vacuum valve during a skin cycle.
056 - Tool - Skin: Upper vacuum time [s]	The time required to create a vacuum in the tool during a skin cycle.
057 - Tool - Skin: Forming compensation start delay [s]	The opening time of the sealer compensation valve during a skin cycle.
058 - Tool - Skin: Upper tool compensation start delay [s]	The opening time of the upper tool compensation valve during a skin cycle.
059 - Tool - Skin: Upper tool compensation time [s]	The opening time of the upper tool compensation valve during a sealing cycle.

060 - Tool - Skin: Forming compensation time [s]	The time required for the upper tool compensation valve to open when sealing the film on the tray during the skin cycle.
061 - Tool - Skin Super: Forming compensation start delay [s]	The delay time regulating opening of the sealer compensation valve during a skin cycle.
062 - Tool - Skin Super: Upper tool compensation start delay [s]	The delay time regulating opening of the upper tool compensation valve during a skin cycle.
063 - Tool - Skin: Pre-heating set point [°C]	The operating temperature of the film preheat resistance during a skin cycle.
064 - Tool - Skin: Pre-heating temperature max allowed deviation [°C]	Maximum deviation of the set-point temperature from the temperature read by the film pre-heat resistance sensor beyond which a machine alarm is generated.
065 - Tool - Skin: Pre-heating 1 offset [°C]	The difference between the temperature read by the sensor and the actual temperature of film pre-heat resistance 1.
066 - Tool - Skin: Pre-heating 2 offset [°C]	The difference between the temperature read by the sensor and the actual temperature of film pre-heat resistance 2.
067 - Tool - Stretch cycle: Compensation start delay [s]	The delay time regulating opening of the tool compensation valve during the stretch cycle.
068 - Tool - Mirabella: Cooling water ON time [s]	The cooling water operation time during "Mirabella" cycle.
069 - Tool - Mirabella: Cooling water OFF time [s]	The cooling water stop time during "Mirabella" cycle.
070 - Pusher Hand 1: Home position [mm]	The start-of-cycle position of pusher arm 1.
071 - Pusher Hand 1: Open position [mm]	The opening position of pusher arm 1.
072 - Pusher Hand 1: SAFE position [mm]	The safety position of pusher arm 1 for tool change-over.
073 - Pusher Hand 1: Middle position [mm]	The intermediate position of pusher arm 1.
074 - Pusher Hand 1: SLIMFRESH stopper lowering position [mm]	The position in which the SLIMFRESH stopper lowering.
075 - Pusher Hand 1: Close position [mm]	The closing position of pusher arm 1.
076 - Pusher Hand 1: Closing threshold [mm]	The closing height of pusher arm 1.
077 -	
078 -	
079 -	

080 - Pusher Hand 2: Home position [mm]	The start-of-cycle position of pusher arm 1.
081 - Pusher Hand 2: Open position [mm]	The opening position of pusher arm 2.
082 - Pusher Hand 2: SAFE position [mm]	The safety position of pusher arm 2 for tool change-over.
083 - Pusher Hand 2: Middle position [mm]	The intermediate position of pusher arm 2.
084 - Pusher Hand 2: SLIMFRESH stopper lowering position [mm]	The position in which the SLIMFRESH stopper lowering.
085 - Pusher Hand 2: Close position [mm]	The closing position of pusher arm 2.
086 - Pusher Hand 2: Closing threshold [mm]	The closing height of pusher arm 2.
087 -	
088 -	
089 -	
090 - Pusher Hands: Closing delay [s]	The delay time regulating closing of the pusher arms.
091 - Pusher Hands: Closing time [s]	The pusher closing time required to perform a work cycle.
092 - Pusher Hands: Delay before opening [s]	The delay time regulating start of opening of the pusher arms.
093 - Pusher Hands: Opening time [s]	The time the pusher arms remain open.
094 - Pusher Hands: SAFE pos. Check time [mm]	The reading time of the sensor detecting correct pusher position for tool change-over.
095 -	
096 -	
097 -	
098 -	
099 - Pusher Hands: Closing timeout [s]	The maximum reading time of the sensor detecting closing of the pusher arms.
100 - Pusher arm: Forward move acceleration time [s]	The delay time regulating start of forward movement of the pusher arms.
101 - Pusher arm: Forward movement time [s]	The duration of forward movement of the pusher arms.
102 - Pusher arm: Forward pause time [s]	Pusher arm pause time in the forward position.
103 - Pusher arm: Backward movement time [s]	Duration of reverse movement of the pusher arms.
104 - Pusher arm: Backward pause time [s]	Pusher arm pause time in the reverse position.
105 - Pusher arm: Pusher/Film unwind speed ratio	The ratio of pusher arm speed to lid unwinder speed.
106 - Pusher arm: Starting position for the film unwind (following) [mm]	Pusher arm position for guiding the sealing film.

107 - Pusher arm: Forward move at constant speed time [s]	the forward move time at constant speed.
108 - Pusher arm: Deceleration speed [%]	the deceleration speed of the pusher arms.
109 -	
110 - Pusher arm: Offset to advance start raise of the tool [mm]	It is the advance position on the pusher arms stroke, respect to the release position, for the raise of the tool.
111 - Pusher arm: Speed setpoint [%]	Pusher arm operating speed in automatic mode.
112 - Pusher arm: Home position [mm]	Pusher arm start-of-cycle position.
113 - Pusher arm: Backward position (ready to catch) [mm]	Pusher arm reverse position (tray pick-up).
114 - Pusher arm: Forward position (release) [mm]	Pusher arm forward position (tray release).
115 - Pusher arm: Acceleration space [mm]	the position of the pusher arms for the replacement of the sealing tool.
116 - Pusher arm: Deceleration space [mm]	the deceleration space of the pusher arms.
117 - Pusher arm: Deceleration time [s]	the deceleration time of the pusher arms.
118 - Pusher arm: Trays detach space [mm]	the trays detach space of the pusher arms.
119 - Pusher arm: Trays detach time [s]	the trays detach time of the pusher arms.
120 - Infeed production speed [pcs/min]	is the quantity of trays processed by the line expressed in units per minute.
121 - SpacerBelt: Number of trays into the tool [nr]	The number of tool cavities (number of trays sealed at each cycle).
122 - SpacerBelt 1: Step length [mm]	The length covered by spacer belt 1 to perform a step for correct tray positioning.
123 - SpacerBelt 1: Last step length [mm]	The length covered by spacer belt 1 to perform the last cycle step for correct tray positioning.
124 - SpacerBelt 1: Start delay [s]	The delay time regulating start of spacer belt 1.
125 - SpacerBelt 1: Speed [%]	Spacer belt 1 operating speed.
126 - SpacerBelt 1: Acceleration [%]	Acceleration of spacer belt 1 during operation.
127 - RESERVED	
128 - SpacerBelt 1: Deceleration [%]	Deceleration of spacer belt 1 during operation.
129 - SpacerBelt 1: Step length for -continuos end- [mm]	The length covered by spacer belt 2 to perform a step for correct tray positioning.

130 - SpacerBelt 2: Step length [mm]	The length covered by spacer belt 1 to perform a step for correct tray positioning.
131 - SpacerBelt 2: Last step length [mm]	The length covered by spacer belt 1 to perform the last cycle step for correct tray positioning.
132 - SpacerBelt 2: Start delay [s]	The delay time regulating start of spacer belt 2.
133 - SpacerBelt 2: Speed [%]	Spacer belt 2 operating speed.
134 - SpacerBelt 2: Acceleration [%]	Spacer belt 2 operating acceleration.
135 - SpacerBelt 2: Deceleration [%]	Spacer belt 2 operating deceleration.
136 -	
137 - SpacerBelt: Pusher arm start delay [s]	The delay time regulating start of the pusher arms.
138 - SpacerBelt: Pause time when the pusher moves [s]	Spacer belt pause to allow movement of the pusher arms.
139 - SpacerBelt: Step length after the pause [mm]	The length of the spacer belt step after the pause.
140 - SpacerBelt: Starting photocell filter [mm]	The minimum reading time of the start photocell to detect the presence of a tray.
141 - SpacerBelt: Starting photocell timeout [s]	The maximum reading time of the spacer belt start photocell beyond which the machine generates an alarm.
142 - SpacerBelt: Pusher start position advance [mm]	The position of the spacer belt for advanced start of the pusher arms.
143 - SpacerBelt: Pusher start position advance on last tray from starting chain [mm]	The position of the spacer belt for advanced start of the pusher arms on last tray from starting chain.
144 - SpacerBelt: Maximum number of trays allowed [nr]	The maximum number of trays on the spacer belt.
145 - SpacerBelt: Number of trays per cycle [nr]	the number of trays to space out at each spacer belt cycle.
146 - SpecerBelt: Tool bypass belt speed [%]	the operating speed of the bypass belt.
147 - SpacerBelt 1: Start delay while line accelerating [s]	The delay time for spacer belt start while line accelerating.
148 - SpacerBelt 2: Start delay while line accelerating [s]	The delay time for spacer belt start while line accelerating.
149 - SpacerBelt: Photocell false pulses filter [s]	The reading time of the photocell for the trays recognition.
150 - SpacerBelt: Semiautomatic 'On Fly catch' step length [mm]	The step length of the spacer belt during "On Fly catch" cycle on semiautomatic mode.
151 - InBelt: Belt speed setpoint [%]	The feed belt operating speed in automatic mode.
152 - InBelt: Belt start delay [s]	The feed belt start delay.
153 - InBelt: Belt stop delay [s]	The feed belt stop delay.

154 - InBelt: Belt reverse time on stop cycle [s]	The reverse time for the inlet belt on stop cycle.
155 -	
156 -	
157 - InBelt: LEFT DP Pacer downstream (exit) piston closing delay [s]	is the delay time that regulates the closing of the piston 1 left of the pacer on the inlet belt.
158 - InBelt: Tray rotation Belt 1 speed setpoint [%]	The Tray rotation Belt 1 operating speed.
159 - InBelt: Tray rotation Belt 2 speed setpoint [%]	The Tray rotation Belt 2 operating speed.
160 - InBelt: Double Piston Pacer start delay at pusher cycle [s]	The delay time regulating start of the synchroniser on the feed belt.
161 - InBelt: DP Pacer downstream (exit) piston opening delay [s]	The delay time regulating opening of synchroniser piston 1 on the feed belt.
162 - InBelt: RIGHT DP Pacer downstream (exit) piston closing delay [s]	The delay time regulating closing of synchroniser piston 1 on the feed belt.
163 - InBelt: DP Pacer upstream (inlet) piston opening delay [s]	The delay time regulating opening of synchroniser piston 2 on the feed belt.
164 - InBelt: DP Pacer upstream (inlet) piston closing delay [s]	The delay time regulating closing of synchroniser piston 2 on the feed belt.
165 - InBelt: DP Pacer inlet tray sensor timeout [s]	The maximum time it takes the sensor to detect a tray at the synchroniser, beyond which a machine alarm is generated.
166 - InBelt: Tray rotation belt Pacer closing delay [s]	the delay time after which the pacer on the tray rotation belt closes.
167 - InBelt: Tray rotation belt Pacer closing time [s]	the time during which the pacer on the tray rotation belt remains closed.
168 - Deviator: Tray passage time [s]	The reading time of the sensor for the tray passage.
169 - Deviator: Photocell timeout [s]	The maximum time it takes the sensor to detect a tray in the deviator device area, beyond which a machine alarm is generated.
170 - Deviator: Home position [°]	The start-of-cycle position of the tray deviator.
171 - Deviator: Right position [°]	The right-hand position of the tray deviator during its cycle.
172 - Deviator: Left position [°]	The left-hand position of the tray deviator during its cycle.
173 - Deviator: Speed [%]	Tray deviator operating speed.
174 - Deviator: Acceleration [%]	Tray deviator acceleration.
175 - Deviator: Deceleration [%]	Tray deviator deceleration.
176 - Deviator: Start delay [s]	The delay time regulating start of the tray deviator operating cycle.
177 - Deviator - Double piston pacer: Opening delay [s]	The delay time regulating opening of the double piston pacer.

178 - Deviator: Number trays to make the deviator move [nr]	The number of trays to deviate for each operating cycle.
179 - Deviator - Pacer: Closing delay [s]	The delay time regulating closing of the pacing device.
180 - Deviator - Pacer: Tray passage time [s]	The time the pacing device remains open to allow a tray to pass.
181 - Deviator - Pacer: Photocell timeout [s]	The maximum time it takes the sensor to detect a tray in the pacing device area, beyond which a machine alarm is generated.
182 - Deviator - Double piston pacer: Closing time [s]	The delay time regulating closing of the double piston pacer.
183 - Deviator: Belt speed setpoint [%]	The deviator belt operating speed.
184 - Deviator: Belt start delay [s]	The deviator belt start delay.
185 - Deviator: Belt stop delay [s]	The deviator belt stop delay.
186 - Deviator: Tray control start delay [s]	The delay time regulating start of the tray control system.
187 - Deviator: Product in the tray - control start delay [s]	The delay time regulating start of the tray product control system.
188 - Deviator: Product in the tray - control time [s]	The time required to check product in the tray.
189 - Deviator: Ejector piston start delay [s]	The delay time regulating start of the bad tray ejection piston.
190 - Deviator: Eject time [s]	the time it takes for the diverter belt to eject a tray.
191 - Film reel unwind: Speed setpoint [%]	The reel unwinder speed in automatic mode.
192 - Film reel unwind: Start delay [s]	The delay time regulating start of the reel unwinder.
193 - Film reel unwind: Stop delay [s]	The delay time regulating stop of the reel unwinder.
194 - Film waste rewind: Speed setpoint [%]	The delay time regulating start of waste winder.
195 - Film: Number of cycle when reel is empty before stopping machine [nr]	Number of cycle when reel is empty before stopping machine.
196 - Film reel auto. change: Speed setpoint [%]	The maximum operating speed of the automatic reel change-over device.
197 - Film reel auto. change: Contrast rollers close position [mm]	The distance between the film sensor and the closing rollers.
198 - Film reel auto. change: Attaching length [mm]	The amount of film to unwind with the sealing rollers closed to ensure perfect joining of the ends of the bi-adhesive film.
199 - Film reel auto. change: Film length from attaching rollers to tool [mm]	The distance between the film sensor and the sealing tool.

200 - Film unwind: Mark to mark distance [mm]	The distance between one reference mark on the sealing film and the next detected by the corresponding sensors.
201 - Film unwind: Length to pull [mm]	The amount of film to unwind for each sealing cycle.
202 - Film unwind: Speed setpoint [%]	The operating speed of the lid unwinder in automatic mode.
203 - Film unwind: Acceleration [%]	Lid unwinder acceleration.
204 - Film unwind: Deceleration [%]	Lid unwinder deceleration.
205 - Film unwind: Operation mode selection	Used to select the lid unwinder operating mode.
206 - Film unwind: Mark search speed [%]	The speed of the sensor reading the reference mark on the sealing film.
207 - Film unwind: Mark offset [mm]	The difference between the film reference mark and the sensor reading.
208 - Film reel auto. change: Length to bring junction over the tool [mm]	The amount of film to unwind to allow the join in the two sealed films to pass through the tool area.
209 - Film unwind: Film reject time [min]	The film reject time cycle of the film unwind device.
210 - ExitBelt 1: Belt speed setpoint [%]	Delivery belt speed in automatic mode.
211 - ExitBelt 2: Belt speed setpoint [%]	Delivery belt speed in automatic mode.
212 - ExitBelt: Photocell timeout for jammed onto the belt [s]	The maximum time it takes the sensor to detect trays on the delivery belt beyond which a machine alarm is generated.
213 - ExitBelt: Delay evacuation time [s]	The delay time regulating actual tray passage on the delivery belt.
214 - ExitBelt: Pusher arm position for stop [mm]	The delay time regulating actual tray passage on the delivery belt.
215 - ExitBelt 2->1: Speed setpoint [%]	2->1 belt speed in automatic mode.
216 - ExitBelt - Transport belt: Speed setpoint [%]	Transport belt speed in automatic mode.
217 - ExitBelt: Stop delay on Downstream consent missing [s]	The delay time for the exit belt stop on downstream consent missing.
218 -	
219 - ExitBelt: Down stream consent missing timeout [s]	The maximum waiting time of the exit belt stop on downstream consent missing.
220 - Line: Start delay [s]	the delay time after which the chain operating cycle starts.
221 - Line: Pause time [s]	the chain dwell time during the operating cycle.

222 - Line: Chain home position [°]	the cycle start position of the chain.
223 - Line: Chain step length [mm]	the chain displacement required to perform one chain step.
224 - Line: Chain speed setpoint [%]	the chain operating speed.
225 - Line: Chain acceleration [%]	acceleration of the chain during operation.
226 - Line: Chain deceleration [%]	deceleration of the chain during operation.
227 - Line: Step per cycle [nr]	the number of steps performed by the chain to coplete an operating cycle.
228 - Line: Tray discharge blower position along the chain [steps]	the position of the blower device to discharge trays that are not suitable for production.
229 - Line: Tray discharge blowing time [s]	the time it takes for the blower to eject the trays not suitable for production.
230 - Line: Missing trays to generate alarm [nr]	the number of trays missing on the chain that generates an alarm.
231 - Line: Tray detect photocell timeout [s]	The maximum time it takes for the sensor to detect the presence of trays of the chain, beyond which the machine generates an alarm.
232 - Line: Trays for cycle [nr]	the number of trays on the chain required for a work cycle.
233 - Line: Vibrator speed [%]	the operating speed of the vibrator on the chain.
234 - Line: Chain stop delay [s]	the delay time it takes for the chain to stop.
235 - Line: Advance on chain stop to move up the stopper [°]	the advance time it takes for the chain to stop due to the ascent of tray stops.
236 - Line: Stopper up time [s]	the time during which the trays on the chain are in the up position.
237 - Line - Buffer belt: Delay to -belt full- [s]	is the delay time that determines signal for the "buffer belt is full."
238 - Line - Buffer belt: Delay to -belt empty- [s]	is the delay time that determines signal for the "buffer belt is empty."
239 - Line: Tray length timeout [s]	The maximum time it takes for the sensor to detect the lenght of trays of the chain, beyond which the machine generates an alarm.
240 - Denester 1: Start delay [s]	the delay time it takes for the denester to start an operating cycle.
241 - Denester 1: Catch time [s]	the time it takes for the denester to catch the trays from the magazine.
242 - Denester 1: Vacuuum stop delay [s]	the delay time it takes for the vacuum cycle in the denester suction cups to stop.
243 - Denester 1: Translator forward time [s]	the time it takes for the tray translator to move forwards.

244 - Denester 1: Delay translator backward [s]	the delay time it takes for the translator to move backwards.
245 - Denester 1: Translator timeout [s]	the maximum time it takes for the sensor to read correct operation of the translator piston.
246 - Denester 1: Catch trays timeout [s]	the maximum time it takes for the sensor to detect correct operation of the tray catching piston.
247 - Denester 1: Selectors open/close time [s]	the time it takes for the denester selectors to open and close.
248 - Denester 1: Trays missing [nr]	the time it takes for the sensor to detect the absence of trays in the denester magazine.
249 - Denester 1: 1st cycle forcing release delay [s]	the delay time it takes for the trays to be forced released.
250 - Denester 1: Blowing time [s]	the time it takes for the blower to eject the trays not suitable for production.
251 - Denester 1: Home position [mm]	the start-of-cycle position for the denester.
252 - Denester 1: Catch position [mm]	the position for catching trays from the magazine.
253 - Denester 1: Catch speed [%]	the denester operating speed when it catches the trays from the magazine.
254 - Denester 1: Catch acc./dec. [%]	acceleration and deceleration of the denester when it catches the trays from the magazine.
255 - RESERVED	
256 - Denester 1: Cups stretch offset [mm]	the displacement of the denester piston to extend a suction cup.
257 - Denester 1: Cups stretch speed [%]	the operating speed of the denester during extension of a suction cup.
258 - Denester 1: Cups stretch acc./dec. [%]	acceleration and deceleration of the denester during extension of a suction cup.
259 - Denester 1: Trays detach offset [mm]	the displacement of the denester piston to detach a tray from the suction cups.
260 - Denester 1: Trays detach speed [%]	the denester operating speed when a tray is detached from the suction cups.
261 - Denester 1: Trays detach acc./dec. [%]	acceleration and deceleration of the denester when a tray is detached from the suction cups.
262 - Denester 1: Release position [mm]	the position of the denester at which it releases a tray onto the chain.
263 - Denester 1: Release speed [%]	The denester operating speed during release of a tray.
264 - Denester 1: Release acc./dec. [%]	Acceleration and deceleration of the denester when it releases a tray.

265 - Denester 1: Batch stopper close cycle [nr]	the number of cycles controlling batch stopper closing.
266 - Denester 1: Delivery area busy delay [s]	The delay time for discharge area busy.
267 - Denester 1: Delivery area free delay [s]	The delay time for discharge area free.
268 - Denester 1: Belt speed setpoint [%]	The belt operation speed.
269 - Denester 1: Pacer close delay [s]	The delay time for the pacer device closing.
270 - Denester 2: Start delay [s]	the delay time it takes for the denester to start an operating cycle.
271 - Denester 2: Catch time [s]	the time it takes for the denester to catch trays from the magazine.
272 - Denester 2: Vacuuum stop delay [s]	the delay time allowed to stop a vacuum cycle in the denester suction cups.
273 - Denester 2: Translator forward time [s]	the time it takes for the tray translator to move forward.
274 - Denester 2: Delay translator backward [s]	the delay time it takes for the translator to retract.
275 - Denester 2: Translator timeout [s]	the maximum time allowed to the sensor for detecting correct operation of the translator piston.
276 - Denester 2: Catch trays timeout [s]	the maximum time allowed to the sensor for detecting correct operation of the tray catching piston.
277 - Denester 2: Selectors open/close time [s]	the time it takes for the denester selectors to open and close.
278 - Denester 2: Trays missing [nr]	the time allowed to the sensor for detecting the absence of trays in the denester magazine.
279 - Denester 2: 1st cycle forcing release delay [s]	the delay time it takes for the trays to be forced released.
280 - Denester 2: Blowing time [s]	the time it takes the blower to eject the trays not suitable for production.
281 - Denester 2: Home position [mm]	the start-of-cycle position for the denester.
282 - Denester 2: Catch position [mm]	the position at which the trays are taken from the magazine.
283 - Denester 2: Catch speed [%]	the denester operating speed when the trays are taken from the magazine.
284 - Denester 2: Catch acc./dec. [%]	acceleration and deceleration of the denester when it catches the trays from the magazine.
285 - Denester 2: Cups stretch offset [mm]	the displacement of the denester piston to extend a suction cup.
286 - Denester 2: Cups stretch speed [%]	the operating speed of the denester during extension of a suction cup.

287 - Denester 2: Cups stretch acc./dec. [%]	acceleration and deceleration of the denester during extension of a suction cup.
288 - Denester 2: Trays detach offset [mm]	the displacement of the denester piston to detach a tray from the suction cups.
289 - Denester 2: Trays detach speed [%]	the denester operating speed when a tray is detached from the suction cups.
290 - Denester 2: Trays detach acc./dec. [%]	acceleration and deceleration of the denester when a tray is detached from the suction cups.
291 - Denester 2: Release position [mm]	the position of the denester at which it releases a tray onto the chain.
292 - Denester 2: Release speed [%]	The denester operating speed during release of a tray.
293 - Denester 2: Release acc./dec. [%]	Acceleration and deceleration of the denester when it releases a tray.
294 - Denester 2: Batch stopper close cycle [nr]	the number of cycles controlling batch stopper closing.
295 - Denester 2: Delivery area busy delay [s]	The delay time for discharge area busy.
296 - Denester 2: Delivery area free delay [s]	The delay time for discharge area free.
297 - Denester 2: Belt speed setpoint [%]	The belt operation speed.
298 - Denester 2: Pacer close delay [s]	The delay time for the pacer device closing.
299 - Denester 2:	
300 - Tool: Cavities 01 temperature offset [°C]	the operating temperature of cavity 1 heating resistance.
301 - Tool: Cavities 02 temperature offset [°C]	the operating temperature of cavity 2 heating resistance.
302 - Tool: Cavities 03 temperature offset [°C]	the operating temperature of cavity 3 heating resistance.
303 - Tool: Cavities 04 temperature offset [°C]	the operating temperature of cavity 4 heating resistance.
304 - Tool: Cavities 05 temperature offset [°C]	the operating temperature of cavity 5 heating resistance.
305 - Tool: Cavities 06 temperature offset [°C]	the operating temperature of cavity 6 heating resistance.
306 - Tool: Cavities 07 temperature offset [°C]	the operating temperature of cavity 7 heating resistance.
307 - Tool: Cavities 08 temperature offset [°C]	the operating temperature of cavity 8 heating resistance.
308 - Tool: Cavities 09 temperature offset [°C]	the operating temperature of cavity 9 heating resistance.
309 - Tool: Cavities 10 temperature offset [°C]	the operating temperature of cavity 10 heating resistance.

310 - Lid Preformed: Catch time [s]	the time it takes for the denester to catch lids from the magazine.
311 - Lid Preformed: Release time [s]	the time it takes to release pre-formed lids in the tool.
312 - Lid Pref./(Super Protr.)/DARFRESH: Catch 1 (backward) position [mm]	the position for catching preformed lids from the 1st row in the lid magazine.
313 - Lid Pref./(Super Protr.)/DARFRESH: Catch 1 (backward) speed [%]	the operating speed of the preformed lid disc for catching lids from the 1st row.
314 - Lid Pref./(Super Protr.)/DARFRESH: Catch 1 (backward) acc./dec. [%]	acceleration and deceleration of the preformed lid disc for catching lids from the 1st row in the magazine.
315 - Lid Preformed: Catch 2 position [mm]	the operating speed of the preformed lid disc for catching lids from the 2nd row in the magazine.
316 - Lid Preformed: Catch 2 speed [%]	the operating speed of the preformed lid disc for catching lids from the 2nd row.
317 - Lid Preformed: Catch 2 acc./dec. [%]	acceleration and deceleration of the preformed lid disc for catching lids from the 2nd row.
318 - Lid Pref./(Super Protr.)/DARFRESH: Release (forward) position [mm]	the position at which the preformed lids are released inside the tool.
319 - Lid Pref./(Super Protr.)/DARFRESH: Release (forward) speed [%]	the operating speed of the preformed lid disc for releasing the lids inside the tool.
320 - Lid Pref./(Super Protr.)/DARFRESH: Release (forward) acceleration [%]	Acceleration of the preformed lid disc for releasing the lids inside the tool.
321 - Lid Pref./(Super Protr.)/DARFRESH: Home position [mm]	the start-of-cycle position for the preformed lid disc.
322 - Lid Pref./(Super Protr.)/DARFRESH: SAFE position [mm]	A safe position preventing the lid disc from hitting the tool during operation.
323 - Lid Preformed: Transport Cups cleaning time [s]	the time it takes for the blower to clean the lid disc transport cups.
324 - Lid Preformed: Tool Cups cleaning time [s]	the time it takes for the blower to clean the tool suction cups.
325 - Lid Pref./DARFRESH: Upper tool lowering advance [mm]	The advance position for the upper tool lowering.
326 - Lid Pref./DARFRESH: Vacuum+forming opening delay (upper tool) [s]	The delay time for "vacuum+forming" tool opening.
327 - Lid Pref./DARFRESH: Plate lids ready timeout [s]	the maximum time allowed to the sensor for detecting correct operation of the lids plate.

328 - Lid Pref./DARFRESH: Plate vacuum off delay [s]	The delay time for plate vacuum closing.
329 - Lid Pref./(Super Protr.)/DARFRESH: Release (forward) deceleration [%]	Deceleration of the preformed lid disc for releasing the lids inside the tool.
330 - Doser 1: Position along the chain [steps]	the number of chain steps defining the position of doser 1.
331 - Doser 1: Start delay [s]	the delay time it takes for doser 1 to start an operating cycle.
332 - Doser 2: Position along the chain [steps]	the number of chain steps defining the position of doser 2.
333 - Doser 2: Start delay [s]	the delay time it takes for doser 2 to start an operating cycle.
334 - Doser 3: Position along the chain [steps]	the number of chain steps defining the position of doser 3.
335 - Doser 3: Start delay [s]	the delay time it takes for doser 3 to start an operating cycle.
336 - Doser 4: Position along the chain [steps]	the number of chain steps defining the position of doser 4.
337 - Doser 4: Start delay [s]	the delay time it takes for doser 4 to start an operating cycle.
338 - Doser 5: Position along the chain [steps]	the number of chain steps defining the position of doser 5.
339 - Doser 5: Start delay [s]	the delay time it takes for doser 5 to start an operating cycle.
340 - MHead 1 - Carriage: Movement start delay [s]	the delay time it takes for the mobile head carriage to start an operating cycle.
341 - MHead 1: Position along the chain [steps]	the number of chain steps defining the position of mobile head.
342 - MHead 1: Dosing start delay [s]	the delay time it takes for the mobile head dosing cycle to start.
343 - MHead 1: Dosing start phase [°]	the step in the mobile head cycle at which the mobile head starts dosing.
344 - MHead 1: Dosing time [s]	The dosing time for the mobile head 1.
345 - MHead 1 - Hopper: Product call delay [s]	The delay time for the product call start in the hopper.
346 - MHead 1 - Carriage: Start position [°]	the space in which the mobile head synchronises its speed with the chain speed.
347 - MHead 1 - Carriage: Pursuing acceleration [%]	acceleration and deceleration of the mobile head to synchronise its speed with the chain speed.
348 - MHead 1 - Carriage: Pursuing stroke [°]	the space in which the mobile head tracks the chain.
349 - MHead 1 - Hopper: Hopper full delay [s]	the delay time it takes for the mobile head to start tracking the chain.
350 - MHead 1 - Carriage: Return start delay [s]	the delay time it takes for the mobile head to start its movement backwards.

351 - MHead 1 - Carriage: Returning speed [%]	the return speed for the mobile head.
352 - MHead 1 - Carriage: Returning acceleration [%]	acceleration of the mobile head during its backward movement.
353 - MHead 1 - Carriage: Returning deceleration [%]	deceleration of the mobile head during its backward movement.
354 - MHead 1 - Carriage: Gearing ratio	the gearing ratio of the chain speed to the mobile head speed.
355 - MHead 1 - Hopper: Fillling timeout [s]	the maximum time it takes for the sensor to detect correct filling of the hopper, beyond which the machine generates an alarm.
356 - MHead 1 - Hopper: Start delay [s]	the delay time it takes the mobile head to start an operating cycle.
357 - MHead 1 - Hopper: Down position time [s]	the time during which the mobile head remains in the down position.
358 - MHead 1 - Hopper: Timeout [s]	the maximum time it takes for the sensor to detect correct operation of the mobile head, beyond which the machine generates an alarm.
359 - MHead 1 - Carriage: Pursuing deceleration [%]	The carriage deceleration during the pursuing movement.
360 - TraysStock: Buffer belt speed [%]	the operating speed of the buffer belt.
361 - TraysStock: Number of trays to load [nr]	the number of trays to load at each cycle.
362 - TraysStock: Batches ready on the buffer belt delay [s]	the delay time it takes for the system to open the tray batch stopper.
363 - TraysStock: Batches stopper closing delay [s]	the delay time it takes the system to cover the tray batch stopper.
364 - TraysStock: Buffer belt backward movement time [s]	the time for the buffer belt backward movement.
365 - TraysStock: -Batch on deliver position- safe time [s]	the safe time for the batch on deliver position.
366 - TraysStock: Nr of cycle before moving back the Pusher [nr]	the number of waiting cycles before the batch pusher retracts.
367 - TraysStock: Pusher speed [%]	the operating speed of the batch pusher.
368 - TraysStock: Pusher backward position [mm]	the retracted position of the batch pusher.
369 - TraysStock: Pusher forward position [mm]	the forward position of the batch pusher.
370 - TraysStock:	
371 - TraysStock: Belt backward positioning stop delay [s]	the delay time for the belt stop during the backward movement.
372 - TraysStock:	
373 - TraysStock:	
374 - TraysStock:	

375 - MHead 2 - Carriage: Movement start delay [s]	the delay time it takes for mobile head carriage to start an operating cycle.
376 - MHead 2: Position along the chain [steps]	the number of chain steps defining the position of the mobile head.
377 - MHead 2: Dosing start delay [s]	the delay time it takes for the moving head to start a dosing cycle.
378 - MHead 2: Dosing start phase [°]	the step in the mobile head cycle where the mobile head starts dosing.
379 - MHead 2: Dosing time [s]	The dosing time for the mobile head 2.
380 - MHead 2 - Hopper: Product call delay [s]	The delay time for the product call start in the hopper.
381 - MHead 2 - Carriage: Sinchronization space [mm]	the space in which the mobile head synchronises its speed with the chain speed.
382 - MHead 2 - Carriage: Pursuing acc/dec [%]	acceleration and deceleration of the mobile head to synchronise its speed with the chain speed.
383 - RESERVED	
384 - MHead 2 - Carriage: Pursuing stroke [mm]	the space in which the mobile head tracks the chain.
385 - MHead 2 - Hopper: Hopper full delay [s]	the delay time it takes for the mobile head to start tracking the chain.
386 - MHead 2 - Carriage: Return start delay [s]	the delay time it takes for the mobile head to start its movement backwards.
387 - MHead 2 - Carriage: Returning speed [%]	the return speed for the mobile head.
388 - MHead 2 - Carriage: Returning acceleration [%]	acceleration of the mobile head during its backward movement.
389 - MHead 2 - Carriage: Returning deceleration [%]	deceleration of the mobile head during its backward movement.
390 - MHead 2 - Carriage: Gearing ratio	the gearing ratio of the chain speed to the mobile head speed.
391 - MHead 2 - Hopper: Fillling timeout [s]	the maximum time it takes for the sensor to detect correct filling of the hopper, beyond which the machine generates an alarm.
392 - MHead 2 - Hopper: Start delay [s]	the delay time it takes the mobile head to start an operating cycle.
393 - MHead 2 - Hopper: Down position time [s]	the time during which the mobile head remains in the down position.
394 - MHead 2 - Hopper: Timeout [s]	it is the maximum reading time of the sensor that notices the correct operation of the mobile head over which the machine enters alarm.
395 - MHead 2:	
396 - Line - DP Pacer: Downstream (exit) piston opening delay [s]	the delay time it takes for the aligner piston 1 to open.
397 - Line - DP Pacer: Downstream (exit) piston closing delay [s]	the delay time it takes for the aligner piston 1 to close.

398 - Line - DP Pacer: Upstream (inlet) piston opening delay [s]	the delay time it takes for the aligner piston 2 to open.
399 - Line - DP Pacer: Upstream (inlet) piston closing delay [s]	the delay time it takes for the aligner piston 2 to close.
400 - Line - DP Pacer: Inlet tray sensor timeout [s]	the maximum time for the sensor to detect correct operation of the aligner, beyond which the machine generates an alarm.
401 - Line - DP Pacer: Trays per batch	the number of trays controlling the operating cycle of the aligner.
402 - Line - DP Pacer: Chain step offset	the displacement of the chain at each step.
403 - Line - DP Pacer: Delivery cam pulse duration [s]	It's the time for the cam pulse window for the charge from the pacer to the line.
404 - Line - Chain load belts: Belt speed [%]	The belt speed operation.
405 - Line - DP Pacer: Feed buffer belt speed [%]	The feed buffer belt speed operation.
406 - Super Protruding: Temperature set point [°C]	the tool operating temperature during a "hi protruding" cycle.
407 - Super Protruding: Temperature max allowed deviation [°C]	maximum deviation from the set point temperature during a "hi protruding" cycle, beyond which the machine generates an alarm.
408 - Super Protruding: Heating 1 offset [°C]	temperature deviation from the value read by the sensor.
409 - Super Protruding: Heating 2 offset [°C]	temperature deviation from the value read by the sensor.
410 - DARFRESH: 'Tool lid OK' vacuum threshold [mBar]	It is the depression threshold for the upper tool to ensure that the portions of film are properly positioned in the tool during the "DARFRESH" cycle.
411 - Lid Pref./(Super Protr.)/DARFRESH: 'Format change' position [mm]	The position to the format change.
412 - Super Protruding: Film heating time [s]	the time it takes the sealing film to preheat during a "hi protruding" cycle.
413 - Super Protruding: Film blowing time [s]	the operating time of the film blower during a "hi protruding" cycle.
414 - DARFRESH: Lid cut time [s]	The time for the lid cutting.
415 - Super Protruding: Dome formation delay [s]	the delay time for the film in the tool cavities to start a forming cycle during a "hi protruding" cycle.
416 - Super Protruding: Dome formation time [s]	the time it takes for the film in the cavities to form during a "hi protruding" cycle.
417 - Super Protruding: -No crinkle- vacuum time [s]	the time it takes for the vacuum valves to open during a "hi protruding" cycle.

418 - Super Protruding: Upper vacuum delay [s]	the delay time it takes for the upper tool vacuum valves to open during a "hi protruding" cycle.
419 - Super Protruding: Upper vacuum time [s]	the time it takes the upper tool vacuum valve to open during a "hi protruding" cycle.
420 - Super Protruding: Lower vacuum delay [s]	the delay time it takes for the lower tool vacuum valves to open during a "hi protruding" cycle.
421 - Super Protruding: Lower vacuum time [s]	the time it takes for the lower tool vacuum valves to open during a "hi protruding" cycle.
422 - Super Protruding: Dome formation compensation time [s]	the time it takes for the compensation valves of the film forming cavities to open during a "hi protruding" cycle.
423 - Super Protruding: Upper compensation time [s]	the time it takes for the upper tool compensation valves to open during a "hi protruding" cycle.
424 - Super Protruding: Lower vacuum delay while tool opening [s]	the delay time it takes for the lower tool vacuum valves to open during tool opening.
425 - Super Protruding: Lower vacuum time while tool opening [s]	the time it takes the lower tool vacuum valves to open during tool opening.
426 - Super Protruding:	
427 - Deviator - Belt 2->1: Speed setpoint [%]	the operating speed of the deviator belt 2→1.
428 - Deviator - Belt 2->1: RIGHT piston opening delay [s]	The delay time for the opening of the right piston on the deviator belt 2→1.
429 - Deviator - Belt 2->1: LEFT piston opening delay [s]	The delay time for the opening of the right piston on the deviator belt 2→1.
430 -	
431 - ExitBelt 4->2: Speed setpoint [%]	the operating speed of the exit belt 4→2.
432 - ExitBelt 4->2: Tray evacuation time [s]	the trays evacuation time of the exit belt.
433 -	
434 -	
435 - Snap on Lid - Chain: Home position [°]	The beginning cycle position of the snap on lid chain.
436 - Snap on Lid - Chain: Step length [°]	The length step of the snap on lid chain.
437 - Snap on Lid - Chain: Speed setpoint [%]	The snap on lid chain speed.
438 - Snap on Lid - Chain: Acceleration [%]	The snap on lid chain acceleration.
439 - Snap on Lid - Chain: Deceleration [%]	The snap on lid chain deceleration.
440 - Snap on Lid - Chain: Steps per cycle [nr]	The number of the steps per cycle.

441 - Snap on Lid - Chain: Advance on chain stop to move up the stopper [°]	The advance position for the snap on lid chain stop to move up the stopper.
442 - Snap on Lid - Chain: Trays stopper up time [s]	The time for the trays stopper up.
443 - Snap on Lid - Chain:	
444 - Snap on Lid - Chain:	
445 - DARFRESH: Forming compensation delay [s]	The delay time for the forming compensation valve opening during the DAR cycle.
446 - DARFRESH: Forming compensation time [s]	The time for the forming compensation valve opening during the DAR cycle.
447 - DARFRESH: Lower vacuum off delay [s]	The delay time for the lower vacuum valve closing.
448 - DARFRESH: Forming vacuum off delay [s]	The delay time for the forming vacuum valve closing.
449 - DARFRESH: Skin Air blowing delay [s]	The delay time for the Skin Air blowing start.
450 - DARFRESH: Skin Air blowing time [s]	duration of the Skin Air blowing.
451 - DARFRESH: Forming (Skin) vacuum 'pause' advance on vacuum position [mm]	It is the advance on the vacuum position of the tool for the temporary closure of the vacuum skin valve.
452 - DARFRESH: Forming (Skin) vacuum 'pause' time while opening lower vacuum [s]	duration of the closure of the vacuum skin valve.
453 - SpacerBelt - 1st step: Step length [mm]	The first step length of the spacer belt.
454 - SpacerBelt - 1st step: Start delay [s]	The first step start delay of the spacer belt.
455 - SpacerBelt - 1st step: Speed [%]	The first step speed of the spacer belt.
456 - SpacerBelt - 1st step: Acceleration [%]	The first step acceleration of the spacer belt.
457 - SpacerBelt - 1st step: Deceleration [%]	The first step deceleration of the spacer belt.
458 - SpacerBelt - 1st step: Step length when the chain accelerating [mm]	The first step length of the spacer belt when the chain of the line is accelerating.
459 - SpacerBelt - 1st step: Step length when the chain decelerating [mm]	The first step length of the spacer belt when the chain of the line is decelerating.
460 - SpacerBelt: Pusher start position advance on last tray from stopping chain [mm]	The start position for the pusher arms start when the chain of the line is stopping.
461 - Film reel unwind: Speed setpoint at reel emptying [%]	The reel emptying speed in automatic mode.
462 - Film reel unwind: Puncher work time [s]	The time for the puncher cycle.

463 - Film reel unwind: Puncher down delay [s]	The delay time for the puncher down motion.
464 - Film reel unwind:	
465 - Denester 1: Batch stopper opening position [°]	The opening position of batch trays stopper.
466 - Denester 1: Lower batch stopper open time [s]	The time the lower batch trays stopper remain open.
467 - Denester 1: Blower start position [°]	the blower start position of the chain or belt.
468 - Denester 1: Batch stopper cycle restart delay [s]	The delay time for the batch trays stopper restart.
469 - Denester 1: Lower batch stopper opening delay [s]	The delay time for the lower batch stopper opening.
470 - Denester 1: Upper batch stopper opening delay [s]	The delay time for the upper batch stopper opening.
471 - Denester 1: Lower selector close delay [s]	The delay time for the lower batch stopper closing.
472 - Denester 1: Obstructed timeout [s]	
473 - Denester 2: Batch stopper opening position [°]	The opening position of batch trays stopper.
474 - Denester 2: Batch stopper open time [s]	The time the lower batch trays stopper remain open.
475 - Denester 2: Blower start position [°]	the blower start position of the chain or belt.
476 - Denester 2: Batch stopper cycle restart delay [s]	The delay time for the batch trays stopper restart.
477 - Denester 2: Lower batch stopper opening delay [s]	The delay time for the lower batch stopper opening.
478 - Denester 2: Upper batch stopper opening delay [s]	The delay time for the upper batch stopper opening.
479 - Denester 2: Lower selector close delay [s]	The delay time for the lower batch stopper closing.
480 - Denester 2: Obstructed timeout [s]	
481 - Doser 1 - Pacer: In piston Close delay [s]	The delay time for the closing of IN piston.
482 - Doser 1 - Pacer: In Piston Open delay [s]	The delay time for the opening of IN piston.
483 - Doser 1 - Pacer: Out Piston Close delay [s]	The delay time for the closing of OUT piston.
484 - Doser 1 - Pacer: Out Piston Open delay [s]	The delay time for the opening of OUT piston.
485 - Doser 1: Hoppers down time [s]	The time for the hoppers in down position.
486 - Doser 1: Hoppers flap opening delay [s]	The delay time for the hoppers flap opening.
487 - Doser 1: Deviator switch delay [s]	The delay time for the switch of deviator.

488 - Doser 1: Doser to Hopper product drop time [s]	The time of fall produced by the doser to the hopper.
489 - Doser 1: Hoppers flap opening time [s]	The time of hopper discharge.
490 - Doser 1: Product call delay [s]	The delay time for the product call start.
491 - Doser 1: Product waiting (from Ishida) timeout [s]	The maximum time of attended of the Ishida doser.
492 - Doser 1: Hopper safe hight delay [s]	The delay time for hopper in safe high position.
493 - Doser 1:	
494 - Doser 6: Position along the chain [steps]	the number of chain steps defining the position of doser 6.
495 - Doser 6: Start delay [s]	the delay time it takes for doser 6 to start an operating cycle.
496 - DARFRESH: Still gripper open delay when cutting [s]	is the delay time that regulates the opening of the fixed gripper for blocking of the film during the cutting operation of the film for the "DARFRESH" cycle.
497 - Super Protruding: Upper tool descent delay [s]	The delay time regulating start of upper tool descent.
498 - Super Protruding: Upper tool lift delay [s]	The delay time regulating start of upper tool lift.
499 - Super Protruding: Tool Border vacuum delay [s]	The delay time regulating the start of the tool border vacuum cycle.
500 - Super Protruding: Shuttle vacuum (heating) delay [s]	The delay time regulating the start of the shuttle vacuum cycle.
501 - Super Protruding: Plate vacuum off time to avoid film crinkle [s]	is the pause time of the vacuum on the shuttle to allow the crinkles on the film just cut to stretch.
502 - DARFRESH: Shuttle plates blow delay [s]	The delay time for the shuttle plates blow start.
503 - DARFRESH: Shuttle plates blow time [s]	The time for the shuttle plates blow start.
504 - Super Protruding: Dome forming vacuum OFF delay [s]	The delay time for the dome forming vacuum stop.
505 - Super Protruding: Dome forming vacuum RECALL delay [s]	The delay time for the dome forming vacuum recall start.
506 - Super Protruding: Dome forming vacuum RECALL time [s]	The time for the dome forming vacuum recall cycle.
507 - Protruding: Lower vacuum valve LEFT delay [s]	is the delay time for the opening of the left lower vacuum valve respect to the right lower vacuum valve.
508 - Super Protruding:	
509 - Protruding: Film heating time on the skin pre-heating plates [s]	duration of the film heating time on the skin pre-heating plates.

510 - Protruding: Film blowing time far from skin pre-heating plates [s]	duration of the far from skin pre-heating plates.
511 - RESERVED	
512 - Tool Axis: Home position [mm]	The sealing tool cycle start position.
513 - Tool Axis: Safe Quote [mm]	The tool safety position.
514 - Tool Axis: Tray Catch position [mm]	The tool tray pick-up position.
515 - Tool Axis: Tray Catch speed [%]	The tool positioning speed to reach the tray pick-up position.
516 - Tool Axis: Tray Catch final speed [%]	The final tool positioning speed to reach the tray pick-up position.
517 - Tool Axis: Tray Catch Acc./Dec. [%]	Tool positioning acceleration/ deceleration to reach the tray pick-up position.
518 - Tool Axis: Tray Check start position [mm]	The tool start position for receiving trays.
519 - Tool Axis: Tray Check start speed [%]	The tool positioning speed to reach the tool start position for receiving trays.
520 - Tool Axis: Tray Check start final speed [%]	The final tool positioning speed to reach the tool start position for receiving trays.
521 - Tool Axis: Tray Check start Acc./Dec. [%]	Tool positioning acceleration/ deceleration to reach the tray start position.
522 - Tool Axis: Tray Check end position [mm]	The tool end position for receiving trays.
523 - Tool Axis: Tray Check end speed [%]	The tool positioning speed to reach the tool end position for receiving trays.
524 - Tool Axis: Tray Check end final speed [%]	The final tool positioning speed to reach the tool end position for receiving trays.
525 - Tool Axis: Tray Check end Acc./Dec. [%]	The tool positioning acceleration/ deceleration to reach the end tray position.
526 - Tool Axis: Closing position [mm]	The sealing tool closing position.
527 - Tool Axis: Closing speed [%]	The tool positioning speed to reach the tool closing position.
528 - Tool Axis: Closing final speed [%]	The final tool positioning speed to reach the tool closing position.
529 - Tool Axis: Closing Acc./Dec. [%]	The tool positioning acceleration/ deceleration to reach the closing position.
530 - Tool Axis: Vacuum On Advance [mm]	The sealing tool advance position to run a vacuum/gas cycle.
531 - Tool Axis: Vacuum position [mm]	The sealing tool position to run a vacuum/gas cycle.

532 - Tool Axis: Vacuum/gas speed [%]	The tool positioning speed to reach the vacuum/gas cycle run position.
533 - Tool Axis: Vacuum/gas final speed [%]	The final tool positioning speed to reach the vacuum/gas cycle run position.
534 - Tool Axis: Vacuum/gas Acc./Dec. [%]	Tool positioning acceleration/ deceleration to reach the vacuum/gas position.
535 - Tool Axis: Sealing position [mm]	The tool position for running a sealing cycle.
536 - Tool Axis: Sealing speed [%]	The tool positioning speed to reach the tray sealing position.
537 - Tool Axis: Sealing Acc./Dec. [%]	Tool positioning acceleration/ deceleration to reach the sealing position.
538 - Tool Axis: Opening position [mm]	The sealing tool opening position.
539 - Tool Axis: Opening speed [%]	The tool positioning speed to reach the tool opening position.
540 - Tool Axis: Opening final speed [%]	The final tool positioning speed to reach the tool opening position.
541 - Tool Axis: Opening Acc./Dec. [%]	Tool positioning acceleration/ deceleration to reach the opening position.
542 - Tool Axis: Film unwind Consent [mm]	The tool position enabling the start of a lid unwinder cycle.
543 - Tool Axis: Adjust position [mm]	The sealing tool pause position during tool descent.
544 - Tool Axis: Adjust speed [%]	The tool positioning speed to reach the pause position during descent.
545 - Tool Axis: Adjust final speed [%]	The final tool positioning speed to reach the pause position during descent.
546 - Tool Axis: Adjust Acc./Dec. [%]	Tool positioning acceleration/ deceleration to reach the pause position during descent.
547 - Tool Axis: Approach position [mm]	Tool approach to the lowest position in a sealing cycle.
548 - Tool Axis: Approach speed [%]	The tool positioning speed to reach the lowest position in a sealing cycle.
549 - Tool Axis: Approach final speed [%]	The final tool positioning speed to reach the lowest position in a sealing cycle.
550 - Tool Axis: Approach Acc./Dec. [%]	Tool positioning acceleration/ deceleration to reach the lowest position in a sealing cycle.
551 - Tool Axis: Low position [mm]	The tool low position during a sealing cycle.

552 - Tool Axis: Low position speed [%]	The tool positioning speed to reach the lowest position in a sealing cycle.
553 - Tool Axis: Low position Acc./Dec. [%]	Tool positioning acceleration/ deceleration to reach the lowest position in a sealing cycle.
554 - Tool Axis: Blade ON position [mm]	The tool position for the blade on operation.
555 - Tool Axis: Blade ON time [s]	The time for the blade on operation.
556 - Tool Axis: Blade ON delay [s]	The delay time for the blade on operation.
557 - Tool Axis: Stroke for DARFRESH needle taking out [mm]	The stroke for needles taking out during DARFRESH cycle.
558 - Tool Axis: Open/Close time for G tool pistons [s]	The open/close time of the tool pistons during the G cycle.
559 - Tool Axis - HYBRIC: Tray catch position [mm]	the tool position for the tray catch at the release of the pushers arms (for HYBRIK cycle only).
560 - Tool Axis: Film preparation consent position [mm]	the tool position for the film preparation consent.
561 -	
562 -	
563 -	
564 -	
565 -	
566 -	
567 -	
568 -	
569 -	
570 -	
571 -	
572 -	
573 -	
574 -	
575 -	
576 -	
577 -	
578 -	
579 -	
580 -	
581 -	
582 -	
583 -	
584 -	
585 -	
586 -	
587 -	
588 -	
589 -	
590 -	

591 -	
592 -	
593 -	
594 -	
595 -	
596 -	
597 -	
598 -	
599 -	
600 - Snap on Lid - Slide: Catch time [s]	The time in catch position of the trays of the snap on lid piston.
601 - Snap on Lid - Slide: Release time [s]	The time in release position of the trays of the snap on lid piston.
602 - Snap on Lid - Slide: Catch 1 (backward) position [mm]	The piston position for the catch of the first tray.
603 - Snap on Lid - Slide: Catch 1 (backward) speed [%]	The piston speed for the catch of the first tray.
604 - Snap on Lid - Slide: Catch 1 (backward) acc./dec. [%]	The piston acceleration/deceleration for the catch of the first tray.
605 - Snap on Lid - Slide: Catch 2 position [mm]	The piston position for the catch of the second tray.
606 - Snap on Lid - Slide: Catch 2 speed [%]	The piston speed for the catch of the second tray.
607 - Snap on Lid - Slide: Catch 2 acc./dec. [%]	The piston acceleration/deceleration for the catch of the second tray.
608 - Snap on Lid - Slide: Release (forward) position [mm]	The piston position for the release of the trays.
609 - Snap on Lid - Slide: Release (forward) speed [%]	The piston speed for the release of the trays.
610 - Snap on Lid - Slide: Release (forward) acc./dec. [%]	The piston acceleration/deceleration for the release of the trays.
611 - Snap on Lid - Slide: SAFE position [mm]	The safe position of the snap on lid tool.
612 - Snap on Lid - Slide: Cups cleaning time [s]	The time for the cups cleaning cycle.
613 - Snap on Lid - Tool: Closing time [s]	The closing time of the snap on lid tool.
614 - Snap on Lid - Tool: Vacuum off delay [s]	The delay time for the vacuum valves closing.
615 - Snap on Lid - Tool: Cups cleaning time [s]	The time for the cups cleaning cycle.
616 - Snap on Lid:	
617 - Snap on Lid:	
618 - Snap on Lid:	
619 - Snap on Lid:	
620 - Snap on Lid - In Belt: speed [%]	The speed of the snap on lid inlet belt.
621 - Snap on Lid - Pacer: Tray present at the pacer delay [s]	The reading time for tray recognition in front of the snap on lid pacer.

622 - Snap on Lid - Pacer: Photocell timeout [s]	The maximum reading time of the pacer start photocell beyond which the machine generates an alarm.
623 - Snap on Lid - Pacer: Close position [°]	The snap on lid pacer closing position.
624 - Snap on Lid - Pacer: Open position [°]	The snap on lid pacer opening position.
625 - Snap on Lid:	
626 - Snap on Lid:	
627 - Snap on Lid:	
628 - Snap on Lid:	
629 - Snap on Lid - Chain: Start delay [s]	The delay time for the snap on lid chain start.
630 - Snap on Lid:	
631 - Snap on Lid:	
632 - Snap on Lid - Tool: Home position [°]	The beginning cycle position of the snap on lid machine.
633 - Snap on Lid - Tool: Close position [°]	The snap on lid tool closing position.
634 - Snap on Lid - Tool: Close speed [%]	The closing speed on the snap on lid tool.
635 - Snap on Lid - Tool: Close acceleration [%]	The closing acceleration on the snap on lid tool.
636 - Snap on Lid - Tool: Close deceleration [%]	The closing deceleration on the snap on lid tool.
637 - Snap on Lid - Tool: Open position [°]	The snap on lid tool opening position.
638 - Snap on Lid - Tool: Open Speed [%]	The opening speed on the snap on lid tool.
639 - RESERVED	
640 - Snap on Lid - Tool: Open acceleration [%]	The opening acceleration on the snap on lid tool.
641 - Snap on Lid - Tool: Open deceleration [%]	The opening deceleration on the snap on lid tool.
642 - Snap on Lid:	
643 - Snap on Lid:	
644 - Snap on Lid:	
645 - Snap on Lid:	
646 - Snap on Lid:	
647 - Snap on Lid:	
648 - Denester 1: Advance low selector opening offset [mm]	The offset position for the low selector early opening on the denester 1.
649 - Denester 1: Batch stoppers opening/closing time [s]	The batch stoppers opening/closing time.
650 - Denester 1 - Deviator Flap: Start delay [s]	
651 - Denester 1 - Deviator Flap: Backward position [mm]	The left position of the deviator flap on the denester 1.

652 - Denester 1 - Deviator Flap: Forward position [mm]	The right position of the deviator flapon the denester 1.
653 - Denester 1 - Deviator Flap: Speed set point [%]	The speed of the deviator flap on the denester 1.
654 - Denester 1 - Deviator Flap: Acceleration [%]	The acceleration of the deviator flap on the denester 1.
655 - Denester 1 - Deviator Flap: Deceleration [%]	The deceleration of the deviator flap on the denester 1.
656 - Denester 2: Advance low selector opening offset [mm]	The offset position for the opening of the low selector.
657 - Denester 2: Batch stoppers opening/closing time [s]	The batch stoppers opening/closing time.
658 - Denester 2 - Deviator Flap: Start delay [s]	The delay time for the deviator flap start.
659 - Denester 2 - Deviator Flap: Backward position [mm]	The backward position of the deviator flap.
660 - Denester 2 - Deviator Flap: Forward position [mm]	The forward position of the deviator flap.
661 - Denester 2 - Deviator Flap: Speed set point [%]	The speed of the deviator flap.
662 - Denester 2 - Deviator Flap: Acceleration [%]	The acceleration of the deviator flap.
663 - Denester 2 - Deviator Flap: Deceleration [%]	The deceleration of the deviator flap.
664 -	
665 -	
666 -	
667 -	
668 -	
669 -	
670 - Meat Line: Delay start [s]	The delay time for the meat line start.
671 - Meat Line - Cuve belt: Set point speed [%]	The speed of the curve belt during the production cycle.
672 -	
673 - Meat Line - Curve belt: Full delay [s]	The delay time for the curve belt filling of trays.
674 - Meat Line - Curve belt: Empty delay [s]	The delay time for the curve belt emptying of trays.
675 - Meat Line - Trays belt: Set point speed [%]	The speed of the trays belt during the production cycle.
676 - Meat Line - Trays belt: Stopper open delay [s]	The delay time for the opening of the trays belt stopper.
677 - Meat Line - Trays belt: Stopper close delay [s]	The delay time for the closing of the trays belt stopper.
678 - Meat Line: Timeout tray presence photocell [s]	The reading time of the photocell for the tray recognition.
679 - Meat Line - Meat belt: Set point speed [%]	The speed of the meat belt during the production cycle.

680 - Meat Line - Meat belt: Frequency time [s]	The frequency time between a meat portion and the following one.
681 - Meat Line: Timeout meat presence photocell [s]	The reading time of the photocell for the meat portion recognition.
682 -	
683 -	
684 -	
685 -	
686 - Overturner - Rotation: Home position [°]	The position for the beginning of the rotation cycle.
687 - Overturner - Belts: BUSY delay [s]	The delay time for the “busy” condition of the belts.
688 - Overturner - Belts: FREE delay [s]	The delay time for the “free” condition of the belts.
689 - Overturner - Rotation: Speed [%]	The speed for the rotation cycle.
690 - Overturner - Rotation: Acceleration [%]	The acceleration for the rotation cycle.
691 - Overturner - Rotation: Deceleration [%]	The deceleration for the rotation cycle.
692 - Overturner - Belts: Speed [%]	The speed for the belts cycle.
693 - Overturner: Photocell timeout [s]	The maximum reading time of the sensor detecting the correct operation of the overturner.
694 - Overturner:	
695 - Overturner:	
696 - Overturner:	
697 - Overturner:	
698 - Shuttle: Mark search correction offset [mm]	It’s the correction offset position for the mark search.
699 - Shuttle: Backward stroke to pause position [mm]	The backward stroke to pause position.
700 - Shuttle: Speed of the back movement to pause position [%]	The backward movement speed.
701 - Shuttle: Tool down time at pause position [s]	The tool down time when in pause position.
702 - Shuttle - ZeroMeno: Film stretch backward stroke [mm]	The shuttle backward travel for film tensioning.
703 - Shuttle: Offset back for the film stretch [mm]	The shuttle forward travel to allow the moving grippers to clamp the film end piece.
704 - Shuttle: Film stretching speed [%]	The shuttle backward speed for film tensioning.
705 - Shuttle - ZeroMeno: Film grabing forward speed [%]	The shuttle forward speed to allow the moving grippers to clamp the film end piece.
706 - Tool: Cavities 11 temperature offset [°C]	The temperature difference between the set point and the value measured by the sensor of cavity 11.

707 - Tool: Cavities 12 temperature offset [°C]	The temperature difference between the set point and the value measured by the sensor of cavity 12.
708 -	
709 - Tool - Skin: Forming compensation BLOW start delay [s]	The delay time for the forming compensation blow start.
710 -	
711 - Tool: Select upper pressure transducer to trace 0=None / 1=Skin Rx / 2=Skin Lx / 3=Upper	Select upper pressure transducer to trace
712 - MHead 1 - Shutter: Start delay [s]	The delay time for the shutter opening on the mobile head 1.
713 - MHead 1 - Shutter: Open time [s]	The opening time of the shutter on the mobile head 1.
714 - MHead 1 - Shutter: Timeout [s]	The maximum reading time of the sensor detecting the shutter movement on the mobile head 1.
715 - MHead 1:	
716 - MHead 1:	
717 - MHead 1:	
718 - MHead 2 - Shutter: Start delay [s]	The delay time for the shutter opening on the mobile head 2.
719 - MHead 2 - Shutter: Open time [s]	The opening time of the shutter on the mobile head 2.
720 - MHead 2 - Shutter: Timeout [s]	The maximum reading time of the sensor detecting the shutter movement on the mobile head 2.
721 - MHead 2:	
722 - MHead 2:	
723 - MHead 2:	
724 - Vertical Store: Catch position (High) [mm]	The position for the of the trays from the store.
725 - Vertical Store: Release position (Low) [mm]	The position for the of the release of the trays on the line.
726 - Vertical Store: Speed [%]	The speed of the vertical store.
727 - Vertical Store: Acceleration [%]	The acceleration of the vertical store.
728 - Vertical Store: Deceleration [%]	The deceleration of the vertical store.
729 - Vertical Store:	
730 - Vertical Store:	
731 - Vertical Store:	
732 - Vertical Store:	
733 - Vertical Store:	
734 - Vertical Store:	
735 - Vertical Store:	
736 - Vertical Store:	
737 - Vertical Store:	

738 - Vertical Store: Home position [mm]	The beginning position of the vertical store cycle.
739 - Vertical Store: Approch position to the release [mm]	The approach position for the release of the trays on the line.
740 - Line - Tray Measure: Level zero [mm]	is the zero level for the measurement of the trays.
741 - Line - Tray Measure: Maximum Value [mm]	is the maximum level for the measurement of the trays.
742 -	
743 -	
744 -	
745 -	
746 -	
747 -	
748 - Micvac - Label dispenser: Label stop [mm]	This parameter refers to the label stop position.
749 - Micvac - Label dispenser: High speed [mm/s]	It's the carrying out speed of the film.
750 - Micvac - Label dispenser: Acceleration [%]	It's the carrying out acceleration of the film.
751 - Micvac - Linear Unit: Label start [mm]	It's the departure position of the MicVac labeller.
752 - Micvac - Linear Unit: Label space [mm]	it is the distance among the labels applied on the film.
753 - Micvac - Linear Unit: Start speed [mm/s]	It's the start speed of the MicVac labeller.
754 - Micvac - Linear Unit: Application speed [mm/s]	It's the MicVac labeller speed during the labels application movement.
755 - Micvac - Linear Unit: Return speed [mm/s]	It's the MicVac labeller speed during the return movement.
756 - Micvac - Linear Unit: Acceleration [%]	It's the MicVac labeller acceleration during the speeds change.
757 - Micvac - Linear Unit: Knife on time [ms]	It's the operation time of the film cutter pistons.
758 - Micvac - Linear Unit: Knife off time [ms]	It's the pause time of the film cutter pistons.
759 - Micvac - Linear Unit: Number of labels	It's the labels number applied on the film to every operation cycle.
760 - Micvac:	
761 - Micvac:	
762 - Micvac:	
763 - Micvac:	
764 -	
765 -	
766 -	
767 -	
768 -	
769 -	
770 -	

771 -	
772 -	
773 -	
774 -	
775 -	
776 -	
777 -	
778 -	
779 -	
780 -	
781 -	
782 -	
783 -	
784 -	
785 - Doser 1: Dosing start pulse length [s]	The dosing start pulse length.
786 - Doser 1: Dosing timeout [s]	The maximum reading time for the correct operation of the doser.
787 - Doser 2: Dosing start pulse length [s]	The dosing start pulse length.
788 - Doser 2: Dosing timeout [s]	The maximum reading time for the correct operation of the doser.
789 - Doser 3: Dosing start pulse length [s]	The dosing start pulse length.
790 - Doser 3: Dosing timeout [s]	The maximum reading time for the correct operation of the doser.
791 - Doser 4: Dosing start pulse length [s]	The dosing start pulse length.
792 - Doser 4: Dosing timeout [s]	The maximum reading time for the correct operation of the doser.
793 - Doser 5: Dosing start pulse length [s]	The dosing start pulse length.
794 - Doser 5: Dosing timeout [s]	The maximum reading time for the correct operation of the doser.
795 - Doser 6: Dosing start pulse length [s]	The dosing start pulse length.
796 - Doser 6: Dosing timeout [s]	The maximum reading time for the correct operation of the doser.
797 - Doser:	
798 - Doser:	
799 - Doser:	
800 - DARFRESH: Heating plate 1 offset [°C]	The operating temperature of heating plate in the cutting area.
801 - DARFRESH: Heating plate 2 offset [°C]	The operating temperature of heating plate in the cutting area.
802 - DARFRESH: Heating plate 3 offset [°C]	The operating temperature of heating plate in the cutting area.
803 - DARFRESH: Heating plate 4 offset [°C]	The operating temperature of heating plate in the cutting area.

804 - DARFRESH: Heating plate 5 offset [°C]	The operating temperature of heating plate in the cutting area.
805 - DARFRESH: Heating plate 6 offset [°C]	The operating temperature of heating plate in the cutting area.
806 - DARFRESH: Heating plate 7 offset [°C]	The operating temperature of heating plate in the cutting area.
807 - DARFRESH: Heating plate 8 offset [°C]	The operating temperature of heating plate in the cutting area.
808 - DARFRESH: Heating plate 9 offset [°C]	The operating temperature of heating plate in the cutting area.
809 - DARFRESH: Heating plate 10 offset [°C]	The operating temperature of heating plate in the cutting area.
810 - DARFRESH: Heating plate 11 offset [°C]	The operating temperature of heating plate in the cutting area.
811 - DARFRESH: Heating plate 12 offset [°C]	The operating temperature of heating plate in the cutting area.
812 - DARFRESH: Heating plate 13 offset [°C]	The operating temperature of heating plate in the cutting area.
813 - DARFRESH: Heating plate 14 offset [°C]	The operating temperature of heating plate in the cutting area.
814 - DARFRESH: Heating plate 15 offset [°C]	The operating temperature of heating plate in the cutting area.
815 - DARFRESH: Heating plate 16 offset [°C]	The operating temperature of heating plate in the cutting area.
816 - DARFRESH: Heating plate 17 offset [°C]	The operating temperature of heating plate in the cutting area.
817 - DARFRESH: Heating plate 18 offset [°C]	The operating temperature of heating plate in the cutting area.
818 - DARFRESH: Heating plate 19 offset [°C]	The operating temperature of heating plate in the cutting area.
819 - DARFRESH: Heating plate 20 offset [°C]	The operating temperature of heating plate in the cutting area.
820 - DARFRESH: Heating plate temperature reached timeout [s]	The maximum time it takes the sensor to detect a heating plate temperature, beyond which a machine alarm is generated.
821 - DARFRESH: Heating plate cycle advance [s]	The advance time regulating heating plates cycle start.
822 - DARFRESH: Still gripper lowering position [mm]	The position in which the still gripper lowering.
823 - SidePlate:	
824 - SidePlate:	
825 - SidePlate:	
826 - SidePlate:	
827 - SidePlate:	
828 - SidePlate:	
829 - SidePlate:	
830 - SidePlate:	

831 - SpacerBelt: Last step length if belt moving [mm]	The length covered by spacer belt 1 to perform the last cycle step if belt moving
832 - SpacerBelt: Last step length when chain accelerating [mm]	The length covered by spacer belt 1 to perform the last cycle step when chain accelerating.
833 - SpacerBelt: Last step length when chain decelerating [mm]	The length covered by spacer belt 1 to perform the last cycle step when chain decelerating.
834 - SpacerBelt - 1st step: Start delay while line accelerating [s]	The length covered by spacer belt 1 to perform the last cycle step while line accelerating.
835 - SpacerBelt - 1st step: Start delay while line decelerating [s]	The length covered by spacer belt 1 to perform the last cycle step while line decelerating.
836 - SpacerBelt 1: Start delay while line decelerating [s]	The delay time regulating start of spacer belt 1 while line decelerating.

6 MACHINE OPERATION

INDEX

6.1 DAILY START UP OPERATIONS

Every day, prior to starting production, the operator must run a series of procedures, and specifically:

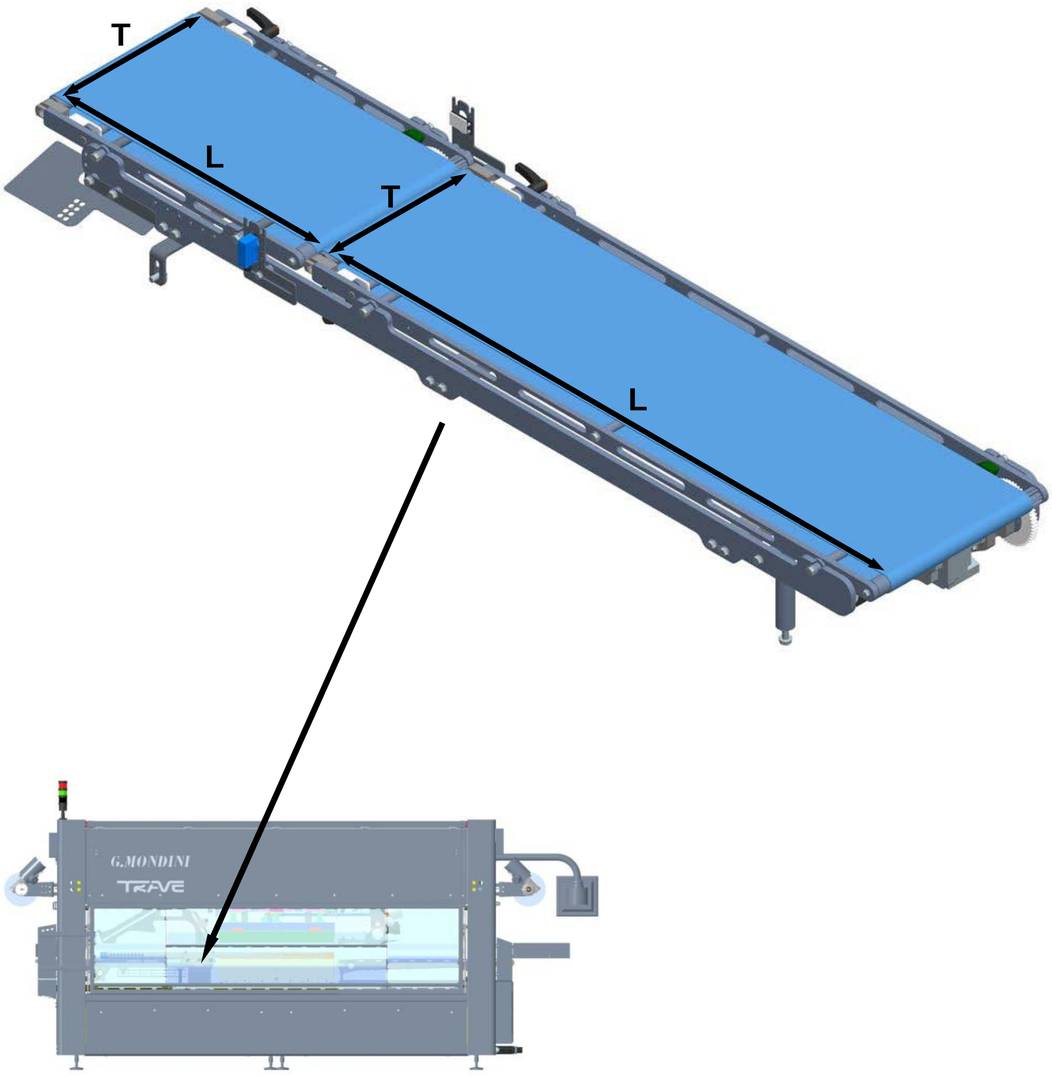
1. Inspection of the short infeed belt
2. Inspection of the tray detection photocell
3. Sealing film assembly operations
4. Check up on the work cycle recipe to be run
5. Check run of the tool assembled on the machine;
6. Checking to ensure that the machine is clean. A wash cycle must also be run.
7. Inspection of safety guards and doors to see that they are closed.

A description on how to proceed with each one of the above points is provided herein as follows:

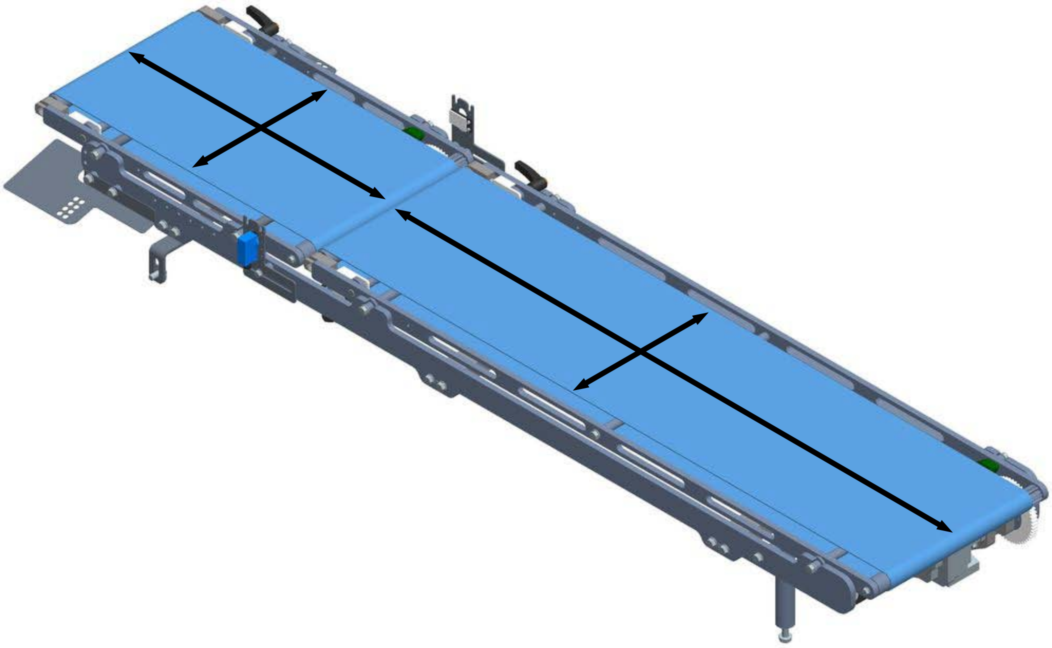
DANGER
Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

6.1.1 Inspection of the short infeed belt

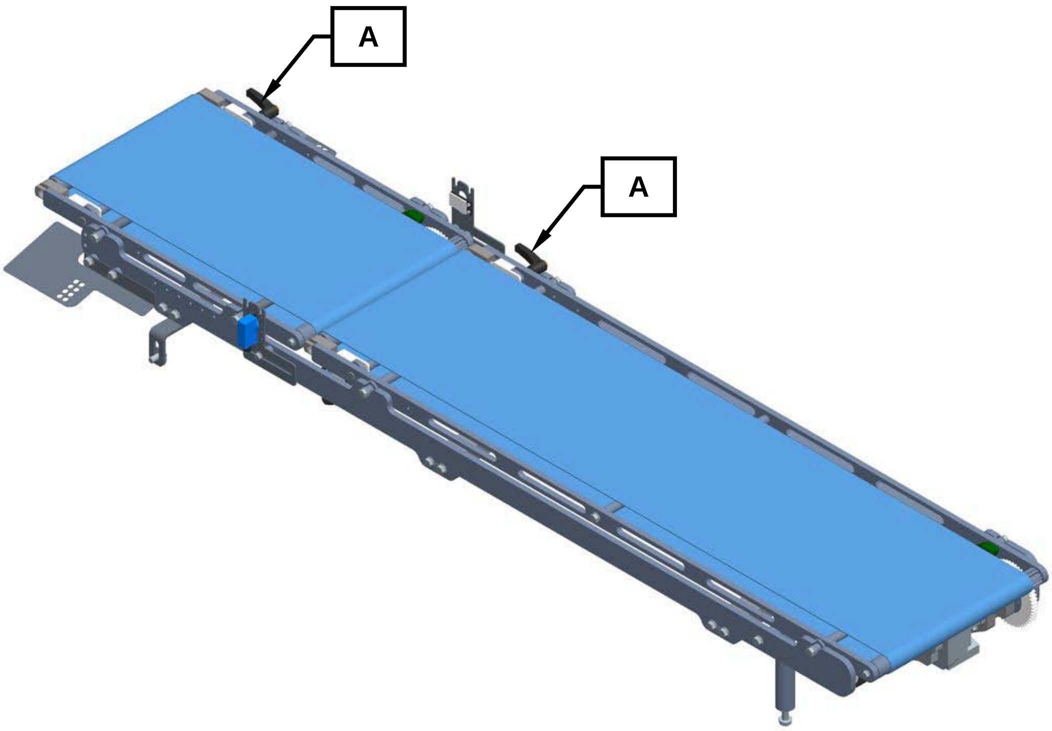
1. Check to ensure that the short infeed belt is assembled and centred correctly and that the tray travels both in longitudinal (**L**) and in transversal (**T**) directions.



If necessary, the operator must see that they are accurately aligned and centred by hand.



2. Use the appropriate lever (**A**) to lock the short infeed belts.

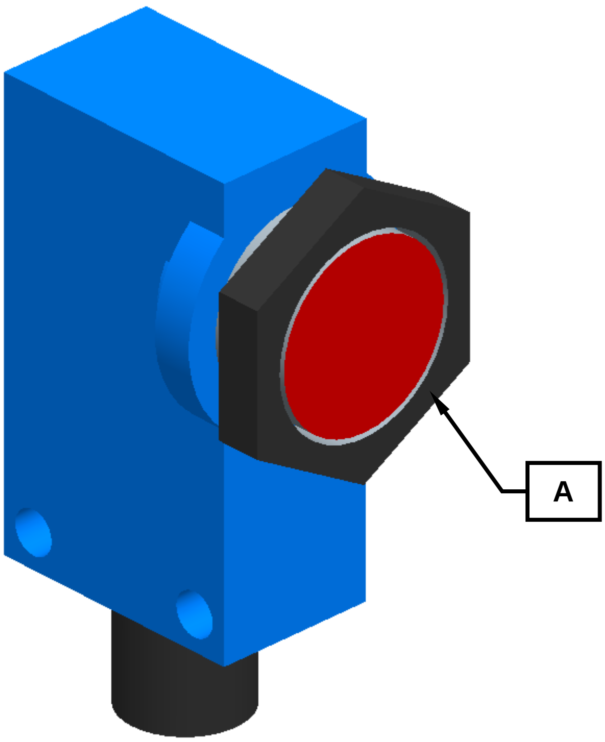


6.1.2 Inspection of the tray detection photocell

1. Remove the safety and protection covering from the photocell (A) checking to ensure that it is neither dirty nor wet (as a consequence of the washing cycle). If necessary, wipe it clean using a soft, dry cloth.

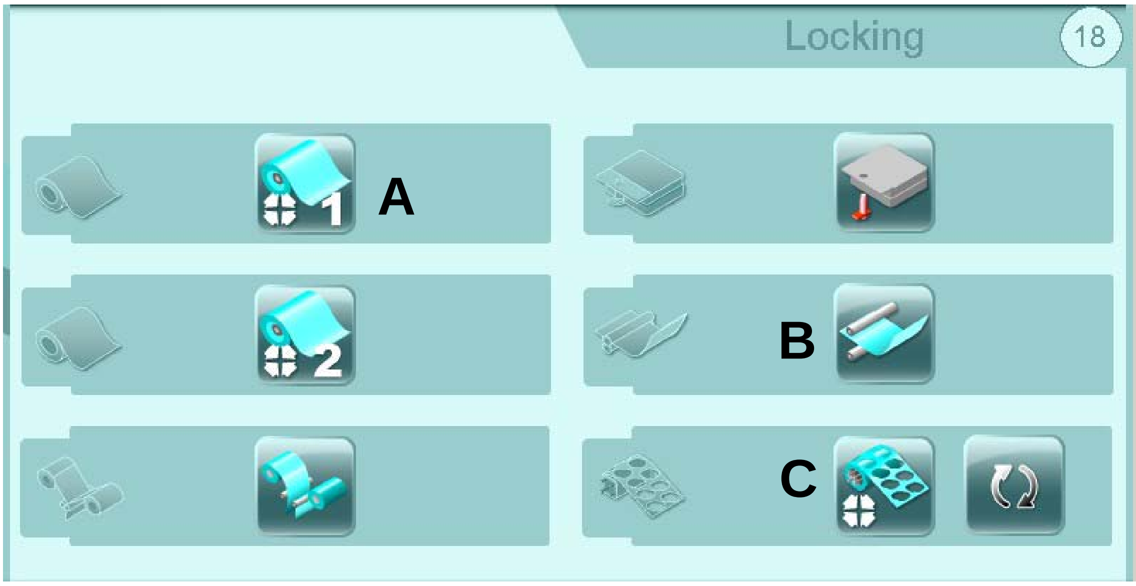
Important!

- NEVER wash the photocells, the reflectors and uncovered motors using high-pressure water jets.
- NEVER wash the photocells, the reflectors and uncovered motors using solvents or thinners. This because solvents or thinners contain chemical substances that will damage the plastic photocell eyes thereby compromising their operation.
- NEVER wash the photocells, the reflectors and uncovered motors using hot water as this will damage the electronic components. This is particularly important to observe after having powered off the line because the photocells are still hot.

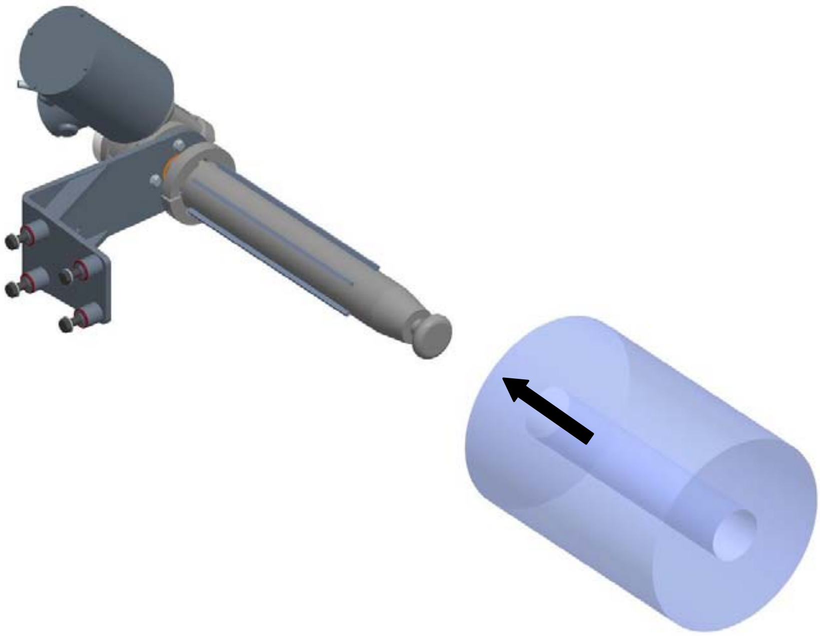


6.1.3 Sealing film assembly operations

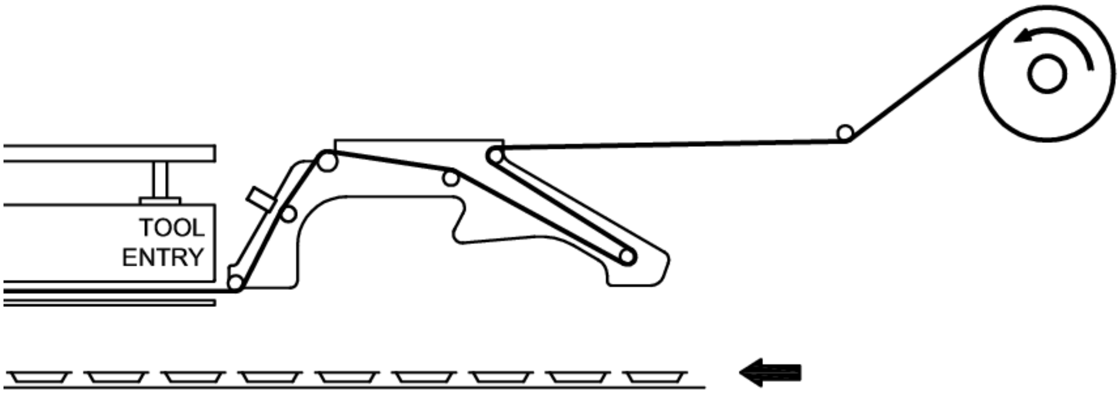
1. Check that the system locking functions on the Lid Unrolling, on the Waste Winding and on the Film Reel are all deactivated by acting on the relative selector keys (**A**), (**B**) and (**C**).

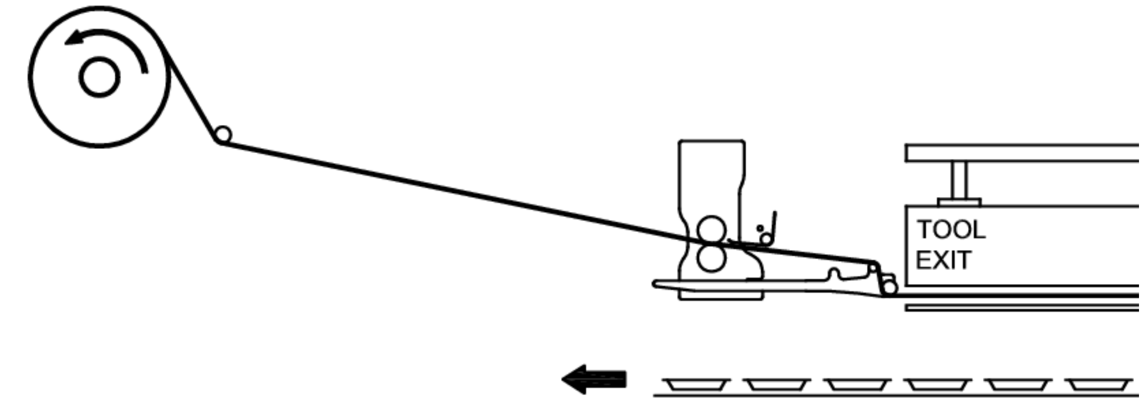


2. Fit on the sealing film reel.

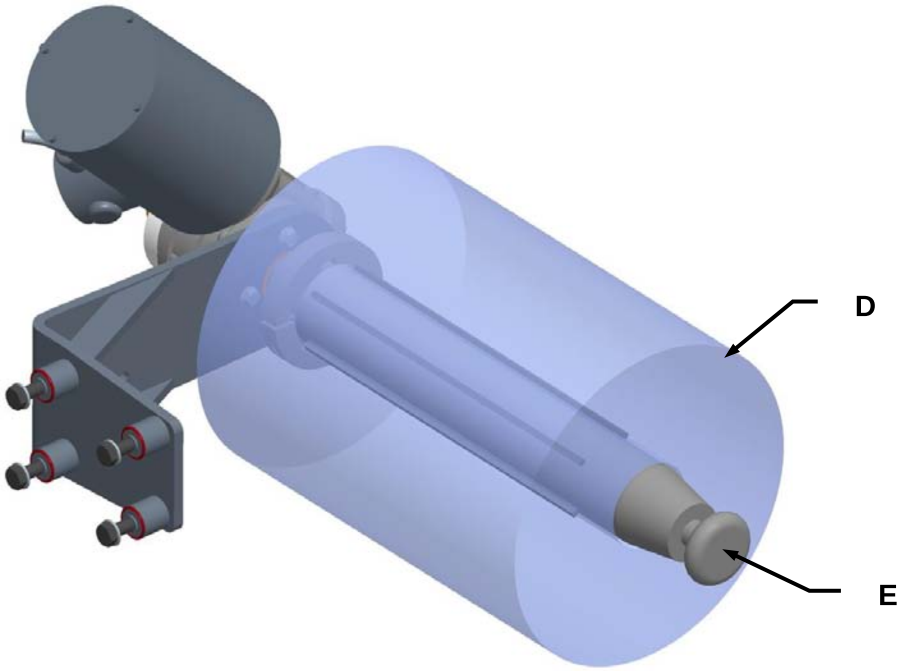


3. Thread the film in through the bobbin unwinder rollers as per the directions indicated by the arrows provided near each one of the rollers.

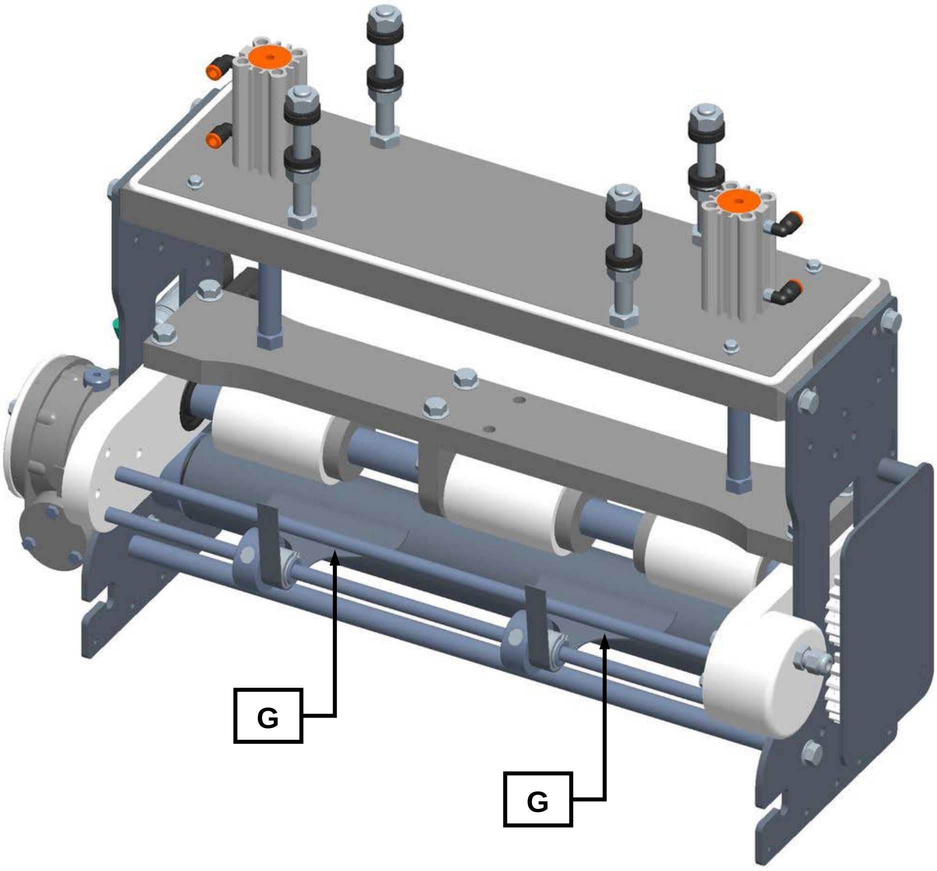




4. Check to ensure that the bobbin (**D**) is correctly centred on the film unwinder system shaft (**E**) and lock it in place using the appropriate button (**F**).



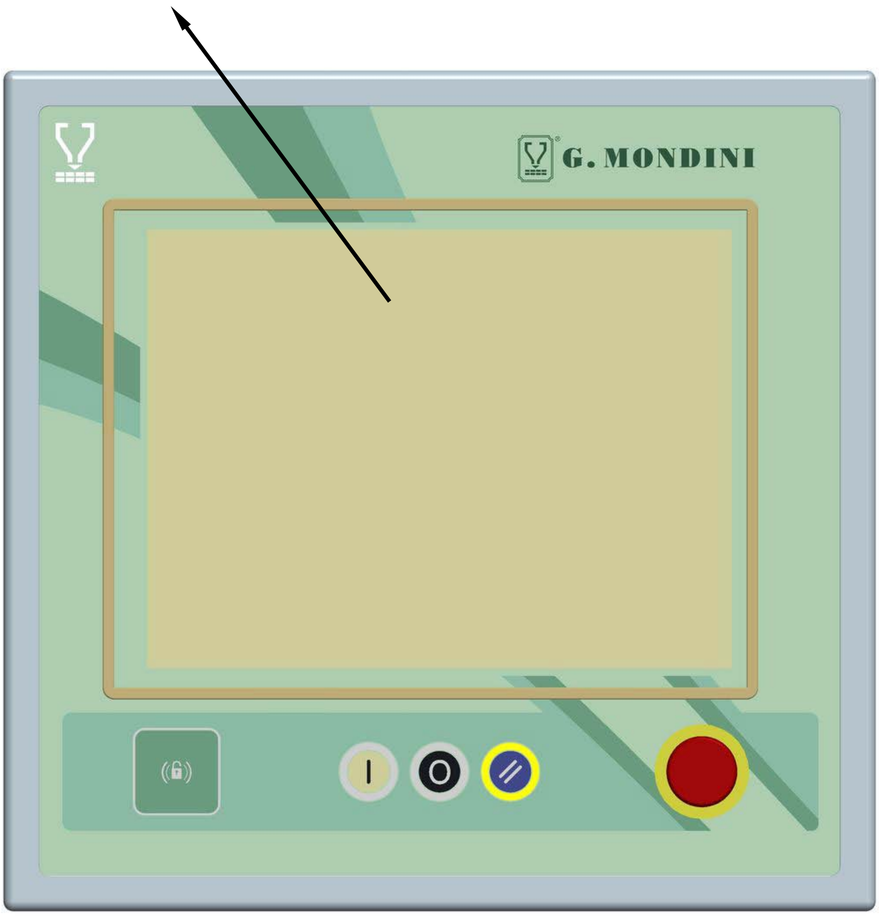
5. Check to see that the sealing film waste blade plates (**G**) are positioned on the relative outer edges. If not, move them so that they are positioned correctly as shown.



6. Push the button (**H**) to block the film unrolling pullers and wind the waste onto the waste winding shaft. To lock the waste push the button (**L**).

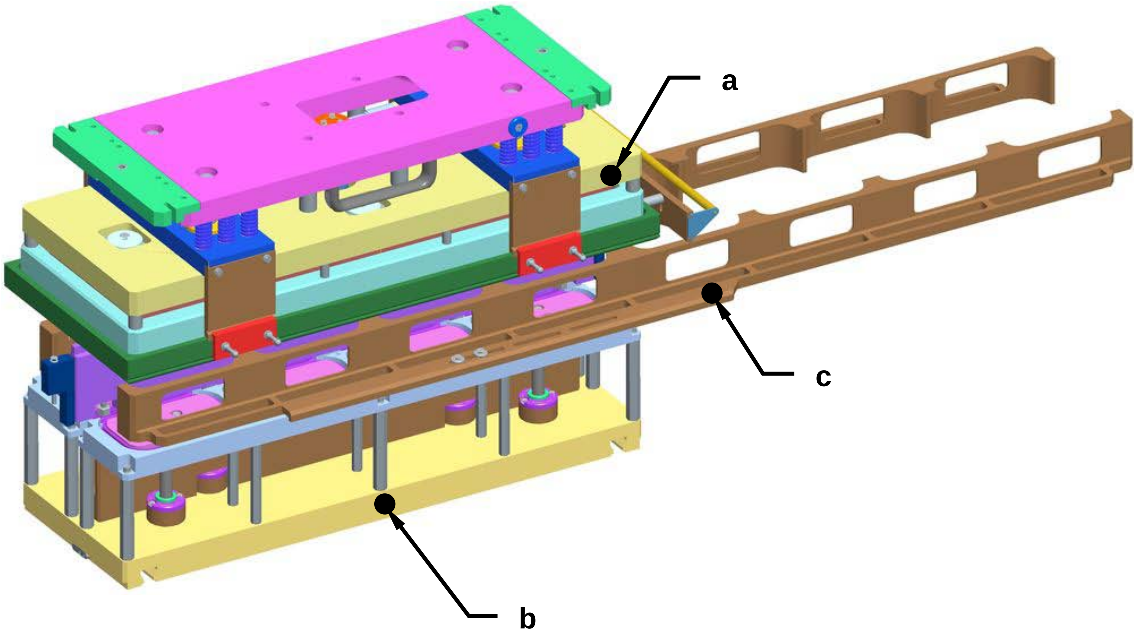
6.1.4 Check up on the work cycle recipe to be run

1. Check the system to ensure that the correct work cycle recipe has been selected for machine operation. To do so, it is necessary to consult the relative “SCREEN PAGE” section in Chapter 5 in this manual.



6.1.5 Check run of the tool assembled on the machine

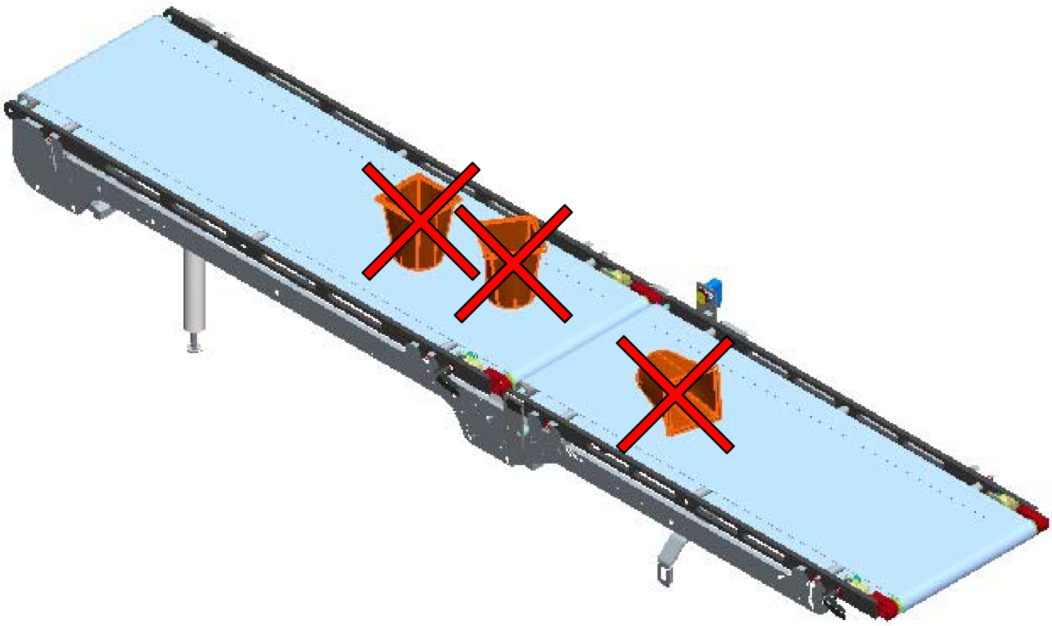
1. Check to ensure that all the tool components belong with one another (i.e. with matching coloured dots and/or numbers).



a. Top tool half
b. Bottom tool half
c. Right and left pusher units

6.1.6 Checking to ensure that the machine is clean - running a wash cycle

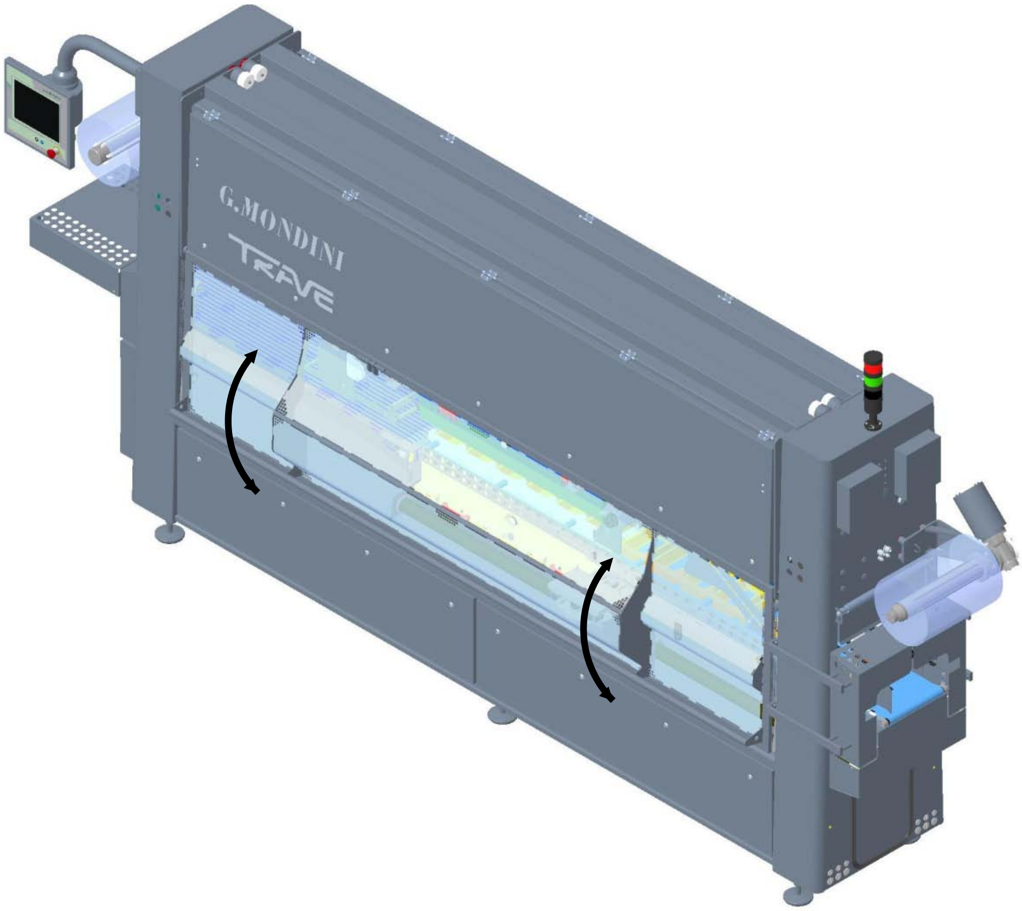
1. Check to ensure that there is nothing lying around on the machine, on the belts or in the vicinity of the tool.



2. Disengage the wash cycle (if selected) by acting on the relative selector key.

6.1.7 Inspection of safety guards and doors to see that they are closed

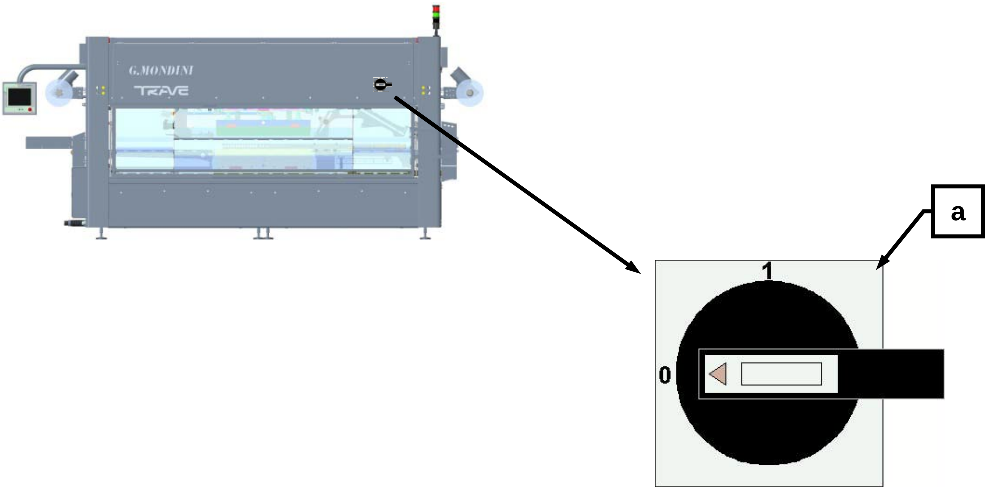
1. Check to make sure that all the sealing machine cover guards and doors are perfectly closed.



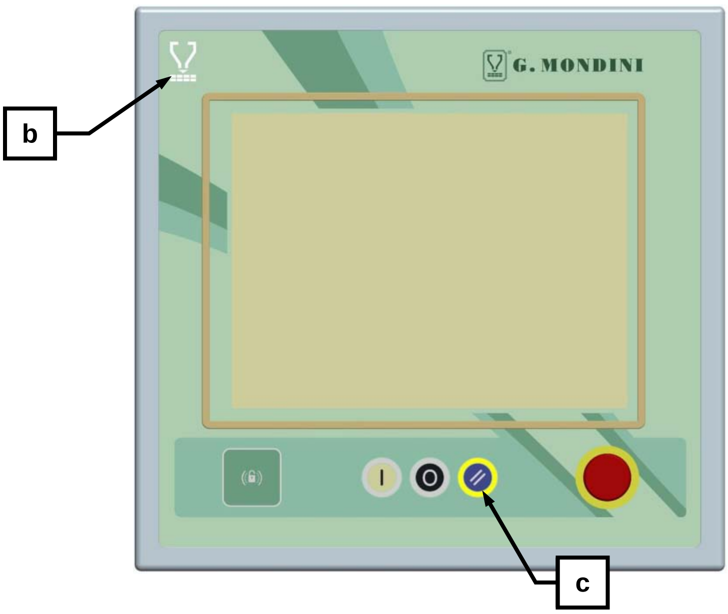
6.2 FIRST-TIME MACHINE START-UP

After having run all the inspection operations as specified and described in the previous pages, proceed as follows for machine start-up.

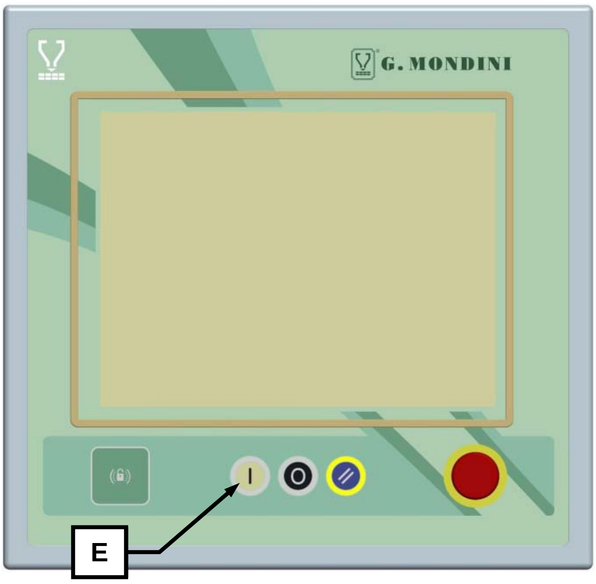
1. Power up the machine's power supply by turning main switch (**A**) into position 1.



2. To check whether the machine has been powered up, check the electric controls panel to see that the white signal light (**B**) is on, meaning that POWER IS CONNECTED.
3. After having checked the machine according to the alarm messages displayed on screen, press the pushbutton (**C**) for ALARM RESET.



4. Switch selector (**D**) to the required machine operation mode, i.e. either to MANUAL or to AUTOMATIC.
5. Press the START pushbutton (**E**) to activate machine operations.
    - Every time a “manual cycle” pushbutton is pressed, with selector (**D**) in MANUAL position, the respective device runs one work cycle.
    - With selector (**D**) in the AUTOMATIC position and pressing the “manual cycle” pushbutton, the machine work cycles are controlled based on the signals sent by the photocells and based on the parameters set into the recipe selected for operation.



6.3 END OF PRODUCTION (DAILY)

Every day, at the end of production, before departing from the machine the operator must perform a set of routine procedures and specifically:

1. Sealing film removal operations
2. Wash cycle activation and covering up of the container detection photocell
3. Unlocking of the short infeed belts
4. Removal of the short infeed belts (only in the event of sterilisation).

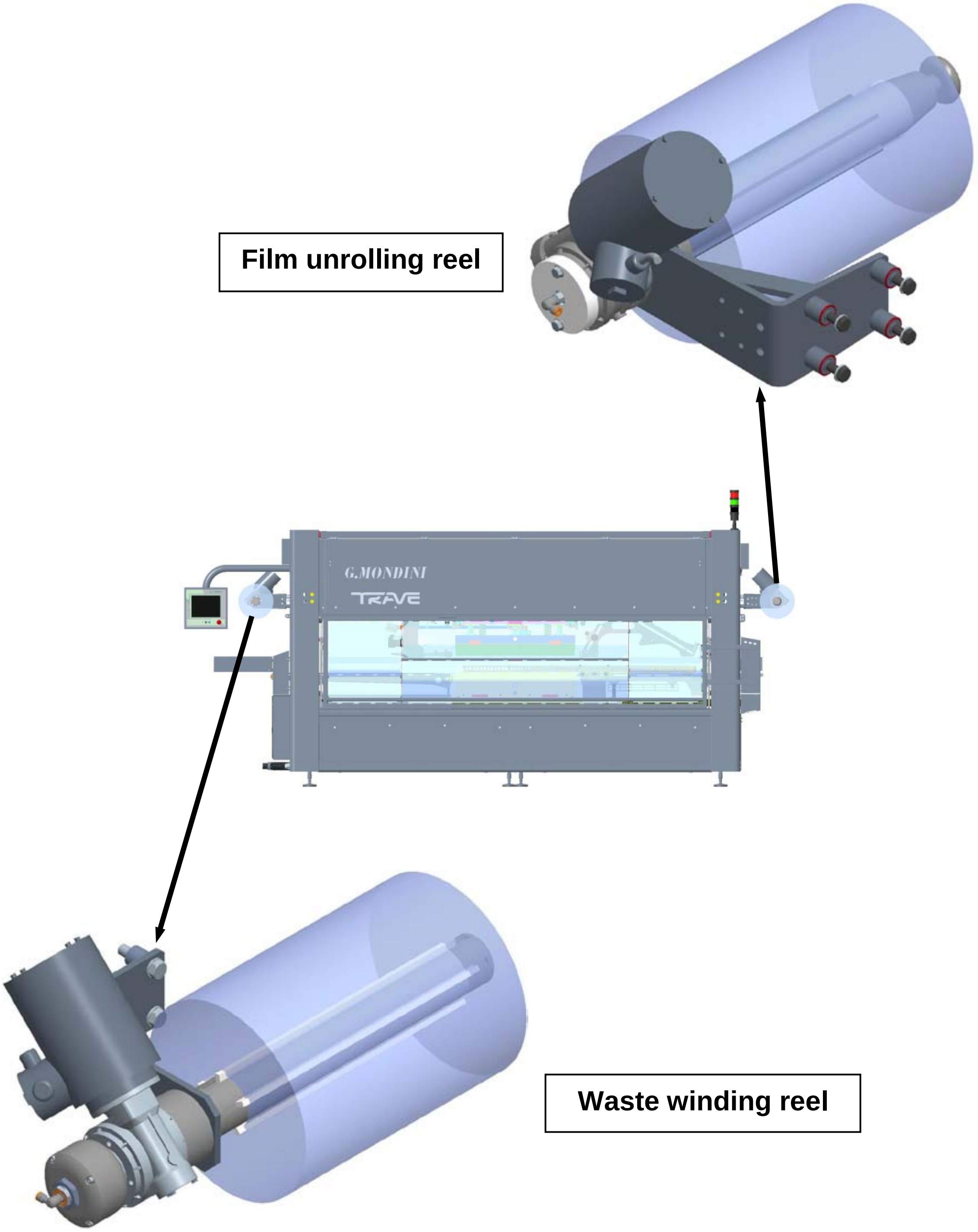
A description on how to proceed with each one of the above points is provided herein as follows:

DANGER
Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

6.3.1 Sealing film removal operations

1. Check that the system locking functions on the Lid Unrolling, on the Film Unrolling and on the Waste Winding are all deactivated by acting on the relative selector keys (**A**), (**B**) and (**C**).

2. Break off the sealing film and remove the film reels as well as on the waste winding reels.

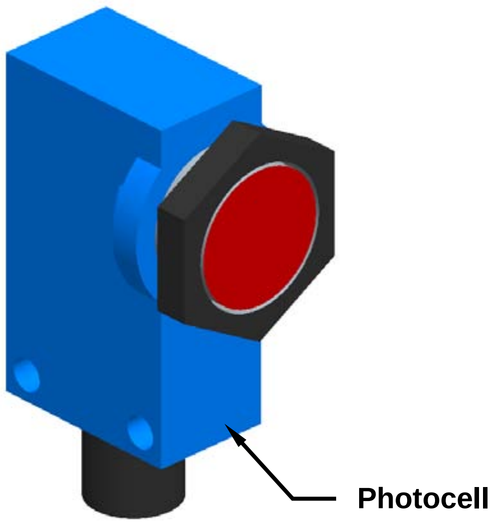


6.3.2 Wash cycle activation + cover-up of the tray detection photocell

1. Activate the wash cycle and cover up all photocells, reflectors and uncovered motors installed in the system (in-line, sealing machines, doser units, etc…) to prevent them from being damaged during the cleaning phases. All the photocells are provided with an **IP 65** or higher protection class to ensure overall safety against contacts with energised machine parts as well as protection against dust and/or water jet penetration hazards. We recommend that, prior to beginning machine washing operations, the above mentioned parts are covered using plastic bags. Should it nevertheless occur that the photocells get wet, we recommend drying them off using a soft, dry cloth so as not to damage or scratch the photocell eyes.

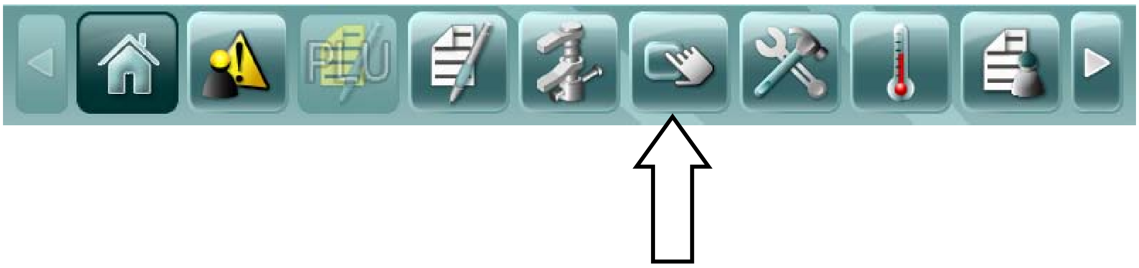
⚠ WARNING!

- It is strictly prohibited to wash the photocells, the reflectors and uncovered motors using high-pressure water jets.
- It is strictly prohibited to wash the photocells, the reflectors and uncovered motors using solvents or thinners. The chemical composition of said products could damage the plastic photocell eyes thereby compromising their operation.
- To prevent damages to the relative electronic systems, it is strongly recommended NOT to wash the photocells using hot water. This is very important especially after the machine has been powered off because the photocell is still hot.

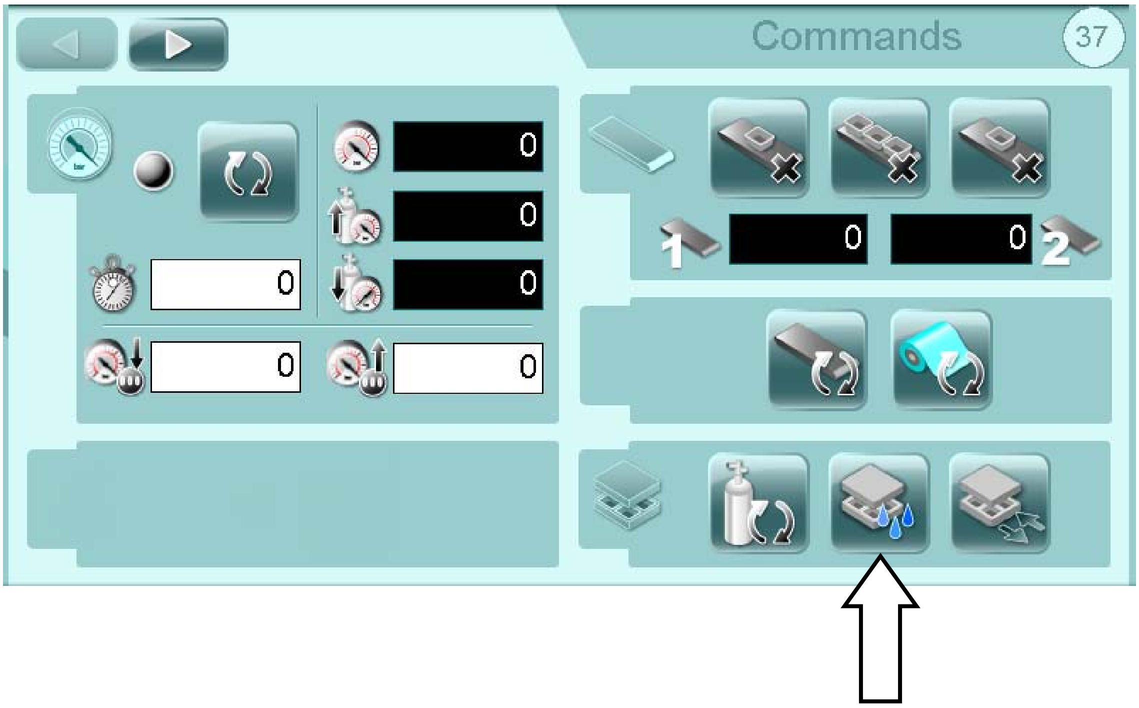


2. Check to make sure that all the machine's safety coverings are firmly closed.

3. From the menu bar access the "Commands" page.



4. From the "Commands" page, activate the ""wash cycle" function by switching the relative selector to "enable".



5. After having checked again that all containers and trays as well as the film bobbins have been correctly removed, press on the “OK” pushbutton key.

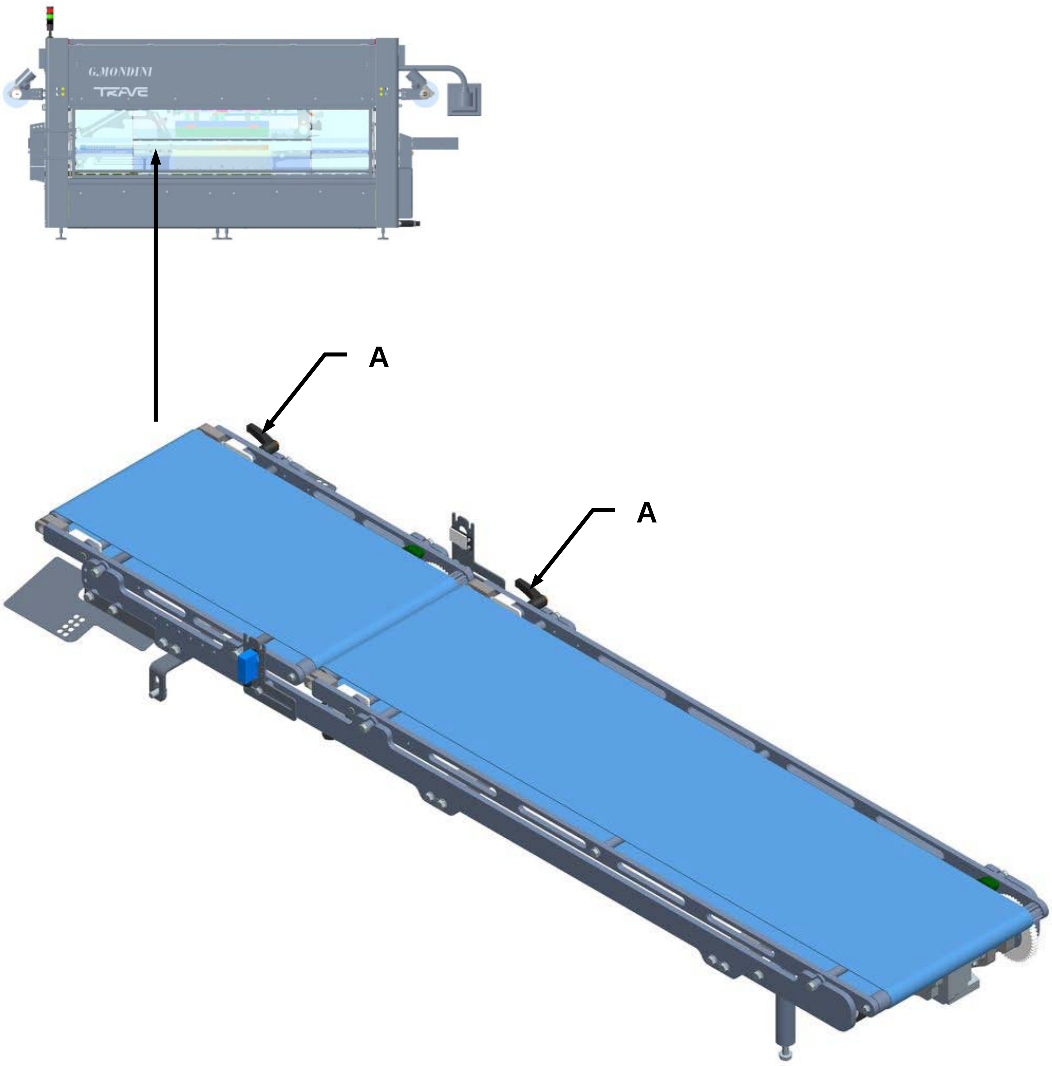
6. Once the “washing cycle” has been enabled and activated, the tool will close up to prevent water and detergents possibly entering the bell frames.

7. Open up the coverings and start the machine washing operations.

WARNING! We strongly recommend using neutral Ph detergents and foams, that combine disinfectant and sanitising properties with grease-removal properties.

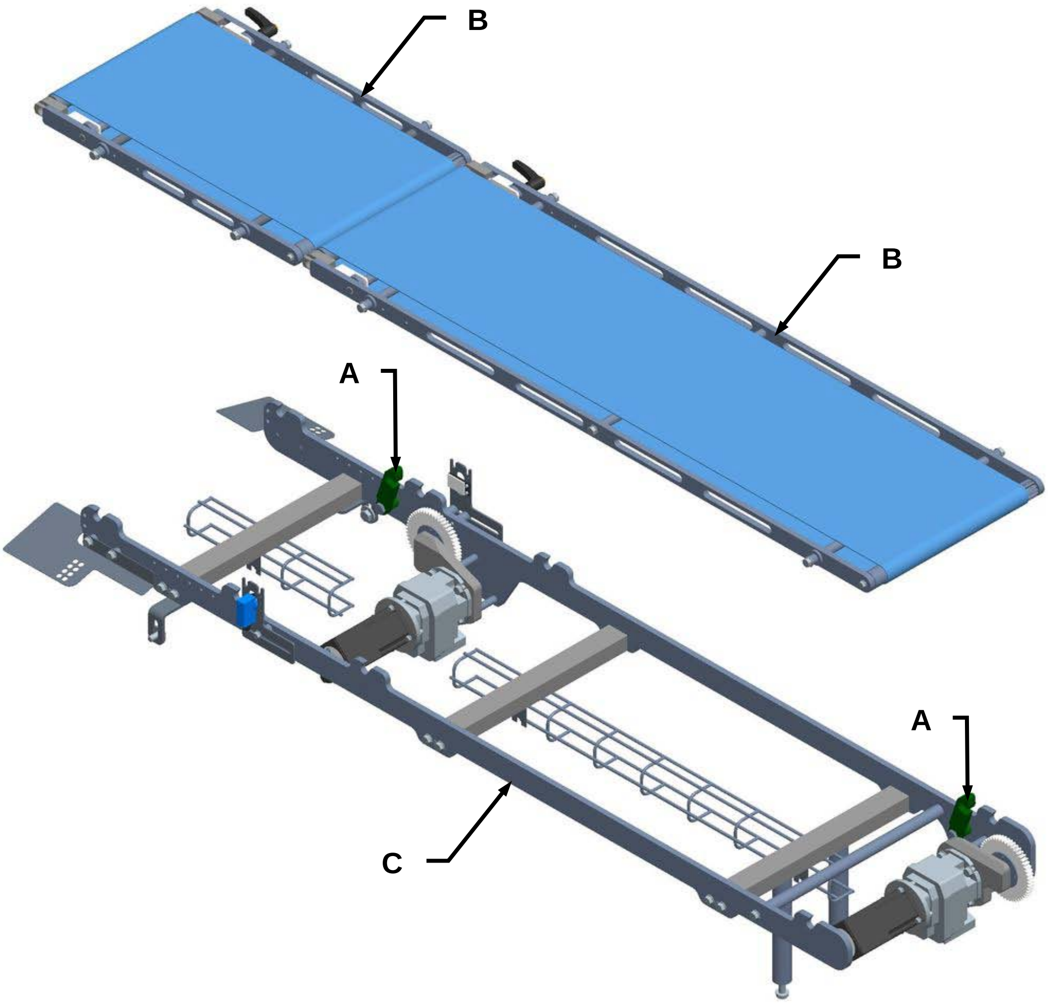
6.3.3 Unlocking the short infeed belt

1. Use the appropriate lever (**A**) to unlock the short infeed belt.

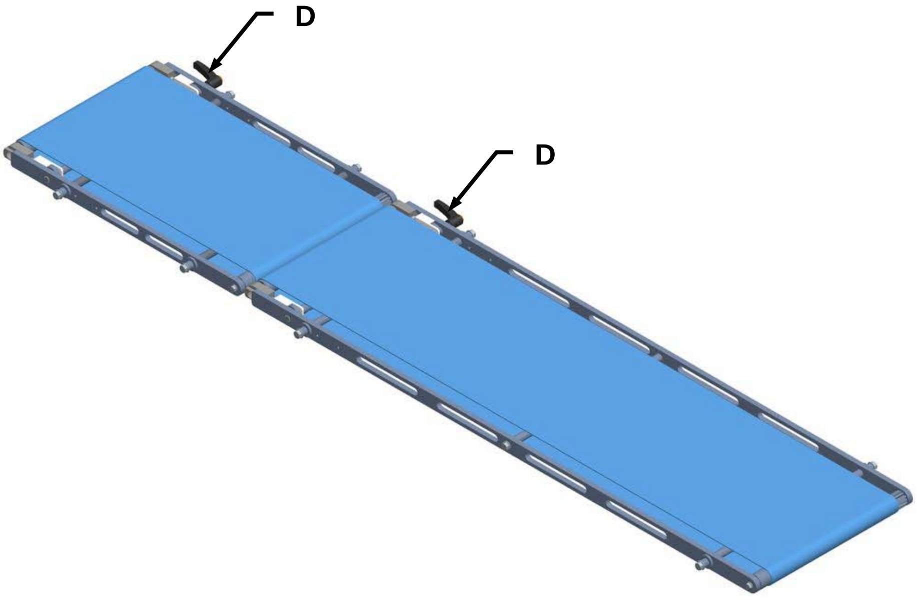


6.3.4 Removal of the short infeed belt (only in the event of sterilisation)

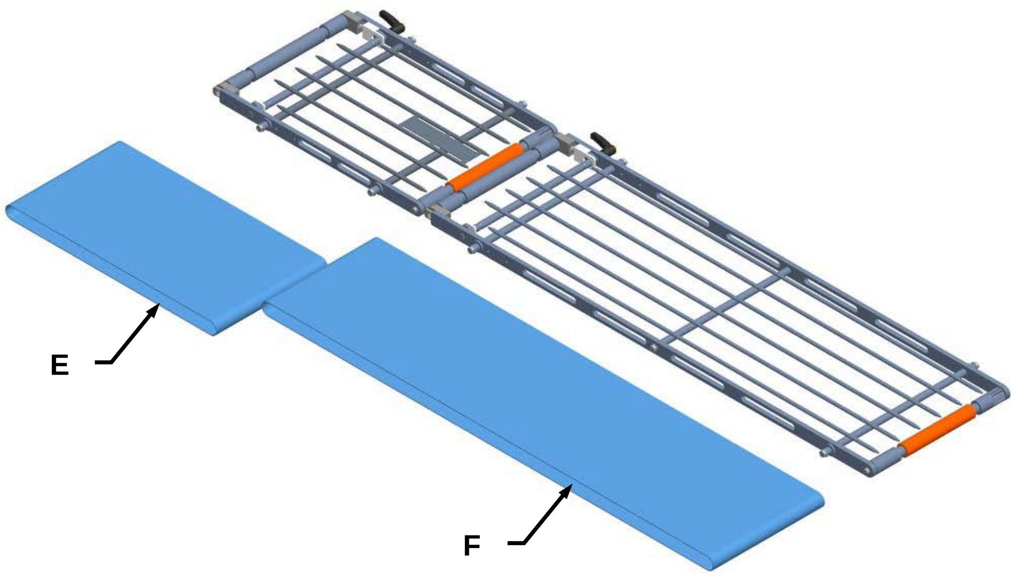
1. Open the hooks (**A**) and remove the belts (**B**) from the support (**C**).



2. Use the appropriate lever (**D**) to unlock the belts.

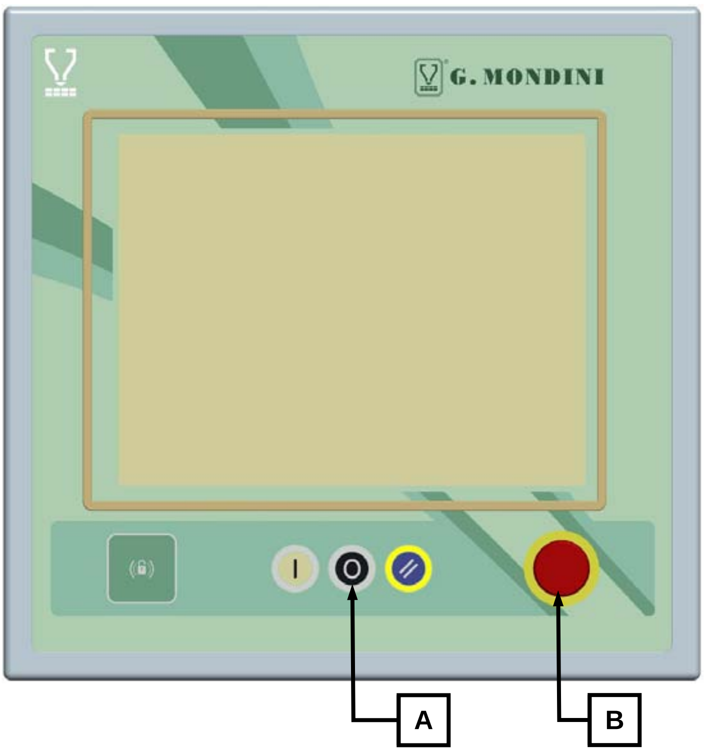


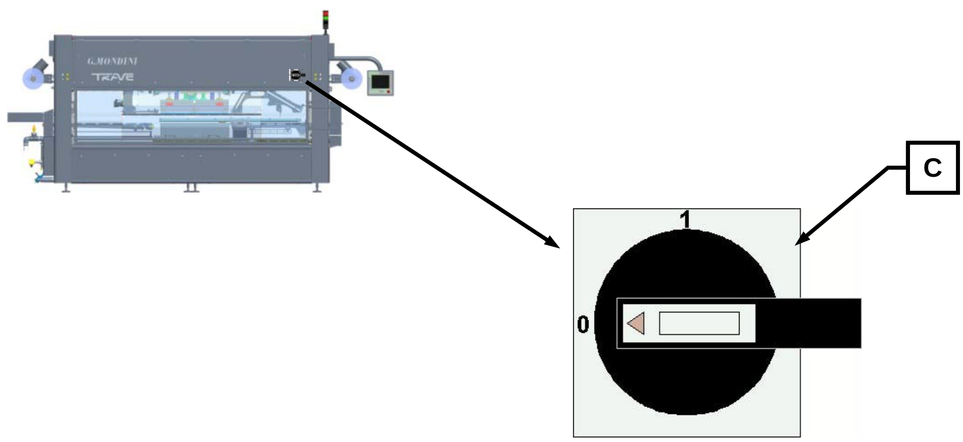
3. Remove the belts (**E**) and (**F**).



6.4 MACHINE CYCLE STOPS

1. To stop the machine work cycle at the end of production it is necessary to press the red "STOP PROGRAM CYCLE" pushbutton (**A**), thereby allowing all the machine elements to travel back into cycle start position.
2. Carry out all the operations involved with "production program end" then switch off the machine by first pressing the red EMERGENCY pushbutton (**B**) then proceed with cutting off the machine's power supply by positioning the main switch (**C**) into position 0.



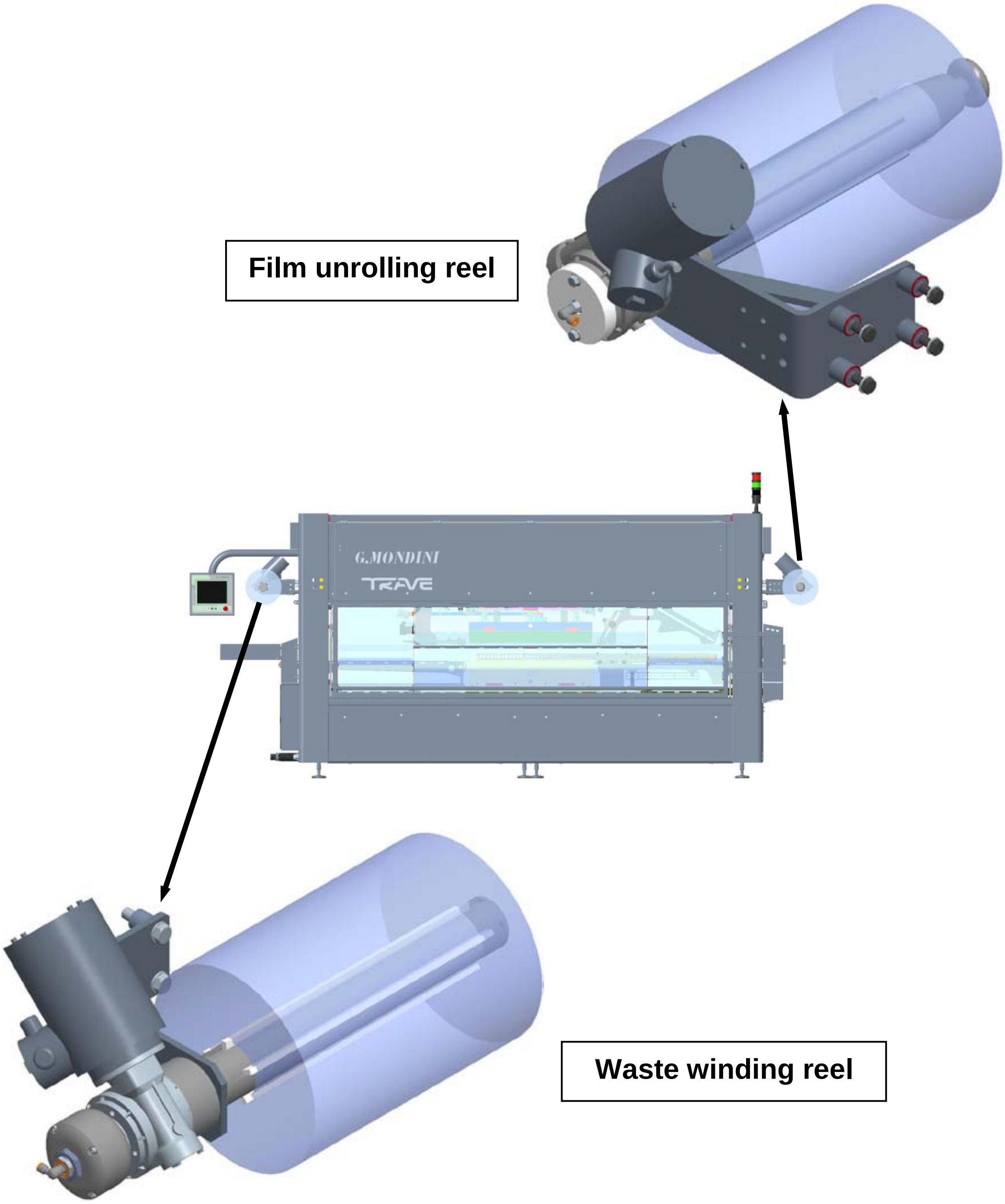


6.5 TRAY FORMAT CHANGES

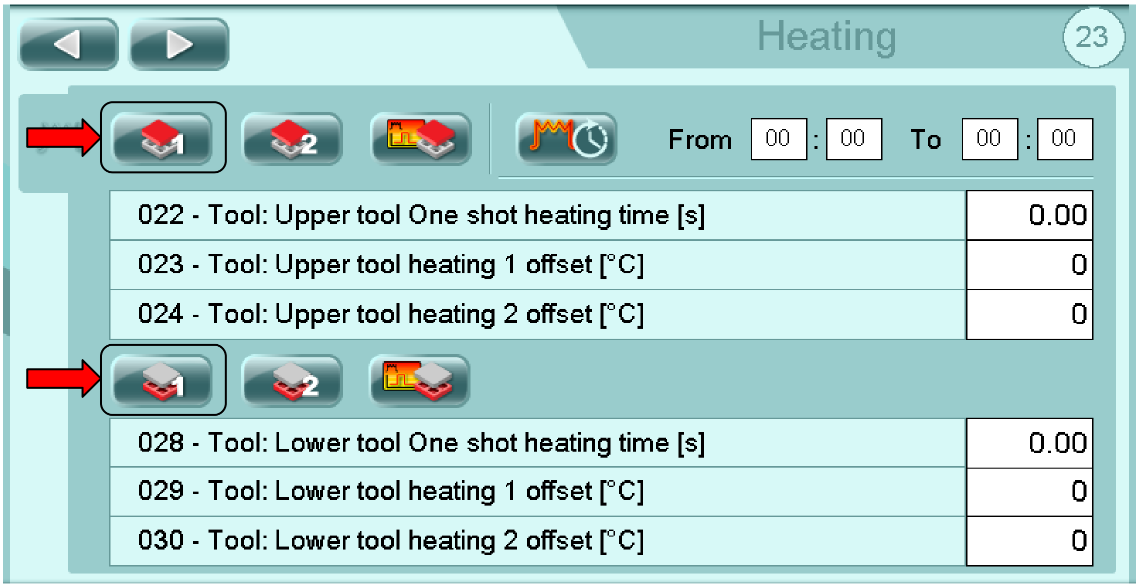
To carry out tray format changes, proceed as follows:

6.5.1 Extraction/removal of the tool

1. Tear off the film at tool level and gather one end onto the film unrolling reel and one end onto the film waste collector shaft. If necessary, extract to remove the two reels.

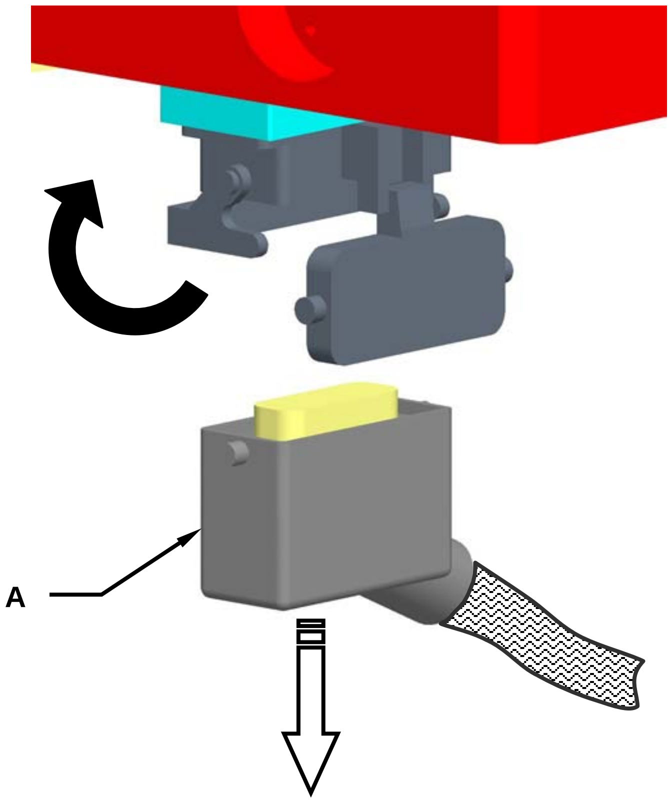


2. Disable the operation of the tool resistances from page 23 of the operator panel.

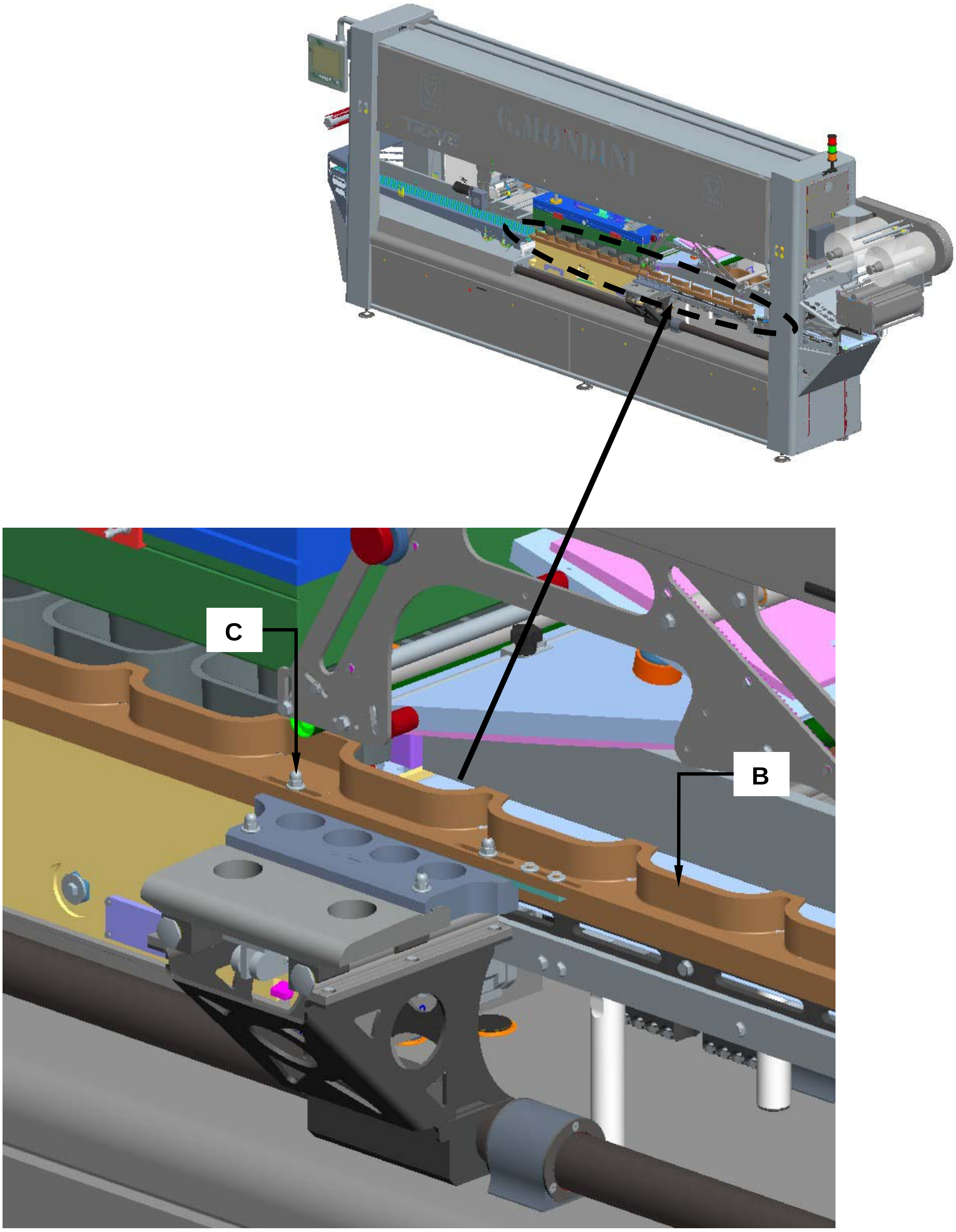


⚠ **WARNING!**	Before disconnecting the plug HARTING you must disable the operation of the resistance of the tool!

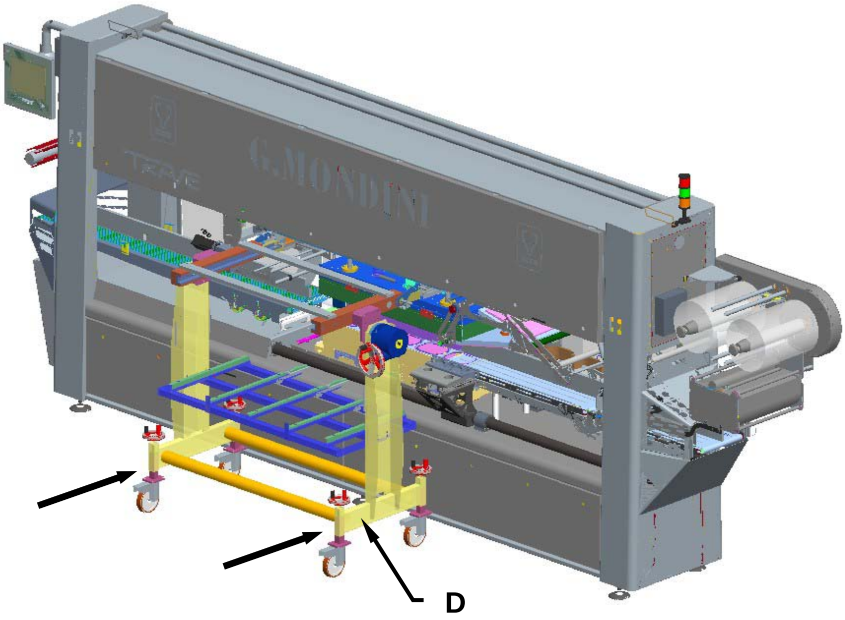
3. Disconnect the tool's electrical connections by disconnecting the HARTING socket (**A**) as illustrated.

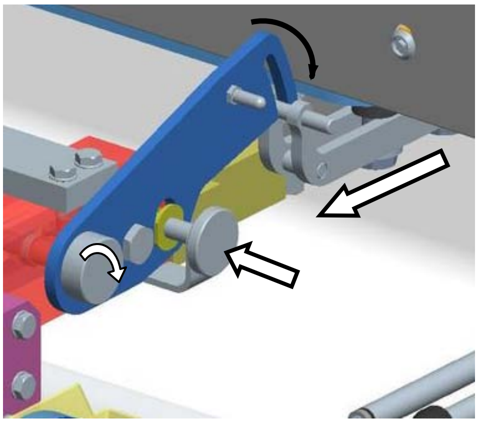


4. Remove the two left and right pusher units (**B**), by acting on the two lock nuts (**C**).

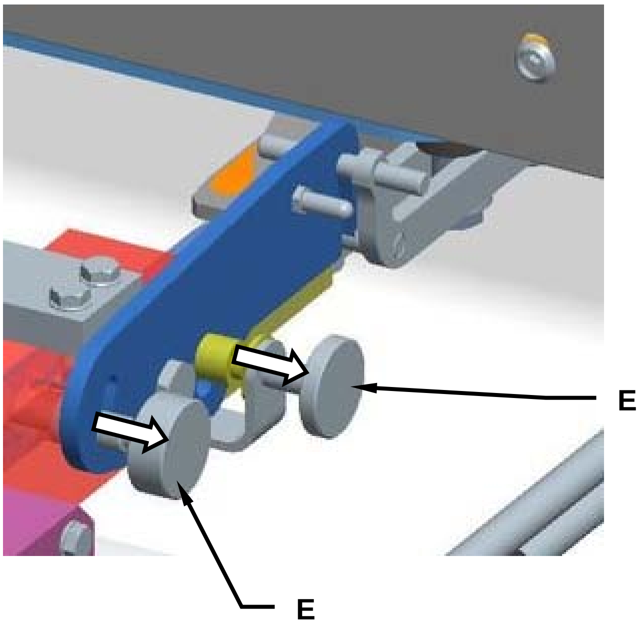


5. Draw the tool removal trolley (**D**) up alongside the machine and hook it up to the machine.



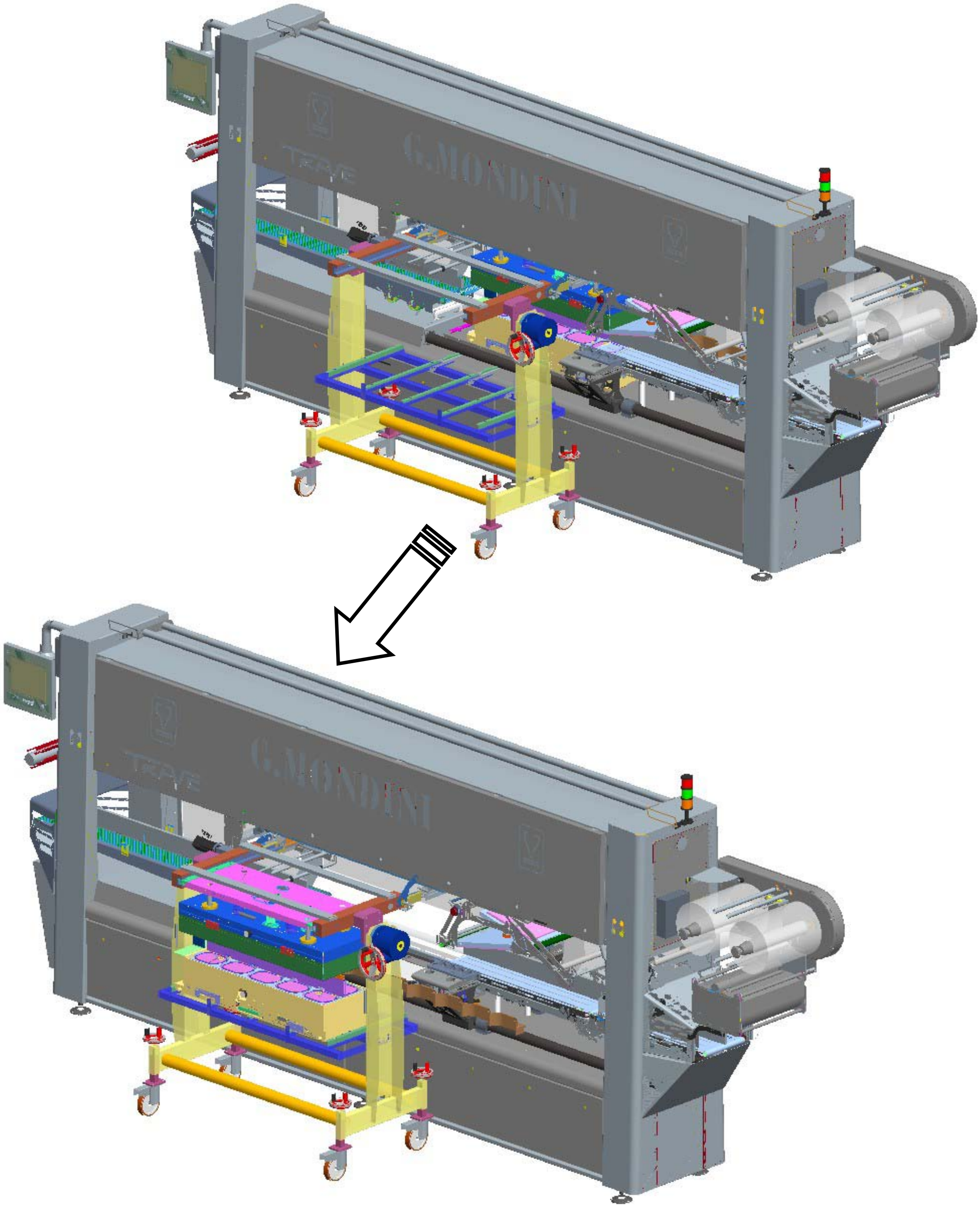


6. Turn out the two lock screws (**E**).

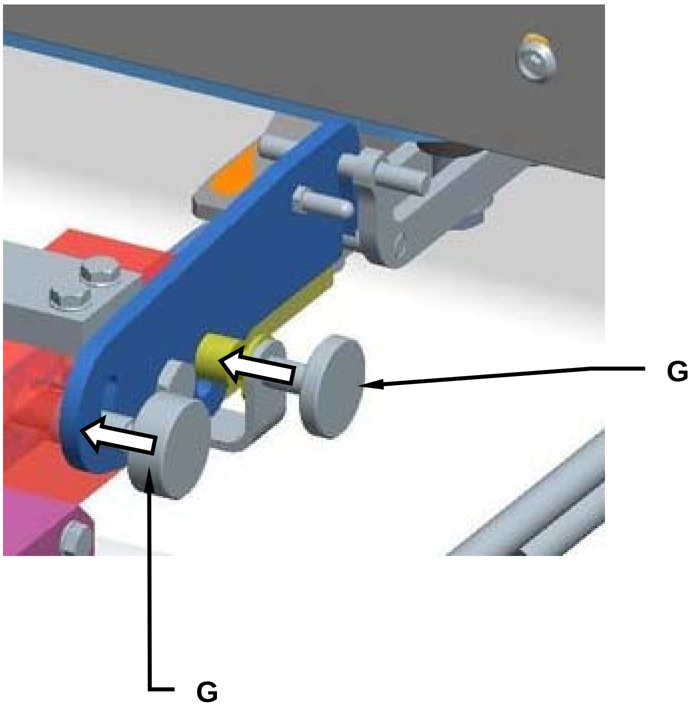


7. Unlock the tool by acting on the “TOOL LOCK” selector switch (**F**).

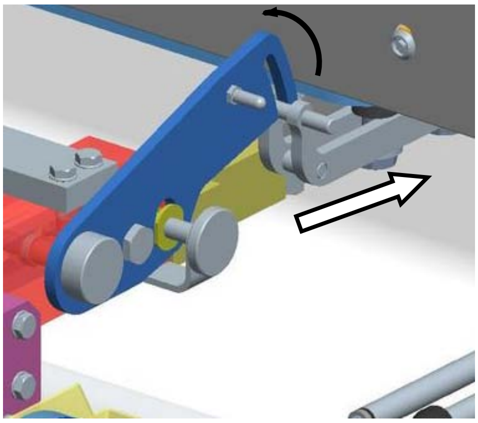
8. Extract the two tool ends (both the bottom and top tool edges) ***using the appropriate handles***.

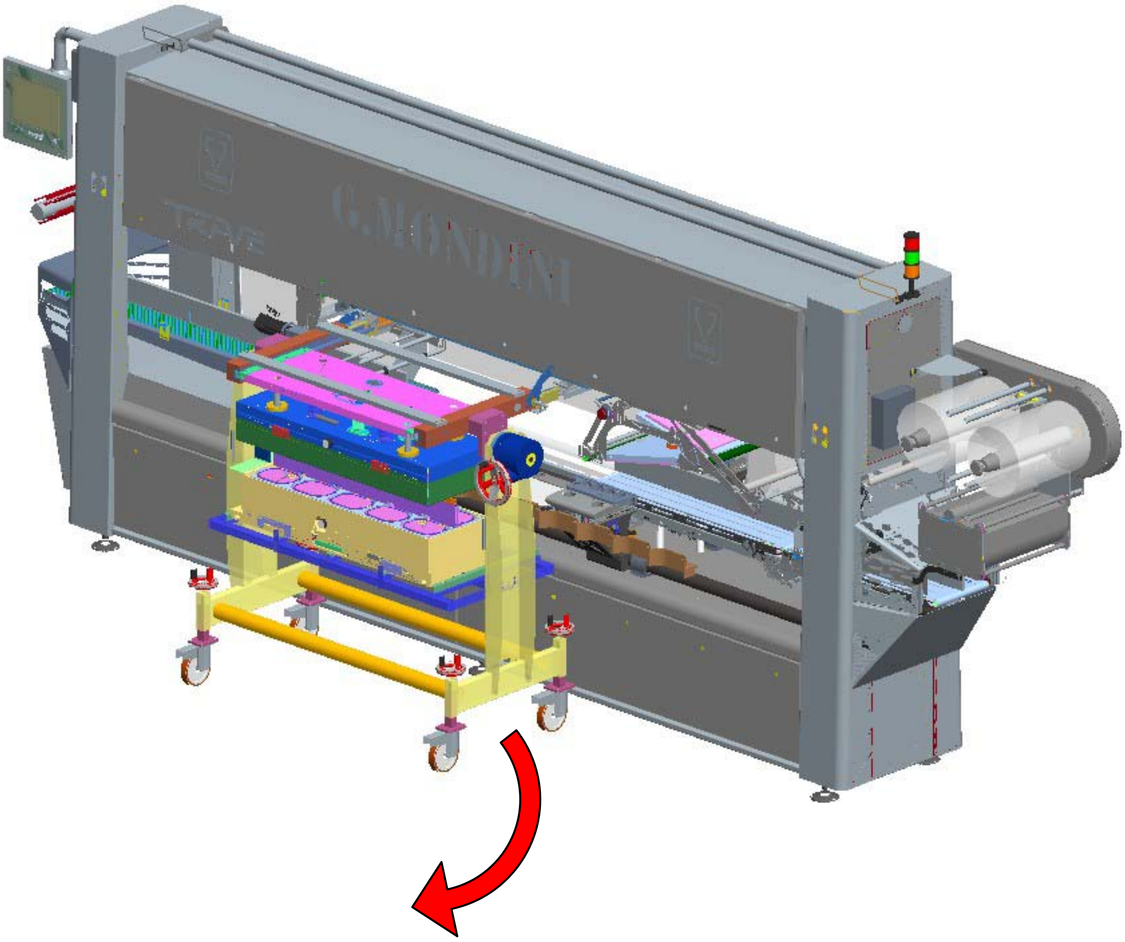


9. Screw in the two lock screws (**G**).



10. Unhook the tool removal trolley from the machine and move it away whilst checking to ensure that the trolley lock pins are locked in correctly.

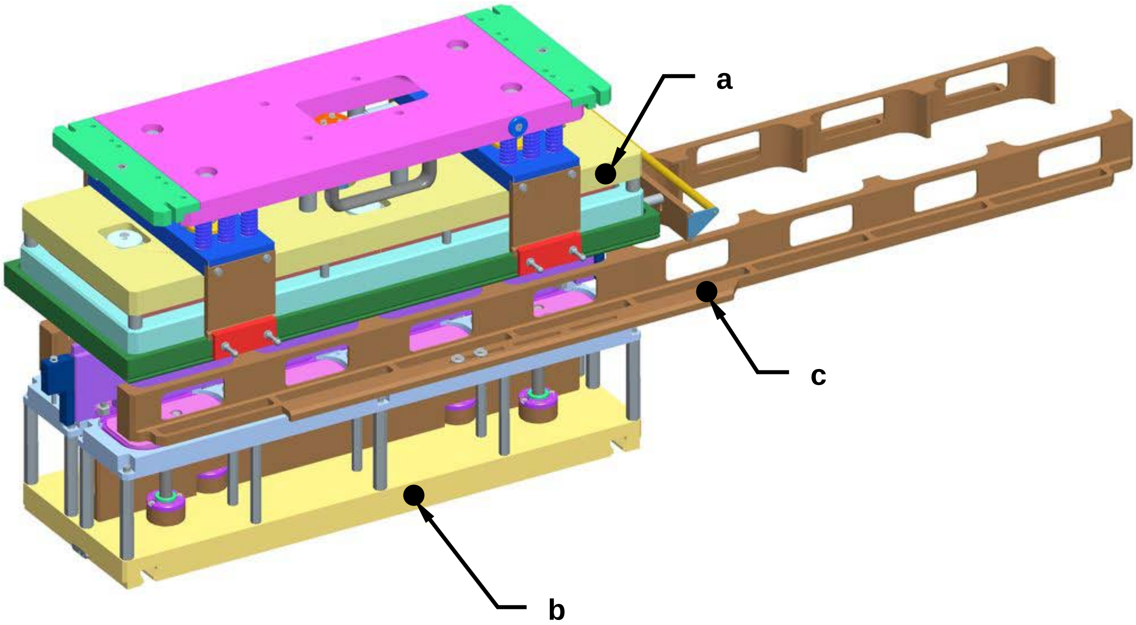




6.5.2 Tool insertion

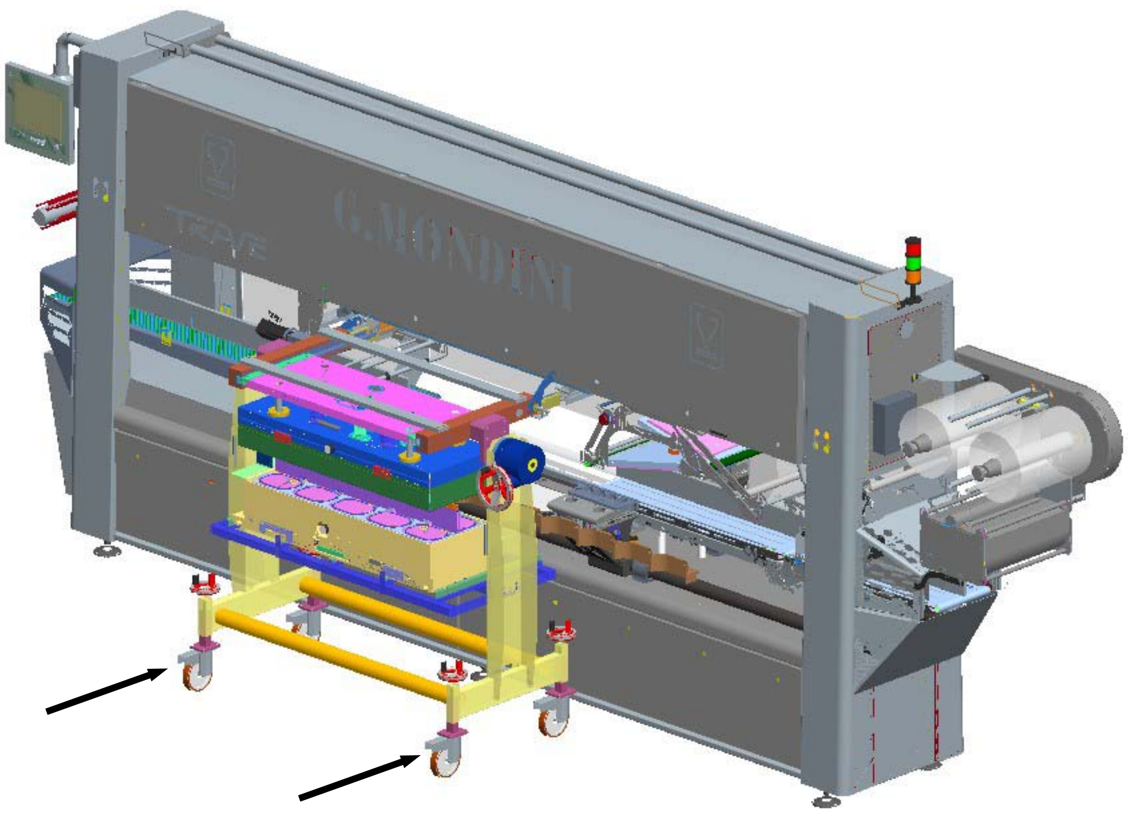
For tool insertion into the sealing machine proceed as follows:

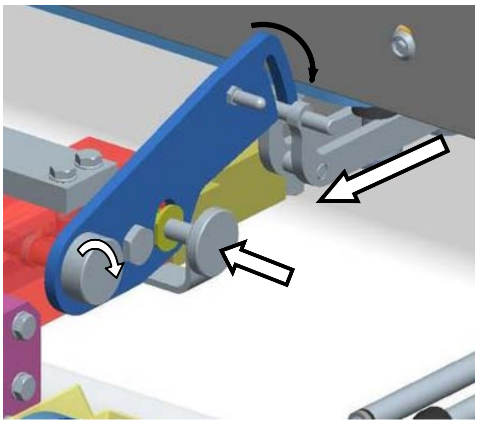
1. Check to see that all tool details and components are of the same type (i.e. the same colour or the same number).



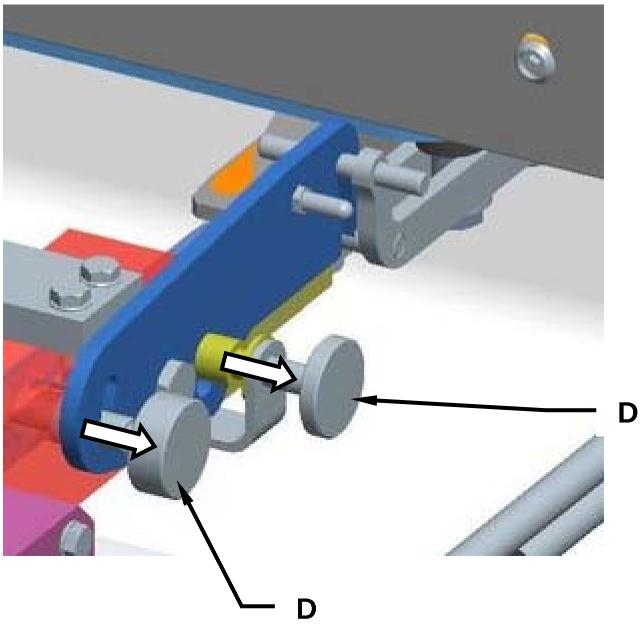
a. Top tool half
b. Bottom tool half
c. Right and left pusher units

2. Position the tool removal trolley up next to the sealing machine and hook it on.

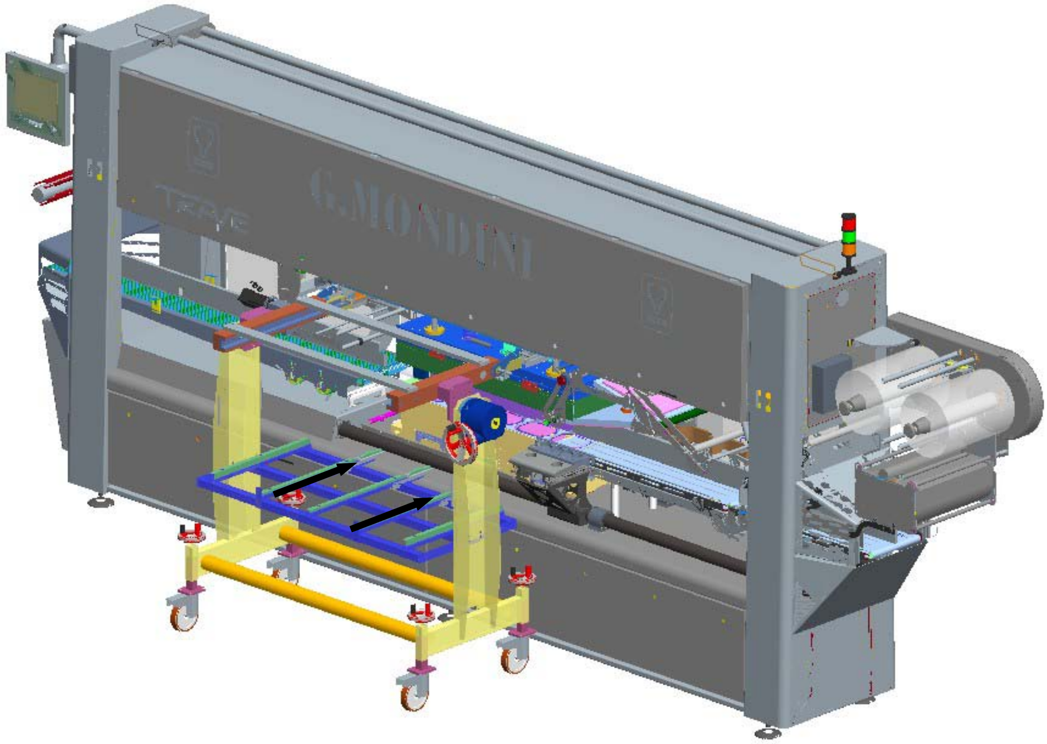




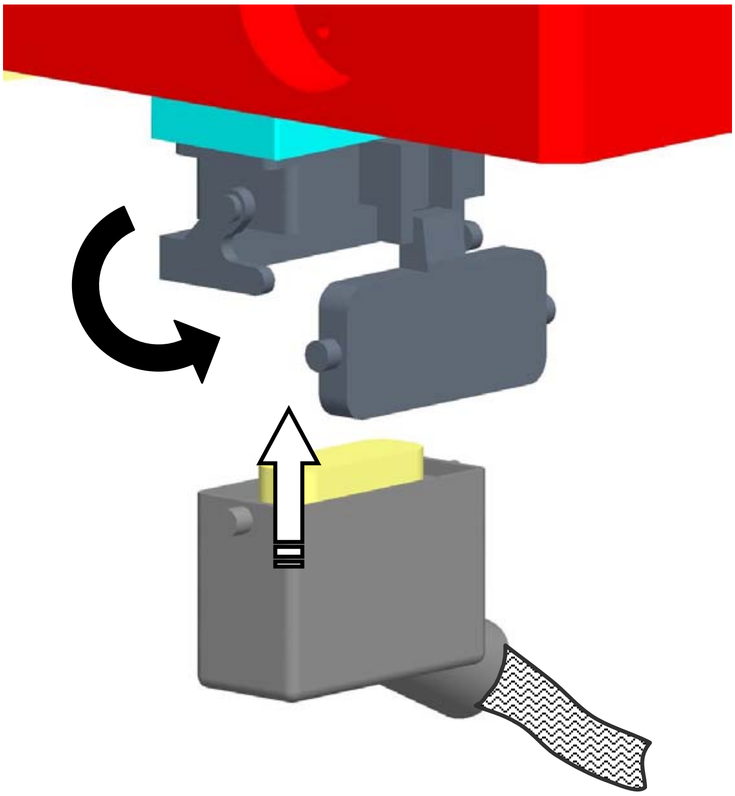
3. Remove the two lock screws (**D**).



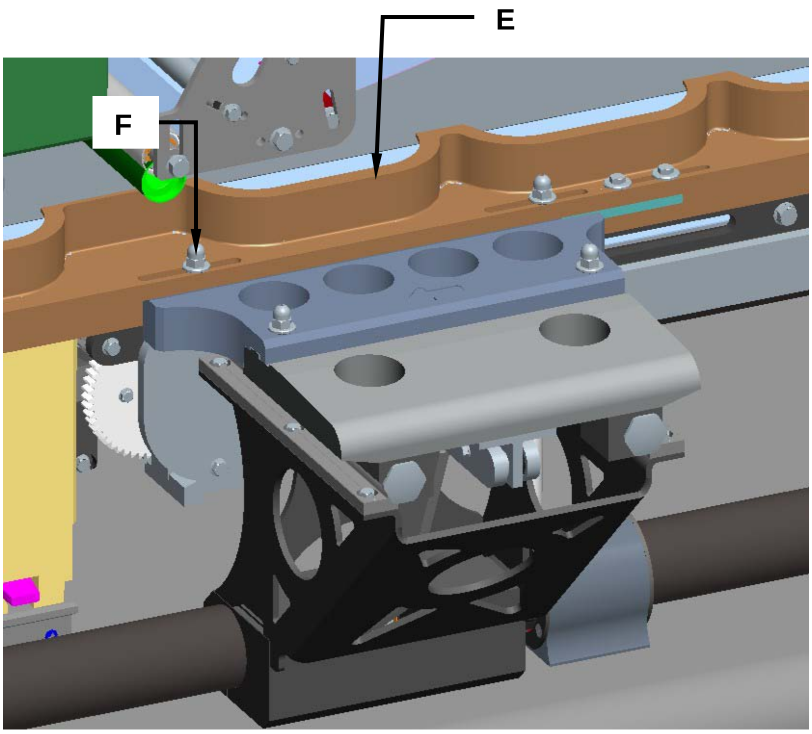
4. Insert the two half tools (both the top and the bottom halves), ***using the appropriate handles,*** and ensure they are accurately housed.



5. Connect the tool HARTING socket up to the connector on the machine.

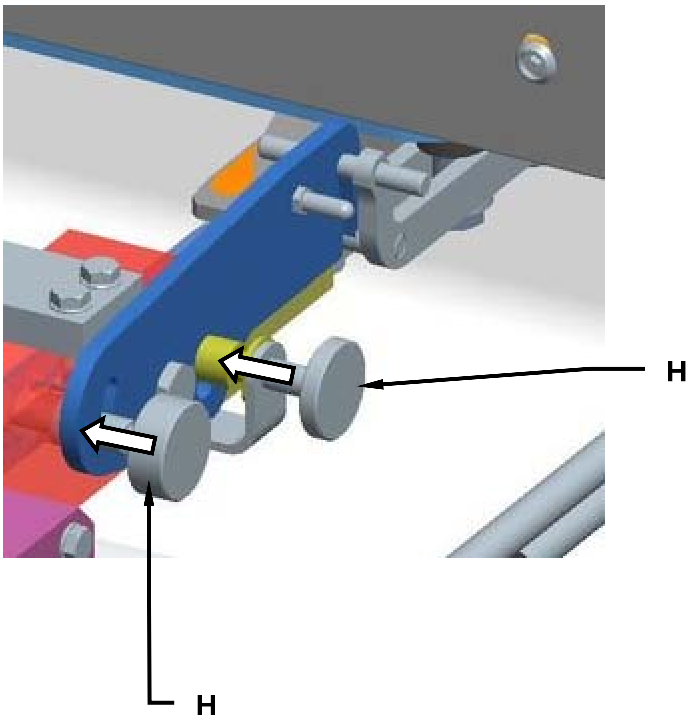


6. Assemble on both the right and the left pusher arms (**E**) and, after ensuring that they are perfectly housed in position, lock them on using the relative lock nuts (**F**).

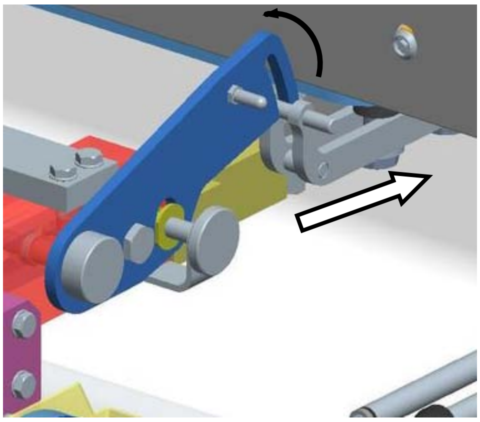


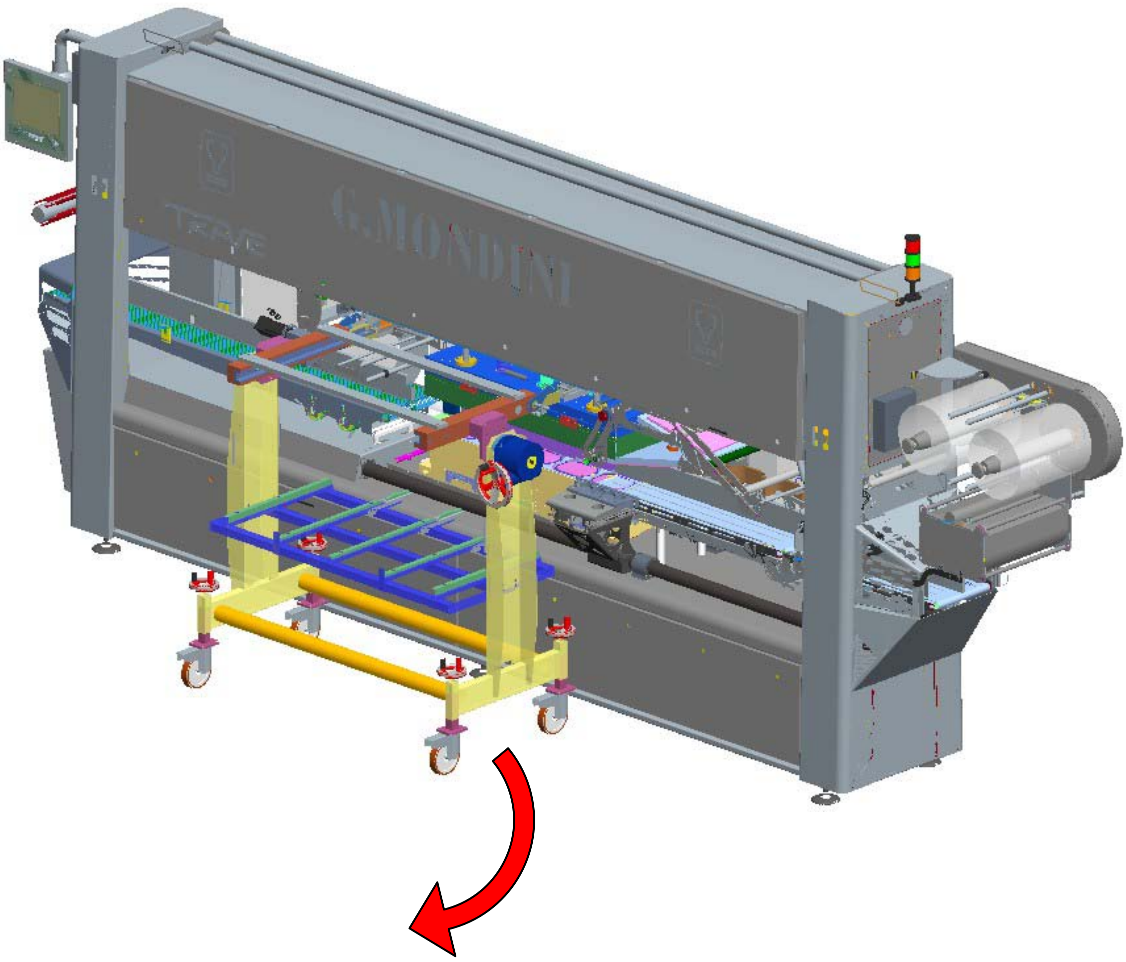
7. Lock the tool by acting on the “TOOL LOCK” selector switch (**G**).

8. Screw in the two lock screws (**H**).

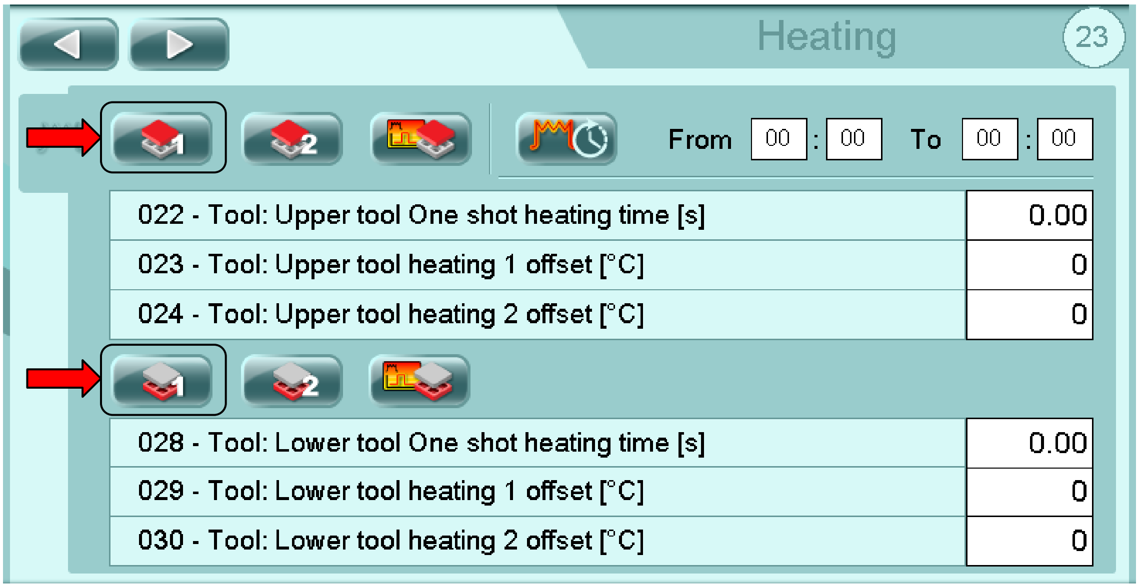


9. Unhook the tool removal trolley from the sealing machine and remove it.

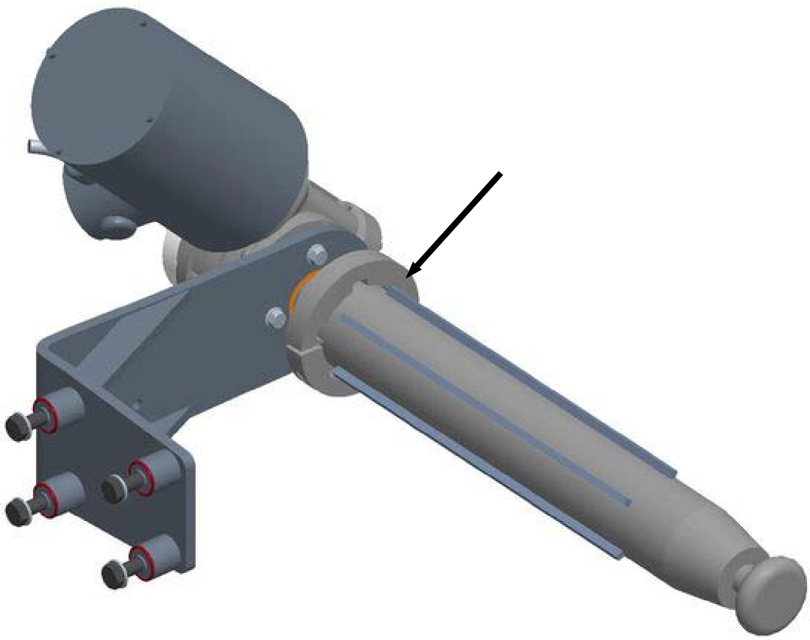




10. Enable the operation of the tool resistances from page 23 of the operator panel.

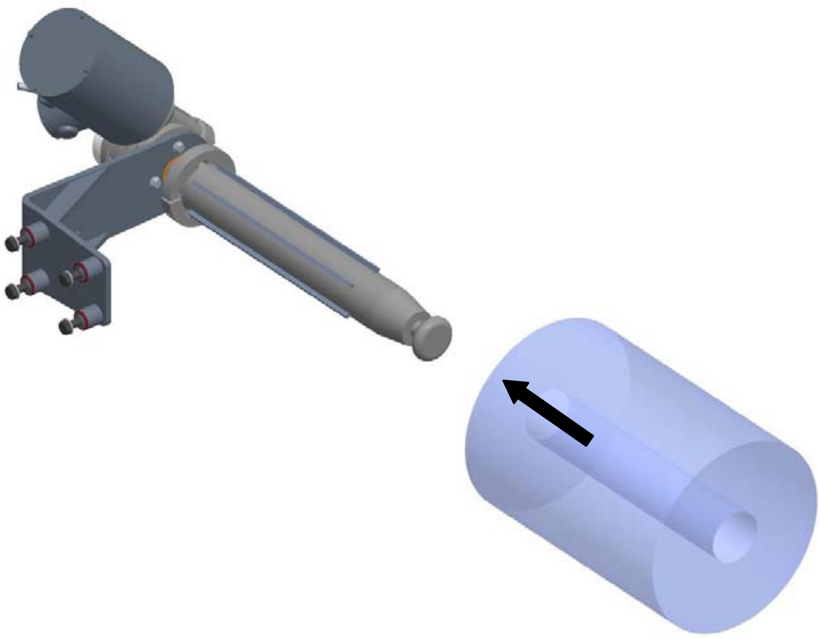


11. Insert spacers on the film unwinder bobbin, as the case may be.

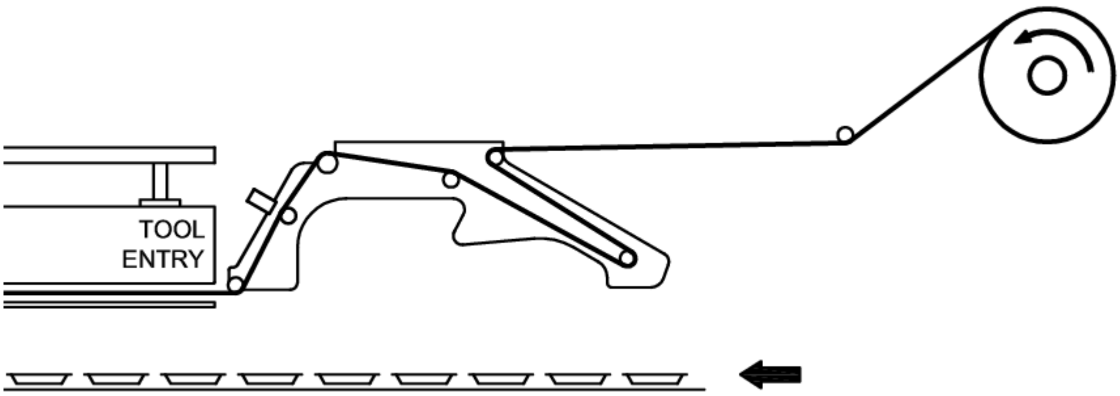


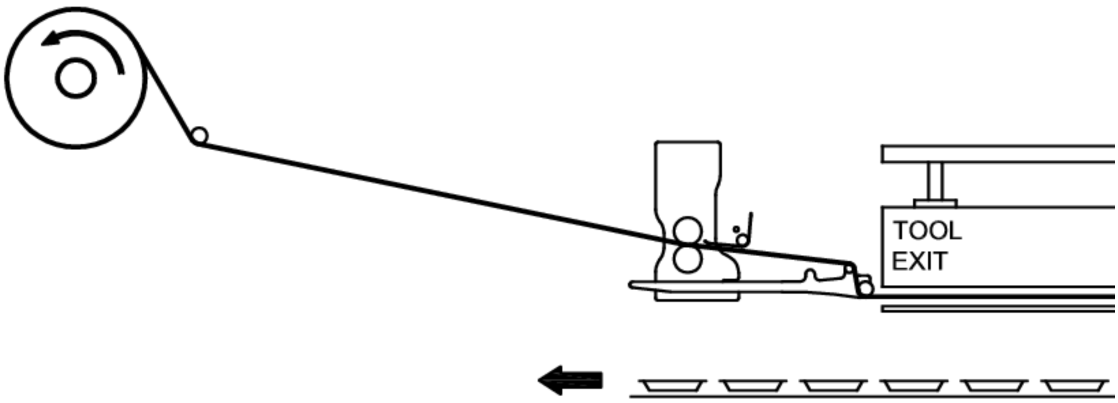
12.Check that the system locking functions on the Lid Unrolling, on the Waste Winding and on the Film Reel are all deactivated by acting on the relative selector keys (**L**), (**M**) and (**N**).

13.Fit on the sealing film reel.

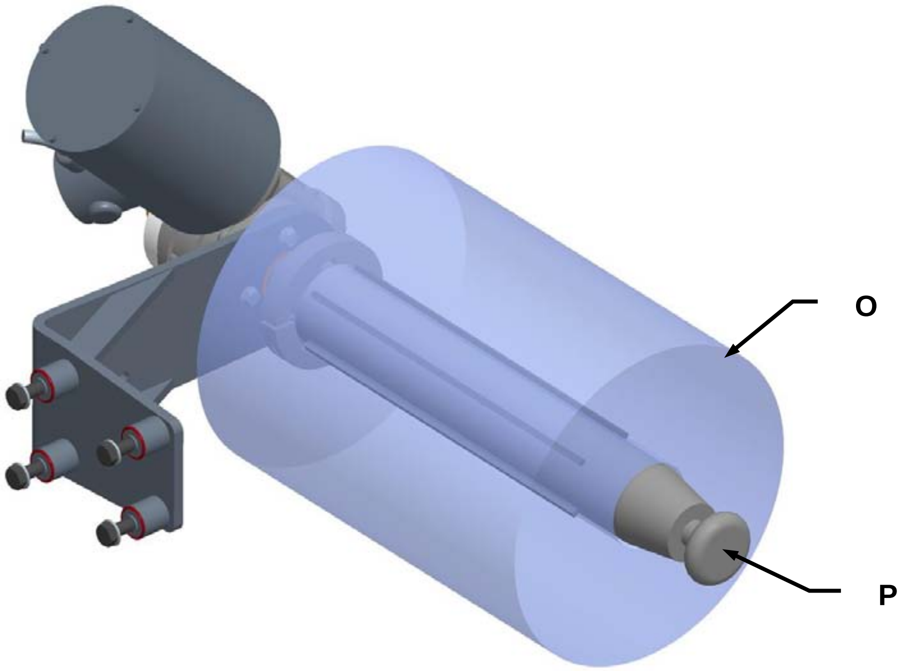


14. Thread the film in through the bobbin unwinder rollers as per the directions indicated by the arrows provided near each one of the rollers.

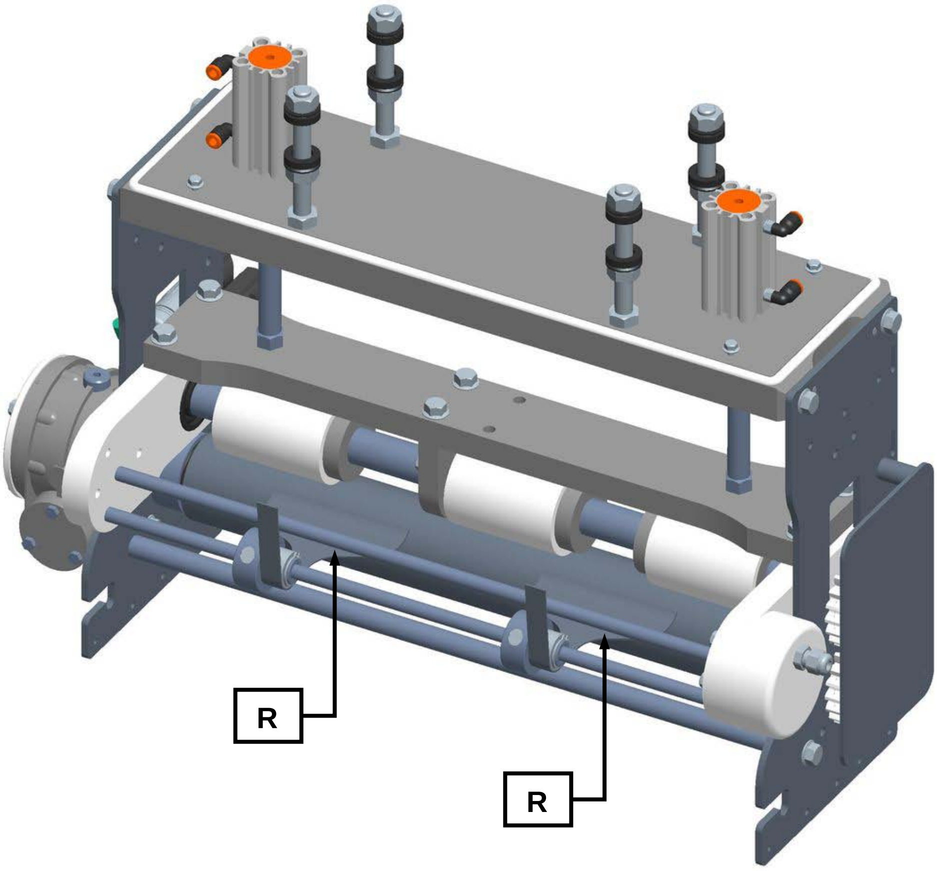




15. Check to ensure that the bobbin (**O**) is correctly centred on the film unwinder system shaft (**P**) and lock it in place using the appropriate button (**Q**).

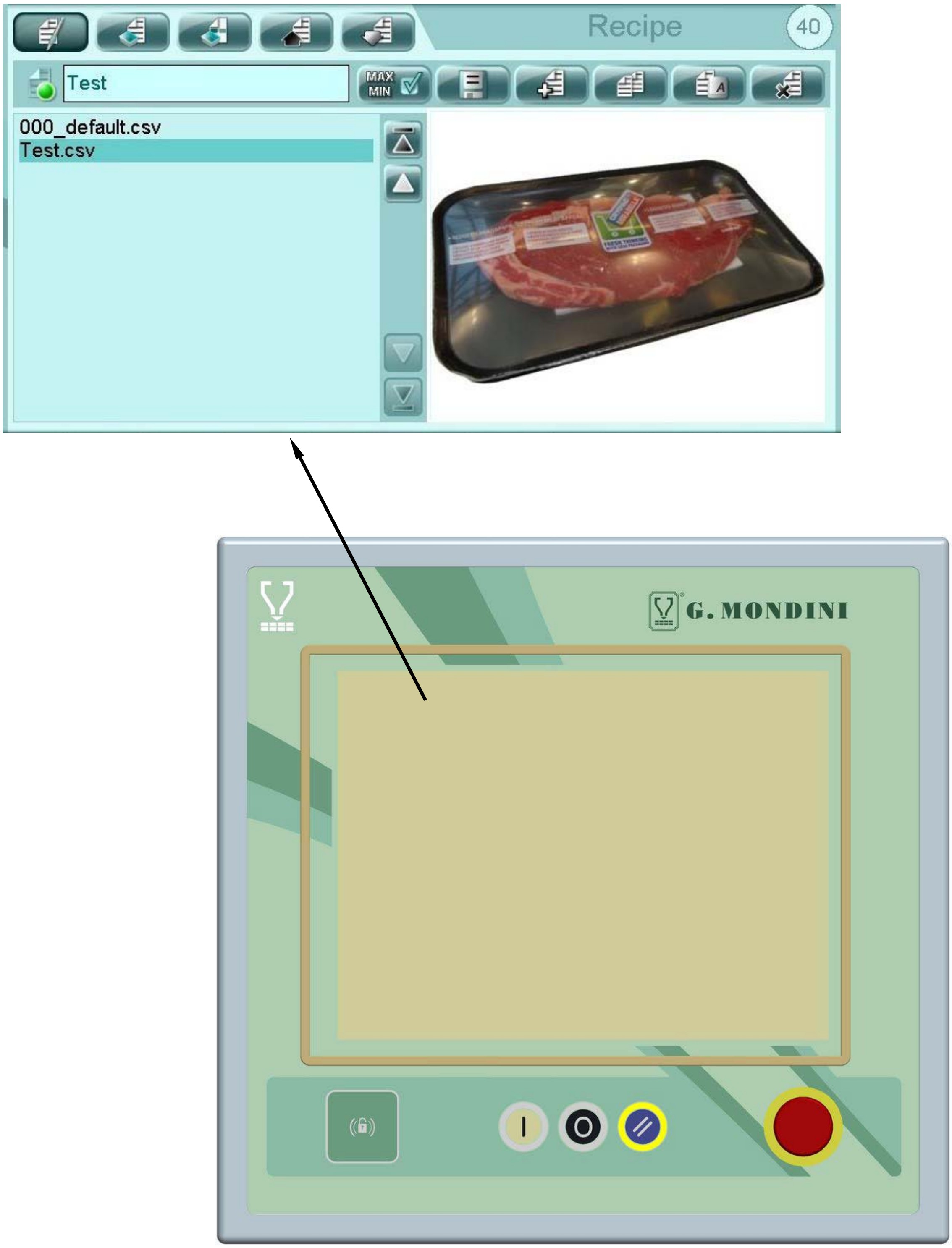


16. Check to see that the sealing film waste blade plates (**R**) are positioned on the relative outer edges. If not, move them so that they are positioned correctly as shown.



17. Rotate selectors (**S**) to block the film unrolling pullers and to wind the waste onto the waste winding shaft. To lock the waste, rotate relative selector (**T**).

18. Go to the operator panel and enter the recipe relative to the tool you have just inserted.



6.6 TOOL CENTERING

TIME REQUIRED:	60/90 min.
EQUIPMENTS:	Workshop equipment.

WARNING!

The clothing of those who run the line or carry out maintenance on it is to conform the safety requirements specified in the EU directive 89/656/EEC and the laws in force in the country where the line is installed.

DANGER!

To avoid mechanical risks, such as dragging, trapping and others, do not wear bracelets, wristwatches, rings or chains during operations in processing cycles and maintenance.

It’s forbidden more than one operator running on the machine at the same time.

It’s obligatory to operate in the presence of a supervisor, who monitors the work phases and intervenes in danger case.

WARNING! Danger of cutting, abrasion and burns.

It's obligatory the use of appropriate personal protective equipment, such as gloves for high temperatures and resistant to cuts. These must be confortable to be able to wear them during all work phases.

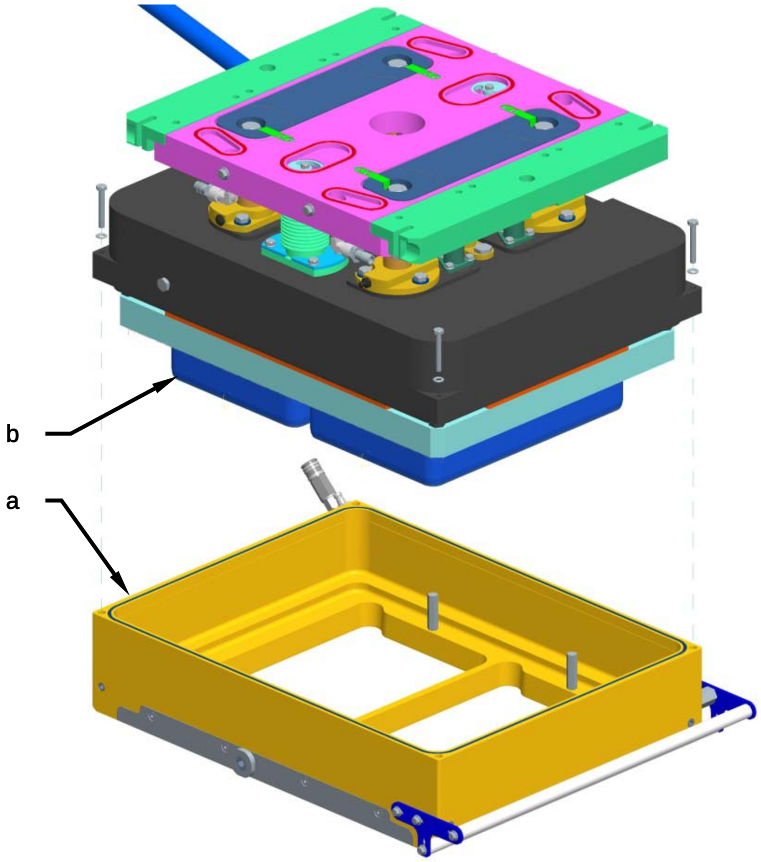
<u>WARNING!</u> The maintainer, before performing these operations must be trained and informed, he have to also make sure of any "off-center" between the bells and note these differences, they are to be considered with the measures to note down for the centering.

WARNING!

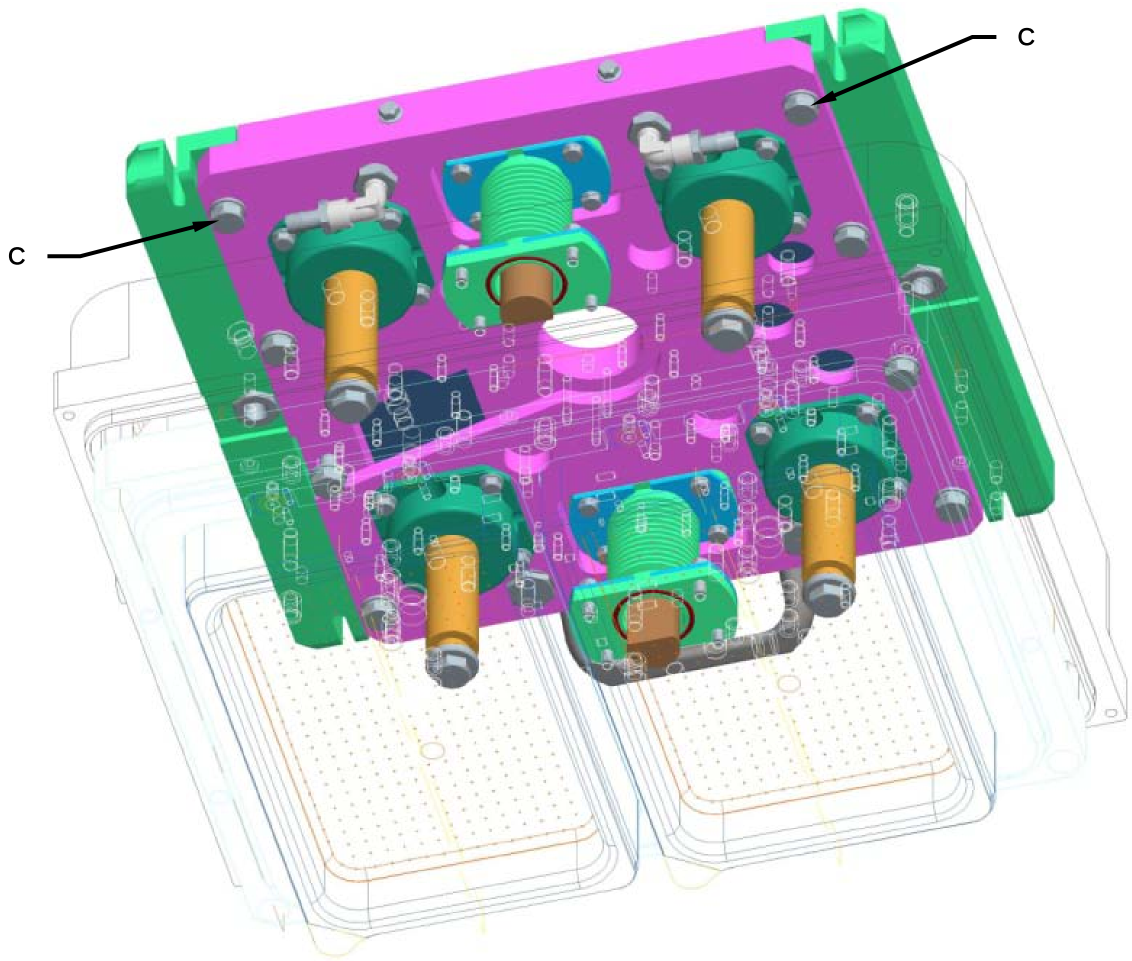
G. Mondini S.p.A. is not responsible for any damage caused by untrained and uninformed maintenance personnel.

For a correct centering operation scrupulously perform the following operations:

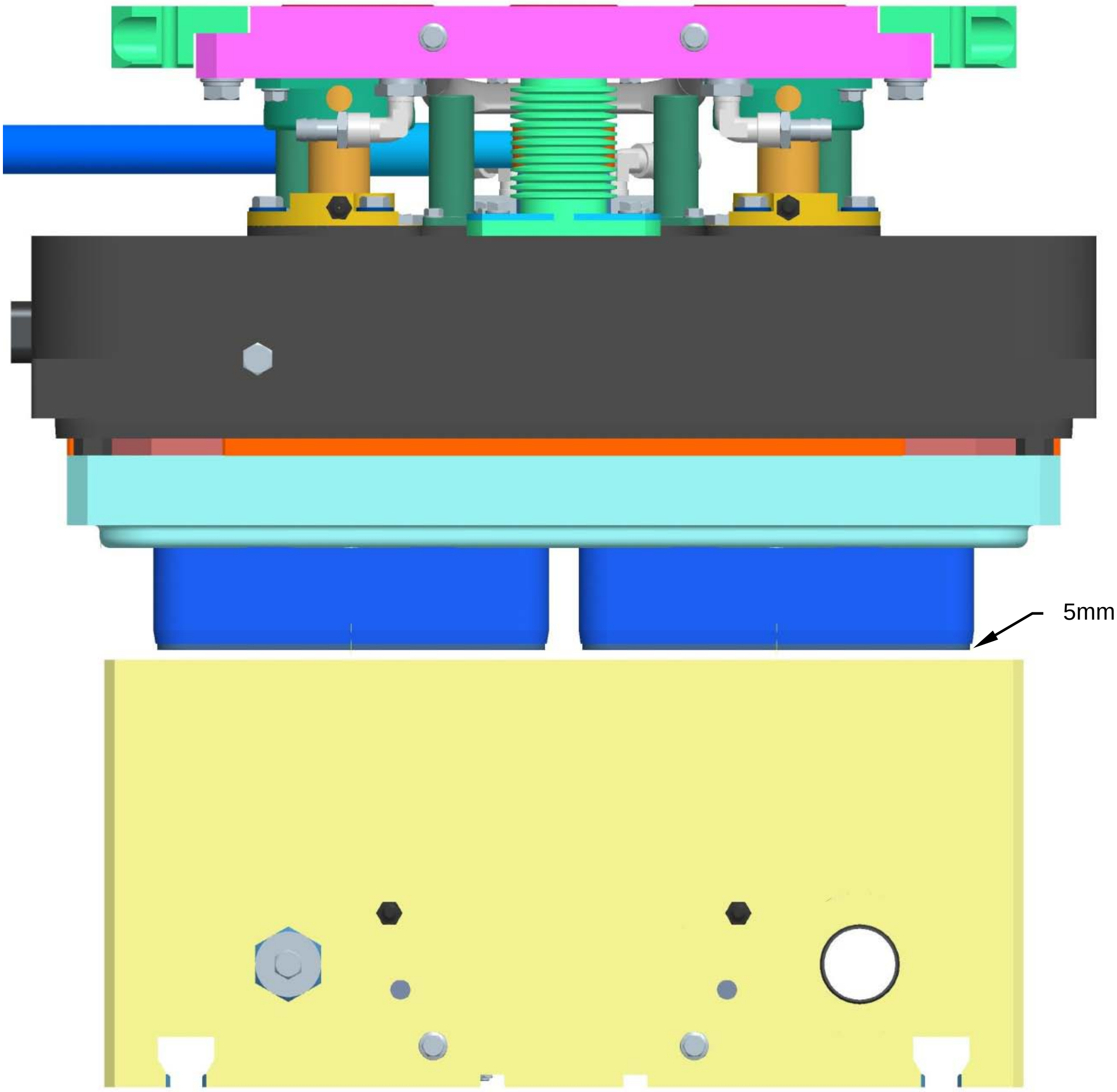
1. Lock the tool and heat up the soldering resistance to the operating temperature.
2. Remove the middle bell (**A**) and the blades (**B**) from the upper tool.



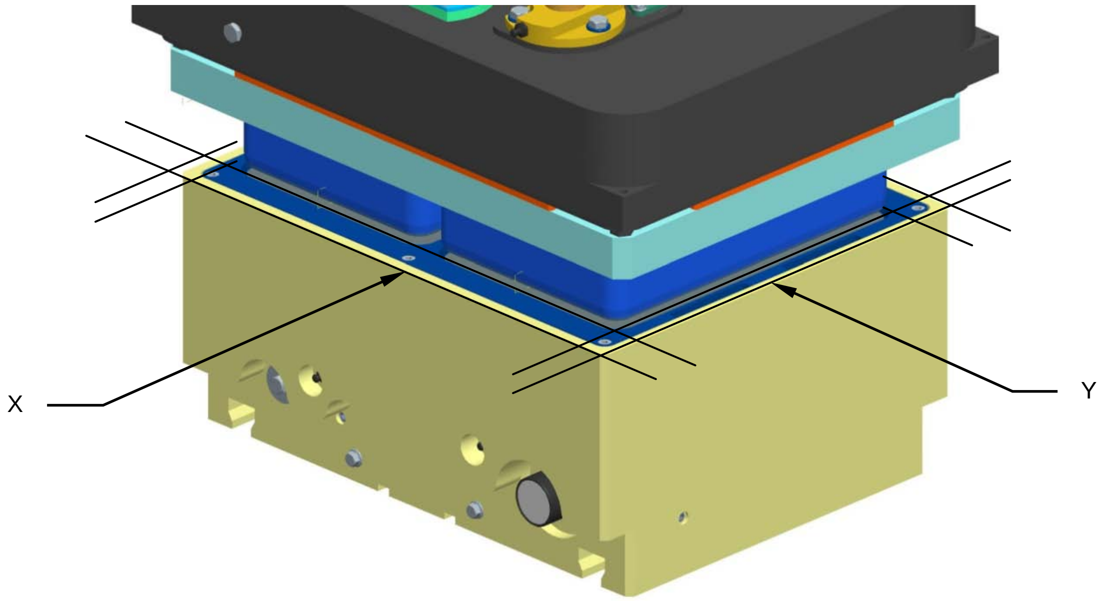
3. Loosen the screws (**C**) of the brackets.



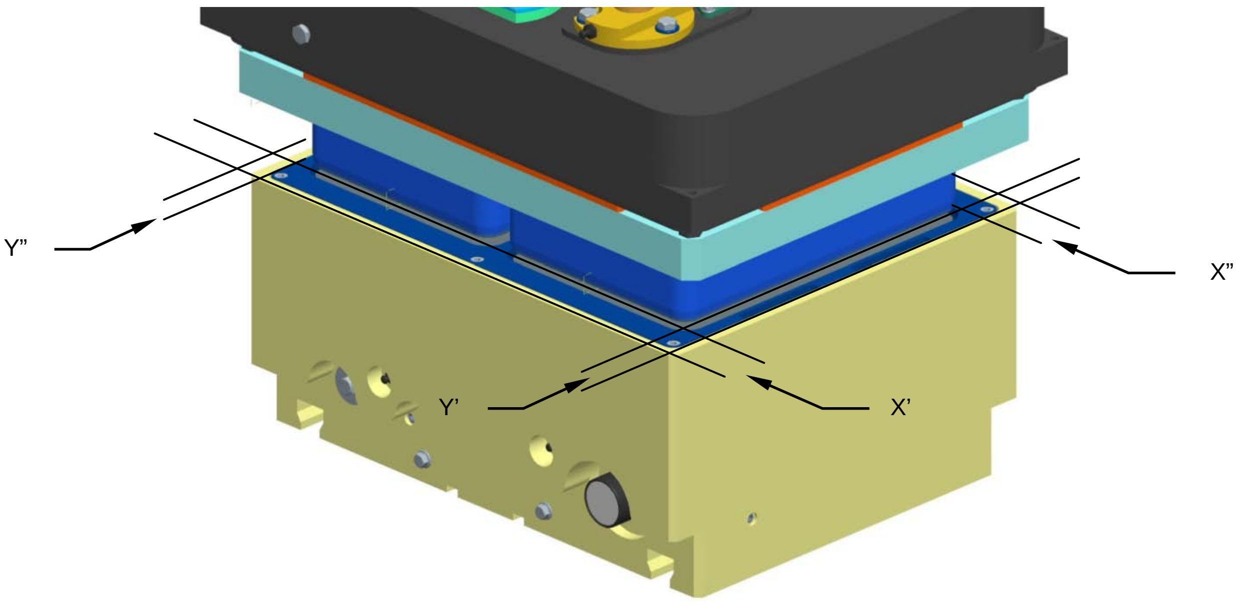
4. In "manual cycle", move the lower tool very slowly, up to a distance of about 5 mm.



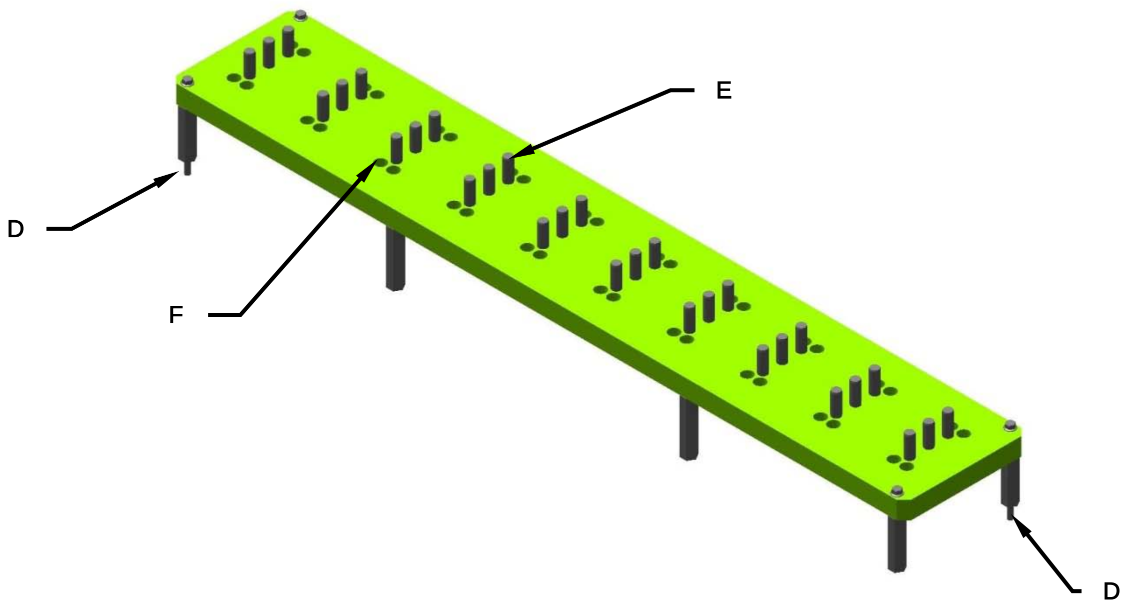
5. Measure with a caliber the distance between the outside of the lower bell and the plate of blades in the directions (**X**) and (**Y**).



6. Move the tool manually until the same dimensions are detected in all directions (**X'=X"; Y'=Y"**).



7. Lock the screws (**C**) of the brackets.
8. Lower the lower tool.
9. Loosen the fixing screws of the solders.
10. Insert the centering plate, on the counter-tool, centering with the appropriate reference pins (**D**).



11. Heat up the soldering resistance to the operating temperature.
12. Move the lower tool very slowly, until the centering pins (**E**) come into contact with the welding elements, centering each other.
13. Through the appropriate holes (**F**), tighten the fixing screws of the welding elements.
14. Retract with the lower tool and remove the centering plate.
15. Reposition the blades (**B**) and the middle bell (**A**).
16. Restore the "automatic cycle".

WARNING! The operations described above must be observed scrupulously, operating with the utmost care and complying with all the safety regulations.

7 DIAGNOSTICS

Malfunctions may occur the sealing cycle that prevent correct operation of the machine.

These malfunctions, referred to as ALARMS, are displayed promptly by the system on synoptic tables showing alarm messages and possible solutions.

ALARM	SOLUTION
0001 - System: At least one E-STOP pressed along the machine	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0002 - System: E-STOP pressed on Operator Panel	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0003 - System: E-STOP pressed at the film reel unwind	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0004 - System: E-STOP pressed on Denester - TraysStock	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0005 - System: E-STOP pressed on deviator	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0006 - System: E-STOP pressed along the feeding line	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0007 - System: E-STOP pressed on feeding line (end)	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0008 - System: Safety guard 1 closing machine OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0009 - System: Safety guard 2 closing machine OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0010 - System: Safety guard 3 closing machine OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0011 - System: Safety guard 4 closing machine OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0012 - System: E-STOP pressed on Moving Head	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0013 - System: E-STOP 1 pressed on the Snap on Lid	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0014 - System: E-STOP 2 pressed on the Snap on Lid	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0015 - System: E-STOP 4 pressed on closing machine exit	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0016 - System: STOP push button pressed on the Operator Panel	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0017 - System: STOP push button pressed at the film reel unwind	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.

0018 - System:	
0019 - System: STOP push button pressed at feeding line	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0020 - System: STOP push button pressed on the Snap On Lid	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0021 - System: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
0022 - System: Safety reset relay FAULT	- Reset the automatic circuit breaker. - Check the current absorption. - Check the power supply at the PLC outputs.
0023 - System: 24Vdc feeding line for the PLC outputs protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption. - Check the power supply at the operator panel.
0024 - System: 24Vdc feeding line for Operator Panel protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0025 - System: Auxiliary socket protection FAULT	- Reset the automatic circuit breaker of the cooling device. - Check the current absorption of the cooling device. - Check the power supply at the terminals of the cooling device.
0026 - System: Auxiliary cooling fan protection FAULT	- Reset the automatic circuit breaker of the cooling device. - Check the current absorption of the cooling device. - Check the power supply at the terminals of the cooling device.
0027 - System: Cabinet Air Conditioner protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0028 - System: Lubrication pump power protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0029 - System: Lubrication grease level FAULT	- Check the level of grease in the automatic greaser tank. - Reset the alarm via the relevant button on the operator panel.
0030 - System: Lubrication pressure FAULT	- Check the level of grease in the automatic greaser tank. - Reset the alarm via the relevant button on the operator panel.
0031 - System: Lubrication cycle ERROR	- Check the level of grease in the tank. - Check sensor operation.

0032 - System: MESSAGE to the operator to ask for lubrication when only manual	- Lubricate the machine via the nipples provided. - Use ROLOIL LITEX- EP/00 grease, referring to the instruction manual. - Effect the operation with the machine switched off. - Reset the counter on the operator panel (maintenance).
0033 - System: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine's requirements. - Check the air supply pipes.
0034 - System: Gas supply pressure FAULT	- Check the gas supply pressure
0035 - System: Gas Analyzer FAULT	- Check the gas mixture - Check the correct operation of the gas analyser.
0036 - System: Ethercat network FAULT	- Check the correct operation of the Ethercat net.
0037 - System: No communication with CVS internal I/O remote	- Check the correct operation of the in/out cards.
0038 - System: SDO messaging for the Remote I/O configuration FAULT	- Check remote I/O remote board assembly.
0039 - System: Recipe data transmission FAULT	- Check the Ethernet connection between the LENZE PLC and the B&R PC. - Repeat recipe transmission
0040 - System: NO communication with the Operator Panel	- Check the communication cable. - Press the emergency button and re-enter the commands. - Power off the machine and restart as usual.
0041 - System: NO consent from Downstream machine	- Restart the downstream line. - To operate the tray sealer with the downstream line stopped, switch the machine to autonomous operation via the selector on the operator panel.
0042 - System: NO consent from the gas mixer	- Check the gas mixer alarm.
0043 - System: Vacuum+Gas safeties bypass ACTIVE	- Check the gas mixer alarm.
0044 - System: New data value will be acepted at machine stop	- - - - -
0045 - System: CVS cycle not completed when new batch ready	- Check that the tray sealer cycle time is shorter than the tray entry time and reduce the sealing- vacuum- gas time. - Check the feed line speed. - Check the distance between trays.
0046 - System: Waiting for the Command Enable	- Await the commands abilitation of the machine.
0047 - System: Servodrives parametrization running	- - - - -

0048 - System: Pusher not in position to catch the batch	- Check the pusher position and press the button to reposition. - Check the position and efficiency of the control sensors. - Check that the motor speed does not exceed the maximum value, reducing if necessary as indicated in the instruction manual.
0049 - System: No communication with Deviator remote	- Check the correct wiring.
0050 - System: X2X communication FAULT	- Check the correct wiring.
0051 - System: Feedline remote power supply protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0052 - System: No communication with the feedline PLC	- Check the correct wiring.
0053 - System: Gas mixer/analyzer power protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0054 - System: Printer not ready	- Turn on or restore the printer.
0055 - System: Weigher not ready	- Turn on or restore the weigher.
0056 - System: X-Ray non ready	- Turn on or restore the X-Ray.
0057 - System: Stacher not ready	- Turn on or restore the stacher.
0058 - System: Digi Vision not ready	- Turn on or restore the Digi Vision.
0059 - System: Digi rejector 1 not ready	- Turn on or restore the Difi rejector 1.
0060 - System: Ravenwood not ready	- Turn on or restore the Ravenwood.
0061 - System: Digi rejector 2 not ready	- Turn on or restore the Digi rejector 2.
0062 - System: Waiting for Snap on Lid consens	- Check the good state and the operation of the Snap on lid.
0063 - System: Snap on Lid on delay	- Turn on or restore the Snap on lid.
0064 - InletBelt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0065 - InletBelt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0066 - InletBelt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0067 - InletBelt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0068 - InletBelt: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0069 - InletBelt: DP Pacer PEC timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0070 - InletBelt:	
0071 - InletBelt:	
0072 - InletBelt:	
0073 - InletBelt:	
0074 - InletBelt:	
0075 - InletBelt:	
0076 - InletBelt:	
0077 - InletBelt:	
0078 - InletBelt:	
0079 - InletBelt:	
0080 - Inlet Tray rotation - Belt 1: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0081 - Inlet Tray rotation - Belt 1: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0082 - Inlet Tray rotation - Belt 1: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0083 - Inlet Tray rotation - Belt 1: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0084 - Inlet Tray rotation - Belt 1: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0085 - Inlet Tray rotation - Belt 2: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0086 - Inlet Tray rotation - Belt 2: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0087 - Inlet Tray rotation - Belt 2: Motor motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0088 - Inlet Tray rotation - Belt 2: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0089 - Inlet Tray rotation - Belt 2: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.

0090 - Inlet Tray rotation: Pacer photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0091 - InletBelt:	
0092 - InletBelt:	
0093 - InletBelt:	
0094 - InletBelt:	
0095 - InletBelt:	
0096 - SpacerBelt: Servodrive power supply line protection FAULT	Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0097 - SpacerBelt: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0098 - SpacerBelt: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0099 - SpacerBelt 2: Servodrive power supply line protection FAULT	Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0100 - SpacerBelt 2: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0101 - SpacerBelt 2: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0102 - SpacerBelt:	
0103 - SpacerBelt:	
0104 - SpacerBelt:	
0105 - SpacerBelt:	
0106 - SpacerBelt: Trays positioning error	- Check tray position below the sealing tool. - Check the control sensors.
0107 - SpacerBelt 2: Trays positioning error	- Check tray position below the sealing tool. - Check the control sensors.
0108 - SpacerBelt:	
0109 - SpacerBelt:	
0110 - SpacerBelt:	
0111 - SpacerBelt: Trays counter wrong set point	- Check the recipe parameters.

0112 - SpacerBelt: Trays inlet onto the belt too fast	- Check that the trays arrive correctly spaced. - Check the feed line speed.
0113 - SpacerBelt: Starting PEC timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0114 - SpacerBelt 2: Trays inlet onto the belt too fast	- Check that the trays arrive correctly spaced. - Check the feed line speed.
0115 - SpacerBelt 2: Starting PEC timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0116 - SpacerBelt: Belt test running	- - - - -
0117 - SpacerBelt - Bypass belt: Mounting error or not plugged in	- Check the correct insertion of the connector.
0118 - SpacerBelt - Bypass belt: Still mounted on the machine	- Check the mounting of the ByPass belt on the machine. - Check the control sensors.
0119 - SpacerBelt - Bypass belt: Inverter power supply FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0120 - SpacerBelt - Bypass belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0121 - SpacerBelt - Bypass belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0122 - SpacerBelt - Bypass belt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0123 - SpacerBelt - Bypass belt: Disconnect switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0124 - SpacerBelt:	
0125 - SpacerBelt:	
0126 - SpacerBelt:	
0127 - SpacerBelt:	
0128 - Pusher Arm: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0129 - Pusher Arm: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0130 - Pusher Arm: Servodrive communication FAULT	- Check the cables of the ETHERCAT network - Switch off and restart.
0131 - Pusher Arm:	
0132 - Pusher Arm: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0133 - Pusher Arm: Axis HOME missing	- Perform a home position search cycle.
0134 - Pusher Arm: Axis HOMING PROCEDURE running	- Perform the HOME cycle.
0135 - Pusher Arm: HOME procedure TIMEOUT	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0136 - Pusher Arm: Axis FWD overtravel	- Check the control sensor.
0137 - Pusher Arm: Axis BWD overtravel	- Check the control sensor.
0138 - Pusher Arm: Axis following error	- Check there are no obstacles to motor or transmission organ operation. - Check encoder connection.
0139 - Pusher Arm: Condition missing while running homing	- Restore the condition to perform the home cycle.
0140 - Pusher: Crashed into the tool!	- Check there are no obstacles to motor or transmission organ operation.
0141 - Pusher:	
0142 - Pusher: Cam syncronization missing!	- Perform home cycles.
0143 - Pusher: Movement settings error! Change the parameters	- Check the transmission organs. - Check the cam parameter setting.
0144 - Pusher Arm: Wrong movement parameters	- Correct the cam parameter setting.
0145 - Pusher Arm: Requested acceleration or speed too high!	- Correct the cam parameter setting.

0146 - Pusher Arm: Hands not opened before moving back! parameters error!	- Correct the cam parameter setting.
0147 - Pusher Arm: Fwd position not reached	- Correct the cam parameter setting.
0148 - Pusher Arm: Bwd position not reached	- Correct the cam parameter setting.
0149 - Pusher: Tool change repositioning - active	- - - - -
0150 - Pusher - Hand 1: HYBRIC prop movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0151 - Pusher - Hand 2: HYBRIC prop movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0152 - Pusher - Hand 1: SLIMFRESH stopper movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0153 - Pusher - Hand 2: SLIMFRESH stopper movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0154 - Pusher:	
0155 - Pusher:	
0156 - Pusher:	
0157 - Pusher:	
0158 - Pusher: Tool MUST be down to allow the cycle	- Lower the tool
0159 - Pusher Arm: Maintenance requested	- Refer to chapter 5 - MAINTENANCE.
0160 - Pusher Hand 1: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0161 - Pusher Hand 1: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0162 - Pusher Hand 1: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0163 - Pusher Hand 1:	

0164 - Pusher Hand 1: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0165 - Pusher Hand 1: Axis HOME missing	- Perform a home position search cycle.
0166 - Pusher Hand 1: Axis HOMING PROCEDURE running	- - - - -
0167 - Pusher Hand 1: HOME procedure TIMEOUT	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0168 - Pusher Hand 1: Axis FWD overtravel	- Check the control sensor.
0169 - Pusher Hand 1: Axis BWD overtravel	- Check the control sensor.
0170 - Pusher Hand 1: Axis out of position	- Check the transmission organs. - Repeat the home cycles.
0171 - Pusher Hand 1: Condition missing while running homing	- Restore the condition to perform the home cycle.
0172 - Pusher Hand 1: Wrong movement parameters	- Correct the cam parameter setting.
0173 - Pusher Hand 1: Requested acceleration or speed too high!	- Correct the cam parameter setting.
0174 - Pusher Hand 1: Close position not reached	- Correct the cam parameter setting.
0175 - Pusher Hand 1: Open position not reached	- Correct the cam parameter setting.
0176 - Pusher Hand 2: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0177 - Pusher Hand 2: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0178 - Pusher Hand 2: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0179 - Pusher Hand 2:	
0180 - Pusher Hand 2: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0181 - Pusher Hand 2: Axis HOME missing	- Perform a home position search cycle.
0182 - Pusher Hand 2: Axis HOMING PROCEDURE running	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0183 - Pusher Hand 2: HOME procedure TIMEOUT	- Check the control sensor.

0184 - Pusher Hand 2: Axis FWD overtravel	- Check the control sensor.
0185 - Pusher Hand 2: Axis BWD overtravel	- Check the transmission organs. - Repeat the home cycles.
0186 - Pusher Hand 2: Axis out of position	- Restore the condition to perform the home cycle.
0187 - Pusher Hand 2: Condition missing while running homing	- Restore the condition to perform the home cycle.
0188 - Pusher Hand 2: Wrong movement parameters	- Correct the cam parameter setting.
0189 - Pusher Hand 2: Requested acceleration or speed too high!	- Correct the cam parameter setting.
0190 - Pusher Hand 2: Close position not reached	- Correct the cam parameter setting.
0191 - Pusher Hand 2: Open position not reached	- Correct the cam parameter setting.
0192 - Film unwind: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0193 - Film unwind: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0194 - Film unwind: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0195 - Film unwind:	
0196 - Film unwind: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0197 - Film unwind: Axis out of position	- Check the transmission organs. - Repeat the home cycles.
0198 - Film waste rewind: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0199 - Film waste rewind: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0200 - Film waste rewind: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0201 - Film waste rewind: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0202 - Film waste rewind: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0203 - Film reel unwind 1: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0204 - Film reel unwind 1: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0205 - Film reel unwind 1: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0206 - Film reel unwind 1: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0207 - Film reel unwind 1: Disconnect switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.

0208 - Film reel unwind 2: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0209 - Film reel unwind 2: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0210 - Film reel unwind 2: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0211 - Film reel unwind 2: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0212 - Film reel unwind 2: Disconnect switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0213 - Film reel unwind 1: Reel almost empty	- Reel of film finished, replace with a new one. - Check the control sensors.
0214 - Film reel unwind 2: Reel almost empty	- Reel of film finished, replace with a new one. - Check the control sensors.
0215 - Film:	
0216 - Film unwind 2: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0217 - Film unwind 2: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0218 - Film unwind 2: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.

0219 - Film unwind 2: Servodrive enable FAULT	
0220 - Film:	
0221 - Film:	
0222 - Film:	
0223 - Film:	
0224 - Film: Waste broken	- Check the notch search photocell. - Check film positioning.
0225 - Film: Register mark missed	- Check the set unwinding length is not excessive. - With printed film, check the approach cycle is performed correctly.
0226 - Film: The machine cycle is waiting for the film unwind cycle	- Reset and repeat the cycle.
0227 - Film: Cycle NOT done	- Reset and repeat the cycle.
0228 - Film unwind: Locking valve OFF	- Activate the relevant pneumatic selector.
0229 - Film waste rewind: Locking valve OFF	- Activate the relevant pneumatic selector.
0230 - Film reel unwind 1: Locking valve OFF	- Activate the relevant pneumatic selector.
0231 - Film reel unwind 1: End of reel	- Reel of film finished, replace with a new one. - Check the control sensors.
0232 - Film reel unwind 2: Locking valve OFF	- Activate the relevant pneumatic selector.
0233 - Film reel unwind 2: End of reel	- Reel of film finished, replace with a new one. - Check the control sensors.
0234 - Film reel unwind: Timeout	- Check the parameters setting.
0235 - Film reel automatic change: Reel change in progress	- - - - -
0236 - Film reel automatic change: Not ready	- Reset the reel-change guard.
0237 - Film: Film end	- Reel of film finished, replace with a new one. - Check the control sensors.
0238 - Film reel automatic change: Film junction detected	- - - - -
0239 - Film reel automatic change: Locking valve OFF	- Activate the relevant pneumatic selector.
0240 - Film reel automatic change: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
0241 - Film reel automatic change: Safety guard OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0242 - Film: Mirabella detector supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0243 - Film: Mirabella detector not ready	- Turn on or restore the "Mirabella detector".
0244 - Film: Mirabella film missing or upset	- Check the correct insertion of the sealing film among the rolls. - Check the control sensor.
0245 - Film - Movable roller: Safety guard OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0246 - Film - Movable roller: Axis high position overtravel	- Check the parameters setting. - Check the control sensors.
0247 - Film:	
0248 - Film: Hollow punch blade not on High position	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0249 - Film: Hollow punch blade movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0250 - Film:	
0251 - Film:	
0252 - Film:	
0253 - Film:	
0254 - Film:	
0255 - Film:	
0256 - Tool: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0257 - Tool: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0258 - Tool: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0259 - Tool: Brake resistor FAULT (chopper)	- Power off the machine. - Check the state of the braking resistance.

0260 - Tool: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0261 - Tool: Axis HOME missing	- Perform a home position search cycle.
0262 - Tool: Axis HOMING PROCEDURE running	- - - - -
0263 - Tool: HOME procedure TIMEOUT	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0264 - Tool: Axis HIGH overtravel	- Check the control sensor.
0265 - Tool: Axis LOWER overtravel	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0266 - Tool: Motor cooling fan protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0267 - Tool: Motor mechanical brake FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0268 - Tool: Upper heating 1 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0269 - Tool: Upper heating 1 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
0270 - Tool: Upper heating 1 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0271 - Tool: Upper heating 2 power line protection FAULT	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0272 - Tool: Upper heating 2 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
0273 - Tool: Upper heating 2 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0274 - Tool: Lower heating 1 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0275 - Tool: Lower heating 1 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.

0276 - Tool: Lower heating 1 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0277 - Tool: Upper cavities heating power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0278 - Tool: Upper cavities heating temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0279 - Tool: SKIN 1 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0280 - Tool: SKIN 1 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0281 - Tool: SKIN 2 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0282 - Tool: SKIN 2 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0283 - Tool: Collision with the pushers!	- Check there are no obstacles to motor or transmission organ operation.
0284 - Tool: Upper exit side not locked	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.
0285 - Tool: Upper inlet side not locked	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.
0286 - Tool: Lower exit side not locked	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.
0287 - Tool: Lower inlet side not locked	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.

0288 - Tool: Locking valve OFF	- Check the tool position - turn the pneumatic selector on the control panel to lock it. - Check control sensor operation and position.
0289 - Tool: Axis wrong position	- Check the transmission organs. - Repeat the home cycles.
0290 - Tool: No cooling water	- Check the correct operation of the cooling water circuit.
0291 - Tool: Vacuum pump motor protection FAULT-Overload	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0292 - Tool: Vacuum valves timeout. Open carter and remove film	- Open carter and remove film.
0293 - Tool: Booster vacuum pump motor protection FAULT-Overload	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0294 - Tool: Liquid separator cycle running	- - - - -
0295 - Tool: Trays collision in the tool	- Remove trays from the tool. - Check photocell operation and position.
0296 - Tool: Vacuum pressure FAULT	- Check vacuum cycle parameter settings. - Check vacuum circuit seal. - Check the pressure transducer.
0297 - Tool: Gas pressure FAULT	- Check gas cycle parameter settings. - Check gas circuit seal. - Check the pressure transducer.
0298 - Tool: Vacuum-gas cycle ERROR	- Check the parameters setting.
0299 - Tool: Sealing force set point not reached	- Check the parameters setting.
0300 - Tool: Maintenance requested	- See chapter 5 - MAINTENANCE.
0301 - Tool: Washing Cycle running	- - - - -
0302 - Tool: Vacuum test running	- - - - -
0303 - Tool - High Oxigen surveillance: Test running	- - - - -
0304 - Tool - High Oxigen surveillance: Upper gas valve closing timeout	- Check the good state of the sensor and the valve.
0305 - Tool - High Oxigen surveillance: Lower gas valve 1A closing timeout	- Check the good state of the sensor and the valve.
0306 - Tool - High Oxigen surveillance: Lower gas valve 1B closing timeout	- Check the good state of the sensor and the valve.
0307 - Tool - High Oxigen surveillance: Lower gas valve 2A closing timeout	- Check the good state of the sensor and the valve.

0308 - Tool - High Oxigen surveillance: Lower gas valve 2B closing timeout	- Check the good state of the sensor and the valve.
0309 - Tool - High Oxigen surveillance: Lower compensation valve closing anomaly	- Check the good state of the sensor and the valve.
0310 - Tool - High Oxigen surveillance: Upper compensation valve closing anomaly	- Check the good state of the sensor and the valve.
0311 - Tool - High Oxigen surveillance: Skin compensation valve closing anomaly	- Check the good state of the sensor and the valve.
0312 - Tool - High Oxigen surveillance: Lower vacuum valve closing timeout	- Check the good state of the sensor and the valve.
0313 - Tool - High Oxigen surveillance: Upper vacuum valve closing timeout	- Check the good state of the sensor and the valve.
0314 - Tool - High Oxigen surveillance: Skin vacuum valve closing timeout	- Check the good state of the sensor and the valve.
0315 - Tool - High Oxigen surveillance: MAIN vacuum valve closing timeout	- Check the good state of the sensor and the valve.
0316 - Tool: No upper cooling water	- Check the correct operation of the cooling water circuit.
0317 - Tool: Sealing force reached before sealing position	- Check the parameters setting.
0318 - Tool - High Oxigen surveillance: Vacuum and gas valves opening conflict	- Check the good state of the sensor and the valve.
0319 - Tool: Film clamp open/close timeout	- Check the parameters setting. - Check photocell operation and position.
0320 - ExitBelt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0321 - ExitBelt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0322 - ExitBelt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0323 - ExitBelt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0324 - ExitBelt: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0325 - ExitBelt 2: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0326 - ExitBelt 2: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0327 - ExitBelt 2: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0328 - ExitBelt 2: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0329 - ExitBelt 2: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.

0330 - ExitBelt 4->2: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0331 - ExitBelt 4->2: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0332 - ExitBelt 4->2: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0333 - ExitBelt 4->2: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0334 - ExitBelt 4->2: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0335 - ExitBelt 4->2: Belts jammed	- Clear the belts of trays. - Check photocell operation and position.
0336 - ExitBelt: Belts jammed	- Clear the belts of trays. - Check photocell operation and position.
0337 - ExitBelt:	
0338 - ExitBelt 2 -> 1: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0339 - ExitBelt 2 -> 1: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0340 - ExitBelt 2 -> 1: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0341 - ExitBelt 2 -> 1: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0342 - ExitBelt 2 -> 1: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0343 - ExitBelt - Transport: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0344 - ExitBelt - Transport: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0345 - ExitBelt - Transport: Motor thermistor FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0346 - ExitBelt - Transport: Inverter communication FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0347 - ExitBelt - Transport: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0348 - ExitBelt:	
0349 - ExitBelt:	
0350 - ExitBelt:	
0351 - ExitBelt:	
0352 - Deviator: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0353 - Deviator: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0354 - Deviator: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0355 - Deviator:	
0356 - Deviator:	
0357 - Deviator: Axis home missing	- Perform a home position search cycle.
0358 - Deviator: Home procedure running	- - - - -
0359 - Deviator: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0360 - Deviator: Max left position	- Check the deviator movement parameters. - Check the sensor. - Repeat the home cycle.
0361 - Deviator: Max right position	- Check the deviator movement parameters. - Check the sensor. - Repeat the home cycle.
0362 - Deviator: Axis following error	- Check the transmission organs. - Repeat the home cycles.
0363 - Deviator: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.

0364 - Deviator - Belt 2->1: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0365 - Deviator - Belt 2->1: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0366 - Deviator - Belt 2->1: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0367 - Deviator - Belt 2->1: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0368 - Deviator - Belt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0369 - Deviator - Belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0370 - Deviator - Belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.

0371 - Deviator - Belt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0372 - Deviator - Belt: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0373 - Deviator:	
0374 - Deviator:	
0375 - Deviator:	
0376 - Deviator: Tray ejected	- - - - -
0377 - Deviator: Photocells timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0378 - Deviator - Belt 2->1: Photocells timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0379 - Deviator:	
0380 - Deviator: Safety guards open (from safety module)	
0381 - Deviator: Safety guard OPEN	- Check the safety control unit in the absence of other guard alarms.
0382 - Deviator - DP pacer: Timeout	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0383 - Deviator - Belt 2->1: Belt jammed	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0384 - Line: Servodrive power supply line protection FAULT	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.

0385 - Line: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0386 - Line: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0387 - Line: Axis home missing	- Perform a home position search cycle.
0388 - Line: Home procedure running	- - - - -
0389 - Line: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0390 - Line: Axis following error	- Check the transmission organs. - Repeat the home cycles.
0391 - Line: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0392 - Line: Tray missing on the feeding line	- Load the trays on the line.
0393 - Line: Motor cooling fan protection FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0394 - Line: Timeout -shift register-PEC way 1	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0395 - Line: Timeout -shift register-PEC way 2	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0396 - Line: Timeout -shift register-PEC way 3	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0397 - Line: Timeout -shift register-PEC way 4	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.

0398 - Line: NO consent from CVS	- Check the good state and the correct operation of the CVS.
0399 - Line: Moving Head Hopper not in position	- Check transmission parts. - Execute the home cycles.
0400 - Line - Vibrator: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0401 - Line - Vibrator: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0402 - Line - Vibrator: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0403 - Line - Vibrator: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0404 - Line - Vibrator: Motor Disconect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0405 - Line: Trays missing at the CVS entrance	- Load the trays. - Check the correct operation of the control sensor.
0406 - Line: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine’s requirements. - Check the air supply pipes.
0407 - Line: Product belt inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0408 - Line: Product belt inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0409 - Line: Product belt motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0410 - Line: Product belt inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0411 - Line: Product belt motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0412 - Line: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
0413 - Line: Line-CVS passage safety guard OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0414 - Line: Error in chain movement parameters	- Check the respective parameters setting.
0415 - Line: Wait restore control	- Await the commands restoration.
0416 - Denester 1: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0417 - Denester 1: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0418 - Denester 1: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0419 - Denester 1:	
0420 - Denester 1: Axis home missing	- Perform a home position search cycle.
0421 - Denester 1: Home procedure running	- - - - -

0422 - Denester 1: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0423 - Denester 1: Axis high position overtravel	- Check the control sensor.
0424 - Denester 1: Axis low position overtravel	- Check the control sensor.
0425 - Denester 1: Axis following error	- Check the transmission organs. - Repeat the home cycles.
0426 - Denester 1: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0427 - Denester 1: Trays missing	- Load the trays inside the denester. - Check the correct operation of the control sensor.
0428 - Denester 1: Traslator timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0429 - Denester 1: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
0430 - Denester 1: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.
0431 - Denester 1: Vacuum pump motor protection FAULT-Overload	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0432 - Denester 1: Vacuum pump NOT connected	- Check the good state and the correct connection of the vacuum pump.
0433 - Denester 1: Safety guard 5 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0434 - Denester 1: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0435 - Denester 1: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0436 - Denester 1: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0437 - Denester 1: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0438 - Denester 1: Tray discharge area obstucted	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0439 - Denester 1 - Belt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0440 - Denester 1 - Belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0441 - Denester 1 - Belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0442 - Denester 1 - Belt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0443 - Denester 1 - Belt: Pacer photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.

0444 - Denester 1: E-STOP 1 pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0445 - Denester 1: E-STOP 2 pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0446 - Denester 1: Error in movement parameters	- Check the recipe parameters.
0447 - Denester 1: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine’s requirements. - Check the air supply pipes.
0448 - Denester 2: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0449 - Denester 2: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0450 - Denester 2: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0451 - Denester 2:	
0452 - Denester 2: Axis home missing	- Perform a home position search cycle.
0453 - Denester 2: Home procedure running	- - - - -
0454 - Denester 2: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0455 - Denester 2: Axis high position overtravel	- Check the control sensor.
0456 - Denester 2: Axis low position overtravel	- Check the control sensor.
0457 - Denester 2: Axis following error	- Check the transmission organs. - Repeat the home cycles.
0458 - Denester 2: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0459 - Denester 2: Trays missing	- Load the trays inside the denester. - Check the correct operation of the control sensor.

0460 - Denester 2: Traslator timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0461 - Denester 2: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
0462 - Denester 2: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.
0463 - Denester 2: Vacuum pump motor protection FAULT-Overload	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0464 - Denester 2: Vacuum pump NOT connected	- Check the good state and the correct connection of the vacuum pump.
0465 - Denester 2:	
0466 - Denester 2: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0467 - Denester 2: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0468 - Denester 2: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0469 - Denester 2: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0470 - Denester 2: Tray discharge area obstucted	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.

0471 - Denester 2 - Belt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0472 - Denester 2 - Belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0473 - Denester 2 - Belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0474 - Denester 2 - Belt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0475 - Denester 2 - Belt: Pacer photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0476 - Denester 2: E-STOP 1 pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0477 - Denester 2: E-STOP 2 pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0478 - Denester 2: Error in movement parameters	- Check the recipe parameters.

0479 - Denester 2: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine's requirements. - Check the air supply pipes.
0480 - Lid Preformed: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0481 - Lid Preformed: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0482 - Lid Preformed: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0483 - Lid Preformed:	
0484 - Lid Preformed: Axis home missing	- Perform a home position search cycle.
0485 - Lid Preformed: Home procedure running	- - - - -
0486 - Lid Preformed: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0487 - Lid Preformed: Fwd position overtravel	- Check the control sensor.
0488 - Lid Preformed: Bwd position overtravel	- Check the control sensor.
0489 - Lid Preformed: Axis following error	- Check the transmission organs. - Repeat the home cycles.
0490 - Lid Preformed: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0491 - Lid Preformed: Lids missing	- Load the Lids inside the denester. - Check the correct operation of the control sensor.
0492 - Lid Preformed: Traslator timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0493 - Lid Preformed: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
0494 - Lid Preformed: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.

0495 - Lid Preformed: Vacuum pump motor protection FAULT-Overload	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0496 - Lid Preformed: Vacuum pump NOT connected	- Check the good state and the correct connection of the vacuum pump.
0497 - Lid Preformed: Crashed into the tool!	- Check the safety quota of the preformed device and that there are no obstacles to the movement of the preformed.
0498 - Lid Preformed: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0499 - Lid Preformed: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0500 - Lid Preformed: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0501 - Lid Preformed: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0502 - Lid Preformed: Lid lost on the transport	- Check the good state of the suckers and the vacuum circuit. - Check the correct operation of the tray selectors.
0503 - Lid Preformed: Lid lost on the tool	- Check the good state of the suckers and the vacuum circuit.
0504 - Lid Preformed: Transport must be at pick up position	- Check there are no obstacles to the movement of the preformed.
0505 - Lid Preformed: The lower skin plate must be completely back!!!	- Position the plate in safety position.
0506 - Lid Prefor./DARFRESH: Hooks for the tool change not in backward position	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0507 - Lid: Safe pos. sensor doesn't meet Axis position	- Check the correct operation of the control sensor.

0508 - Lid Preformed: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine's requirements. - Check the air supply pipes.
0509 - Shuttle: Timeout of the vacuum on the shuttle	- Check the good state of the sensor and the valve.
0510 - Lid Preformed:	
0511 - Lid Preformed:	
0512 - Doser 1: Not supplied	- Check the power supply and the correct wiring of the device.
0513 - Doser 1: Not ready	- Check the good state of the doser 1.
0514 - Doser 1: Dosing phase too long!!!	- Check the cycle time of the doser 1.
0515 - Doser 2: Not ready	- Check the good state of the doser 2.
0516 - Doser 2: Dosing phase too long!!!	- Check the cycle time of the doser 2.
0517 - Doser 3: Not ready	- Check the good state of the doser 3.
0518 - Doser 3: Dosing phase too long!!!	- Check the cycle time of the doser 3.
0519 - Doser 4: Not ready	- Check the good state of the doser 4.
0520 - Doser 4: Dosing phase too long!!!	- Check the cycle time of the doser 4.
0521 - Doser 5: Not ready	- Check the good state of the doser 5.
0522 - Doser 5: Dosing phase too long!!!	- Check the cycle time of the doser 5.
0523 - Doser 6: Not ready	- Check the good state of the doser 6.
0524 - Doser 6: Dosing phase too long!!!	- Check the cycle time of the doser 6.
0525 - Doser:	
0526 - Doser:	
0527 - Doser:	
0528 - Doser: Hoppers lifter movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0529 - Doser: Hoppers flap movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0530 - Doser: Product deviator movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0531 - Doser: Waiting product ready from ISHIDA doser	- - - - -
0532 - Doser: Fault or product wait too long !!! Run 'HOPPERS RESTORE' cycle	- Run 'HOPPERS RESTORE' cycle

0533 - Doser: Product into the hoppers timeout!!! Run 'HOPPERS RESTORE' cycle	- Run 'HOPPERS RESTORE' cycle
0534 - Doser 2: Not supplied	- Check the power supply and the correct wiring of the device.
0535 - Doser 1: E-STOP pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0536 - Doser 2: E-STOP pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0537 - Doser:	
0538 - Doser:	
0539 - Doser:	
0540 - Doser:	
0541 - Doser:	
0542 - Doser:	
0543 - Doser:	
0544 - MHead 1: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0545 - MHead 1: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0546 - MHead 1: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0547 - MHead 1:	
0548 - MHead 1: Axis home missing	- Perform a home position search cycle.
0549 - MHead 1: Home procedure running	- - - - -
0550 - MHead 1: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0551 - MHead 1: Fwd position overtravel	- Check the control sensor.
0552 - MHead 1: Bwd position overtravel	- Check the control sensor.
0553 - MHead 1: Axis following error	- Check the transmission organs. - Repeat the home cycles.

0554 - MHead 1: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0555 - MHead 1:	
0556 - MHead 1:	
0557 - MHead 1:	
0558 - MHead 1: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.
0559 - MHead 1:	
0560 - MHead 1:	
0561 - MHead 1: Collision with the chain	- Check the correct movement of the hopper.
0562 - MHead 1: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0563 - MHead 1: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0564 - MHead 1: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0565 - MHead 1: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0566 - MHead 1 - Doser: E-STOP pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0567 - MHead 1 - Doser: Safety guard OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0568 - MHead 1 - Doser: NOT connected	- Check the correct wiring of the mobile head 1.
0569 - MHead 1 - Doser: Not ready	- Check the good state and the correct operation of the mobile head 1.

0570 - MHead 1: Dosing phase too long	- Check the dosing time of the mobile head 1.
0571 - MHead 1 - Hopper: Movement timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
0572 - MHead 1 - Hopper: Fillling timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
0573 - MHead 1: Product missing	- Load the product. - Check the correct operation of the control sensor.
0574 - MHead 1 - Shutter: Movement timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
0575 - MHead 1:	
0576 - TraysStock - Buffer belt: Inverter power supply line protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0577 - TraysStock - Buffer belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0578 - TraysStock - Buffer belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0579 - TraysStock - Buffer belt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0580 - TraysStock - Buffer belt: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.

0581 - TraysStock - Translator: Inverter power supply line protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0582 - TraysStock - Translator: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0583 - TraysStock - Translator: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0584 - TraysStock - Translator: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0585 - TraysStock - Translator: Motor Disconect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0586 - TraysStock - Translator: Fwd position overtravel	- Check the control sensor.
0587 - TraysStock - Translator: Bwd position overtravel	- Check the control sensor.
0588 - TraysStock - Translator: Fwd movement locked	- Check there are no obstacles to the trays translator movement.
0589 - TraysStock - Translator: Anomaly position transductor	- Check the correct operation and the connections of the transductor.
0590 - TraysStock - Translator: Timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0591 - TraysStock: Load photocell error	- Check the correct load of the trays.
0592 - TraysStock: Stocks missing	- Load the trays

0593 - TraysStock: Shutter flap timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0594 - TraysStock: Trays batches sequence ERROR	- Check the respective parameters setting. - Check the control sensor.
0595 - TraysStock - Translator: Batches translating ERROR	- Check the respective parameters setting. - Check the control sensor.
0596 - TraysStock: Belt positioning timeout	- Check the respective parameters setting. - Check the control sensor.
0597 - TraysStock - Translator: Not in position	- Check the respective parameters setting. - Check the control sensor.
0598 - TraysStock: Belt positioning ERROR	- Check the respective parameters setting. - Check the control sensor.
0599 - TraysStock:	
0600 - TraysStock:	
0601 - TraysStock:	
0602 - TraysStock:	
0603 - TraysStock:	
0604 - TraysStock:	
0605 - TraysStock:	
0606 - TraysStock:	
0607 - TraysStock:	
0608 - MHead 2: Servodrive power supply line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0609 - MHead 2: Servodrive FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0610 - MHead 2: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0611 - MHead 2:	
0612 - MHead 2: Axis home missing	- Perform a home position search cycle.
0613 - MHead 2: Home procedure running	- - - - -
0614 - MHead 2: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0615 - MHead 2: Fwd position overtravel	- Check the control sensor.

0616 - MHead 2: Bwd position overtravel	- Check the control sensor.
0617 - MHead 2: Axis following error	- Check the transmission organs. - Repeat the home cycles.
0618 - MHead 2: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0619 - MHead 2:	
0620 - MHead 2:	
0621 - MHead 2:	
0622 - MHead 2: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.
0623 - MHead 2:	
0624 - MHead 2:	
0625 - MHead 2: Collision with the chain	- Check the correct movement of the hopper.
0626 - MHead 2: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0627 - MHead 2: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0628 - MHead 2: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0629 - MHead 2: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0630 - MHead 2 - Doser: E-STOP pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0631 - MHead 2:	
0632 - MHead 2 - Doser: NOT connected	- Check the correct wiring of the mobile head 2.
0633 - MHead 2 - Doser: Not ready	- Check the good state and the correct operation of the mobile head 2.

0634 - MHead 2: Dosing phase too long	- Check the dosing time of the mobile head 2.
0635 - MHead 2 - Hopper: Movement timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
0636 - MHead 2 - Hopper: Fillling timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
0637 - MHead 2:	
0638 - MHead 2:	
0639 - MHead 2:	
0640 - Tool: Upper heating 1 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
0641 - Tool: Upper heating 2 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
0642 - Tool: Lower heating 1 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
0643 - Tool: Horizontal skin heating PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
0644 - Tool: Vertical skin heating PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
0645 - Tool: Telescopic guide not in position	- Check the telescopic guide position. - Check the correct operation of the control sensors.
0646 - Tool: Upper section not on HIGH position	- Check the parameters setting. - Check sensors operation and position.
0647 - Tool: Upper section movement timeout	- Check the timeout setting. - Check any sensors.
0648 - Tool: Vacuum pump motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0649 - Tool: Vacuum pump 2 motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0650 - Tool: Vacuum pump 3 motor protection FAULT-Overload	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.

0651 - Tool: Vacuum pump 3 motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0652 - Tool: Lower heating 2 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0653 - Tool: Lower heating 2 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
0654 - Tool: Lower heating 2 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0655 - Tool: Lower heating 2 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
0656 - System: Cycle can't start! JOG movement still active	- The machine is in JOG modality.
0657 - System: Exit chain BUSY! Pusher can't start	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0658 - System: Printer power supply protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0659 - System: Weigher power supply protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0660 - System: Antistaic device power supply protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0661 - System: Antistaic device not ready	- Check the antistatic device.
0662 - System: Gas Analyzer - Remove connection pipe	- Remove connection pipe.
0663 - System: CPU I/O ERROR	- Reboot the machine.
0664 - Line - DP Pacer: Photocell 1 timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0665 - Line - DP Pacer: Photocell 2 timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.

0666 - Line - DP Pacer: Trays min level photocell 1 timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0667 - Line - DP Pacer: Trays min level photocell 2 timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0668 - Line: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0669 - Line: Trays stopper timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0670 - Line:	
0671 - Line:	
0672 - Super Protruding: Heating 1 power line protection FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check the power supply at the PLC exit .
0673 - Super Protruding: Heating 1 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0674 - Super Protruding: Heating 1 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
0675 - Super Protruding: Heating 2 power line protection FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check the power supply at the PLC exit .
0676 - Super Protruding: Heating 2 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
0677 - Super Protruding: Heating 2 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
0678 -	
0679 -	
0680 -	
0681 -	
0682 - Super Protruding: Lock between Cut Group and Plates still closed	- Lock Cut Group and Plates still. - Check the correct operation of the control sensors.

0683 -	
0684 -	
0685 -	
0686 -	
0687 - DARFRESH: Film preparation not done	- Make the film preparation.
0688 - DARFRESH: Cutting blades not on HIGH position	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0689 - DARFRESH: Format change running	- - - - -
0690 - DARFRESH: Timeout of the Lids presence in the tool	- Removing portions of film from the tool plates using a suitable tool. **DO NOT PLACE HANDS IN WORKING PLATES AREA AND IN THE BLADES!** - Check any sensors.
0691 - DARFRESH: Film preparation clamp CLOSED	- Open the clamp whit the button.
0692 - DARFRESH: Film sheets not ready on the plates	- Check the correct insertion of the film. - Check the correct operation of the control sensors.
0693 - DARFRESH: Plates movement timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0694 - DARFRESH: Plates not in position to cut	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0695 - DARFRESH: Plates lids timeout	- Check the piston position. - Check there are no obstacles to the piston movement. - Check the correct operation of the control sensors.
0696 - DARFRESH: Locking valve OFF	- Close locking valve whit the button.
0697 - DARFRESH: Compressed air for pneumatic movement OFF	- Check for proper connection of pneumatic fittings. - Switch compressed air whit the button.
0698 - DARFRESH: Film not ready	- Check the correct insertion of the sealing film among the rolls. - Check the control sensor. - Reset and repeat the cycle.
0699 - DARFRESH: Shuttle pre-heating cycle running	- Notice message.
0700 - DARFRESH: Shuttle pre-heating cycle halted before finishing	- Notice message.
0701 -	
0702 -	

0703 - Lid Preformed: Vacuum pump 2 motor protection FAULT-Overload	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0704 - Line - Feeding buffer belt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0705 - Line - Feeding buffer belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0706 - Line - Feeding buffer belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0707 - Line - Feeding buffer belt: Inverter communication FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0708 - Line - Feeding buffer belt: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0709 - Line: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0710 - Line: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0711 - Line - Chain load belts: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0712 - Line - Chain load belts: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0713 - Line - Chain load belts: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0714 - Line - Chain load belts: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0715 - Line - Chain load belts: Motor Disconnect Switch OPENED	- Make sure no one is working on the machine and reactivate the knife switch.
0716 -	
0717 -	
0718 -	
0719 -	
0720 - Line: Timeout PEC at CVS entrance way 1	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0721 - Line: Timeout PEC at CVS entrance way 2	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.

0722 - Line: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0723 - Line: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0724 - Line: Safety guard 5 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0725 - Line: EMPTYING mode active	- - - - -
0726 -	
0727 - Line: Not enough trays on chain to complete batch loading	- Load the trays on the line.
0728 - ExitBelt:	
0729 - ExitBelt:	
0730 - ExitBelt:	
0731 - ExitBelt:	
0732 - ExitBelt:	
0733 - ExitBelt:	
0734 - ExitBelt:	
0735 - ExitBelt:	
0736 - ExitBelt:	
0737 - ExitBelt:	
0738 - ExitBelt:	
0739 - ExitBelt:	
0740 - ExitBelt:	
0741 - ExitBelt:	
0742 - ExitBelt:	
0743 - ExitBelt:	
0744 - Tool - High Oxigen surveillance: Test cycle request	- Run the valve test cycle.
0745 - Tool - High Oxigen surveillance: RESET push button Anomaly	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0746 - Tool - High Oxigen surveillance: SafetyRelayAnomaly	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.

0747 - Tool - High Oxigen surveillance: TestRelayAnomaly	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0748 - Tool - High Oxigen surveillance: Lower vacuum command anomaly	- Check the good state of the sensor and the valve.
0749 - Tool - High Oxigen surveillance: Lower vacuum opening timeout	- Check the good state of the sensor and the valve.
0750 - Tool - High Oxigen surveillance: Upper vacuum command anomaly	- Check the good state of the sensor and the valve.
0751 - Tool - High Oxigen surveillance: Upper vacuum opening timeout	- Check the good state of the sensor and the valve.
0752 - Tool - High Oxigen surveillance: SKIN vacuum command anomaly	- Check the good state of the sensor and the valve.
0753 - Tool - High Oxigen surveillance: SKIN vacuum opening timeout	- Check the good state of the sensor and the valve.
0754 - Tool - High Oxigen surveillance: Lower gas valve 1A command anomaly	- Check the good state of the sensor and the valve.
0755 - Tool - High Oxigen surveillance: Lower gas valve 1A opening timeout	- Check the good state of the sensor and the valve.
0756 - Tool - High Oxigen surveillance: Lower gas valve 1B command anomaly	- Check the good state of the sensor and the valve.
0757 - Tool - High Oxigen surveillance: Lower gas valve 1B opening timeout	- Check the good state of the sensor and the valve.
0758 - Tool - High Oxigen surveillance: Lower gas valve 2A command anomaly	- Check the good state of the sensor and the valve.
0759 - Tool - High Oxigen surveillance: Lower gas valve 2A opening timeout	- Check the good state of the sensor and the valve.
0760 - Tool - High Oxigen surveillance: Lower gas valve 2B command anomaly	- Check the good state of the sensor and the valve.
0761 - Tool - High Oxigen surveillance: Lower gas valve 2B opening timeout	- Check the good state of the sensor and the valve.
0762 - Tool - High Oxigen surveillance: MAIN vacuum valve command anomaly	- Check the good state of the sensor and the valve.
0763 - Tool - High Oxigen surveillance: MAIN vacuum valve opening timeout	- Check the good state of the sensor and the valve.

0764 - Tool - Vacuum: Circuit tight FAULT	- Check the good state of the circuit.
0765 - Tool - Vacuum: Main valve tight FAULT	- Check the good state of the valve.
0766 - Tool - Gas: Circuit tight FAULT	- Check the good state of the circuit.
0767 - Tool - Gas: Main valve tight FAULT	- Check the good state of the valve.
0768 - Tool - High Oxigen surveillance:	
0769 - Tool - High Oxigen surveillance:	
0770 - Tool: G shape border bender out of position	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0771 - Tool - Skin Plate: Movemnt timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0772 - Tool - Skin Plate: Clamp timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0773 - Tool: It's not allowed to select two gas at the same time	- Select a gas type.
0774 - Tool: Gas change requested	- Change the gas type.
0775 - Tool: G shape collar opening timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0776 - Snap on Lid - Slide: Servodrive power supply line protection FAULT	Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0777 - Snap on Lid - Slide: Servodriver FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0778 - Snap on Lid - Slide: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0779 - Snap on Lid - Slide:	
0780 - Snap on Lid - Slide: Axis home missing	- Perform a home position search cycle.
0781 - Snap on Lid - Slide: Home procedure running	- Perform the HOME cycle.

0782 - Snap on Lid - Slide: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0783 - Snap on Lid - Slide: Fwd position overtravel	- Check the control sensor.
0784 - Snap on Lid - Slide: Bwd position overtravel	- Check the control sensor.
0785 - Snap on Lid - Slide: Axis following error	- Check there are no obstacles to motor or transmission organ operation. - Check encoder connection.
0786 - Snap on Lid - Slide: Axis movement timeout	- Check there are no obstacles to motor or transmission organ operation. - Check encoder connection.
0787 - Snap on Lid - Slide: Lid store empty	- Load the lids inside the denester. - Check the correct operation of the control sensor.
0788 - Snap on Lid - Slide: Lid catcher movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0789 - Snap on Lid: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
0790 - Snap on Lid - Slide: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.
0791 - Snap on Lid - Slide: Vacuum pump motor protection FAULT-Overload	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0792 - Snap on Lid: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine's requirements. - Check the air supply pipes.
0793 - Snap on Lid: Collision between Tool and Slide	- Check there are no obstacles to motor or transmission organ operation.
0794 - Snap on Lid: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0795 - Snap on Lid: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

0796 - Snap on Lid: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0797 - Snap on Lid: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0798 - Snap on Lid - Slide: Lid lost	- Check the good state of the suckers and the vacuum circuit. - Check the correct operation of the tray selectors.
0799 - Snap on Lid - Tool: Lid lost	- Check the good state of the suckers and the vacuum circuit. - Check the correct operation of the tray selectors.
0800 - Snap on Lid - Tool: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.
0801 - Snap on Lid - Exit belt: Photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0802 - Snap on Lid - Tool: Driver FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0803 - Snap on Lid - Tool: Driver communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0804 - Snap on Lid - Tool: Home missing	- Perform a home position search cycle.
0805 - Snap on Lid - Tool: Homing running	- Perform the HOME cycle.
0806 - Snap on Lid - Tool: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0807 - Snap on Lid - Tool: Movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.

0808 - Snap on Lid - Chain: Servodrive power supply line protection FAULT	Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
0809 - Snap on Lid - Chain: Servodriver FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.
0810 - Snap on Lid - Chain: Servodrive communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
0811 - Snap on Lid - Chain: Axis home missing	- Perform a home position search cycle.
0812 - Snap on Lid - Chain: Home procedure running	- Perform the HOME cycle.
0813 - Snap on Lid - Chain: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0814 - Snap on Lid - Chain: Axis following error	- Check there are no obstacles to motor or transmission organ operation. - Check encoder connection.
0815 - Snap on Lid - Chain: Axis movement timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0816 - Snap on Lid - Chain: Trays missing	- Load the trays on the line.
0817 - Snap on Lid - Chain: Not ready	- Turn on or restore the snap on lid machine.
0818 - Snap on Lid - Chain:	
0819 - Snap on Lid - Chain:	
0820 - Snap on Lid - Chain:	
0821 - Snap on Lid - Chain:	
0822 - Snap on Lid - Chain:	
0823 - Snap on Lid: Emptying cycle running	- - - - -
0824 - Snap on Lid - Exit belt: Power supply protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0825 - Snap on Lid - Exit belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0826 - Snap on Lid:	
0827 - Snap on Lid:	

0828 - Snap on Lid - Exit belt 1: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0829 - Snap on Lid - Exit belt 1: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0830 - Snap on Lid - Exit belt 1: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0831 - Snap on Lid - Exit belt 1:	
0832 - Snap on Lid - Exit belt 2: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0833 - Snap on Lid - Exit belt 2: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0834 - Snap on Lid - Exit belt 2: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0835 - Snap on Lid - Exit belt 2:	
0836 - Snap on Lid:	
0837 - Snap on Lid:	
0838 - Snap on Lid:	
0839 - Snap on Lid:	
0840 - Snap on Lid:	
0841 - Snap on Lid:	
0842 - Snap on Lid:	
0843 - Snap on Lid:	

0844 - Snap on Lid:	
0845 - Snap on Lid:	
0846 - Snap on Lid:	
0847 - Snap on Lid:	
0848 - Snap on Lid:	
0849 - Snap on Lid:	
0850 - Snap on Lid:	
0851 - Snap on Lid:	
0852 - Snap on Lid:	
0853 - Snap on Lid:	
0854 - Snap on Lid:	
0855 - Snap on Lid:	
0856 - Snap on Lid:	
0857 - Snap on Lid:	
0858 - Snap on Lid:	
0859 - Snap on Lid:	
0860 - Snap on Lid:	
0861 - Snap on Lid:	
0862 - Snap on Lid:	
0863 - Snap on Lid:	
0864 - Snap on Lid:	
0865 - Snap on Lid:	
0866 - Snap on Lid:	
0867 - Snap on Lid:	
0868 - Snap on Lid:	
0869 - Snap on Lid:	
0870 - Snap on Lid:	
0871 - Snap on Lid:	
0872 - Meat Line - Curve belt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0873 - Meat Line - Curve belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0874 - Meat Line - Curve belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.

0875 - Meat Line - Curve belt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0876 - Meat Line - Trays belt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0877 - Meat Line - Trays belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0878 - Meat Line - Trays belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0879 - Meat Line - Trays belt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0880 - Meat Line - Meat belt: Inverter power supply protection FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0881 - Meat Line - Meat belt: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.

0882 - Meat Line - Meat belt: Motor thermistor FAULT	- Reset the motor circuit breaker. - Check motor current absorption. - Check there are no obstacles to motor operation. - Check the power supply at the motor terminals.
0883 - Meat Line - Meat belt: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0884 - Meat Line: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine’s requirements. - Check the air supply pipes.
0885 - Meat Line: Aspirator power supply protection FAULT	- Reset the automatic circuit breaker. - Check the current absorption.
0886 - Meat Line:	
0887 - Meat Line:	
0888 - Meat Line - Meat belt: Photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0889 - Meat Line - Trays belt: Photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0890 - Meat Line: Meat frequency too high	- Check the recipe parameters.
0891 - Meat Line: Tray not ready for the meat load	- Check the recipe parameters.
0892 - Meat Line: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
0893 - Meat Line: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0894 - Meat Line - Trays belt: Buffer photocell timeout	- Remove the tray in front of the photocell. - Check that the set alarm time is compatible with the current tray type. - Check operation on the photocell and retro- reflector.
0895 - Meat Line:	

0896 - Meat Line:	
0897 - Meat Line - Meat belt: E-STOP pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
0898 - Meat Line:	
0899 - Meat Line:	
0900 - Meat Line:	
0901 - Meat Line:	
0902 - Meat Line:	
0903 - Meat Line:	
0904 - Overturner - Rotation: Servo power supply protection fault	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.
0905 - Overturner - Rotation: Servodriver FAULT	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.
0906 - Overturner - Rotation: Servodriver communication FAULT	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.
0907 - Overturner - Rotation: Enabling error	
0908 - Overturner - Rotation: Home missing	
0909 - Overturner - Rotation: Home procedure running	
0910 - Overturner - Rotation: Home procedure timeout	
0911 - Overturner - Rotation: Fwd position overtravel	
0912 - Overturner - Rotation: Bwd position overtravel	
0913 - Overturner - Rotation: Following error	
0914 - Overturner - Rotation: Movement timeout	
0915 - Overturner - Rotation:	
0916 - Overturner - Rotation:	
0917 - Overturner - Belt 1: Inverter power supply protection FAULT	
0918 - Overturner - Belt 1: Inverter FAULT	
0919 - Overturner - Belt 1: Motor thermistor FAULT	
0920 - Overturner - Belt 1: Inverter communication FAULT	

0921 - Overturner - Belt 1:	
0922 - Overturner - Belt 1:	
0923 - Overturner - Belt 2: Inverter power supply protection FAULT	
0924 - Overturner - Belt 2: Inverter FAULT	
0925 - Overturner - Belt 2: Motor thermistor FAULT	
0926 - Overturner - Belt 2: Inverter communication FAULT	
0927 - Overturner: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0928 - Overturner: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0929 - Overturner: Cycle in delay	
0930 - Overturner: Cycle can't start! Entrance photocell busy	
0931 - Overturner: Safety guards open (from safety module)	
0932 - Overturner: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0933 - Overturner: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
0934 - Overturner: Photocell timeout	
0935 - Overturner: Evacuation timeout	
0936 - Denester 1: Lubrification cycle needed	- Make the lubrification cycle.
0937 - Denester 1: Piston revision needed	- Make the piston revision.
0938 -	
0939 -	
0940 - Denester 1 - Deviator Flap: Right side of the -delivery belt- full or not working	
0941 - Denester 1 - Deviator Flap: Left side of the -delivery belt- full or not working	

0942 - Denester 1 - Deviator Flap: Following Error	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0943 - Denester 1 - Deviator Flap: Movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
0944 - Denester 1 - Deviator Flap: Servo power supply protection fault	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.
0945 - Denester 1 - Deviator Flap: Servodriver FAULT	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.
0946 - Denester 1 - Deviator Flap: Servodriver communication FAULT	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.
0947 - Denester 1 - Deviator Flap: Home Missing	- Perform a home position search cycle.
0948 - Denester 1 - Deviator Flap: Home procedure running	- - - - -
0949 - Denester 1 - Deviator Flap: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
0950 - Denester 1 - Deviator Flap: Fwd position overtravel	- Check the control sensor.
0951 - Denester 1 - Deviator Flap: Bwd position overtravel	- Check the control sensor.
0952 - Denester 2: Lubrification cycle needed	- Make the lubrification cycle.
0953 - Denester 2: Piston revision needed	- Make the piston revision.
0954 -	
0955 -	
0956 -	
0957 -	
0958 -	
0959 -	
0960 -	
0961 -	
0962 -	
0963 -	
0964 -	
0965 -	
0966 -	

0967 -	
0968 - Tool - High Oxigen surveillance: SKIN vacuum 2 closing timeout	- Check the good state of the sensor and the valve.
0969 - Tool - High Oxigen surveillance: SKIN vacuum 2 opening timeout	- Check the good state of the sensor and the valve.
0970 - Tool - High Oxigen surveillance: SKIN vacuum 2 command anomaly	- Check the good state of the sensor and the valve.
0971 - Tool - High Oxigen surveillance: Upper gas valve opening timeout	- Check the good state of the sensor and the valve.
0972 - Tool - High Oxigen surveillance: Upper gas valve command anomaly	- Check the good state of the sensor and the valve.
0973 - Tool: Vaccum pump speeding up!	- - - - -
0974 - Tool - Vacuum Pump 1: Compressed air supply missing	- Check for proper connection of pneumatic fittings. - Switch compressed air whit the button.
0975 - Tool - Vacuum Pump 1: Cooling water supply missing	- Check the correct operation of the cooling water circuit. - Check for proper connection of fittings. - Switch water circuit whit the button.
0976 - Tool - Vacuum Pump 1: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0977 - Tool - Vacuum Pump 1: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0978 - Tool - Vacuum Pump 1: Cleaning phase	- - - - -
0979 - Tool - Vacuum Pump 1: Speeding up ...	- - - - -
0980 -	
0981 -	
0982 -	
0983 -	

0984 - Tool - Vacuum Pump 2: Compressed air supply missing	- Check for proper connection of pneumatic fittings. - Switch compressed air whit the button.
0985 - Tool - Vacuum Pump 2: Cooling water supply missing	- Check the correct operation of the cooling water circuit. - Check for proper connection of fittings. - Switch water circuit whit the button.
0986 - Tool - Vacuum Pump 2: Inverter FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0987 - Tool - Vacuum Pump 2: Inverter communication FAULT	- Check the error code on the inverter display. - Check the automatic circuit breaker on the inverter power supply. - Check motor absorption. - Check there are no obstacles to motor operation. - Check the inverter power supply.
0988 - Tool - Vacuum Pump 2: Cleaning phase	- - - - -
0989 - Tool - Vacuum Pump 2: Speeding up …	- - - - -
0990 -	
0991 -	
0992 -	
0993 -	
0994 -	
0995 -	
0996 -	
0997 -	
0998 -	
0999 -	
1000 - Vertical Store: Servo power supply protection fault	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.
1001 - Vertical Store: Servodriver FAULT	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.
1002 - Vertical Store: Servodriver communication FAULT	- Check the error code on the drive display. - Check the automatic circuit breaker on the drive power supply. - Check the drive power supply.

1003 - Vertical Store:	
1004 - Vertical Store: Home missing	- Perform a home position search cycle.
1005 - Vertical Store: Home procedure running	- Perform the HOME cycle.
1006 - Vertical Store: Home procedure timeout	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
1007 - Vertical Store: High position overtravel	- Check the control sensor.
1008 - Vertical Store: Low position overtravel	- Check the control sensor.
1009 - Vertical Store: Following error	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
1010 - Vertical Store: Movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
1011 - Vertical Store: Compressed air supply pressure FAULT	- Check the air supply pressure, which must not be below 5.5 bar. - Check the pressure is sufficient to meet the machine’s requirements. - Check the air supply pipes.
1012 - Vertical Store: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
1013 - Vertical Store: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1014 - Vertical Store: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1015 - Vertical Store: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1016 - Vertical Store: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

1017 - Vertical Store:	
1018 - Vertical Store:	
1019 - Vertical Store: E-STOP 1 pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
1020 - Vertical Store: E-STOP 2 pressed	- Release the emergency button by turning clockwise. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the main control panel.
1021 - Vertical Store:	
1022 - Vertical Store:	
1023 - Vertical Store: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.
1024 - Vertical Store: Upper Shutter flap movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
1025 - Vertical Store: Lower Shutter flap movement timeout	- Check the correct movement of the piston. - Check the pneumatic low regulators. - Check the timeout parameter setting. - Check the trays are correctly lined up.
1026 - Vertical Store:	
1027 - Vertical Store:	
1028 - Vertical Store:	
1029 - Vertical Store:	
1030 - Vertical Store:	
1031 - Vertical Store:	
1032 - MicVac - Dispenser: Not Ready	- Turn on the MicVac - Dispenser.
1033 - MicVac - Dispenser: Label missing	- Replace the labels roller. - Check the correct operation of the control sensors.
1034 - MicVac - Dispenser: Timeout	- Check the correct operation of the control sensors.
1035 - MicVac:	
1036 - MicVac:	
1037 - MicVac - Linear Unit: Not Ready	- Turn on the MicVac - Linear Unit.
1038 - MicVac - Linear Unit: Timeout	- Check the correct operation of the control sensors.
1039 - MicVac - Linear Unit: Label Roll Empty	- Replace the labels roller. - Check the correct operation of the control sensors.
1040 - MicVac - Linear Unit: Reference Not OK	- Check the MicVac - Linear Unit.
1041 - MicVac:	
1042 - MicVac:	

1043 - MicVac:	
1044 - MicVac:	
1045 - MicVac: Data saving timeout	- Check the MicVac cables. - Restart the MicVac.
1046 - MicVac: New data saving running	- Wait.
1047 - MicVac: Power supply protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1048 - MicVac: Safety guards open (from safety module)	- Check the safety control unit in the absence of other guard alarms.
1049 - MicVac: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1050 - MicVac: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1051 - MicVac: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1052 - MicVac: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1053 - Micvac:	
1054 - Micvac:	
1055 - Micvac:	
1056 - Micvac:	
1057 - Micvac:	
1058 - Micvac:	
1059 - Micvac:	
1060 - Micvac:	
1061 - Micvac:	
1062 - Micvac:	
1063 - Micvac:	
1064 - Tray Lifter:	
1065 - Tray Lifter: Servodriver FAULT	- Check motor current absorption. - Check there are no obstacles to motor operation. - Check the cables linking the motor and servo (power and encoder). - Check the motor temperature.

1066 - Tray Lifter: Servodriver communication FAULT	- Check the cables of the ETHERCAT network. - Switch off and restart.
1067 - Tray Lifter:	- Perform a home position search cycle.
1068 - Tray Lifter: Home missing	- - - - -
1069 - Tray Lifter: Home procedure running	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
1070 - Tray Lifter: Home procedure timeout	- Check the control sensor.
1071 - Tray Lifter: High position overtravel	- Check the control sensor.
1072 - Tray Lifter: Low position overtravel	- Check the transmission organs. - Repeat the home cycles.
1073 - Tray Lifter: Following error	- Check there are no obstacles to motor operation. - Check the timeout setting. - Check any sensors.
1074 - Tray Lifter: Movement timeout	- Perform a home position search cycle.
1075 - Tray Lifter:	
1076 - Tray Lifter:	
1077 - Tray Lifter: Safety guards open (from safety module)	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1078 - Tray Lifter: Cycle in delay	- Check that the set cycle time is compatible with the rest of the machine.
1079 - Tray Lifter:	
1080 - Tray Lifter:	
1081 - Tray Lifter:	
1082 - Tray Lifter: Safety guard 1 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1083 - Tray Lifter: Safety guard 2 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1084 - Tray Lifter: Safety guard 3 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.

1085 - Tray Lifter: Safety guard 4 OPEN	- Check that the guards are closed and the microswitches are active. - Reset the alarm by pressing the memory reset button. - Check operation of the safety relay inside the control panel.
1086 - Tray Lifter:	
1087 - Tray Lifter:	
1088 - Tray Lifter:	
1089 - Tray Lifter:	
1090 - Tray Lifter:	
1091 - Tray Lifter:	
1092 - Tray Lifter:	
1093 - Tray Lifter:	
1094 - Tray Lifter: Error in chain movement parameters	- Check the parameters setting.
1095 - Tray Lifter:	
1096 - Doser 1: Dosing timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
1097 - Doser 2: Dosing timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
1098 - Doser 3: Dosing timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
1099 - Doser 4: Dosing timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
1100 - Doser 5: Dosing timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
1101 - Doser 6: Dosing timeout	- Check there are no obstacles to motorisation operation. - Check the timeout setting. - Check any sensors.
1102 -	
1103 -	
1104 -	
1105 -	
1106 -	
1107 -	
1108 -	
1109 -	
1110 -	
1111 -	
1112 -	
1113 -	

1114 -	
1115 -	
1116 -	
1117 -	
1118 -	
1119 -	
1120 -	
1121 -	
1122 -	
1123 -	
1124 -	
1125 -	
1126 -	
1127 -	
1128 - DARFRESH: Heating plate 1/2/3/4 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1129 - DARFRESH: Heating plate 1 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1130 - DARFRESH: Heating plate 1 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1131 - DARFRESH: Heating plate 1 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1132 - DARFRESH: Heating plate 2 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1133 - DARFRESH: Heating plate 2 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1134 - DARFRESH: Heating plate 2 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1135 - DARFRESH: Heating plate 2 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1136 - DARFRESH: Heating plate 3 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1137 - DARFRESH: Heating plate 3 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1138 - DARFRESH: Heating plate 3 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.

1139 - DARFRESH: Heating plate 3 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1140 - DARFRESH: Heating plate 4 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1141 - DARFRESH: Heating plate 4 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1142 - DARFRESH: Heating plate 4 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1143 - DARFRESH: Heating plate 4 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1144 - DARFRESH: Heating plate 5/6/7/8 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1145 - DARFRESH: Heating plate 5 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1146 - DARFRESH: Heating plate 5 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1147 - DARFRESH: Heating plate 5 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1148 - DARFRESH: Heating plate 6 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1149 - DARFRESH: Heating plate 6 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1150 - DARFRESH: Heating plate 6 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1151 - DARFRESH: Heating plate 6 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1152 - DARFRESH: Heating plate 7 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1153 - DARFRESH: Heating plate 7 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.

1154 - DARFRESH: Heating plate 7 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1155 - DARFRESH: Heating plate 7 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1156 - DARFRESH: Heating plate 8 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1157 - DARFRESH: Heating plate 8 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1158 - DARFRESH: Heating plate 8 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1159 - DARFRESH: Heating plate 8 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1160 - DARFRESH: Heating plate 9/10/11/12 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1161 - DARFRESH: Heating plate 9 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1162 - DARFRESH: Heating plate 9 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1163 - DARFRESH: Heating plate 9 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1164 - DARFRESH: Heating plate 10 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1165 - DARFRESH: Heating plate 10 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1166 - DARFRESH: Heating plate 10 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1167 - DARFRESH: Heating plate 10 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1168 - DARFRESH: Heating plate 11 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.

1169 - DARFRESH: Heating plate 11 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1170 - DARFRESH: Heating plate 11 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1171 - DARFRESH: Heating plate 11 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1172 - DARFRESH: Heating plate 12 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1173 - DARFRESH: Heating plate 12 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1174 - DARFRESH: Heating plate 12 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1175 - DARFRESH: Heating plate 12 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1176 - DARFRESH: Heating plate 13/14/15/16 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1177 - DARFRESH: Heating plate 13 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1178 - DARFRESH: Heating plate 13 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1179 - DARFRESH: Heating plate 13 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1180 - DARFRESH: Heating plate 14 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1181 - DARFRESH: Heating plate 14 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1182 - DARFRESH: Heating plate 14 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1183 - DARFRESH: Heating plate 14 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.

1184 - DARFRESH: Heating plate 15 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1185 - DARFRESH: Heating plate 15 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1186 - DARFRESH: Heating plate 15 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1187 - DARFRESH: Heating plate 15 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1188 - DARFRESH: Heating plate 16 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.
1189 - DARFRESH: Heating plate 16 SCR FAULT	- Reset the alarm by pressing the memory reset button. - Check actuator power supply.
1190 - DARFRESH: Heating plate 16 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1191 - DARFRESH: Heating plate 16 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1192 - System: Metal detector not ready	- Restart the machine. - Call the assistance.
1193 -	
1194 -	
1195 -	
1196 -	
1197 -	
1198 -	
1199 -	
1200: DARFRESH: Heating plate 17 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1201: DARFRESH: Heating plate 17 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1202: DARFRESH: Heating plate 18 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1203: DARFRESH: Heating plate 18 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.

1204: DARFRESH: Heating plate 19 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1205: DARFRESH: Heating plate 19 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1206: DARFRESH: Heating plate 20 temperature ERROR	- Check the alarm setting. - Check the thermoregulator is on. - Check operation of the thermal resistance and the heating resistances.
1207: DARFRESH: Heating plate 20 PT100 broken or misconnected	- Connect the connector. - Check the correct operation of the PT 100.
1208: DARFRESH: Heating plate 17/18/19/20 power line protection FAULT	- Reset the motor circuit breaker. - Check actuator current absorption. - Check the power supply at the terminals of the drive unit.

2034 - Panel: Texts too long	- Reduce the length of the texts in the panel.
2035 - Panel:	
2036 - Panel:	
2037 - Panel:	
2038 - Panel:	
2039 - Panel: Recipe sync error doser (ishida)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2040 - Panel: Recipe sync error printer (domino)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2041 - Panel: Recipe sync error weigher (digi)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2042 - Panel: Recipe sync error x-ray (loma)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2043 - Panel: Recipe sync error stacker (ulma)	- Missed alignment with third parts. - Check the connection, the working and the good state of the third parts.
2044 - Panel: Recipe fry date not accepted	Check Ethernet connection between PLC LENZE and PC B&R.
2045 - Panel: Max number of recipes reached	- Cancel the recipes not used for recovering free spaces.
2046 - Panel: Max alarm story log almost reached	- Cancel the alarm story list.
2047 - Panel: PLC ip cannot be reached check connection	

8 MAINTENANCE

INDEX

8.1 SAFETY PRECAUTIONS

8.1.1 Introduction

The personnel in charge of the machine operation and maintenance must be well trained and qualified and placed into positions that the customer requires, with an extensive knowledge of the safety and accident prevention standards and regulations in force.

Non-authorised personnel must strictly stay out of the work area during machine operations.

The safety and accident prevention warnings and precautions specified in this section must be accurately observed at all times during machine operation and maintenance, in order to avoid and prevent personal injuries as well as machine damages.

Said warnings and precautions are consistently referred to and further detailed throughout the Manual whenever there is a procedure involving injury or damage hazards, by way of **WARNING** and **DANGER** notes.

1. The **WARNING** notes normally precede an operation that, if not correctly performed, may incur in machine damages.
2. The **DANGER** notes normally precede an operation that, if not correctly performed, may lead to personal injuries.

DANGER
Before servicing the machine or performing any other intervention, always press the emergency button associated with the opening of the moving guards to disconnect the machine from the power supply. Check operation of this safety device regularly.

8.2 DANGER NOTES

- High voltage can cause death by electrocution. Operate at all times with maximum carefulness and in total compliance with the accident prevention standards and regulations in force in the country the machine is installed in.
- Prior to carrying out any operation at all on the machine, always ensure that all the main and secondary power and feed-in supplies have been totally cut-off. Hang specific warning signs such as “**MACHINE MAINTENANCE OPERATIONS UNDERWAY – DO NOT SWITCH ON THE POWER MAINS**” in the vicinity of the mains switch.
- During operations, moving machine parts all constitute a hazard and may lead to serious personal injuries. Strictly avoid coming into contact with said moving parts. Prior to starting any intervention at all on the machine, always ensure that there is no way that it can be started up by any one of the other components attached to it.
- Never remove the safety guards or cut off the protection devices installed on the machine.
- Failure to provide machine grounding may lead to serious personal injuries. Always make sure that the machine is accurately grounded and that all grounding operations are compliant with regulations in force.
- Avoid use of hazardous solvent products.
- Always ensure, prior to starting the machine up, that the personnel in charge of maintenance is at a safe distance from the machine and that no tools or materials have been left in the vicinity of the machine itself.
- Always use accident prevention and safety equipment during machine maintenance operations.
- Machine installation must always be updated and maintained according to all the accident prevention provisions in force. All machine moving parts and drives must be covered to prevent accidental contacts.
- Always check to make sure that all the machine safety guards and protection devices have been correctly closed prior to starting machine up.

- Extended overloading or failures may lead to electric motor and electrical equipment overheating, with the subsequent issue of hazardous fumes; in these cases, immediately power the machine down and do not approach the smoking areas until the smoke has been dispersed via adequate ventilation. Avoid inhaling the fumes forming internally to the machine devices during repair and maintenance interventions.
- Should the machine ever catch fire, never use water jets. Cut off all the power and feed-in supplies and use CO_2 fire extinguishers.
- Be very careful not to mix air and hydraulic oil when the system is under pressure. This to avoid explosive atmosphere hazards.
- Always ensure that the systems are all totally depressurized, prior to proceeding with any maintenance operation on the relative components.
- Keep well away from any discharge tap or hole during all system depressurising operations.
- Accurately inspect all the fittings and make sure there is no dust, oil, dirt or wear and tear on the threads, prior to proceeding with connecting them up.
- Make sure that all fittings and joints are accurately closed and tightened prior to pressurising the plant systems, after maintenance operations.
- Do not handle hydraulic fluid in the presence of electric sparks and/or naked flames.
- Always keep well away from any component that can be activated or moved by pressure, if the pressure contents have not been totally discharged from the plant systems. Do not wear objects that can tangle in with the machine and relative devices and act as conductors (chain necklaces, bracelets, etc.)
- **Ensure that all tools and equipment for use are always in perfect working conditions** and, where mandatory, that they are provided with **insulating handles and grips**. Check to ensure that the insulation on test equipment cables and conductors does not show the minimum sign of wear and tear, and/or damage.

8.3 WARNING NOTES

- Prior to putting the machine back into operation after a breakdown, it must be accurately inspected and controlled for evidence of any possible further damages.
- When using a ground detector to check any electrical device insulation, be sure to check that all the electronic control equipment is disconnected, to avoid relative component damages.
- Always use perfectly dry air during air pressure cleaning operations, with a pressure that does not go above 2 kg/cm^2.
- Always use tools and equipment that is in perfectly kept conditions and appropriate for the operations that need to be performed; the use of unsuitable tools and equipment may cause some serious damages.
- Any repair operations that may be possibly required, must be conducted in clean working environments, as free of dust as possible. Protect all the connection lights with plastic cover caps and keep dismantled pieces with machined surfaces accurately covered, until they are reassembled back onto the machine.
- Make sure that the lubrication system is working and distributing correctly at all times. Lubrication failures may seriously damage the sealing machine components.
- Never act on the adjustments and positioning of the overstroke or limit micro switches, unless expressly required to eliminate an anomaly. Tampering with them can lead to serious machine damages.
- Prior to proceeding with the reassembly of any assembly unit, always spread a thin layer of oil on the internal parts and coupling surfaces. Replace all gaskets with original spares prior to component reassembly.

= **Note:**
Some of the machine components weigh over 20 kg. It is therefore necessary to use appropriate handling equipment for disassembly and reassembly operations thereof. Never work on said parts by yourself. G.MONDINI S.p.A. cannot be held in any way liable for damages incurring to persons and/or objects due to failure to observe this rule and/or any other safety-related rule and provision.

8.4 QUALIFICATION OF PERSONNEL ASSIGNED TO MACHINE MAINTENANCE OPERATIONS

8.4.1 General Duties

In order to meet the ever-rising need for qualification in the field of entirely automatic operated manufacturing system maintenance, personnel assigned to maintenance duties is required:

– to be knowledgeable of the standards and regulations in force on accident prevention during the performance of operations on motor-driven machines and to be capable of implementing them;

– to have read and understood the section on “Accident Prevention Safety” in the FOREWORD chapter;

– to be knowledgeable of the fundamental construction and operation principles of special component manufacturing systems

– to be capable of using and consulting the manufacturing documentations as well as the automatic sealing machine documents;

– to being responsible for independent decision-making, in as far as interventions on entirely automatic manufacturing systems are concerned;

– to be willing to adapt to any technological change made to the machine;

– to acknowledge production process irregularities and/or anomalies and, whenever necessary, to go ahead with any measure, as required.

8.4.2 Qualified Personnel duties and tasks

The make-up and qualification of the personnel teams specified in the maintenance plans, are as recommended by G.MONDINI S.p.A.

The different operations can, if necessary, be also performed by personnel with equal or higher qualifications, provided they have attended the relative training courses.

The professional job descriptions for personnel required to operate on the machine are:

8.4.2.1 Technician in charge of machine

Typical tasks:

Control and upkeep of production quality on entirely automatic manufacturing systems, and specifically:

– implementation and assessment of the user's guide and diagnostics system results
– use of the machine under standard operating conditions (in automatic work mode) and restoration of operations after emergency stop switch interventions
– the provision (upon request, as the case may be) of the pieces to be processed and of the assessment tools
– where necessary, quality control and implementation of provisions necessary for the maintenance of the nominal operation quality.
– cleaning of certain machine parts (rest-on and lock-on elements and parts)

cooperation for the performance of the following activities:

– maintenance activities
– troubleshooting and repairs

Performance of regular inspections/verifications, particularly:

– inspection of tubing and pipe tightness
– inspection to check lubrication efficiency
– inspection of the protection devices for any wear and tear
– inspection of the flexible pipe and tube systems, for any wear and tear
– inspection for any visible oil leaks around machine moving parts
– inspection for the absence of any foreign objects or foreign matter in the machine operation radius.
– inspection to see that all indicator and signal lamps are working
– work pressure and delivery level inspections involving the hydraulic, pneumatic and lubrication systems.

Technical knowledge required:

- knowledge on the use and operation of the **TRAVE Sealing Machine**
- knowledge of the lubricants used in the machine and the hazards connected with use thereof
- machine breakdown logic research methods and assessment of the results
- organisation skills for the management and control of the measures necessary to restore the machine back into its functional production status.
- professional experience on automatic manufacturing and operation systems (automatic handling systems, element handling systems, etc.)
- ground knowledge of pneumatic, hydraulic and electric control and adjustment techniques.

Qualifications required:

- Complete industrial mechanic education and training, with specific skills in the technical automated system sectors
- Instruction and training on the Sealing machine are warranted by G.MONDINI S.p.A.

8.4.2.2 Technician in charge of lubrication

Typical tasks:

Lubricant tank filling and emptying operations, to be performed at regular intervals on entirely automatic manufacturing systems:

- inspection of the remaining lubricant levels
- lubricant tank cleaning and replacement of the lubricant tank contents
- top-up of the lubricant reserves depleted by grease pumps, oiling devices, oil canisters, etc.
- replacement of lubricants that are too old or used.

Technical knowledge required:

- Knowledge on lubricant qualities and the greases required for the various operations
- independent working skills according to pre-established lubrication plans
- knowledge of the correct spent lubricant scrapping methods as per provisions on environmental protection.

Qualifications required:

These operations can be performed by qualified personnel, that has undergone a sufficiently long training period on the machine.

8.4.2.3 Mechanical Maintenance Technician

Typical tasks:

Performance of preventive maintenance operations, revision and, where necessary, repairs on mechanical assembly units pertaining to entirely automatic manufacturing systems and specifically:

- drive belt replacement
- inspection of machine moving parts
- repairs on mechanical unit assemblies

Technical knowledge required:

- good knowledge of mechanical, pneumatic and hydraulic installations;
- knowledge on numeric control systems implemented in the devices
- knowledge on the fundamental principles of electric control and adjustment techniques
- inspection results assessment skills and decision making skills on necessary action
- knowledge on issuing revision reports
- knowledge on measurement and testing methods for the determination of the effective machine/handling status.

Qualifications required:

- Complete industrial mechanic education and training, with specific skills in the technical automated system sectors.
- Experience in automated handling system maintenance operations. Instruction and training on the machine are warranted by G.MONDINI S.p.A.

8.4.2.4 Electric/electronic Maintenance Technician

Typical tasks:

Performance of preventive maintenance operations, revision and, where necessary, repairs on electric and electronic assembly units pertaining to entirely automatic manufacturing systems and specifically:

- backup battery replacement operations
- analysis of microprocessor controlled device failures
- converter replacement and adaptation for three-phase servo motors.

Technical knowledge required:

- knowledge on three phase servo motors and relative adaptation to converter
- knowledge on research and repair methods of control system failures, performed via diagnostics systems, of computerised control systems and/or similar equipment.

Qualifications required:

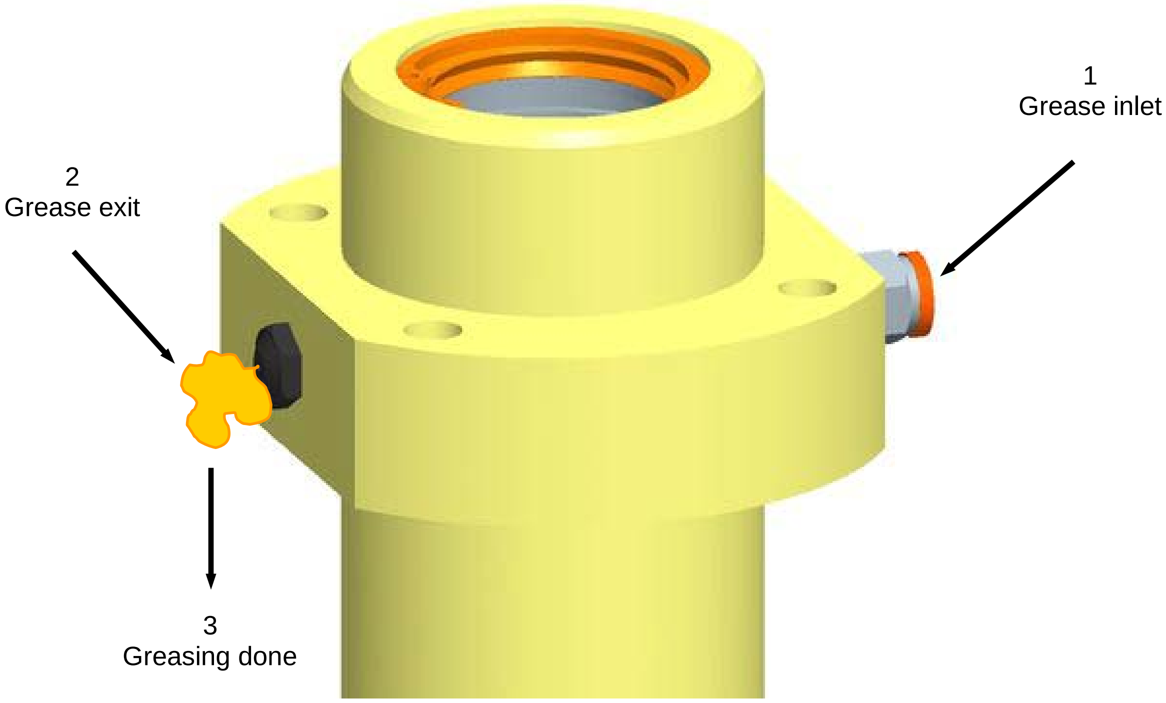
- Complete industrial electronic technician education and training, with specific skills in the technical apparatus sectors
- Experience in machine tools maintenance operations. Instruction and training on the machine are warranted by G.MONDINI S.p.A.

8.5 LUBRICATION SYSTEM

There are two ways that machine lubrication operations are to be performed in, one is via the centralised greasing system (using either a manual or an automatic grease pump), that the machine is equipped with, whilst the other is via manual lubrication using a manual grease lubrication pump.

To ensure proper lubrication of the machine is recommended to carry out the greasing operations when corresponding message appears on the operator panel.

To ensure that the greasing operations are performed correctly it is necessary to check that there is grease coming out of the bleed valves assembled on the part being greased (see the figure). This confirms the fact that greasing has been performed successfully.



A special non-toxic grade of grease formulated for use in the foodstuff industry is used when assembling and testing the sealer at our factory. This grease is:

- **NILS FOOD NLGI 1** for use in the foodstuff industry

Manufacturer	NAME	GENERAL FEATURES
NILS	FOOD NLGI 1	Nils FOOD, white-yellow in colour, odourless and tasteless; it has been specifically formulated to prevent any physiological problems when it accidentally comes into contact with food. Grease in the FOOD line complies with the FDA 21 CFR 178.3570 guidelines.

This grease can be compared with NSF H1 non-toxic grease, with an NLGI 1 viscosity index, available from different manufacturers.

For what concerns lubricating oil of the clutch blocks our Company uses the following oil in the assembly phases:

- **TOTAL Fluide ATX** oil.

This product is similar to other numerous brands available from the trade that offer the same technical features (the products listed below may be marketed under other brands in other countries).

COMPANY	NAME	GENERAL FEATURES
TOTAL	FLUIDE ATX	Dexron II D, Ford Mercon, MB 236.6, CAT TO 2
AGIP	ATF II D	
ELF	MAGIC G 2	
ESSO	A.T.F. D	
Q8	AUTO 14	
SHELL	DONAX TA	
TAMOIL	ATF II D	

8.5.1 Centralised lubrication system (manual or automated)

The **Closing machine TRAVE** is supplied with a centralised lubrication system that is either manual or automated.

We can identify two principal areas that are lubricated via the centralised greasing system (automated or manual) and specifically:

- the sealing screw and vertical tool motion area;
- the pusher arms motion assembly.

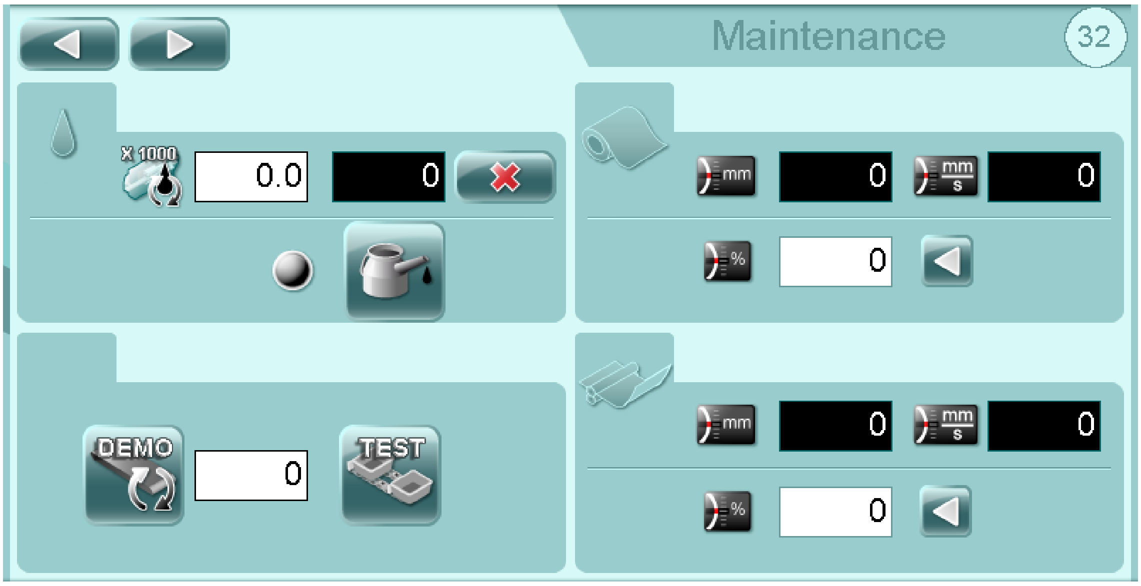
IMPORTANT!

For correct machine greasing and to maintain adequate lubrication of all machine components that do need lubrication, it is appropriate to carry out checks on the tank levels at regular intervals and to regularly inspect that the lubrication system works appropriately.

Lubrication of the parts connected to the (centralised and automated) lubrication system is controlled directly via the operator panel.

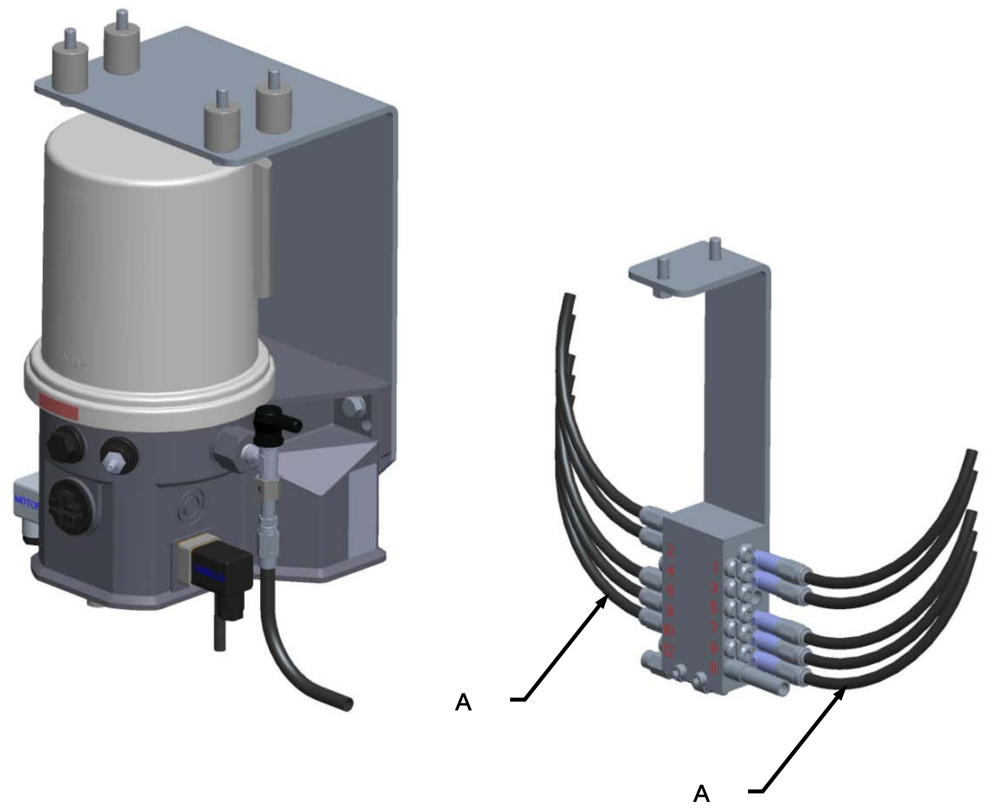
After having entered a “Maintenance” level password, the operator panel’s Main Menu will display the “Maintenance icon: . Selecting this icon will access the Maintenance menu pages that also provide parameters for programmed lubrication functions. Said pages can be accessed and viewed using the arrow keys in the top left hand corner of the display screen.

A description of these menu pages is provided in Chapter 5.



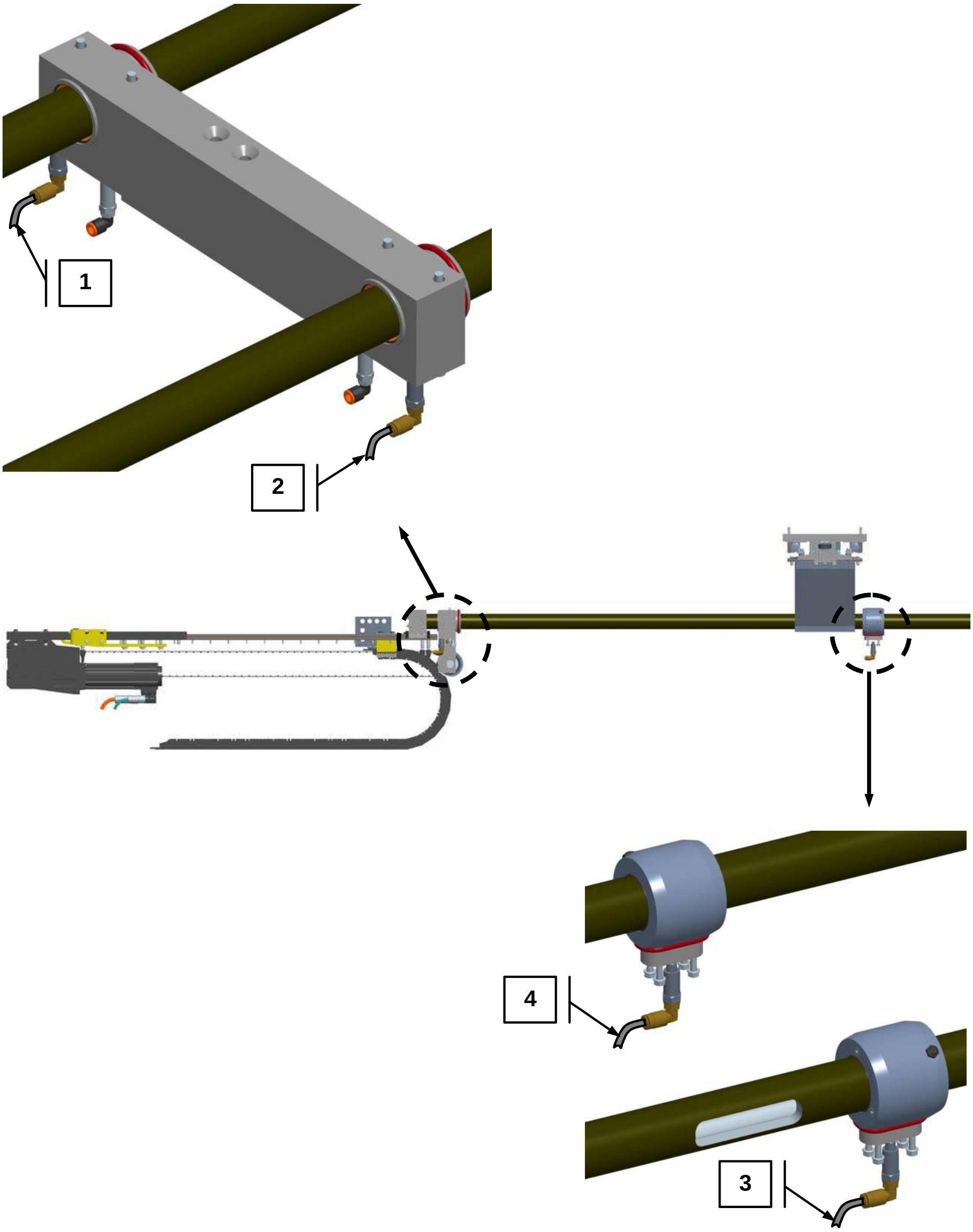
It is hereby recommended that the grease pump is checked at regular intervals for correct operations and that all greasing points are therefore correctly reached by the lubricant.

To check the efficiency of the centralised lubrication plant, it is necessary to check that all the tubes (**A**) are not clogged and that the grease reaches the lubrication point efficiently. This inspection can be performed by detaching all the tubes leading to the greasing points one tube at a time, and after a couple of pump cycles, by checking to see that the grease has actually flowed out of the tube. After inspection of each tube as above, reattach the tube that was checked to restore circuit tightness.

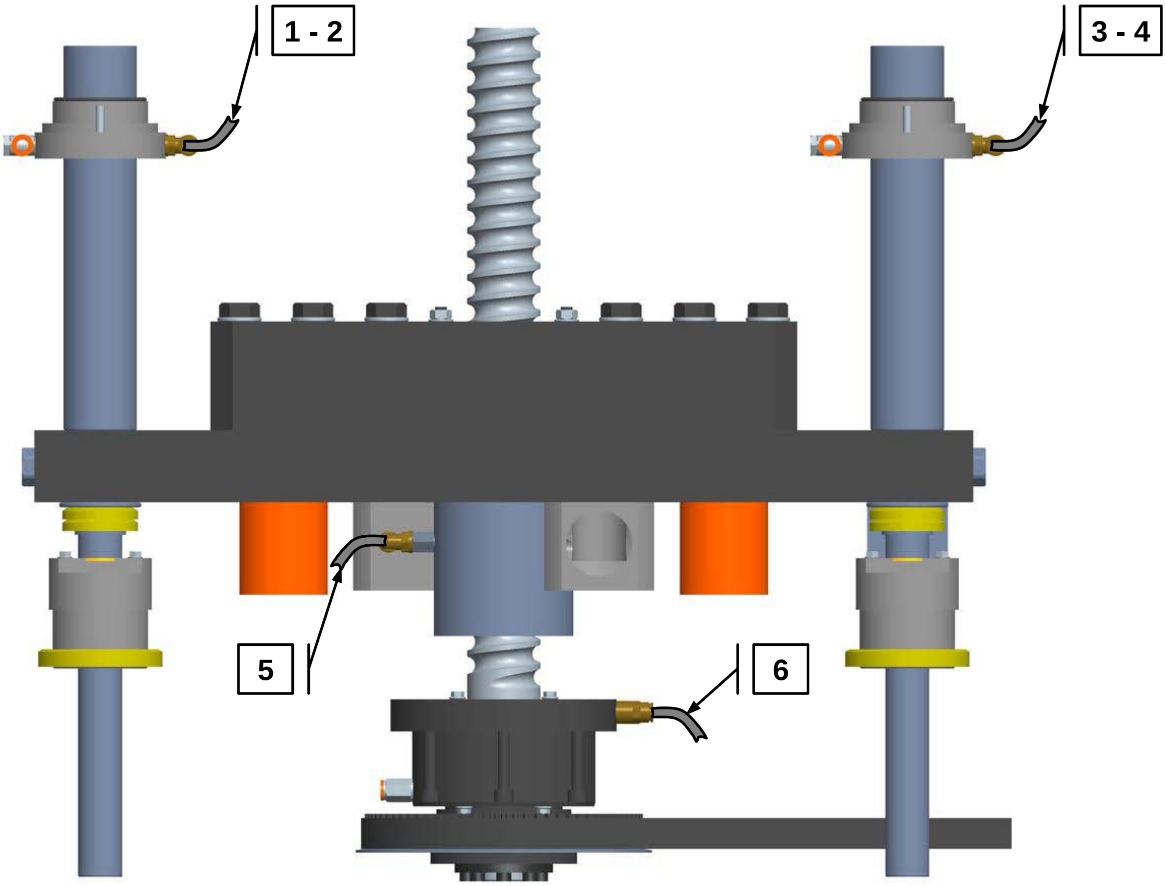


The greasing points that are reached by lubrication here are as follows:

1. The pusher unit areas are provided with **4** points that are reached by the lubrication grease.



2. The Screw-operated sealing area is provided with **6** points that are reached by the lubrication grease.



8.5.2 Manual lubrication

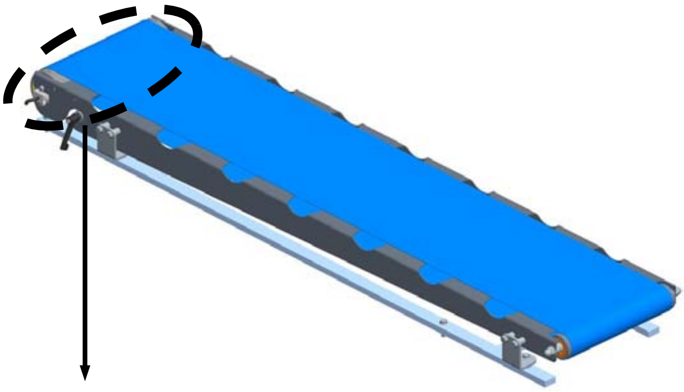
DANGER
Prior to carrying out any operation at all on the machine, always ensure that all the main and secondary power and feed-in supplies have been totally cut-off. Hang specific warning signs such as "MACHINE MAINTENANCE OPERATIONS UNDERWAY – DO NOT SWITCH ON THE POWER MAINS" in the vicinity of the mains switch.

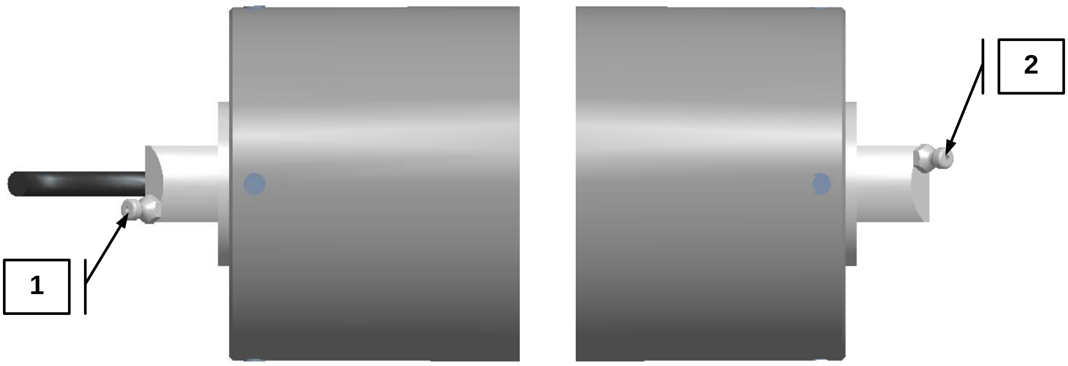
Furthermore, there are machine greasing points that require manual maintenance, such as the welding tool, the motor drums, plus all the other drive and motion elements (drive chains, conveyor belts, etc.).

To ensure correct machine greasing operations, we recommend that the operations listed herein are performed at the beginning of each work day and after each machine washing cycle.

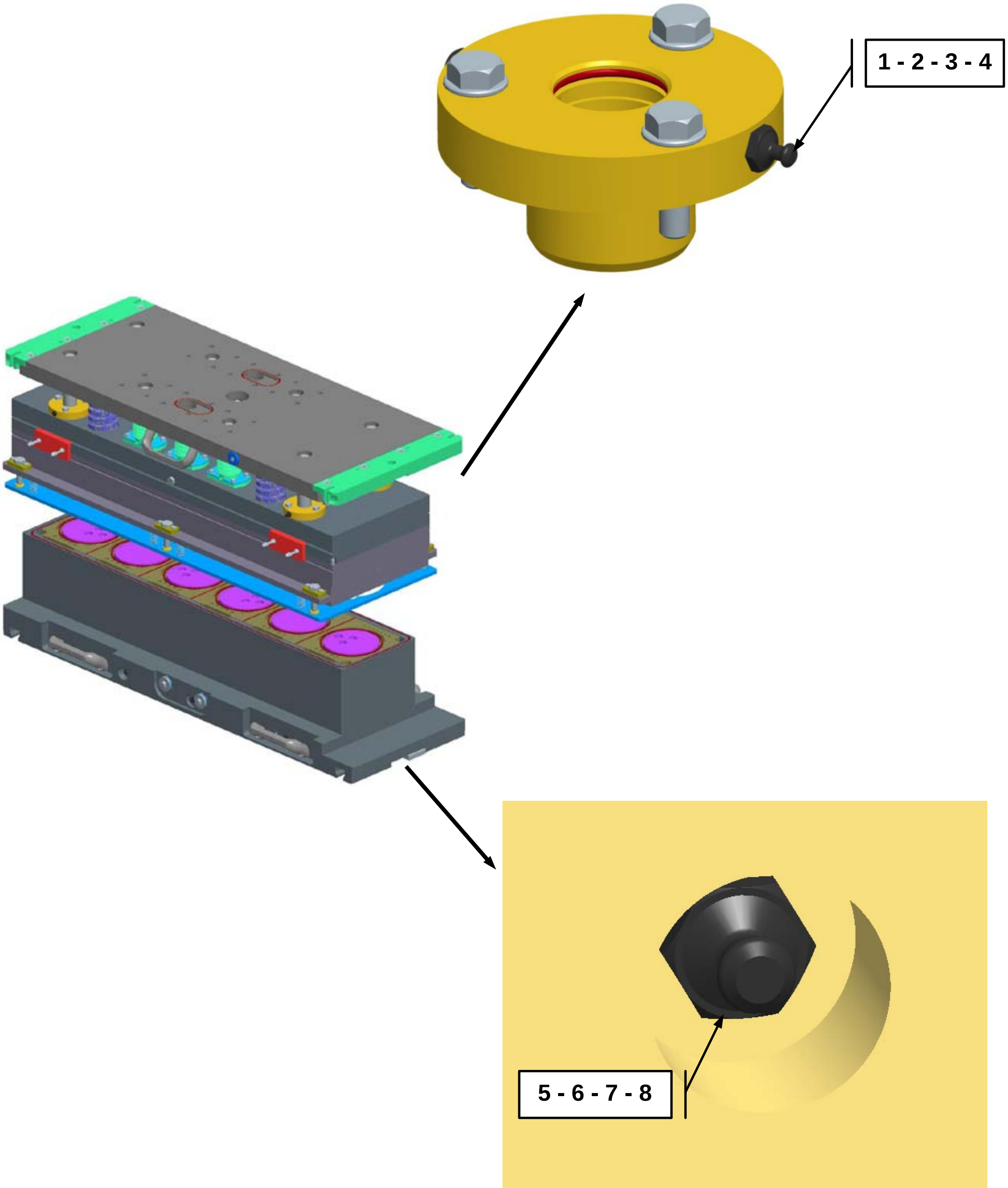
The unit assemblies requiring manual lubrication operations are:

1. The exit belt area, that has been provided with **2** greasing points that require manual lubrication.





2. The sealing tool area, that has been provided with **8** greasing points that require manual lubrication.



8.6 MACHINE CLEANING

Like most of the G. MONDINI machines, the **TRAVE** tray sealer has been designed to package food and food derivatives under all possible conditions.

This production requires full compliance with the proper rules of hygiene in order to prevent the occurrence of any possible inconveniences.

Cleaning and hygiene in the workplace in which these machines operate are thus of critical factor.

It is very important to clean the machine immediately at the end of the day or on completion of a work cycle to prevent proliferation of bacteria due to the presence of any residues left on the machine.

Before cleaning the machine, it is important to ensure that it is in a safe condition so as to avoid incurring dangerous situations or hazards.

The machine can be cleaned using high-pressure jets (at a maximum pressure of 40 bar and a maximum temperature of 50°C) on most of the surface of the machine. Great care must be taken not to point high-pressure jets against components that are delicate or not covered by suitable guards.

It is important to periodically check that the guard gaskets are perfectly sealed to prevent water from falling into the machine and damage any mechanical parts, motors and the wiring system.

IMPORTANT! Do not point the high-pressure water jet directly to mechanical parts, motors and wiring systems. Make sure all the guards are properly closed and periodically check the guard gaskets are perfectly tight.

As stated at the beginning of this section, the machine needs to be cleaned not only to remove dust, foreign bodies, and dirt or product residues but also to sanitize and disinfect parts that might come into contact with the packaged products.

Numerous detergents available from the trade are not harmful nor dangerous to the health of humans, the environment and the machine itself and they can be easily removed by rinsing off the machine.

It is up to the end user to arrange for the operators in charge of cleaning the machine to wear suitable personal protection equipment and be in possession of suitable tools, and to inform them of the risks deriving from the use of cleaning products and equipment.

The concentration of detergents/disinfectants and the operating temperature must correspond to the values stated in the specifications and the material safety sheets and be compatible with the materials of which the machine is made of.

It is advisable to use mild detergents and foaming agents that combine disinfecting and degreasing properties.

8.6.1 Table of recommended products

Most of the packaging systems designed and manufactured by G. MONDINI S.p.A. consist of light alloys, such as aluminium, and painted surfaces that may be etched by the chemical products used. It is therefore advisable to use as much as possible safe cleaning solutions. The use of aggressive chemicals for an effective cleaning affects the integrity of the materials used on G. MONDINI S.p.A. systems.

The table below provides a list of different recommended chemicals that have been tested on G. MONDINI S.p.A. equipment for all types of application in order to prevent all potential contamination under all possible weather conditions.

All cleaning products must comply with local requirements. Locally approved products may differ from this list.

If any of the recommended products is not available on the market, please feel free to contact G. MONDINI S.p.A. and jointly identify any other equally compatible products.

Application	Type of product	Trade name
Descaling	Acid SMS	Aluwash VA3
External foam cleaning	Alkaline SMS	Diverfoam SMS HD VF22
	Alkaline Chlorinated SMS	Diverfoam SMS Chlor VF18
	Acid SMS	NP Freefoam VF11
Manual cleaning	Mid alkaline	Fatsolve VF21
	Daily cleaning and disinfection	Delladet VS2
	Neutral	Shureclean Plus VK9
Manual disinfection by immersion	Surfactant	Suredis VT1
	Surfactant	Divosan QC VT50
Manual disinfection by spraying	Alcohol-based solution	Divodes FG VT29
	Alcohol-based solution	Alcosan VT10
Manual disinfection by paper wipes	Alcohol-based solution	Divodes FG Wipes VT75
Hands wash	Sanitizing soap	Soft Care Sensisept H34
Hands disinfection	Alcoholic solution	Soft Care Med H5

8.6.2 Cleaning protocol

Cleaning equipment

Products	Concentr.
Foam	Follow the label directions
General cleaner	Follow the label directions
Descaling agent	Follow the label directions
Disinfectant	Follow the label directions

Test kits required

Water temperature

max. 50°C

Water pressure

max. 40 bar

Washing frequency

Daily

Washing methods

Manual and foaming

Personal protection equipment

Eye protection	✓
Safety gloves	✓
Safety boots	✓
Protective suit	✓
Hard cap	✓

Tools and equipment

Safety precautions

Always wear personal protective equipment. Never add water to chemicals. Always add chemicals to water. Immediately report any accidents!

Cleaning Protocol

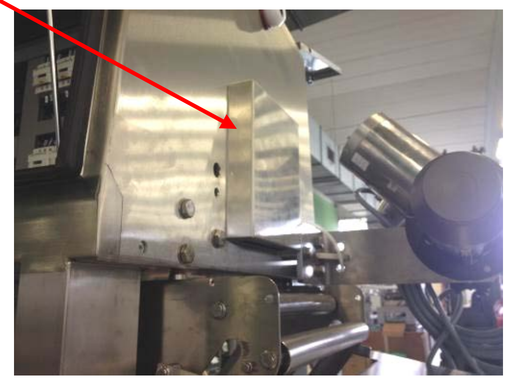
Immediately after production:

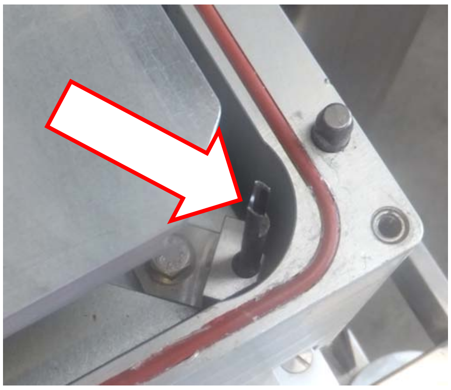
1. Activate all the stops required (disconnect the power and compressed air supply) and lock all the component parts before cleaning. Put in place the appropriate warning signs indicating that cleaning is in progress.
2. Clear the area involved of any packaging and processing materials.
3. Pre-clean the area with a shovel, brush and suitable non-abrasive tools. Place food waste in bags/containers that are suitable for disposal.
   Notes. Do not use compressed air to remove any dust from the equipment. This will cause dissemination of dust and microorganisms all around and on the equipment to be cleaned. If the use of compressed air is required, air must be filtered and applied at the lowest possible pressure.

During disinfection:

4. Make sure that the procedures under points 1 to 3 above have been properly performed.
5. Clean the switchboards and water-sensitive components by hand, and cover everything with disposable plastic bags.
6. Rinse the equipment that is sensitive to high pressures, the supporting structures and the surfaces in contact with the floor, using plain water at a medium-low pressure (not exceeding 40 bar). Do not rinse at a high pressure. Take great care and prevent water and chemicals from getting into contact with sensitive surfaces.
7. Rinse the floor, collect any contaminated food and put them into bins for disposal. Do not dispose of waste fluids in the drain system. They must be gathered and stored for final disposal.
8. Remove the equipment, such as belts and conveyors, moulding tools, valves and metering pipes (see details further on in this protocol) and position them in designated areas or on storage carts, and clean them separately (do not place them on the floor).
9. Manually clean the areas of the moulding tools in contact with the machine. Cover any vacuum and air ducts that have remained open due to the removal of equipment, using disposable plastic bags to prevent water from entering the pneumatic circuits.
10. Clear the machine of any deposits of dirt. Thoroughly rinse the area and remove contaminated food from the collecting bins.
11. Inspect the area and rinse again, if necessary. Wash up the floor and collection bins again.
12. Manually clean the equipment and sensitive surfaces that have not been washed in step 5 above.
13. Apply the cleaning solution or foaming cleanser on all washable surfaces of the component parts and floors, both under and around the system. Leave the solution to activate for the time needed (see the datasheet for details). Rub your hands vigorously and wash them thoroughly, if necessary.
14. Rinse with fresh water. Gather and remove all traces of cleaning solutions or contaminants.
15. Carefully check all the surfaces and make sure they are properly clean. If it is not so, apply the cleaning solution again and, if necessary, clean by hand.
16. If cleaning solutions have been re-applied, rinse thoroughly to remove any traces of contaminants and solution.
17. Repeat steps 14 and 15 if necessary.
18. Apply one of the recommended disinfectants or a similar one. When using a disinfectant other than the one recommended, it may be necessary to rinse the surfaces thoroughly.
19. Clean and dry the photoelectric cells using a soft non-abrasive cloth and a suitable disinfectant.
20. Remove and dispose of the plastic bags used to protect the electrical panels, the water-sensitive surfaces and the open vacuum and air ducts.
21. Remount the equipment as specified in the maintenance plan, taking care to keep it clean and disinfected.

See below for further details on cleaning.

Details	Photographs
Clean the control panels and water-sensitive components manually and protect them with a plastic bag. Do not spray water, cleaning solutions or disinfectants directly on panels or other electrical components – they must be protected against water. Example: Operator panel, electric motors and electrical connectors.	
Do not spray water, cleaning solutions or disinfectants directly on the detection points of the machine. The photoelectric cells must be cleaned using a non-abrasive soft cloth and be disinfected with specific alcohol-based products. Example: sensors, photoelectric cells Switchboard ventilation	 
Remove all additional equipment as specified in the maintenance plan (see also the details shown in this protocol) and position it on dedicated surfaces or storage carts for cleaning (do not place on the floor). Examples: Conveyor belts	

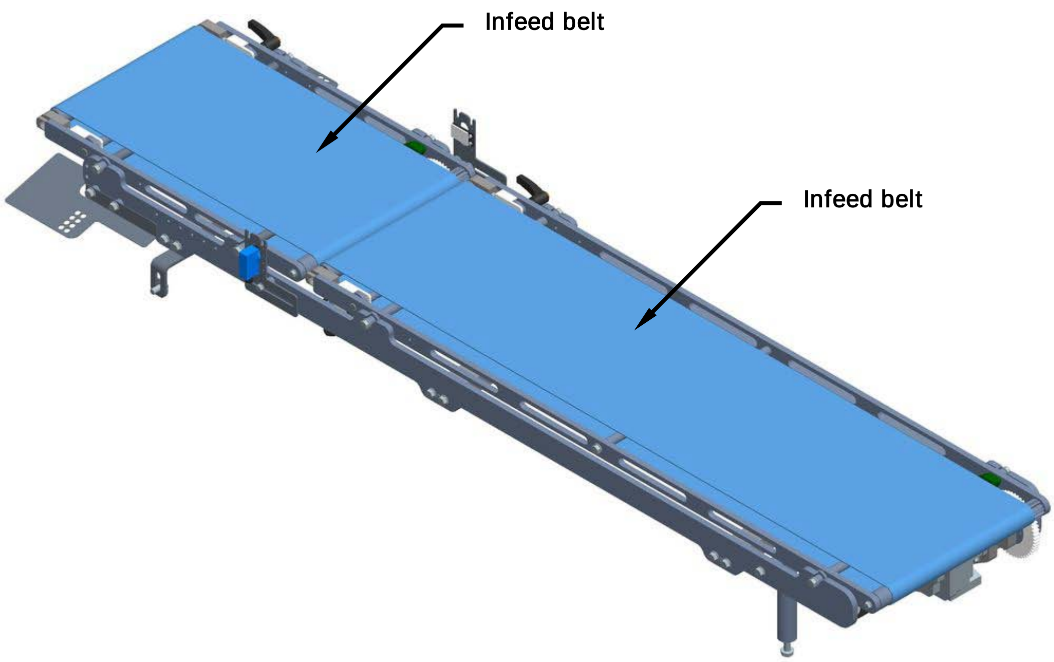
Cleaning the tool: The moulding tool must be removed from the machine, cleaned and disinfected manually, using recommended chemicals. Use a soft cloth and non-abrasive brushes, taking care of water-sensitive component parts. It must be disinfected using one of the recommended alcohol-based products.	
Disinfecting the Darfresh® tool during production: The lower Darfresh® tool consists of needles that perforate the tray and can therefore become contaminated with the product. Cleaning and disinfecting the needles during production would be difficult. We therefore suggest to disinfect them using one of the alcohol-based sprays.	 

8.6.3 Washing the infeed belts

The infeed belts are components of the sealing machine that require special care in terms of cleaning and disinfection. Here the trays are not yet sealed and there is the risk for the food coming into contact with the belt.

It is advisable to wash and disinfect the belt on a daily basis, using the most suitable solution among those recommended to prevent the onset of bacteria proliferation in the organic product exposed to the air for long, which may affect the hygiene of the machine.

Daily cleaning of the infeed belts must be effected with the belts removed from the machine. For extraordinary cleaning, the infeed belts must be removed completely, following the procedure described in subsection 8.11.2 of this handbook.



8.6.4 Cleaning the sealing tool

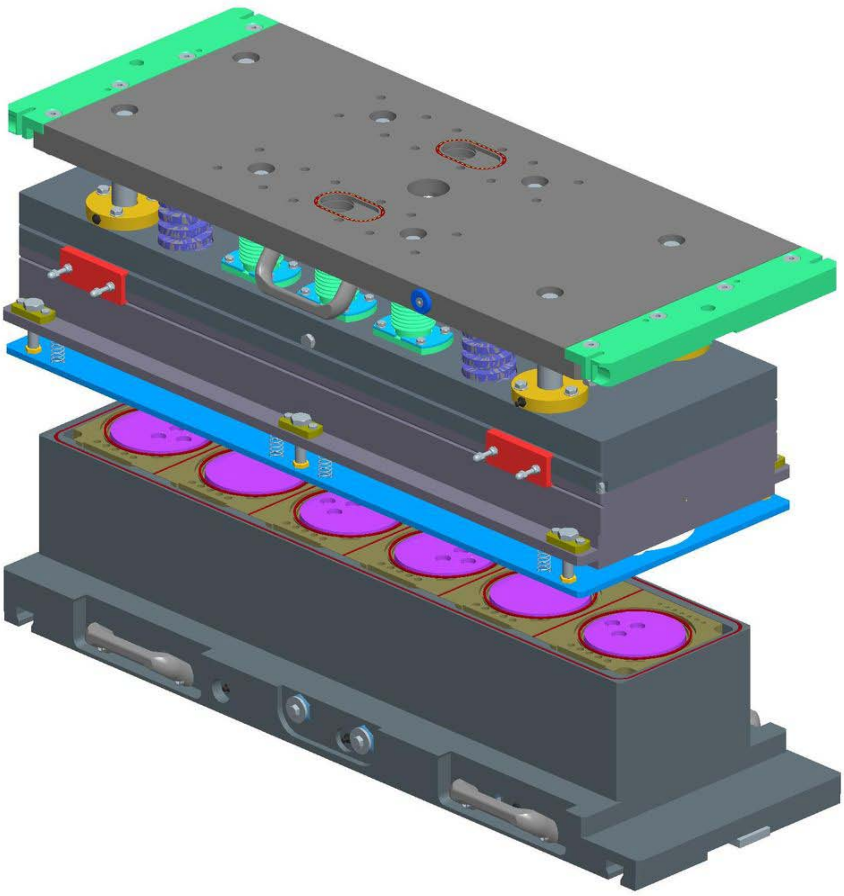
The sealing tool is another area of the machine that can easily get dirty and therefore requires special cleaning.

When entering the sealing tool, the tray is still open, filled with the metered product. During heat-sealing, the filling can spill out from the tray or even the tray can tip over in the tool.

In consideration of this and, more importantly, as a result of the fact that the tool comes into contact with the tray edge and the sealing film, it is extremely important to keep it clean and perfectly disinfected.

We recommend removing the sealing tool every day at the end of the production cycle. Leave it to cool down and clean it separately from the rest of the sealing machine.

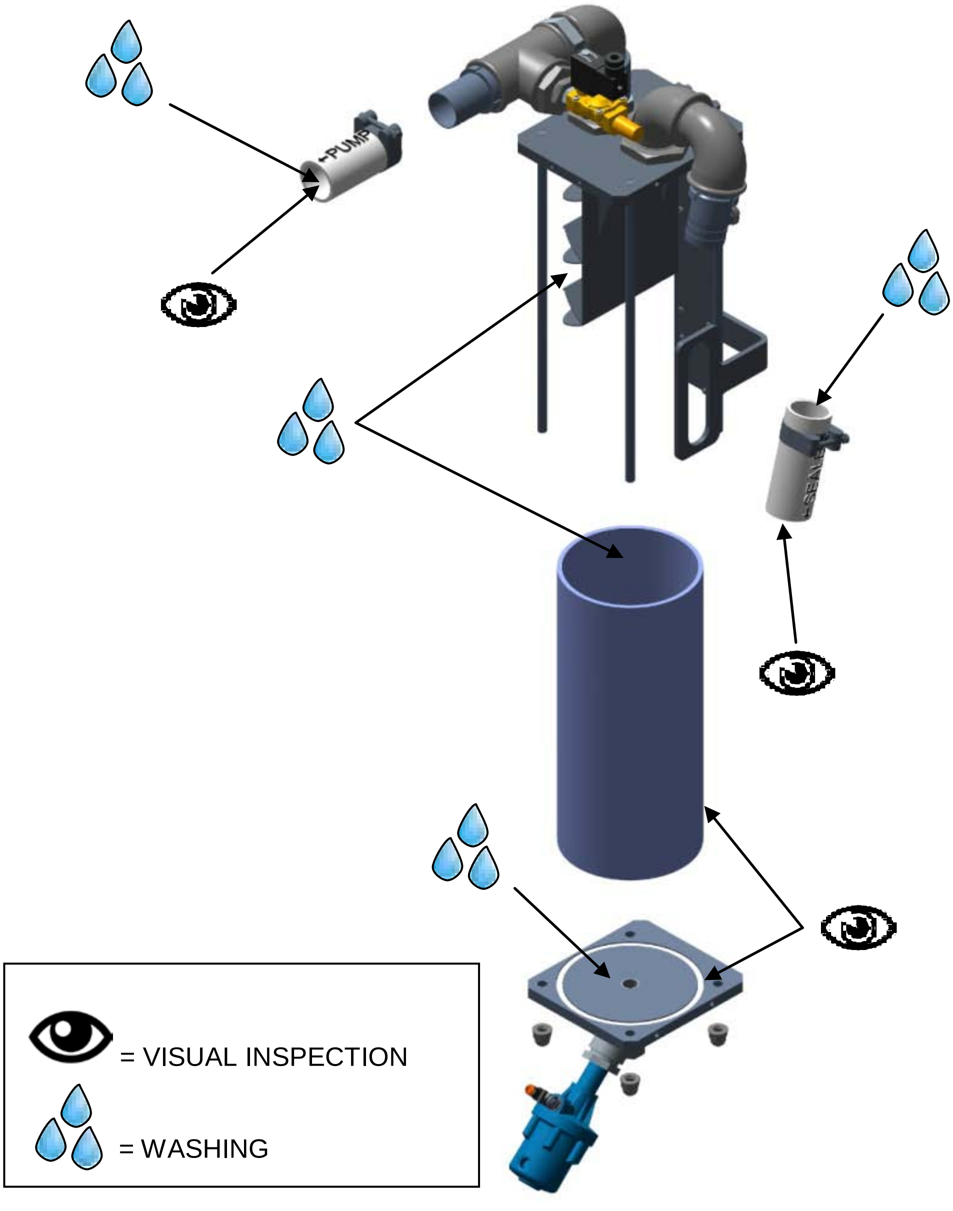
After removing the various parts (counter-plate, intermediate plate, insulators) and accessing the internal structures, it is advisable to clean with a brush and remove any residue deposited on the tool. When all traces of product have been removed, blow the inside of the tool with compressed air at the lowest possible pressure, clean with a soaked soft cloth and sanitize with the recommended alcohol-based products. Then remount the tool back in position.



8.6.5 Cleaning the liquid separator

Proceed as follows to clean the inside of the liquid separator:

- Disconnect the liquid separator;
- Remove the separator and the hose pipes;
- Wash the separator and pipes with running water (**max. 3 bar**), using a cleaning product chosen from among those recommended;
- Disinfect the area using any of the recommended products;
- Remount the separator and pipes;
- Check that the hoses and seals and not damaged, and replace them if necessary.



8.7 TIGHTENING TORQUES FOR NUTS AND BOLTS

For selection of the tightening torque class, consult table 5.2 and establish the bolt to be used as specified on table 5.3 and considering the traction force for each single bolt, as specified in table 5.4.

Table 5.2

TIGHTENING TORQUE CLASS	APPLICATIONS	TOLERANCES REFERRED TO NOMINAL TORQUE
I	EXTREMELY STRESSED: for extremely strained connections, subject to vibrations, impacts, high traction strengths or jarring movements, reducers, racks, connecting or piston rods, traverse motion stops, alternate loads and drive element components in general, dynamic hanging loads. INSPECTION TO BE PERFORMED EVERY 6 MONTHS USING A TORQUE WRENCH.	±5%
II	STRESSED: for uses not subject to excessive vibration, yet nevertheless requiring good locking safety (frames, static hanging loads, precision-locking, pallets, rollers, clamp and clench supports). INSPECTION REQUIRED ONCE A YEAR (USE OF A TORQUE WRENCH IS RECOMMENDED)	+5% -15%
III	LOW STRESS: for low stress applications or of secondary importance (blocks and plugs, references, dais and footboards, fencing) INSPECTION REQUIRED AT EVERY MACHINE INTERVENTION THAT MAY INFLUENCE TORQUE FEATURES, BASED ON THE TYPE OF OPERATION THAT IS PERFORMED.	+5% -35%

Table 5.3

THREAD	NUT AND/OR BOLT WRENCH SIZE	TIGHTENING TORQUE CLASS (Nm)		
		III	II	I
		SCREW STRENGTH CLASS		
		8,8	10,9	12,9
		NUT STRENGTH CLASS		
		8	10	12
M4	7	2,3	3,3	4
M5	8	4,8	6,8	8
M6	10	8	11,2	13,6
M8	13	20	28	32,8
M10	17	39,2	55,2	66,4
M12	19	68,8	96	116
M14	22	108	152	184
M16	24	168	236	284
M18	27	232	324	388
M20	30	328	464	552
M22	32	440	624	744
M24	36	568	800	960
M27	41	840	1000	1440
M30	46	1160	1600	1920

Table 5.4

THREAD	NUT AND/OR BOLT WRENCH SIZE	TIGHTENING TORQUE CLASS (Nm)		
		III	II	I
		SCREW STRENGTH CLASS		
		8,8	10,9	12,9
		NUT STRENGTH CLASS		
		8	10	12
M4	7	3120	4360	5240
M5	8	5080	7160	8560
M6	10	7200	10080	12080
M8	13	13200	18560	22320
M10	17	20960	29520	35440
M12	19	30640	43200	51600
M14	22	42000	59200	70800
M16	24	58400	81600	98400
M18	27	70400	99200	118400
M20	30	91000	128000	153600
M22	32	112800	159200	191000
M24	36	131000	184000	220800
M27	41	172000	241600	290400
M30	46	209600	294400	253600

8.8 COG BELT TENSIONING

The correct tensioning of the cog belts is essential, especially when servomotors are used.

If the belt is too taut. it causes early wear of the load-supporting bearings. If the belt is too loose, it causes incorrect positioning of the load being handled and irregular readings of the control system in the servomotors.

The belts can be tensioned in either of the following ways:

1. deflection is measured at the centre of the free section of the belt by applying a specific weight. The value should correspond exactly to the design values.
2. A properly calibrated tension-frequency metre is used to measure the oscillation of the belt and determine the desired tension.

The belts requiring a precise tensioning value are those making up:

- the pushers: forward and backward, open and close movements.
- Darfresh® on-tray / Zero system; forward and backward movements.

8.8.1 Belt tensioning tables

TYPE OF BELT AND SEALING MACHINE		Total force (N)	Single Force (N)	Type of belt	WITH SONIC 507 C TENSION METER				WITH WEIGHT	
					Mass (g/m)	Belt width (mm)	Pulley centre distance (mm)	**Frequency (Hz)**	Weight (Kg)	**Deflection (mm)**
Pushers Forward-Backward	TRAVE 340	-	-	-	-	-	-	**-**	-	**-**
	TRAVE 350	631	310	GT3-8	5,8	20	872	**30**	3,5	**17**
	TRAVE 367	631	315,5	GT3-8	5,8	20	1016	**25**	1,8	**10**
	TRAVE 384	631	315,5	GT3-8	5,8	20	1188	**22**	1,8	**12**
	TRAVE 1000	2209	1104,5	GT3-8	5,8	30	1396	**29**	10,5	**28**
	TRAVE 1200/590XL	2209	1104,5	GT3-8	5,8	30	1512	**27**	10,5	**30**
	TRAVE 1400	1894	947	Polychain 8	4,7	21	1872	**28**	8	**29**
Pushers Open - Close	TRAVE 1000/1200/590XL (stroke 100)	373	186,5	GT3-5	4,1	15	210	**130**	2	**4**
	TRAVE 1000/1200/590XL (stroke 160)	373	186,5	GT3-5	4,1	15	285	**185**	2	**6**
	TRAVE 1000/1200/590XL (stroke 200)	373	186,5	GT3-5	4,1	15	310	**90**	2	**6**
	TRAVE 1400 (stroke 160)	435	217,5	GT3-5	4,1	15	260	**115**	2,3	**5**
	TRAVE 1400 (stroke 200)	435	217,5	GT3-5	4,1	15	310	**75**	2,7	**6**
Darfresh® on Tray / Zero	TRAVE 340	631	315,5	GT3-8	5,8	20	624	**42**	3,6	**12**
	TRAVE 350	572	286	GT3-8	5,8	20	880	**28**	3,5	**18**
	TRAVE 367	600	300	GT3-8	5,8	20	1080	**24**	3,5	**22**
	TRAVE 384	600	300	GT3-8	5,8	20	1280	**20**	3,5	**26**
	TRAVE 1000	1578	789	GT3-8	5,8	20	1272	**33**	7,5	**25**
	TRAVE 1200/590XL	1578	789	GT3-8	5,8	20	1512	**28**	7,5	**30**
	TRAVE 1400	950	470	GT3-8	5,8	20	1672	**19**	5	**33**

8.9 PREVENTIVE MAINTENANCE

The following table lists the order of the preventive maintenance specifications designed for the closing machine **TRAVE**, complete with the time intervals for each intervention.

D	Daily
W	Weekly
M	Monthly
Q	Quarterly
Y	Yearly
WH	Working hours
WC	Work cycles
OR	On request

Frequency		Unit Assembly	Element	Maintenance	Description at section
First intervention	Repeated every				
M	D	Feed Belt	Feed Belts	Inspection of good working conditions	8.11.1
M	W	Feed Belt	Servomotors and reducers	Inspection of good working conditions	8.11.3
M	D	Feed Belt	Cogwheels	Inspection of good working conditions	8.11.5
M	D	Delivery Belt	Polycord belts	Inspection of good working conditions	8.12.1
M	W	Delivery Belt	Motor drum and rollers	Inspection of good working conditions	8.12.3

Frequency		**Unit Assembly**	**Element**	**Maintenance**	**Description at section**
First intervention	Repeated every				
M	W	Film unwinder	Motor and reducer	Inspection of good working conditions	8.13.1
M	W	Film unwinder	Expanding shaft	Inspection of good working conditions	8.13.3
M	W	Film drive rollers and control	Servomotor and reducer	Inspection of good working conditions	8.14.1
M	W	Film drive rollers and control	Drive roller and rubber-coated shaft	Inspection of good working conditions	8.14.3

Frequency		**Unit Assembly**	**Element**	**Maintenance**	**Description at section**
First intervention	Repeated every				
M	W	Winding Shaft	Motor and reducer	Inspection of good working conditions	8.15.1
M	W	Pushers	Servo-reducer	Inspection of good working conditions	8.16.1
M	D	Pushers	Cog-belt	Inspection of good working conditions	8.16.3
M	W	Pushers	Servomotor, reduction gear and cog-belt	Inspection of good working conditions	8.16.5

Frequency		Unit Assembly	Element	Maintenance	Description at section
First intervention	Repeated every				
M	W	Screw sealing cam	Servomotor	Inspection of good working conditions	8.17.1
M	W	Screw sealing cam	Cog-belt	Inspection of good working conditions	8.17.3
M	D	Screw sealing cam	Sealing screw	Inspection of good working conditions and tension	8.17.5
M	W	Screw sealing cam	Columns	Inspection of good working conditions	8.17.7
M	W	Screw sealing cam	Bearings	Inspection of good working conditions	8.17.9
M	W	Screw sealing cam	Bushings	Inspection of good working conditions	8.17.11

8.10 REACTIVE MAINTENANCE

The following table lists the order of the reactive mechanical maintenance specifications designed for the **TRAVE closing machine**.

Unit Assembly	Element	Maintenance	Description at section
Feed Belt	Feed Belts	Replace	8.11.2
Feed Belt	Servomotors and reducers	Replace	8.11.4
Feed Belt	Cogwheels	Replace	8.11.6
Delivery Belt	Plycord belts	Replace	8.12.2
Delivery Belt	Motor drum and rollers	Replace	8.12.4

Unit Assembly	Element	Maintenance	Description at section
Film unwinder	Motor and reducer	Replace	8.13.2
Film unwinder	Expanding shaft	Replace	8.13.4
Film drive rollers and control	Servomotor and reducer	Replace	8.14.2
Film drive rollers and control	Drive roller and rubber-coated shaft	Replace	8.14.4

Unit Assembly	Element	Maintenance	Description at section
Winding Shaft	Motor and reducer	Replace	8.15.2
Pushers	Servo-reducer	Replace	8.16.2
Pushers	Cog-belt	Replace	8.16.4
Pushers	Servomotor, reduction gear and cog-belt	Replace	8.16.6

Unit Assembly	Element	Maintenance	Description at section
Screw sealing cam	Servomotor	Replace	8.17.2
Screw sealing cam	Cog-belt	Replace	8.17.4
Screw sealing cam	Sealing screw	Replace	8.17.6
Screw sealing cam	Columns	Replace	8.17.8
Screw sealing cam	Bearings	Replace	8.17.10
Screw sealing cam	Bushings	Replace	8.17.12

8.11 FEED BELT

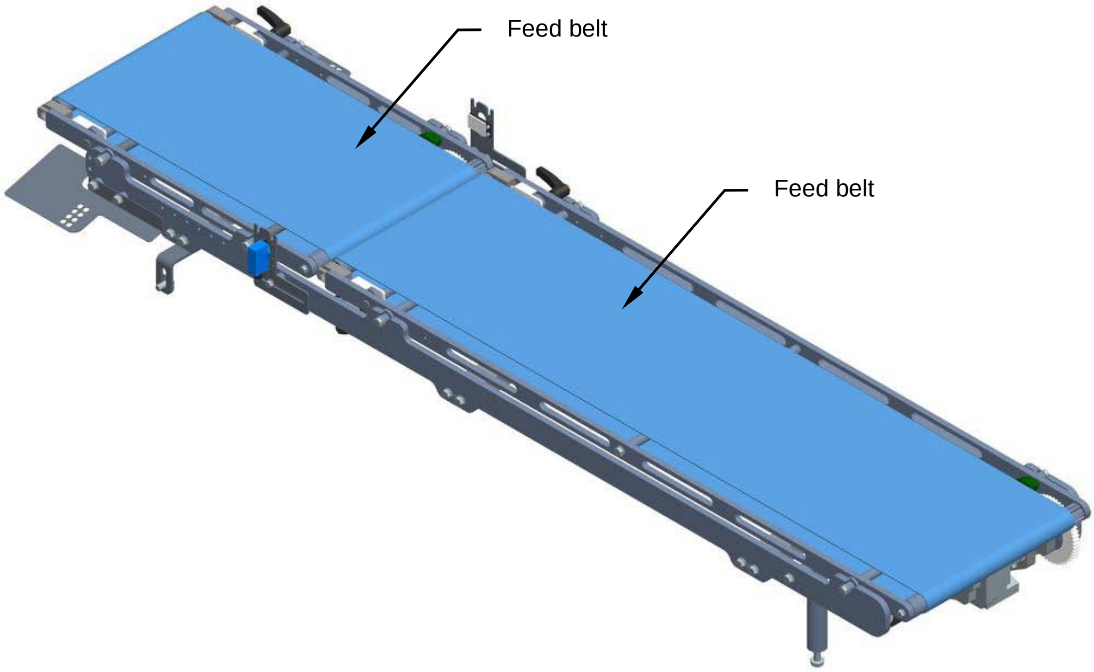
8.11.1 Checking feed belts condition

Machine status	Power sources activated
Equipment	Standard workshop tools

Visually check the condition of the feed belts during machine operation. Make sure they are centred, runs smoothly and shows no signs of wear.

Check the tension of the belts, which determines correct movement of the elements being conveyed. Make sure the belts remains in contact with the rollers.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

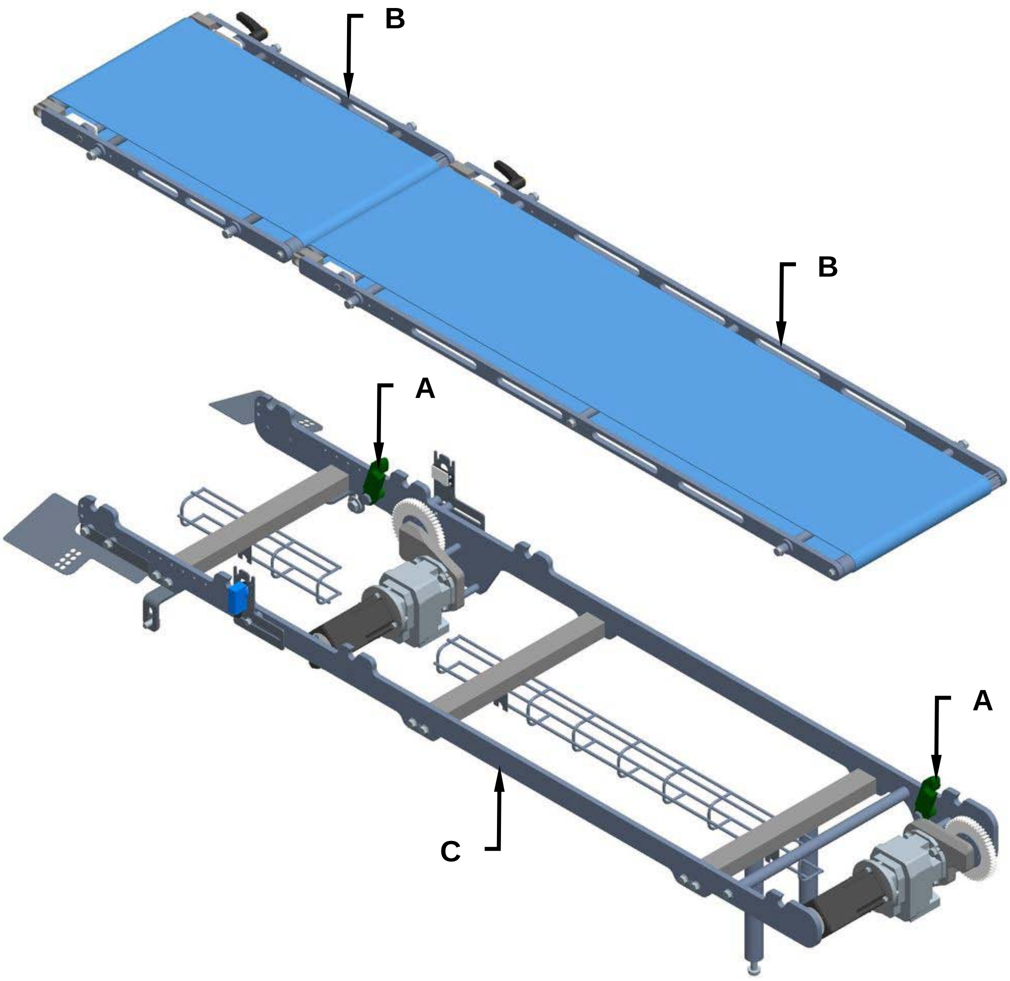


8.11.2 Replacing the feed belt

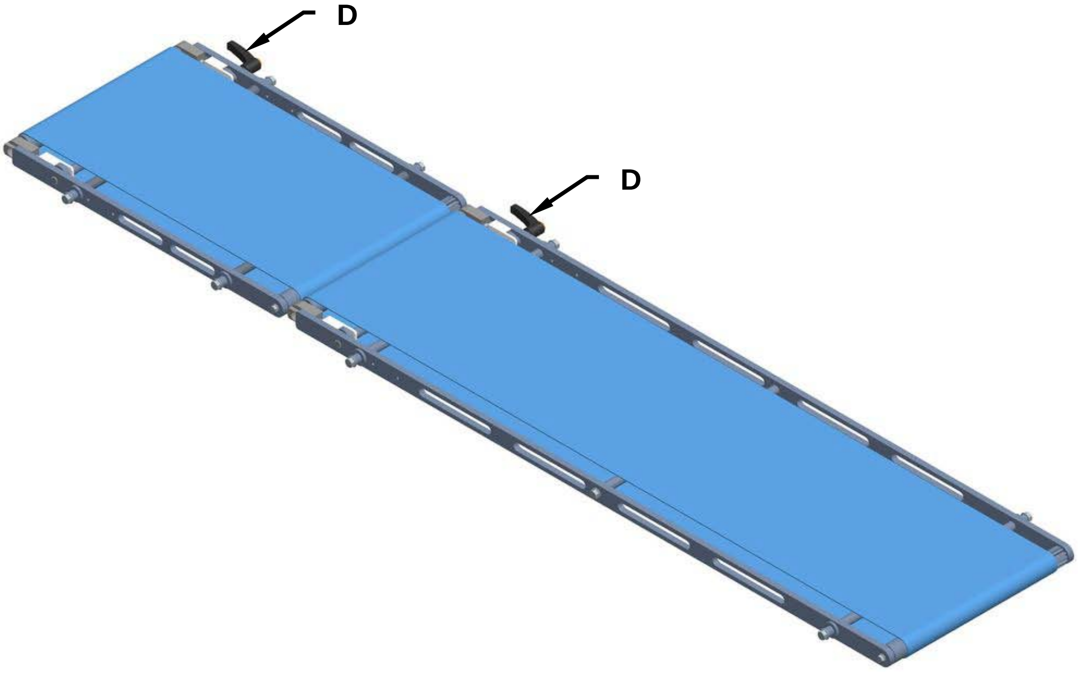
Machine status	Power sources deactivated
Equipment	Standard workshop tools

Proceed as follows to replace the feed belt:

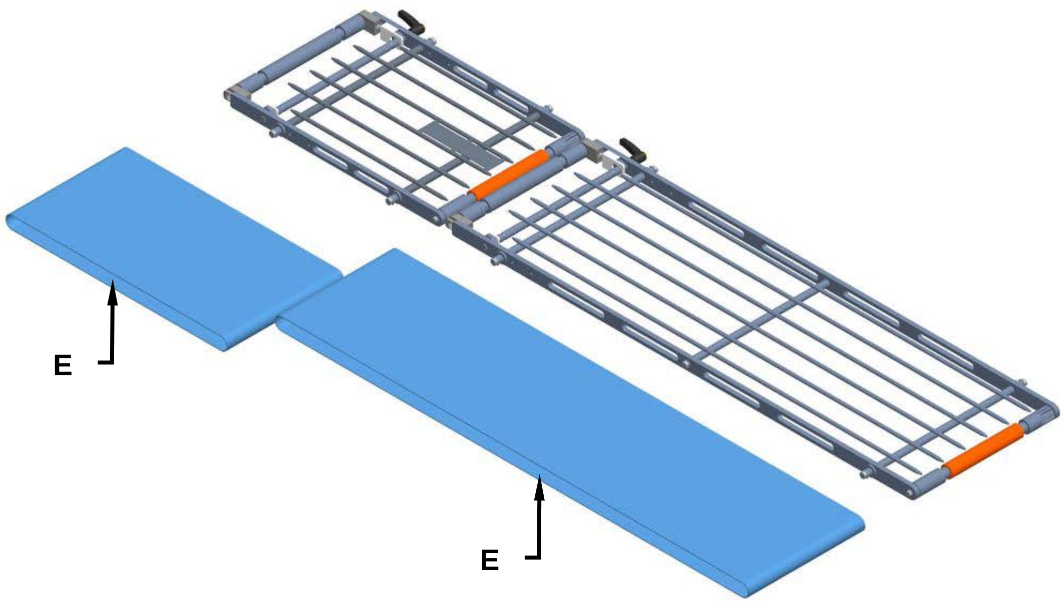
1. Disconnect the power supply.
2. Open the hooks (**A**) and remove the belts (**B**) from the support (**C**).



3. Use the appropriate levers (**D**) to unlock the belts.



4. Remove and replace the belts (**E**).



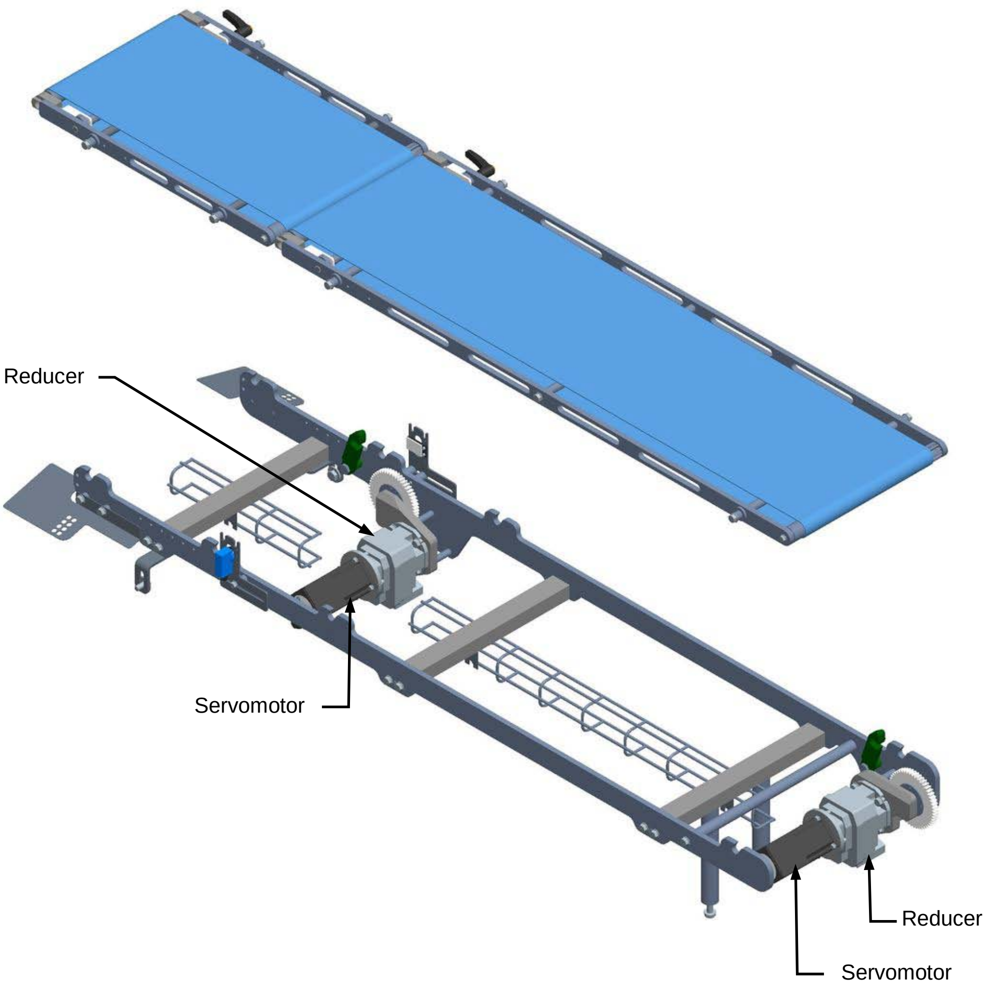
2. Reassemble all the parts back on again by reverse procedure.

8.11.3 Checking the servomotors and reducers condition

Machine status	Power on
Equipment	Standard workshop tools

Visually check the condition of the servomotors and reducers during machine operation. Check that the servomotors turn in the right direction.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

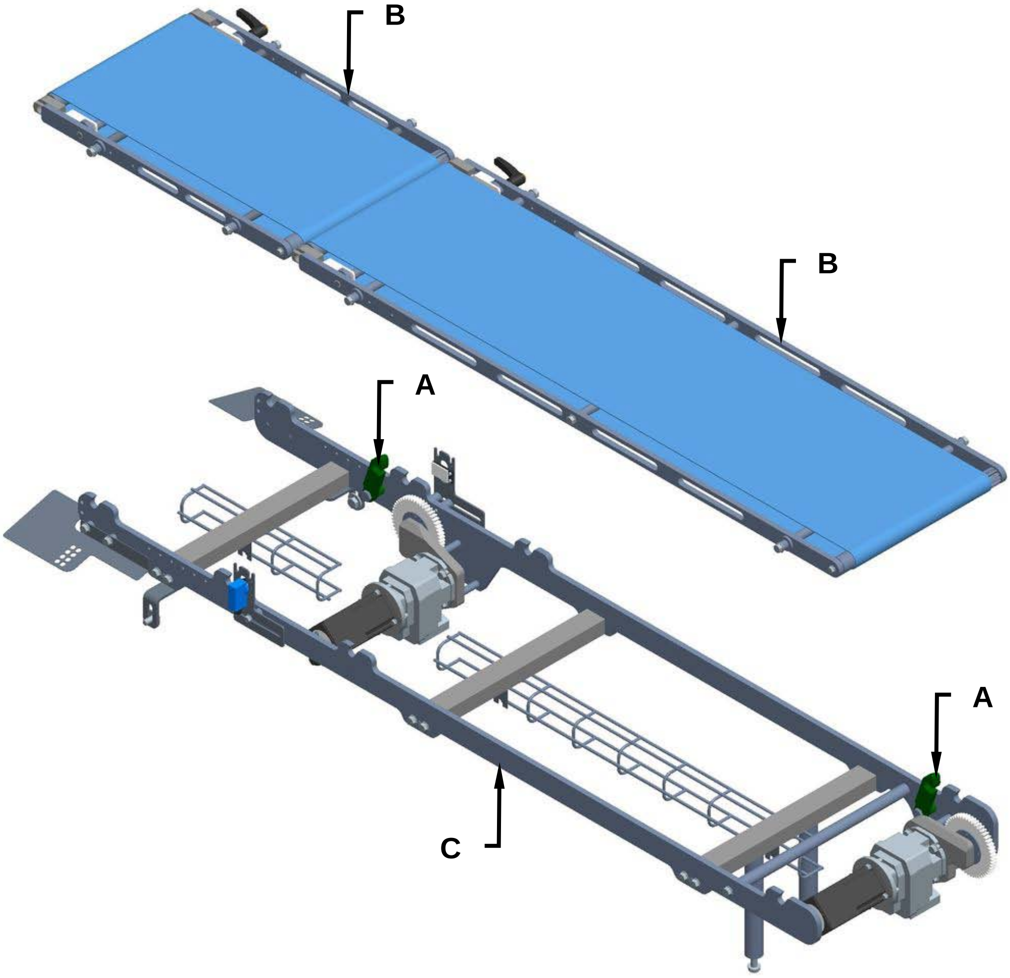


8.11.4 Replacing the servomotors and reducers

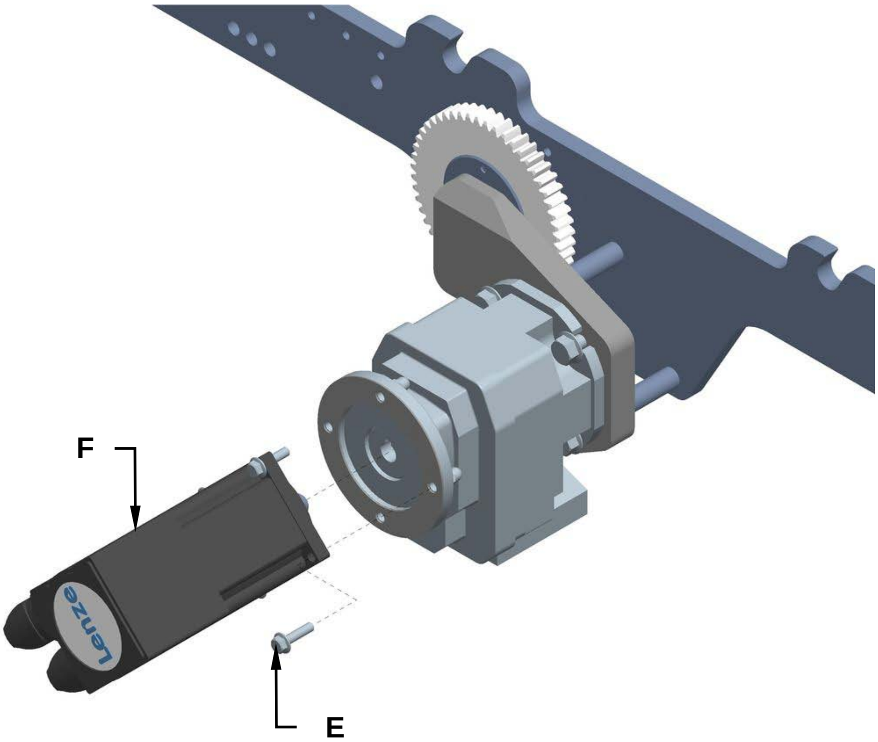
Machine status	Power sources deactivated
Equipment	Standard workshop tools

Proceed as follows to replace the motor, servomotors and reducers:

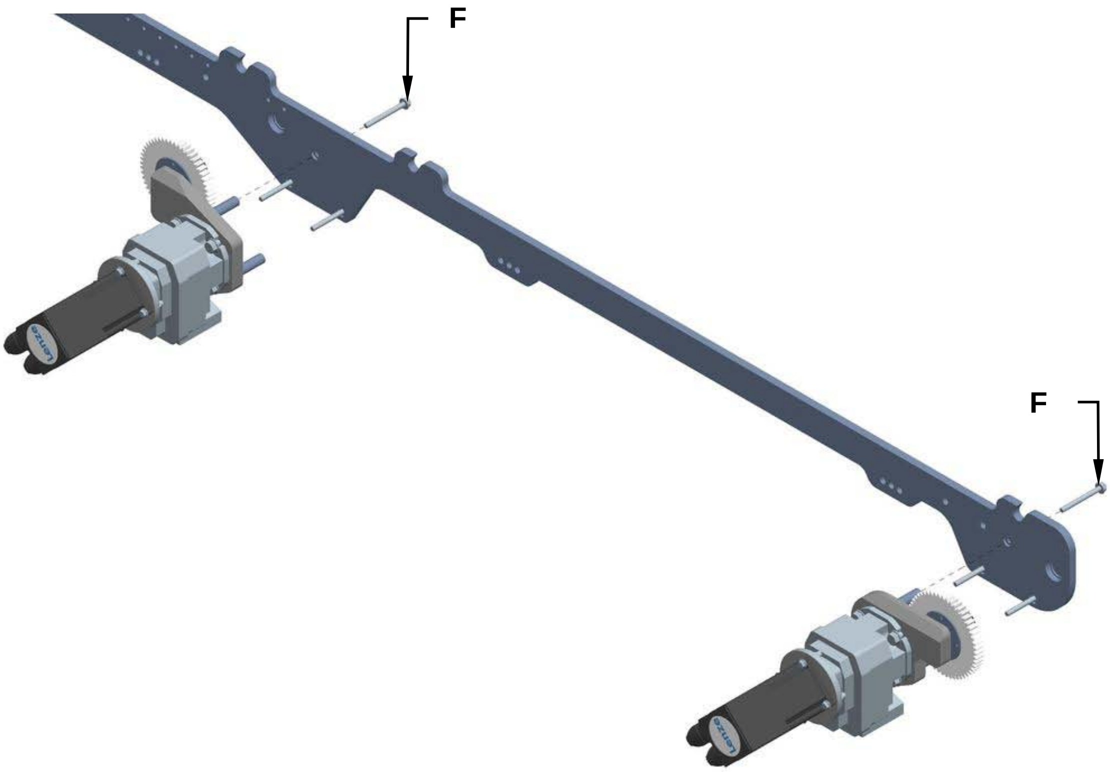
1. Disconnect the power supply.
2. Open the hooks (**A**) and remove the belts (**B**) from the support (**C**).



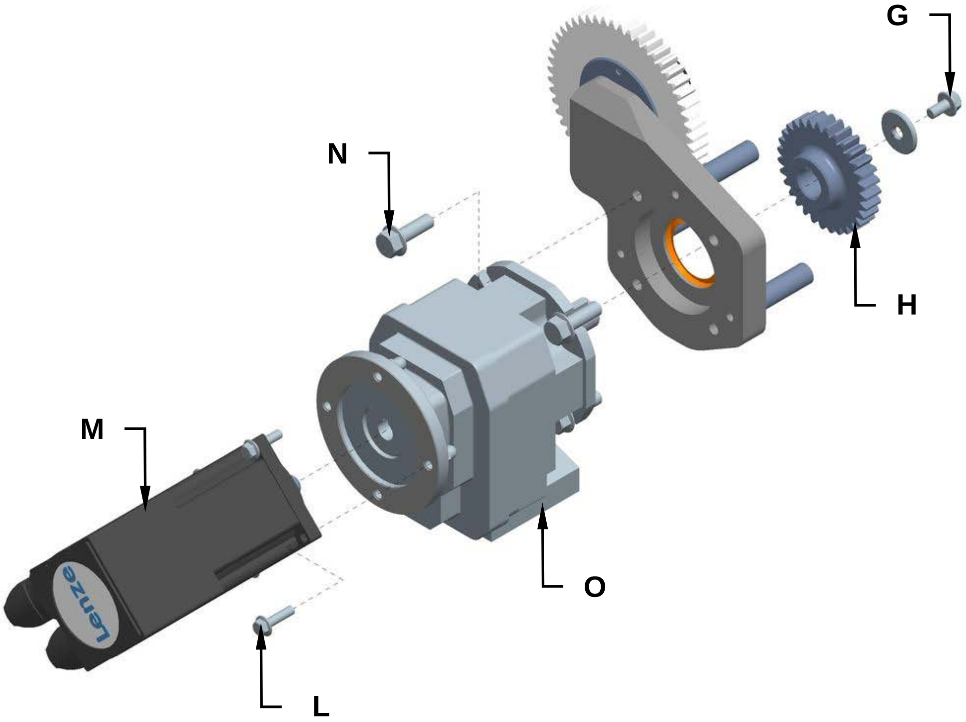
3. Unscrew the screws (**D**) and replace the servomotors (**E**).

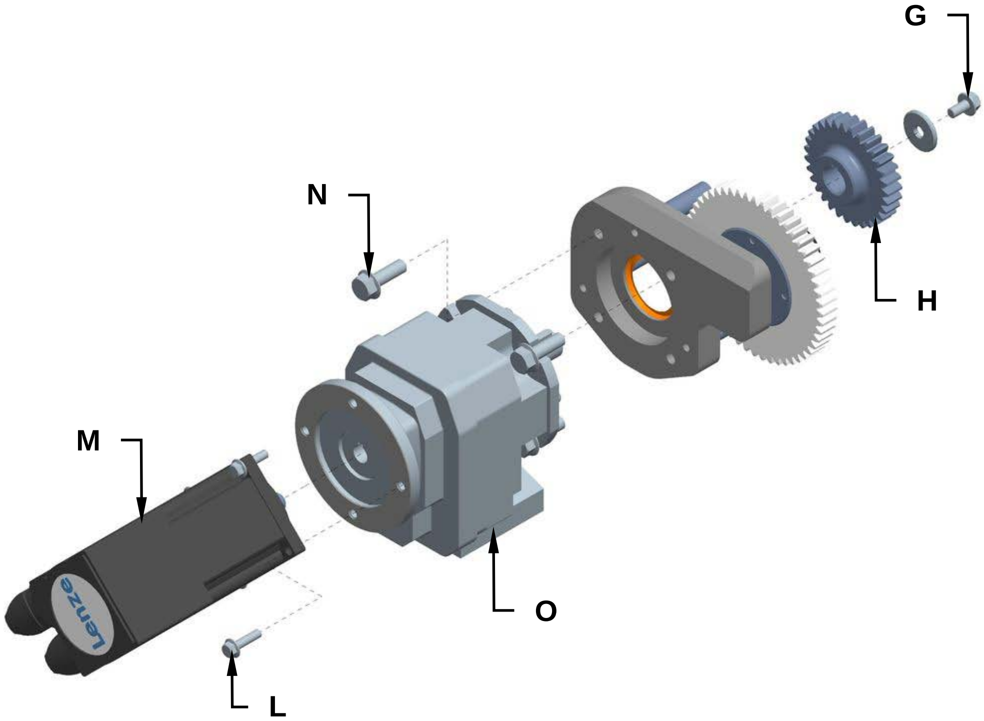


4. To replace the reducers, unscrew the screws (**F**) and remove the entire servomotor assemblies.



5. For both assemblies, unscrew the screw (**G**) and remove the cogwheel (**H**).
6. Unscrew the screws (**L**) and remove the servomotors (**M**).
7. Unscrew the screws (**N**) and replace the reducers (**O**).





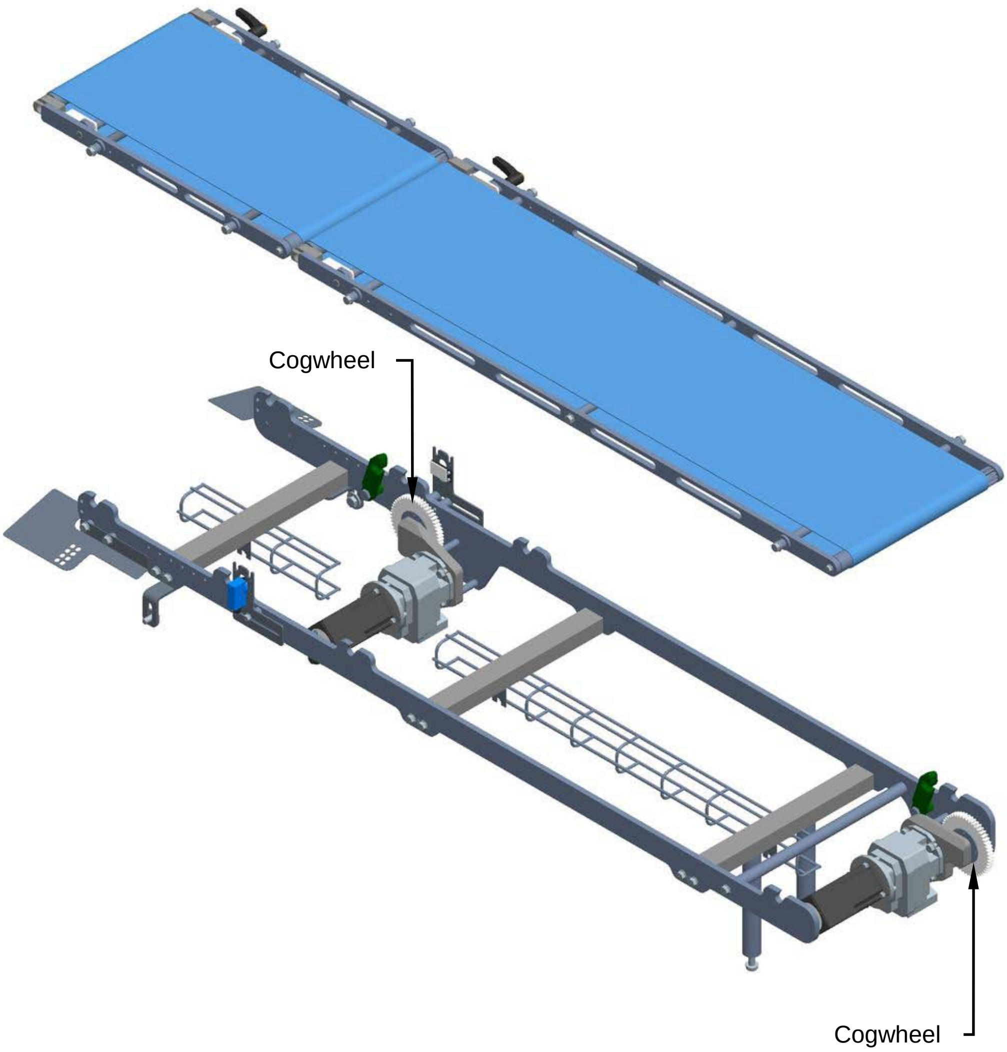
8. Remount the new assemblies, following the above procedure in the reverse order.

8.11.5 Checking the cogwheels condition

Machine status	Power sources activated
Equipment	Standard workshop tools

Visually check the condition of the cogwheels during machine operation.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

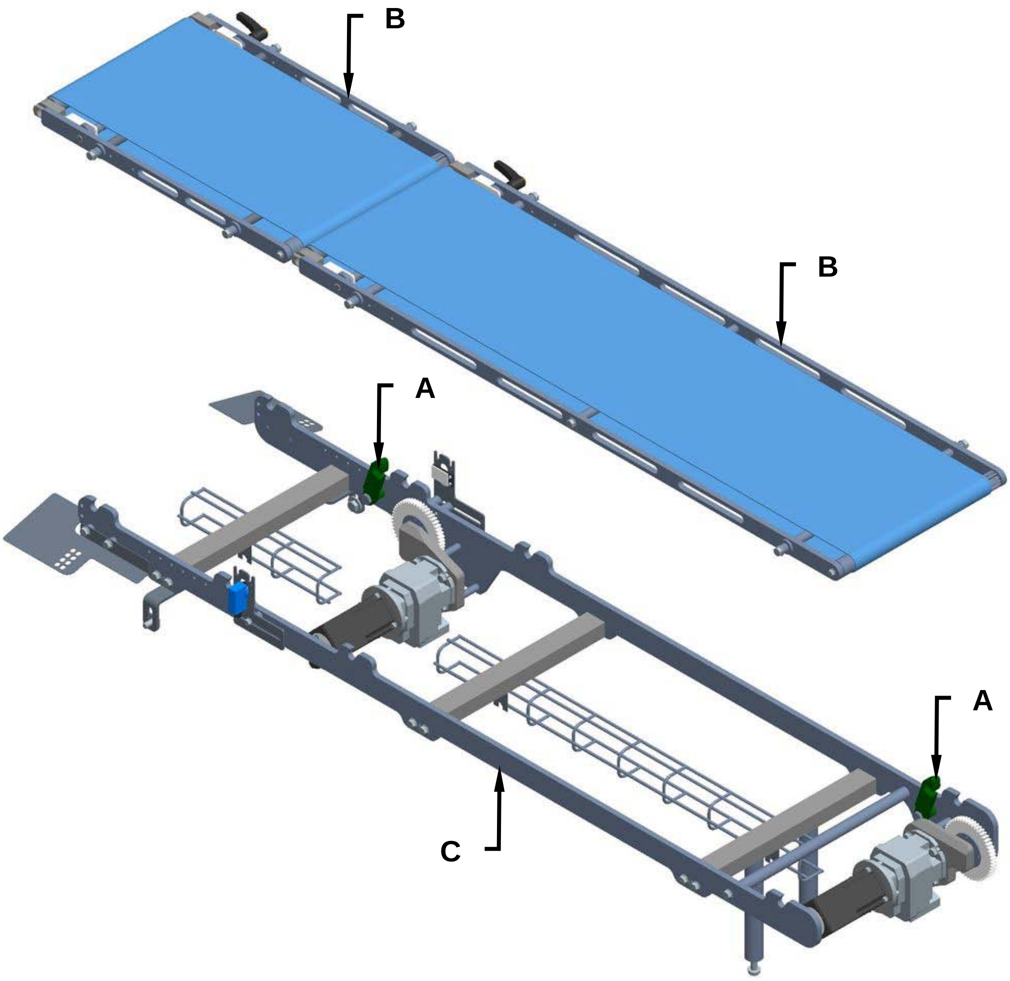


8.11.6 Replacing the cogwheels

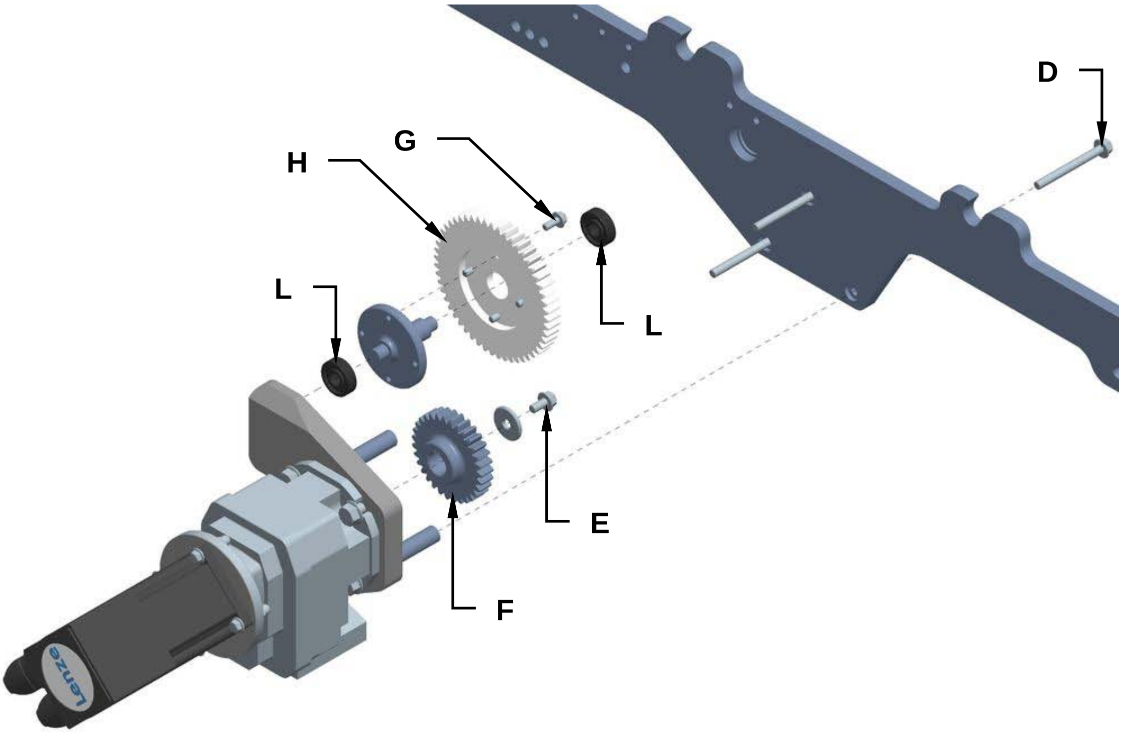
Machine status	Power sources deactivated
Equipment	Standard workshop tools

Proceed as follows to replace the cogwheels:

1. Disconnect the power supply.
2. Open the hooks (**A**) and remove the belts (**B**) from the support (**C**).



3. Unscrew the screws (**D**) to remove the servomotor and reducer assembly.
4. Unscrew the screw (**E**) and replace the cogwheel (**F**).
5. Unscrew the screws (**G**) and replace the cogwheel (**H**) and the bearings (**L**).



6. Mount the new cogwheel, following the above procedure in the reverse order.

8.12 DELIVERY BELT

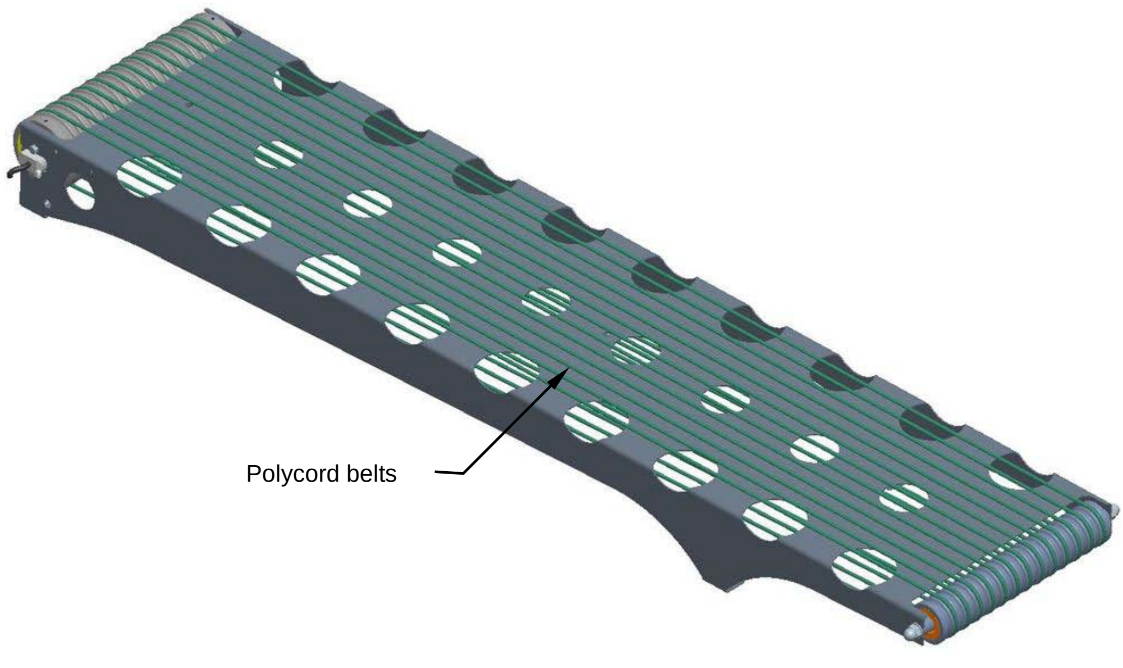
8.12.1 Checking the Polycord belts

Machine status	Power sources activated
Equipment	Standard workshop tools

Visually check the Polycord belt condition during machine operation. Make sure they are centred, run smoothly and show no signs of wear.

Check the tension of the belt, which determines correct movement of the elements being conveyed. Make sure the belt remains in contact with the rollers.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

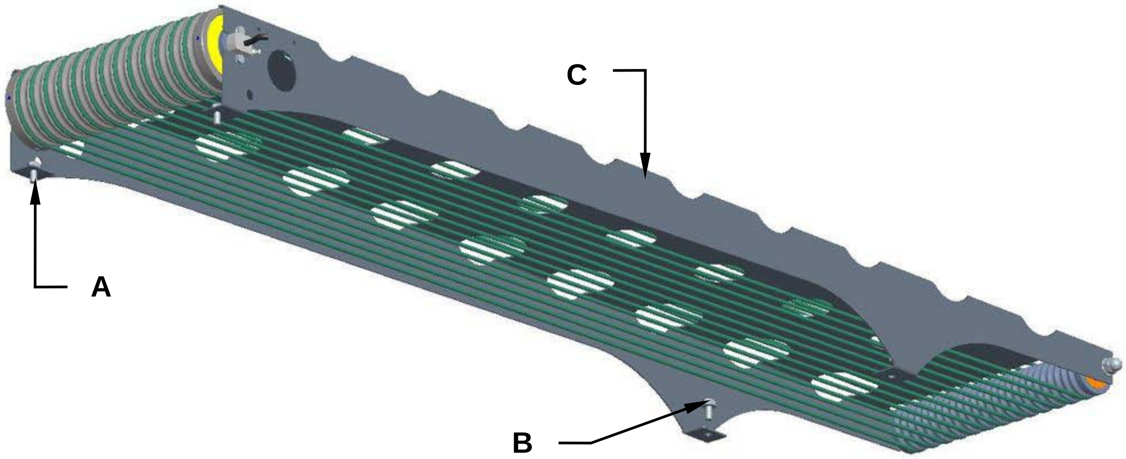


8.12.2 Replacement of the Polycord Belts

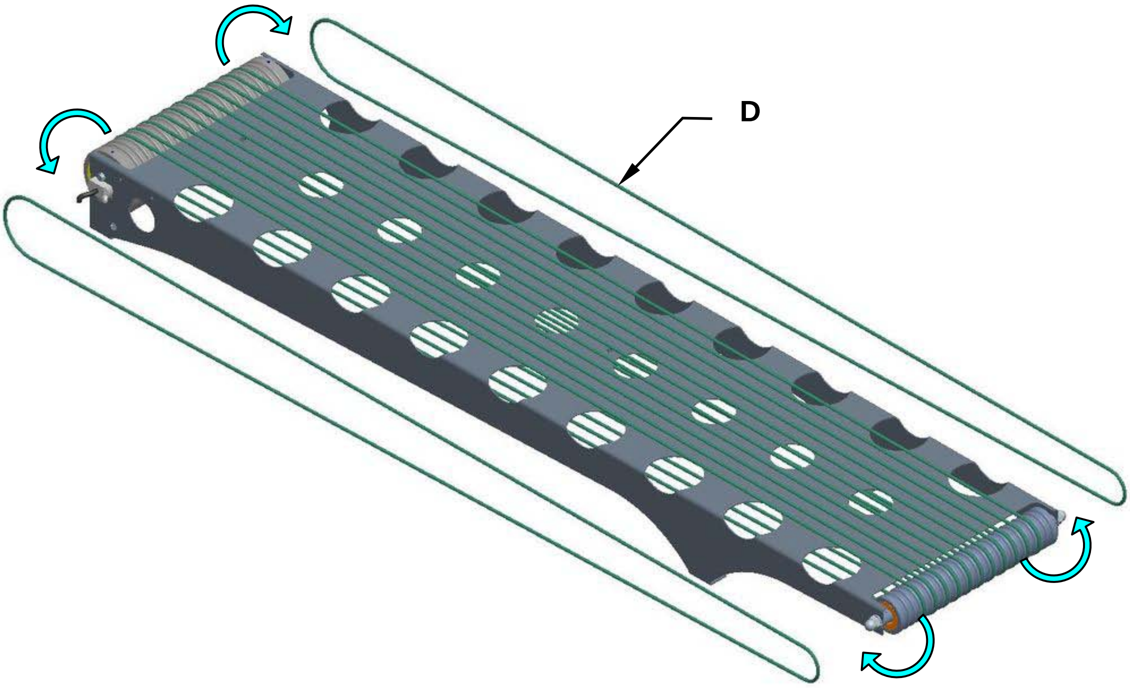
Machine status	Power ON.
Equipment	Standard workshop tools

For Polycord belt replacement, proceed as follows:

1. Unscrew the screws (**A**) and (**B**), remove the exit belt from the closing machine.



2. Remove the Polycord belts (**D**) from the drum motor and the roller.



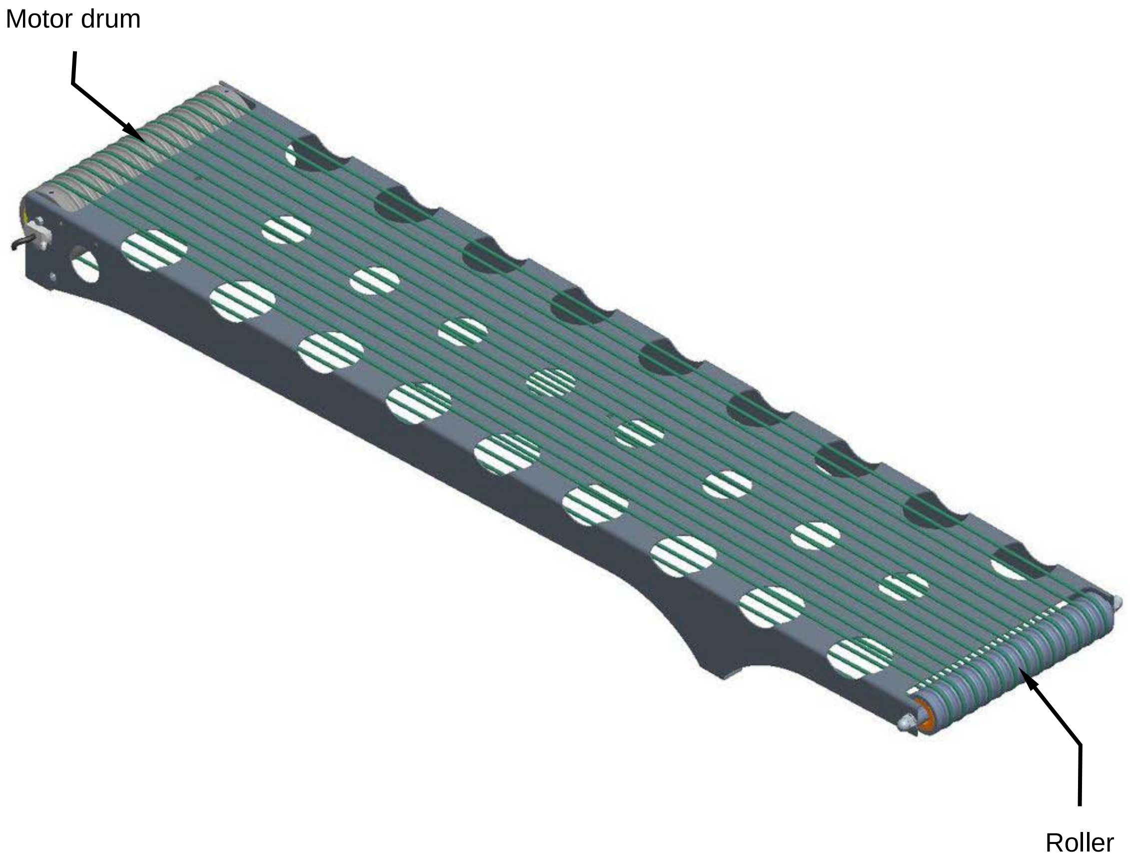
3. Replace the Polycord belts and assemble all the components following the preceding procedure to bashful.

8.12.3 Checking the Motor Drum and Roller good working conditions

Machine status	Power ON.
Equipment	Standard workshop tools

During standard machine operation cycles, carry out a visual inspection to ensure that the motor drum and roller are in good working conditions. Check to ensure that direction the motor drum is turning in, is correct.

Should any anomalies be found, please warn the person responsible/people in charge of setting the unit group back to optimised working conditions.

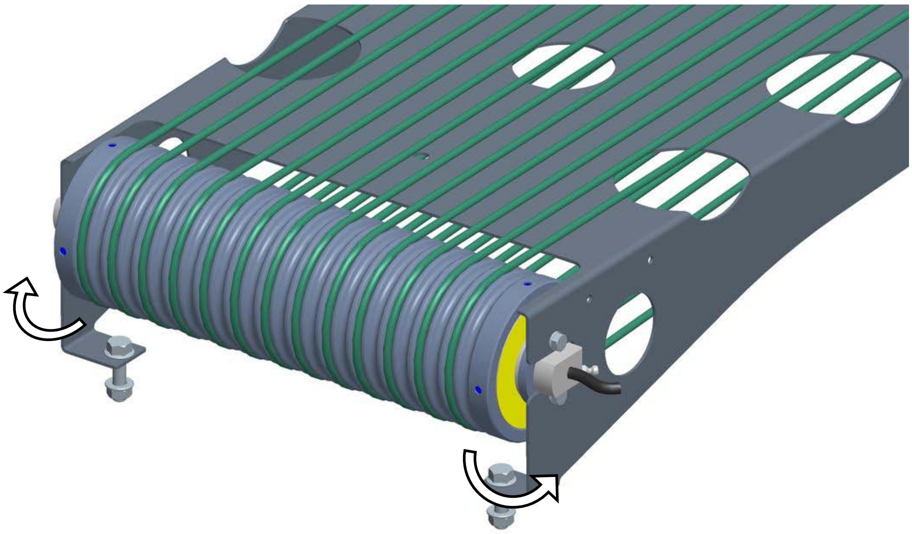


8.12.4 Motor drum and roller replacement

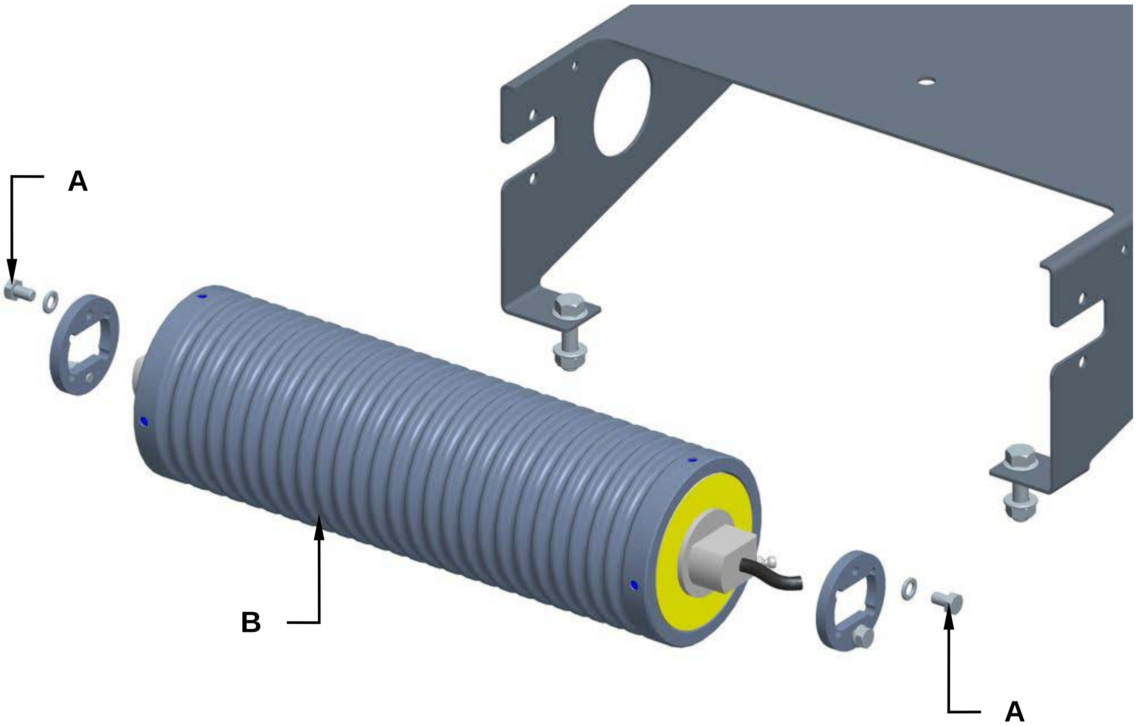
Machine status	Power ON.
Equipment	Standard workshop tools

For motor drum replacement, proceed as follows:

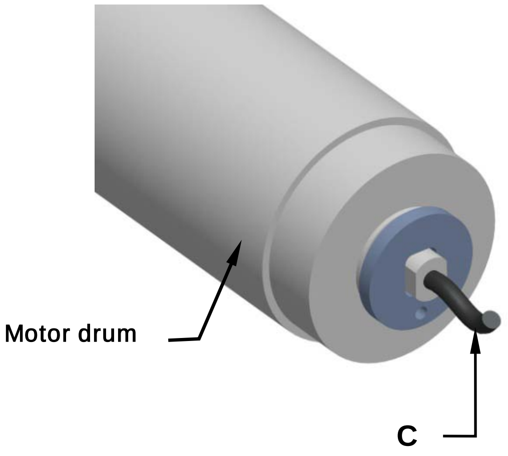
1. Remove the Polycord belts from the drum motor.

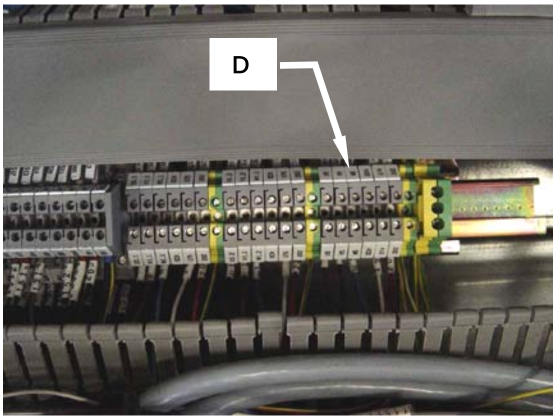


2. Unscrew the screws (**A**) and replace the drum motor (**B**).



3. Disconnect the electric cable (**C**) from the terminal block (**D**) located in the electric control cabinet by consulting and using the code number specified both on the electric cable and on the terminal block. Slip the electrical cable out on the motor drum side and after having assembled on the new motor drum, slip its electric cable back on in the same way the replaced cable was on.

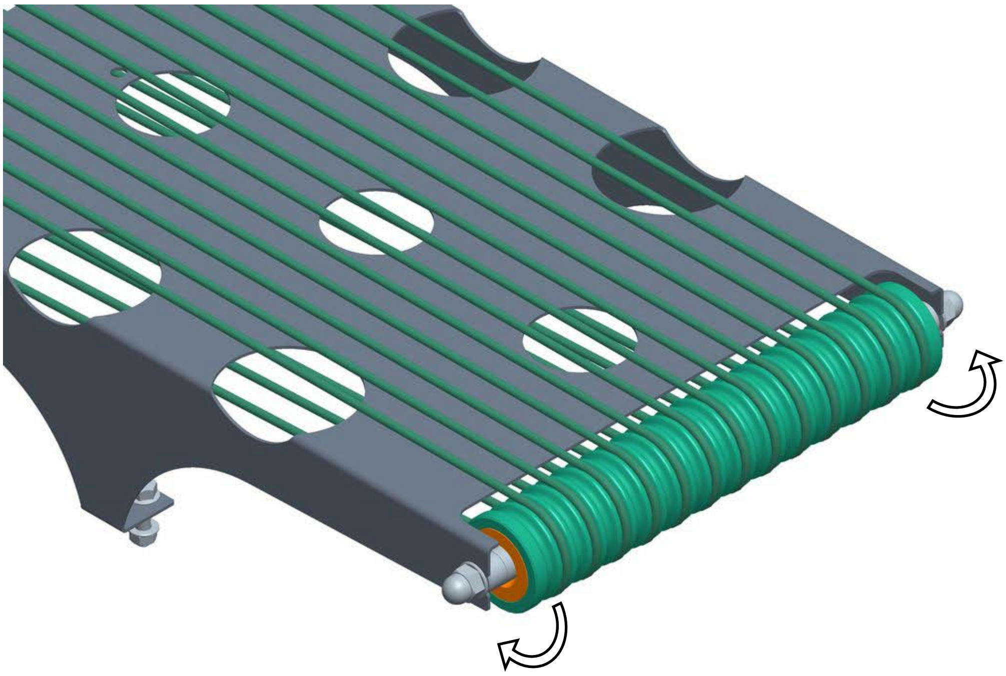




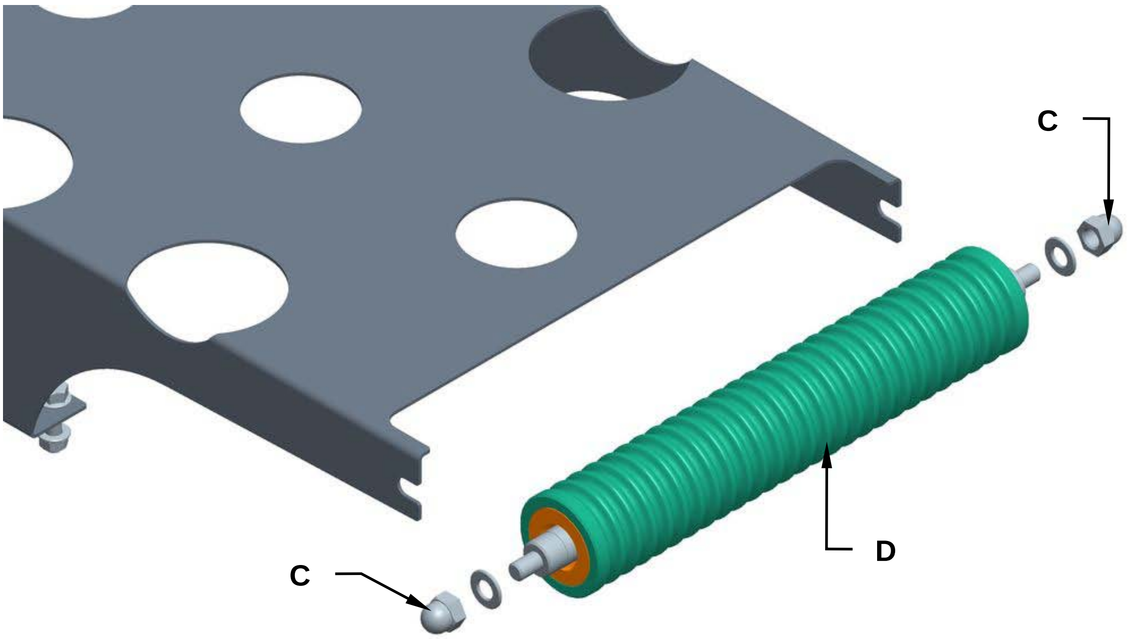
4. Reassemble all the parts back on again by reverse procedure.

To replace the roller, proceed as follows:

1. Remove the Polycord from the roller.



2. Unscrew the nuts (**C**) and replace the roller (**D**).



3. Assemble all the components following the preceding procedure to bashful.

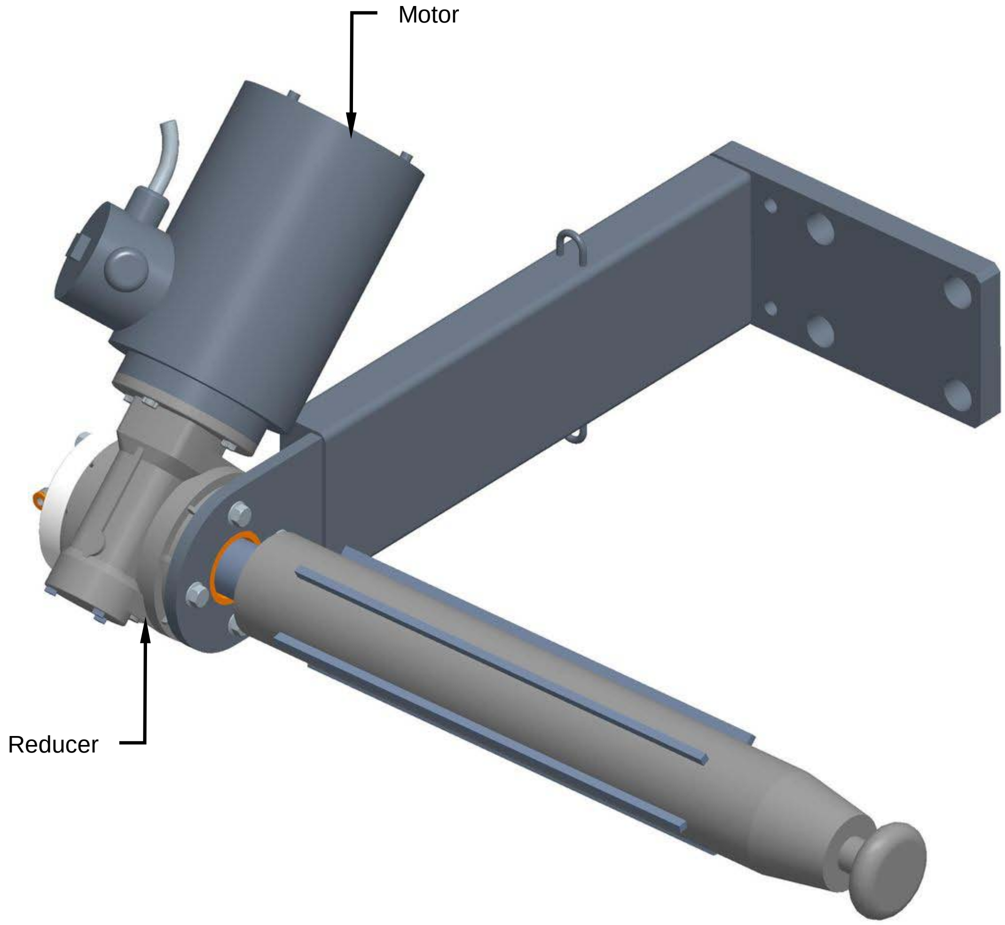
8.13 FILM UNWINDER

8.13.1 Checking the motor and reducer

Machine status	Power on
Equipment	Standard workshop tools

Visually check the condition of the motor and reducer during machine operation. Check that the motor turn in the right direction.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

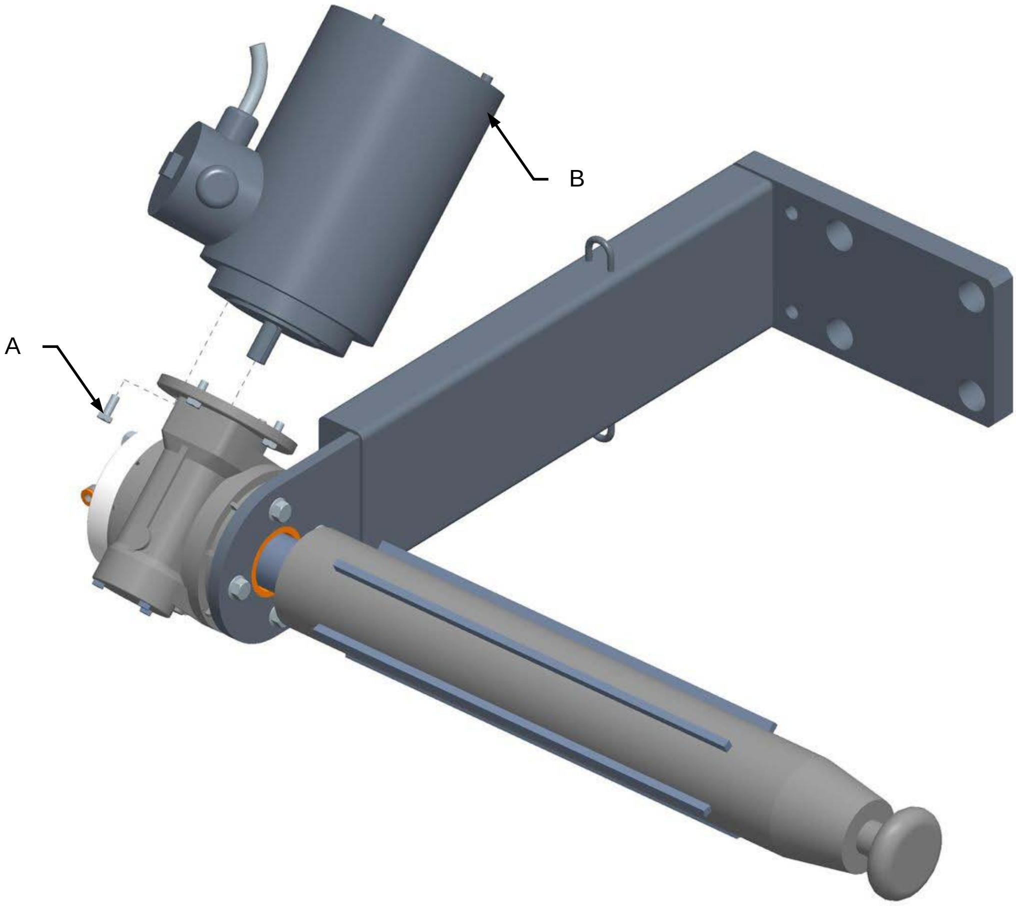


8.13.2 Replacing the motor and reducer

Machine status	Power sources deactivated
Equipment	Standard workshop tools

Proceed as follows to replace the motor:

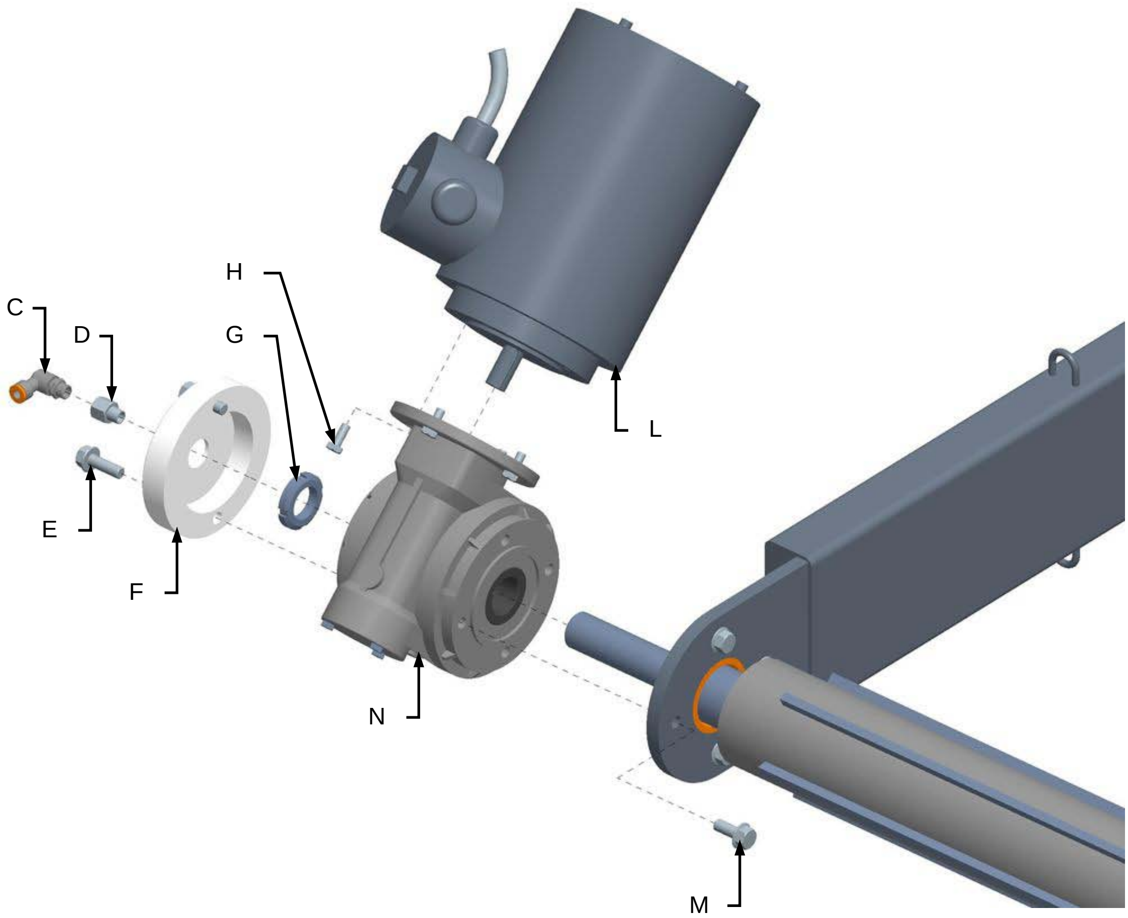
1. Disconnect the power supply.
2. Unscrew the screws (**A**) and replace the motor (**B**).



3. Remount all the components in the reverse order.

Proceed as follows to replace the reduction gears:

1. Disconnect the electric power and compressed air supply.
2. Unscrew the fitting (**C**) and screws (**D**).
3. Unscrew the screws (**E**) and remove the cover (**F**).
4. Unscrew the ring nut (**G**).
5. Unscrew the screws (**H**) and remove the motor (**L**).
6. Unscrew the screws (**M**) and replace reduction gear (**N**).



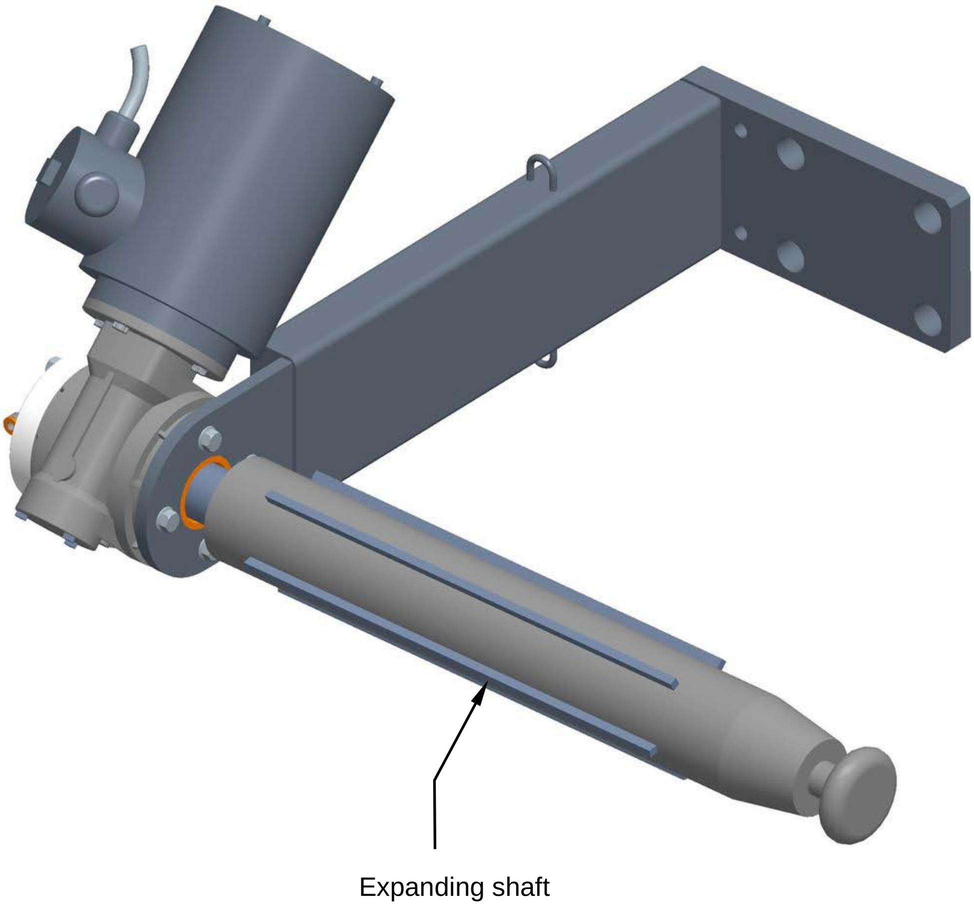
7. Remount all the components in the reverse order.

8.13.3 Checking the expanding shaft condition

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the expanding shafts.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

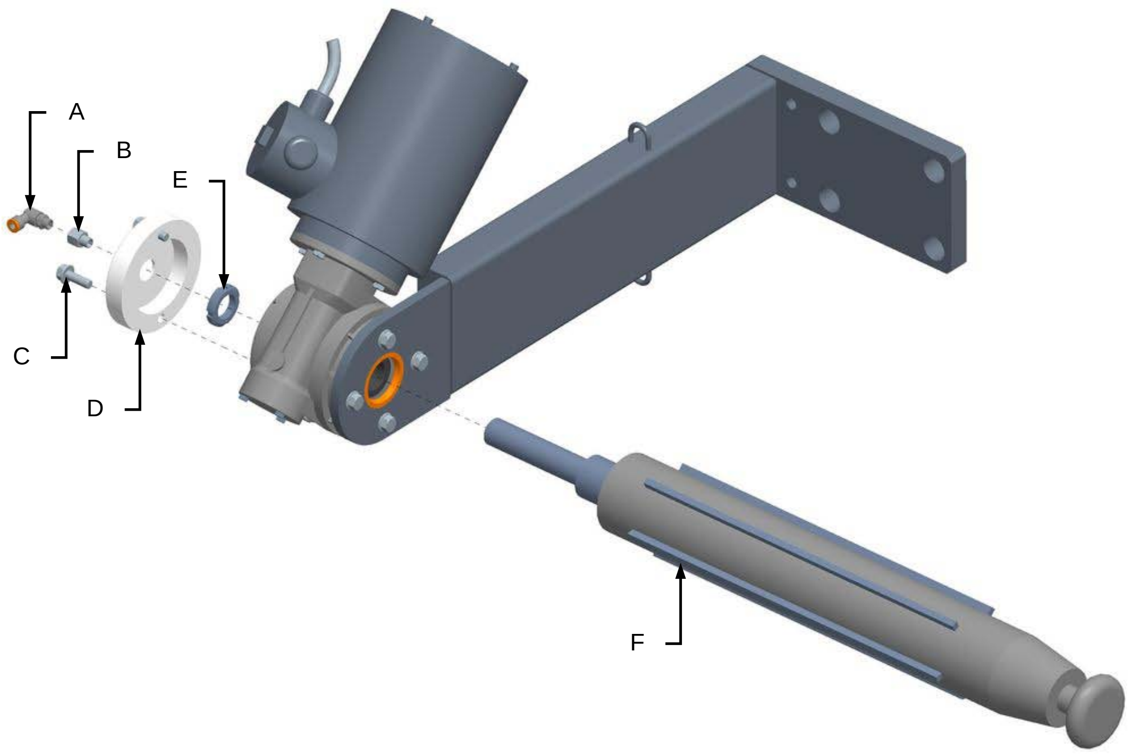


8.13.4 Replacing the expanding shafts

Machine status	Power off
Tools	Normal workshop equipment

Proceed as follows to replace the expanding shafts:

1. Disconnect the electric power and compressed air supply.
2. Unscrew the fitting (**A**) and (**B**).
3. Unscrew the screws (**C**) and remove the cover (**D**).
4. Unscrew the ring nut (**E**).
5. Replace the expanding shaft (**F**).



6. Remount all the components in the reverse order.

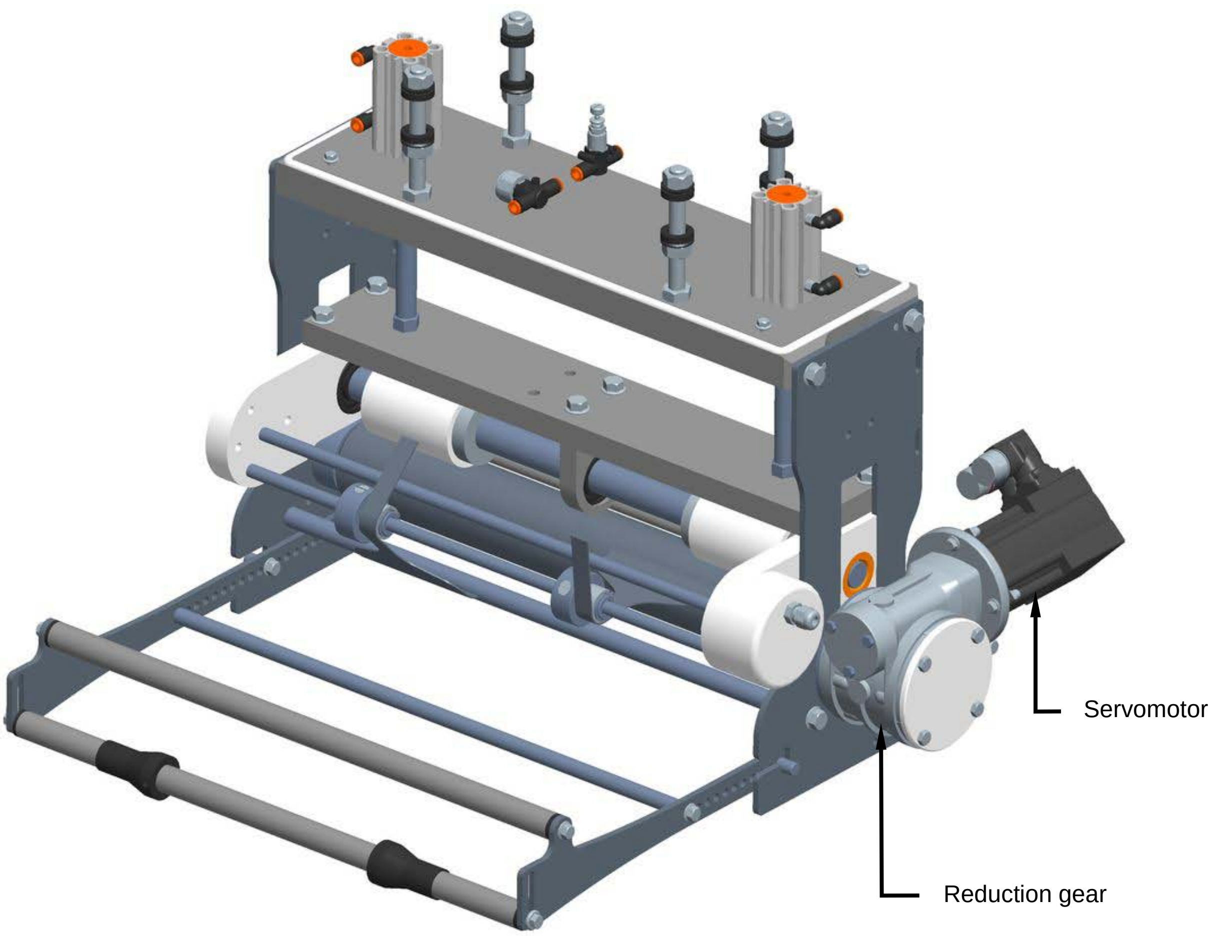
8.14 FILM DRIVE ROLLERS AND CONTROL

8.14.1 Checking the servomotor and reduction gear

Machine status	Power sources activated
Equipment	Standard workshop tools

With the machine running a normal cycle, visually check the condition of the servomotor and reduction gear. Make sure the servomotor turns in the right direction.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

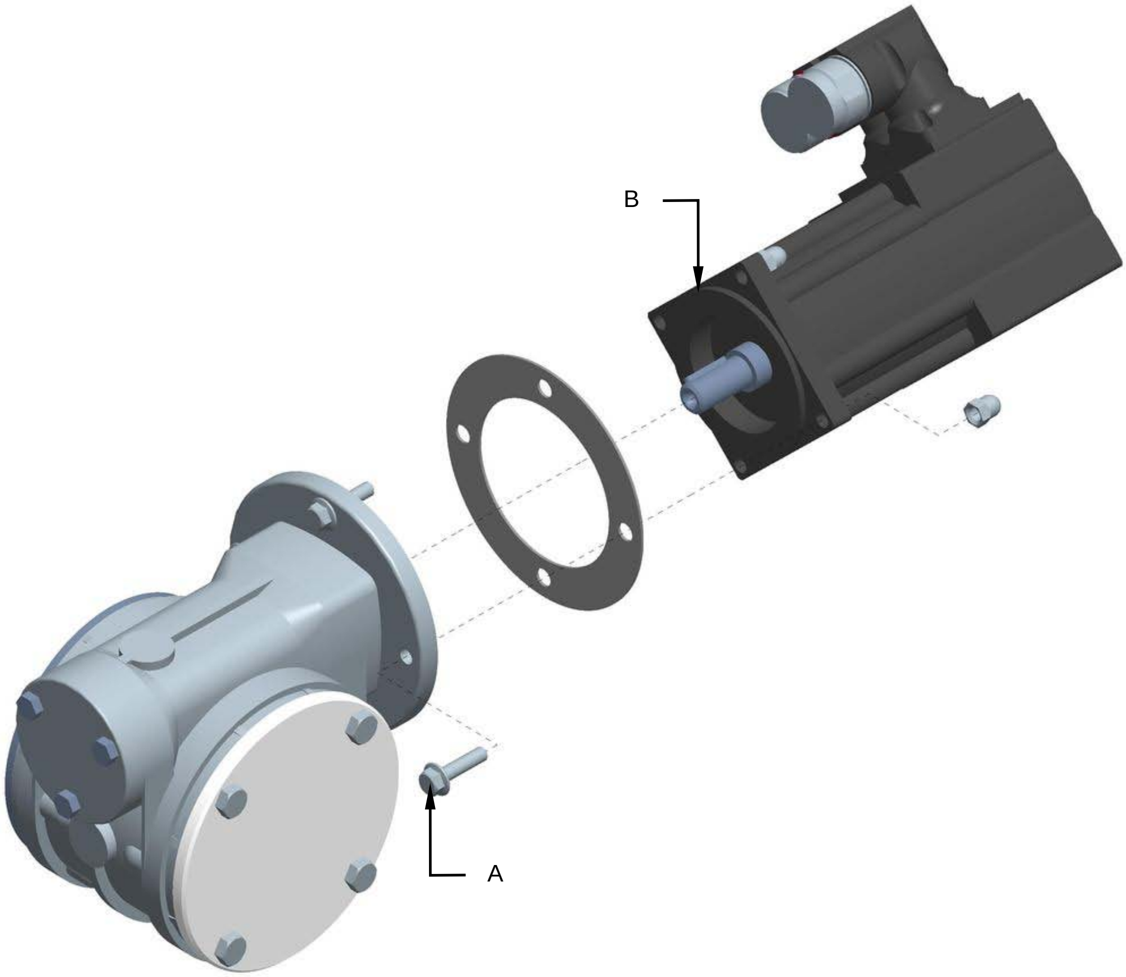


8.14.2 Replacing the servomotor and reduction gear

Machine status	Power sources deactivated
Equipment	Standard workshop tools

Proceed as follows to replace the servomotor:

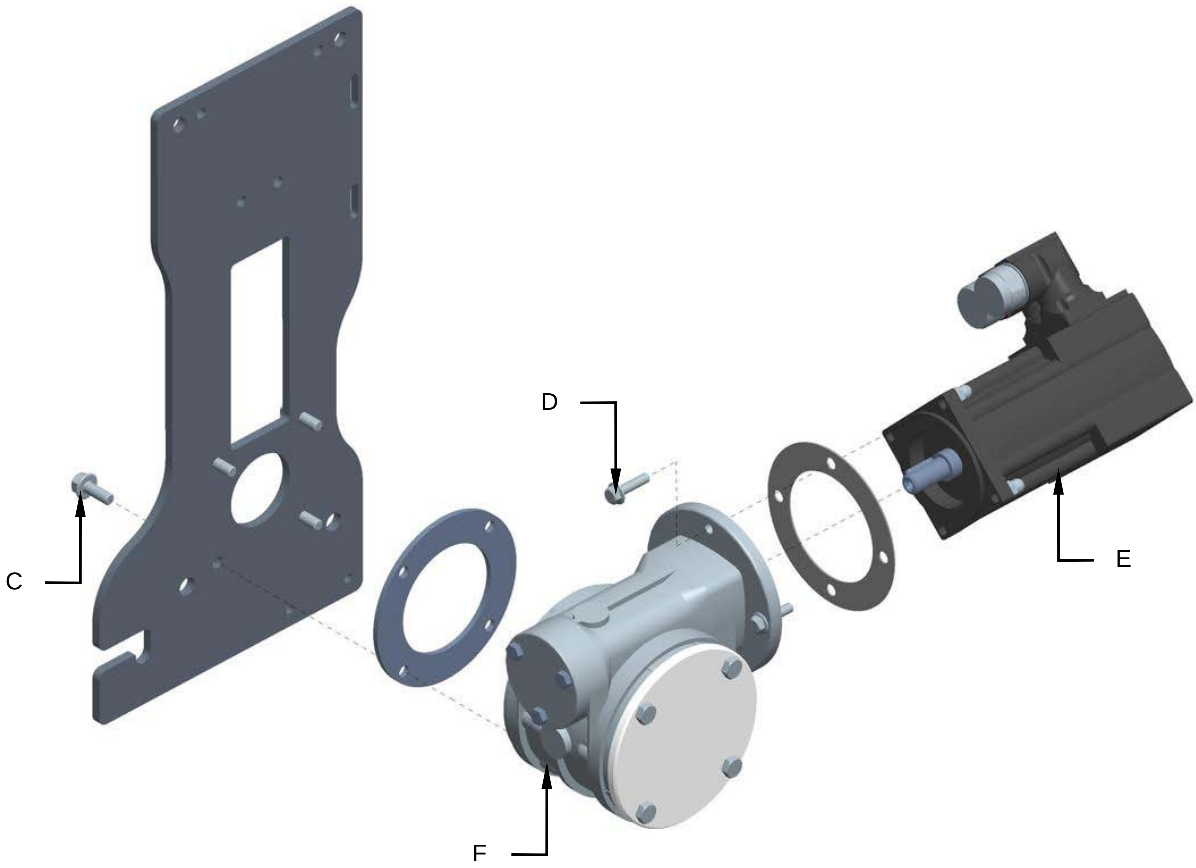
1. Disconnect the servomotor power sources.
2. Unscrew the screws (**A**) and remove the servomotor (**B**).



3. Remount all the components in the reverse order.

Proceed as follows to replace the reduction gear:

1. Disconnect the servomotor power sources.
2. Unscrew the screws (**C**).
3. Unscrew the screws (**D**) and remove the servomotor (**E**).
4. Replace the reduction gear (**F**).



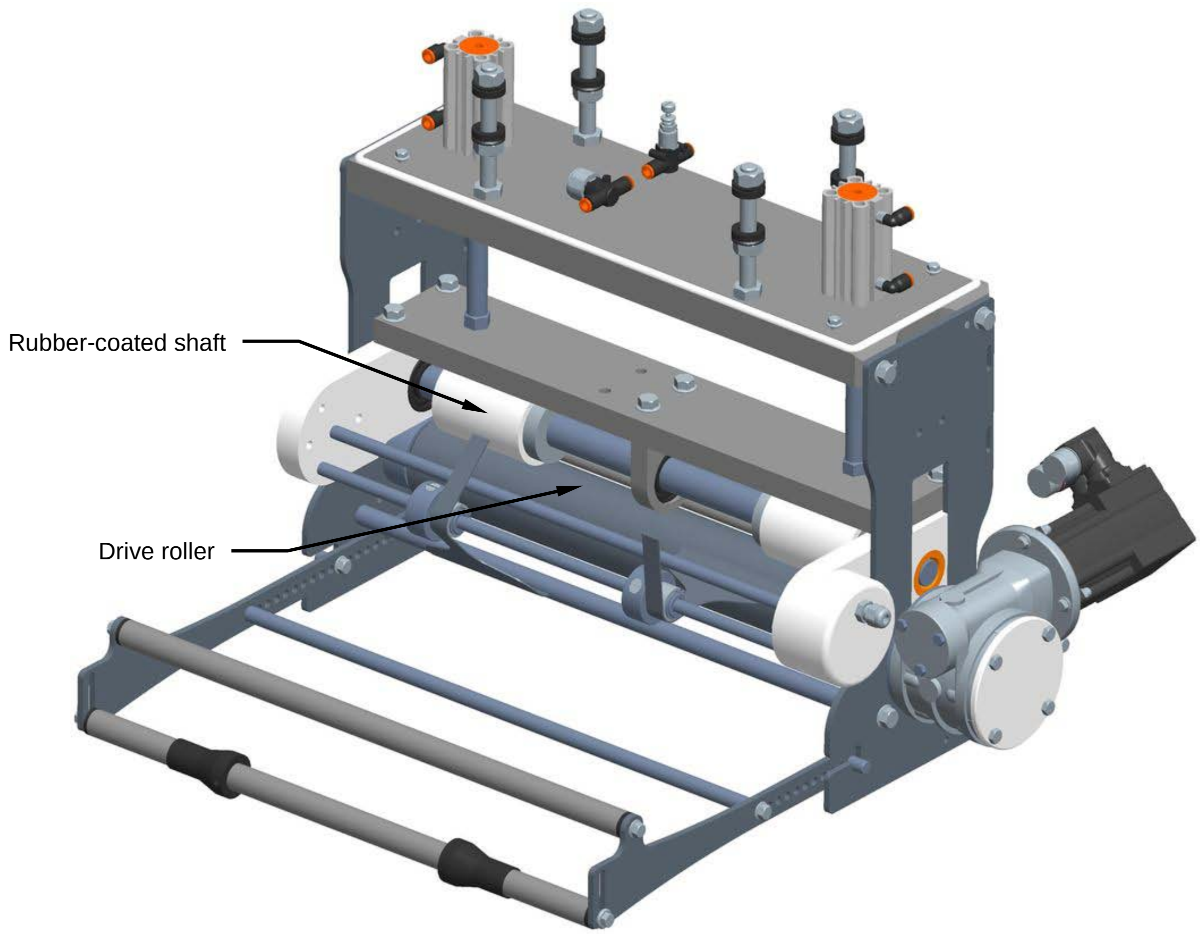
5. Remount all the components in the reverse order.

8.14.3 Checking the drive roller and the rubber-coated shaft

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the drive roller and the rubber-coated shaft. Make sure their level of wear is not such as to affect film integrity.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

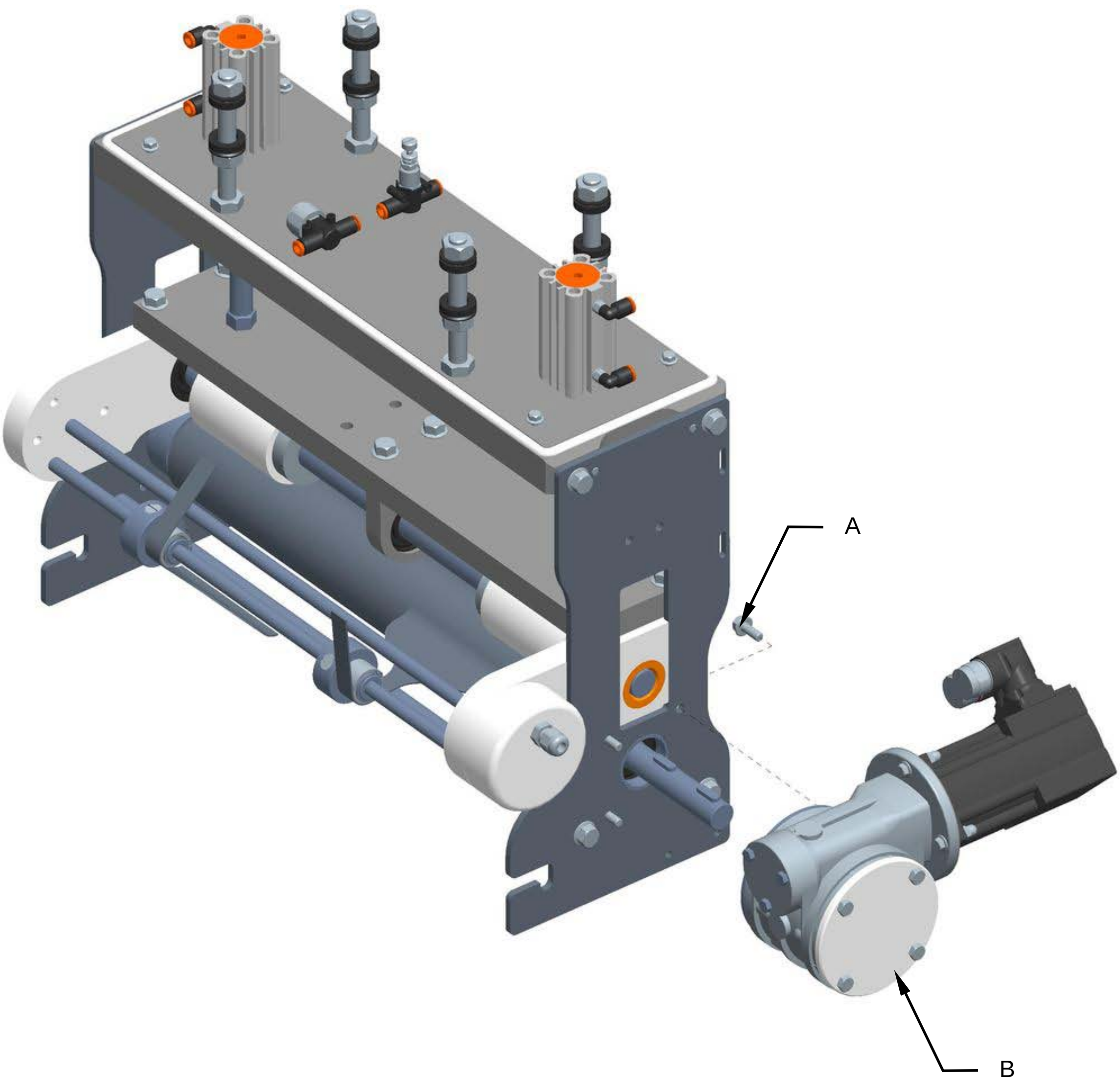


8.14.4 Replacing the drive roller and rubber-coated shaft

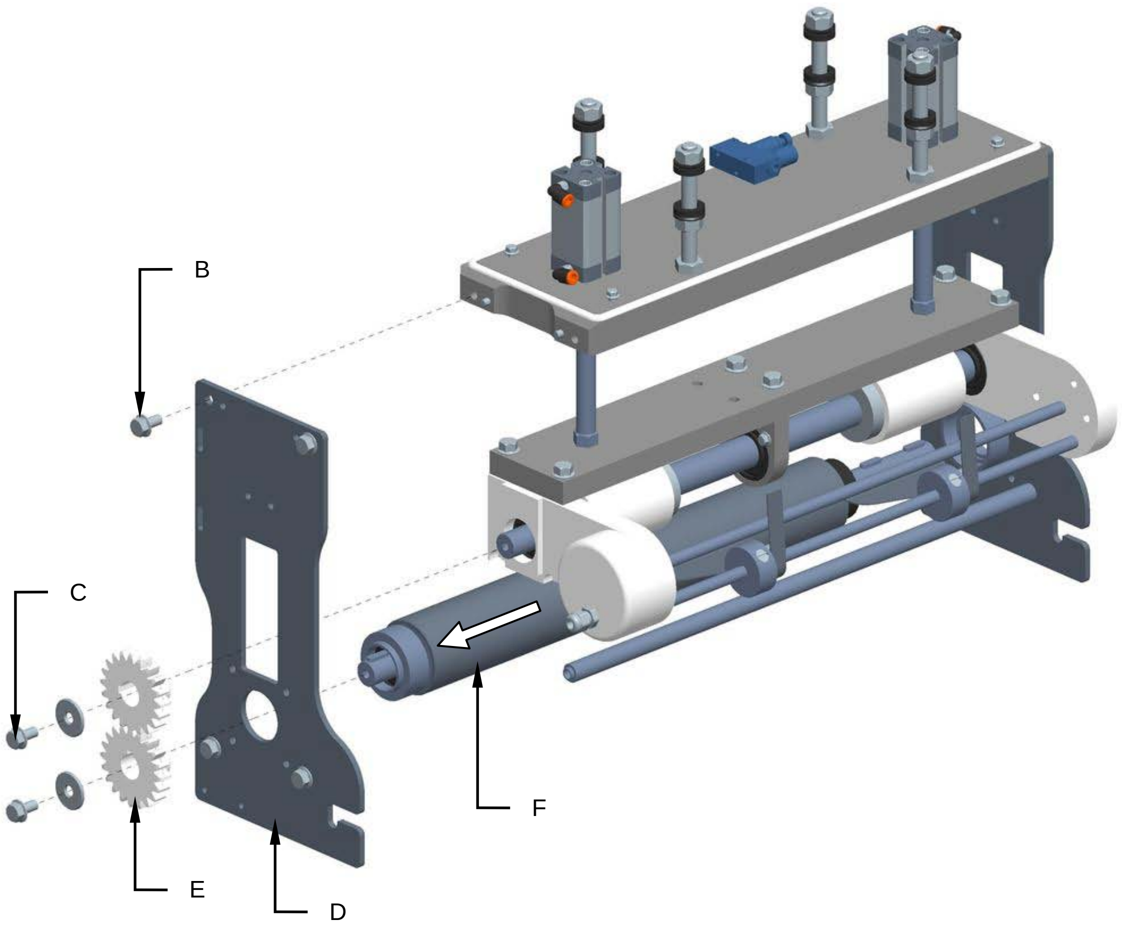
Machine status	Power off
Tools	Normal workshop equipment

Proceed as follows to replace the drive roller and rubber-coated shaft

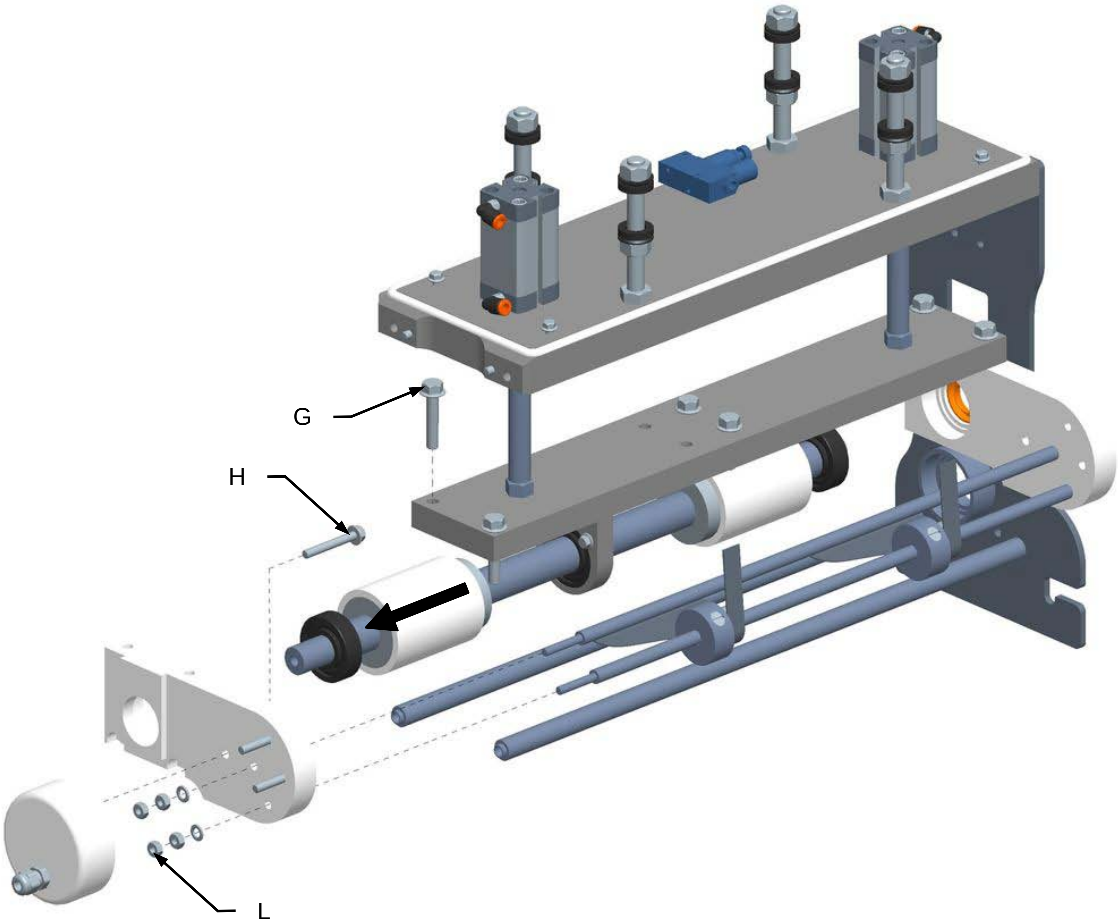
1. Unscrew the screws (**A**) and remove the servomotor and reduction gear group (**B**).



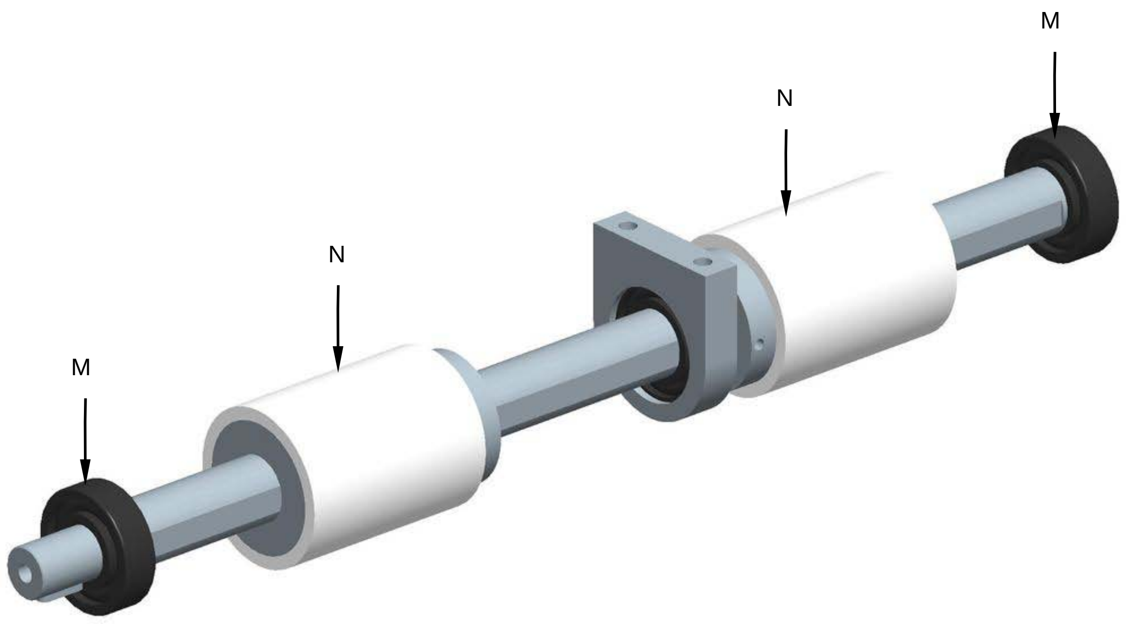
2. Unscrew the screws (**B**) e (**C**) and remove the shoulder (**D**).
3. Remove the gears (**E**).
4. Extract the drive roller and replace it (**F**).



5. Unscrew the screws (**M**) and (**N**).
6. Unscrew the nuts (**L**).
7. Extract the rubber-coated shaft.



8. Remove the bearings (**M**).
9. Remove and replace both rubber-coated shafts (**N**).



10. Remount all the components in the reverse order.

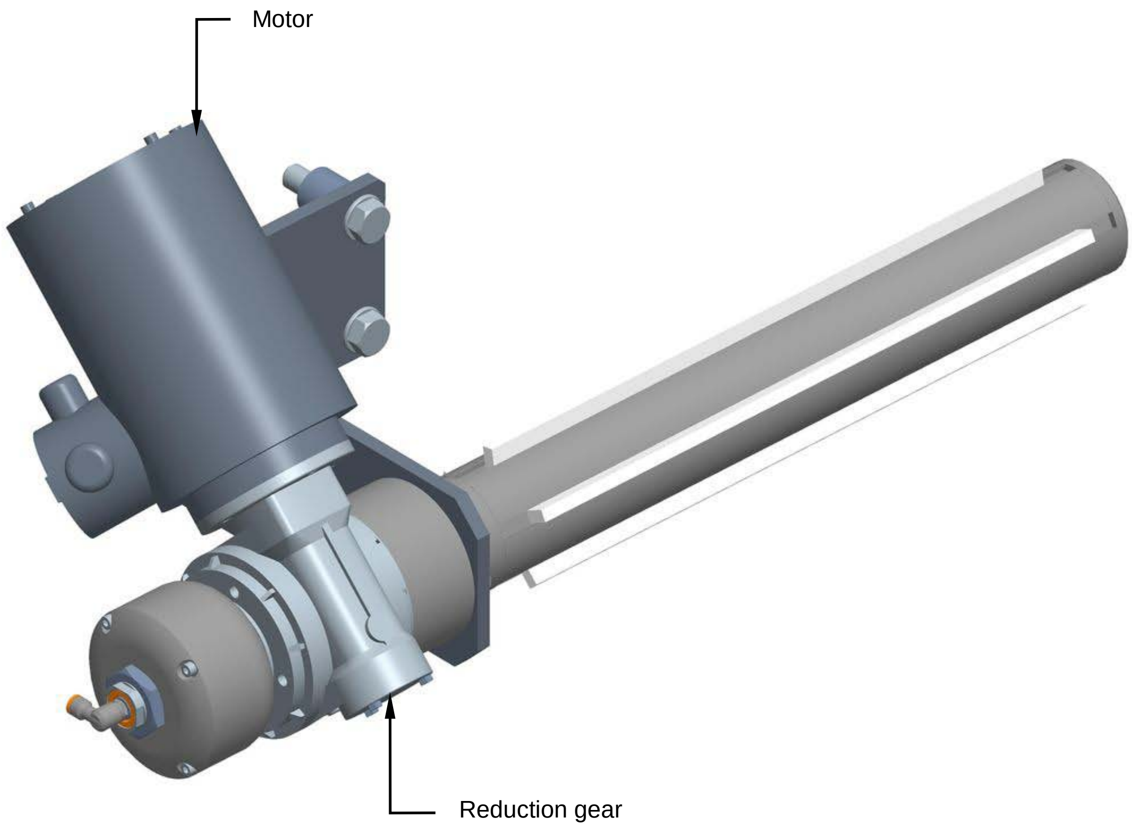
8.15 WINDING SHAFT

8.15.1 Checking the motor and reduction gear

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the motor and Reduction gear. Make sure the motor runs in the correct direction.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

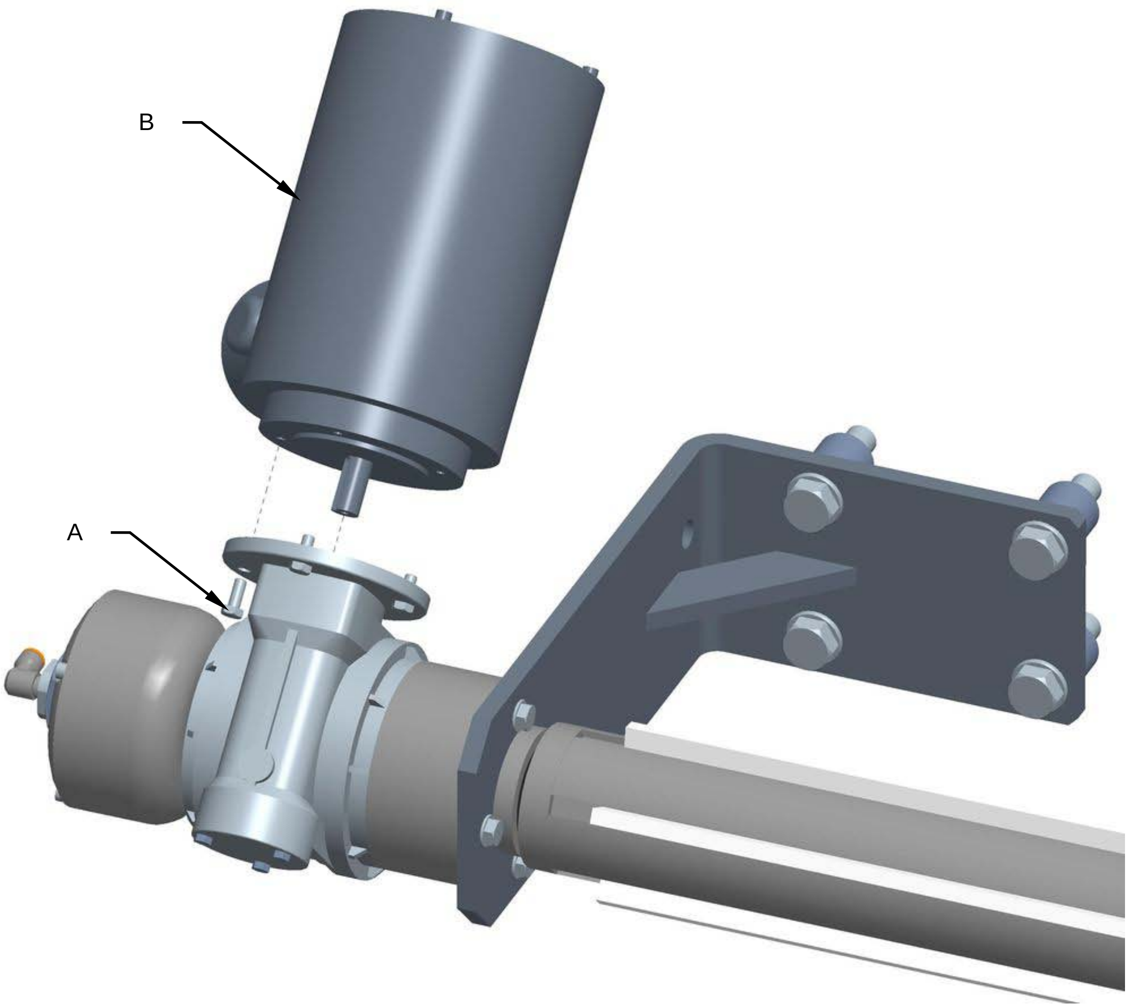


8.15.2 Replacing the motor and reduction gear

Machine status	Power off
Tools	Normal workshop equipment

Proceed as follows to replace the motor and reduction gear:

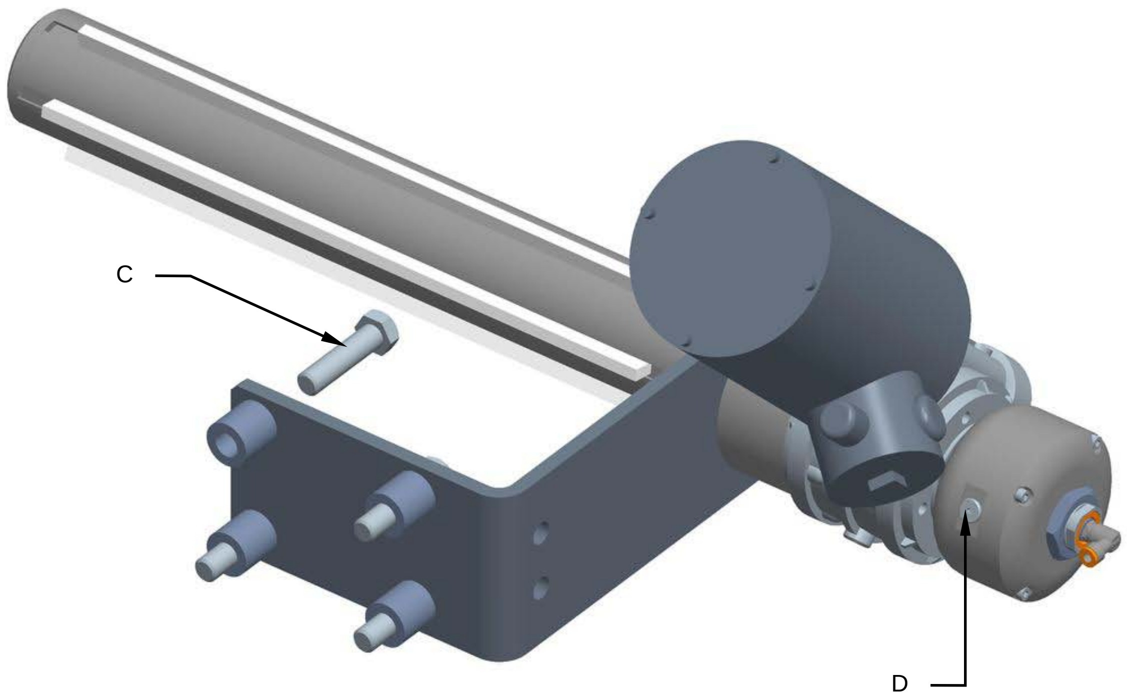
1. Disconnect the electric power and compressed air supply.
2. Unscrew the screws (**A**) and replace the motor (**B**).



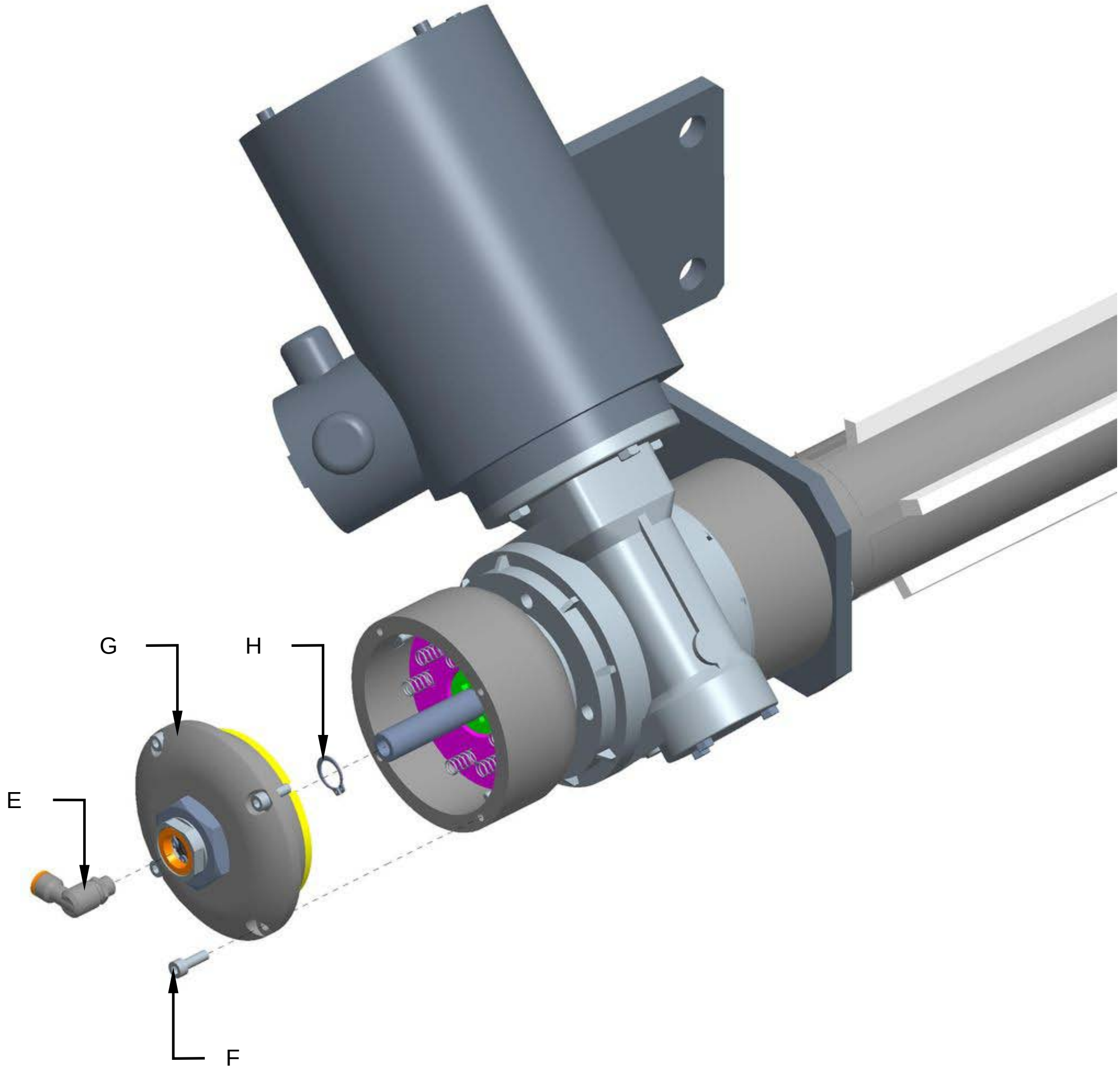
3. Remount all the components in the reverse order.

Proceed as follows to replace the reduction gear:

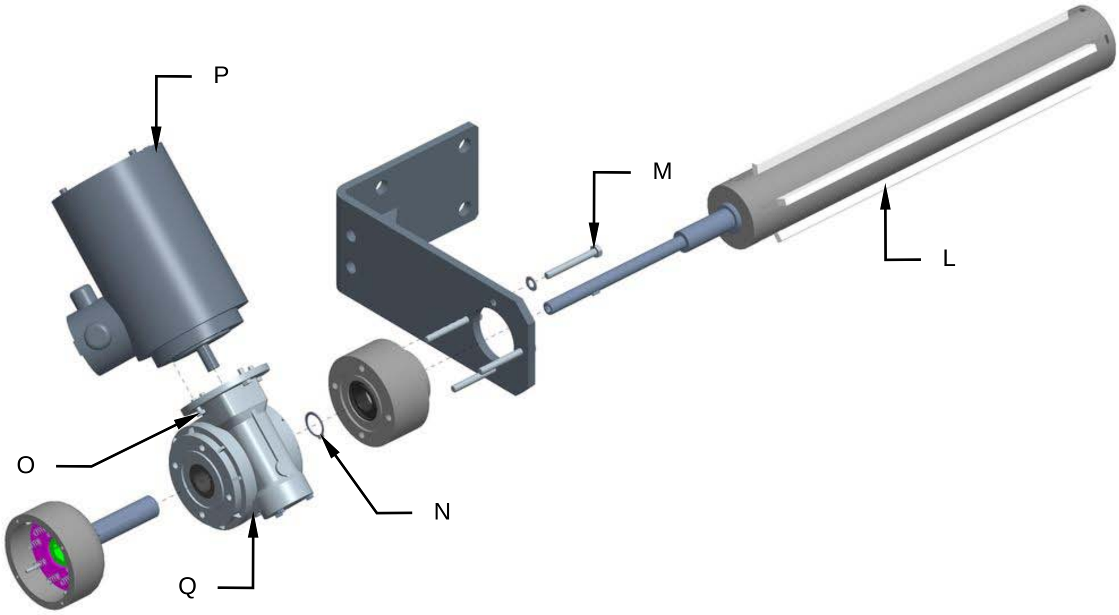
1. Disconnect the electric power and compressed air supply.
2. Unscrew the screws (**C**) and remove the reduction gear from the machine.
3. Unscrew the cap (**D**) and drain the oil from the clutch.



4. Remove the fitting (**E**).
5. Unscrew the screws (**F**) and remove the clutch cover (**G**).
6. Remove the snap ring (**H**).



7. Remove the expanding shaft (**L**).
8. Unscrew the screws (**M**).
9. Remove the snap ring (**N**).
10. Unscrew the screws (**O**) and remove the motor (**P**).
11. Remove and replace the reduction gear (**Q**).



12. Remount all the components in the reverse order.

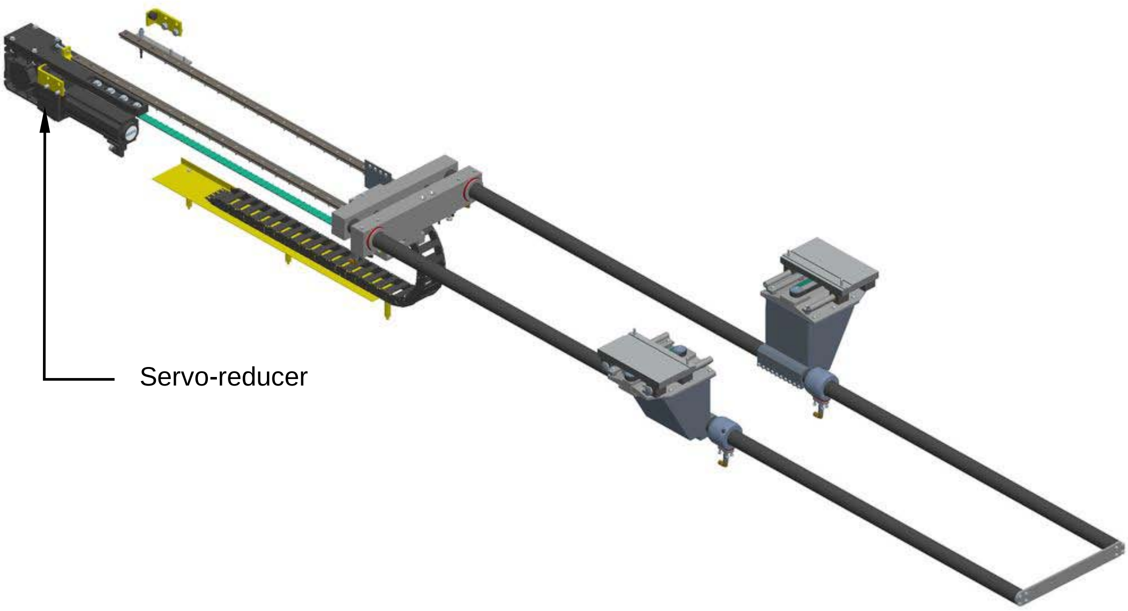
8.16 PUSHERS

8.16.1 Checking the servo-reducer condition

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the servo-reducer. Make sure the servo-reducer turns in the correct direction.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

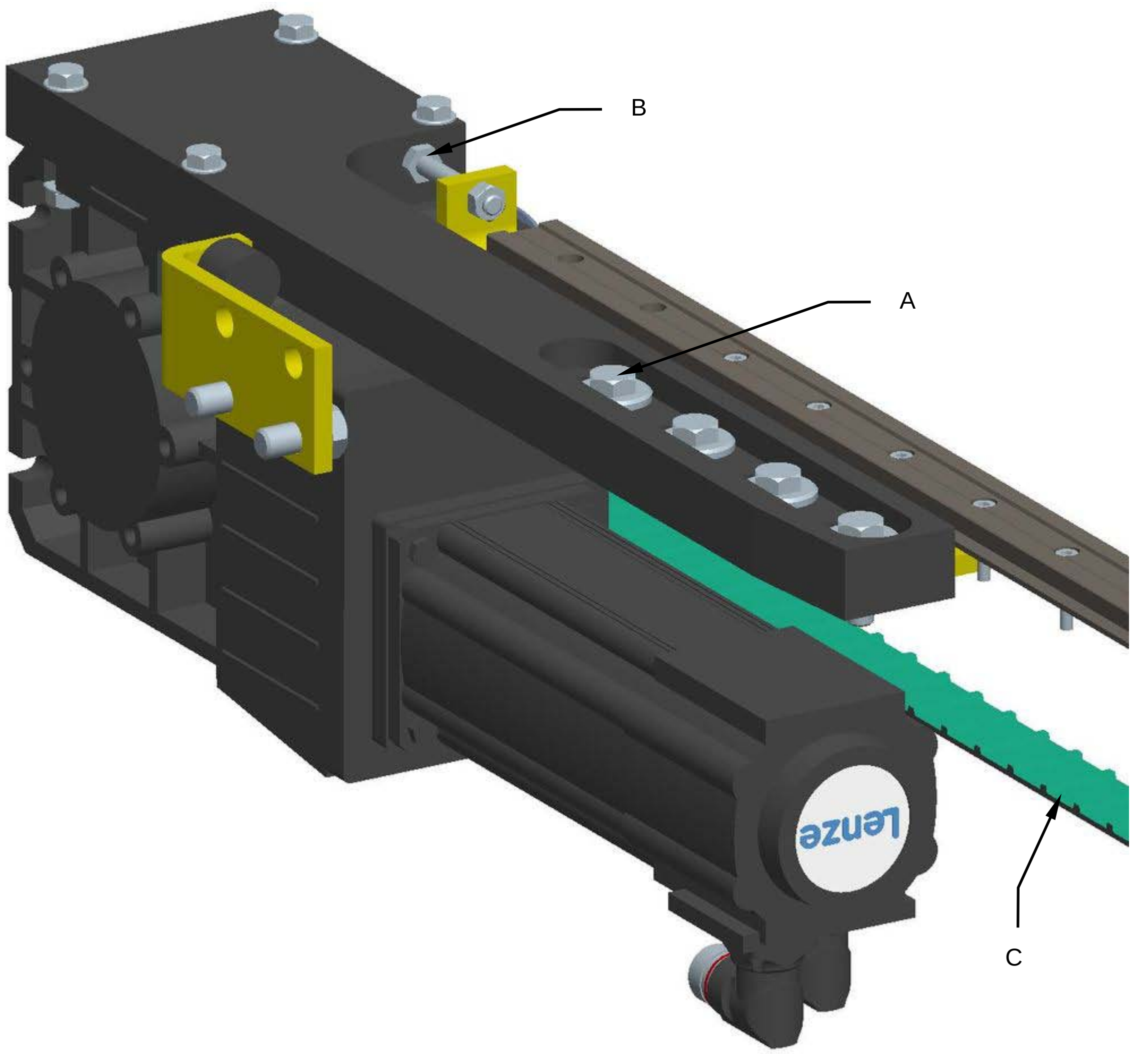


8.16.2 Replacing the servo-reducer

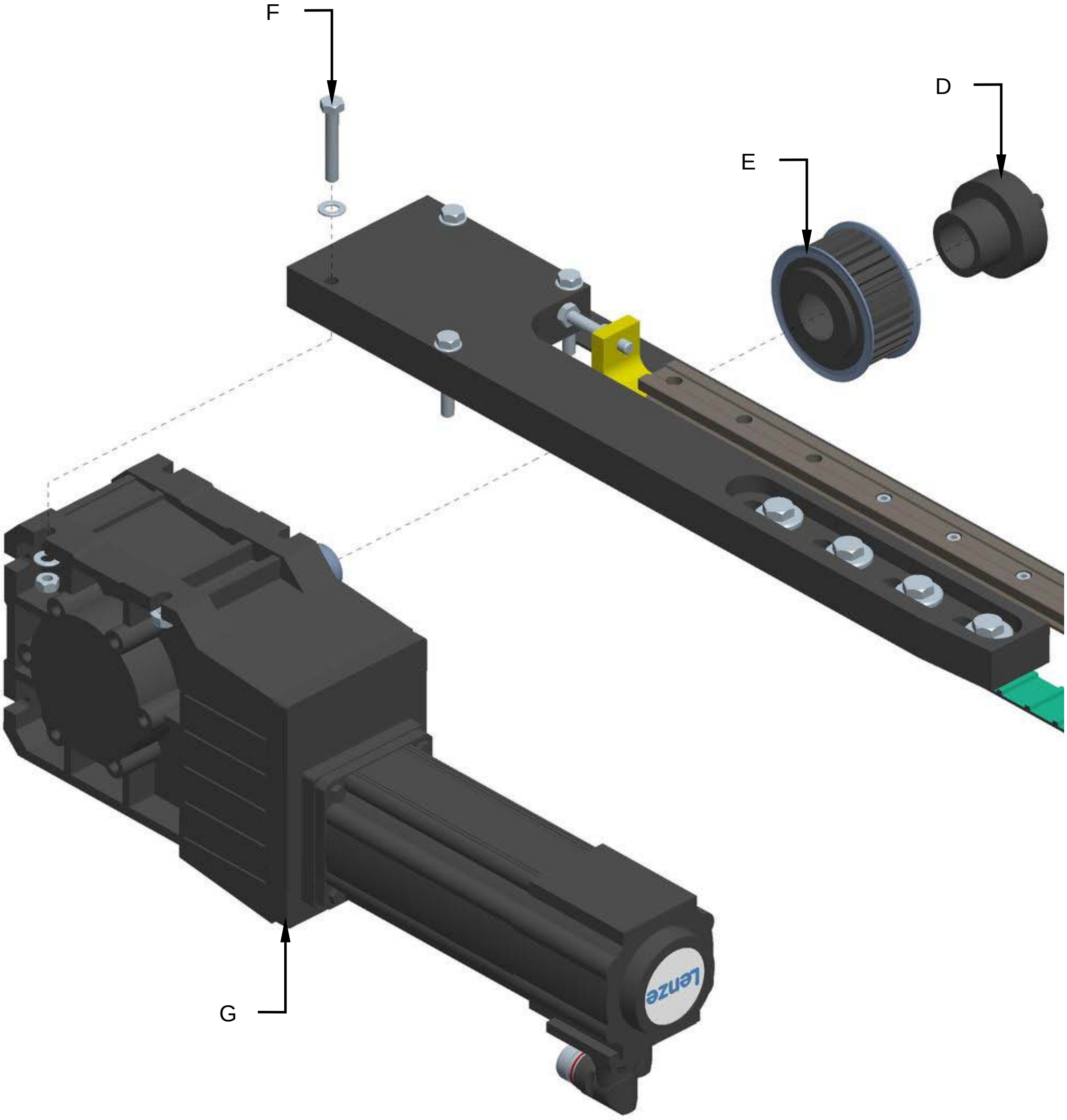
Machine status	Power off
Tools	Normal workshop equipment

Proceed as follows to replace the servo-reducer:

1. Open the lower guard of the machine and disconnect the servo-reducer power supply.
2. Loosen the screws (**A**) and tighten the screw (**B**) to reduce the tension of the belt (**C**).



3. Remove the Tollok component (**D**) and the pulley (**E**).
4. Unscrew the screws (**F**) and replace the servo-reducer (**G**).



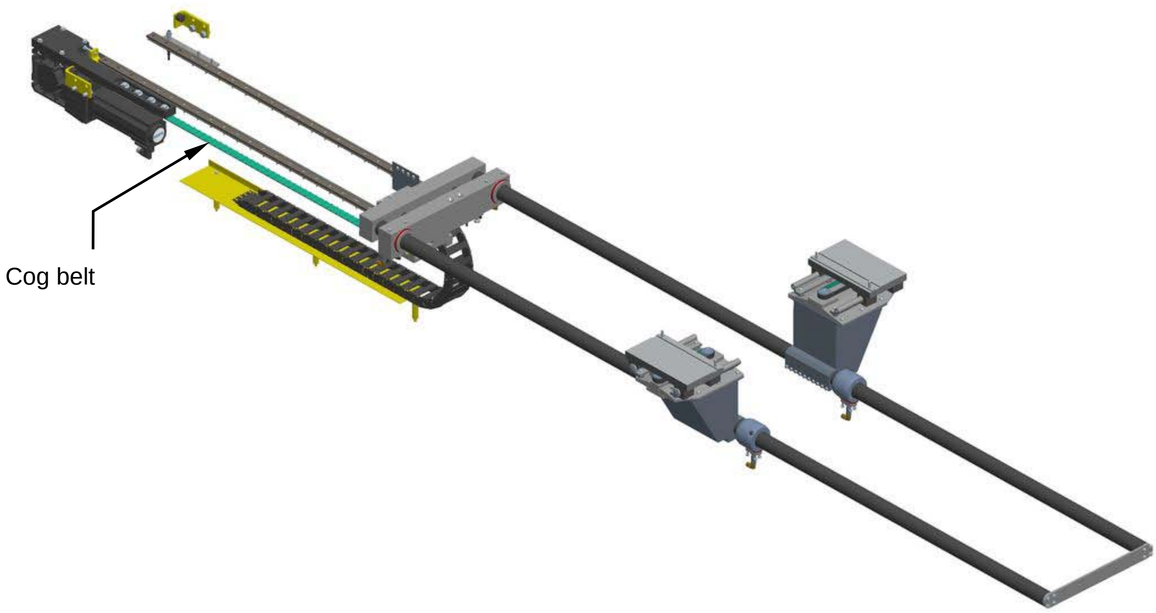
5. Remount all the components in the reverse order.

8.16.3 Checking the cog belt condition and tension

Machine status	Power off
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the cog belt.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

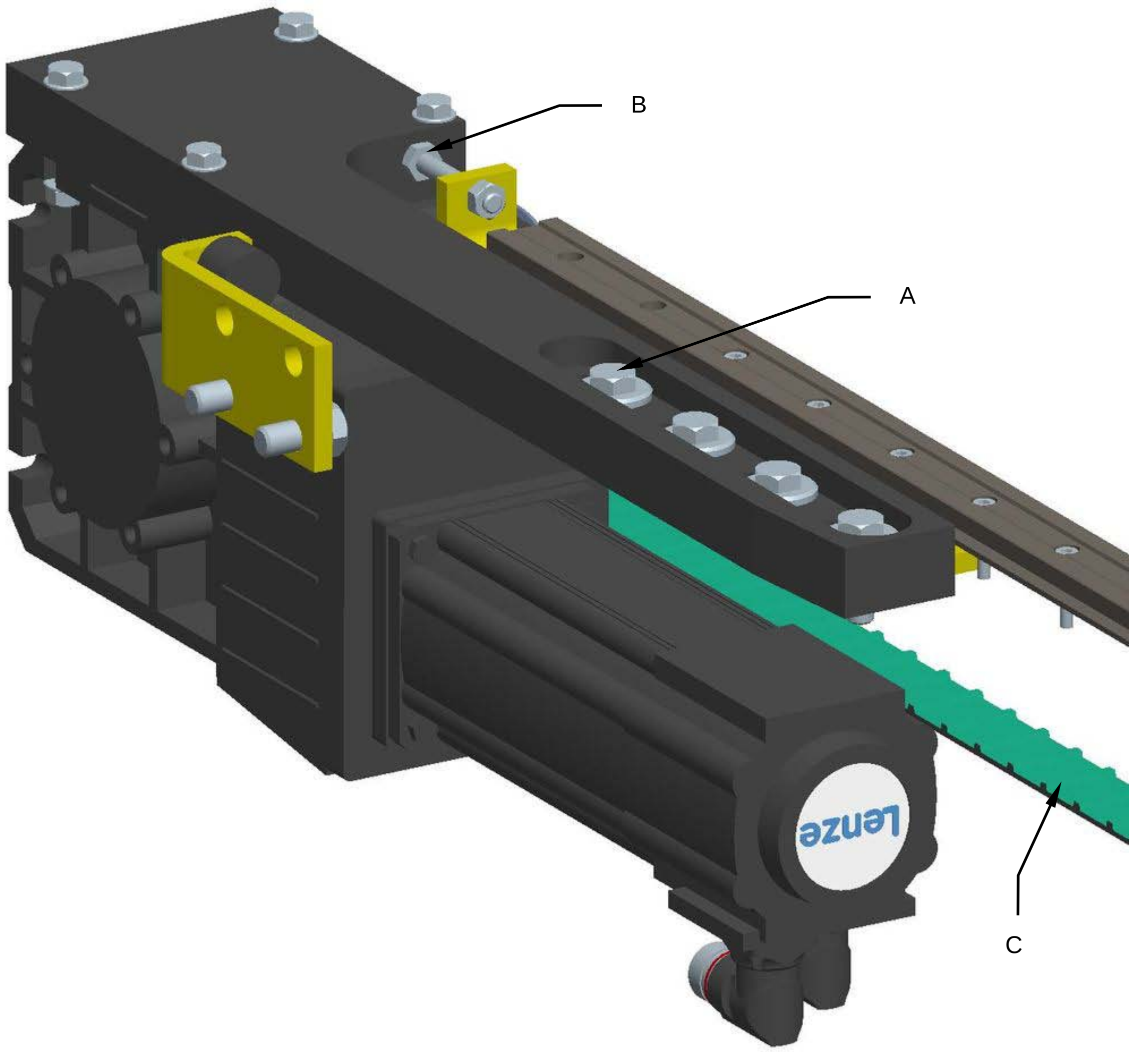


8.16.4 Replacing the cog belt

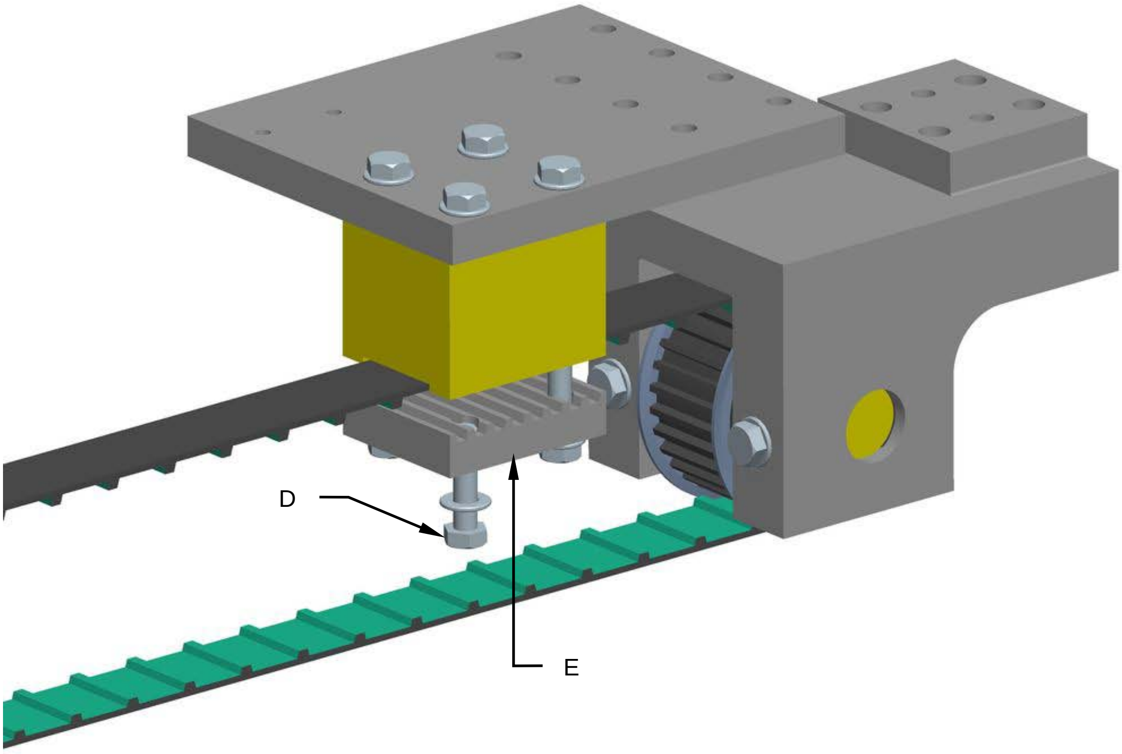
Machine status	Power off
Tools	Normal workshop equipment

Proceed as follows to replace the cog belt:

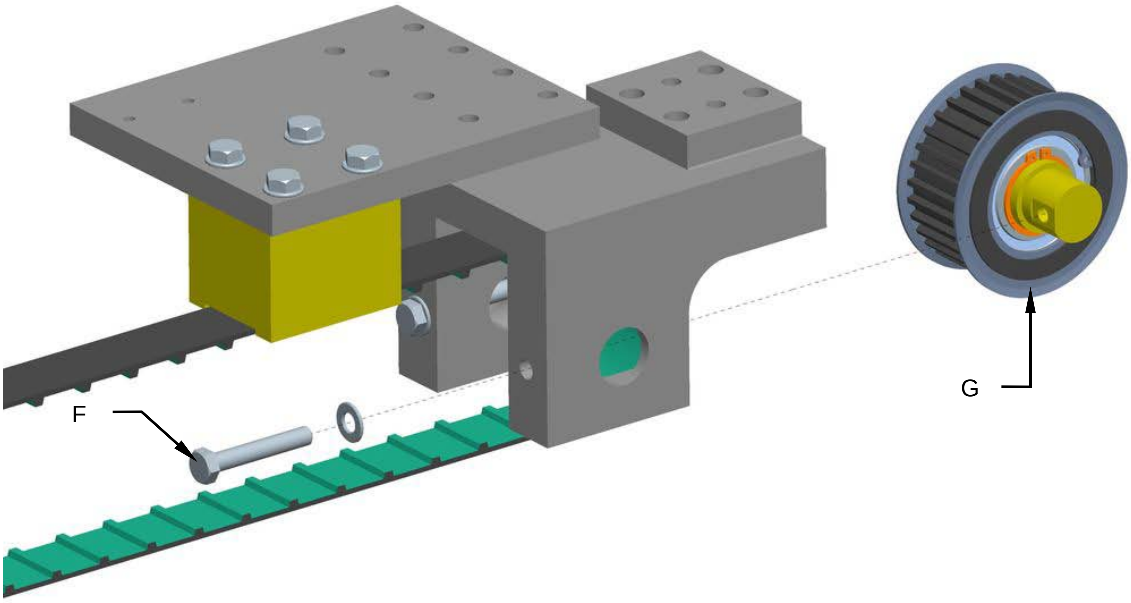
1. Loosen the screws (**A**) and tighten the screw (**B**) to reduce the tension of the belt (**C**).



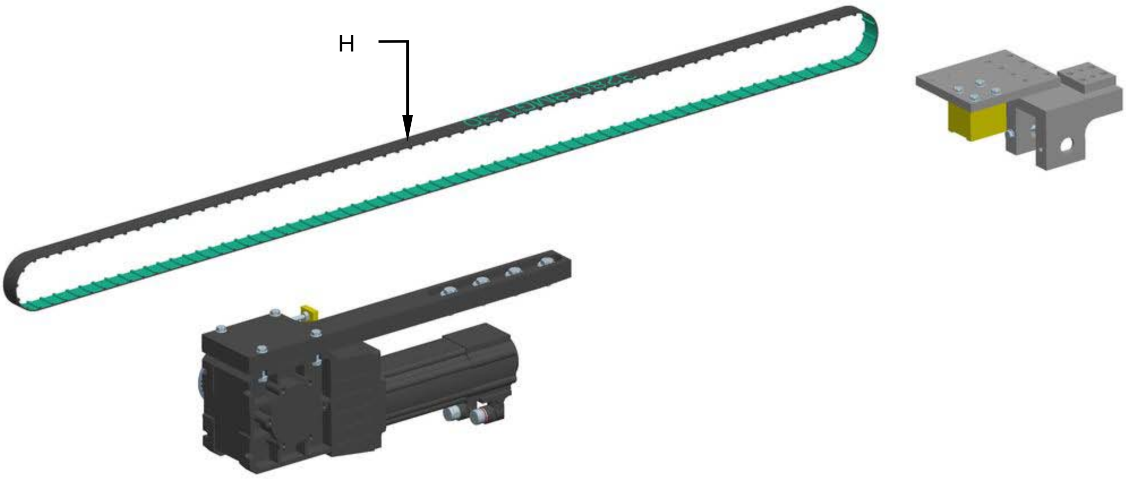
2. Unscrew the screws (**D**) and remove the bracket (**E**).



3. Unscrew the screws (**F**) and remove the pulley (**G**).



4. Remove and replace the cog belt (**H**).



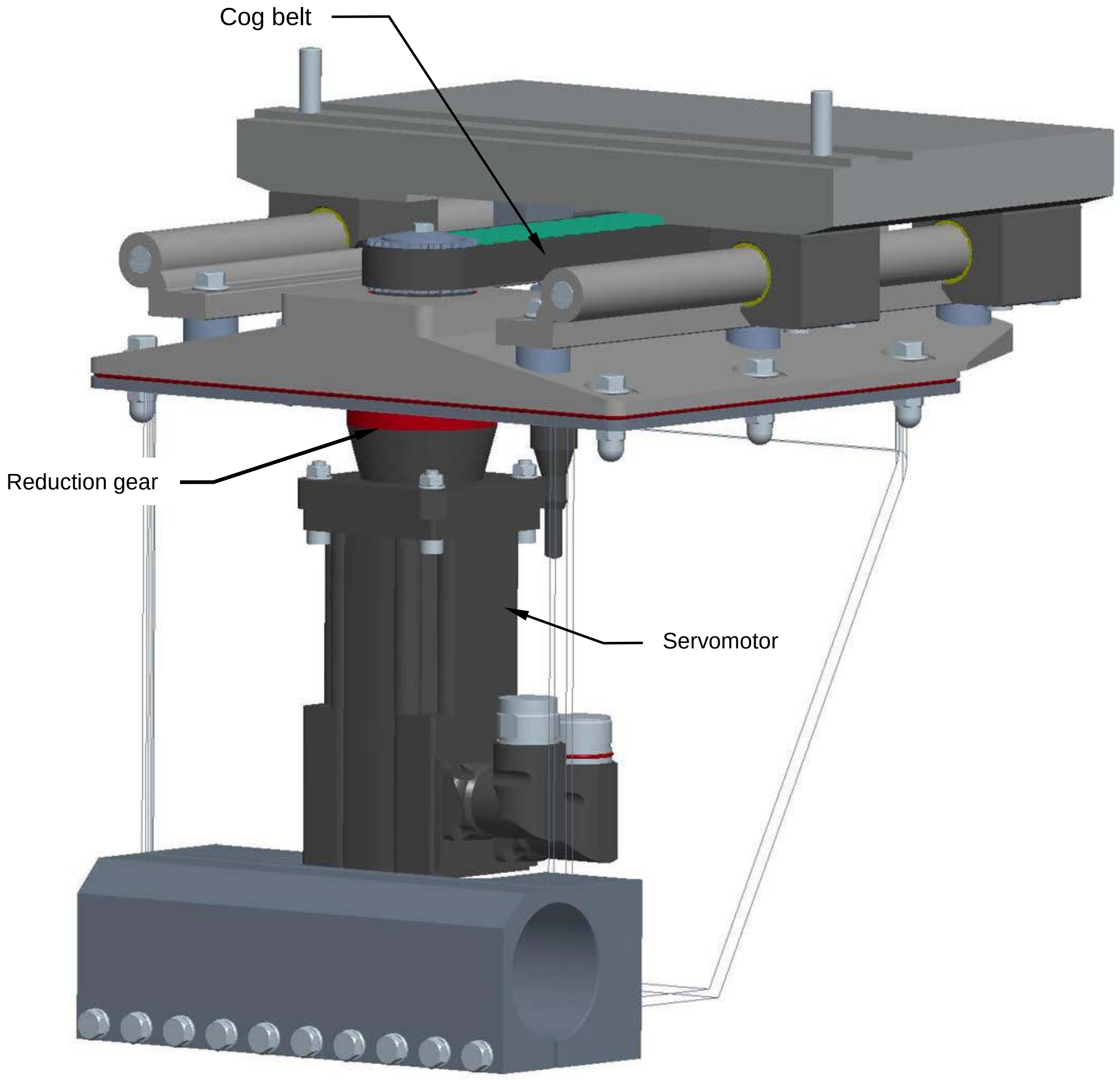
5. Remount all the components in the reverse order.

8.16.5 Checking the servomotor, reduction gear and cog belt condition

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the servomotor, reduction gear and cog belt.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

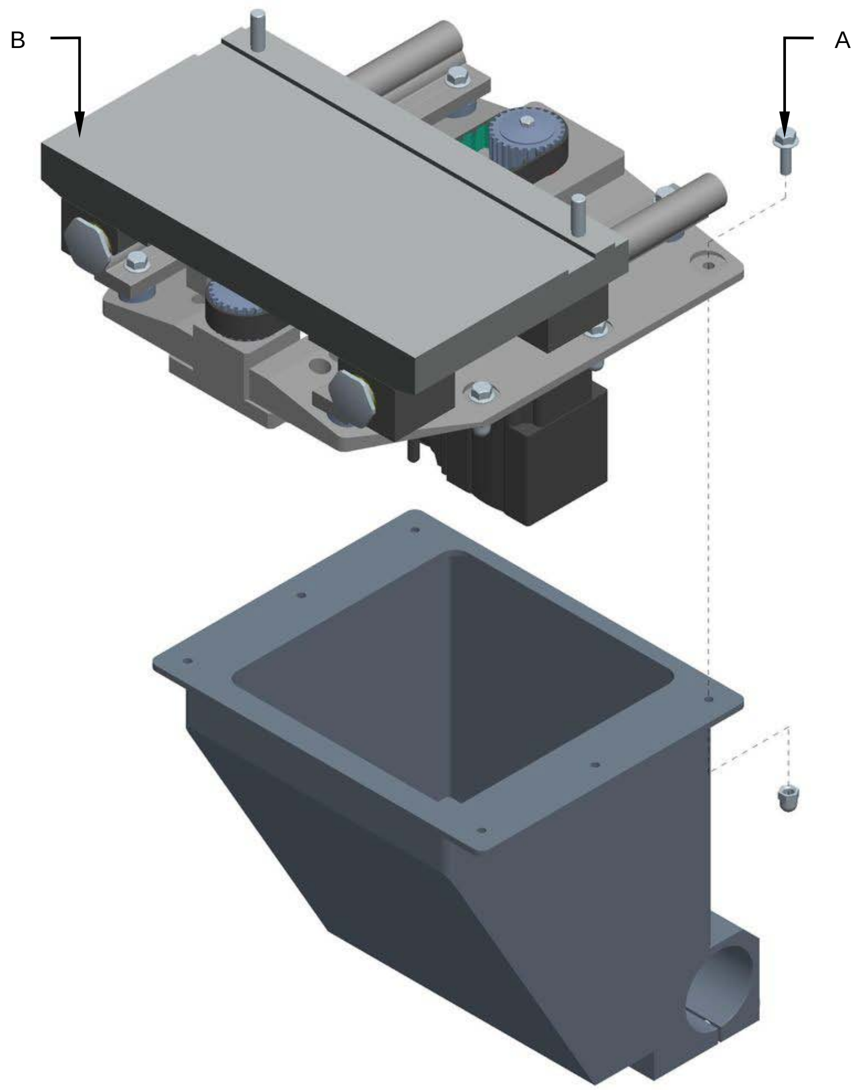


8.16.6 Replacing the servomotor, reduction gear and cog belt

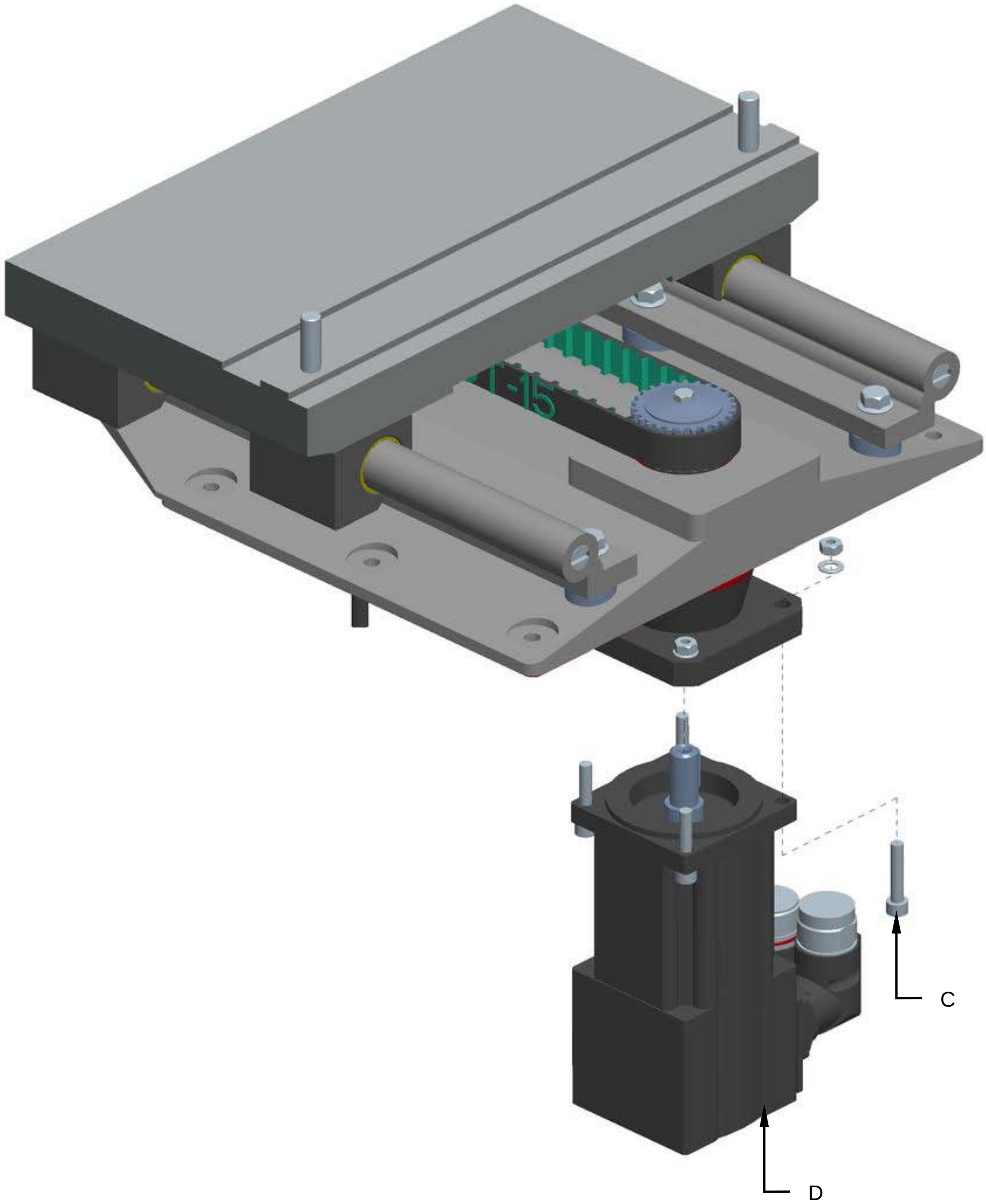
Machine status	Power off
Tools	Normal workshop equipment

Proceed as follows to replace the servomotor:

1. Disconnect the power supply to the servomotor.
2. Unscrew the screws (**A**) and remove the entire assembly (**B**).



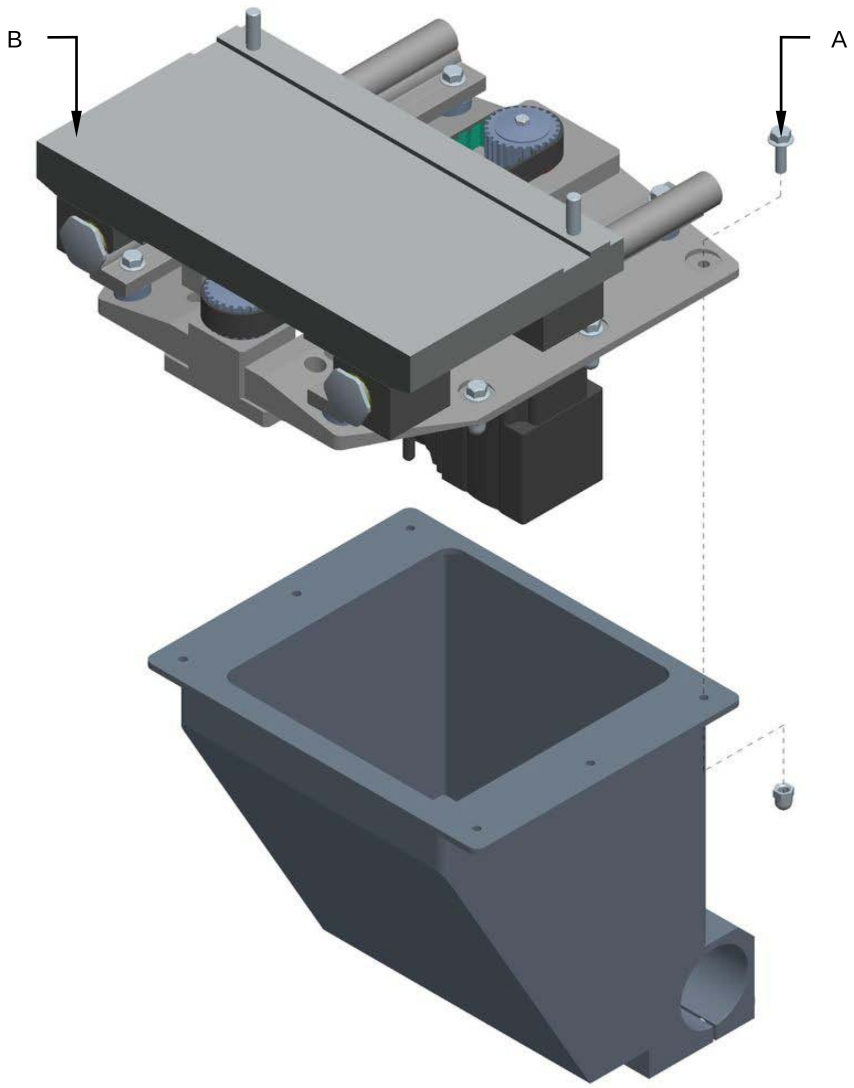
3. Unscrew the screws (**C**) and replace the servomotor (**D**).



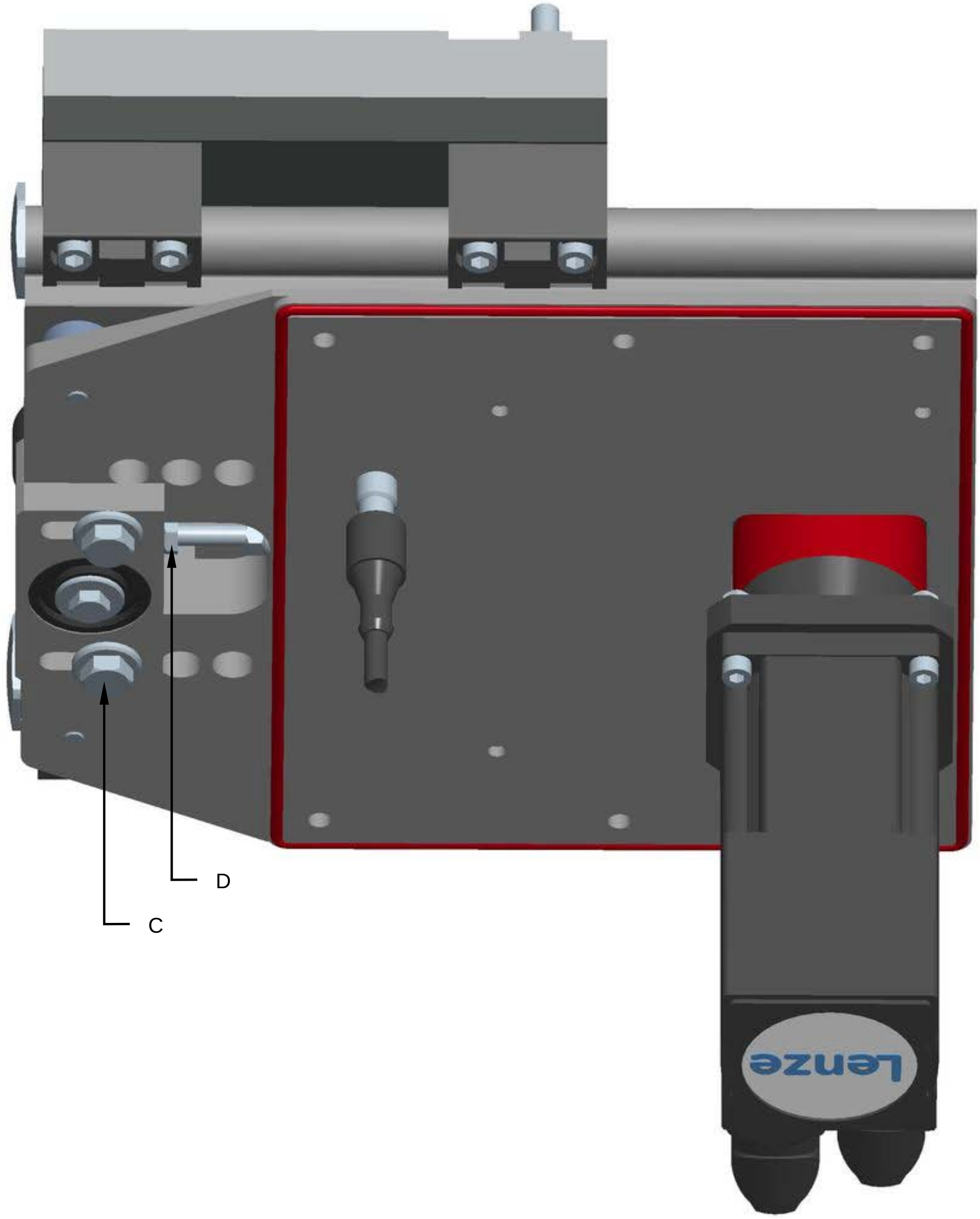
4. Remount all the components in the reverse order.

Proceed as follows to replace the reduction gear:

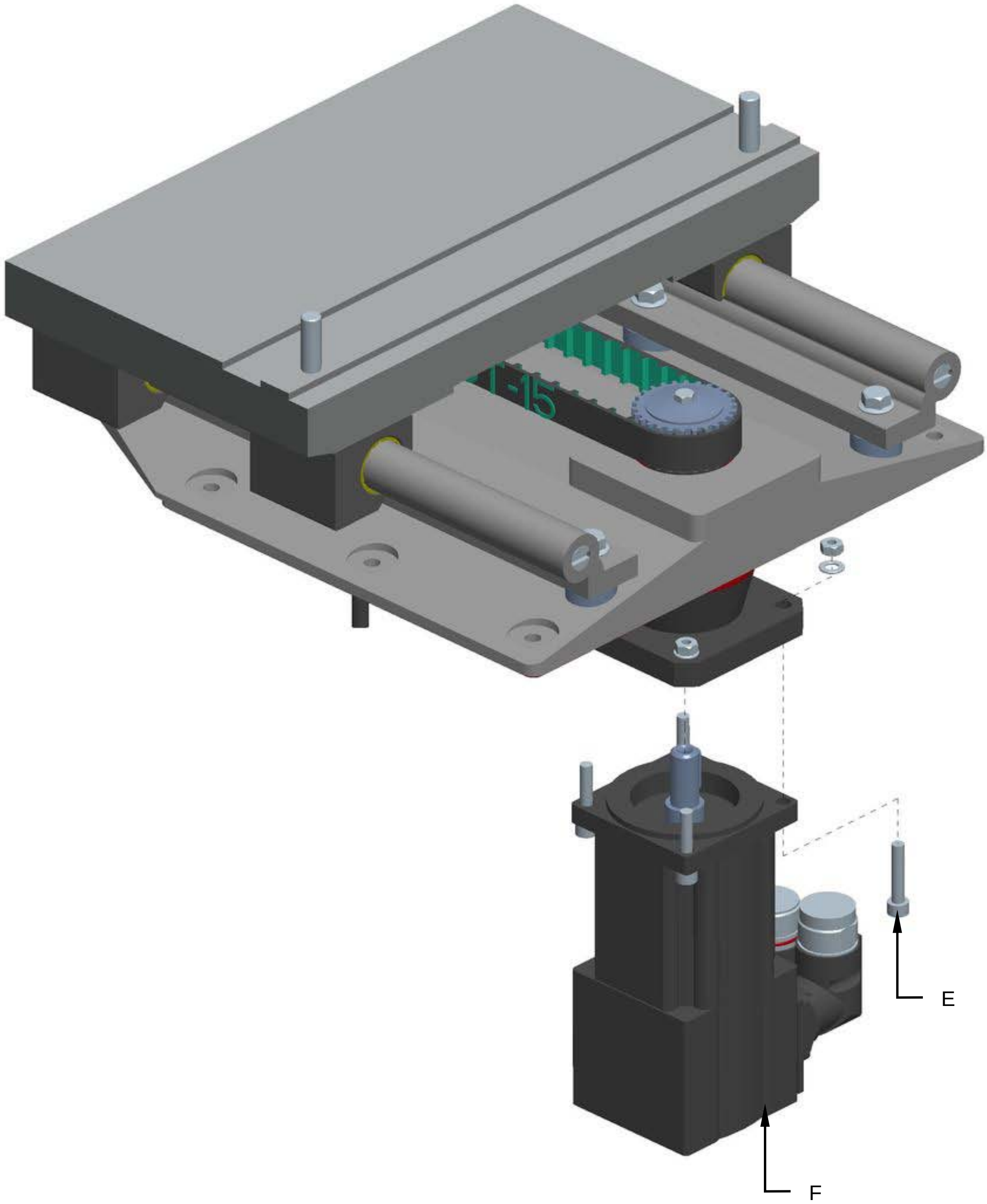
5. Disconnect the power supply to the servomotor.
6. Unscrew the screws (**A**) and remove the entire assembly (**B**).



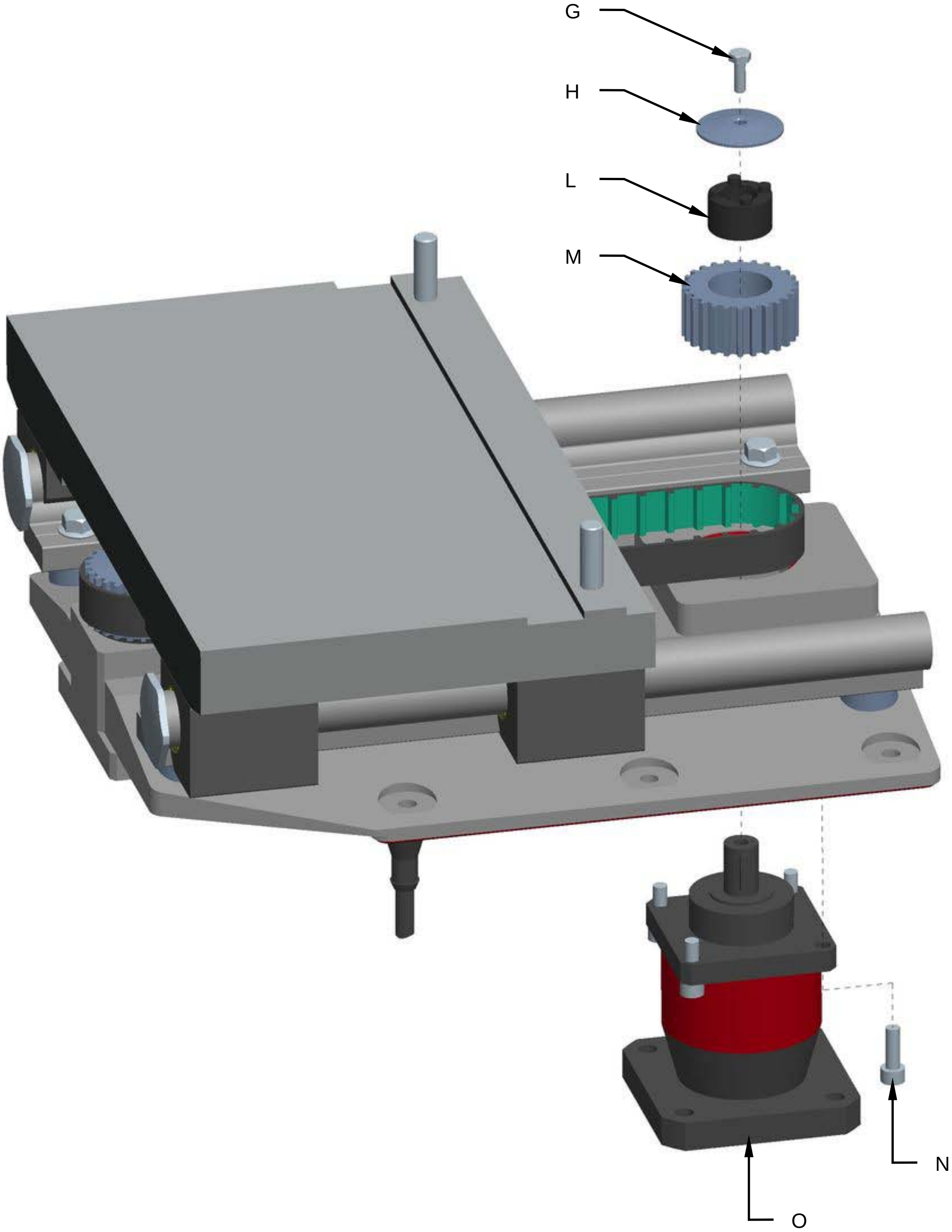
7. Loosen the screws (**C**) and tighten the screw (**D**) to reduce the tension of the belt.



8. Unscrew the screws (**E**) and remove the servomotor (**F**).



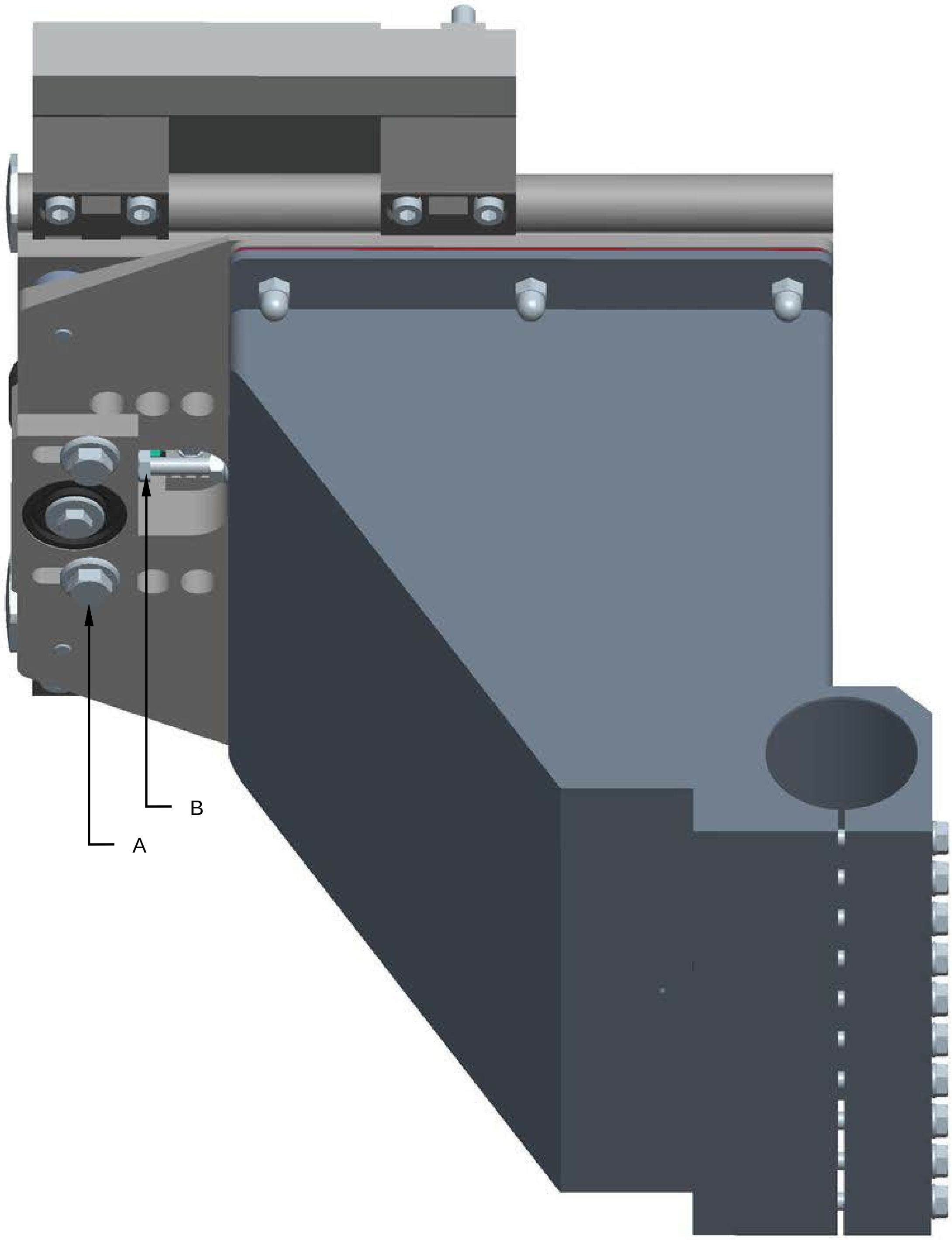
9. Unscrew the screw (**G**) and remove the cap (**H**).
10. Unscrew the Tollok component (**L**) and remove the pulley (**M**).
11. Unscrew the screws (**N**) and replace the reduction gear (**O**).



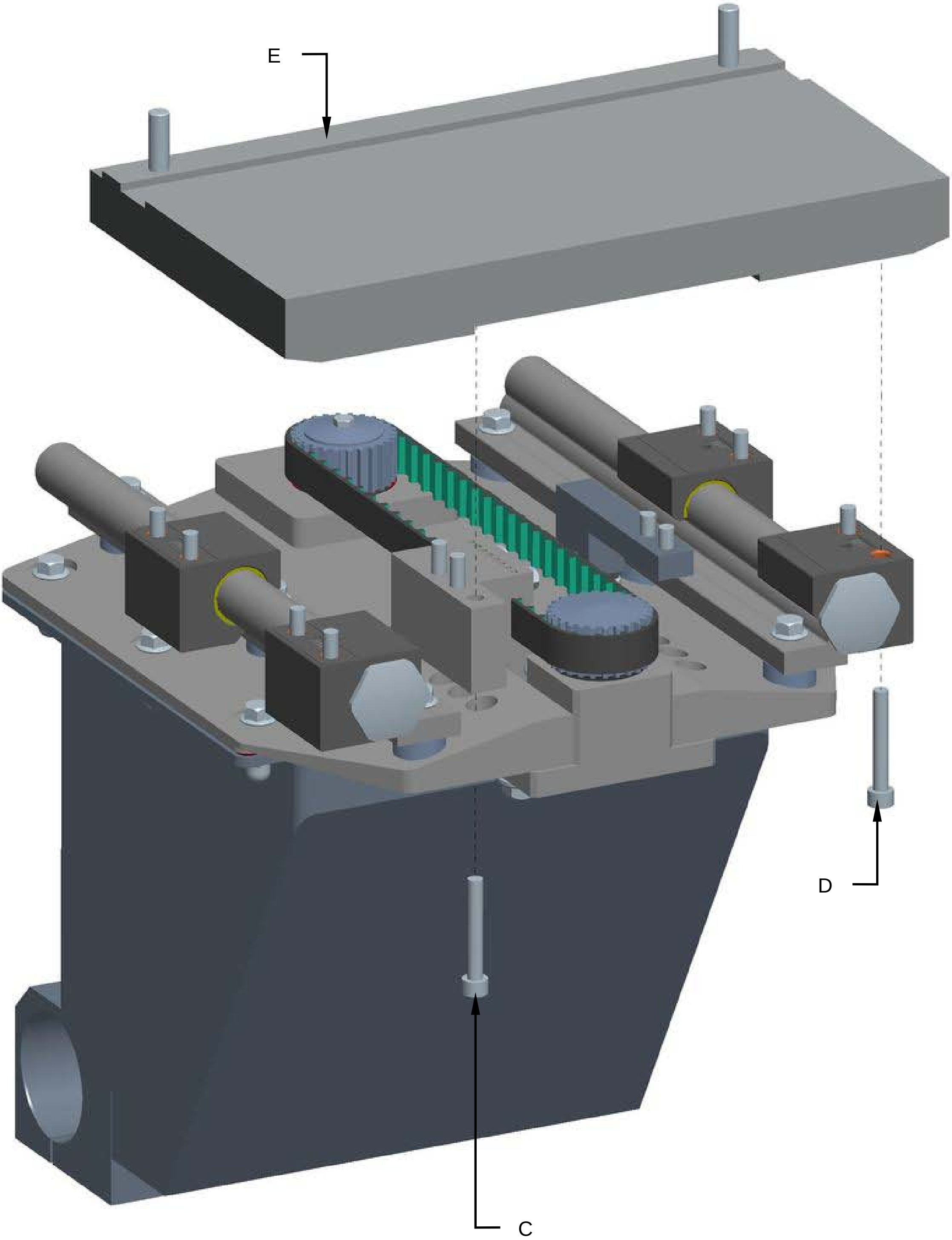
12. Remount all the components in the reverse order.

Proceed as follows to replace the cog belt:

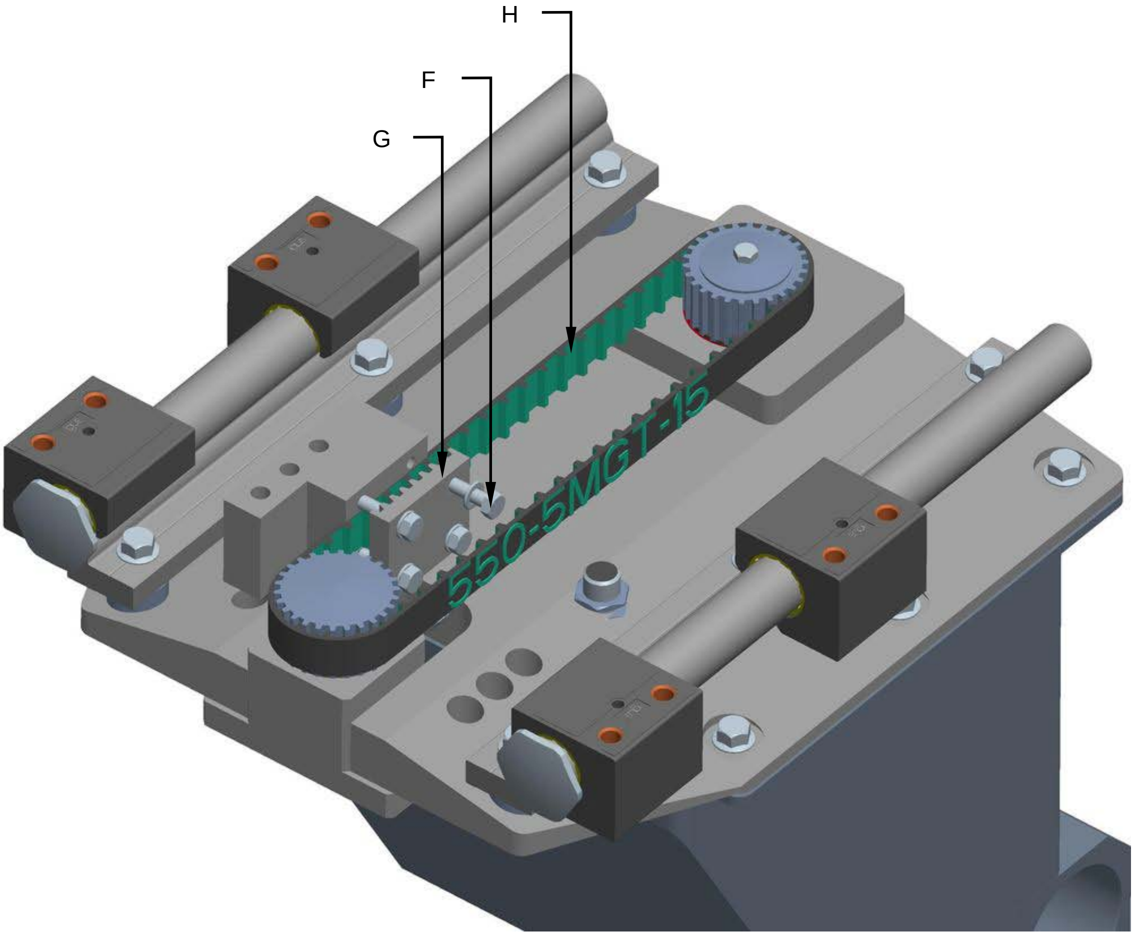
13. Loosen the screws (**A**) and tighten the screw (**B**) to reduce the tension of the belt.



14. Unscrew the screws (**C**) and (**D**) to remove the plate (**E**).



15. Unscrew the screws (**F**) and remove the bracket (**G**).
16. Remove and replace the cog belt (**H**).



17. Remount all the components in the reverse order.

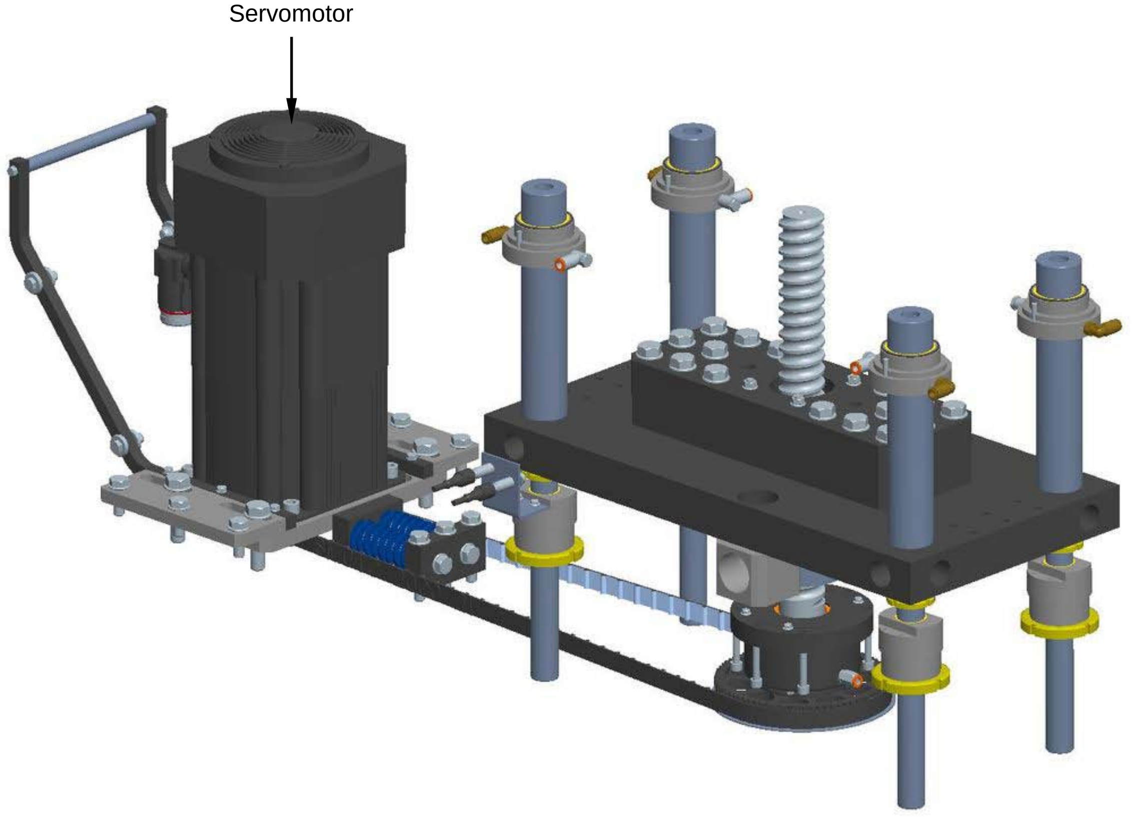
8.17 SCREW SEALING CAM

8.17.1 Checking the servomotor condition

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the servomotor. Make sure the servomotor turns in the right direction.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

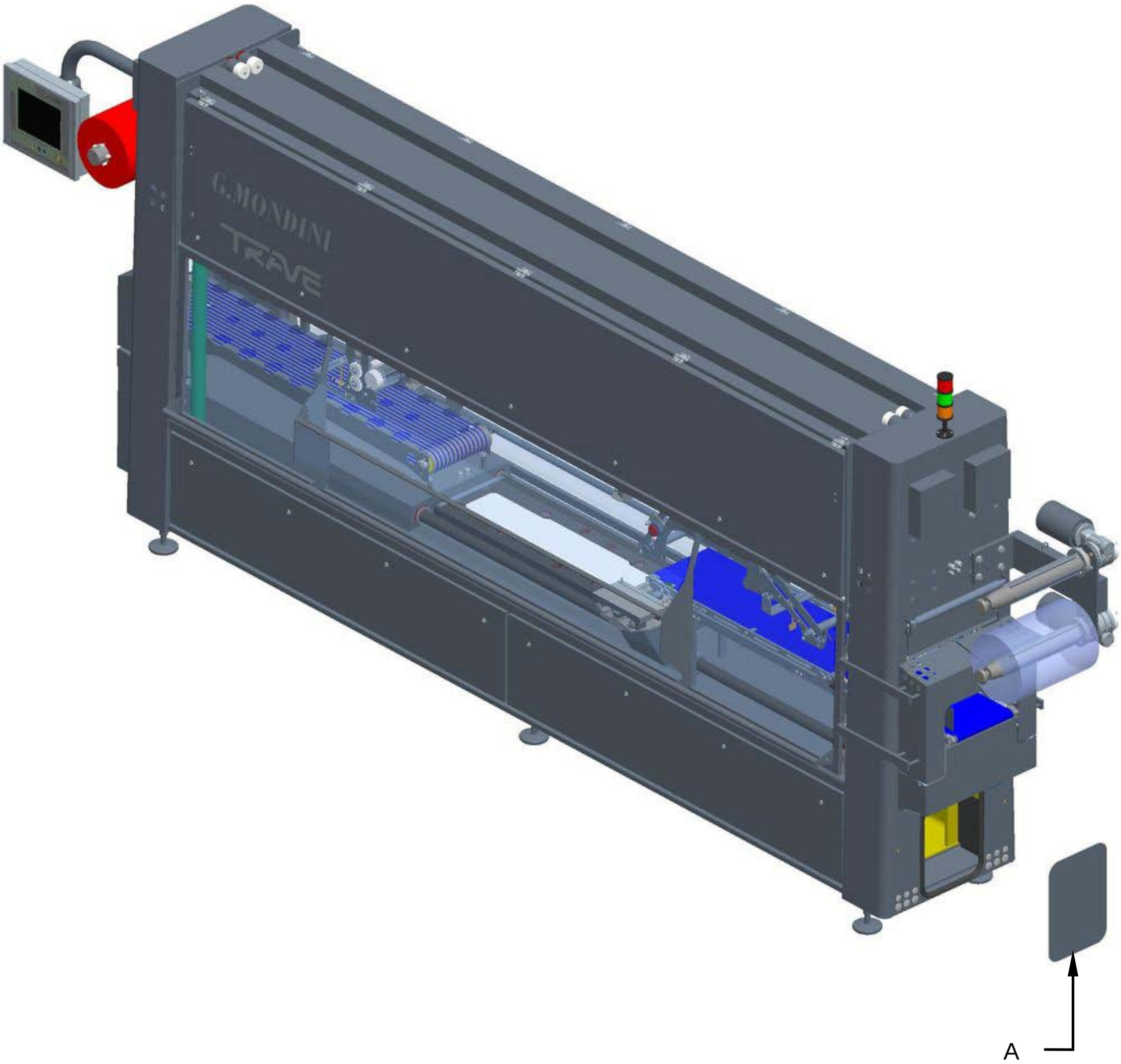


8.17.2 Replacing the servomotor

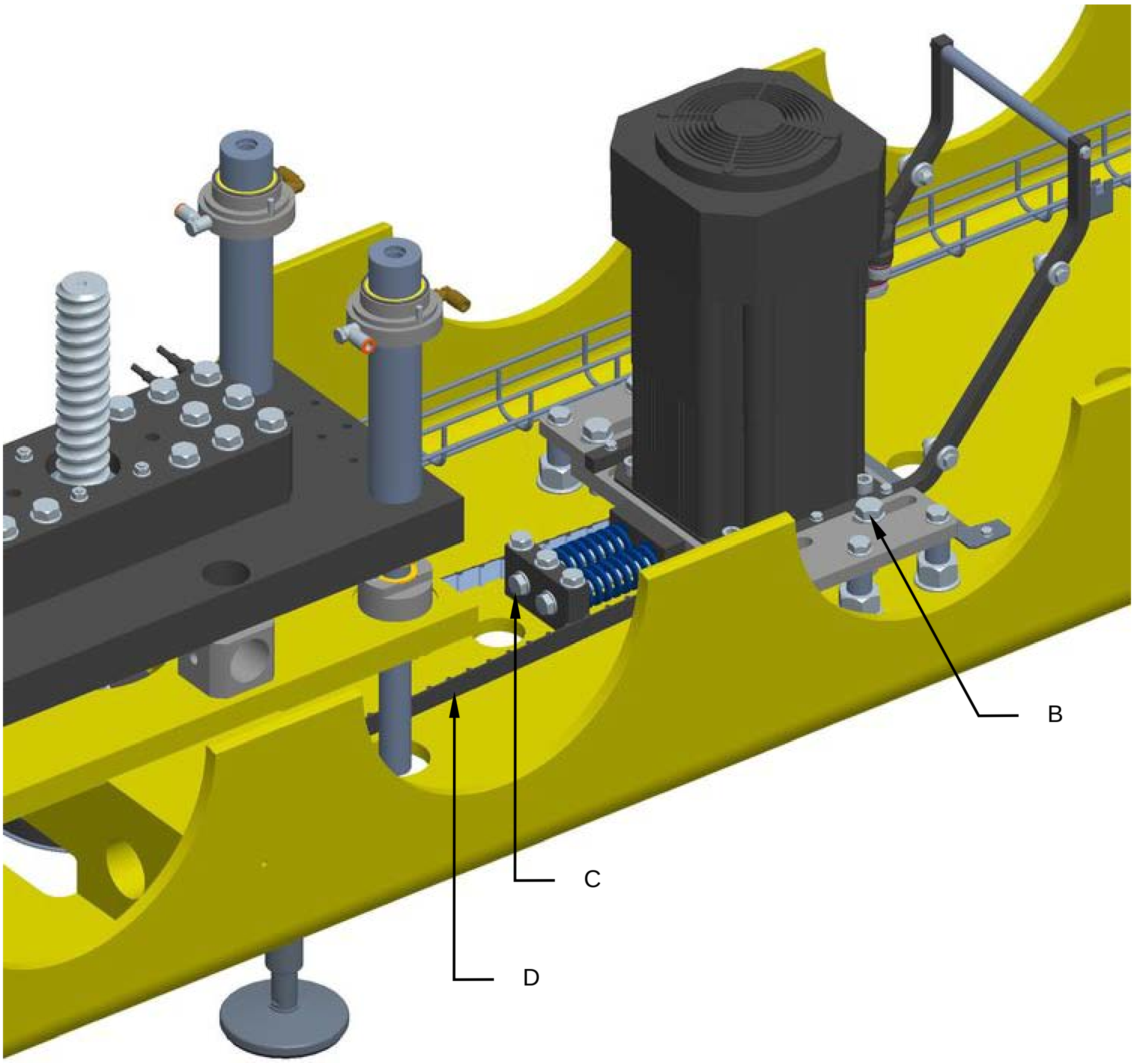
Machine status	Power off
Tools	Normal workshop equipment, Steel supporting blocks

Proceed as follows to replace the servomotor

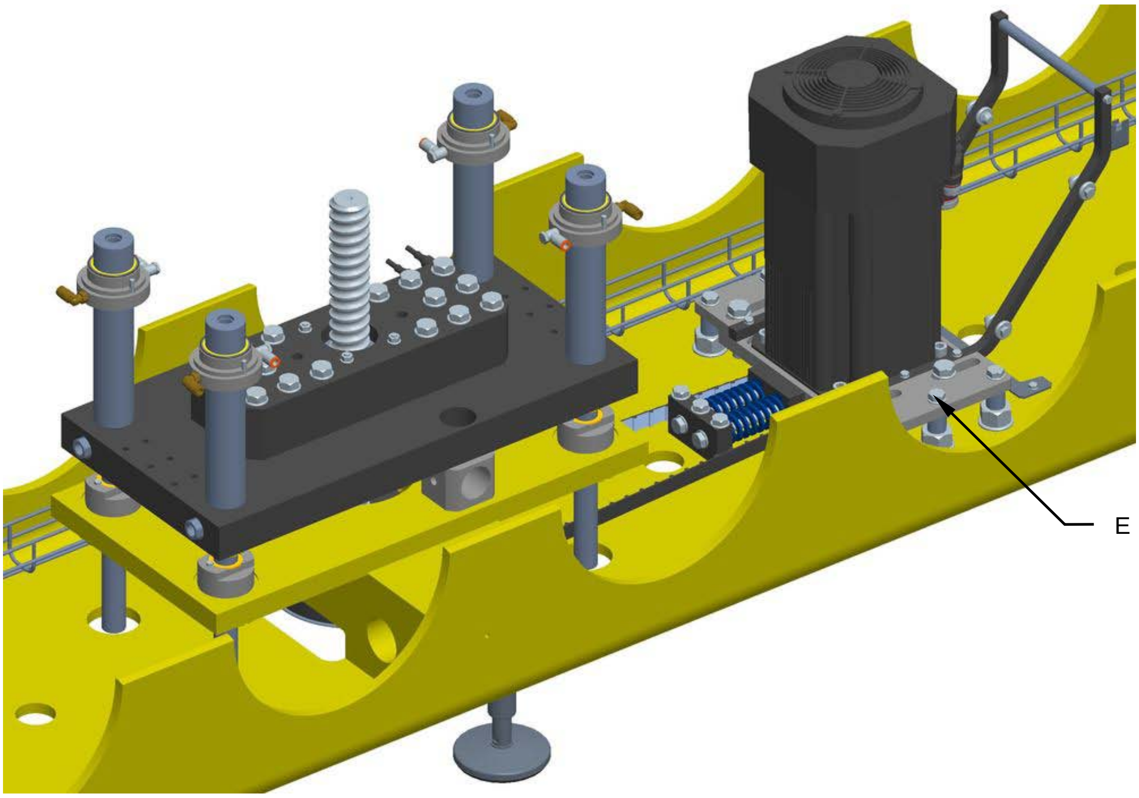
1. Disconnect the power to the servomotor.
2. Open the guard (**A**) and remove all the components in order to free the servomotor area.



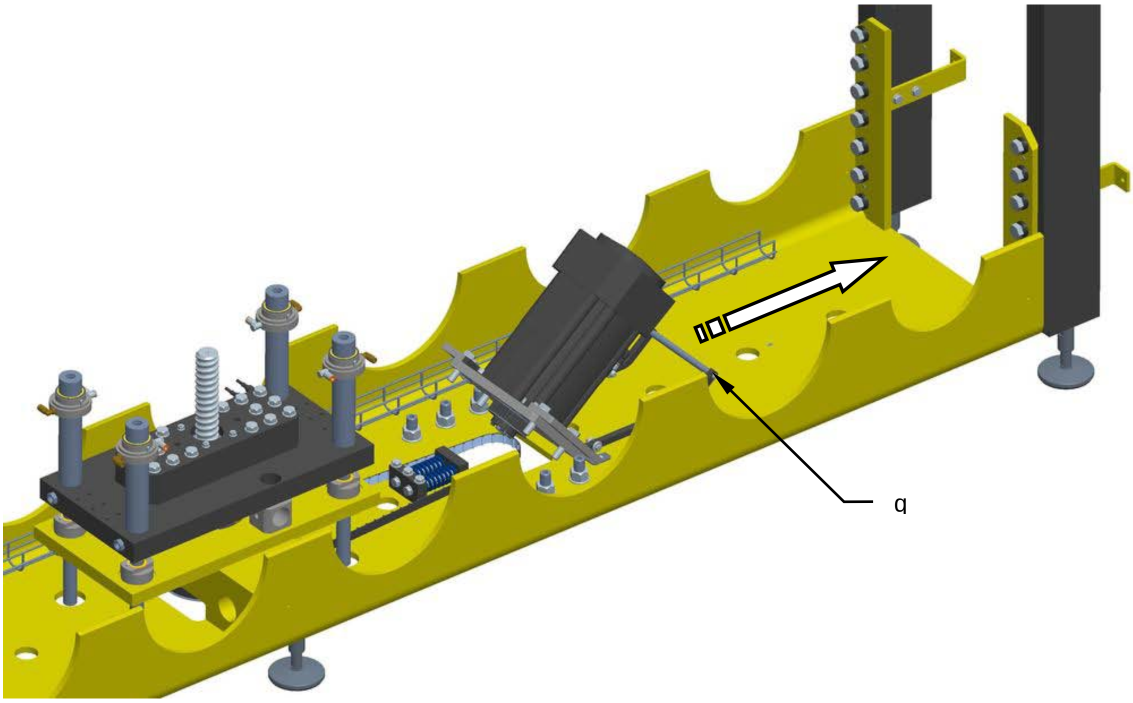
3. Loosen the screws (**B**), tighten the screws (**C**) to reduce the tension of the belt (**D**).



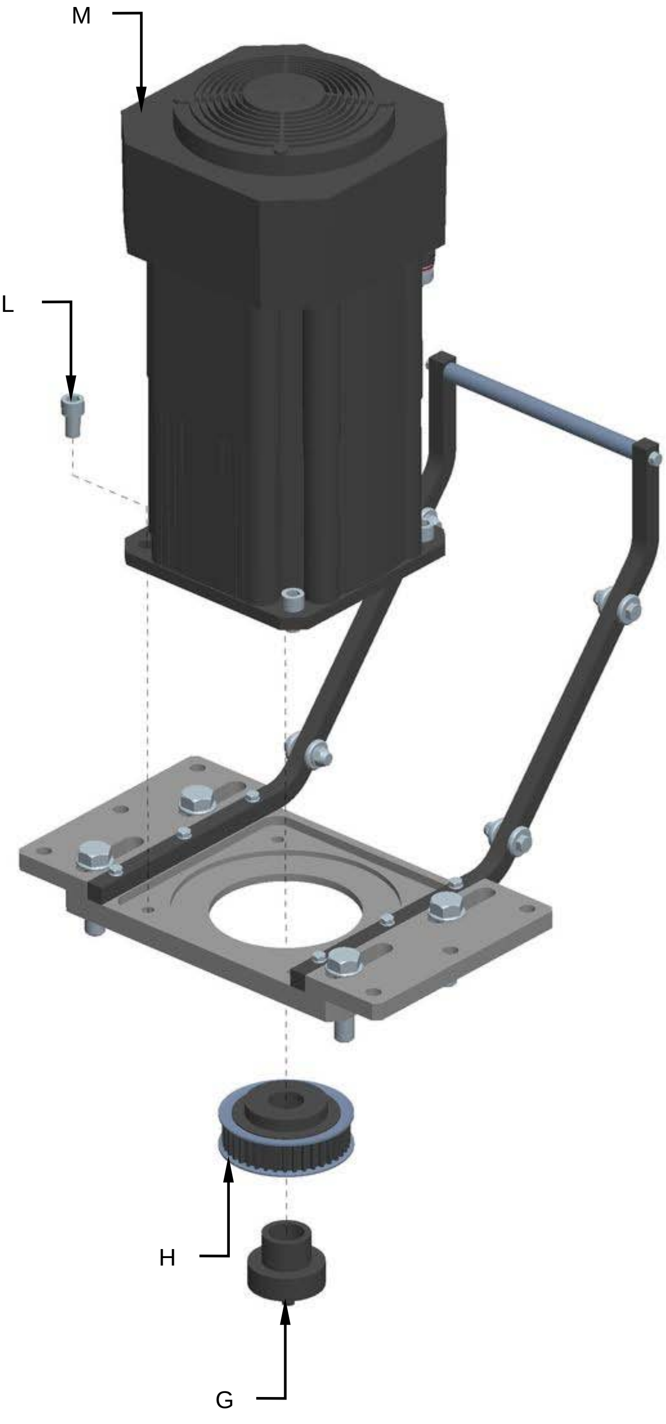
4. Unscrew the screws (**E**).



5. Grasp the handle (**F**), tilt the servomotor and pull it out.



6. Unscrew the Tollok component (**G**) and remove the pulley (**H**).
7. Unscrew the screws (**L**) and replace the reduction gear (**M**).



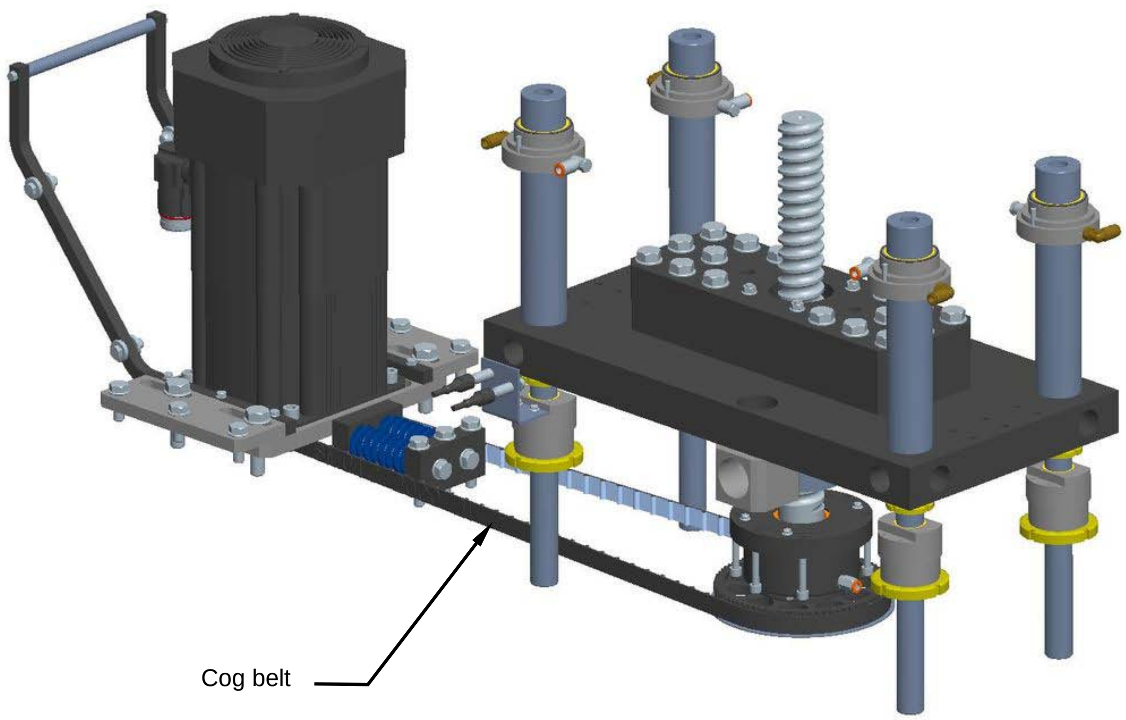
8. Remount all the components in the reverse order.

8.17.3 Checking the cog belt condition and tension

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check that the cog belt is not worn. Make sure it is correctly tensioned and moves smoothly.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

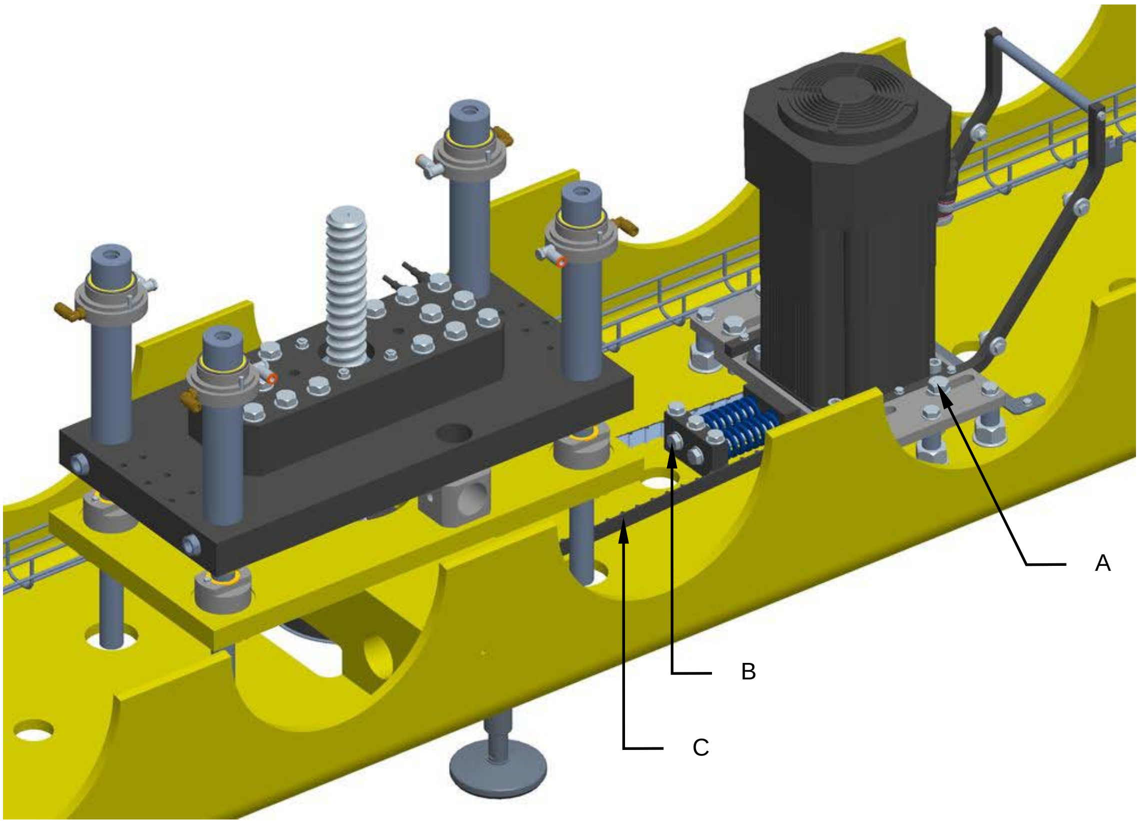


8.17.4 Replacing the cog belt

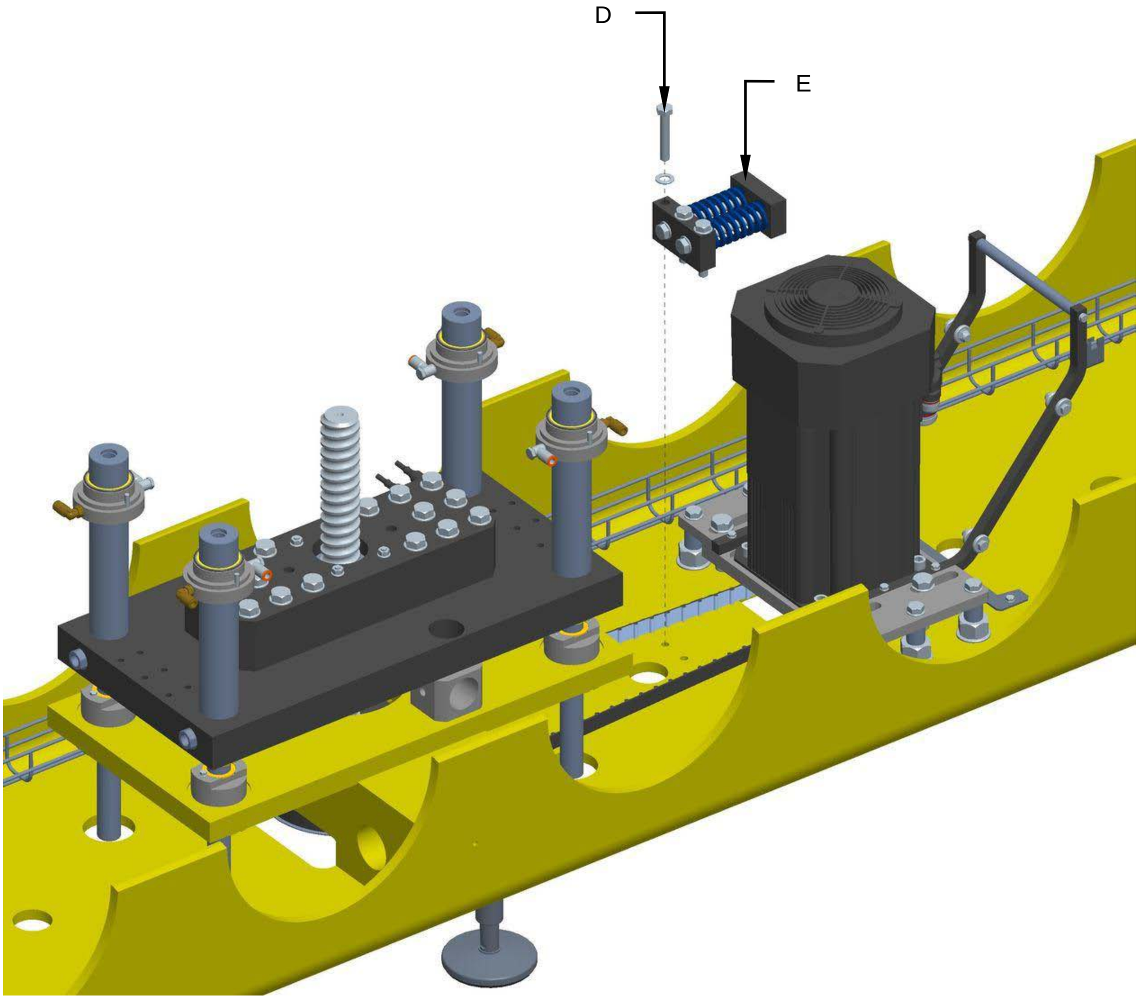
Machine status	Power off
Tools	Normal workshop equipment, Steel supporting blocks

Proceed as follows to replace the cog belt:

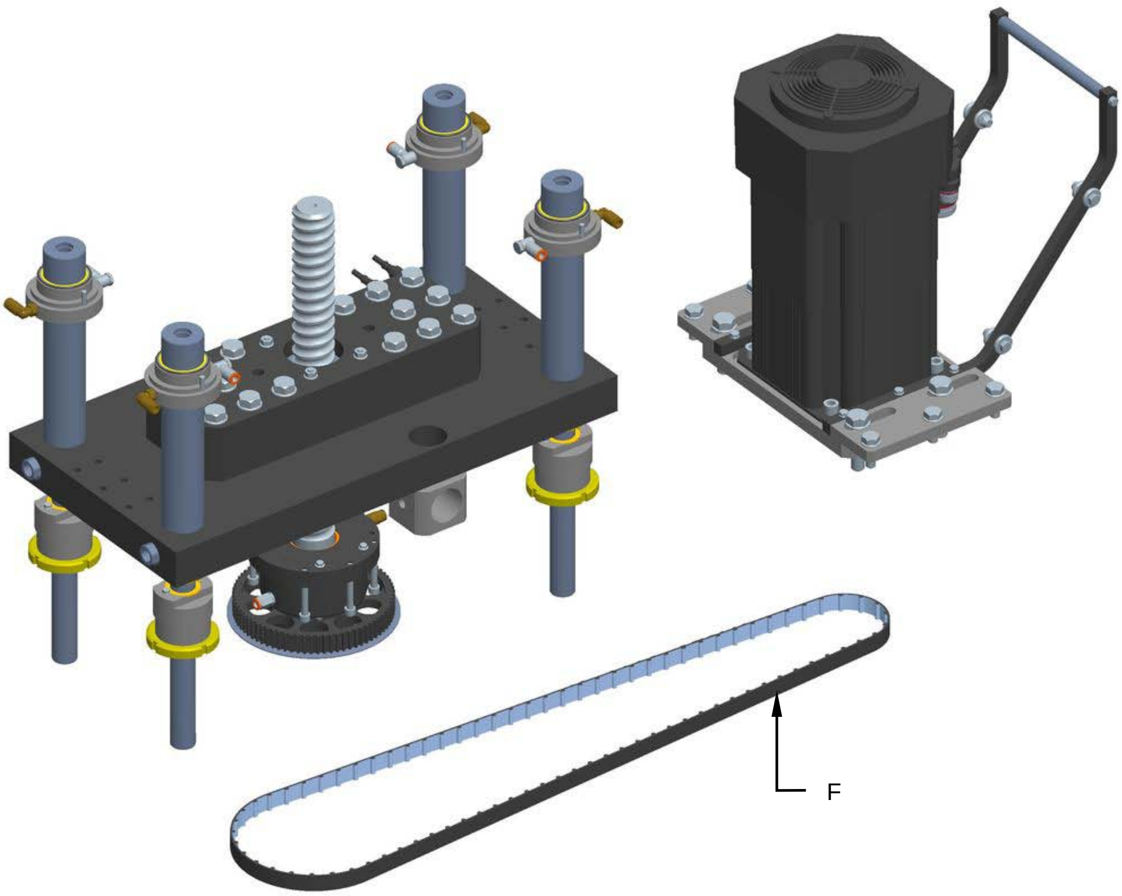
1. Disconnect the power and compressed air supply.
2. Loosen the screws (**A**), tighten the screws (**B**) to reduce the tension of the belt (**C**).



3. Unscrew the screws (**D**) and remove the belt tensioner (**E**).



4. Remove and replace the cog belt (**F**).



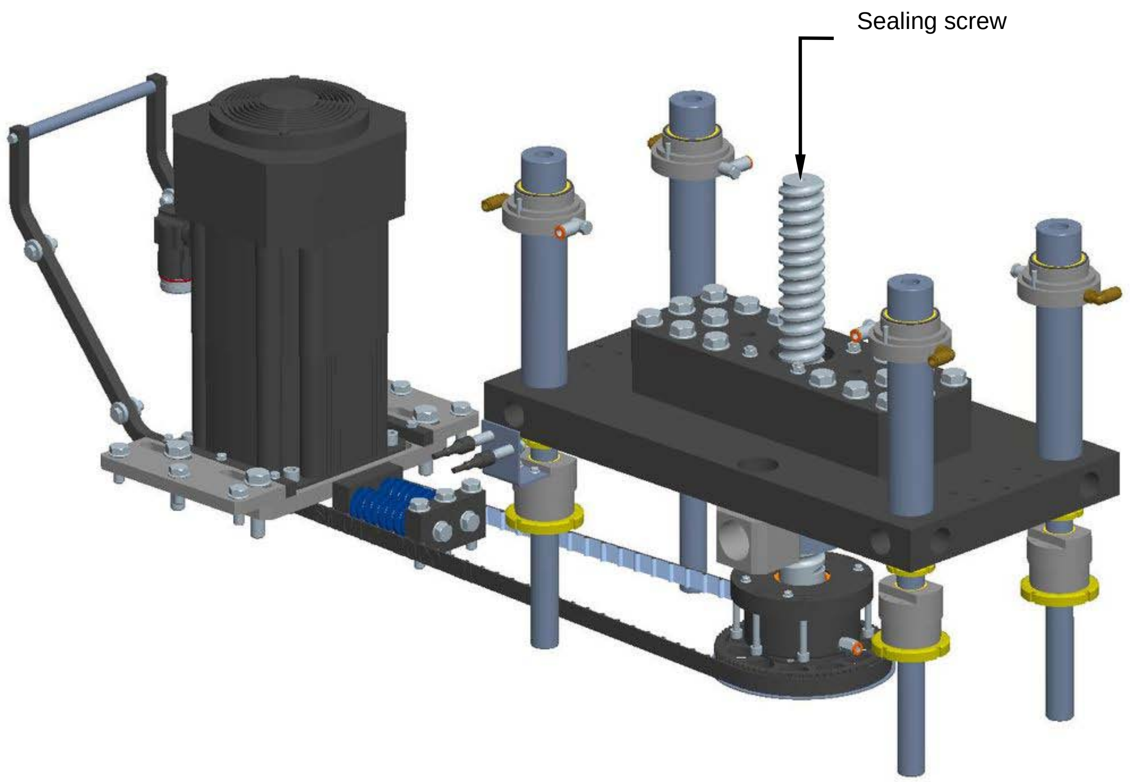
5. Remount all the components in the reverse order.

8.17.5 Checking the sealing screw condition

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the sealing screw.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

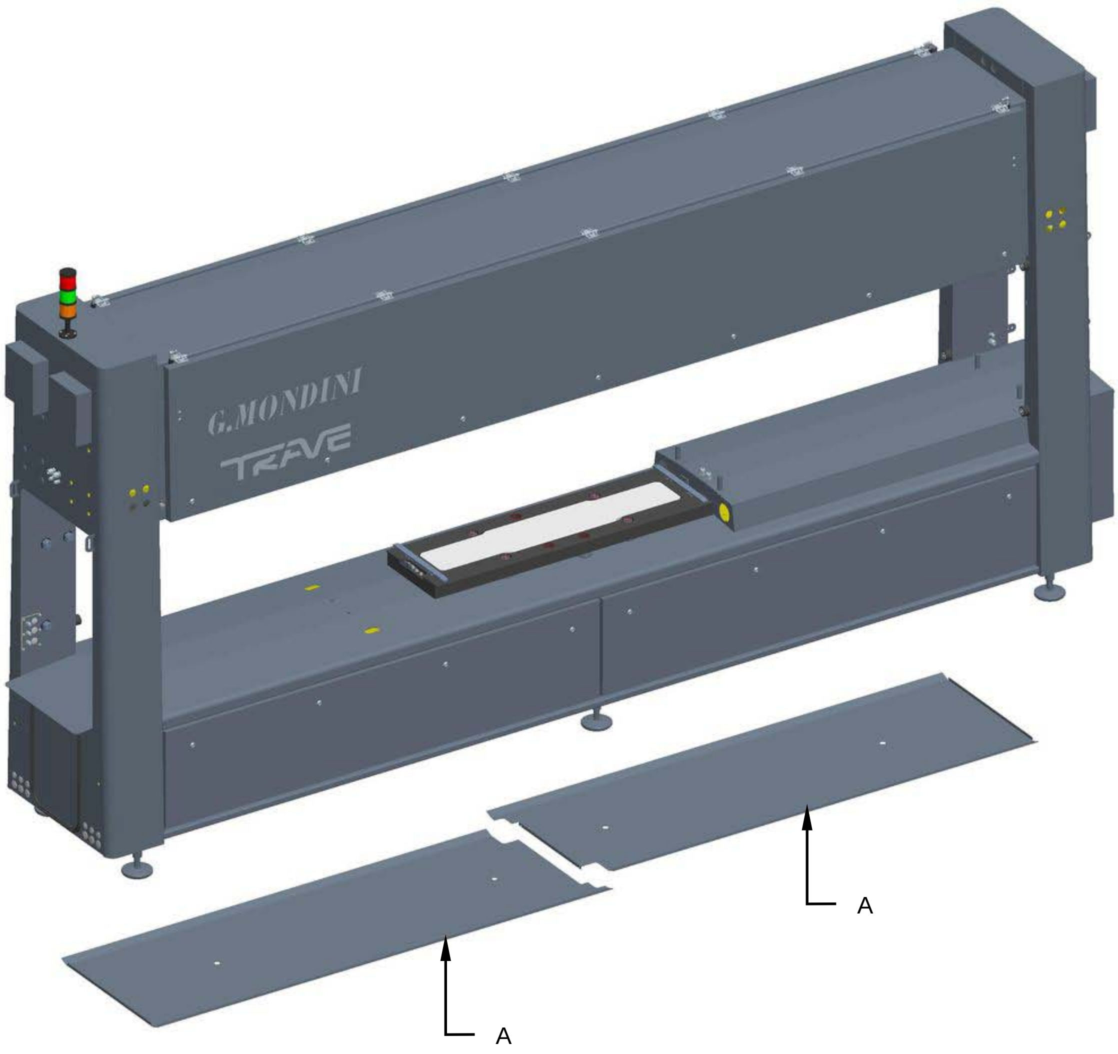


8.17.6 Replacing the sealing screw

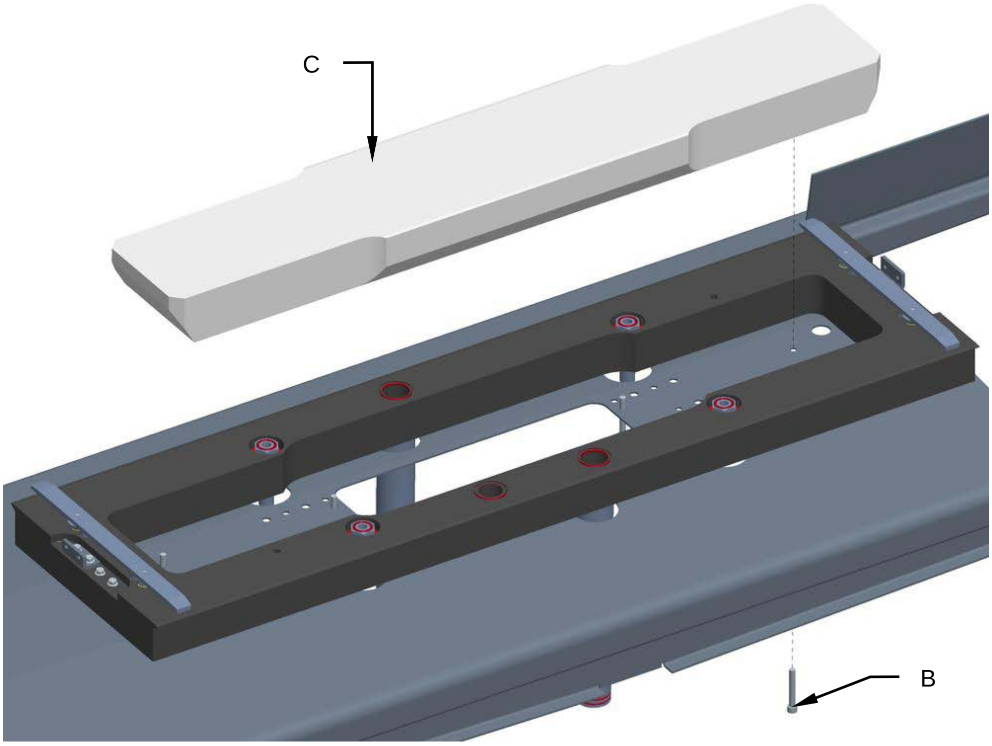
Machine status	Power off
Tools	Normal workshop equipment, Steel supporting blocks

Proceed as follows to replace the sealing screw:

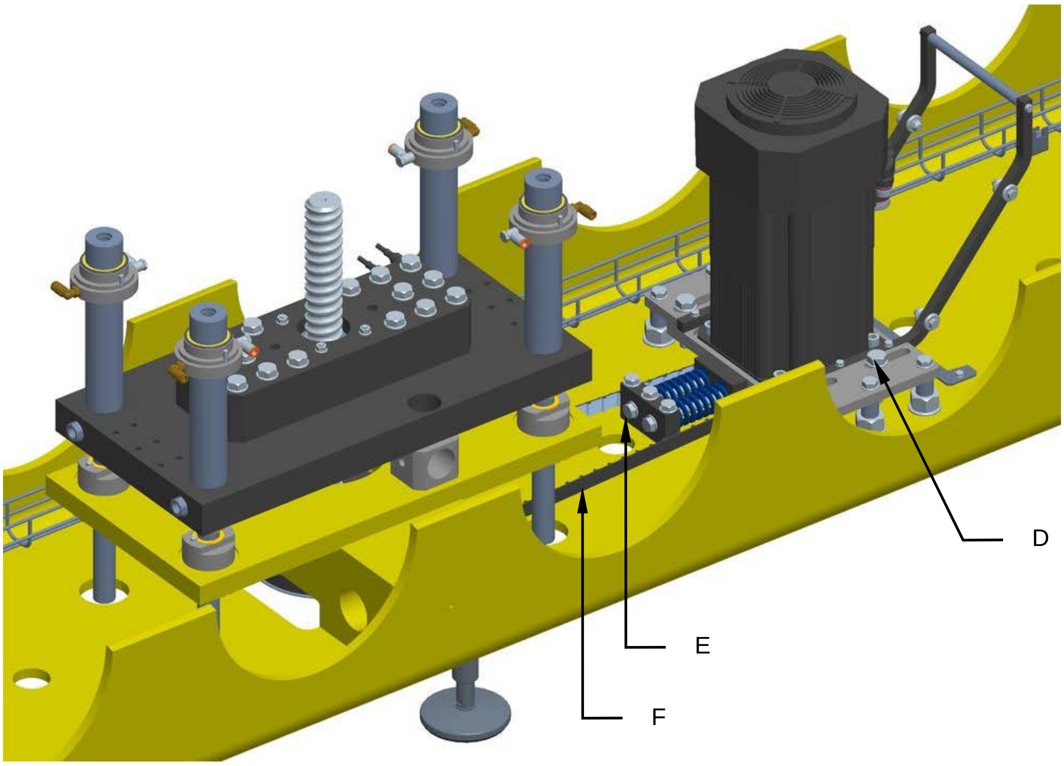
1. Disconnect the power and compressed air supply.
2. Remove the lower guards (**A**).



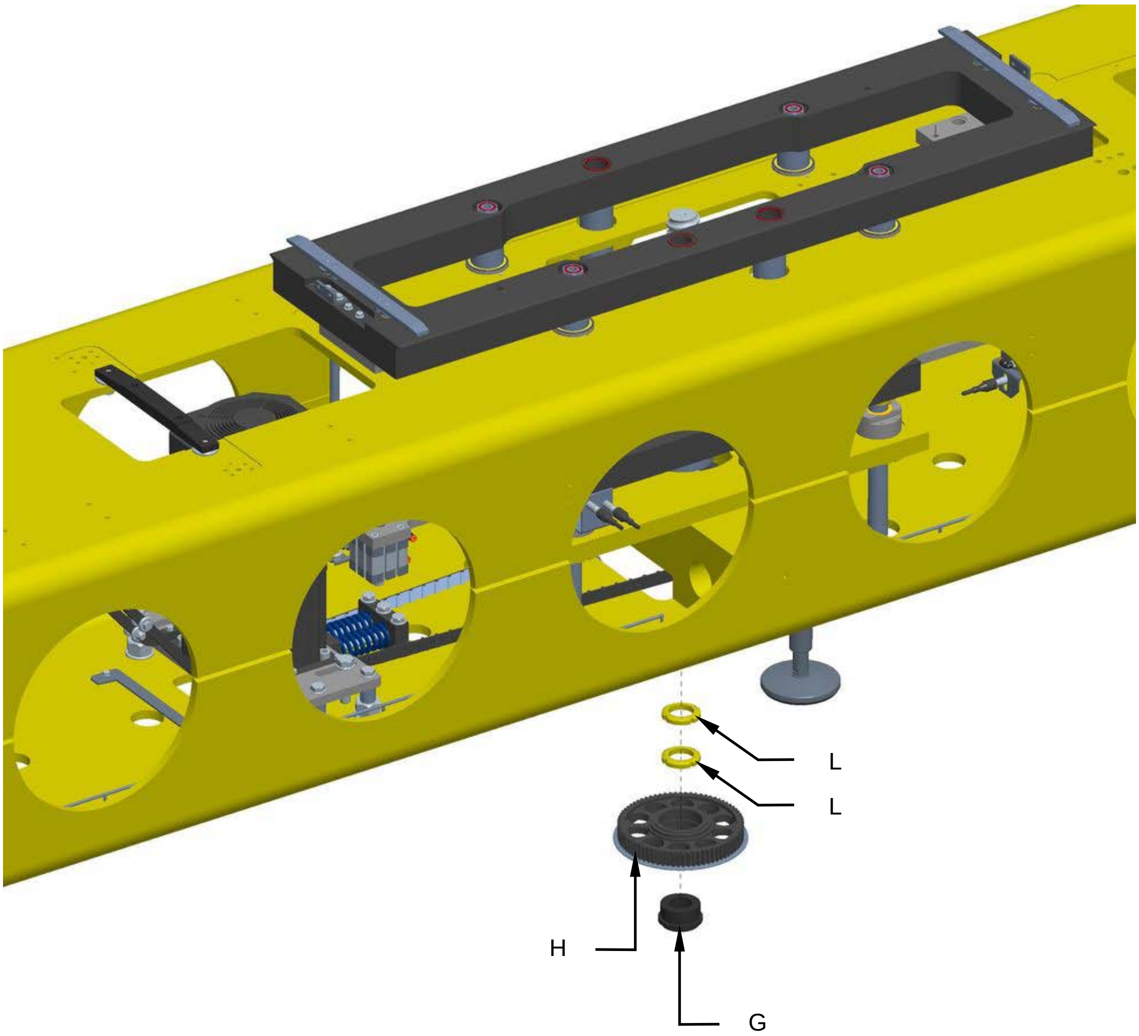
3. Unscrew the screws (**B**) and remove the central element (**C**).



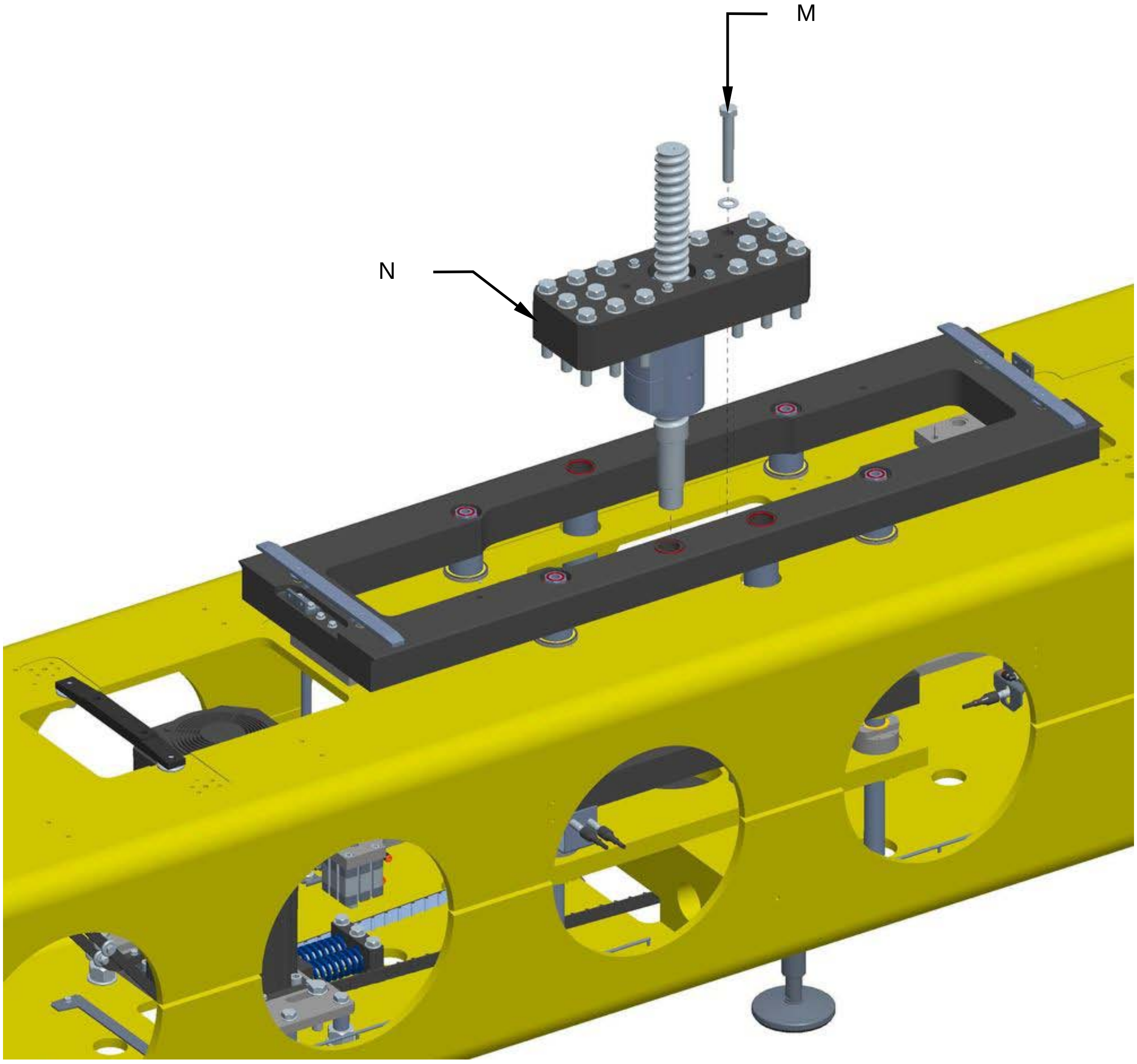
4. Loosen the screws (**D**), tighten the screws (**E**) to reduce the tension of the belt (**F**).



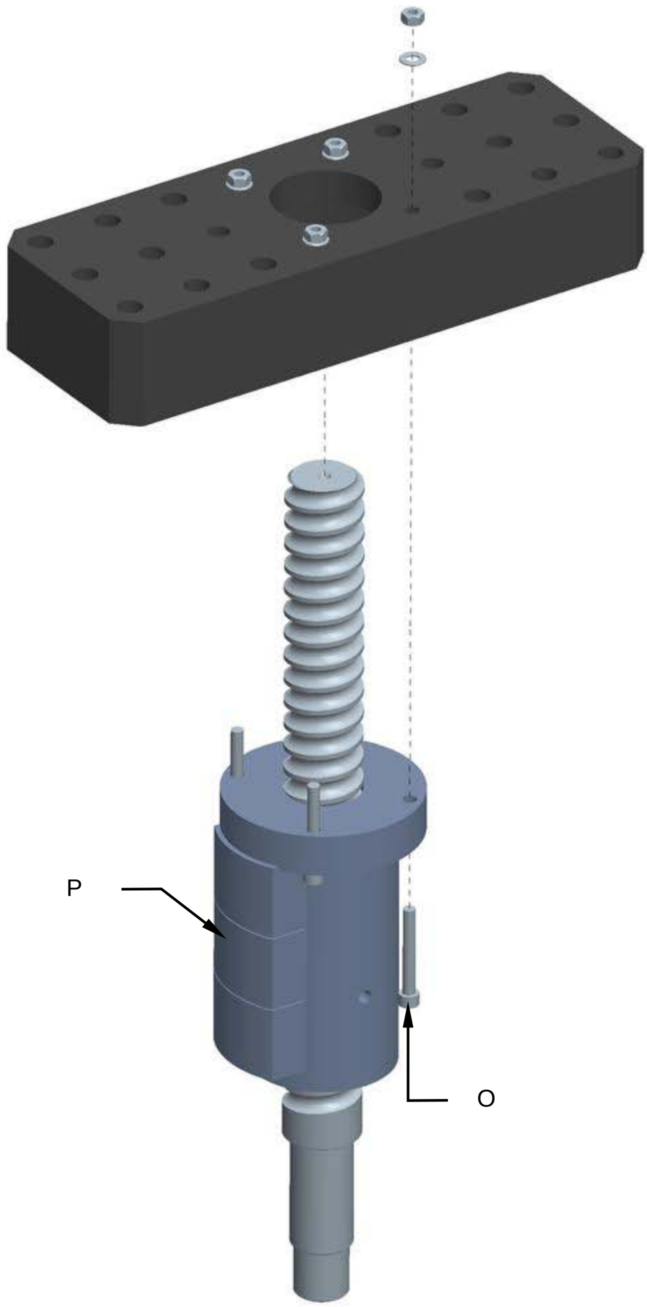
5. Unscrew the Tollok component (**G**) and remove the pulley (**H**).
6. Unscrew the ring nuts (**L**).



7. Unscrew the screws (**M**) and remove the screw assembly (**N**).



8. Unscrew the screws (**O**) and replace the sealing screw (**P**).



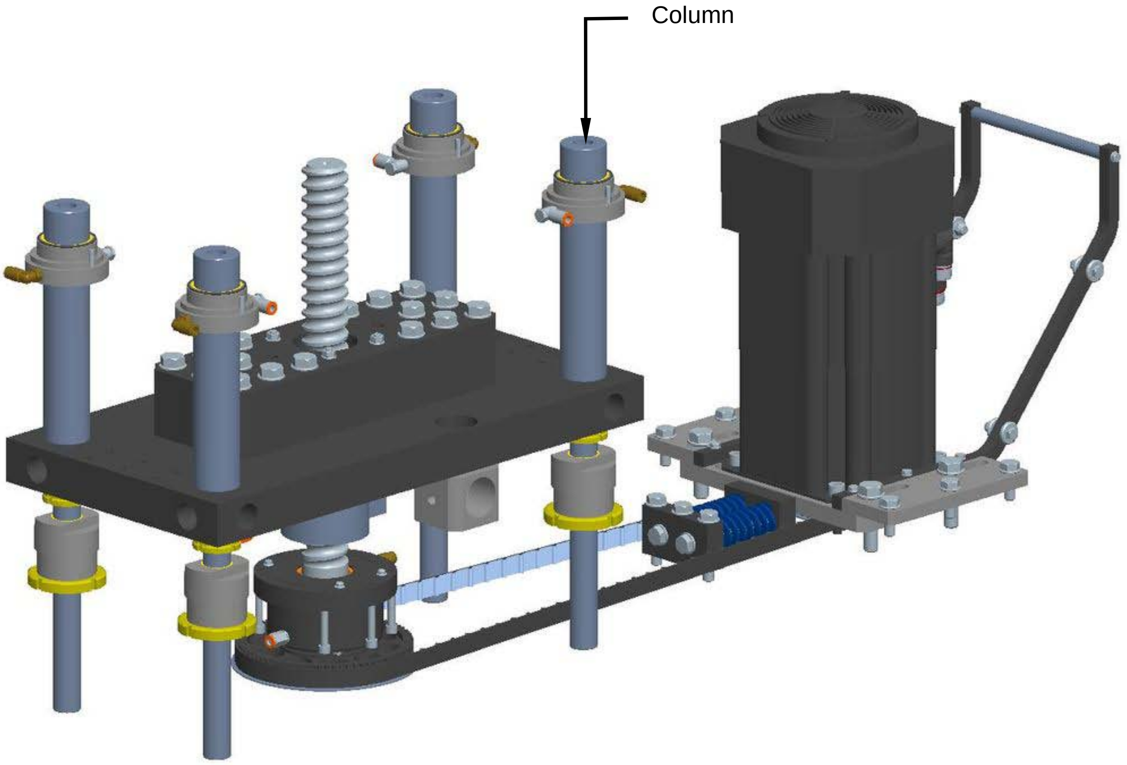
9. Remount all the components in the reverse order.

8.17.7 Checking the column condition

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the columns.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

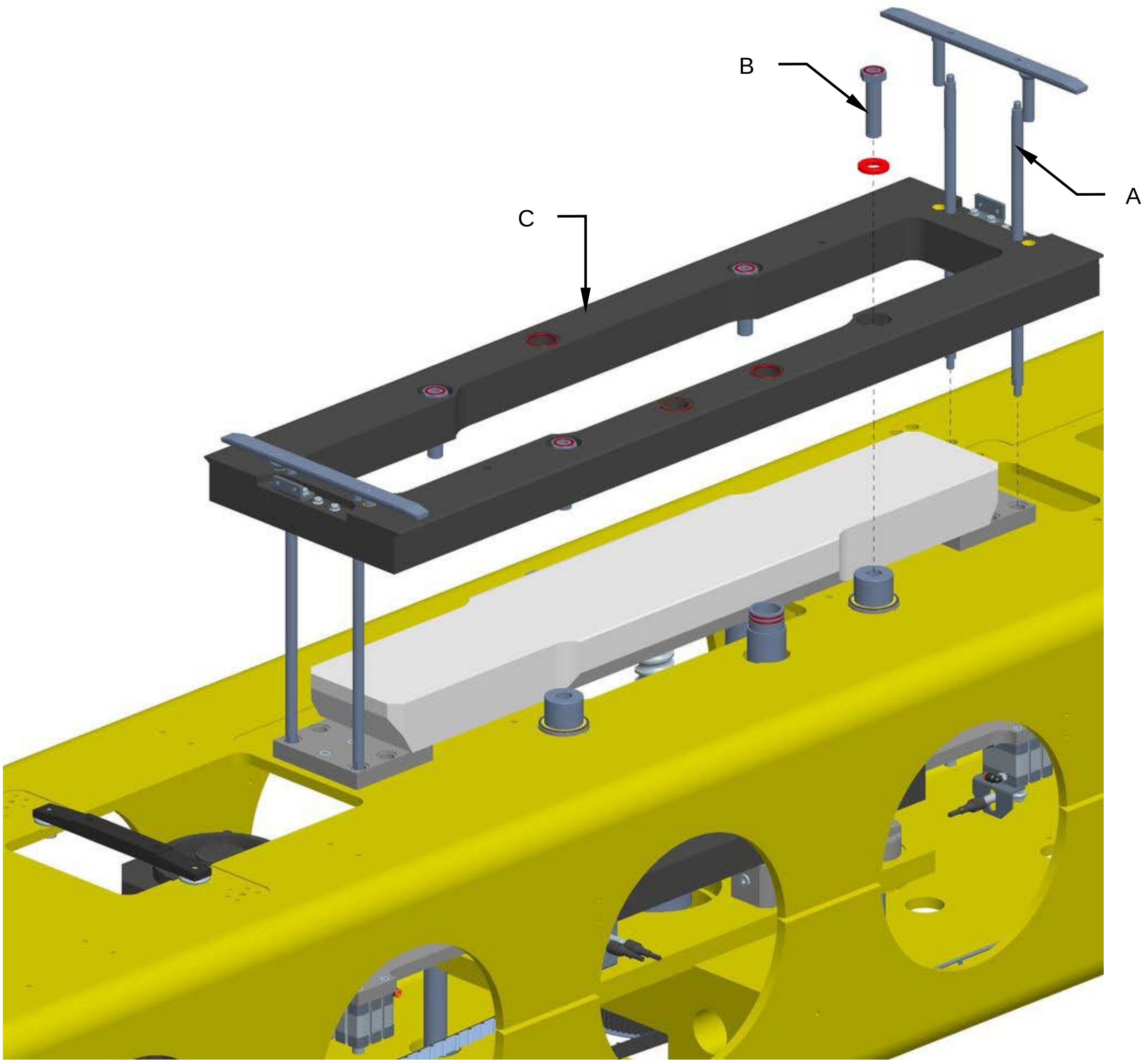


8.17.8 Replacing the columns

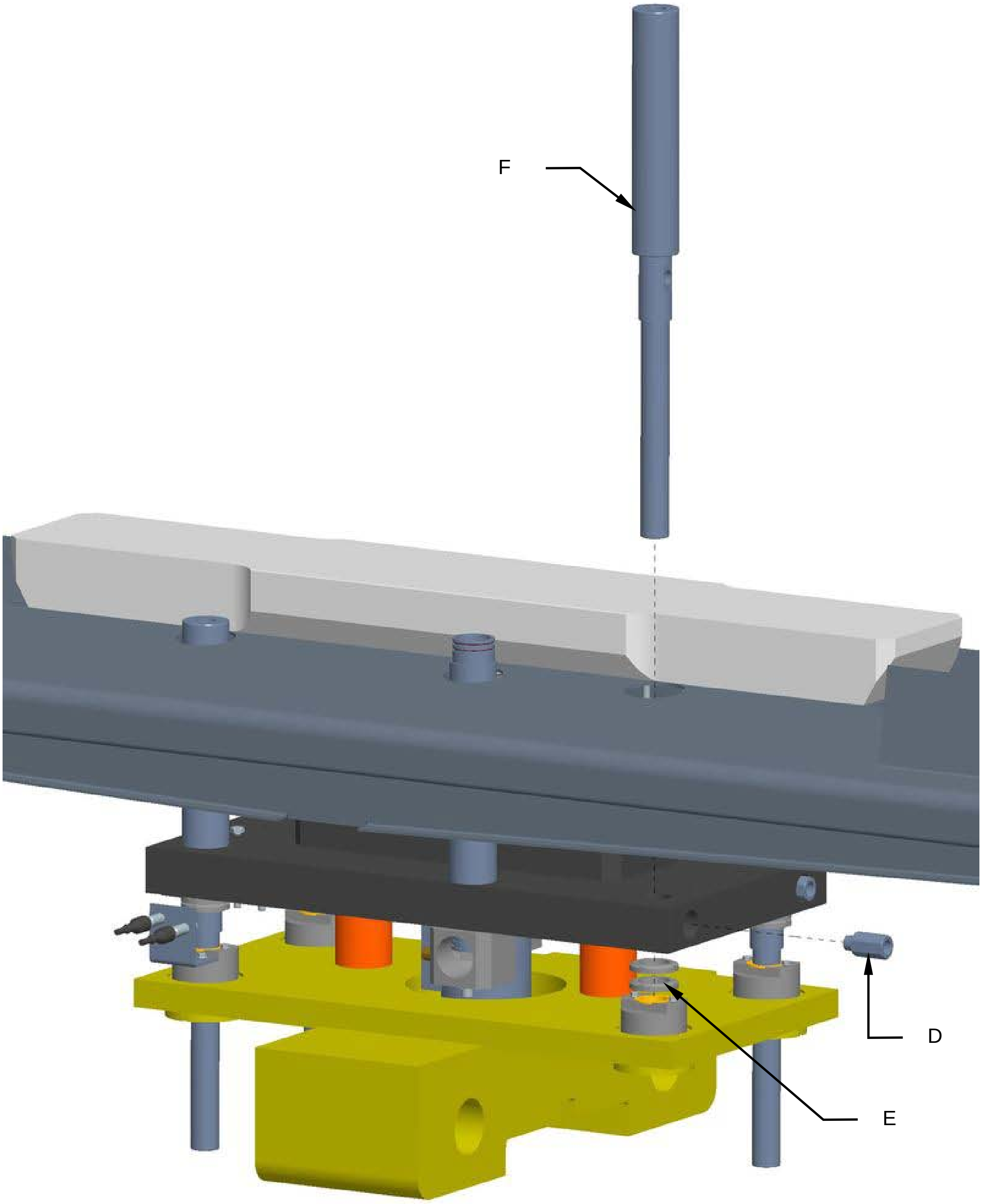
Machine status	Power off
Tools	Normal workshop equipment, Steel supporting blocks

Proceed as follows to replace the columns:

1. Disconnect the power and the compressed air supply.
2. Unscrew the rod extensions (**A**).
3. Unscrew the screws (**B**) and remove the moving plate (**C**).



4. Unscrew the extension (**D**).
5. Unscrew the ring nuts (**E**) and replace the column (**F**).



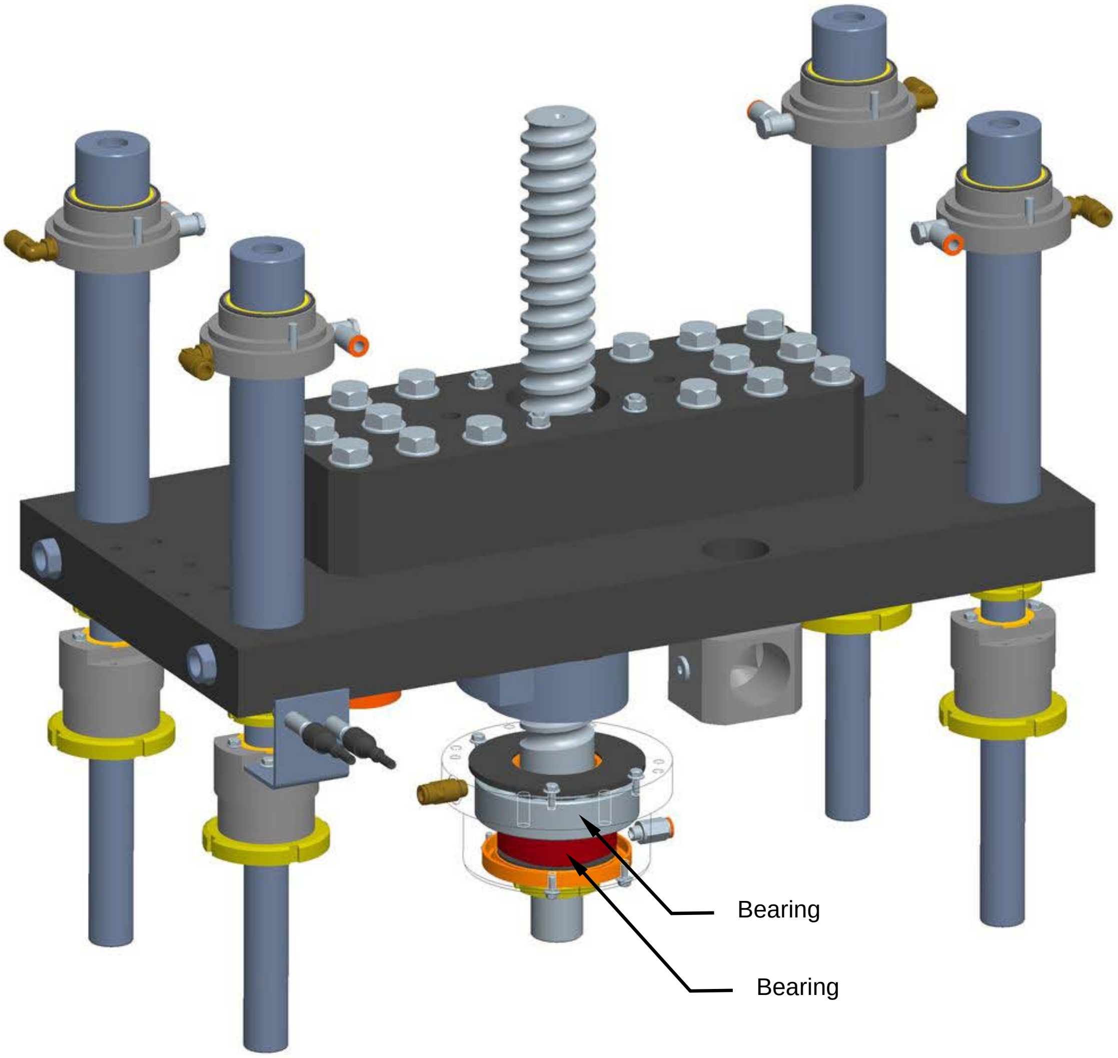
6. Remount all the components in the reverse order.

8.17.9 Checking the bearings condition

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the bearings.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

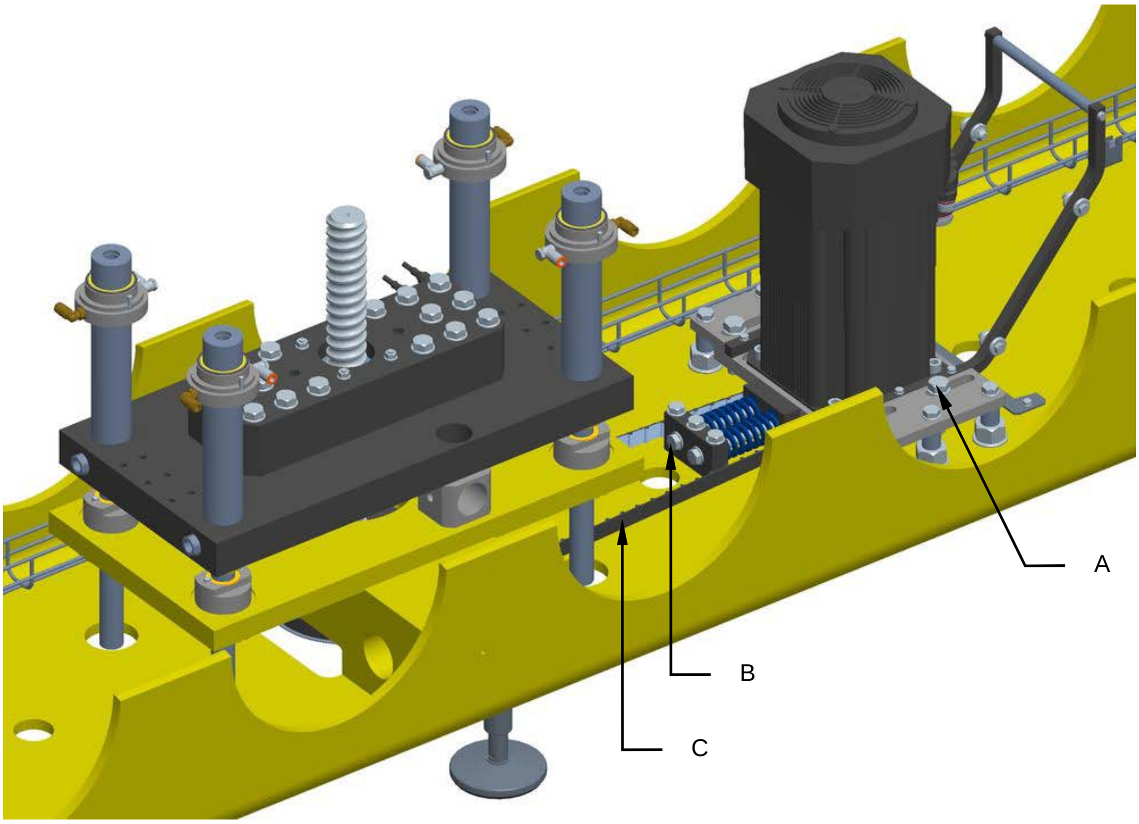


8.17.10 Replacing the bearings

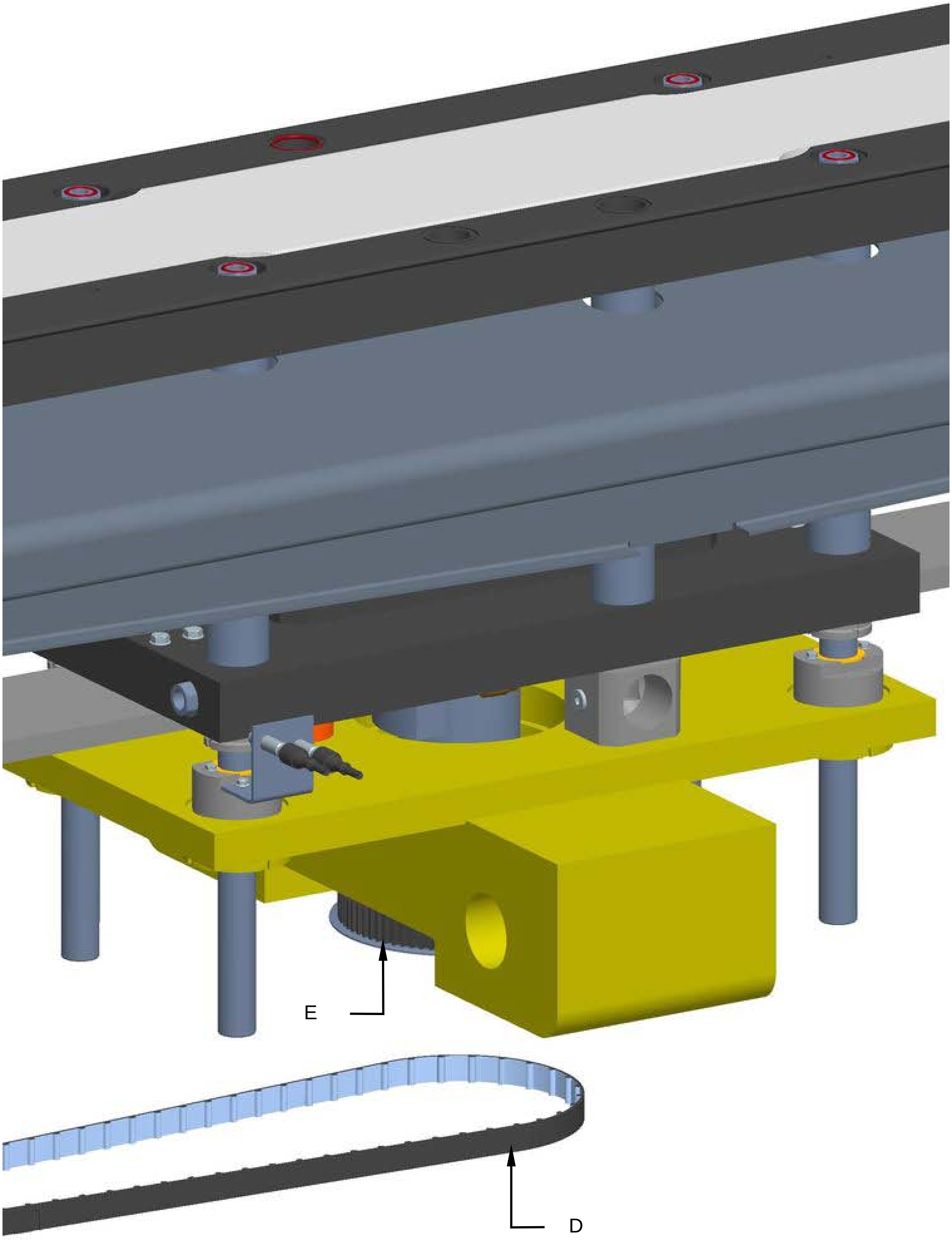
Machine status	Power off
Tools	Normal workshop equipment, steel supporting blocks

Proceed as follows to replace the bearings:

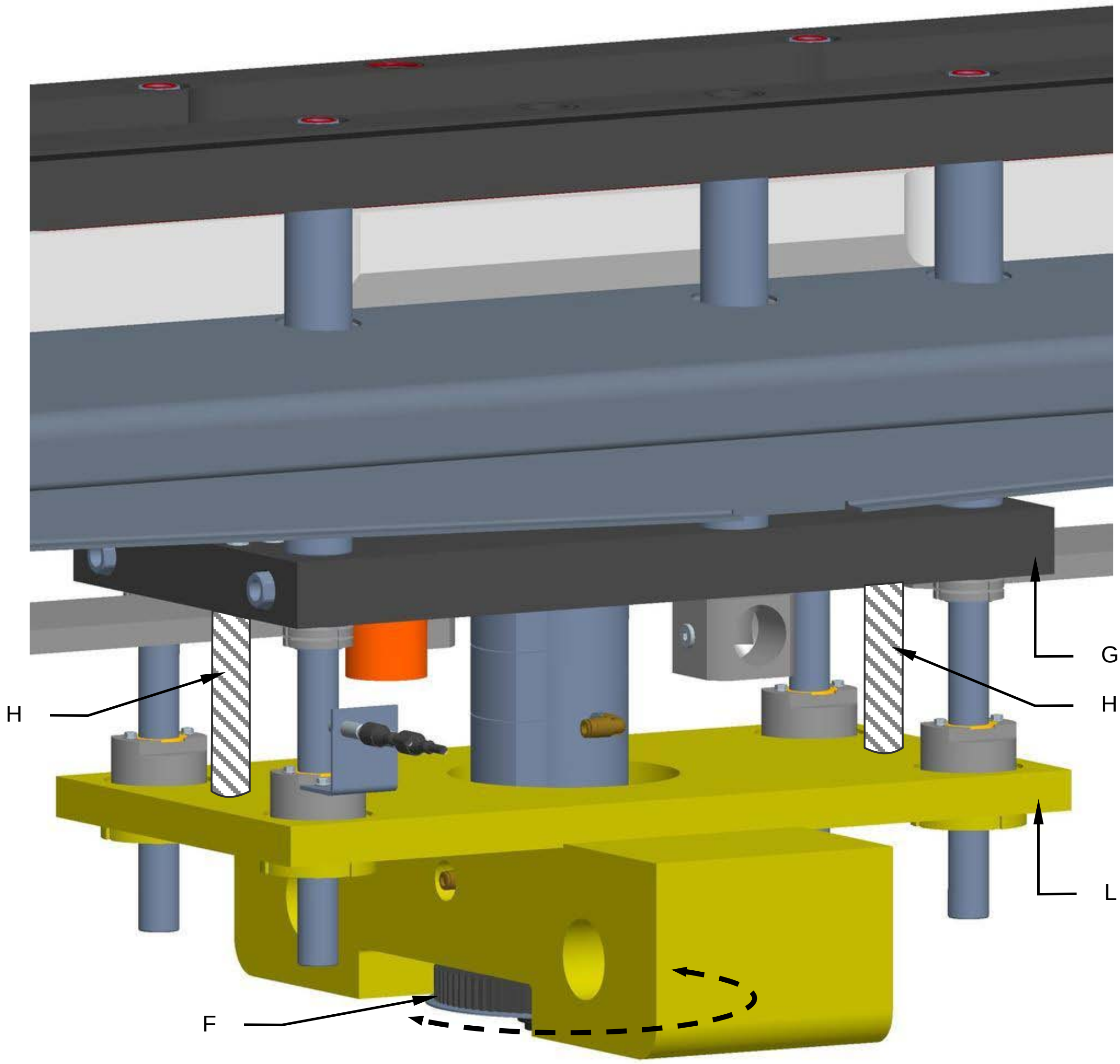
1. Disconnect the power and the compressed air supply.
2. Loosen the screws (**A**), tighten the screws (**B**) to reduce the tension of the belt (**C**).



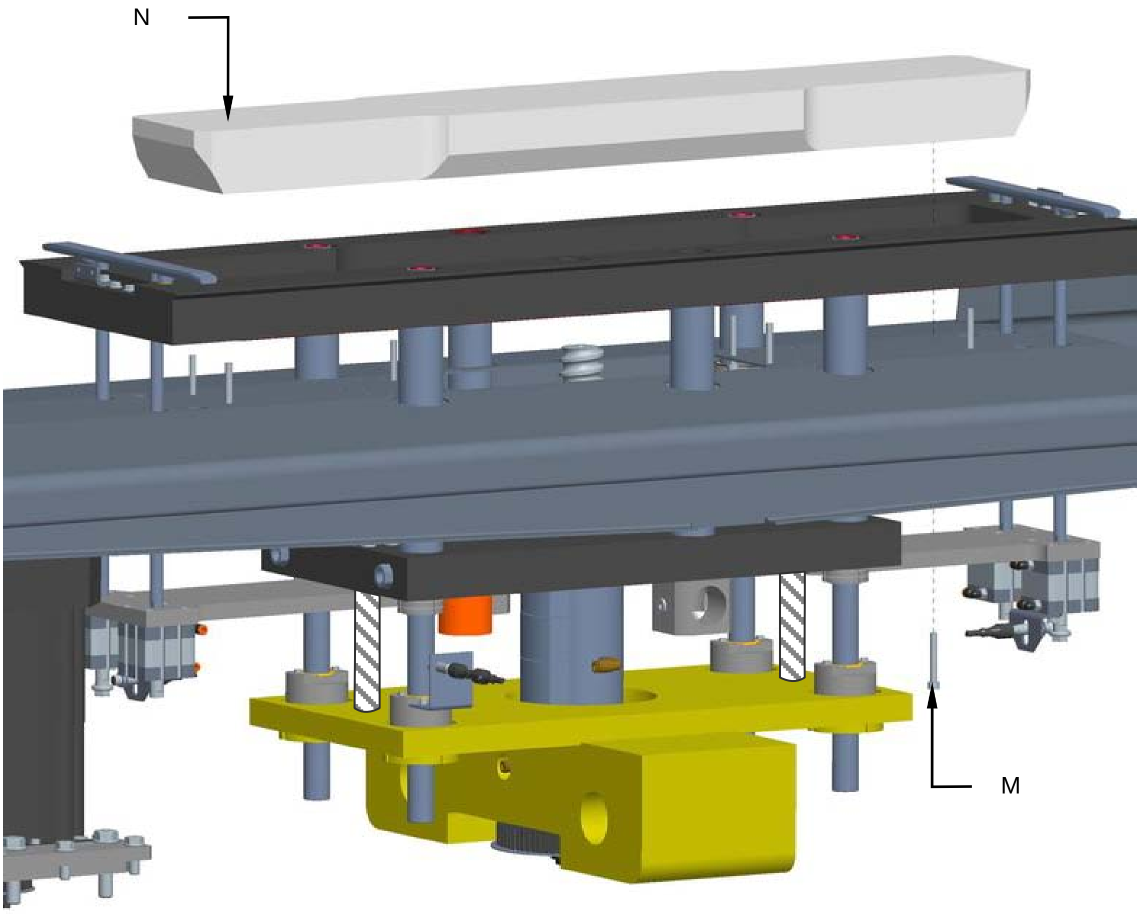
3. Remove the cog belt (**D**) from the pulley (**E**).



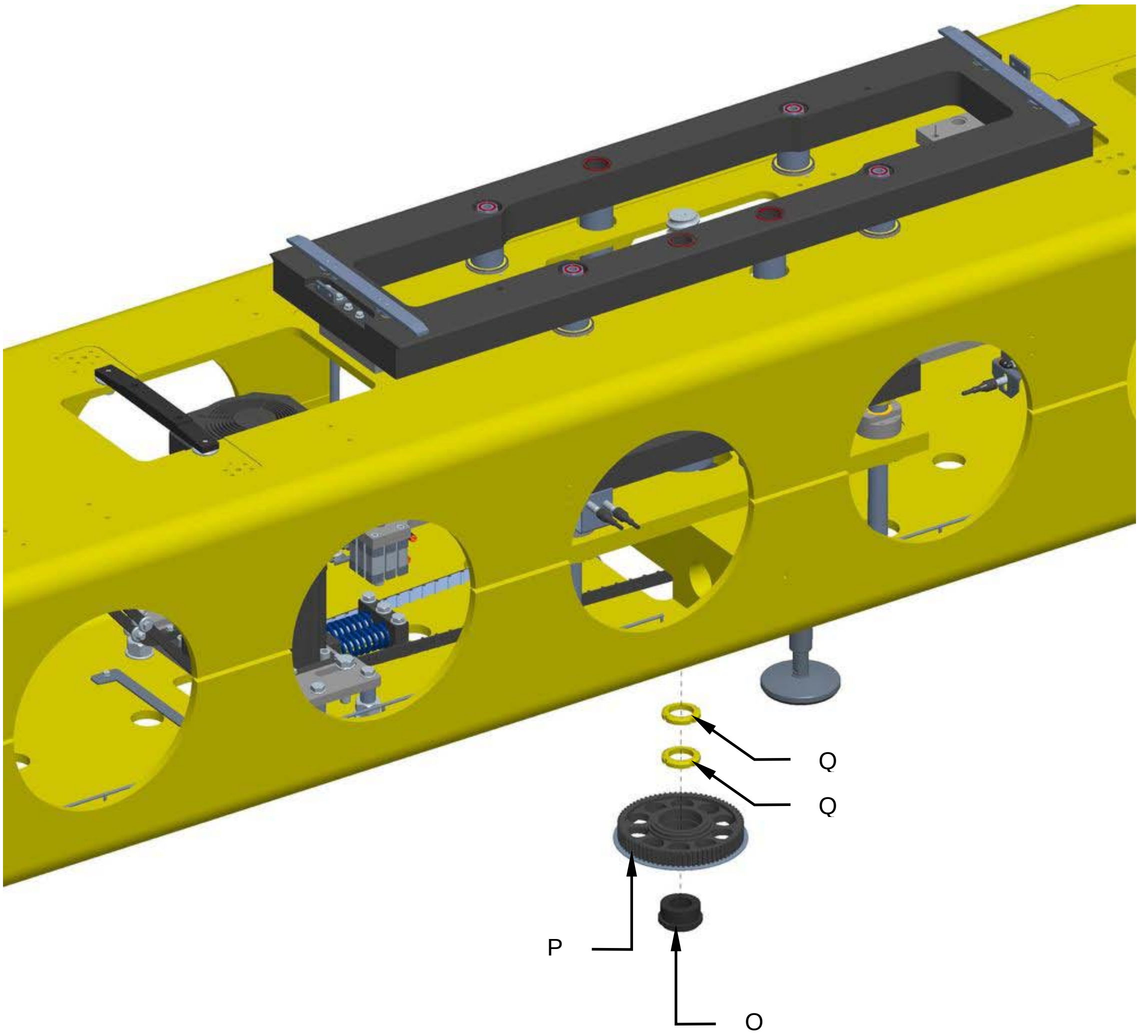
4. Rotate the pulley manually (**F**) so as to raise the plate (**G**).
5. Insert the supports (**H**) between the plate (**G**) and the plate (**L**) so as to create enough room for maintenance interventions.



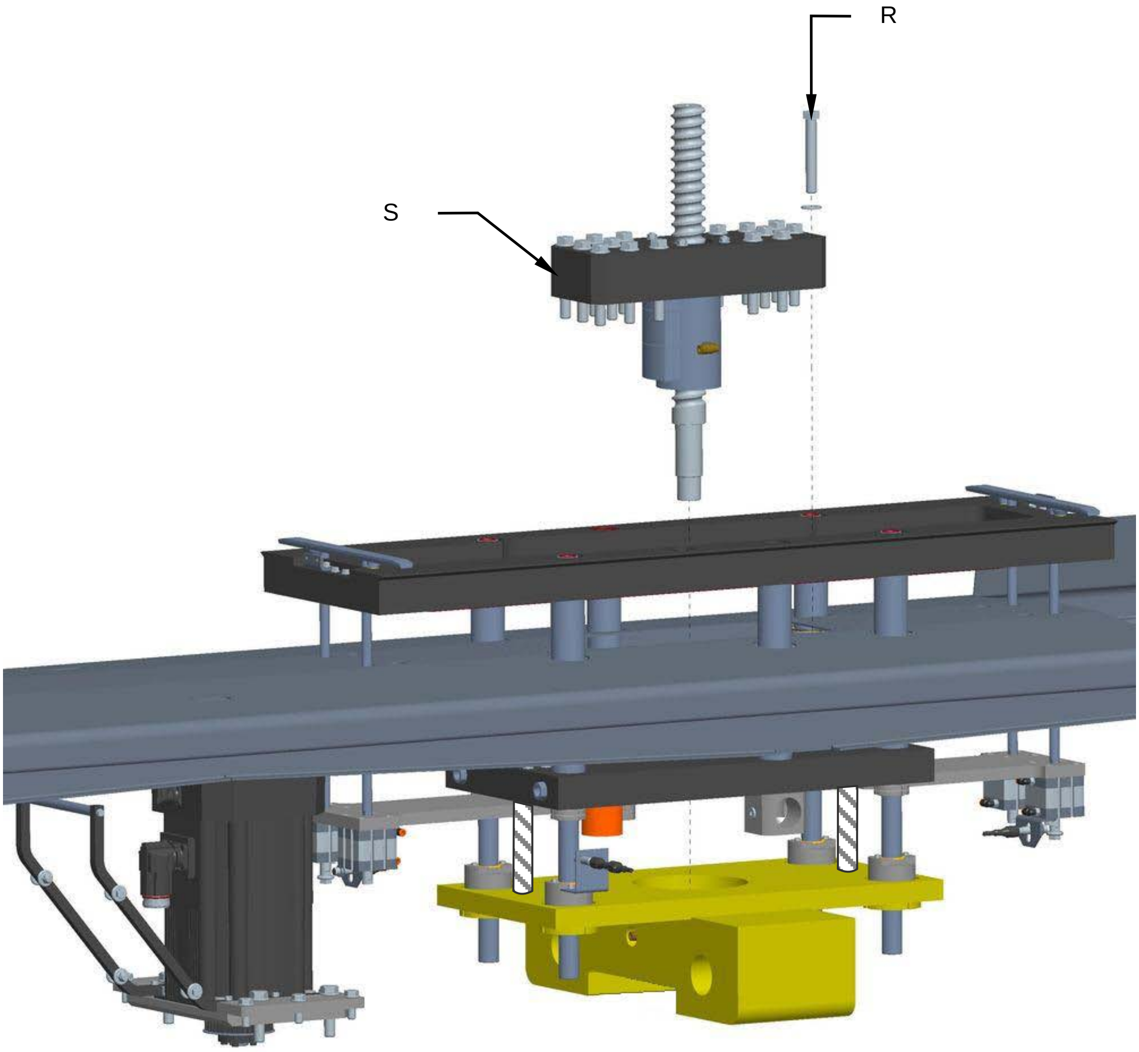
6. Unscrew the screws (**M**) and remove the central element (**N**).



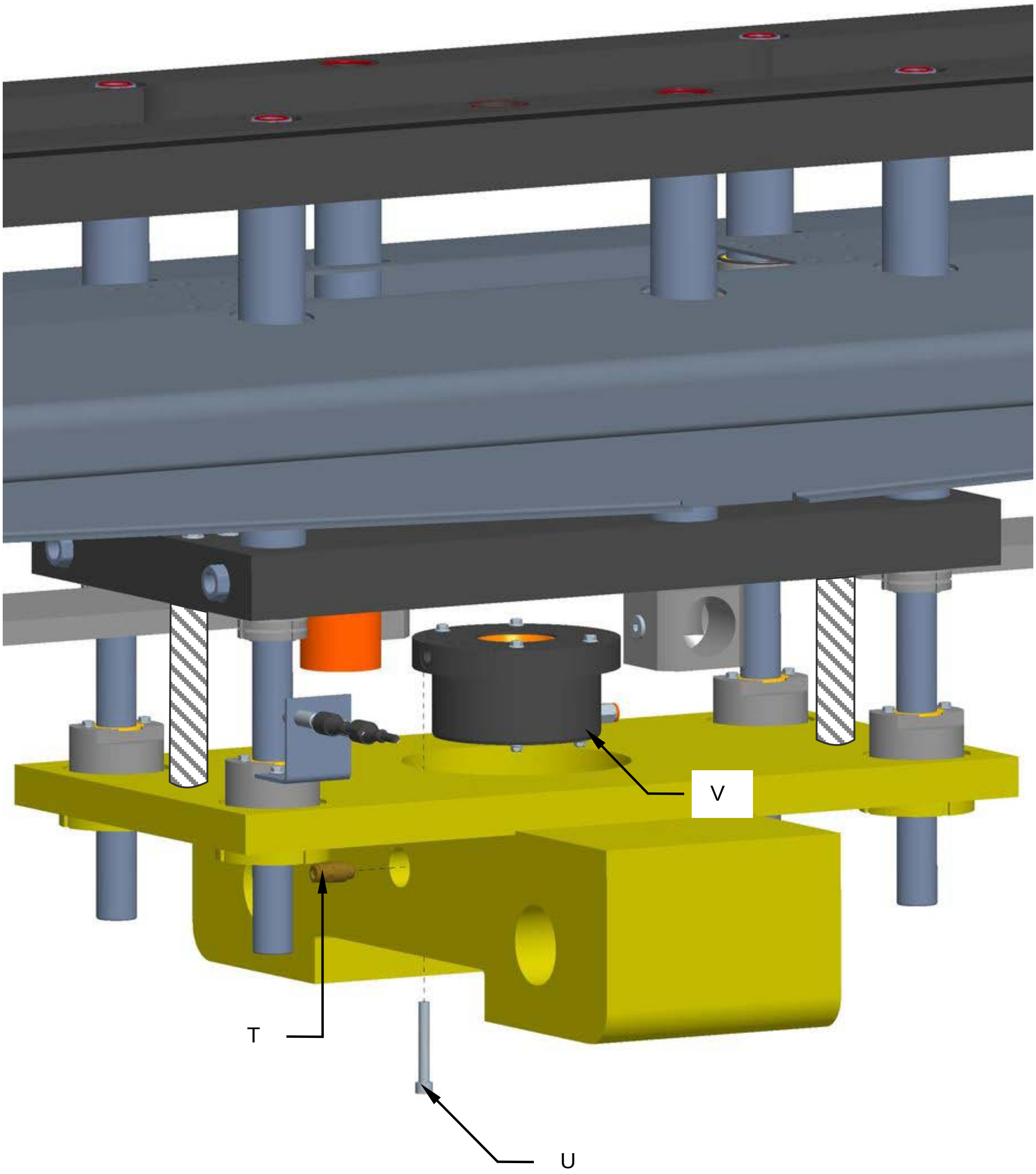
7. Unscrew the Tollok component (**O**) and remove the pulley (**P**).
8. Unscrew the ring nuts (**Q**).



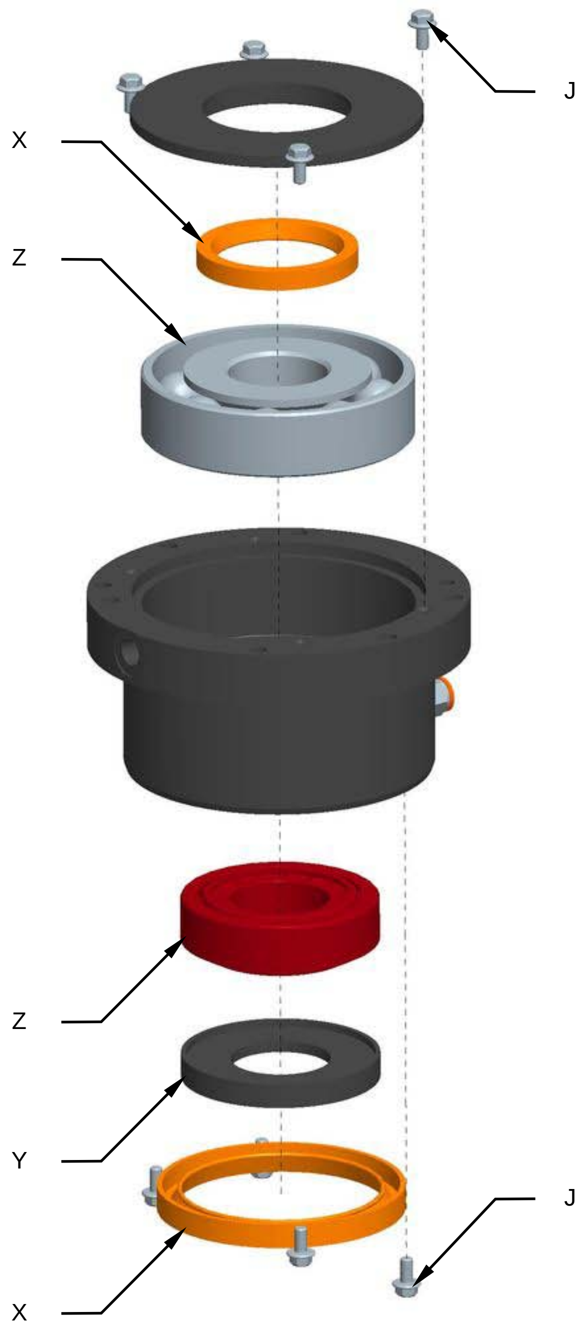
9. Unscrew the screws (**R**) and remove the screw assembly (**S**).



10. Remove the fitting (**T**).
11. Unscrew the screws (**U**) and remove the flange (**V**).



12. Unscrew the screws (**J**).
13. Remove the gaskets (**X**) and the spacer (**Y**).
14. Replace the bearings (**Z**).



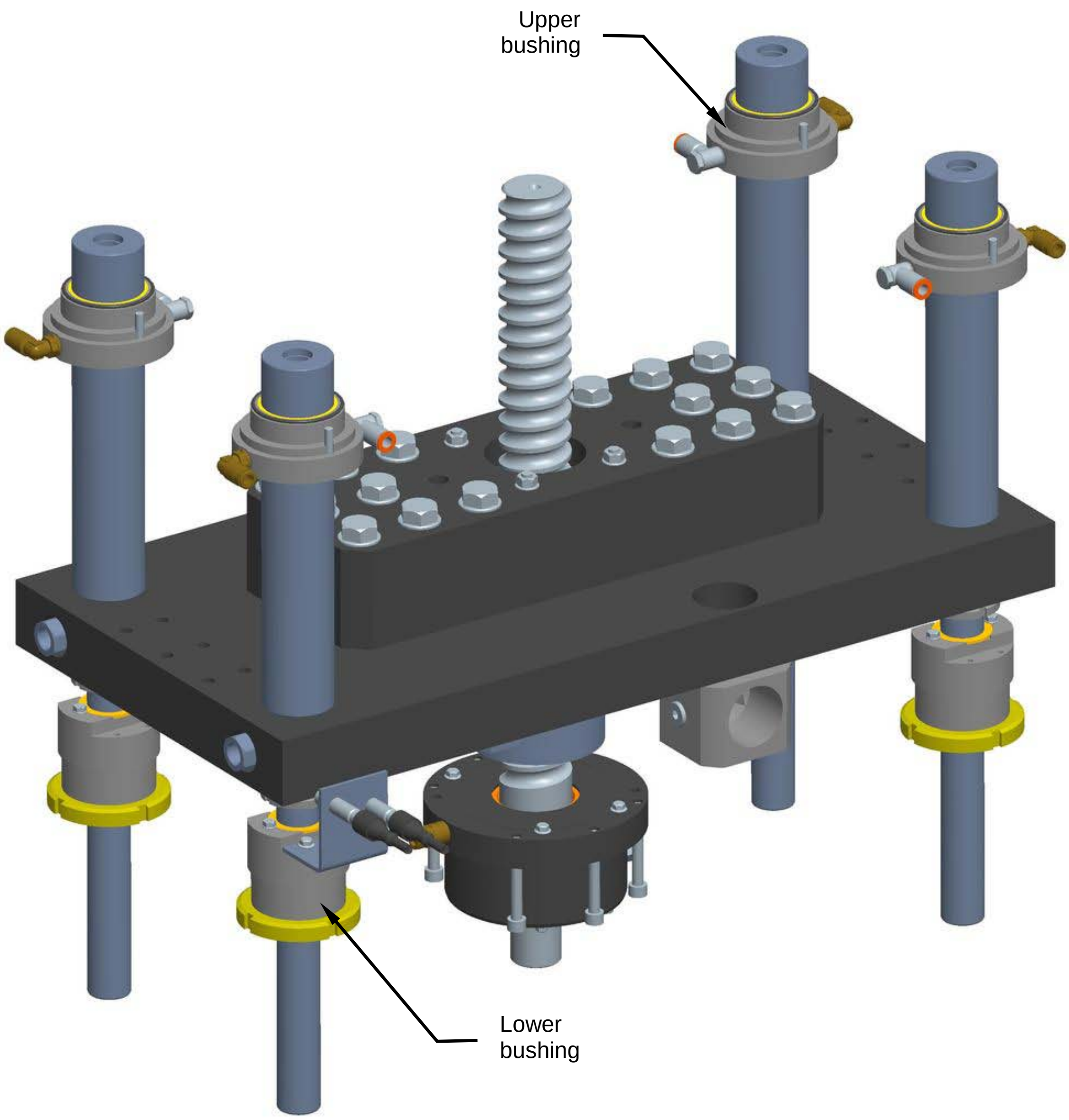
15. Remount all the components in the reverse order.

8.17.11 Checking the bushing condition

Machine status	Power on
Tools	Normal workshop equipment

With the machine running a normal cycle, visually check the condition of the condition of the upper and lower bushings.

If any problems are encountered, inform the person in charge, who will restore correct operation of the assembly.

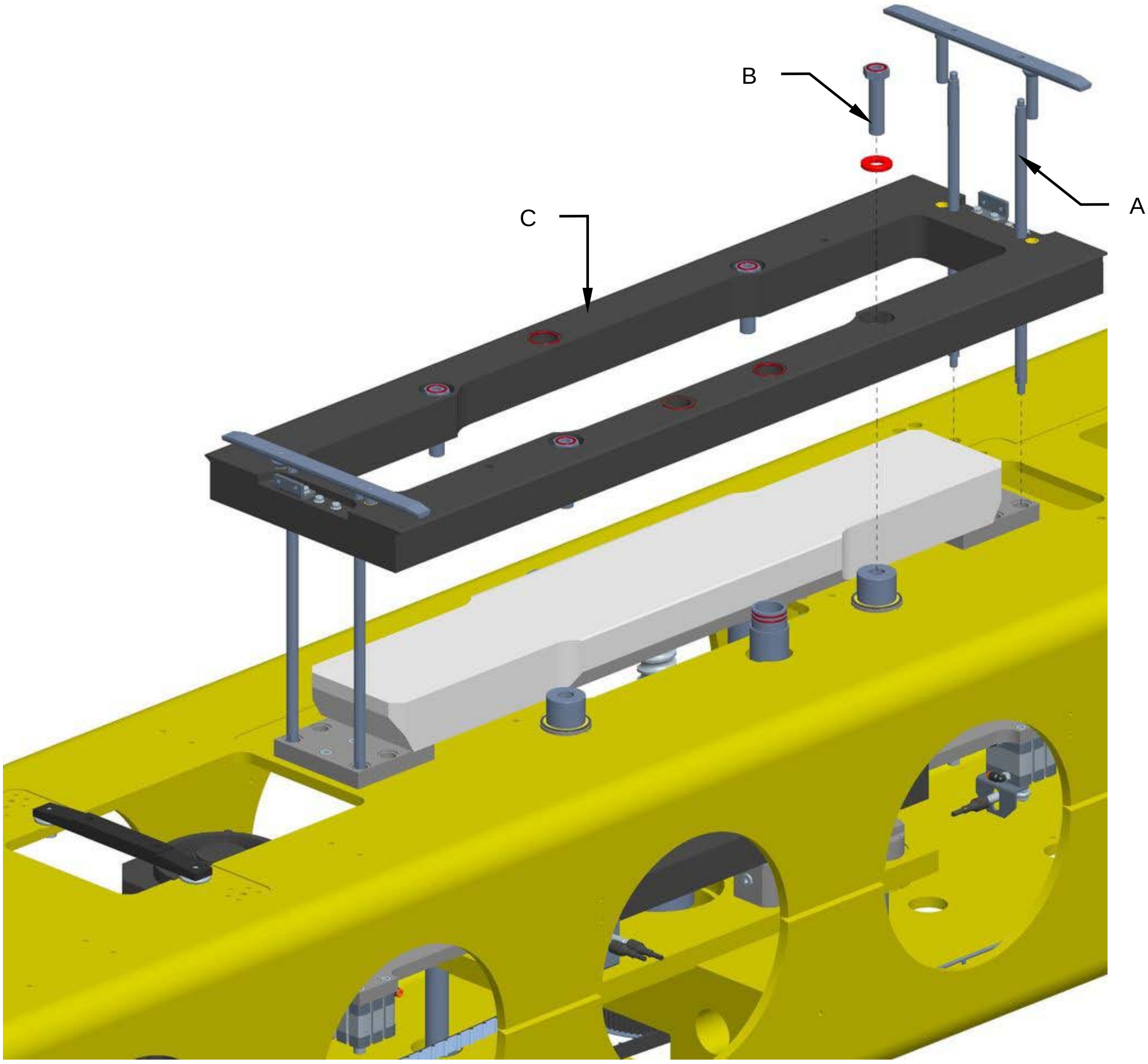


8.17.12 Replacing the bushings

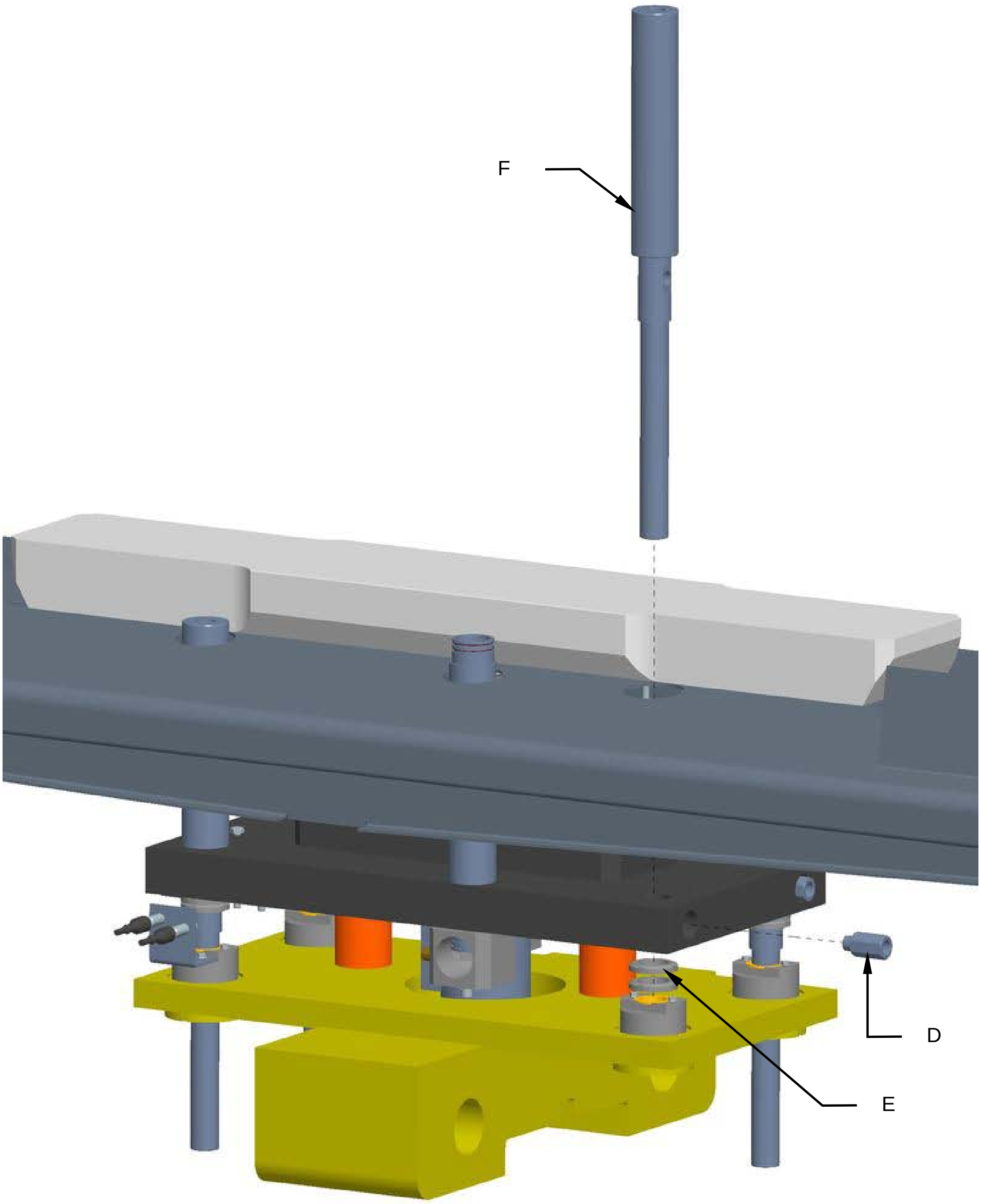
Machine status	Power off
Tools	Normal workshop equipment, Steel supporting blocks

Proceed as follows to replace the upper and lower bushings:

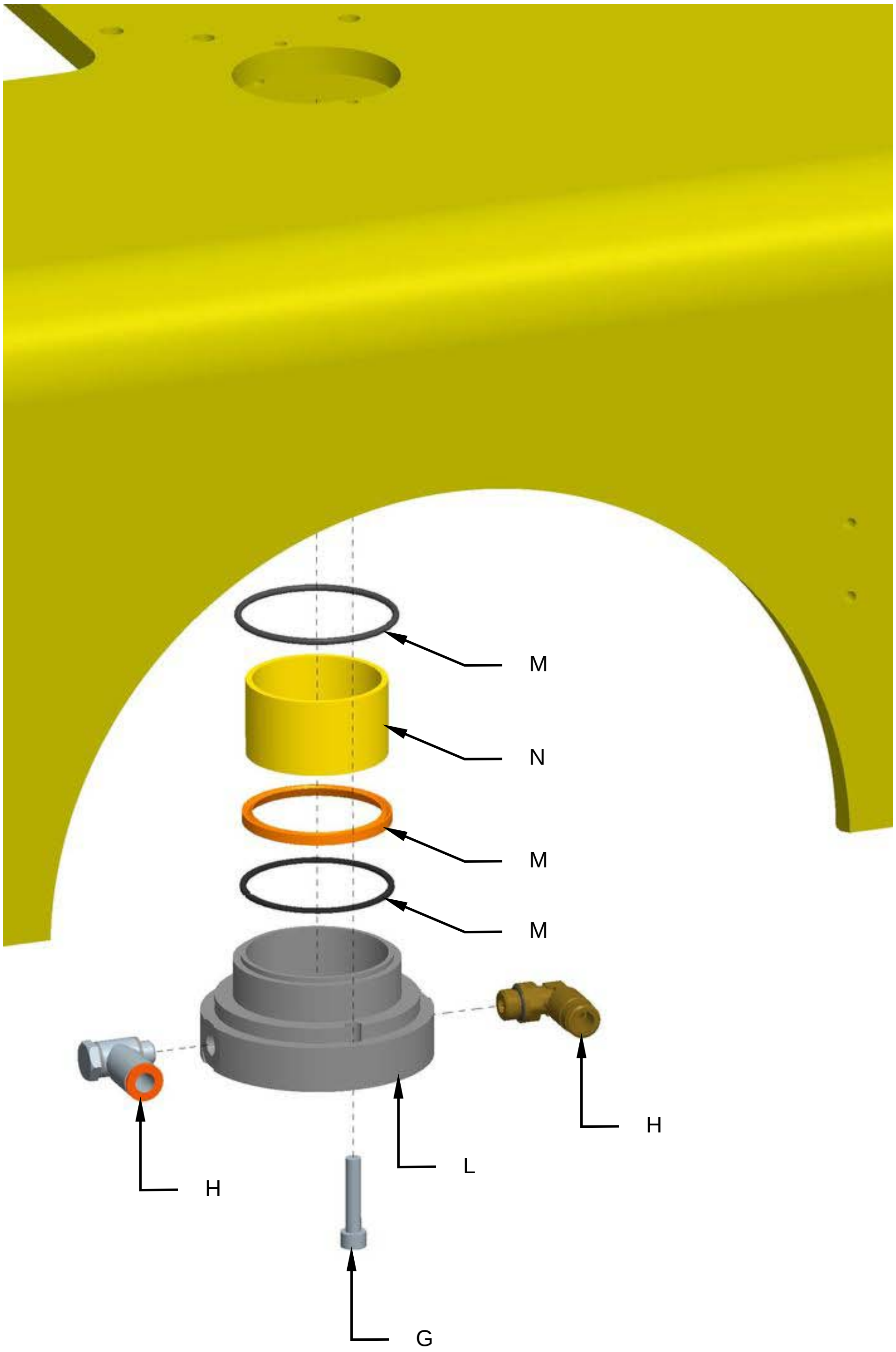
1. Disconnect the power and the compressed air supply.
2. Unscrew the rod extensions (**A**).
3. Unscrew the screws (**B**) and remove the moving plate (**C**).



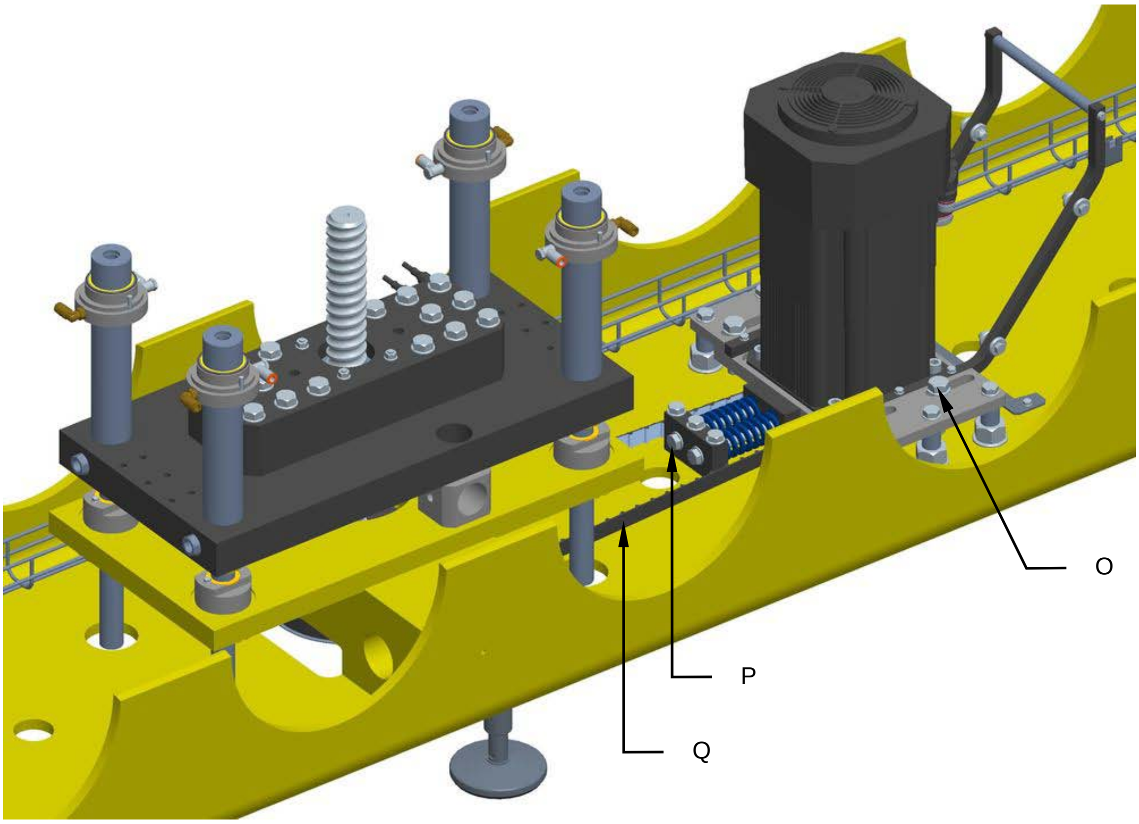
4. Unscrew the extension (**D**).
5. Unscrew the ring nuts (**E**) and remove the column (**F**).



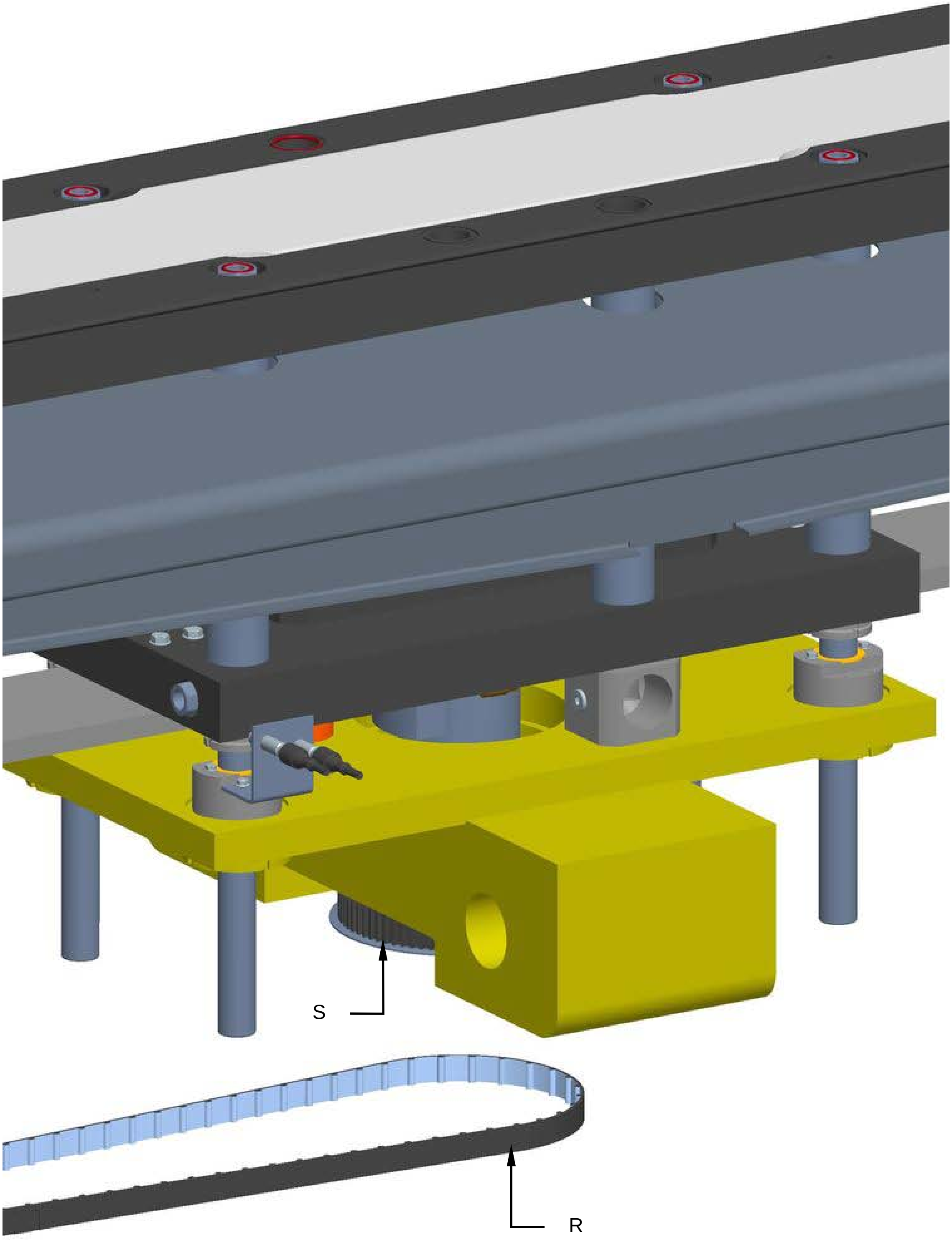
6. Unscrew the screws (**G**) and remove the fittings (**H**).
7. Replace the flange (**L**), the gaskets (**M**) and the bushing (**N**).



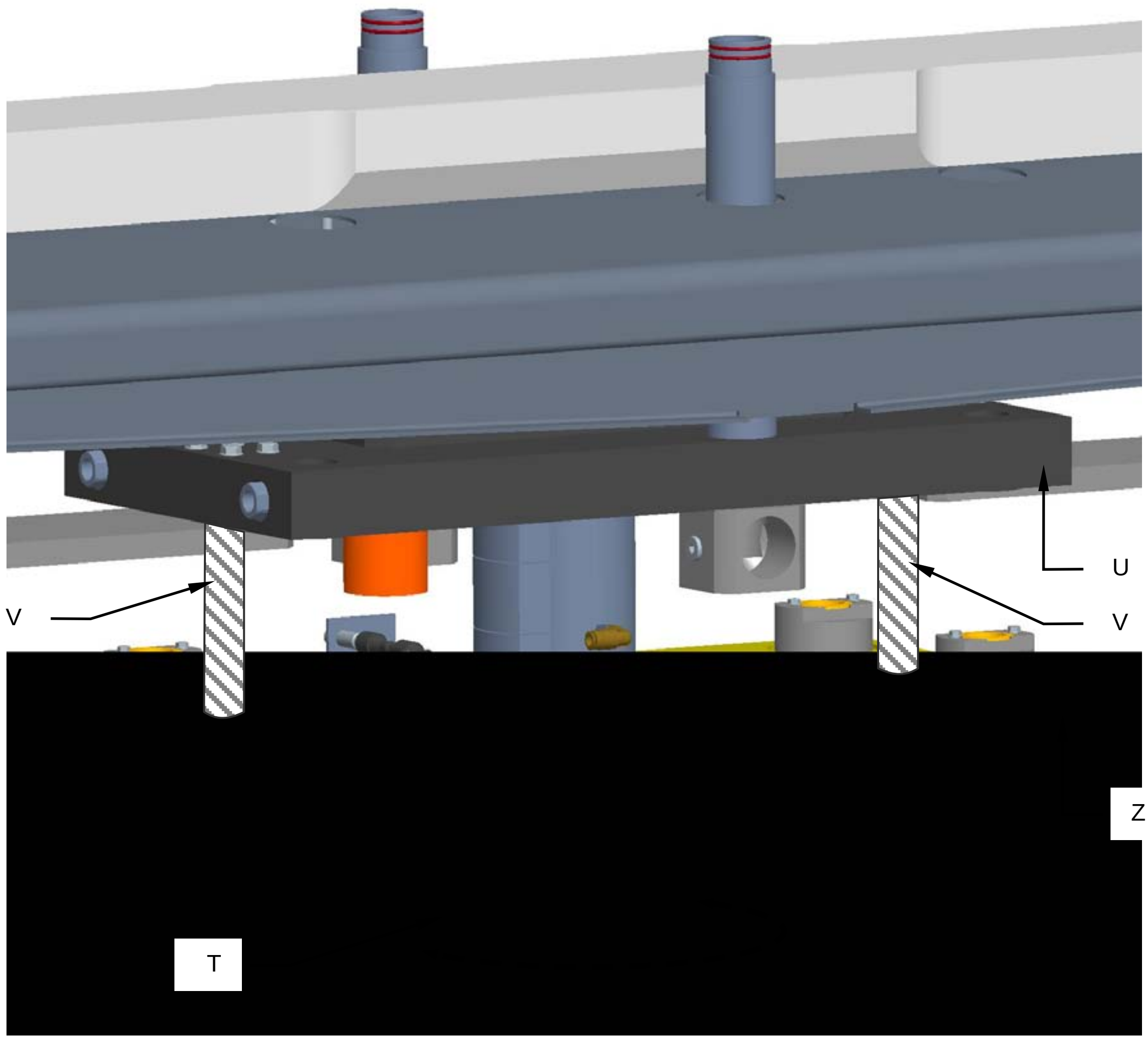
8. Loosen the screws (**O**), tighten the screws (**P**) to reduce the tension of the belt (**Q**).



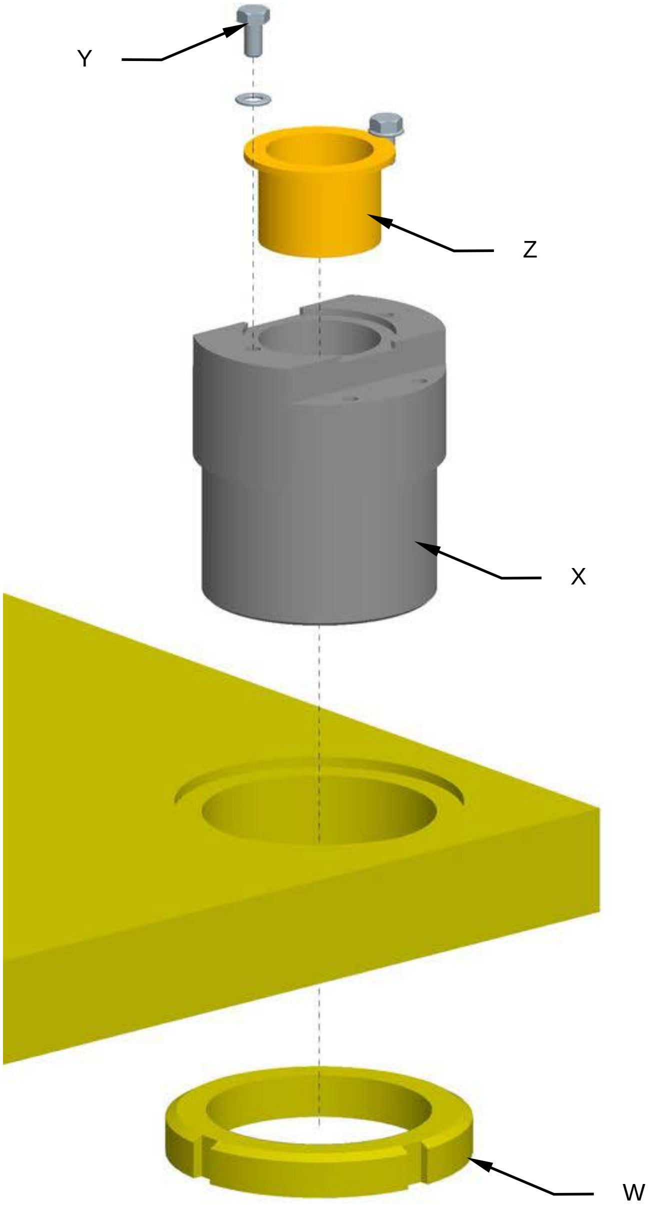
9. Remove the cog belt (**R**) from the pulley (**S**).



10. Rotate the pulley by hand (**T**) so as to raise the plate (**U**).
16. Insert the supports (**V**) between the plate (**U**) and the plate (**Z**) so as to create enough room for maintenance interventions.



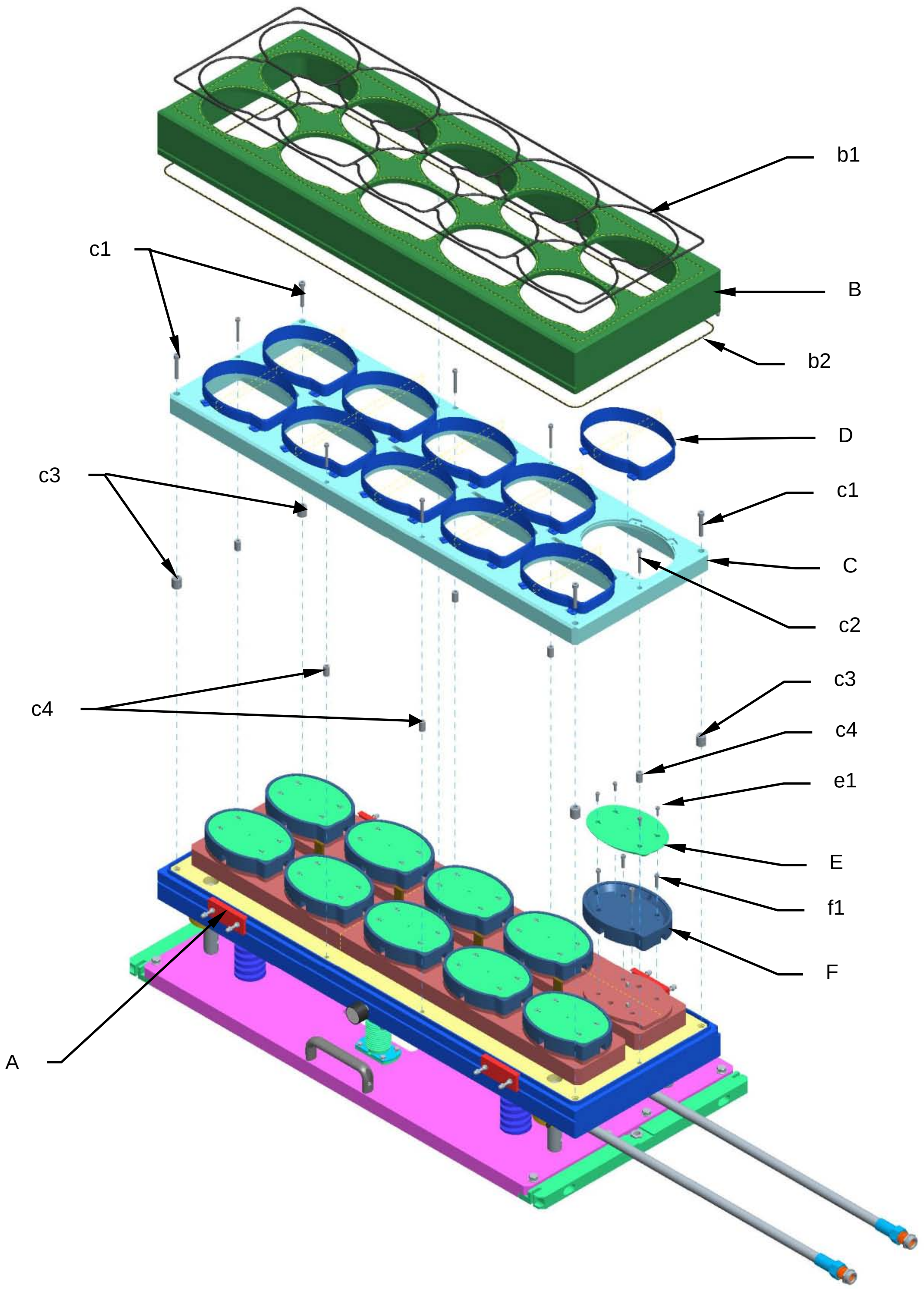
17. Unscrew the ring nut (**W**) and replace the lower flange (**X**).
18. Unscrew the screws (**Y**) and replace the bushing (**Z**).



19. Remount all the components in the reverse order.

8.18 UPPER TOOL MAINTENANCE

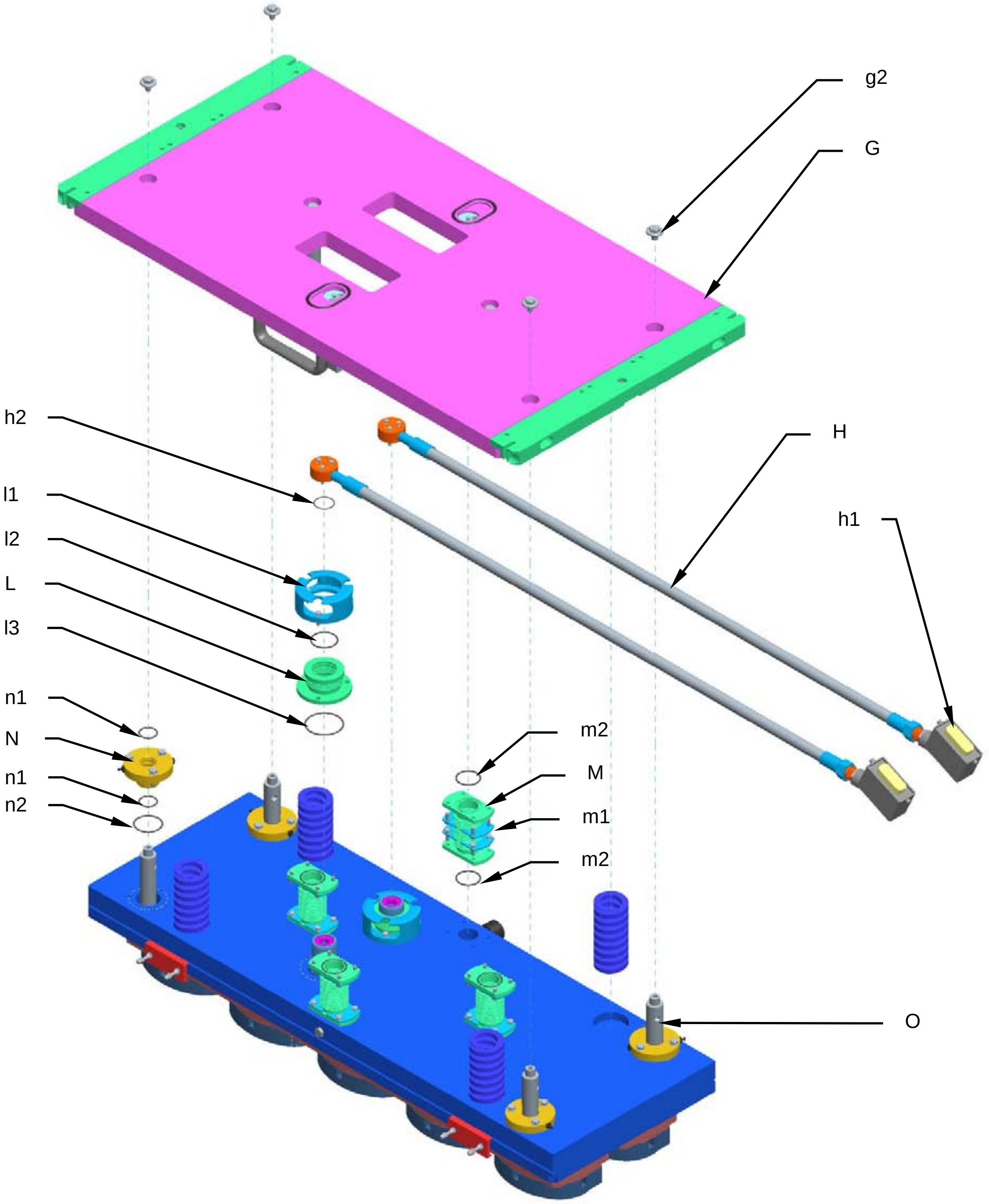
8.18.1 Intermediate cover, die cutter and sealer disassembly and cleaning sequence



1. Loosen the self-locking nuts in the bracket (**A**) until the bracket opens.
2. Remove the intermediate cover (**B**) and check the gaskets (**b1** and **b2**). Replace any damaged ones.
3. Unscrew the screws (**c1** and **c2**) to loosen the die cutter holders (**C**), taking care not to hurt yourself on the die cutters (**D**). Remove the spacers (**c3** and **c4**).
4. Clean and check the die cutters (**D**). If any damaged ones need to be replaced, unscrew them carefully from the plate (**C**).
5. Clean and check the insulation (**E**) and the sealer (**F**). Unscrew the screws (**e1**) if any damaged insulation needs to be replace, and screws (**f1**) if the sealer needs to be replaced.

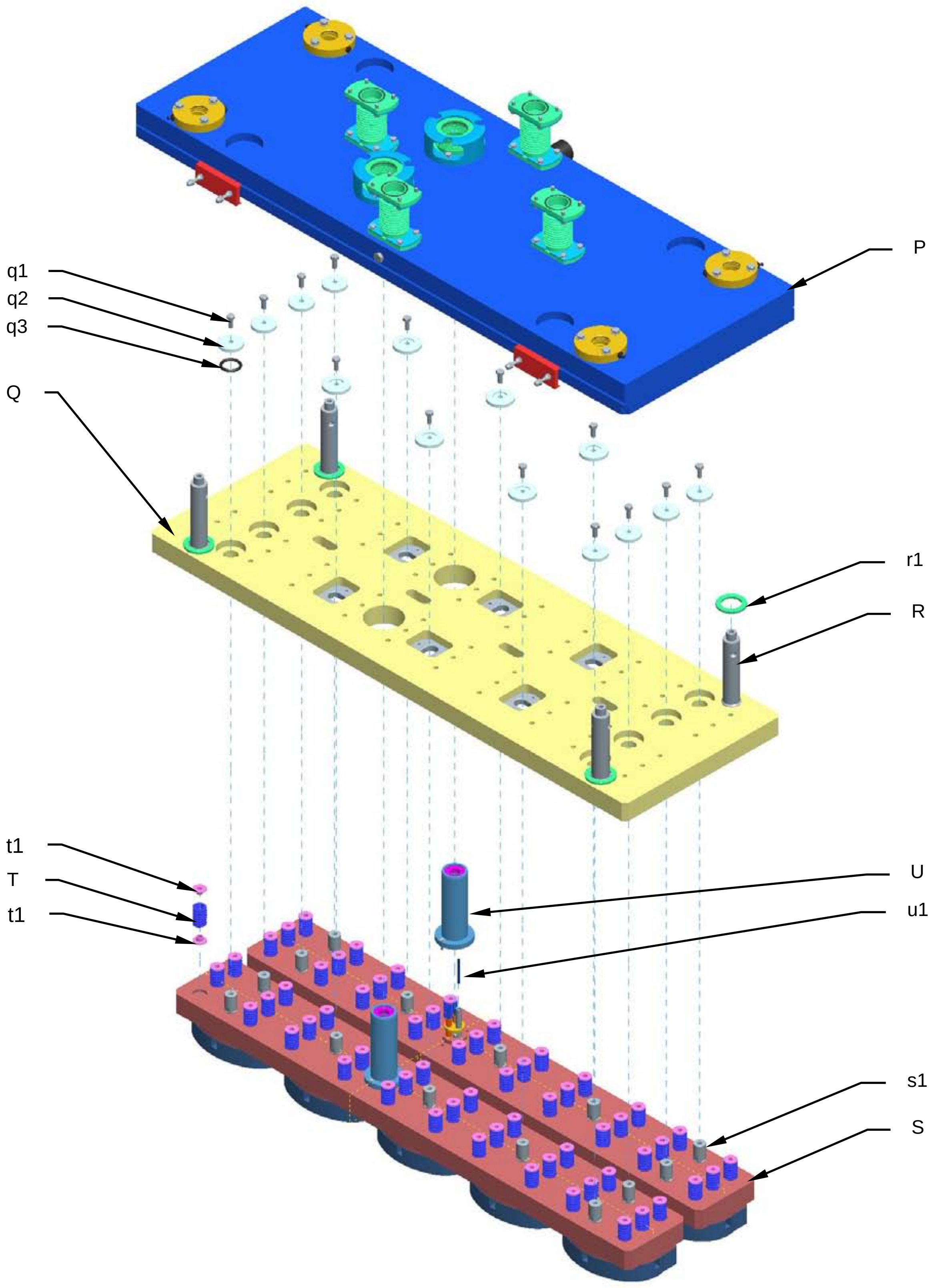
   Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

8.18.2 Upper plate and blower disassembly and cleaning sequence



1. Remove the plate (**G**) by unscrewing the fixing screw (**g1**) and the screws in the upper brackets (**m1**) that lock the blowers (**M**) to the plate.
2. Remove the sheaths (**H**) by disconnecting the power cables from the connectors (**h1**) and pulling them out of the sheaths. Check the O-ring (**h2**) and the sheath for wear and replace if necessary.
3. Check the blower (**L**) and the O-ring (**l2**). If they need to be replaced, remove the bracket (**l1**). Once the blower has been freed you can also check the O-ring (**l3**).
4. Check the blowers (**M**) and the O-rings (**m2**). If they need to be replaced, remove the lower brackets.
5. Check the grease nipples of the bushes (**N**) and make sure they are not clogged. They must allow the grease to lubricate the column. Also check the state of the O-rings (**n1**). If they need to be replaced, remove the bush and this will allow you to check the O-ring (**n2**).
6. Remove and check the springs (**O**).
7. Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

8.18.3 Internal plate and resistance disassembly and cleaning sequence

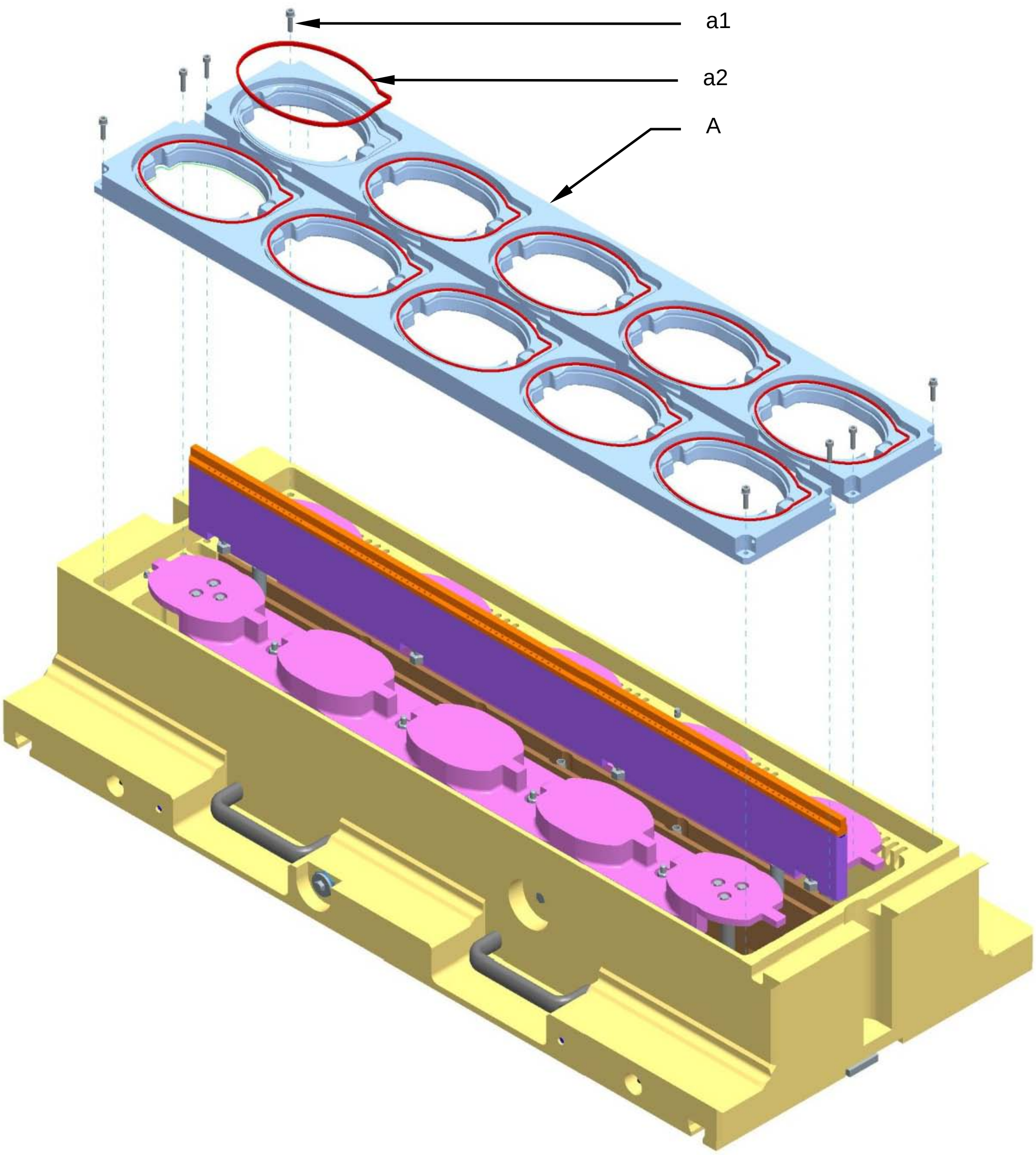


1. Remove the upper cover (**P**).
2. Check the columns (**R**) and the shock-absorbers (**r1**).
3. Remove the internal plate (**Q**) by unscrewing the screws (**q1**) in the washers (**q2**). Check the O-rings (**q3**), and replace if necessary.
4. Check the pins (**s1**) of the resistance (**S**) and make sure they are tight. If necessary, replace them or apply a threadlocker.
5. Check all the springs (**T**) and replace is necessary. Reposition with the flanges (**t1**) well lubricated with high-pressure/temperature grease.
6. To replace the PT100 probe (**u1**), remove the sleeve (**U**) and unscrew the probe-locking screw. Check the state of the sleeve (**U**) O-ring.
7. Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

It is good practice to check the tightness of the sealing tool screws at least once a month.

8.19 LOWER TOOL MAINTENANCE

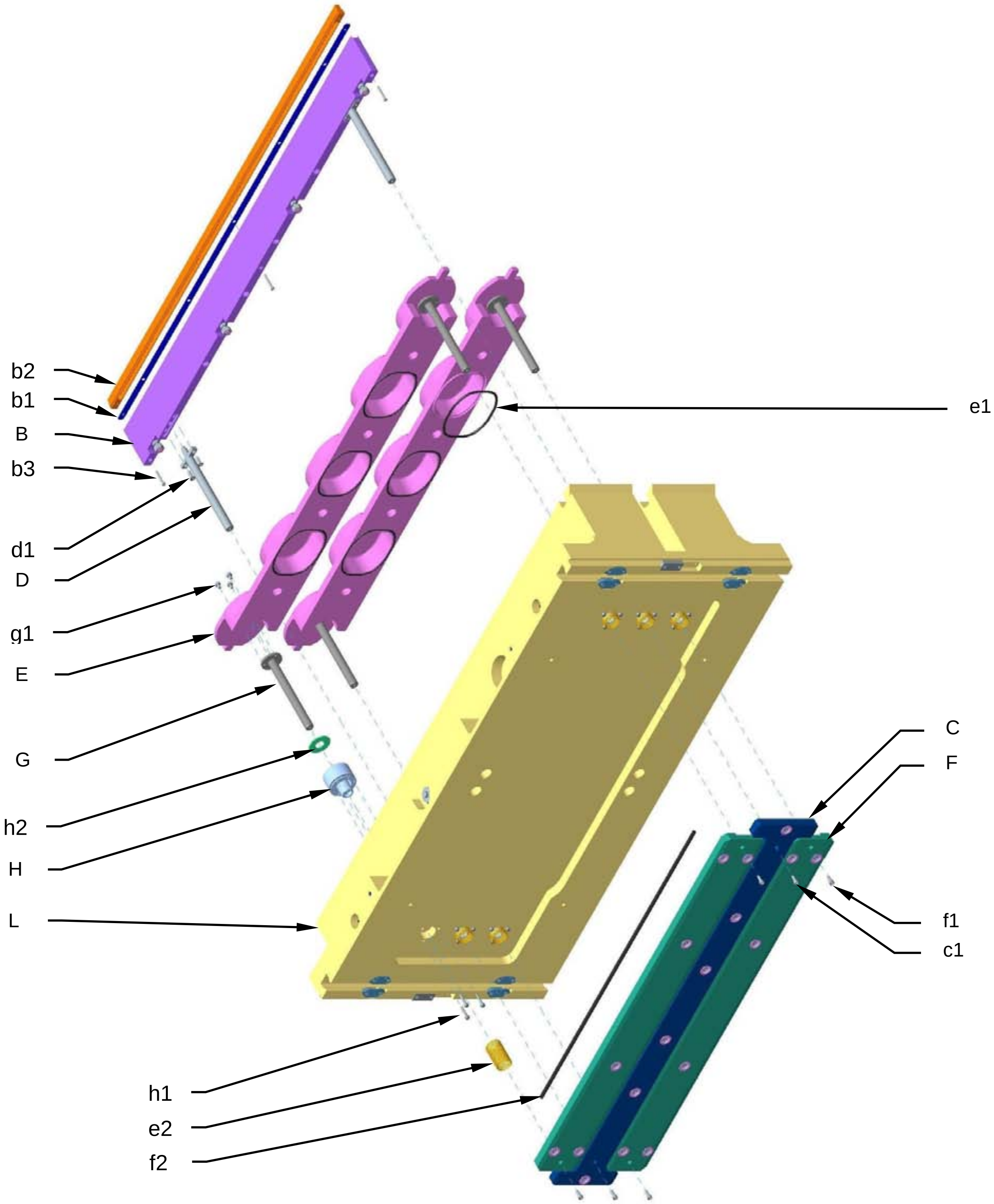
8.19.1 Counter-tool assembly and cleaning sequence



1. Unscrew the securing screws (**a1**) and remove the counter-tool (**A**). Check the gasket (**a2**) and slot and replace any worn parts.

2. Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

8.19.2 Plate and central guide assembly and cleaning sequence



1. Unscrew the screws (**c1**) of the pin plate (**C**) to remove the centre guide (**B**). Check that the holes in the central distributor (**b2**) are free. If necessary, unscrew the screws (**b3**) and clean the holes, checking the gasket (**b1**) as well.
2. Check the columns (**D**) in the centre guide. If they are not in perfect condition, unscrew the securing screws (**d1**) and replace them.
3. Unscrew the securing screws (**f1**) in the pin plate (**F**) and remove the tray-raising plate (**E**). Check the state of the gasket (**e1**) and the springs (**e2**).
4. Check the gaskets (**f2**) of the pin plates (**C** and **F**).
5. Check the state of the columns (**G**) of the tray-raising plate and replace if necessary by unscrewing the screws (**g1**).
6. Check the gaskets and guide bush (**H**). If they need to be replaced, unscrew the screws (**h1**) and remove the bush. Also check the state of the shock-absorber (**h2**).
7. Check the grease nipples of the lower cover (**L**) and make sure they are not clogged. They must allow the grease to lubricate the columns (**D** and **G**). Also make sure all the vacuum and gas passages are clear of obstruction and clean if necessary
8. Tighten all the screws using a torque wrench, keeping to the torques for each diameter, as shown in the attached table (**8.3**).

It is good practice to check the tightness of the sealing tool screws at least once a month.

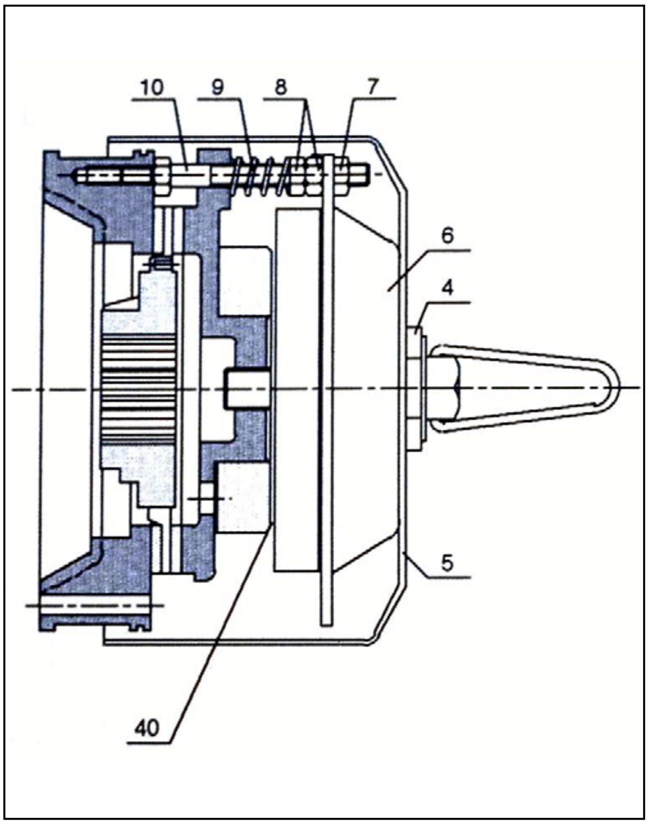
8.20 TABLE OF PRE-LOADS AND TORQUES FOR STAINLESS STEEL SCREWS

		Pre-load in kN								Torque in Nm							
Friction coeff. >		0.1	0.12	0.14	0.16	0.18	0.20	0.30	0.40	0.1	0.12	0.14	0.16	0.18	0.20	0.30	0.40
M4	50	1.38	1.33	1.27	1.22	1.17	1.12	0.90	0.74	0.8	0.9	1.0	1.1	1.2	1.3	1.5	1.6
	70	2.97	2.85	2.73	2.62	2.50	2.40	1.94	1.60	1.7	2.0	2.2	2.3	2.5	2.6	3.0	3.3
	80	3.97	3.80	3.64	3.49	3.34	3.20	2.59	2.13	2.3	2.6	2.9	3.1	3.3	3.5	4.1	4.4
M5	50	2.26	2.18	2.09	2.00	1.92	1.83	1.49	1.22	1.6	1.8	2.0	2.1	2.2	2.4	2.8	3.2
	70	4.85	4.66	4.47	4.29	4.11	3.93	3.19	2.62	3.4	3.8	4.2	4.6	4.9	5.1	6.1	6.6
	80	6.47	6.22	5.96	5.72	5.48	5.24	4.25	3.50	4.6	5.1	5.6	6.1	6.5	6.9	8.0	8.8
M6	50	3.20	3.07	2.94	2.82	2.70	2.59	2.09	1.73	2.8	3.1	3.5	3.7	4.0	4.1	4.8	5.3
	70	6.85	6.57	6.31	6.05	5.79	5.54	4.49	3.70	5.9	6.7	7.4	7.9	8.4	8.8	10.4	11.3
	80	9.13	8.77	8.41	8.06	7.72	7.39	5.98	4.93	8.0	9.1	9.9	10.5	11.2	11.8	13.9	15.0
M8	50	5.86	5.63	5.40	5.18	4.96	4.75	3.85	3.17	6.8	7.6	8.4	9.0	9.6	10.1	11.9	12.9
	70	12.6	12.1	11.6	11.1	10.6	10.2	8.25	6.80	14.5	16.3	17.8	19.3	20.4	21.5	25.5	27.6
	80	16.7	16.1	15.4	14.8	14.2	13.6	11.0	9.1	19.3	21.7	23.8	25.7	27.3	28.7	33.9	36.8
M10	50	9.32	8.96	8.60	8.27	7.91	7.58	6.14	5.05	13.7	15.4	16.7	18.1	19.3	20.3	24.0	26.2
	70	20.0	19.2	18.4	17.7	16.9	16.2	13.1	10.8	30	33	36	39	41	44	51	56
	80	26.6	25.6	24.6	23.6	22.6	21.7	17.5	14.4	39.4	44	47.8	51.6	55.3	58	69	75
M12	50	13.6	13.1	12.6	12.0	11.6	11.1	9.00	7.38	23.3	26.0	28.9	30.8	32.8	34.8	41.0	44.6
	70	29.1	28.1	26.9	25.8	24.8	23.7	19.2	15.8	50	56	62	66	70	74	88	96
	80	38.8	37.4	35.9	34.4	33.0	31.6	25.6	21.1	67	74	82	88	94	100	117	128
M14	50	18.7	17.9	17.3	16.5	15.8	15.2	12.3	10.1	37.1	41.7	45.6	49	52	56	66	71
	70	40.6	38.5	37.0	35.4	34.0	32.6	26.4	21.7	79	89	98	105	112	119	141	152
	80	53.3	51.3	49.3	47.3	45.3	43.3	35.2	29.0	106	119	131	140	150	159	188	204
M16	50	25.7	24.7	23.8	22.8	21.9	20.9	17.0	14.0	56	63	70	75	81	86	102	110
	70	55.0	52.9	50.9	48.9	46.8	44.9	36.4	30.0	121	136	150	162	173	183	218	237
	80	73.3	70.6	67.9	65.2	62.4	59.8	48.6	40.0	161	181	198	217	231	245	291	316
M18	50	32.2	31.0	29.8	28.5	27.3	26.2	21.2	17.5	81	91	100	108	115	122	144	156
	70	69.0	66.4	63.8	61.2	58.6	56.2	45.5	37.5	174	196	213	232	246	260	308	334
	80	92.0	88.5	85.0	81.6	78.1	74.9	60.7	50.1	232	261	285	310	329	346	411	447
M20	50	41.3	39.8	38.3	36.7	35.2	33.8	27.4	22.6	114	128	142	153	164	173	205	223
	70	88.6	85.4	82.0	78.7	75.4	72.4	58.7	48.1	244	274	303	328	351	370	439	479
	80	118	114	109	105	101	96.5	78.3	64.6	325	366	404	438	467	494	586	639
M22	50	51.6	49.8	47.9	46.0	44.1	42.3	34.3	28.3	154	174	191	208	222	234	279	303
	70	61.5	59.3	57.0	54.7	52.5	50.3	40.9	33.7	182	206	227	247	263	279	332	361
	80	148	142	137	131	126	121	98.2	80.9	437	494	545	593	613	670	797	866
M24	50	59.6	57.4	55.1	52.9	50.7	48.6	39.4	32.6	197	222	243	264	282	298	354	385
	70	70.9	68.3	65.6	63.0	60.4	57.9	47.0	38.8	234	264	290	314	336	355	421	458
	80	170	170	157	151	145	139	113	93.1	561	634	696	754	806	852	1010	1099
M27	50	75.6	72.9	70.1	67.3	64.5	61.9	50.2	41.5	275	311	344	377	399	421	503	548
	70	90.0	86.8	83.4	80.1	76.9	73.7	59.8	49.4	328	371	410	444	475	502	599	652
M30	50	91.9	88.6	85.2	81.7	78.4	75.2	61.0	50.3	374	423	467	506	540	571	680	740
	70	104	105	101	97.3	93.3	89.5	72.6	59.9	445	503	556	602	643	680	809	881

¶

All the stainless steel screws, nuts and bolts used by G.MONDINI S.p.A. are class 70. We only guarantee this type. Use only spares obtained from G.MONDINI S.p.A.

8.21 CLUTCH-BRAKE UNIT ASSEMBLY (COEL)



- **Air gap adjustment**

Air gap 40 (i.e. the distance between the two magnetic nuclei of the electromagnet and of the mobile anchor) must amount to three tenths of a millimetre.
A first inspection after the machine's first week of work would be appropriate, with subsequent inspections after 30 days because, due to wear by the brake disk gaskets, the air gap tends to increase. It is very important that the air gap does not go over 0.5 mm.
To bring the air gap back to the size it should be, it is necessary to act on the couple of nuts (7-8) that lock on the electromagnet, in order to have it travel towards the mobile anchor. Once done, accurately check the nut torques.

- **Adjustment of the braking torque**

The braking torque is proportional to the compression of springs 9, that can be adjusted by acting on nuts 8, i.e. unscrewing them to decrease and screwing them to increase torque. The spring compression must be uniform at all times.

- **Electromagnet replacement**

Remove screw 4, remove the guard 5, remove the six magnet terminals, remove the three nuts 7, then slide the electromagnet 6 out of the columns 10.
Rearrange the new electromagnet onto the columns, being very careful that when reconnecting the terminals, the relative colours all correspond with one another.
Lock in nuts 7-8 and check to make sure that the electromagnet works correctly.

- **Brake disk replacement**

Remove screw 4, remove the guard 6, and remove the three nuts 7 without detaching the terminals. Remove nuts 8 and spring 9. Fix on the new brake disk.

9 DISPOSAL

9.1 DISPOSAL

The final user shall provide for the disposal of and elimination of the materials making up the component parts of the machine in accordance with EU and local applicable regulations.

If the machine or its parts are to be scrapped , take the necessary safety precautions to avoid any risks associated with the disposal of industrial equipment.

In particular, great care must be taken at the following stages:

- Removal of the machine from the working area
- Transportation and handling
- Dismantling
- Material sorting

IMPORTANT

For material sorting, recycling or disposal, reference must be made to National and regional Laws on industrial solid waste management.

If the machine is to be scrapped, the and user shall take particular care in disposing of the following materials in accordance with local regulations.

- Makrolon, PVC, Plexiglas or other material used for guards.
- Plastic used for compressed air pipes.
- Sheathed electric cables.
- Rubber belts
- Any toxic and corrosive substances.

If the materials used for normal operation, such as lubricants, grease and water condensate, are not disposed of properly in compliance with the applicable laws and regulations, there may be residual risks for:

- Environmental pollution
- Intoxication of personnel involved in disposal operations

9.2 DEMOLITION

Demolition may involve the same residual risks envisaged for installation of the system and connection to the power mains.

It is thus recommended to perform these operations with great care.

A SEALING TOOL TROLLEY CPS/R

INDEX

A.1 GENERAL DESCRIPTION

The CPS/R type die holder is a manual device used for die insertion and removal in closing machines.

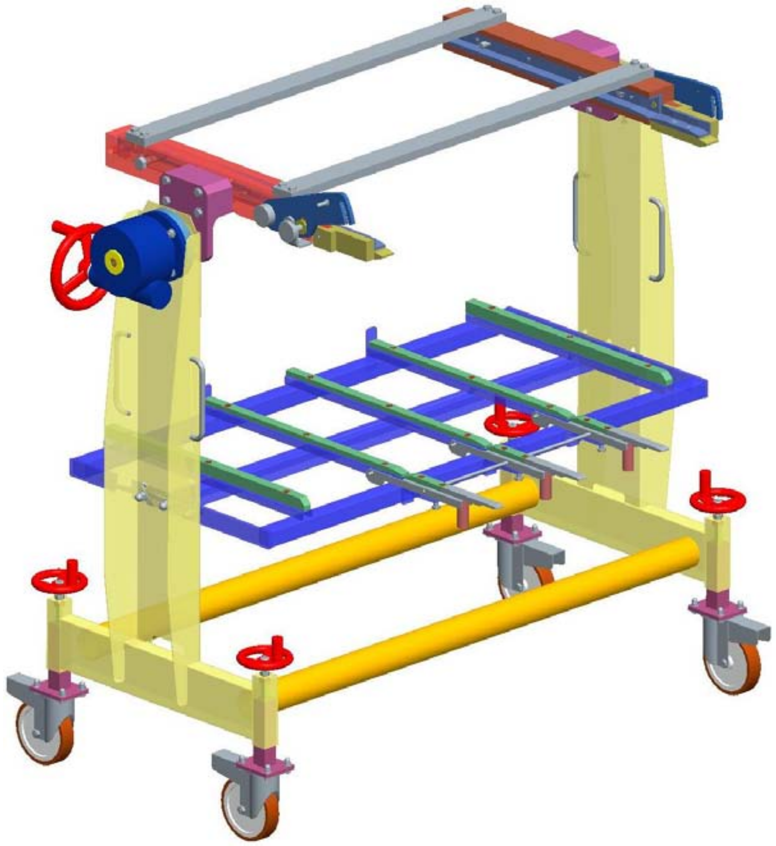
The trolley is made up of the following:

- Stainless steel framework and plastic slides to ease the sliding of parts.

- Gearbox for manual tipping.

- A bottom stand.

The die holder only function is to extract or position sealing dies inside the working area without performing unusual actions by using unsuitable devices.
The proper use of the die holder ensures maximum safety in handling sealing dies that, depending on the type of the closing machine, may often have significant sizes and weights.
The figure below shows the CPS/R trolley.



A.2 LIST OF HAZARDS

A.3.1) Upper section of the trolley

A.3.2) Rotation device

A.3.3) Locking pins

A.3.4) Castor wheels

A.3.5) Sealing tool

A.3 RISK ANALYSIS

A.3.1 Upper section of the trolley

The upper holder is the part that, once attached to the closing machine, supports and eases extraction and protects the sealing upper die.

- Solution: Scrupulously observe the indications on the anchorage of the holder before releasing the die from the retaining pistons.

- Residual risks: None provided that the instructions are followed to the letter.

A.3.2 Rotation Device

The rotation device, connected to the upper holder, supports, protects and overturns the upper sealing die extracted from the working area, allowing better maintenance (if necessary). If the sealing tool is not secured properly, it may fall off and cause the operator very serious injury.

- Solution: Scrupulously observe the indications on the anchorage of the holder before releasing and handling the upper die.

- Residual risks: None provided that the instructions are followed to the letter.

A.3.3 Locking pins

The trolley is fitted with numerous mechanical locking devices allowing the sealing tool to be removed from the work area in complete safety. Before performing of these operations, please read this manual and identify and check the holder parts and safety areas before using it to take a die.

- Solution: Follow these recommendations before attempting to move the sealing tool.

- Residual risks: None provided that the instructions are followed to the letter and don't approach the hands to the pivots.

A.3.4 Castor wheels

The die holder is equipped with wheels specifically sized to ease handling under load. They have mechanical brakes acting as an added safety measure while the sealing tool is being loaded. In case the brakes are not actuated, a sudden moving back of the holder when loading a part of die could cause serious and deep injuries to the users' limbs.

- Solution: Always apply the brakes before loading or unloading the trolley and put on special wear safety shoes.

- Residual risks: None provided that the instructions are followed to the letter.

A.3.5 Sealing tool

The sealing die seals a thermoplastic film on the container thanks to very high temperatures obtained by means of an electrical resistance.
Moreover, depending on the type of the closing machine, the die may have a significant weight and, therefore, it is important to take into account these variables when removing it.

- Solution: The sealing tool stands on runners to make it easier to extract. The electric resistance supply must always be switched off before removing the sealing tool.

- Residual risks: The die weight is always a factor to be taken into account and its temperature, if just used, is still high and may cause deep and serious burns. For this reason, it is advisable to always wear suitable protective gloves.

WARNING!

The sealing die reaches high temperatures and, during removal, could be still hot and cause serious burns and injuries. Wear safety gloves in all working steps.

A.4 SAFETY DEVICES

The safety of G. MONDINI S.p.A. die extraction and transport holders is guaranteed thanks to technical and mechanical devices tested over time. All the safety devices on these machines have been thoroughly tested and fall into two categories:

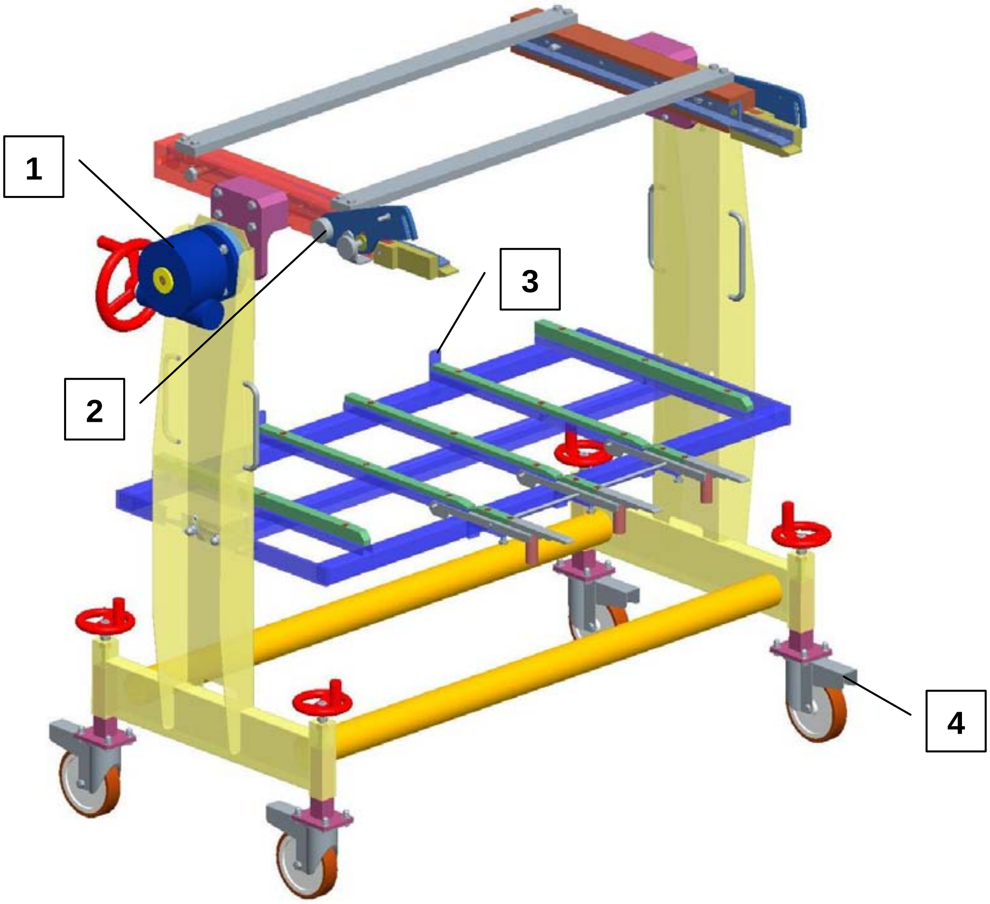
1) Direct safety devices.
2) Indirect safety devices.

A.4.1 Direct safety devices

Direct safety systems used by us are active elements, such as safety microswitches, emergency stop buttons and general switches specific to industrial equipment.
However, since the die holder is an exclusively mechanical operating device, no direct safety systems are used for it.

A.4.2 Indirect safety devices

Indirect safety systems are mechanical passive elements (guards, pins, hooks, etc.) that, though independent from the operator's will, prevent the operator from accidentally coming into contact with risk areas. Each mechanical device is equipped with these safety systems in hazard areas and, if it is the case, in manual working areas. The safety devices on the trolley are shown in the diagram and explained below:



A.4.2.1 Safety Device no. 1

This safety system refers to the gear unit connected to the rotation device of the holder upper part.
The manually operated gear unit ensures absolute stability, thanks to slow rotation, preventing uncontrolled movements of the upper extraction holder.
In case the rotation handle is accidentally rotated, the use of a high gear ratio in the gear unit prevents sharp rotation, ensuring slow and safe movement.

A.4.2.2 Safety Device no. 2

This safety system comprises locking knobs, which create a mechanical lock between the front and the rear part of the upper tool holder plate.
This lock prevents the die from coming out the slides even if the die is in horizontal or vertical position.
Moreover, a mechanical hook, once attached to the closing machine upper plate, firmly constrains the extraction holder to the machine framework when removing the upper die.

A.4.2.3 Safety Device no. 3

This safety system concerns the rear locking device. It is connected to the framework of the lower slide bed and halts, as a sort of limit switch, the lower die during removal.

A.4.2.4 Safety Device no. 4

This is a brake on the castors of the trolley. After constraining the holder to the machine, the wheel brakes shall be always actuated, preventing moving back caused by the die weight or the safety system (2) wrong connection, thus avoiding any injuries or die falling during removal.

Never remove, replace or modify the safety systems.

Modification to any of the safety devices could also cause a malfunction and bring the entire system to a halt.

IMPORTANT NOTE

Do not modify any parts of the system nor remove or bypass any protection system without consulting G. MONDINI S.p.A. before. Failure to observe these warnings shall RELIEVE the company from any LIABILITY. The Manufacturer cannot be held liable for:

A) THE USER'S SAFETY

B) EFFICIENT OPERATION of the machine supplied.

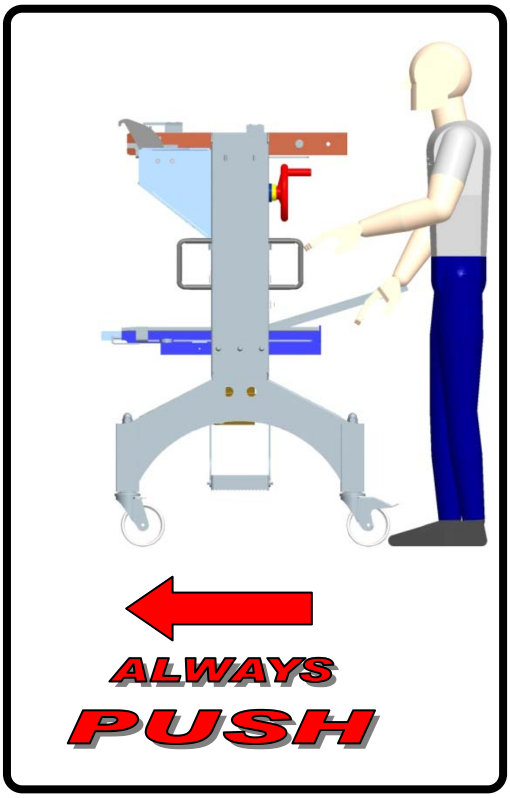
IMPORTANT:

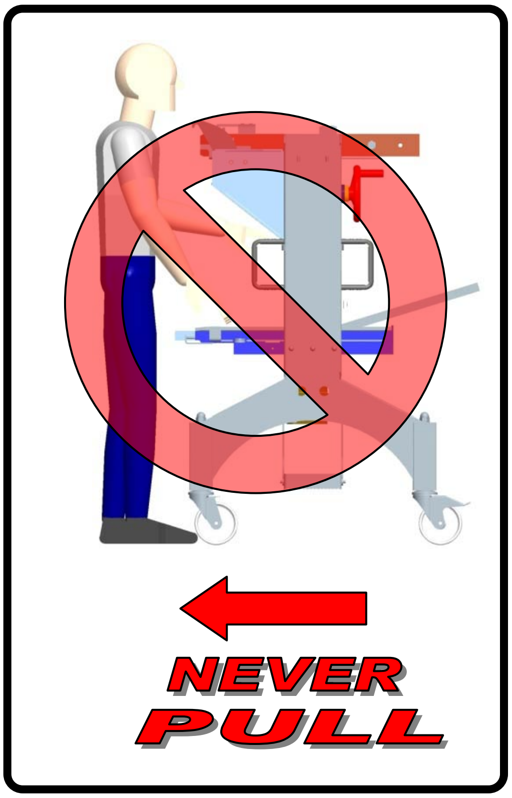
If the machine is to be connected to an existing one, unless agreed otherwise, the user must provide adequate protection in the area of connection,G. MONDINI S.p.A. declines all liability if failure to do so causes damage or injury.

WARNING!

When moving or shifting the tool carrier trolley, the operator must strictly PUSH it and NEVER pull it. This to prevent any possible crushing hazards occurring to the operator, due to the trolley possible capsizing and/or bucking.

WHEN MOVING OR SHIFTING THE TROLLEY, ALWAYS PUSH IT. NEVER PULL IT!





A.5 OPERATING INSTRUCTIONS

Like everything made by MONDINI, the CPS/R trolley is delivered fully tested, so only a few simple operations need to be carried out before it can be put into use.

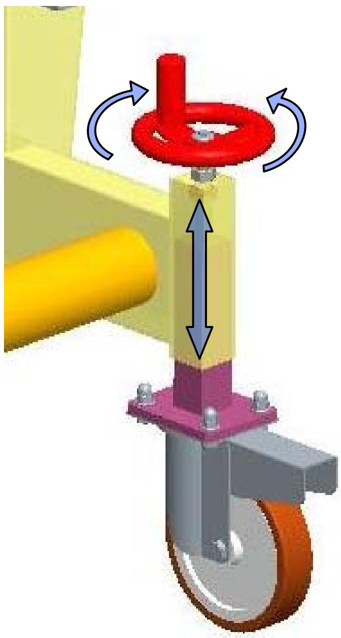
After unpacking, put the equipment in optimum working conditions.
By using a level, make sure that the holder is in its proper horizontal working position, otherwise use the height adjustment devices above the handling wheels.

This device has the only function to extract or insert in the right position the desired sealing die, with no risks.

IMPORTANT!

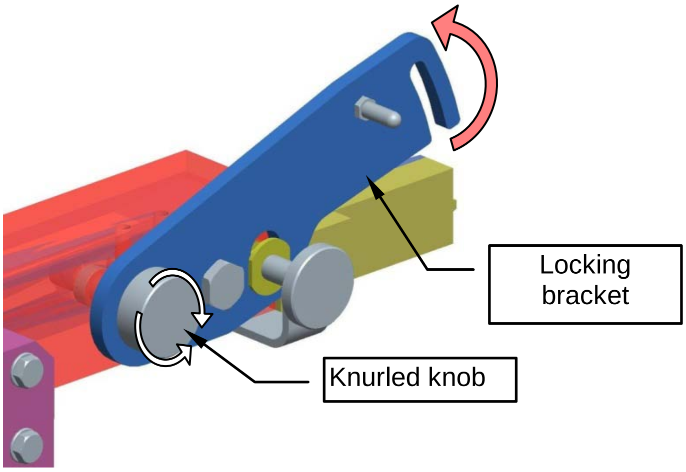
Before beginning the sealing tool extraction/insertion procedure it is necessary to check that:

- all the four wheels of the trolley are contemporarily on the floor;
- the trolley slide guides are to the same height of the sealing machine slide guides;
- the trolley is in a correct horizontal position.

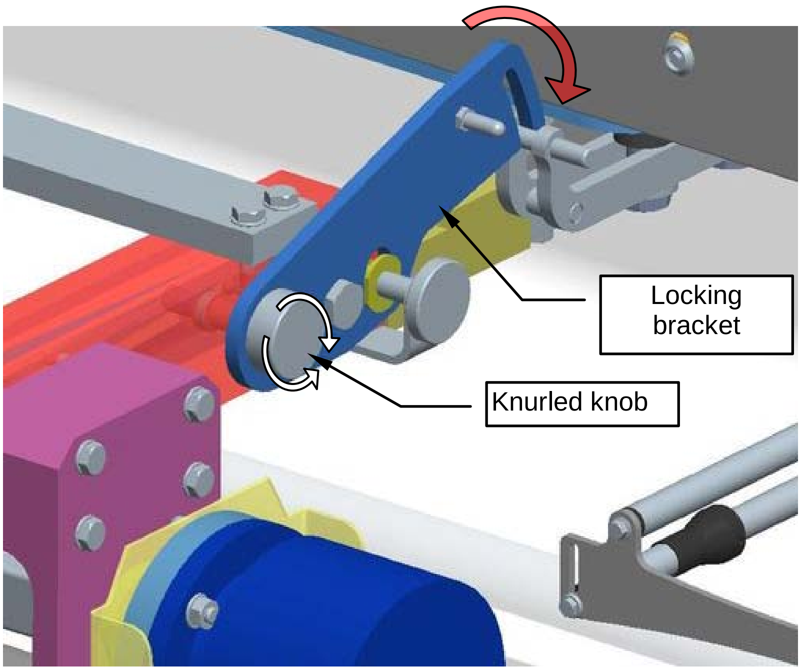


A.5.1 How to remove a tool

1. Loosen the knurled handwheel and release the retaining brackets. Lift the brackets and tighten the handwheel to lock them.

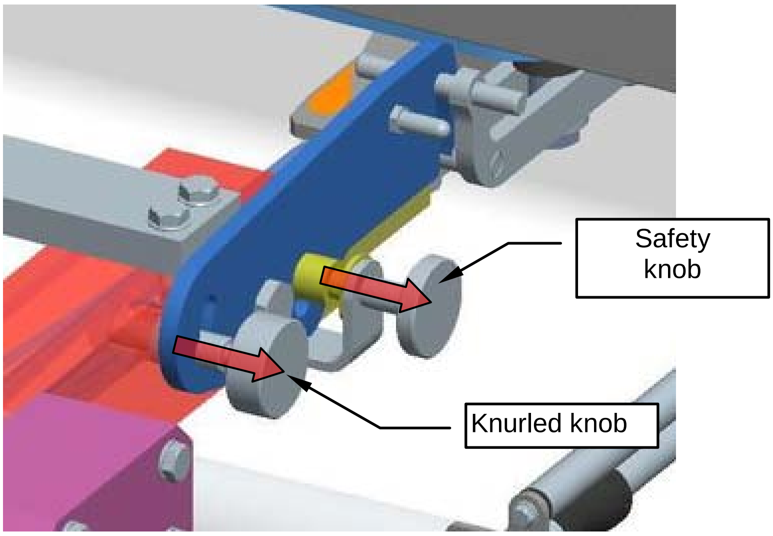


2. Move the die extraction holder closer to the closing machine. Extract the hooks from the closing machine, unscrew the knurled handwheel and attach the holder by lowering the retaining brackets inside the slots.

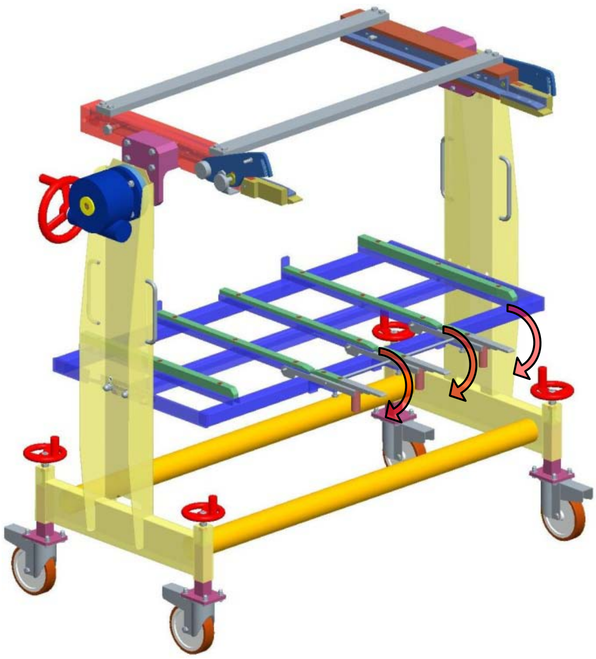


3. Apply the brake of the holder wheels.

4. Make sure that the brackets are at end stop, fully unscrew the knurled handwheel by fully retracting the pin inside the slide and extract the safety pin.



5. Lower the lower slides so that the part below properly leans on the machine lower plate. The guides are correctly positioned when they are horizontally aligned with the direction of extraction of the lower tool. If necessary, adjust the position on the lower knobs.



6. Remove the die:

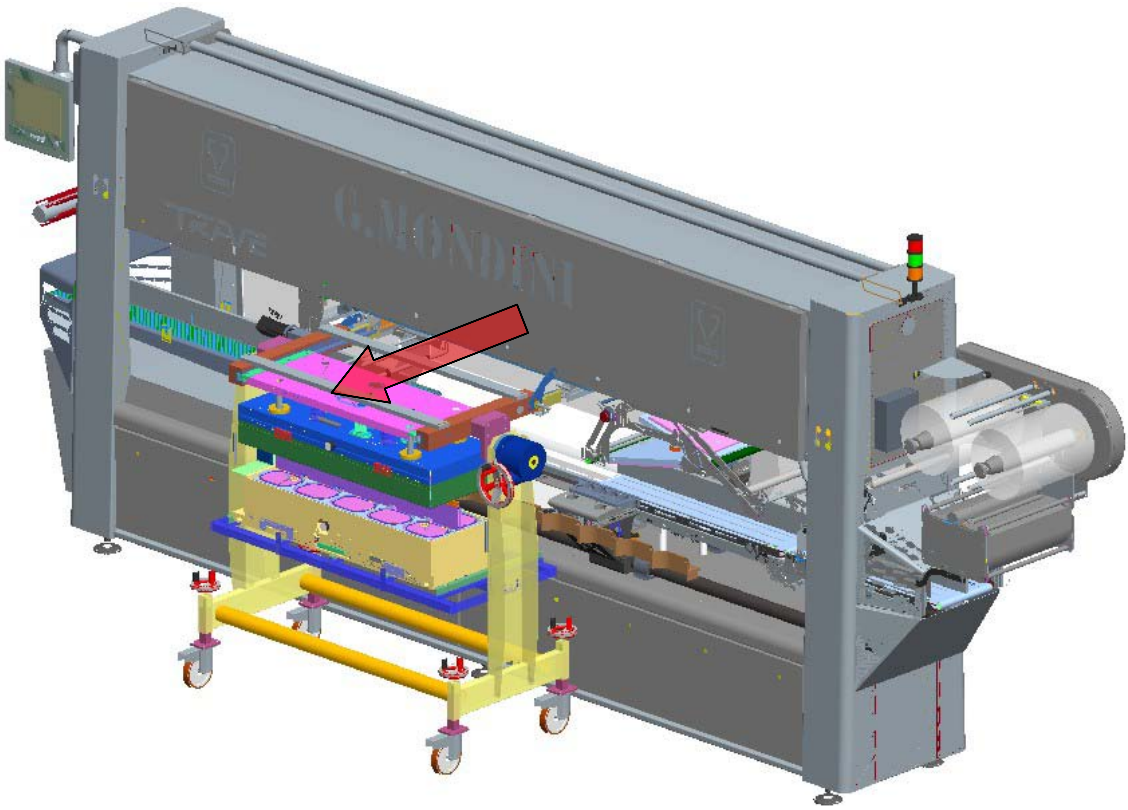
- Remove any residual film from inside the tool by hand.

- Disable the die resistances from the operator panel and disconnect the electrical connector of the sealing resistance.

> WARNING!
> **Before disconnecting the plug HARTING you must disable the operation of the resistance of the tool!**

- Operate the automatic unlocking selector of the die from the operator panel.

> WARNING!
> **The die unlocking selector releases both the upper and lower parts of the die. Always use extreme caution.**

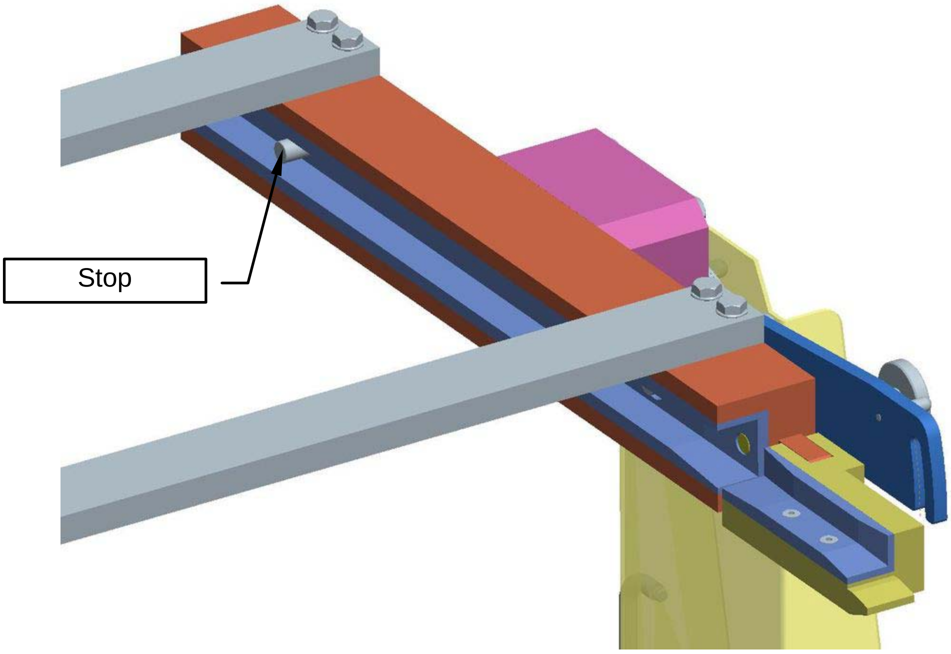


WARNING!

The sealing die reaches high temperatures and, during removal, could be still hot and cause serious burns and injuries. Wear safety gloves in all working steps.

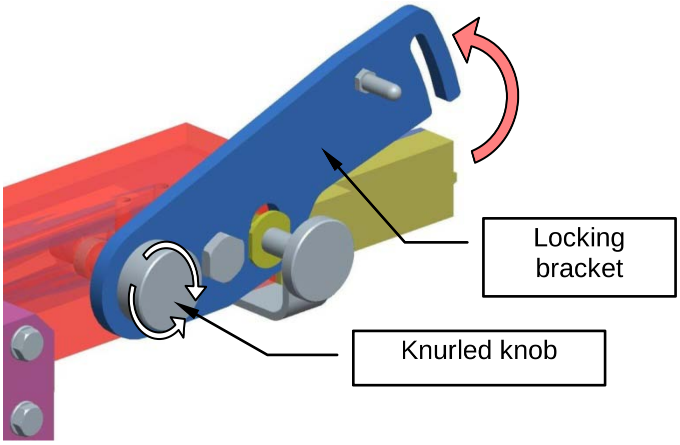
7. Take the lower die by the handles and drag it until it stops to the catches on the slides.

8. Take the upper die by the handles and drag it until it stops to the catches on the slides.



9. Screw the knurled handwheel and insert the safety pin, preventing the upper die from moving during transport.

10. Lift the lower slides and place them in their fully raised position.



11. Raise the upper plate anchoring brackets and tighten the knurled knob.

ATTENTION!

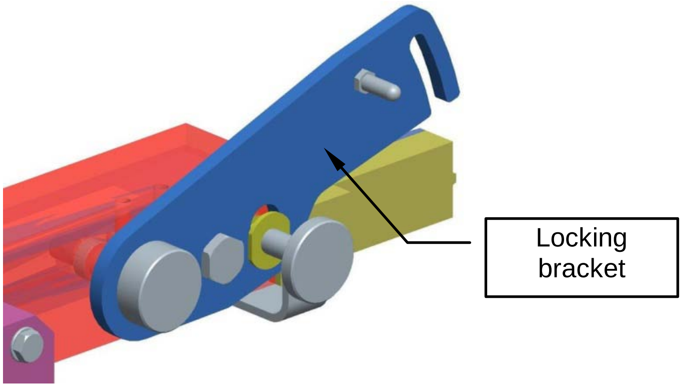
When the trolley is uncoupled from the machine, the operator is in complete safety only if the two knurled knobs have been tightened. In case of failure to perform this procedure, G. MODINI S.p.A. accepts no responsibility for any damages.

12. Release the brake of the holder wheels and take the die to the storage area.

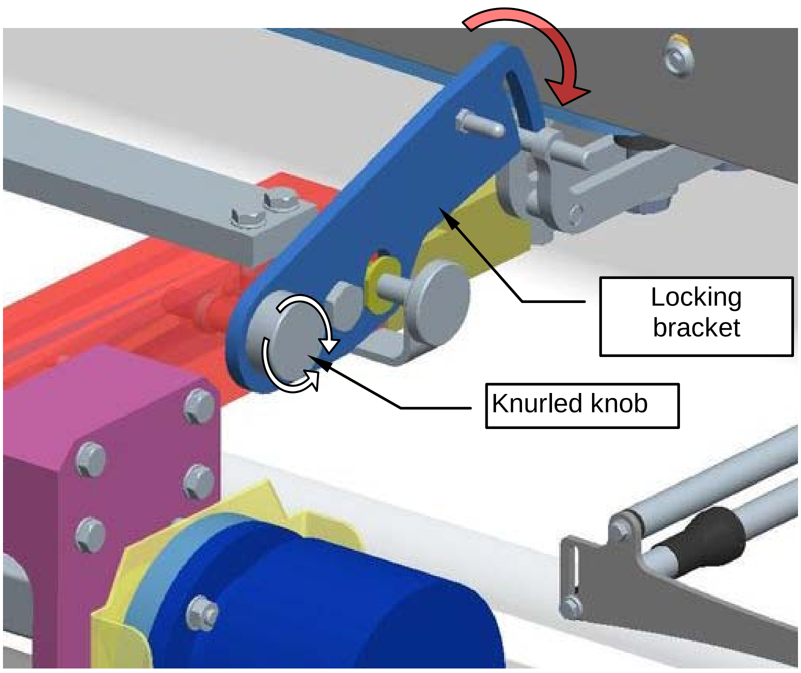
A.5.2 How to insert a tool

1. Take the die holder with the new die and go near the closing machine on the insertion side.

2. Loosen the knurled handwheel and release the retaining brackets. Lift the brackets and tighten the handwheel to lock them.

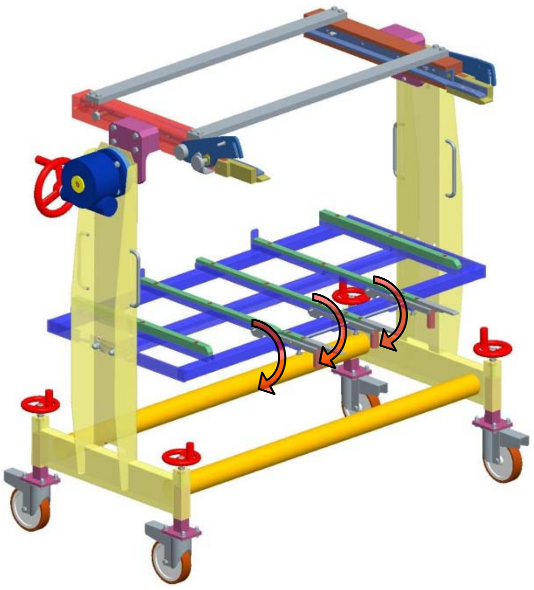


3. Move the die extraction holder closer to the closing machine. Extract the hooks from the closing machine, unscrew the knurled handwheel and attach the holder by lowering the retaining brackets inside the slots.

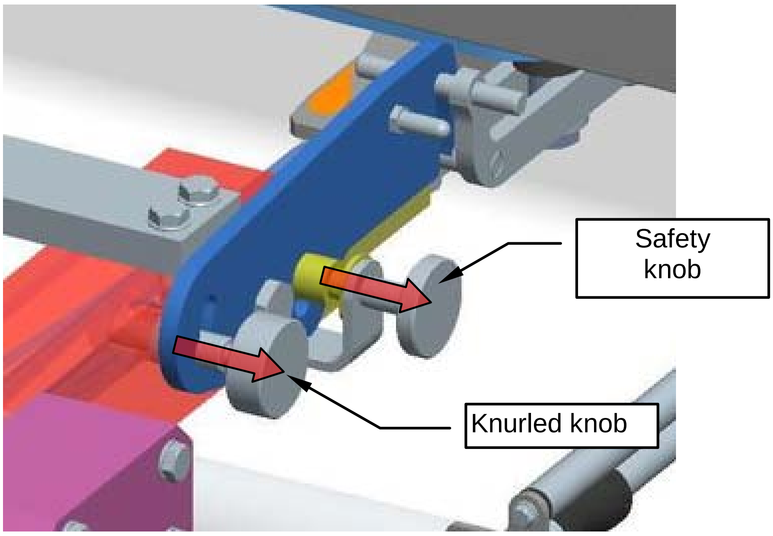


4. Apply the brake of the holder wheels.

5. Lower the lower slides so that the part below properly leans on the machine lower plate.

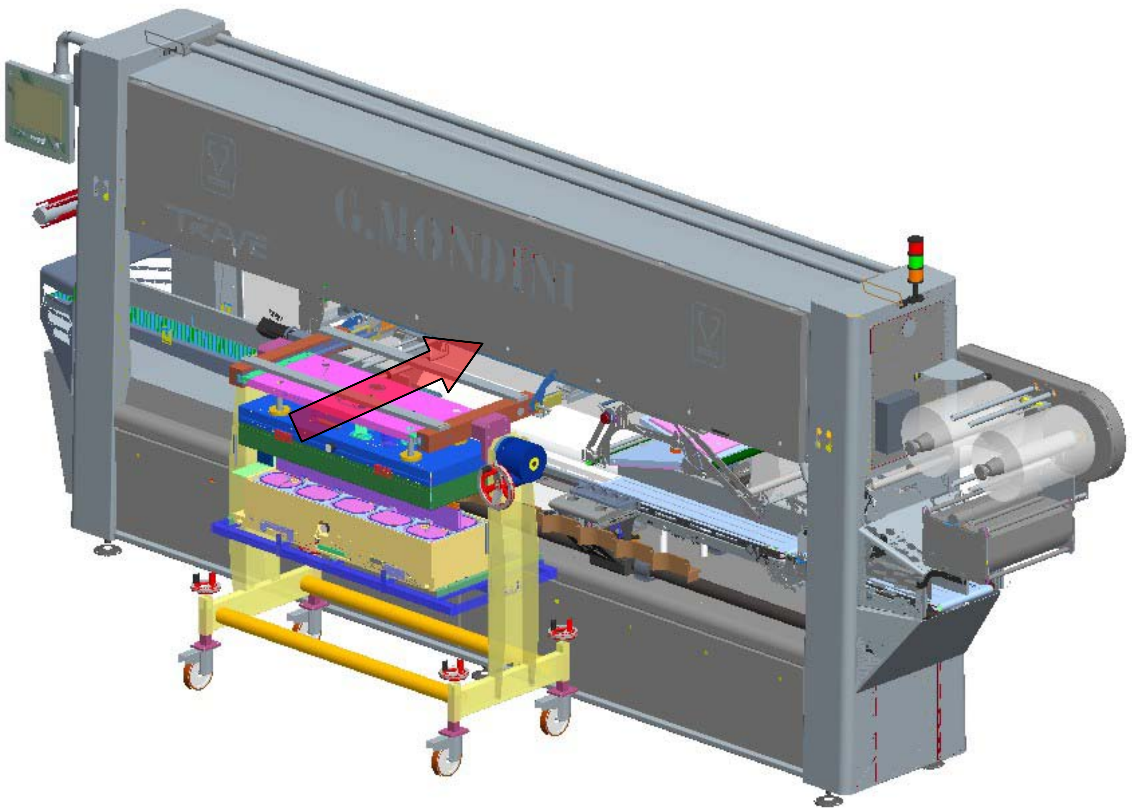


6. Make sure that the brackets are at end stop, fully unscrew the knurled handwheel by fully retracting the pin inside the slide and extract the safety pin.



7. Take the upper die by the handles and drag it to the most advanced position on the slides of the machine (the position is given by specific catches).

8. Take the lower die by the handles and drag it to the most advanced position on the slides of the machine (the position is given by specific catches).



9. Operate the automatic locking selector of the die from the operator panel.

10. Insert the electric connector for tool heating.

ATTENTION!
Before connecting the HARTING socket, make sure that the operation of the die resistance has been enabled!

11. Enable the operation of the die resistances from the operator panel.

12. Lift the lower slides. Release the trolley from the machine and resume production as instructed in the operator handbook.

A.6 MALFUNCTIONS - SOLUTIONS

The sealing tool trolley will only malfunction if parts are wrongly adjusted or not kept in proper working order.

Malfunction	Possible Solutions
The safety pins are difficult to pull out	Check for dirt or encrustation on the outer surface of the pin. Remove any dirt and grease the safety pins.
The lower guides are not aligned	Check the position of the height adjusting pins below the guides and adjust as necessary. Check for dirt or other material accumulated under the guides. Remove any dirt and grease the guides.
The castors stick or do not move at all	Check for obstructions and clean thoroughly. Check that the brake is not engaged. Clean and lubricate the castors to allow the bearings to move freely.
The sealing tool does not slide on the guides	Check there are no foreign bodies and clean accurately. Check that the plastic slides are not damaged and the fixing screws have not come out preventing sliding. In case of serious wear, replace the part concerned. Check that the safety pins have been properly extracted.

A.7 MAINTENANCE - CLEANING

A.7.1 Maintenance

The trolley does not require any particular maintenance. The only point to bear in mind is to handle the guides and safety pins with care. Never hit them with a hammer or similar tool if they do not work.

Lubricate the bearings and moving parts periodically. Keep clean all the safety devices used when replacing and handling the sealing tools.

A.7.2 Cleaning

In case there is residual product or dirt from die handling on the holder, comply with the recommended procedures and products described in the cleaning protocol.

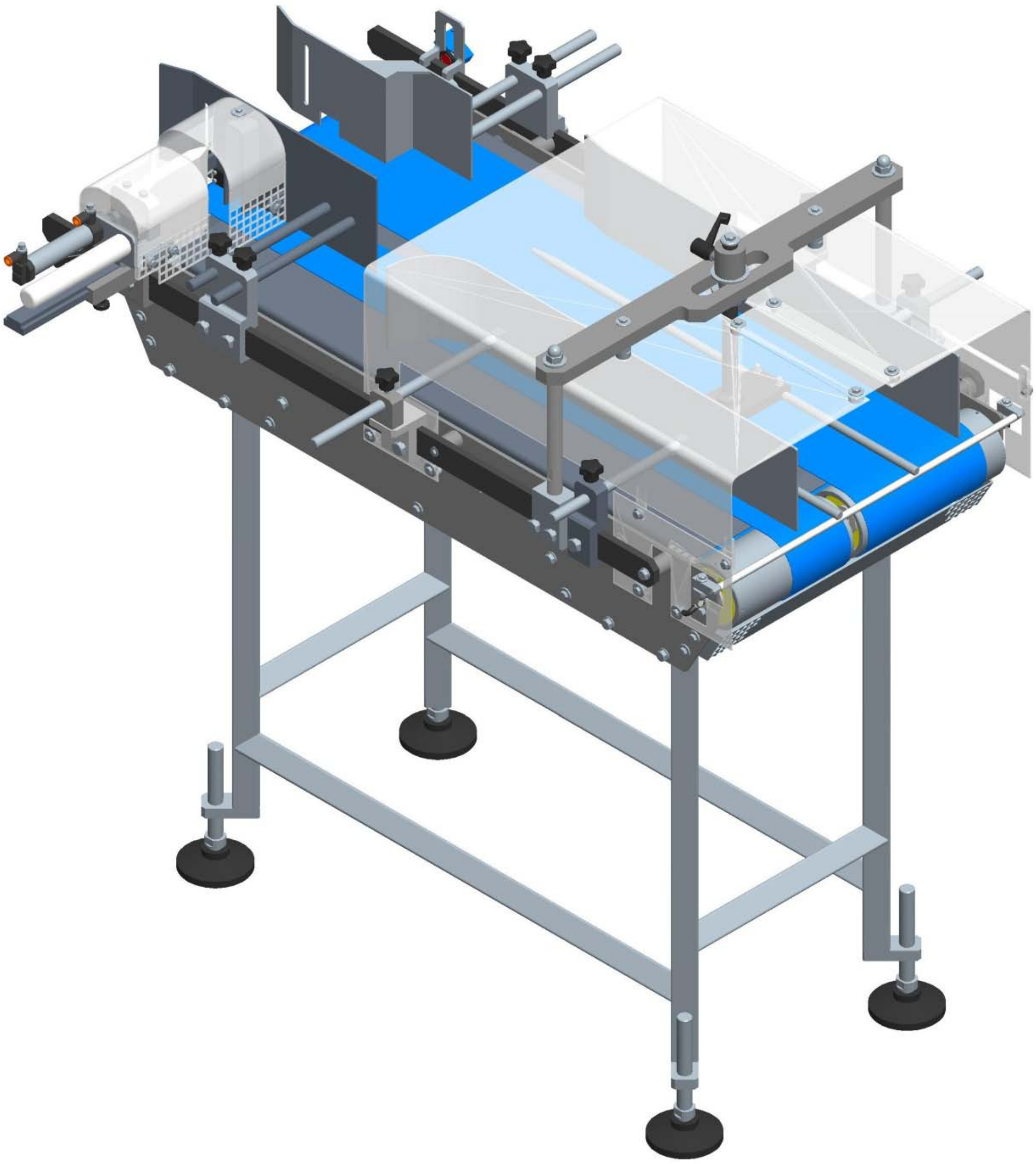
	Wear safety shoes.
	Use anti-cut and burn protection gloves.
	See the use and maintenance handbook.
	This symbol indicates the direction of rotation of the part .
	**Warning!** Danger of limbs being crushed.
	**Warning!** Hot surface hazard.
	Constant hazard pay attention when handling.

B TRAYS ROTATION BELT

INDEX

B.1 GENERAL DESCRIPTION

This belt is made by a inox steel structure where a belts operates moved by two drum motors. This belt has a side guide that allows the trays rotation modifying the advancement mode.



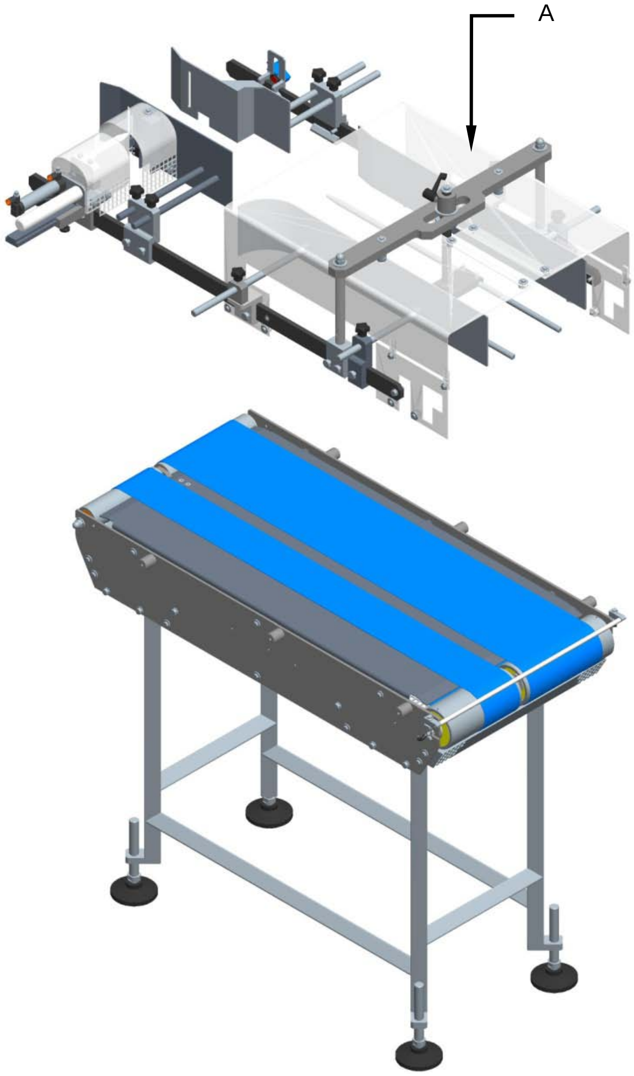
B.2 MAINTENANCE

B.2.1 Belts replacement

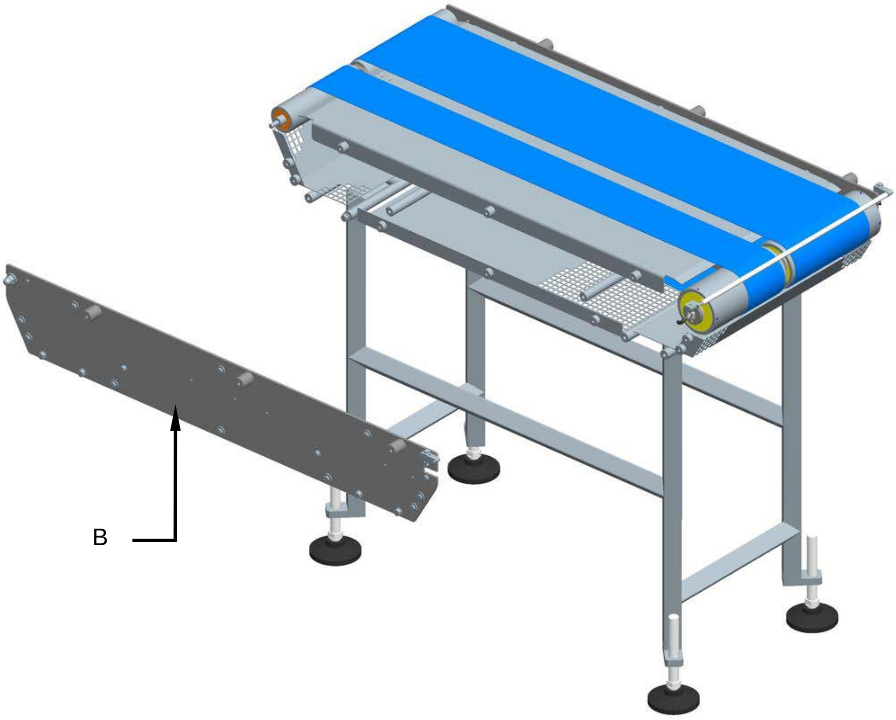
Machine status	Power sources deactivated
Equipment	Standard workshop tools

Proceed as follows to replace the Polycord belts:

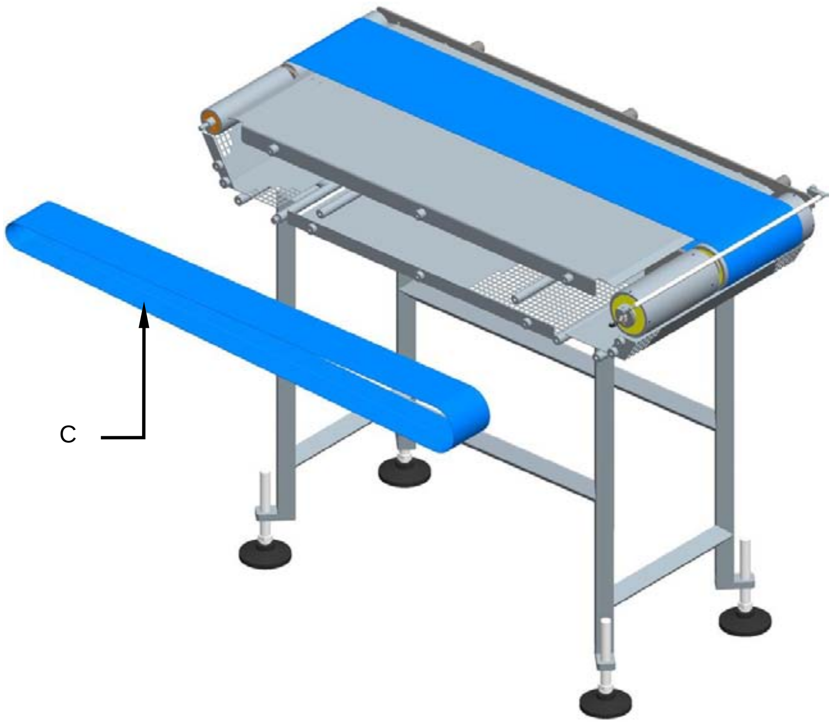
1. Disconnect the power supply of the belt.
2. Remove the guides group (**A**).



3. Remove the support (**B**).



4. Remove and replace the belt (**C**).



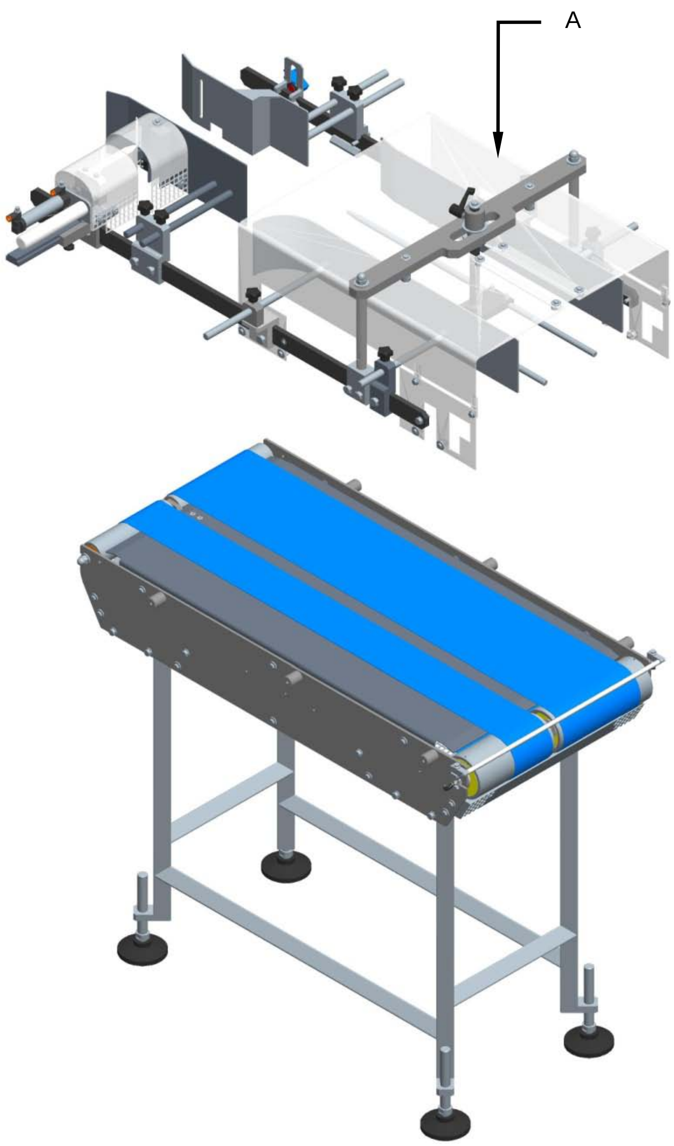
5. Reassemble following the preceding procedure to bashful.

B.2.2 Motor drums replacement

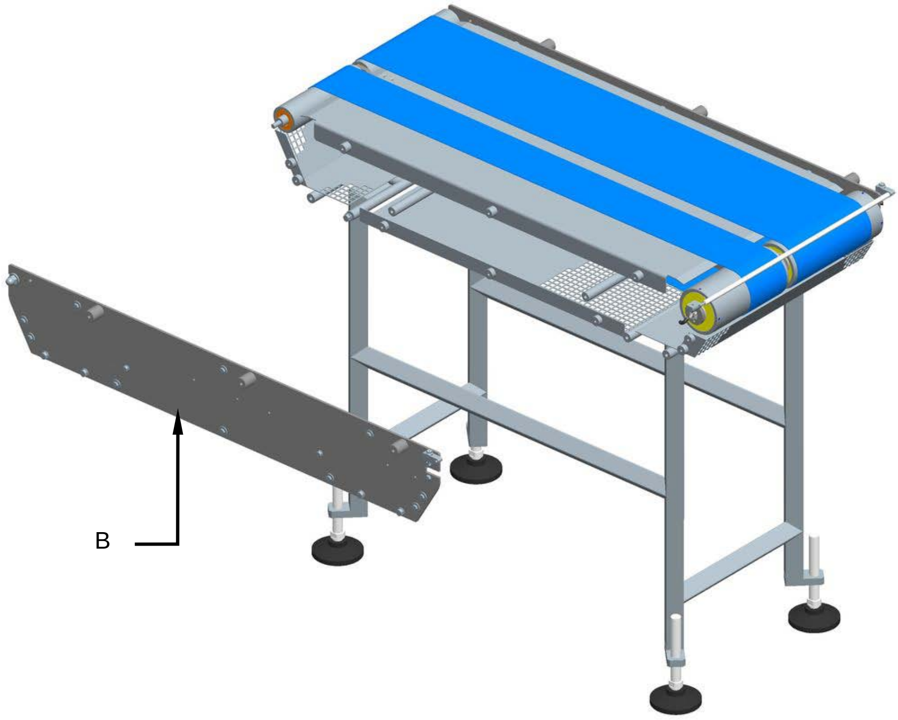
Machine status	Power sources deactivated
Equipment	Standard workshop tools

Proceed as follows to replace the motor drums:

1. Disconnect the power supply of the belt.
2. Remove the guides group (**A**).



3. Remove the support (**B**).



4. Disconnect the electrical wire and replace the drum motor (**C**).

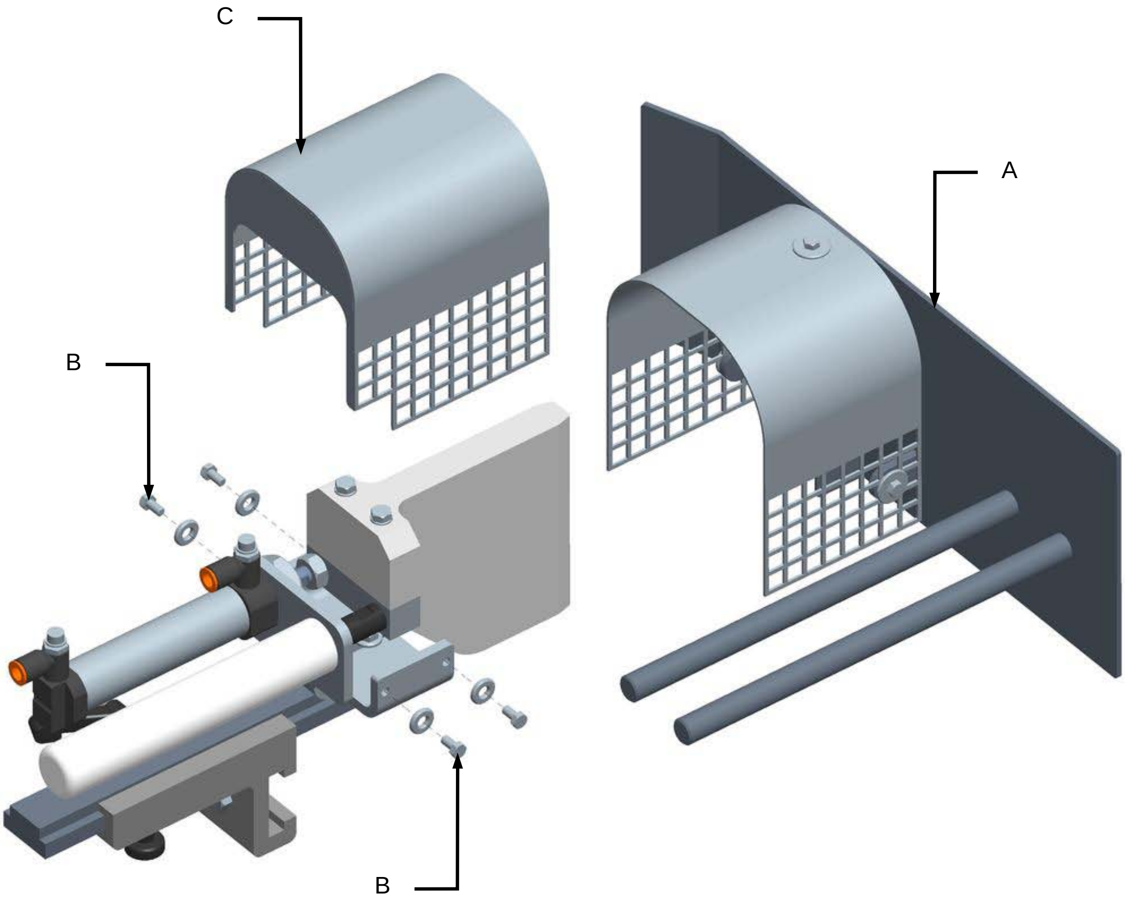
5. Reassemble following the preceding procedure to bashful.

B.2.3 Stop cylinder replacement

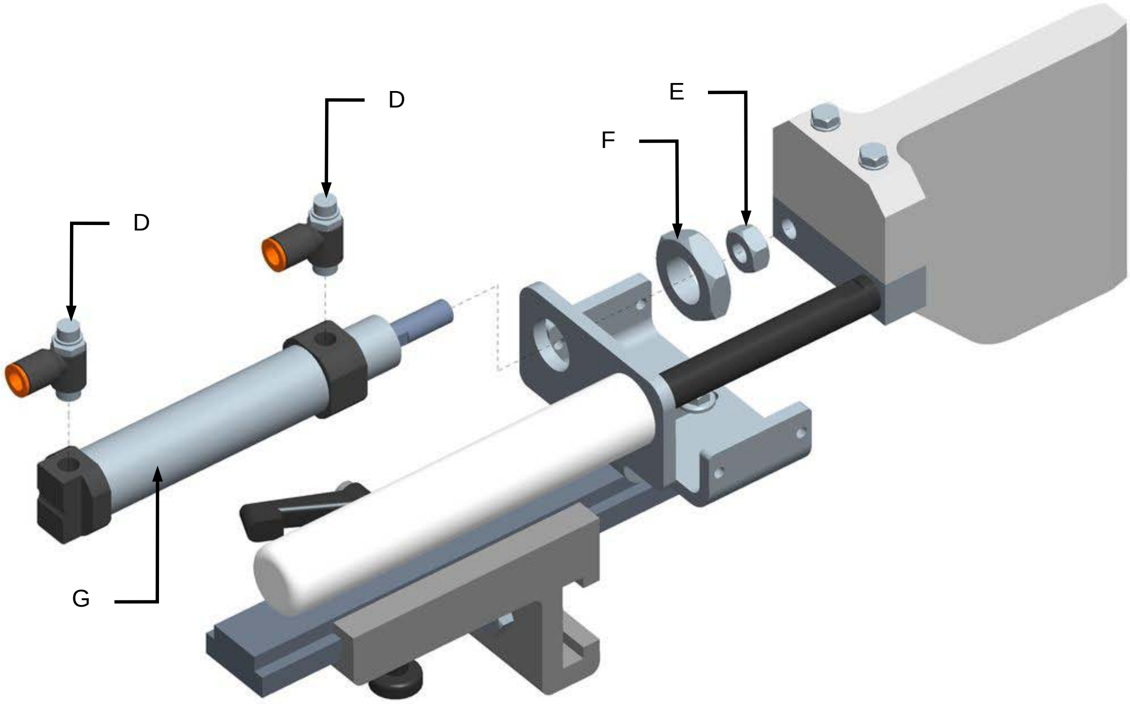
Machine status	Power totally disconnected
Equipment	Standard workshop tools

For stop cylinder replacement, proceed as follows:

1. Shut off the compressed air circuit.
2. Remove the guide (**A**).
3. Unscrew the screws (**B**) and remove the protection (**C**).



4. Remove the joint fittings (**D**).
5. Unscrew the nuts (**E**) and (**F**).
6. Remove and replace the cylinder (**G**).



7. Reassemble all parts back on again by reverse procedure.